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EUROPEAN SIMULATION AND MODELLING CONFERENCE 2026

Preface

Dear participants,

It is our great pleasure to welcome you to Celebrating 40 years ESM® in Europe held jointly with GAME-ON®2026 in Metz, France. ESM® and GAME-ON® aim to bring together researchers, academics, engineers, practitioners, professionals and the well-established EUROSIS scientific community to share their latest findings, exchange ideas, and discuss emerging challenges and opportunities in the field of modelling and simulation, and, in the field of simulation and artificial intelligence in computer games.

The contributions collected in these proceedings reflect the diversity and quality of the research presented during the conference. They cover a broad range of topics, including computational modelling, data and knowledge representation, artificial intelligence and machine learning, autonomous and multi-agent systems. These approaches are investigated across a wide variety of scientific and industrial applications from smart manufacturing, supply chains, logistics and operations, autonomous, robotic and human-centric systems to energy, infrastructure and environmental systems and healthcare, safety and emergency management, illustrating the multidisciplinary and rapidly evolving nature of the field of ESM®. GAME-ON® will once again bring us together to explore the community's favourite topics, ranging from game design, player interaction, virtual reality, and immersive technologies to interactive entertainment and multiplayer games, as well as immersive games and serious games.

This year, we have 57 accepted presentations, plus 3 keynote presentations and 3 invited talks, 2 workshops and 1 tutorial over the course of an intense two-and-a-half-day program.
For the three keynote speakers, we are honoured to announce: Joël Colloc, Normandie University, Le Havre, France, who will talk about "INDICIA: An Assessment Indicator for Healthcare AI Systems that focuses on the Caregiver-Patient Relationship and Algorithmic Transparency", Hoai An Le Thi, Lorraine University, Metz, France, who will explore "Beyond LLMs: Structured World Models via DCA", and Rémi Bonidal, ArcelorMittal, Maizieres les Metz, France, who will present "Multiscale Numerical Simulation for Steelmaking Process Decarbonization and Optimization". Concerning the three invited talks, we are happy to introduce to you: Vladimir Koltchanov, Anylogic Europe, Fontainebleau, France, who will discuss about "Multi-Method Simulation Software AnyLogic. Current Features and Road Map", Peter Mitic, University College London, London, United Kingdom, who will focus on "Sentiment and Reputation: a New Paradigm", and Marcin Choinski, Institute of Information Technology, Warsaw University of Life Sciences, Warsaw, Poland who will share his research on "A Continuous-Time SIS Criss-Cross Model of Co-Infection in a Heterogeneous Population".

We would like to thank all contributors for sharing their research, ideas, and expertise. We are particularly grateful to the reviewers for their thoughtful and rigorous evaluation of the submissions. We are also grateful to the keynote and invited speakers, partners, and all participants for their commitment and enthusiasm. We also sincerely wish to acknowledge Philippe Geril and the organizing team; whose professionalism and dedication made this event possible and all those who contributed to the preparation and success of the conference.

Preface

Last but not least, please allow us to thank our local hosts, “Arts et Métiers Institute of Technology Campus of Metz” and “Ecole Nationale d’Ingénieurs de Metz - ENIM” in Metz, France, for their great organization.

We hope that these proceedings will continue the conversations initiated during the conference and foster new ideas, collaborations, and scientific advances well beyond the event itself.

We are honoured and delighted to welcome you all to Metz and wish you a most enjoyable stay.

Sophie Hennequin and Daniel Roy, ENIM
Ecole Nationale d’Ingénieurs de Metz, Lorraine University

and

Jelena Petronijevic, Paul Stief and Tian Zhang
Arts et Métiers Institute of Technology

ESM® ’2026 Editors

Contents

Preface ..... **XI**
Scientific Programme **1**
Author Index **359**

Keynote

INDICIA: An Assessment Indicator for Healthcare AI Systems that focuses on the Caregiver-Patient Relationship and Algorithmic Transparency
Joël Colloc **5**

Invited Talk

A Inherently Discrete-Time *SIS* of Epidemic Model Spread in a Heterogeneous Population with a Health Function projecting a Closed Unite Interval into Itself
Marcin Choíński **19**

Analytical and Numerical Techniques

Pies with Gradient-Driven Knot Refinement for 2D Stress Analysis
Agnieszka Boltuć and Eugeniusz Zieníuk **29**

A Hybrid PIES-PINN Approach for solving Two-Dimensional Orthotropic Potential Problems
Eugeniusz Zieníuk and Krzysztof Szerszeń **35**

Influence of Selected IFPIES Parameters on Solutions in the Domain
Andrzej Kuźelewski and Eugeniusz Zieniuk **40**

Efficient 2D Elastoplastic Analysis using PIES with Global TELLES Transformation for handling weakly Singular Domain Integrals
Agnieszka Boltuć and Eugeniusz Zieníuk **46**

Advanced Modeling and Algorithmic Techniques for Self-Similar IP Network Traffic Simulation and Performance Evaluation
Izabella V. Lokshina and Cees J. M. Lanting **54**

A SIS–Type Continuous-Time Model of Tuberculosis Spread in a Heterogeneous Population with a Linear and a Bilinear Transmission Function
Marcin Choíński and Sylwia Stachowiak **61**

Contents

Optimization Methodology

Direct Versus Indirect Methods in Singular optimal Control: A Structural Benchmark
Mariusz Bodzioch**.....71**

Gradient-Based Switching-Time Optimization for Singular Optimal Control: Locality, Admissibility and PMP Diagnostics
Mariusz Bodzioch**.....77**

Robust Set-Based Design by Monte-Carlo Simulation and Decision Tree
Jean-Yves Dantan **.....81**

Simulation Tools

Quantum Kernels without the Quantum Computer: Demonstrating a Simulation-Based Stylometry Pipeline for Attributing AI-Generated Text
Kalin Kopanov and Tatiana Atanasova **.....91**

A Probabilistic Risk-Ranking Tool for Assessing Food Chain and Environmental Risks of Nitrification Inhibitors in Grazed Pastoral Systems
Vaibhav Bhatia, Patrick Forrestal, Surinder Saggar, Dumsane T. Matse, Marta Alfaro and Enda Cummins **.....96**

AI in Simulation

How a Small Fine-Tuned Model can rival a Large General-Purpose Model in Culinary Tasks
Jan K. Krajewski, Aleksandra Hinc, Leszek J. Chmielewski and Arkadiusz Orłowski**.....101**

Set Transformer for Bundle Recommendation: Modeling Bundles as Sets
Suman Kumar, Aakash Swami and Siddhartha Sarkar**.....106**

Conservative Simulation-Guided Design Exploration using Uncertainty-Aware Graph Neural Network Surrogates
Pancho Dachkinov **.....111**

Machine Learning

A Machine Learning Approach for Decision Support in Infertility Risk Assessment
Dawid Zamojski, Adam Pudelko, Emilia Morawiec, Anna Bednarska-Czerwinska and Michal Marczyk **.....119**

Multi-Agent Simulation of Phishing using a Metamodel Integrating Machine Learning for Detection
Moussa Koïta, Mohamed Y. Maize and Oumar Y. Maïga **.....125**

Sequential Linearization for Physics-Informed Extreme Learning Machines: A Benchmark-Motivated Fix for Nonlinear PDE Failures
Karol Struniawski **.....132**

Physics-Informed Extreme Learning Machines for Steady-State Amplified Spontaneous Emission: A Feasibility Study
Karol Struniawski **.....137**

Large Language Models

Dynamic Super-Resolution Strategy for Efficient Scene Text Legibility Classification
Seweryn Kalisz, Michał Marczyk and Rafał Fagas**.....145**

Cross-Lingual Agreement between Text and Speech Emotion Recognition: Towards Robust Multimodal Perception in Social Robots
Kamil Skowroński, Adam Galuszka and Eryka Probierz **.....153**

Exploring OODA Loops through Categorical Formalisation and Executable Aspect Semantics
Ian Oliver and Pekka Kuure **.....161**

Parallel Computing

A Design-Time Archetype Lens for HLA Object Models
Okan Topçu **.....171**

Beyond Single Stream Testing: A Large-Scale Statistical Evaluation of Modern Pseudo-Random Number Generators
Théau Wartel, Loïc Yon, David R.C.Hill and Benjamin Antunes **.....176**

Autonomous Systems

Pathplanning in Dynamic Environments
Matthias Becker and Joel Lienhard**.....187**

Model for Visual Terrain Relative Navigation
Paweł Śliwa**.....192**

Contents

A Particle Swarm Optimization Algorithm for Mobile Robot Path Planning with Hermite Smoothing
Artur Babiarz **.....197**

Development of a Low-Cost Open-Hardware throwable Mobile Robot for Remote Reconnaissance
Krzysztof Jaskot and Dawid Prokop**.....202**

Deep Reinforcement Learning vs. PID Control: A Comparison of Autonomous Navigation of a Blimp in 3D Environment
Tomasz Grzejszczak, Michał Lasak and Maja Rudnicka **.....205**

Engineering Simulation

Learning Strategy for XGBOOST Based Energy Consumption Forecasting of an Injection Molding Machine
Mohammed Charraoui, Rosa Abbou, Alexandre Sava and Julien Launay **.....211**

A Three Degree of Freedom Mathematical Model Representing the Apparent Mass and Fore-and-Aft Cross Axis Apparent Mass of the Human Body exposed to Vertical Vibration
Naser Nawayseh**.....219**

Man-in-The-Loop Simulation

Towards Social Automation: Mathematical Modelling, Simulation and Control of Human Centric Systems
Eryka Probierz **.....229**

Mixed-Model Simulation to Enable Human-Aware Decision-Making in Reconfigurable Manufacturing Systems
Tian Zhang, Jelena Petronijevic, Maroua Ratby, Mustapha Douache, Rémy Balland and Némo Grandidier**.....234**

Agent-Based Simulation of Human Robot Interaction in Assisted Living Environments: Trust, Emotion and Adaptive Robot Behaviour
Eryka Probierz, Anita Gałuszka, Adam Gałuszka and Kamil Skowroński **.....242**

Manufacturing Optimization Simulation

Holonic Control Architecture for Reconfigurable Manufacturing Systems
Sophie Hennequin, Adegoke Adesanmi, Daniel Roy and Ashraf Faraj**.....249**

Using Optimisation Tool for the Design Improvement of Production Systems – An Integrative View of Mathematical Modelling and Design Process
Paul Stief, Meisam Pour Massahian Tafti, Xavier Godot, Ali Siadat, Jean-Yves Dantan, Juan-Camilo Gomez Echeverri and Josselin Schumacker **254**

Toward Deep Energy Neural Networks as Surrogates for Mass-Spring Models of Textiles
Sylvain Chabanet, Atal-Anil Kumar, Oriane Dermy and Thibaut Raharijaona **258**

Simulation in Smart Manufacturing

Towards Smart Industry: A Modular Framework for Adaptive Monitoring of WAAM Systems
Soukaina Sadiki, Sandra Chevret, Paul Stief and Alexandre Sava **265**

Scalability Assessment of Reconfigurable Manufacturing Systems under New Product Integration
Leila Khalili Doost Maklavani, Paul Stief, Jean-Yves Dantan and Nathalie Klement **269**

Vision-Language Models in Smart Industry: Benchmarking OVOD Architectures for Real-World Deployment
Tomasz Strzoda **273**

Industry 5.0 in Production Management Courses: Integration of Operator's Well-Being using Simulation-Based Decision-Making
Jelena Petronijevic, Tian Zhang, Alain Etienne, Jean-Yves Dantan and Ali Siadat **278**

Simulation in Logistics

Modelling of Potential Symbiosis Networks based on a Fuzzy Cognitive Map: A Nigeria Case
Adegoke Adesanmi, Daniel Roy and Sophie Hennequin **283**

GRADES – A New Graph-Based System Dynamics and Discrete Event Simulation for Business Sector Supply Chains: Part 1: Agriculture
Asharab A. Haider, Johannes A. Koller, Alison Moldovan-Mauer and Dirk Beinert **286**

Tackling Transaction Data Scarcity with a Multi-Paradigm Simulation Approach for Supply Chain Networks
Asharab A. Haider, Johannes A. Koller, Alison Moldovan-Mauer and Dirk Beinert **290**

Contents

Ontology-based Market Agents: Towards a Dynamic Instantiation of Supply Chain Simulations
Lasse Hammer, Andreas Schäfer, Patrick Westphal, Lorenz Bühmann, Minu Lee, Arlena Wellßow and Eric MSP Veith **.....300**

Simulation and Facility Layout Design for Intelligent Management in Fast Food Restaurants: Reducing Service Times and Maximizing of Operational Efficiency through Discrete-Event Simulation
Estebin Chuquillanqui-Mantari and Rodolfo Chávez-Castillo **.....307**

Emergency Management Simulation

Potential of Artificial Intelligence in Scenario Creation in the Context of Simulations and Training in Disaster Management
Daniel Chovanec, Jozef Ristvej, Jozef Kubás and Vladimír Andrassy **.....313**

A Multi-Sector Network Model for Assessing Critical Infrastructure Resilience
Piotr Stąpor and Ryszard Antkiewicz **.....318**

Optimisation and Stochastic Simulation Methods to Reduce Uncertainty in Assessing the Impact of EDT on Military Operations
Andrzej Najgebauer, Ryszard Antkiewicz, Sławomir Wojciechowski and Dawid Maślanik **.....326**

When Wildfire Breaks the Grid: A Validated Extreme-Event Testbed for Infrastructure Resilience
Eric M.S.P. Veith and Vivian Sultan **.....332**

Mapping the Next Line of Wildfire Defense: GIS-Based REFCL Prioritization in Southern California
Vivian Sultan, Eric MSP Veith, Aristotle Ochoa and Carlos Rodriguez **.....340**

Simulation in Energy Environments

A Calibrated Sub-Stations Building Simulator for District-Heated Residential Buildings and Evaluation of Control Strategies
Christoffer Hansson and Gideon Mbiydzenyuy **.....347**

An Iterative Co-Simulation Pipeline for Coupled Models
Amy Liffey, Khoa Nguyen, Sandro Luh and René Schumann **.....352**

Scientific Programme

Keynote

INDICIA: AN ASSESSEMENT INDICATOR FOR HEALTHCARE AI SYSTEMS THAT FOCUSES ON THE CAREGIVER-PATIENT RELATIONSHIP AND ALGORITHMIC TRANSPARENCY

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KEYWORDS
AI assessment method, Healthcare ethics, Fuzzy vector space object model, European AI regulation, AI autonomy

Abstract

Today, many applications of artificial intelligence (AI) are offered in medicine with applications more or less respectful of the care relationship and personal data of patients. There are also many different models, methods, and tools that claim to be artificial intelligence, sometimes qualified from "weak AI" to "strong AI." Several frameworks have already been developed to assess the ethical risks of the applications of AI algorithms in many medical specialities which are themselves very numerous and that relies on different decision processes. Thus, the potential use cases of artificial intelligence in healthcare are very numerous and complex. It is necessary to take into account the types of health applications or devices, the actors involved with their respective roles: developers, suppliers and potential users: more or less specialized and experimented caregivers involved in the caregiver-patient relationship. The degree of autonomy of the systems, data sources, artificial intelligence algorithms used and quality and software engineering criteria, legal benchmarks: General Data Protection Regulation (GDPR) and Artificial Intelligence Act (AI Act) must be considered. therefore propose a new indicator called INDICIA that evaluates the different benefits/risks of proposed medical systems using AI according to a method that results from experience with AI and different approaches to AI in health. In the first part, the proposed approach takes into account the ethics of AI (6 axes): (technological, semantic, acceptability, complexity, population, actors) aWelong with the quality technical specifications related to software engineering (axis 8) and European legal regulations (axis 9: GDPR and AI Act). The ethics of AI in health must necessarily be based on medical ethics and the caregiver-patient relationship (axis 7): general references from the french National Ethics Advisory Committee (Comité National Consultatif d'Ethique) and specific references to adapt it to each patient using a field ethics method published in 2008. The method is based on the caregiver-patient relationship where privacy and consent that ensure the trust necessary for the practice of medicine are shared, while respecting medical confidentiality in a context where health data stored in the institutions is increasingly shared, disclosed or hacked. The INDICIA indicator is developed using a time fuzzy vector space object model (FVSOM), whose component objects define the axes of our model in order to show the evolution of (benefits/risks) of available AI tools. AI applications and devices in health are evaluated according to criteria expressed by the attributes of these objects. These guidelines originate from models conceived by scholars and experts in AI. INDICIA is showing the strengths and weaknesses of each category of AI models with some examples.The modularity of our approach offers the possibility to reuse the first seven axes of the indicator with some software engineering and legal milestones (axes 8 and 9), in order to adapt it and to further propose general AI ethics guidelines.

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INTRODUCTION

The computer and medical information systems of the 1970s and 1980s already posed ethical problems. Yet the foundations of artificial intelligence (AI) date back to the 1950s, with expert systems in the USA such as MYCIN (identification of bacterial infections) which already proposed Buchanan and Shortliffe (1984) and then in France Colloc (1985a),Fieschi (1981),Gascuel (1985),Pierre Lebeux (1983). Since 2020, artificial intelligence has experienced a new era that is revolutionizing all human activities, particularly the practice of medicine, with the advent of multi-agent systems and deep learning (extension of neural networks). and currently autonomous AI or generative AI with "Chatbots" like ChatGPT Pratim Ray Partha (2023). The method for assessing the ethical risks of using artificial intelligence in medicine described in this article is aimed at both "city doctors" and "countryside doctors" as underlined by Nicole Léry in Colloc et al. (1994), Colloc and Léry (1997).

History of Medical Ethics

Since the 1980's, doctors, whether they work in hospitals or in offices in the city or the countryside, have come together to reflect on their practices. They wonder about the possible harmful effects on the life of each patient who must necessarily consent and adhere to the acts performed on his body and/or his psyche. Moreover, the supply of care is constrained by the patient's personal means, his insurance, the solidarity of society... Reflection on medical ethics appears as soon as difficult questions emerge in the care relationship figure 1:axis 7. Medical research is constantly proposing new technologies as new pathologies emerge. Thus, ethical reflection is dynamic and must continually adapt. There are two categories of medical ethics: the one, institutionalized by the National Consultative Council on Ethics (CCNE) for life sciences and health, which generally reflects on major general problems Delfraissy and Ruault (2023) and field ethics, which feeds on the experiences of practitioners confronted with singular questions and is close to clinical medicine Colloc et al. (1994), Colloc and Léry (1997), Léry and Colloc (2008), Colloc and Léry (2008). This need led Doctors Nicole and Louis Léry to offer in Lyon, as early as the 1980's, a medical ethics consultation, first at the Edouard Herriot Hospital, then at the Vinatier Hospital as part of the "*Santé Ethique et Liberté*" (SEL): a non-profit organisation.

Our first works in AI applied to medical ethics

After completing one of the first expert systems for medical antibiotic therapy Colloc (1985a) during my medical thesis, I became interested in the ethical issues raised by the use of computer science in medicine during my studies in forensic medicine: how to protect the privacy and patients' data protected by the medical secrecy Colloc (1985b). With Nicole Léry, as part of our research on AI applied to medicine, we proposed a decision support system in medical ethics based on the "relational flower" 9 and the "decisional spiral" 10 (see below) designed by Nicole Léry, whose steps we have formalized and provided an example of use regarding vaccination Colloc et al. (1994), Colloc and Léry (1997). Clinical reasoning which is based on semiology was quite close to Case-Based Reasoning (CBR) and multi-agent systems (MAS), around which communities of AI researchers have been formed Bichindaritz and Marling (2010),Marling et al. (2014). After the death of Nicole, our collaboration continued with Louis Léry until his disappearance in 2009 Léry and Colloc (2008), Colloc and Léry (2008). AI in medicine has since undergone considerable improvement, enriched by new models, methods, and tools: applications, platforms, connected objects, telemedicine, made possible by the development of the Internet of Things (IoT) and the use of data stored in data centers. In this context, we propose an ethical

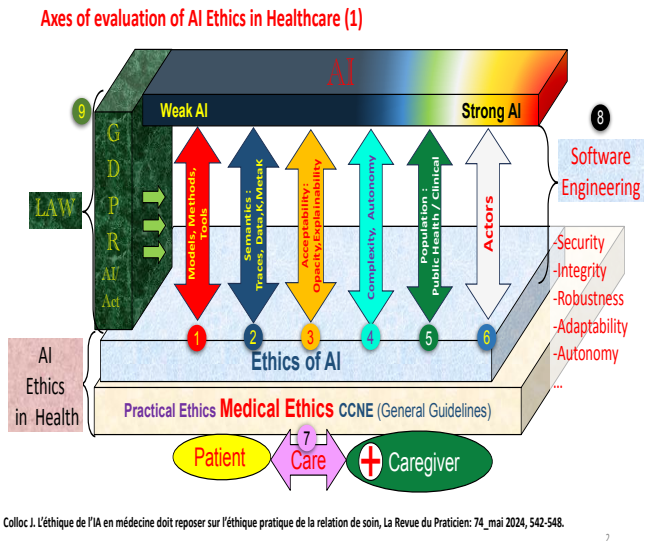

Figure 1: Method for Assessment of AI Systems in Healthcare

evaluation method of the benefits and risks of this new AI healthcare offer Colloc (2020b) which is based on the care relationship (caregiver-patient) and the clinical medical ethics base initially proposed with Nicole and Louis Léry. In order to allow medical practitioners to assess by themselves, simply, the ethical relevance of the new AI applications in health that are proposed to them, we provide a set of criteria on Figure 1, inspired from our previous work Colloc (2024), which shows the 6 axes previously mentioned that revolve between: At the top: the artificial intelligence layer groups weak AI systems with strong AI, some aspects of which are still only prospects. At the bottom, we find the ethics of AI in health, which are based on two layers: the ethics layer of AI, which in the field of health must necessarily be based on the medical ethics layer, which brings together aspects related to ethics and the decision spiral that we will describe later (Léry-Colloc model) Léry and Colloc (2008), Colloc and Léry (2008) which is based on axis 7: the patient-caregiver relationship. On the left are the legislative references in particular the European (General Data Protection Regulation (GDPR), liability...). In France, the GDPR supplements the Data Protection Act of the law of the January 6, 1978 that created the National commission on Informatics and Liberty "*Commission Nationale Informatique et Liberté (CNIL)*"; it requires users to compile a register for processing patients' personal data, which concerns all areas in Figure 1. The main problem is Suppliers outside the European union bypass legislation by requiring users to accept terms and conditions: Terms of Service (TOS) of their software that require them to share their data. On the right: we find software engineering criteria (security, integrity, robustness, adaptability, autonomy, reproducibility...) proposed by Meyer (1988). The model described on figure 1 includes the 7 axes described below plus 2 more axes: Law and Software engineering that complete the method of AI evaluation. Table 1 lists and

defines the main AI models and tools used in medicine and their major uses. Table 2 describes the necessary attributes of AI models which are used to build the objects of the INDICIA indicator.
Table 3 lists and compares the capabilities of AI models according to their types (symbolic S, deep-learning DL, and analog A), listed in Table 1. Specific combinations of techniques can be found that may contradict these general trends. In addition, some systems such as CMAS,MADSS,RMAS (see tables 1 and 2) can combine and cooperate several models .

AXES OF THE AI EVALUATION METHOD

Axis 1: Models, Methods, Tools

The axis 1 provides a non-exhaustive list of AI technologies, models and types of application which are briefly described in Table 1. The acronyms (1st column) are reused to index the necessary attributes: head of column in Table 2 used to compute the INDICIA indicator.

Axis 2: Semantics: Digital Traces, Data, Knowledge, Meta-knowledge

The digital trace is the result of a user's action, left intentionally or not on the net (click on a link in a website) Mille (2013). His interpretation gives him meaning and transforms it into a sign that informs about his preferences, emotions, interests... Interpretation transforms the trace into data when the user is identified. The data is a characteristic or property of a user, a patient, an actor, an object and it is stored on digital media. Knowledge is conduct to be maintained, methods deemed effective in solving a problem or describing situations; it is independent of patients and therefore anonymous. Meta-knowledge determines how to choose the knowledge to solve a problem (examples: steps of a diagnosis, therapeutic strategies). Traces=4 and data=3 harms the privacy much more than knowledge=2 and Meta-knowledge=1 that are anonymous.The higher the semantic score, the less privacy is ensured (see table 4).

Axis 3: Acceptability, Opacity, Explainability

Axis 3 describes the degree of acceptability and opacity, the "*Explainability of algorithms*" on Table 2. The opacity column [1-4] defines to the user how the algorithm is computing data and knowledge. In Static Black Boxes (SBB) (opacity=3), only inputs and outputs are known. These algorithms are approximators. In Dynamic Black Boxes (DBB): the system is able to modify itself (opacity=4). In white boxes, decision steps are well described (opacity=1) by an (Unified Modeling Language) UML activity diagram or a Business Process Model and Notation chart (BPMN) or MERISE MCT:(*Modèle Conceptuel de Traitement*) based on Petri nets. Gray boxes

Table 1: Axis 1 defines AI models and main health applications Types: S=Symbolic, A=Analogic, DL=Deep Learning Colloc (2020a;b;c;d)

Acronym	Types	AI Models	Brief definition and scope
DT	S	**D**ecision **T**ree	Alternatives tree: if (conditions) Then (actions) defining a decision process
EA	A	**E**volutionary **A**lgorithms	Design, artificial invention, new artistic forms: artificial image and music, new objects, products, molecules.
GA	A	**G**enetic **A**lgorithms	Genetic encoding of solutions. Mutation→crossing → competition → new generation → select best solution → test → loop until adequate solution.
BOTS	S,A	Chat**Bots**	A computer robot agent (autonomous or not) with voice and/or text interface (NLP) dialogue with users using a knowledge base. Avatar expressing emotions on an artificial face: telepresence, social robots.
DLNNC	DL	**D**eep **L**earning **N**eural **N**etwork with **C**onvolution	Pattern recognition, signals, images (radiology, MRI, dermatology), search and classification of publications in the Internet.Patient profiling for genomic (see VLDB)
TFVSOM	S,A	**T**emporal **F**uzzy **V**ector **S**paces **O**bject **M**odel	Analogical reasoning: fuzzy temporal extension of object models to model the clinical condition and their diseases. Analogy and temporal CBR
IoT	S,DL	**I**nternet **o**f **T**hings	Internet of Things WIFI connected signal sensors: acquisition and interpretation of sounds, images, physiological signals (heart rate and ECG), temperature, blood sugar. Monitoring medication intake...
FL	S	**F**uzzy **L**ogic	Extension of production rule systems (PRS) where the fuzzy rule predicates in [0,1]. Transition between having or not a characteristic. Fuzzy induction reasoning on imprecise and uncertain thresholds.
BN	S	**B**ayesian **N**etworks	Decision support systems combining statistics using the De Bayes theorem defining conditional probabilities.
PRS	S	**P**roduction **R**ules **S**ystem	Expert decision support systems: deductive reasoning based on formal logic and an inference engine triggering Boolean rules (see DT).
ONTO	S	**Onto**logies and meta-knowledge	Ontologies represent semantic relationships between data and clinical objects (diseases, syndromes...). A meta-model combines the semantics of objects and multi-agent clinical models (see CMAS and TFVSOM).
CBR	A	**C**ase **B**ased **R**easoning	Analog learning by similarity of cases already encountered (distance), Database of clinical cases or situations. Storage and reuse of clinical experience. Learning through experience of successes and failures.
CMAS MADSS	S,A, DL	**C**ognitive **M**ulti-**A**gent **S**ystems	Distributed AI, cooperation of cognitive agents that implement clinical decision models and steps e.g.: Clinical Decision Model CDM(Δ,Π,Θ,SΘ) Δ:Diagnosis, Π:Prognosis, Θ:Therapy, SΘ:Therapy Follow-up.
RMAS	S	**R**eactive **M**ulti-**A**gent **S**ystems	Ant algorithms, collective intelligence based on task coordination. Simulation of complex systems: transport, supply-chain, logistics, care pathway.
NLP LLM	S	**N**atural **L**anguage **P**rocessing and **L**arge **L**anguage **M**odels	Machine translation, Interface and voice control, Automatic dictation, Morphosyntactic parsers, Automatic knowledge ontologies, Electronic Document Management (EDM). Use in chatbots to develop the verbal flow. (LLM rely on GAI.
VLDB	S, DL	**V**ery **L**arge **D**ata **B**ase	Automatic semantic indexing by agents browsing HTML links (bots crawler). Big Data, profiling of patients (genomes), Customers Relationship Management (CRM).
GAI	DL,A	**G**enerative **A**rtificial **I**ntelligence	Reinforcement algorithms based on Big Data and Tensor calculation libraries like Tensorflow (R) and Keras (R) (see Deep Learning with Convolution)

Table 2: Attributes of AI models defined in Table 1

Name Model Type	Learning S/NS/T	Knowledge Sources	CpX	Auton Level [1-10]	MetaK	Opacity [0,4]	Algorithm Explained	Results Justification Transparency	Time Modeling in Knowledge
DT	S	Knowledge Expert	6	1	No	0	Yes	Yes: Alternatives	No or difficult
EA	NS	Genetic Encoding	8	1	Yes	2	Yes	Yes: Generations	No
GA	NS	Genetic Encoding	8	3	No	3	Yes	Yes: Generations	No
BOTS	S/NS	CMAS	7	5	Yes	2-3	As per models	Model dependant	Model dependant
DLNNC	T	VLDB, Image DB	7	3	No	3	Yes	No: Black Box	No
TFVSOM	S	Knowledge Expert	7-8	4	Yes, object diagrams	0	Yes, time vectors	Yes objects relationships	Yes
IoT	S	see CMAS and/or RMAS	6	5	see CMAS	2	Yes	see CMAS and/or RMAS	Yes
FL	S	Knowledge Expert	6	1	No	1	Yes	List of triggered Rules	No or difficult
BN	S	Knowledge Expert	6	1	No	1	Yes	Yes: but weakly	No
PRS	S	Knowledge Expert	6	1	Rules packages	1	Yes	List of triggered Rules	No or difficult
ONTO	S	Knowledge Expert	6-7	1	Yes	1	Graph of links	Graph Path	Before / After links
CBR	S/NS	Semantic Indexation	7-8	1	Yes	2	Distances between cases	Yes: indexing of cases	Cases Timestamp
CMAS MADSS	S/NS	Agents Models	8-9	6	Yes Reflexive	2-3	K reflexive agents	Logging	Message Timestamp
RMAS	NS	Stigmergy	7	6	Yes	2	Reinforce useful Path	Yes: graph Path	Evolution of agents
NLP LLM	S/NS	see ONTO and DLNC	6-7	7	As per model	3	Yes	As per models	Yes verbal flow
VLDB	S/NS	DB Federation	6-7	3	Yes	1	Yes	Queries or NoSQL	Possible but weakly
GAI	T	VLDB, Image DB	7	3	No	3-4	Yes	No Black box	No

Abbreviations:
S: Supervised,**NS**:Non-Supervised,**T**:Training;
CXComplexity Level [1-9]; **Auton**:Autonomy level[1-10],
MetaK:Metaknowledge; **DB**:Database, **K**:Knowledge.

Table 3 Capabilities of AI models
Colloc (2024)

Types of Models described in the table 1 / System capabilities	*Deep Learning* DL	Symbolic Systems S	Analogic Systems A
The system can be satisfied with modest training datasets	No	Yes	Yes
The algorithm and its steps can be explained to users	No	Yes	Yes
The results or predictions of the system are easily interpretable and justifiable	No	Yes	Yes
The system naturally exploits the composition of real objects and the relationships between them.	No	Yes	Yes
The system can perform generalization to domains far from the one being processed	No	Yes	Yes
The system adapts to situations that evolve without significant human intervention	Yes	No	Yes
The system is robust against noisy or ambiguous data. Adversarial Examples introduced knowingly to prevent the training of the system are excluded by the system.	Yes	No	Yes
Useful data representations are automatically discovered from the raw data	Yes	No	Yes
The system can integrate temporality into information processing.	No	No	Yes
The system may take into account the criteria of consistency and relevance	No	Yes	Yes

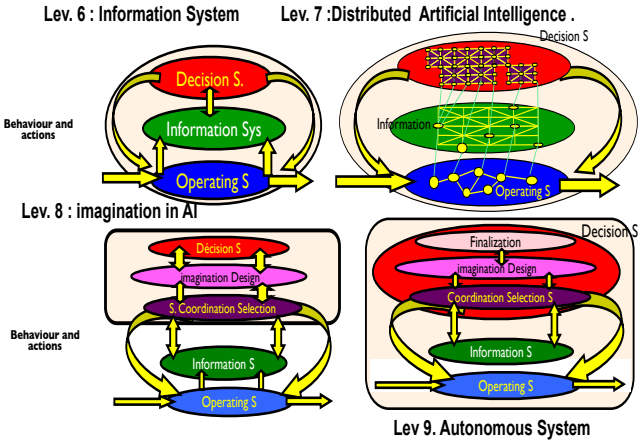

Figure 2: Le Moigne's System Complexity and Autonomy Hierarchy, Le Moigne (1990),pp 58-54.

are intermediate: (opacity=2) some parts of the algorithm are displayed and others are hidden. (opacity=3) means that only superficial and general architecture of the algorithm are shown but the main functions are inaccessible or difficult to understand.the greater the opacity, the more inaccessible the algorithm is (table 4).

Axis 4: Complexity and Autonomy

Axis 4 evaluates the degree of complexity [1-9] and autonomy level [1-9], using the Le Moigne's system classification figure 2 and 3 in order to assess if the AI algorithm can be explained to the user, and if the results can be justified (table 2 linked to table 1). It determines whether the AI device is driven by a caregiver or if it is an autonomous device. Who is the end user is. Levels 1 to 5 define a self-regulated system. Level 6 is the O.I.D (Operating, Information, Decision) model of a classical information system implemented by a database. Level 7 has a system that coordinates the activities of software distributed over a network at all times. This is called distributed AI. Level 8 is equipped with a system of imagination that conceives new possible decisions: the system alone is capable of innovating and inventing new problem-solving methods in medicine (diagnosis, treatments for example). At level 9, the system is finalized; it can decide on its decision and determine the positive and negative aspects of its past actions for itself and its environment. He can be endowed with emotions and even an artificial psyche. He becomes autonomous, able to set his goals like a robot that would be close to possessing human thought. This type of AI system exists in the form of partial prototypes. The level are renumbered as follow $[6-9] \rightarrow [1-4]$ (see Table 4).

Axis 5: Target Population

Axis 5 concerns the target population: does the AI system have a public health purpose based on population statistics? with what sample size? Is the data collection population the same as that of the target population Figure 4 where the system based on the data will be used by end-users with the same purpose or not? Is it an individual clinical AI as part of the care relationship? How are traces, data and knowledge are left by the patient or the physician on the net when he uses the AI medical software figure 5? How these traces are collected and the data inferred without the patient's consent? The population (P) is defined according to its cardinality:

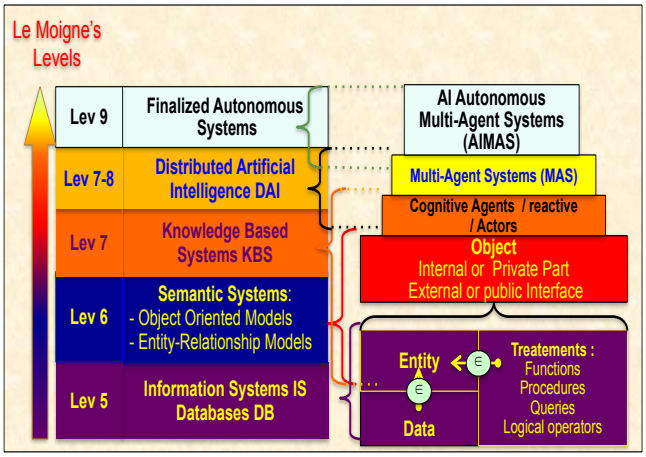

Figure 3: AI Systems Hierarchy according to Le Moigne's Classification only Levels [5-9]

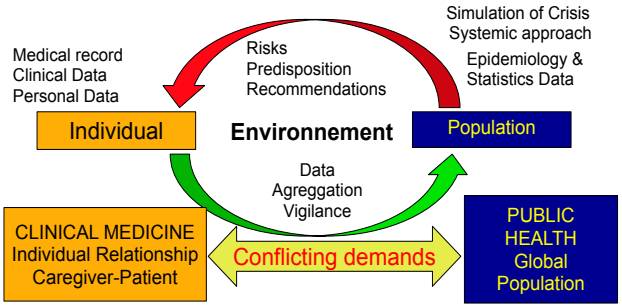

Figure 4: Public Health and Clinical Medicine Conflicts

Only one patient (P=1), Local or a few cases (P=2), Regional (P=3), National (P=4) International (P=5), (see table 4). There are many orphan diseases (approximately 8000 in the world) where few cases (less than one in 2000 people) are recorded and it is not appropriate to carry out statistics with the scope to diagnosing them. For public health AI systems the larger the population the more appropriate the knowledge base is. For clinical AI system, the larger the population, the less specific the knowledge is. It is mandatory to verify if the patient belongs to the same population.

Axis 6: Actors

Axis 6 concerns the actors, their roles, motivations, and means according to the Quintilian approach (a tool used to describe a situation from all angles) which answers the following questions (7W): Who does What, Why, With what, hoW, Where, and When? Axis 6 requires classifying the actors and their respective roles relative to the system: companies and organisations and natural persons such as investors (Type=5), suppliers (Type=4), experts(Type=3), other caregiver users (Type=2), Physician users(Type=1) and Patient users (Type=1) (see Table 4). Figures 6,7,8 describe the

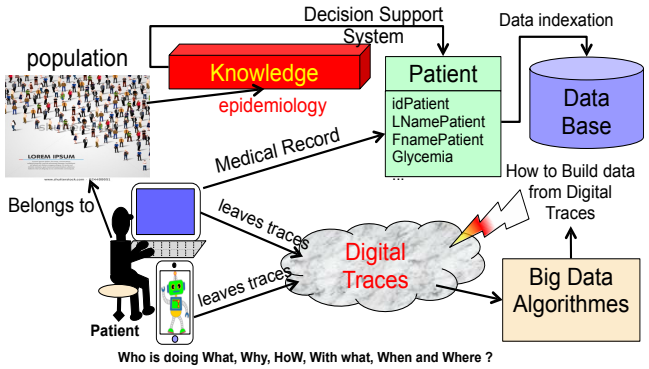

Figure 5: Traces, Data left by the Patient on the Net

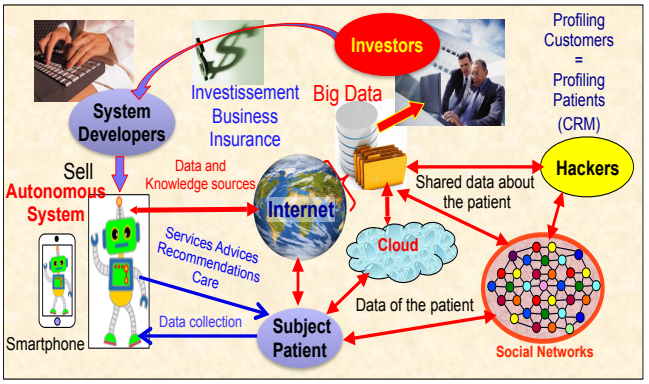

Figure 6: Autonomous Chatbot AI targets Patient's Health Data, the Doctor disappears from the Care Relationship

role of stakeholders relative to the patient data sharing when AI is used. Figure 6 describes the situation where an autonomous AI system is fed with data from the Internet and Big Data to develop advice for the patient whose system establishes the profile without the intermediary of a caregiver. Figure 7 describes a decision support system where a consortium of experts produces a knowledge base for healthcare professionals who treat a patient. We suggests a new architecture "Health Knowledge Hub (HKH)" shown figure 8 which shares an anonymous knowledge base created by hospital or city caregivers using knowledge from national health organisations like *Haute Autorité de Santé*(HAS)(French Health Autority), the *Agence Nationale de Sécurité du Médicament et des produits de santé* French National Agency for Medicines and Health Products Safety (ANSM), the *Institut National de Recherche et de Sécurité* (INRS) National institute of research and security (risks at work). Then the knowledge is delivered to AI providers that can use it to build AI decision support systems. This solution is much more safer for patient's data than the Health Data Hub that shares pseudonymized data that can be re-identified by Internet search engines (especially for orphan dis-

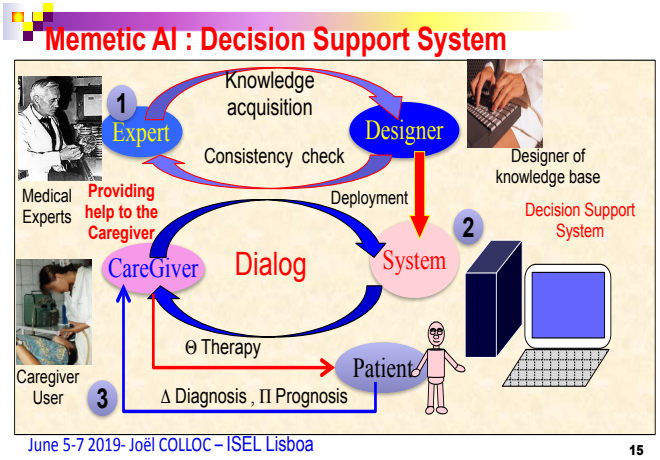

Figure 7: AI Decision Support System (DSS): the Care Relationship is respected, the Caregiver is the End-user

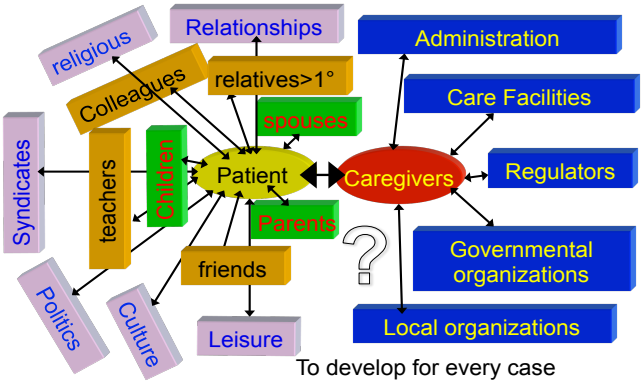

Figure 9: Flower of Family and socio-professional Relationships of the Patient

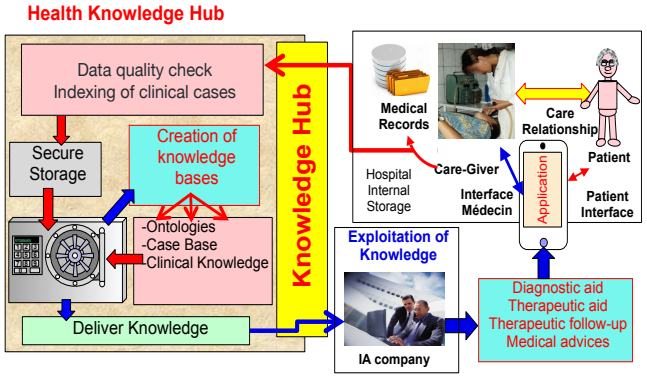

Figure 8: Health Knowledge Hub

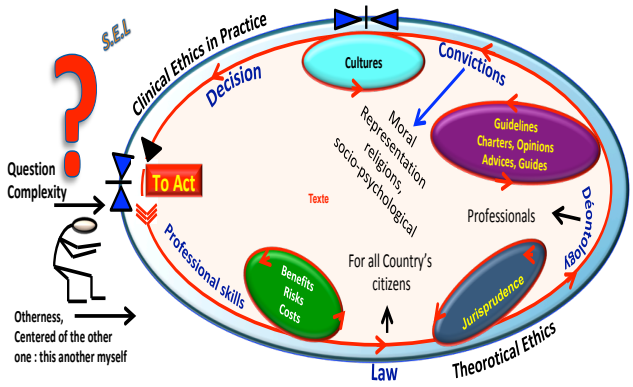

Figure 10: Spiral Decision Process in medical Ethics Léry and Colloc (2008)

eases).The HKH protects the caregiver relationship, patients data stored in local databases and delivers anonymous knowledge, verified by field clinical experts, that concerns the same population and territory as the medical end-user and his patient. The main disadvantage is the lack of doctors trained in AI technologies but the situation is enhancing.

Axis 7:The Care-giver Relationship

Axis 7. Recently, the development of autonomous vehicles has led computer scientists to propose an AI ethic. However, when AI is used in medicine, the patient must remain at the center of the care relationship, and the ethics of AI in medicine must be based on legal benchmarks, notably accountability, the GDPR, and medical ethics. Axis 7 represents the care relationship between the patient and the caregivers, which is based on the relational flower of the patient and the caregiver which corresponds to all the people and organisations with whom the patient and the caregiver have a relationship Figure 9 Colloc and Léry (1997),Léry and Colloc (2008),Colloc and Léry (2008),Colloc (2015). Figure 10 describes the spiral decision process to decide the best action to take concerning a patient in difficult situations. The technical skills of caregivers (know-how), the equation (benefits, risks, costs) of actions taken, the law, case law, ethics, advice from the National Advisory Committee on Ethics (CCNE), charters, and then beliefs, are examined in turn. The beliefs and according to the patient's abilities, his consent when possible and with the members of the patient's relational flower a collegial decision is made on the practical conduct to be taken. The journey of the entire cycle for a patient enriches the experience (Ethics consultations) and knowledge the system's case base for future deliberations in similar cases: Case Based Reasoning system (CBR)Kolodner (1983),Aamodt and Plaza (1994),Colloc and Sybord (1997),Shen et al. (2018b). AI cannot escape the rules of medical practice and when the practitioner uses AI, he engages his responsibility in the same way as when he uses another tool, surgical for example. In the case where the patient is a direct user of an autonomous AI system, the responsibility of the designer of a defective AI system can be sought. The axis 7 is mainly used during the utilisation of the AI system in order to know who is granted to access his records and

for what purpose. The same hierarchy of actors as in Axis 6 is used but the patient can grant or deny the access of his data to some types of actors.

ETHICAL RISK ASSESSMENT FOR AI SYSTEMS IN MEDICINE

A recent paper that reports cases of clinical failure Mui et al. (2026) justifies the interest of this work. INDICIA is an indicator evaluating the potential ethical risk of AI technologies that relies on the 9 axes of the model Figure 1. The AI models are listed and described on Tables 1. The model attributes are defined in Table 2. Axis 2, meta-knowledge and knowledge are less sensitive than patient data and traces. Axis 3 evaluates the transparency of algorithms (white boxes) as more acceptable than opaque and dynamic black boxes whose evolution and purposes are not controlled. Axis 4 renumbers Le Moigne's classification levels from 5 to 9 from 1 to 5. Axis 5 according to the population data used. Axis 6 describes the users. The most at risk are those outside the care relationship, but also paradoxically the patient alone (5) when confronted with autonomous chatbots whose relevance they do not know how to check. INDICIA evaluates the ethical risk on axes 2 to 6. Axis 7 assess how the caregiver-patient relationship is protected: actors and their interfaces are considered. Axis 8: Software Engineering evaluates the AI system architecture and life cyle. That can be difficult because most of the time, the necessary information and data concerning the models really used in the system are hidden and unavailable. We use a set of attributes proposed by Meyer Meyer (1988): Security, Integrity, Robustness, Adaptability and Autonomy to approximately score the software quality. Some informations are available on the net in scientific publications and benchmarks. Axis 9 concern the last update of GDPR EuropeanUnion (2016) and the recent AI-Act European-Union (2024) and the CNIL guidelines and recommendations CNIL (2026). Most of the time, the Term of Service (TOS) of the AI software is very complicated and tries to circumvent GDPR or make the rights so difficult to apply that the user either must give up to his rights if he really wants to use the system or renounce to use it. The GDPR assessment relies on a questionnaire to be completed by the patient that relies on our field experience Yaméogo et al. (2025)Yaméogo and Colloc (2025). The AI-act defines forbidden AI applications and behaviours, it also defines requirements which are mandatory to respect.

FVSOM: FUZZY VECTOR SPACE OBJECT MODEL

This section provides an abstract of the FVSOM model which is fully described in previous works Colloc (2020c), Colloc et al. (2022). Multidisciplinary ontologies are also needed to design the indicator and to provide a synthesis of complex objects involved in each domain Shen et al. (2015), Shen (2015),Shen et al. (2018b), Shen et al. (2018a),Yameogo (2020), in chemical applications Colloc et al. (2022). The model FVSOM is a fuzzy vector space object models which is mainly based on object composition where compound objects are composed of sub-objects which are themselves either compound or simple. A simple object has only qualifier attributes (has-a) that define characteristics and behaviour of this simple object. The model is used to define INDICIA a colour indicator that describe the nine axe objects used to assess an AI system.

Qualifier Attribute Descriptor QAD

A Qualifier Attribute Descriptor (QAD) describes each attribute x_i according its meaning, unit, and definition set D of possible values. x_i is simply called x in the following. The QAD allows to define a function $h : D \rightarrow [-1, 1]$ to normalise the definition set $D = [Vmin, Vmax]$ where $Vmin$ is the minimum value and $Vmax$ is the maximum value of the attribute x. VC is the central value which can be the average, the median or $VC = (Vmin + Vmax)/2$. The normalization function h is used to map the attribute definition domain $D = [Vmin, Vmax]$ on the set $[-1, 1]$. It is also possible to weight attributes with a scalar $\alpha_i.\overrightarrow{f(x_i)}$ to give them more or less importance.

$$h : D = [Vmin, Vmax] \rightarrow [-1, 1]/$$
$$h(x) = \begin{cases} x = Vmax & h(x) = -1 \\ x \in]Vmin, Vmax[& h(x) = \dfrac{2.(VC - x)}{(Vmax - Vmin)} \\ x = Vmin & h(x) = 1. \end{cases} \quad (1)$$

The colour red corresponds to $Vmin \rightarrow -1$ and colour green is $Vmax \rightarrow +1$ as depicted on figures 12,13 and 15. A sigmoid function $\overrightarrow{fx_i}$ defines the vector of each attribute x_i figure 15. The sum of the vectors of the attributes of an object defines the resultant vector that represents this object and so on, along the composition hierarchy, to define the vectors that corresponds to compound objects figure 11. Thus, all attributes x_i of sub-objects O_1, O_2, O_3 are bottom-up inherited (left arrow) by the compound object $O_{\bigoplus[1..3]}$ as depicted figure 11.
The figure 11 shows that each attributes is associated with a sigmoid function equation 2 that define a vector according to its definition domain of each attribute $a_i \in A$ (the set of all attributes) is described by its Qualifier Attribute Descriptor (**QAD**). Its sigmoid function shown figure 15 computes the colour according the value of the attribute a_i as defined in equation 2. The function h is the normalisation function previously defined

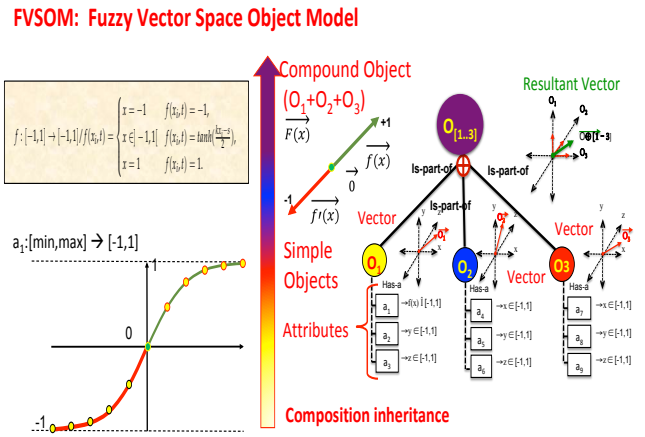

Figure 11: Vectors of compound object computed from simple objects and attributes with FVSOM

equation 1.

$$\forall a_i \in A, D = [Vmin, Vmax], h(a_i) : \ D \rightarrow \ [-1, 1]$$
$$f : [-1, 1] \rightarrow [-1, 1] / f(h(a_i)) =$$
$$\begin{cases} h(a_i) = -1 & f(h(a_i)) = -1, \\ h(a_i) \in]-1, 1[& f(h(a_i)) = tanh(\pi.h(a_i)), \\ h(a_i) = 1 & f(h(a_i)) = 1. \end{cases} \quad (2)$$

All attributes are described in the same manner with a QAD corresponding to its unit and its definition set $D = [Vmin, Vmax]$. In turn a vector is computed for each attribute, each simple object (sum of the vectors of attributes) and then each compound object (the sum of vectors of component objects) figure 11. That allows to compare objects of the same type at each time and to dynamically assess the objects of the knowledge domain with the sigmoid evaluation function. Our model allows to define any kind of attributes Colloc (2020c), Colloc et al. (2022). In this application, attributes are mainly static but can be updated or computed with different variables. The next section shows the definition of objects of the main axes of the method of assessment according figure 1 and the object composition shown figure 14.

COLOUR INDICATOR INDICIA

The colour indicator INDICIA Fuzzy Object Coloured nine Axes Score) designed in this work is based on 9 main axes of the method for assessment of AI systems in healthcare shown on figure 1. The main advantage of our approach is that each axe is a risk object defined by a vector function which value $\in [-1, +1]$, where a negative value shows major drawback (the colour is red) while a positive value indicates that benefits are greater than risks are when the colour is green. When risks and benefits are balanced, the value is approximately null (the axes colour is white). All intermediate values $\in [-1, +1]$ defines a spectrum of colours as shown figure 12. Each risk object is composed of sub-objects and their attributes and defined by their respective vectors. Each AI system is evaluated by a colour indicator for each axes and a global risk value is computed as the resultant of the 9 object vectors which is also displayed by a colour according to its value. Another great advantage of our indicator, compared to the others, is that at a glance, the user is able to see and understand precisely what are advantages and drawbacks of each AI system compared to other models or AI methods used in medicine and it globally assess the risk for the patient. That avoids the black box effect. The motivation is to give back the customers the needed information to choose AI applications according to the patient-caregiver relationship. The challenge is important because some handful of multinational companies that are stakeholders concentrates all the power of deciding what AI product should be sold on the health market with the first scope to do profit and without caring if the use of the software is benefit to the patient or not.

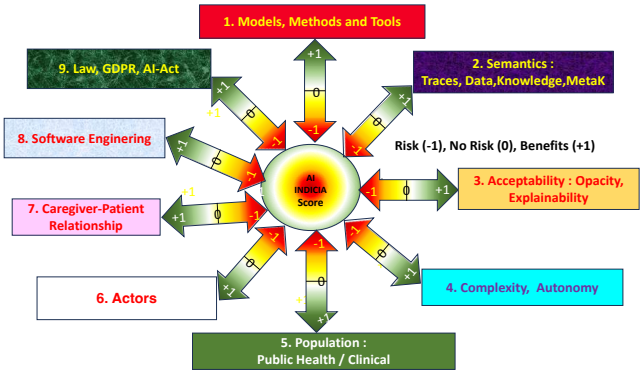

Figure 12: INDICIA: nine axes colour indicator

INDICIA OBJECT COMPOSITION HIERARCHY

In this section, the previous nine axis describes objects that are components of the AI INDICIA Score Object simply called (INDICIA) that is represented by a vector which is the sum of the nine axes vectors previously described $\overrightarrow{INDICIA} = \sum_{i=1}^{i=9} \overrightarrow{O_i}$. The object composition hierarchy is shown figure 14. Each simple object is represented by the sum of its attribute vectors whose definition domain is described by a qualifier attribute descriptor (QAD) and the corresponding sigmoid function figure 15 that gives the appropriate colour of each axis component object. The advantage of our approach is that it provides an assessment of the system at first glance with strength and weakness of the AI system. INDICIA gives also explanation on these objects and their attributes according to the assessment method pre-

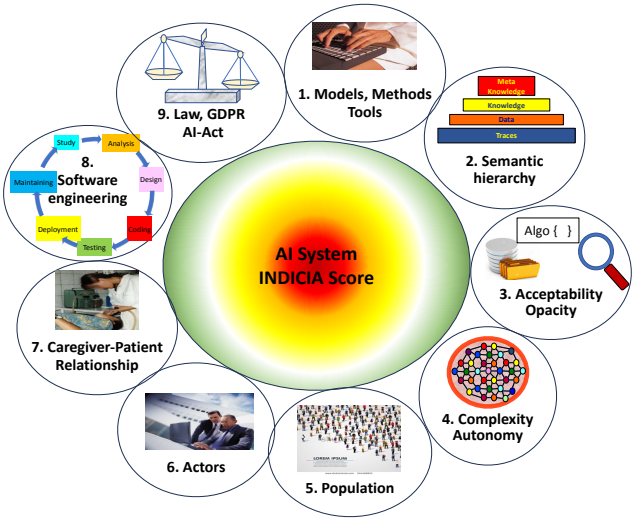

Figure 13: INDICIA Colour Indicator

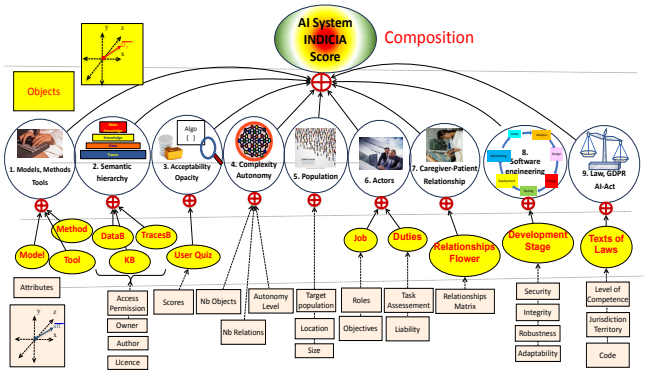

Figure 14: INDICIA Object Composition Hierarchy

sented in the first part of the article and depicted on figure 1. To define INDICIA objects, it is necessary to reuse data from benchmarks and evaluations of different models, methods and tools already used in medicine. These models are listed and attributes defined in tables 1 and 2. We also use texts of law to define mandatory attributes of the GDPR and AI-act with the help of CNIL's Guidelines and recommendations (three objects: Texts of Laws) and their attributes figure 14.

INDICIA EXAMPLE

A clinicians in order to report morbidity or mortality events Mui et al. (2026). Five error cases with AI LLM Claude Opus 4.7 are reported. The LLM provided standard answers without taking into account specific patient's clinical situation (Drug-drug interactions, contraindications, inappropriate antibiotic dose for a baby). The LLM has no clinical reasoning capabilities, its answer is only based on its stochastic model. According to the previous definitions axis 2 to 6 and the values of table 4, we try to define the maximum value for each axis. The object vector AIClinical is shown in equation: 3 and a general object can be computed for each model listed in tables 1 to 4.

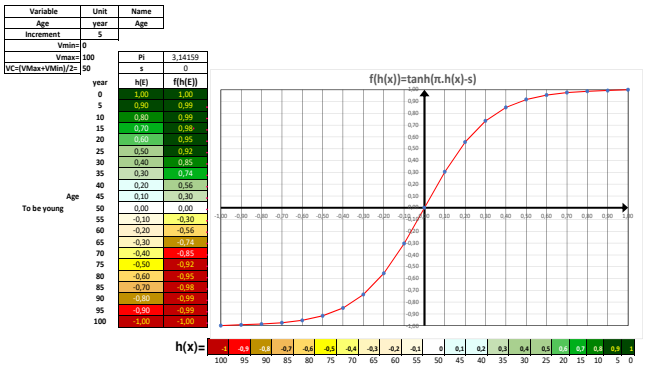

Figure 15: Fuzzy Function to compute Attribute Colour

Tableau 1

1. Acronym	2. Semantic Privacy [1-4]	3.Opacity [1-4]	4. Complexity [1-4]	5. Population [1-5]	6.Actors [1-5]	V(x)	h(V(x))	tanh(PI.h(V(x)))
DT	3	0	1	3	1	1,5	1,00	1,00
EA	3	2	3	2	3	2,625	0,00	0,00
GA	3	3	3	2	3	2,875	-0,22	-0,60
BOTS	2	4	3	5	5	3,5	-0,78	-0,99
DLNNC	2	4	4	5	5	3,75	-1,00	-1,00
TFVSOM	3	0	2	3	3	2	0,56	0,94
IoT	1	4	2	5	5	3	-0,33	-0,78
FL	2	2	1	3	3	2	0,56	0,94
BN	2	1	1	3	1	1,5	1,00	1,00
PRS	3	1	1	2	1	1,625	0,89	0,99
ONTO	3	0	1	4	3	1,875	0,67	0,97
CBR	3	2	1	4	3	2,375	0,22	0,60
CMAS MADSS	3	2	2	3	4	2,625	0,00	0,00
RMAS	2	2	2	3	3	2,25	0,33	0,78
NLP LLM	2	2	2	5	5	2,75	-0,11	-0,34
VLDB	2	1	1	5	5	2,25	0,33	0,78
GAI	1,2	4	4	5	5	3,55	-0,82	-0,99
					Max=	3,75		
					Min=	1,5		
					VC=	2,625		
					Min-Max	2,25		

$$\overrightarrow{AIClinical} = \frac{1}{8}(2.\overrightarrow{Semantic}+2.\overrightarrow{Opacity}+2.\overrightarrow{Complexity} + \overrightarrow{Population} + \overrightarrow{Actors}). \quad (3)$$

Table 4 provides approximate static values for axes 1 to 6 and vectors computed according to equations 1, 2 and 3. Axis 7 is given by the end-user choice (caregiver and/or patient), axis 8 must be computed according to the developer information concerning the AI software life cycle (which is rarely disclosed), and axis 9 values depend on the general terms and conditions of use of the software proposed by the supplier (TOS) compared to current laws (GDPR, AI-Act).

CONCLUSION AND FUTURE WORKS

Artificial intelligence (AI) is increasingly offering health applications on smartphones, computers, telemedicine, and the Internet of Things (IoT). Transdisciplinary, AI must respect software engineering (reliability, robustness, security), knowledge obsolescence, law, and ethics. Today, a wide variety of algorithms, more or less opaque,

process personal data in AI clinical decision support system inviting oneself into the care relationship, infringing the medical confidentiality, and therefore destroying the trust needed for medical practice. Doctors and hospital or private caregivers are wondering about the benefits, risks, and costs of AI for the patient, the care relationship, deontology, and medical ethics. Based on 30 years of experience in AI and medical ethics, we trust in AI based on clinical knowledge elaborated by clinicians during their medical practice. INDICIA uses AI to evaluate AI practices and their applications in clinical medicine and to assess the ethical risks of AI. Future works need to get data from benchmarks, end-users forums, scientific publications in order to test INDICIA on actual cases and compute the component objects of axis 7 to 9 with more specific actual attributes of AI applications.

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BIOGRAPHY

Joël Colloc earned his M.D. at the medical faculty of Lyon and a speciality degree of forensic medicine with a degree of clinical toxicology. He received a MSc. degree of IT from the Business School of Lyon (IAE) and a MSc. degree of computer sciences from the engineering school INSA of Lyon. He served as forensic physician at the Edouard Herriot Hospital in the neurological emergency department to cure drug addicted people, medical ethics and developed drug and addiction database. He went on to earn his Ph.D. in computer sciences at the INSA of Lyon. As Hospital assistant at the laboratory of medical computer science and he taught IT at the medical faculty. He was elected as associate professor in computer sciences at IAE of Lyon and he earned his accreditation to supervise researches in sciences at the Lyon 1. He is a Le Havre Normandy University professor in computer sciences since 2003. His main research topics concern e-health and particularly: fuzzy vectorial spaces (FVS), multi-agent clinical decision support systems (MADSS) and knowledge bases, Case Based Reasoning, ontologies, nervous system modelling and cognitive sciences and AI applications in medicine and human sciences. His human sciences researches try to conciliate the ethics of using Big Data in epidemiological studies, autonomous systems and robots and keeping ethics use of AI in order to improve clinical decision in medicine while preserving the patient-caregiver relationship, the privacy and the freewill choice of the patients.

Invited Talk

AN INHERENTLY DISCRETE–TIME SIS MODEL OF EPIDEMIC SPREAD IN A HETEROGENEOUS POPULATION WITH A HEALTH FUNCTION PROJECTING A CLOSED UNIT INTERVAL INTO ITSELF

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KEYWORDS

SIS model, health function, heterogeneous population, inherently discrete–time system.

ABSTRACT

We introduce and analyze the inherently discrete–time system describing the dynamics of epidemic spread, which is constructed independently on its continuous–time counterpart. The system concerns a heterogeneous population in which we distinguish two subpopulations that differ in the risk of infection. The novelty of our paper lies in introducing the so–called health function, which replaces the commonly used disease transmission function and projects a closed unit interval into itself. Moreover, we assume that each parameter describing given phenomena has a different value for each subpopulation, what is not met in most papers. We first identify the stationary states in the system. Then we investigate local and global stability of the disease–free equilibrium. Finally we complement the analytical results with simulations depicting a real–world case of tuberculosis spread and illustrating various epidemic scenarios.

INTRODUCTION

In deterministic modeling of epidemic spread continuous–time systems were first proposed Murray (1962) and are still mostly used Sticha et al. (2024). However, discrete–time systems become increasingly common. In most cases, such systems are based on the discretization of their continuous–time counterpart. To do this, one can use the explicit Euler scheme, Runge–Kutta methods Ashgi et al. (2021) or non–standard discretization scheme Zinihi et al. (2026). The disadvantage of this approach is the possibility of obtaining negative values of variables, which in biological modeling should not happen. Moreover, the discretization leads to obtaining a step size of discretization, whose interpretation is not clear in the context of epidemiological modeling Singh et al. (2025). An alternative technique is introducing inherently discrete–time systems, which are constructed autonomously, independent of the continuous–time equivalents. Thanks to such inheritance, we omit nuances arising from variables' negativity and from the appearance of the step size of discretization. An exemplary two–dimensional inherently discrete–time model is presented in Castillo-Chavez D. and (2001), where the authors address the existence and global stability of the system's equilibria. Moreover, they indicate possibility of chaos occurrence. Paper Martcheva (2015) also introduces a two–dimensional system and the author focuses on the local stability of stationary states. An unusual concept for applying inheritance to the diffusion model appears in a very recent paper Zhuo et al. (2026), where the authors establish the threshold dynamics in terms of the basic reproduction number.

In this paper we introduce and analyze an inherently discrete–time model of epidemic spread in a heterogeneous population. By a heterogeneous population we mean a population comprising two subpopulations that differ in infection risk: a high–risk (HS) and a low–risk (LS) subpopulation. In each subpopulation we consider two groups: healthy and infected people. Introducing more groups in a heterogeneous population can lead to a lack of explicit analytical results. Our model is a type of SIS *susceptible–infected–susceptible* model, in which after recovery an individual does not gain immunity and becomes susceptible again. Since our model is relatively computationally complex, we make an assumption that the disease is transmitted among HS, LS and from HS to LS. We neglect pathogen transmission from LS to HS, since for many diseases people from HS tend to aggregate, for example homeless communities in the context of tuberculosis or highly sexually active people regarding sexually transmitted diseases. In many epidemiological models for a heterogeneous population authors assume that parameters concerning each subpopulation are the same. This assumption facilitates calculations but actually leads to the case of epidemic spread in a homogeneous population. We want to emphasize that in our paper we assume that each parameter concerning given phenomena in a particular subpopulation has a different value.

In a continuous–time system for the description of disease transmission one mostly uses a transmission

function based on the mass–action law or the so–called standard incidence function. Their formula yields a number of new infections proportional to the multiplication of healthy and infected individuals. A crucial aspect of these functions, however, is the possibility of yielding negative variables in discrete–time models. For this reason in inherently discrete–time systems one introduces the so–called health function (HF), which corresponds to the probability of remaining healthy. In Martcheva (2015) this function reads $G = \left(\beta \frac{I(t)}{N(t)}\right)$, where β is transmission coefficient and $I(t)$, $N(t)$ are the size of the infected group and the whole population at time t, respectively. In our paper we modify the domain of HF to reflect the possibility of contracting the pathogen in HS from both LS and HS. We provide the precise assumptions for this function in the next section.

Our analytical results are complemented by numerical simulations of a real–world case of tuberculosis (TB) spread in the Warmian–Masurian province in Poland. As LS and HS we consider the non–homeless and homeless communities in this region, respectively. Considering population heterogeneity is crucial, as homeless people transmit TB more efficiently than non–homeless individuals and are considered TB super–spreaders Silva et al. (2026). Thanks to active detection campaigns among the homeless, the TB incidence in the Warmian–Masurian province dropped not only in homeless communities but also across the general population Romaszko et al. (2016). Such a widespread drop suggests investigating epidemic spread while accounting for population heterogeneity. Further we also perform simulations to forecast the local stability of each obtained stationary state of our system.

Our paper is a continuation of work presented in Choiński (2023) and Choiński (2024). Paper Choiński (2023) investigates a model in which the arguments of HF are the sum of the prevalence of infected individuals from both subpopulations. For such proposed function we have $G(x) : [0, 2) \to [0, 1]$. In Choiński (2024) the arguments of HF depend on the actual size of the infected community and for this function we have $G(x) : [0, \infty) \to [0, 1]$. In this paper we modify HF so that we obtain $G(x) : [0, 1] \to [0, 1]$, which evidently projects a closed unit interval into itself. This property is very useful for modelling an invariant epidemiological system in which it is convenient to consider the proportion of groups in the population instead of the actual sizes of groups.

The paper is organized as follows. Firstly we provide a heuristic description and a mathematical form of our model. Later we identify stationary states in the system and then address the local and global stability of the disease–free stationary states. Then we conduct numerical simulations of our model and conclude our work.

DESCRIPTION OF A MODEL AND ITS BASIC PROPERTIES

Let us first describe the terms appearing in our model. To each variable and parameter we assign subscripts $i = 1$ and $i = 2$, respectively, regarding LS and HS. If no number is indicated for i, then we suppose that $i \in \{1, 2\}$. By S_1 and S_2 we denote the sizes of the susceptible groups in a particular subpopulation at the n–th node of a discrete timescale. Similarly, I_1 and I_2 refer to the sizes of the infected groups. Intuitively, $N_i = S_i + I_i$ mean the size of each subpopulation. By S_i^+, I_i^+ and N_i^+ we denote the sizes of the given group at the $n + 1$–st node of the discrete timescale. The constant recruitment to each subpopulation is denoted by C_i. Probabilities of survival till the next unit of timescale, recovery and disease–related death are denoted by r_i, γ_i and α_i, respectively. Coefficients β_i are related to pathogen transmission within the given subpopulation. Moreover, we introduce coefficient β concerning disease transmission from HS to LS. Clearly we assume that $r_i, \gamma_i, \beta_i, \beta \in (0, 1)$. Moreover, it is reasonable to assume that $\alpha_i \in (0, r_i)$. As stated in the introduction, disease transmission is determined by the health function G. For function G we impose the following properties:

$$G(x) : [0, 1] \to [0, 1], \quad G(0) = 1, \quad G(1) = 0$$
$$G'(x) < 0, \quad G''(x) > 0.$$

Our proposed model reads

$$S_1^+ = C_1 + r_1 S_1 \mathcal{G} + (r_1 - \alpha_1)\gamma_1 I_1, \tag{1a}$$
$$I_1^+ = r_1 S_1 (1 - \mathcal{G}) + (r_1 - \alpha_1)(1 - \gamma_1) I_1, \tag{1b}$$
$$S_2^+ = C_2 + r_2 S_2 \mathbb{G} + (r_2 - \alpha_2)\gamma_2 I_2, \tag{1c}$$
$$I_2^+ = r_2 S_2 (1 - \mathbb{G}) + (r_2 - \alpha_2)(1 - \gamma_2) I_2, \tag{1d}$$

where

$$\mathcal{G} = G\left(\frac{\beta_1 I_1 + \beta I_2}{N_1 + N_2}\right), \quad \mathbb{G} = G\left(\beta_2 \frac{I_2}{N_2}\right). \tag{2}$$

It is obvious that if an initial condition of system (1) is non–negative, then the next iterations of the system always produce non–negative values of the variables.

In the further part of paper we often relate to values of $G'(0)$. Let us state a lemma for the special case of G, where $G'(0)$ appears.

Proposition 1. *If $G(x)$ is a surjection, then $G'(0) < -1$.*

Proof. We consider a linear function $G^* : [0, 1] \to [0, 1]$ with formula

$$G^*(x) = ax + b, \quad a, b \in \mathbb{R},$$

for which $G^*(0) = 1$ and $G^*(1) = 0$. For instance we have $a = -1$ and $b = 1$, providing $G^*(x) = -x+1$. It is easy to see that functions $G(x)$ and $G^*(x)$ intersect in $x = 0$ and $x = 1$. Since $G'(x) < 0$ and $G''(x) > 0$, $G(x) < G^*(x)$ for $x \in (0, 1)$. Moreover, $(G^*)'(0) = -1$, what finally gives $G'(0) < -1$. □

Observe that in Martcheva (2015) one assumes that the domain of the function G, recalled in the introduction, is $[0, \infty)$ and examples of functions for such a domain are proposed. However, the infinite domain is contrary to the biological meaning of variables $I(t)$ and $N(t)$. Therefore, we restrict the domain of G to $[0, 1]$. Since $G(x) : [0, 1] \to [0, 1]$, our system is especially useful for analyzing epidemic spreads with a constant population size, where the groups are expressed as proportions. Exemplary health functions addressing our model are $G(x) = \frac{2}{x+1} - 1$ and $G(x) = \frac{1}{2}x^2 - \frac{3}{2}x + 1$.

STATIONARY STATES

In this section we identify stationary states of the form (S_1, I_1, S_2, I_2) that appear in our system. To obtain these states, in System (1) we equate

$$S_1^+ + S_1, \quad I_1^+ = I_1, \quad S_2^+ = S_2, \quad I_2^+ = I_2. \tag{3}$$

Adding (1a) and (1b) or (1c) and (1d) under condition (3) yields

$$N_i = C_i + r_i S_i + (r_i - \alpha_i) I_i, \tag{4}$$

what can be converted to

$$N_i = \frac{C_i - \alpha_i I_i}{1 - r_i} \tag{5}$$

or

$$S_i = \frac{C_i - \sigma_i I_i}{1 - r_i}, \tag{6}$$

where $\sigma_i := 1 - r_i + \alpha_i > 0$.

For a general case $I_i, S_i \neq 0$ from Eq. (6) we get positivity of S_i and I_1 if

$$I_i < \frac{C_i}{\sigma_i} \tag{7}$$

and

$$S_i < \frac{C_i}{1 - r_i}, \tag{8}$$

respectively.

Taking $I_i = 0$ immediately provides the always existing disease–free stationary state

$$E_{df} := (S_1^*, 0, S_2^*, 0),$$

where

$$S_1^* = \frac{C_1}{1 - r_1}, \quad S_2^* = \frac{C_2}{1 - r_2}. \tag{9}$$

Observe that at E_{df} for both expressions of the HF from (2) we have $\mathcal{G} = \mathbb{G} = G(0)$.

In a latter part of our work we often use a notation

$$\kappa_i := 1 - (r_i - \alpha_i)(1 - \gamma_i). \tag{10}$$

Naturally, $\kappa_i \in (0, 1)$.

We now focus on the existence of stationary states that reflect the presence of the disease in at least one subpopulation. Investigation of such states relies on our reasoning from our previous work in Choiński (2023) and Choiński (2024). Therefore, in this paper we provide only final statements.

Observe that the dynamics of Eqs. (1c)–(1d) is independent on Eqs. (1a)–(1b). Therefore, one can analyze Eqs. (1c)–(1d) solely. For a case when $I_2 > 0$ we obtain a proposition:

Proposition 2. *If for any equilibirum of System* (1) $I_2 > 0$*, then there exists a unique pair* $(S_2, I_2) = (\overline{S}_2, \overline{I}_2)$*, where* $\overline{I}_2 > 0$ *is a solution of*

$$\frac{\kappa_2(1 - r_2)}{r_2} \frac{I_2}{C_2 - \sigma_2 I_2} = 1 - G\left(\beta_2(1 - r_2)\frac{I_2}{C_2 - \alpha_2 I_2}\right),$$

and $\overline{S}_2 = \frac{C_2 - \sigma_2 \overline{I}_2}{1 - r_2}$*. The pair* $(\overline{S}_2, \overline{I}_2)$ *exists if*

$$-G'(0) \geq \frac{\kappa_2}{\beta_2 r_2}. \tag{11}$$

Now we focus on the whole four–dimensional system (1). We start with the case where $I_1 > 0$ and $I_2 = 0$. We obtain another stationary state.

Proposition 3. *System* (1) *provides an equilibrium*

$$E_1 := \left(\widehat{S}_1, \widehat{I}_1, \frac{C_2}{1 - r_2}, 0\right), \tag{12}$$

where $\widehat{S}_1, \widehat{I}_1 > 0$ *and* $\widehat{I}_1$ *is a solution of*

$$\frac{\kappa_1(1 - r_1)}{r_1} \frac{\widehat{I}_1}{C_1 - \sigma_1 \widehat{I}_1} = 1 - G\left(\frac{\beta_1(1 - r_1)\widehat{I}_1}{C_1 + \frac{C_2}{1 - r_2}(1 - r_1) - \alpha_1 \widehat{I}_1}\right) \tag{13}$$

and $\widehat{S}_1$ *has a form*

$$\widehat{S}_1 = \frac{C_1 - \kappa_1 \widehat{I}_1}{1 - r_1}. \tag{14}$$

E_1 *exists if*

$$-G'(0) \geq \frac{\kappa_1}{\beta_1 r_1}. \tag{15}$$

We are also able to find an endemic stationary states in which every coordinate is positive.

Proposition 4. *In System* (1) *there is positive (endemic) stationary state*

$$E_e := \left(\overline{S}_1, \overline{I}_1, \overline{S}_2, \overline{I}_2\right),$$

where $\overline{I}_1$ *is a solution of*

$$\frac{\kappa_1(1 - r_1)}{r_1} \frac{\overline{I}_1}{C_1 - \sigma_1 \overline{I}_1} = 1 - G\left(\frac{(1 - r_1)(\beta_1 \overline{I}_1 + \beta \overline{I}_2)}{C_1 + \frac{C_2}{1 - r_2}(1 - r_1) - \alpha_1 \overline{I}_1}\right) \tag{16}$$

and $\overline{S}_1 = \frac{C_1 - \kappa_1 \overline{I}_1}{1 - r_1}$*. This state exists if* (11) *and*

$$-G'\left(\frac{\beta \overline{I}_2}{\overline{S}_2 + \overline{I}_2}\right) \geq \frac{\kappa_1}{\beta_1 r_1}. \tag{17}$$

STABILITY OF DISEASE–FREE STATIONARY STATE

Now we focus on both local and global stability of the disease–free equilibrium E_{df}. Let us start from the local stability.

Theorem 1. *The state E_{df} of System* (1) *is locally stable if*

$$-G'(0) < \min\left(\frac{\kappa_1}{r_1\beta_1} \cdot \left(\frac{C_1}{1-r_1}+\frac{C_2}{1-r_2}\right), \frac{\kappa_2}{r_2\beta_2}\right). \tag{18}$$

Proof. The eigenvalues of the Jacobian matrix of system (1) evaluated at E_{df} read

$$\lambda_1 = r_1, \quad \lambda_2 = -r_1\frac{\beta_1}{S_1^*+S_2^*}G'(0)+1-\kappa_1,$$
$$\lambda_3 = r_2, \quad \lambda_4 = -r_2\beta_2 G'(0)+1-\kappa_2.$$

From the meaning of r_i we get $|\lambda_{1,3}|<1$. Condition $|\lambda_2|<1$ can be expressed as two separated inequalities

$$\kappa_1 - 2 < -r_1\frac{\beta_1}{S_1^*+S_2^*}\widehat{G}'(0) \tag{19}$$

and

$$-G'(0) < \frac{\kappa_1}{r_1\beta_1}\cdot(S_1^*+S_2^*). \tag{20}$$

Since $\kappa_1 \in (0,1)$, we reach negativity of the left–hand side of Ineq. (19). Contrarily, its right–hand size is positive. Therefore, Ineq. (19) is always true and only Ineq. (20) is important.

We cope with conditions $|\lambda_4|<1$ analogously as with $|\lambda_2|<1$. We obtain the inequality

$$-G'(0) < \frac{\kappa_2}{r_2\beta_2}. \tag{21}$$

Combining Ineqs. (20) and (21) and using notation from (9) yield Ineq. (18) from the theorem. □

Now we focus on the global stability of state E_{df}. In the proof of the following theorem we will use notations $\mathcal{G}$ from (2) and S_1^*, S_2^* from (9).

Theorem 2. *The state E_{df} of System* (1) *is globally stable if*

$$-G'(0) < \kappa_1\delta_1, \tag{22}$$

where

$$\delta = \frac{C_1 r_1\beta_1}{(1-r_1)(\frac{C_1}{1-r_1}+\frac{C_2}{1-r_2})}.$$

Proof. Substituting (5) in (1b) gives

$$I_1^+ = r_1\frac{C_1-\sigma_1 I_1}{1-r_1}(1-\mathcal{G})+(r_1-\alpha_1)(1-\gamma_1)I_1.$$

Function G, because of its properties, can be assessed in a way

$$\mathcal{G} \geq G\left(\frac{\beta_1 I_1+\beta I_2}{S_1^*+S_2^*}\right), \tag{23}$$

what gives

$$I_1^+ \leq r_1\frac{C_1-\sigma_1 I_1}{1-r_1}(1-\mathcal{G})+(r_1-\alpha_1)(1-\gamma_1)I_1.$$

We replace the above inequality with the equality

$$Y_1^+ = r_1\frac{C_1-\sigma_1 Y_1}{1-r_1}(1-\mathcal{G})+(r_1-\alpha_1)(1-\gamma_1)Y_1,$$

where we introduce the new variable Y_1. It is clear that we treat I_1 and Y_1 in function $\mathcal{G}$ interchangeably. We define a function $\Theta : \left[0, \frac{C_1}{\sigma_1}\right) \to \left[0, 1+\frac{C_1}{\sigma_1}\right)$ such that

$$\Theta(Y_1) = r_1\frac{C_1-\sigma_1 I_1}{1-r_1}(1-\mathcal{G})+(r_1-\alpha_1)(1-\gamma_1)I_1.$$

We interpret $\Theta(Y_1)$ as a so–called reproduction function for the infected individuals from HS. The set of iterates of Θ is equivalent to the sequence produced by Eq. (1b). Differentiating once and twice Θ with respect to I_1, we obtain

$$\Theta'(Y_1) = -r_1\frac{\sigma_1}{1-r_1}+r_1\frac{\sigma_1}{1-r_1}\mathcal{G}-\frac{C_1-\sigma_1 I_1}{1-r_1}\cdot r_1\mathcal{G}'\frac{\beta_1}{S_1^*+S_2^*}+(r_1-\alpha_1)(1-\gamma_1),$$

and

$$\Theta''(Y_1) = r_1\frac{\sigma_1}{1-r_1}G'+\frac{r_1\sigma_1}{1-r_1}\frac{\beta_1}{S_1^*+S_2^*}G' - r_1\frac{C_1-\sigma_1 I_1}{1-r_1}\left(\frac{\beta_1}{S_1^*+S_2^*}\right)^2 G''.$$

After some simplifications we obtain

$$\Theta'(0) = -\frac{C_1}{1-r_1}\frac{r_1\beta_1}{S_1^*+S_2^*}G'(0)+1-\kappa_1. \tag{24}$$

Ineq. (22) implies

$$-\frac{G'(0)}{\delta_1} < \kappa_1 \quad \Longrightarrow \quad -\frac{G'(0)}{\delta_1}+1-\kappa_1 < 1.$$

We get that the condition (22) yields $\Theta'(0)<1$ and a fixed point $Y_1=0$ is locally stable under Θ–iteration. See that the sign of Θ' and Θ'' is not dependent explicitly on $Y_1 \in \left[0, \frac{C_1}{\sigma_1}\right)$. Let us recall that $G'<0$ and $G''>0$. Therefore, $\Theta''<0$. Considering additionally $\Theta'(0)<1$ provides $\Theta'(Y_1)<1$ and $\Theta(Y_1)<Y_1$. A sequence $\left(Y_n^{(1)}\right)_{n=0}^{\infty}$, constituted by the next iteration of Y_1, is therefore strictly decreasing and bounded below by zero giving $\lim_{n\to\infty} Y_n^{(1)} = 0$. We aim to analyze the inequality, so we get that the sequence $\left(I_n^{(1)}\right)_{n=0}^{\infty} \leq \left(Y_n^{(1)}\right)_{n=0}^{\infty}$ is also therefore strictly decreasing and because of the definition of I_1 bounded below by zero. Naturally we get

$$\lim_{n\to\infty} I_n^{(1)} = 0, \quad \lim_{n\to\infty} S_n^{(1)} = S_1^*. \tag{25}$$

Reasoning for obtaining

$$\lim_{n\to\infty} I_n^{(2)} = 0, \quad \lim_{n\to\infty} S_n^{(2)} = S_2^*$$

is similar to argumentation yielding (25). Finally, we obtain

$$\lim_{n\to\infty}\left(S_n^{(1)}, I_n^{(1)}, S_n^{(2)}, I_n^{(2)}\right) = E_{df} \qquad (26)$$

yielding the global stability of E_{df}. □

NUMERICAL SIMULATIONS

In this section we present numerical simulations of the system (1) for the tuberculosis epidemic mentioned in the introduction. As HS and LS we consider the non–homeless and homeless communities, respectively, of the Warmian–Masurian province in Poland.

We first fit the iterated system's values to real data on the epidemic in years 2000–2025. In Table 1 one can find the values of the system's parameters used for the simulations.

Table 1: The parameters values of System (1). Their unit is year^{-1}.

Name	Process	Value
C_1	Inflow into the LS	9783.49
C_2	Inflow into the HS	52.64
γ_1, γ_2	Recovery	0.8993
μ_1, μ_2	Survival	0.9904
α_1, α_2	Disease–related death	0.0899
β_1	Transmission in LS	0.4239
β	Transmission from HS to LS	0.999
β_2	Transmission in HS	0.2864

The values of coefficients C_i, μ_i were estimated according to the data from statistical office of Poland (2024). We obtained γ_i and α_i according to statistics from of Public Health Poland (2024). One must estimate parameters β_i and β since there is no medical definition of transmission coefficient. We conducted such estimations in the Matlab software using the so–called *lsqcurvefit* function Mathworks (2023), based on the Gauss–Newton algorithm Lai et al. (2015). For numerical computations we take the initial condition

$$(S_1, I_1, S_2, I_2) = (1428469 - 3903907157), \quad (27)$$

that relates to the sizes of particular groups in the Warmian–Masurian province in 2000, and we chose HF

$$G(x) = \tfrac{2}{x+1} - 1. \qquad (28)$$

To find the most appropriate values of β_i and β, that minimize the matching error, we simulate our model and compare the resulting values with the real data. Observe that the values of β are nearly equal to 1. It suggests that homeless people are considered as super–spreaders, who infect non–homeless communities very efficiently and largely contribute to TB transmission in the general population.

Let us visualize the obtained computations. The simulated values and real data for the variable I_1, representing infected non–homeless individuals, are depicted in Figure 1.

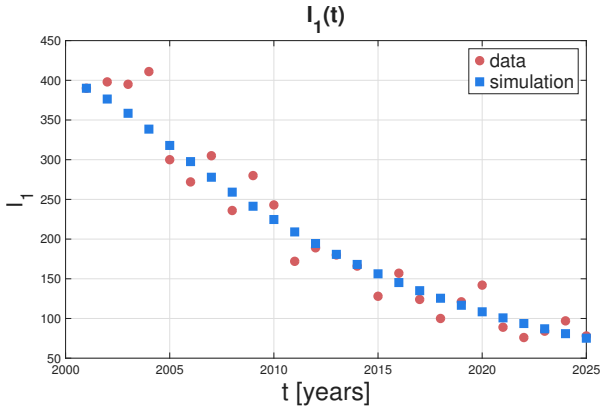

Figure 1: Simulated and real values of infected non–homeless people int the Warmian–Masurian province in years 2000–2025.

Let us now solve our model numerically for additional nodes on the discrete time scale in the context of possible local stability of the obtained stationary states. Since for each reached equilibrium coordinates S_1 and S_2 are positive, we focus only on variables I_1 and I_2. For the rest of this section we take the initial condition (27) and the function (28). We take numbers from Table (1) and conduct consecutive 90 iterations of System (1). The results of simulations are shown in Figures 2–3. These figures suggest that for the given set of parameters' value we reach local stability of the disease–free state E_{df}.

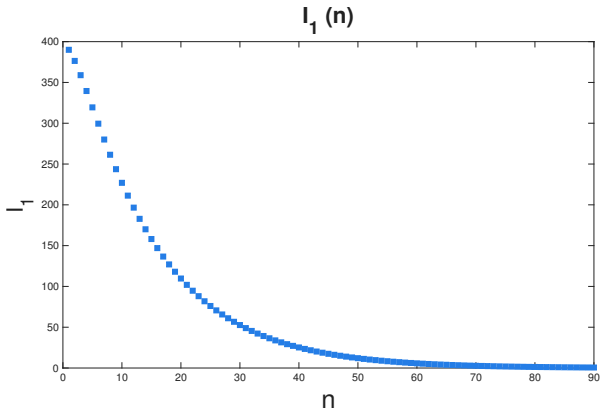

Figure 2: Plots of $I_1(n)$ suggesting local stability of E_{df}.

Now we impose $\beta_1 = 0.5$, maintaining other parameters' numbers, and execute 400 iterations of System (1). Figure 4 provides the obtained values of I_1. Since the dynamics of the epidemic in HS is independent of the dynamics of LS, a change of β_1 does not modify the behavior of I_2, which is converging to 0. Therefore, the modified set of parameters' values results in reaching local stability of state E_1, for which disease is present only in LS.

Let us maintain $\beta_1 = 0.5$ and consider $\beta_2 = 0.5$. Results of performing 350 iterations of System (1) are

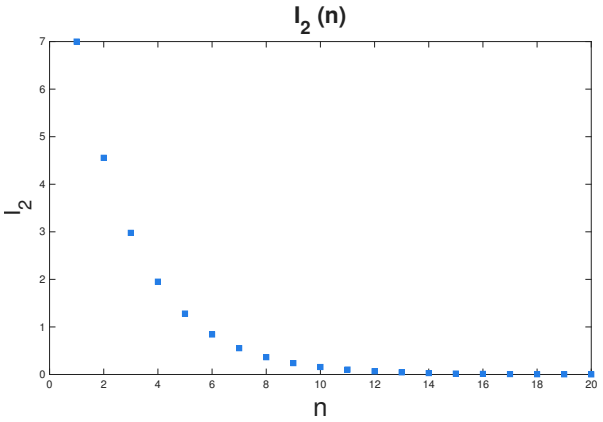

Figure 3: Plots of $I_2(n)$ for local stability of E_{df}. Because of the fast convergence of variable I_2 to 0, we depict only the first 20 consecutive iterations of System (1).

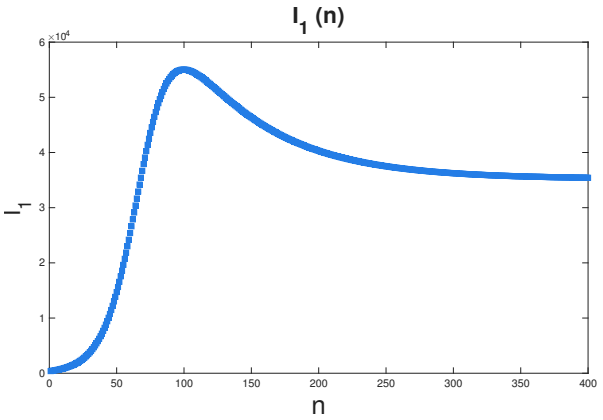

Figure 4: Plots of $I_1(n)$ indicating local stability of E_1.

depicted in Figures (5)–(6). The obtained plots imply local stability of the endemic state E_e for the revised set of parameter values.

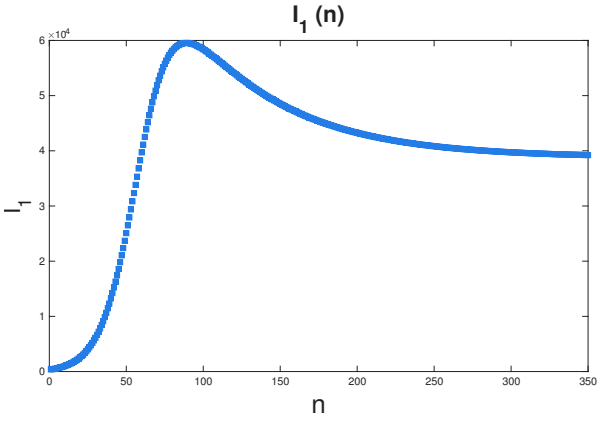

Figure 5: Plots of $I_1(n)$ yielding local stability of E_e.

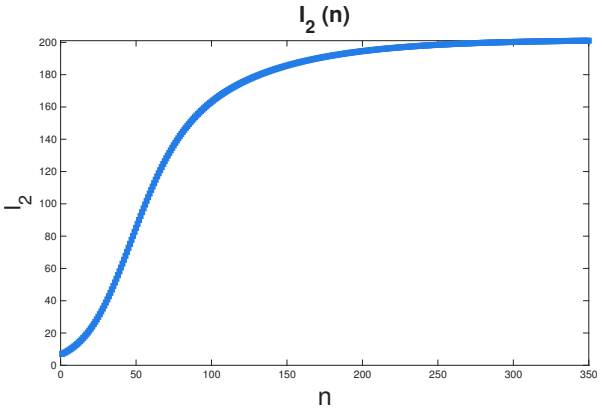

Figure 6: Plots of $I_2(n)$ suggesting local stability of E_e.

DISCUSSION

In this paper we have introduced a four–dimensional SIS–type system of epidemic dynamics. The system concerns the dynamics of a heterogeneous population with two subpopulations: low–risk (LS) and high–risk (HS) differing in the infection risk. Our model is inherently a discrete–time system, meaning it was constructed independently of its continuous–time counterpart. The novelty of our work lies in applying the so–called health function that projects closed the unit interval into itself and in considering different parameter values for each subpopulation. In our model there are three stationary states: disease–free (E_{df}), with disease present only in LS (E_1) and endemic (E_e), with disease persisting in both LS and HS. For state E_{df} we obtain separate conditions that guarantee its local and global stability. Later we fitted our simulated model to real data on the spread of TB in the Warmian–Masurian province in Poland. Then, thanks to the next simulations, it transpired that it is enough to modify only two parameters, which reflect disease transmission in the sole LS and HS, to achieve local stability of each obtained stationary state. We want to emphasize that the universal structure of our model allows us and the other researchers to apply our model to the dynamics of various infectious diseases.

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Analytical and Numerical Techniques

PIES WITH GRADIENT-DRIVEN KNOT REFINEMENT FOR 2D STRESS ANALYSIS

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Adaptive knot refinement, B-spline functions, Parametric integral equation system (PIES), 2D elasticity

ABSTRACT

The paper presents the parametric integral equation system (PIES) with adaptive knot refinement for B-spline basis functions. They are employed to approximate solutions due to their excellent local control properties. The proposed approach is based on knot selection guided by solution derivative analysis. Two knot refinement strategies are explored: relocation - shifting existing knots closer to the singularities, and insertion - adding new knots. Both approaches are effective because adaptation is reduced to modifying the approximation series while leaving the modeled geometry unchanged. Proposed strategies are demonstrated through numerical examples, which indicate that the adaptive B-spline PIES significantly improves accuracy.

INTRODUCTION

The need for greater accuracy and computational efficiency in boundary value elasticity problems drives continuous advancements in the methods used to solve them. The most popular are the finite element method (FEM) (Zienkiewicz 1977, Liu and Quek 2003, Ameen 2005) and the boundary element method (BEM) (Aliabadi 2002, Gao and Davies 2002). They divide a continuous problem into a specified number of elements and use the same mathematical functions to describe the geometry and variability of the solution field. The difference between them is that, in FEM, the solution is approximated directly over the entire domain. In contrast, BEM formulates the problem in terms of boundary integrals, requiring discretization only of the boundary.

The parametric integral equation system (PIES) (Zieniuk and Bołtuć 2006, Bołtuć and Zieniuk 2011, Zieniuk et al. 2014, Bołtuć 2017) is obtained by analytically modifying the boundary integral equation (BIE) (Aliabadi 2002, Gao and Davies 2002), which involves incorporating generalized parametric curves (Farin 2002, Salomon 2006) into its mathematical formulation. The geometry is then modeled using these curves, which improves the efficiency of the design. For straight segments, linear approximation is sufficient, whereas cubic curves accurately capture complex, curved shapes.

However, the most critical aspect is to provide accurate, efficient methods for approximating solutions. PIES can use any method, e.g., the collocation method (Gottlieb and Orszag 1977), which also allows the choice of basis functions. In previous studies (Zieniuk and Bołtuć 2006, Bołtuć and Zieniuk 2011, Zieniuk et al. 2014, Bołtuć 2017), orthogonal or Lagrange polynomials were used, but their global nature hinders the accurate modeling of local features (e.g., singularities). For this reason, B-spline functions (Piegl and Tiller 1997) are applied, whose properties allow practical application in problems with high fluctuations (Bołtuć and Zieniuk 2025). The biggest issue affecting the accuracy is the selection of knots that are necessary to define the B-spline function. They influence the non-singularity of the system of equations being solved and the local control of the solutions. For example, their density can be increased in areas characterized by high solution gradients or singular behavior (de Boor 1973).

This paper proposes an adaptive knot-refinement for 2D elastic problems solved using PIES. Refinement in PIES is efficient because it concerns only the approximation of solutions, without affecting the geometry modeling. This reduces to manipulating the parameters of the approximating series. For each segment, the solution's derivative is computed to identify singularities. It is reduced to calculating the derivative of the B-spline function in the approximation series and using it for individual segments. Two knot-refinement mechanisms are proposed: relocating existing knots and inserting new ones. The paper presents examples demonstrating improvements in solution accuracy.

PIES FORMULA

The mathematical model of PIES for 2D linear elasticity is formulated by mapping the classical BIE into a parametric domain as follows (Zieniuk and Bołtuć 2006)

$$0.5\boldsymbol{u}_l(\bar{s}) = \sum_{j=1}^{n} \int_{s_{j-1}}^{s_j} \left\{ \mathbf{U}^*_{lj}(\bar{s},s)\boldsymbol{p_j}(s) - \mathbf{P}^*_{lj}(\bar{s},s)\boldsymbol{u_j}(s) \right\} J_j(s)ds, \tag{1}$$

where $l, j = 1, 2, ..., n$, $s_{l-1} \leq \bar{s} \leq s_l$, $s_{j-1} \leq s \leq s_j$ and n is the number of segments. PIES is defined in the parametric reference system with parameter s. The Jacobian matrix for the transformation to that system is given by $J_j(s)$, $\boldsymbol{p_j(s)}$ and $\boldsymbol{u_j(s)}$ represent the solution vectors on segment j and $\bar{s}$ is a collocation point. The kernels $\mathbf{U}^*_{lj}(\bar{s}, s)$ and $\mathbf{P}^*_{lj}(\bar{s}, s)$ in (1) represent a parametric transformation of the standard BIE kernels. Their full form is shown in (Zieniuk and Bołtuć 2006).

APPROXIMATION OF THE SHAPE AND THE SOLUTIONS

The shape in PIES is modeled by parametric curves (Salomon 2006, Farin 2002). Different types and degrees are used depending on the problem. To define a polygonal geometry, curves of the first degree are adequate. For curvilinear shapes, third-degree curves are most commonly used, as they are sufficient to approximate most shapes and are numerically stable. Comparing this modeling method to classical BEM or FEM clearly demonstrates its advantages. The number of curves used in PIES depends on the complexity of the shape and is not directly related to the accuracy of the solution approximation. This allows the geometric definition to remain constant while the solution is refined. In the element-based methods, the solution's accuracy is inherently tied to the mesh density. To achieve higher precision, these methods require more elements. Moreover, any modification of the boundary shape typically necessitates remeshing.

The displacement and traction fields in PIES are approximated by the following expressions

$$\boldsymbol{u}_j(s) = \sum_{k=0}^{M-1} \boldsymbol{u}_j^{(k)} N_{k,q}(s), \quad \boldsymbol{p}_j(s) = \sum_{k=0}^{M-1} \boldsymbol{p}_j^{(k)} N_{k,q}(s), \tag{2}$$

where $\boldsymbol{u}_j^{(k)}$, $\boldsymbol{p}_j^{(k)}$ are degrees of freedom, M is the number of degrees of freedom on segment j and $j = 1, 2, ..., n$. $N_{k,q}(s)$ are B-spline basis functions (Piegl and Tiller 1997) of degree q with knots v_k, which are sorted in non-decreasing order, and their role is to partition the parameter space.

An important advantage of PIES is the fact that, due to the separation of the shape approximation from the solutions, any basis functions can be used in (2). However, as described in (Bołtuć and Zieniuk 2025), B-splines offer several advantages over global approximation functions, such as Chebyshev polynomials, which were used most frequently in the PIES method. The key feature is their local character, which gives better control over the solution. Due to the ability to densify knots in places where stress concentrations are expected, B-splines allow for exact capture of changes.

The biggest challenge with the B-spline function is selecting appropriate knots, as their arrangement significantly affects the final results. There are specific properties known that work in exceptional cases, such as (de Boor 1973): densification of knots near singularities or increasing the multiplicity of knots by repeating the same knot value. However, these are general guidelines and are not sufficient to obtain the most accurate solutions.

ADAPTIVE KNOT REFINEMENT

The Schoenberg-Whitney theorem (Schoenberg and Whitney 1953) states that for a B-spline collocation system of equations to be non-singular, the i-th collocation point must lie within the support of the i-th B-spline basis function. This condition is guaranteed by placing the collocation points in the Greville abscissas. However, when solving PIES, the predefined set of collocation points is used (Zieniuk and Bołtuć 2006, Bołtuć and Zieniuk 2011). Therefore, the reverse strategy is applied here, and initial knots are generated based on the collocation points by averaging them in groups of q (Piegl and Tiller 1997)

$$v_{k+q} = \frac{1}{q} \sum_{i=k}^{k+q-1} \bar{s}_i. \tag{3}$$

It should also be checked whether the knot obtained by (3) does not overlap with any collocation point and, if so, move it slightly. The resulting initial knots account only for the basic mechanism of ensuring good conditioning and non-singularity. To obtain an adapted knot distribution, the solution must be examined once to determine its second derivative, which serves as an indicator of regions with high solution variation and potential singularities (de Boor 1973, Yeh et al. 2020). For this purpose, the second derivatives of the approximating series (2) are obtained

$$\boldsymbol{u}_j''(s) = \sum_{k=0}^{M-1} \boldsymbol{u}_j^{(k)} N_{k,q}''(s), \quad \boldsymbol{p}_j''(s) = \sum_{k=0}^{M-1} \boldsymbol{p}_j^{(k)} N_{k,q}''(s), \tag{4}$$

where $N_{k,q}''(s)$ are derivatives of B-spline functions (Piegl and Tiller 1997) and $u_j^{(k)}$, $p_j^{(k)}$ are obtained by solving PIES (1) with knots arranged using (3).

Knowing where the second derivative is greatest or changes rapidly, adaptive refinement of the knot vector can be applied.

Relocating Knots

Step 1 First, the measure w_i is calculated for each analyzed point i on the boundary segment j using (4)

$$w_i = \sqrt{(u_j''(s_i))^2 + (p_j''(s_i))^2}. \tag{5}$$

The maximum value of w_i must be significantly greater than its typical value, represented by the median $\tilde{w}$. If

it is not met, the refinement is terminated due to the lack of a significant singularity.

Step 2 Then w_i are normalized $\bar{w}_i = w_i / \sum w_i$ to obtain the shares throughout segment j and the cumulative weight C is calculated and used as a mapping function to distribute the knots (de Boor 1973, Yeh et al. 2020)

$$C_0 = 0, C_i = \sum_{o=1}^{i-1} \bar{w}_i. \tag{6}$$

Step 3 The outermost $q+1$ knots on both ends remain fixed. For each of the others v_k, the interval $C_i \leq v_k \leq C_{i+1}$ is searched for. The new knot position v_k^{new} is calculated based on

$$v_k^{new} = \alpha v_k + (1-\alpha) v_k^a, \tag{7}$$

where α defines the relative contribution between the initial and adaptive knot location (1 means no adaptation), v_k is an initial knot location, v_k^a is a knot location with adaptation obtained by inversion of the cumulative weight C performed by linear interpolation between discrete points (de Boor 1973, Yeh et al. 2020)

$$v_k^a = s_i + \frac{v_k - C_i}{C_{i+1} - C_i}(s_{i+1} - s_i). \tag{8}$$

Step 4 Displaced knots should remain in proper relationship with collocation points. They can be modified to remain between the collocation points they were initially, or collocation points can be redetermined by formula (3) using new knots instead of $\bar{s}_i$.

Step 5 If a strong local singularity exists, the nearest knot (v_k^s) is pulled toward such point using formula (7) (with v_k^s and α_p). If w_i is constant over an interval (e.g., $q = 2$), its endpoint is selected.

Inserting Knots

Step 1 The same as Step 1 and 2 of Relocating Knots.

Step 2 Based on C (6), intervals $[s_i, s_{i+1}]$ are determined where it exhibits significant growth (the difference between C_i and C_{i+1} is significantly greater than the median).

Step 3 The number of new knots ($cNew$) in a singular interval is determined based on the intensity of the singularity and the width of the interval with the maximum accepted number of knots.

Step 4 New knots are inserted evenly in the designated interval according to the following formula

$$v_k^{new} = s_i + \frac{k}{cNew + 1}(s_{i+1} - s_i). \tag{9}$$

Step 5 Inserting new knots increases the number of collocation points. To maintain proper matrix conditioning, collocation points in the refined regions are readjusted using the Greville abscissas of the updated knots. A flowchart summarizing the steps followed in the proposed approach is presented in Fig. 1.

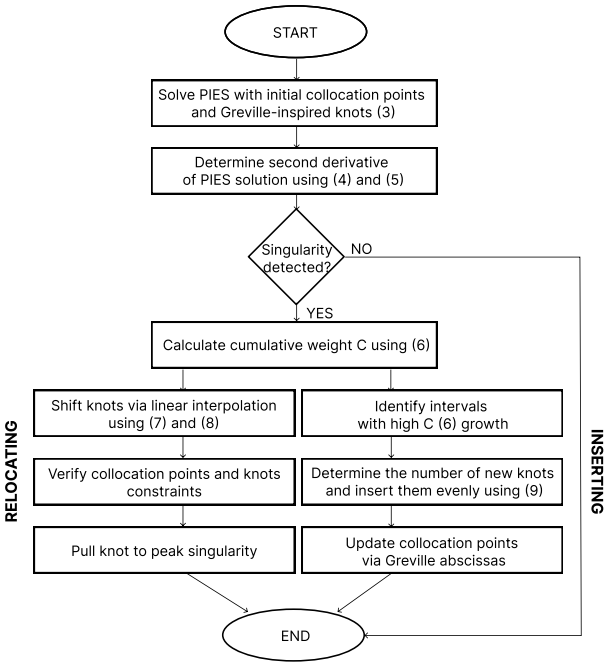

Figure 1: Steps of adaptive knot refinement.

VALIDATION

Case Study 1

The first example concerns a plate with a circular hole with a radius $a = 1m$ and a side length $20m$ subjected to a uniform tension of magnitude $S = 1$ MPa in the x-direction. Fig. 2 shows the problem under consideration and the analytical solution is presented in (Timoshenko and Goodier 1970).

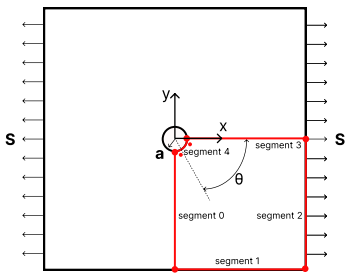

Figure 2: The plate with a hole under a uniform tension.

Due to symmetry, only a quarter of the plate is modeled using five Bézier curves: 4 cubic and 1 linear. A plane stress state is assumed with the following material properties $E = 200 \times 10^9 MPa$ and $v = 0.33$. In the approximation series (2), $M = 9$ is adopted, and the collocation points are distributed according to the roots of the Chebyshev polynomial of the first kind (Abramowitz and Stegun 1972). Then two methods for generating initial knots are considered (Fig. 3): uniform and Greville-inspired (3), both with B-spline functions with $q = 2$.

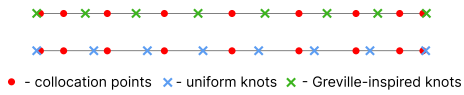

Figure 3: Initial knots and collocation points.

Before applying the adaptive strategies, the stresses σ_θ near the hole were evaluated at 50 boundary points (spaced every 0.01). The initial Greville-inspired knots yielded an approximately 1.5 times lower average relative error ($\varepsilon_{\sigma_\theta} = 3.92\%$) compared to uniform knots ($\varepsilon_{\sigma_\theta} = 6.25\%$). Although Greville-inspired placement improves accuracy, it remains problem-independent and ignores local stress concentrations. To address this, knots are adaptively refined using solution derivatives evaluated at 11 points per segment. The distribution of w_i (5) across all boundary segments is shown on a logarithmic scale in Fig. 4.

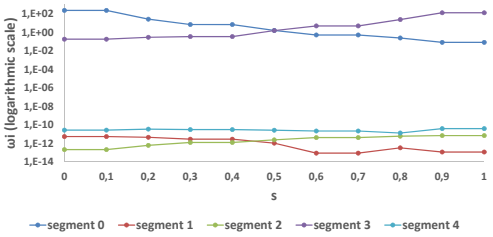

Figure 4: Distribution of w_i as a function of s.

As shown in Fig. 4, the navy blue and purple lines with high slopes near $s = 0$ and $s = 1$, respectively, indicate a sharp change in gradients. The updated knot distribution after applying relocation ($\alpha = 0.6$) is presented in Fig. 5. Unchanged knots on segments 1, 2, and 4 reflect near-zero w_i values, whereas knots on segments 0 and 3 were automatically shifted toward the highest increase of w_i (at $s = 0$ and $s = 1$ respectively).

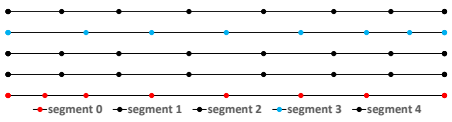

Figure 5: Distribution of knots after relocation.

To analyze the impact of the relocation parameters, subsequent tests evaluated both α (Step 3) and α_p (Step 5). Fig. 6 illustrates the knot distribution on segment 0 (adjacent to the hole) for $\alpha = \{0.4, 0.6, 0.8\}$. Then, using $\alpha = 0.6$, the extreme knot was pulled toward the singularity using various values of α_p, with the resulting stresses σ_θ and corresponding knot vectors shown in Fig. 7.

The influence of α on the distribution is clearly visible in Fig. 6. The smaller α value, the closer the knots get

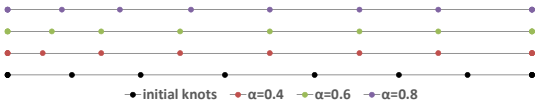

Figure 6: Knots distribution for different values of α.

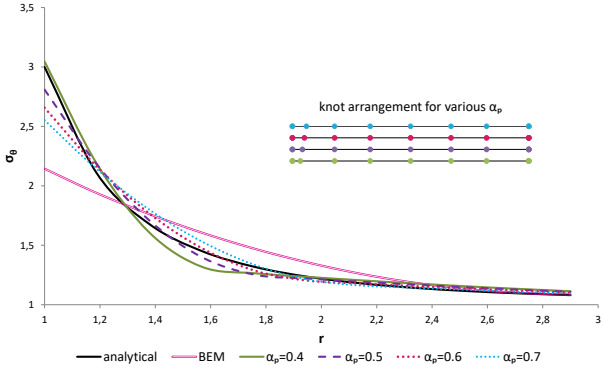

Figure 7: Solutions σ_θ depending on α_p.

to the beginning of segment 0. The average relative error $\varepsilon_{\sigma_\theta}$ for $\alpha = \{0.4, 0.6, 0.8\}$ is 1.14%, 1.68%, 2.55%, respectively. As shown in Fig. 7, the most accurate results are obtained for $\alpha_p = 0.5$ and 0.6. The $\varepsilon_{\sigma_\theta}$ is about 1% in both cases.

Fig. 7 also compares the proposed relocation strategy against classical BEM (35 quadratic elements, 150 DOFs, BEASY Mechanical Design). Near the singularity ($1 \leq x < 3$), PIES with $\alpha_p = \{0.5, 0.6\}$ achieves an average relative error of approximately 3% with only 90 DOFs, compared to 8% for BEM. Without adaptive relocation, the PIES error rises to approximately 10%, proving that proper knot placement - rather than B-splines alone - is essential for singularity capturing.

The relocation algorithm is also applied for B-spline functions with $q = 3$ and $q = 4$, assuming $\alpha = 0.6$ and $\alpha_p = 0.5$. The results confirm the trend observed in the above analysis: solution errors are reduced by almost fivefold compared to the initial non-adaptive placement (for $q = 3$ from 4.67% to 0.83% and for $q = 4$ from 4.47% to 0.80%).

The last test concerns inserting new knots in the singularity region. As a result of the proposed algorithm on segments 0 and 3, three knots are added evenly, near the beginning and end of the segment, respectively. The average relative error of the solutions obtained after inserting new knots is equal to 0.54%. It should be remembered that inserting knots requires increasing the number of collocation points and rearranging them.

Case Study 2

The second example involves the tension of a restrained rectangular plate with dimensions $a = 3m$ and $b = 1m$, shown in Fig. 8. A material in plane strain conditions

is assumed with $E = 1MPa$ and $v = 0.3$. The left side of the plate is completely fixed and $S = 1MPa$.

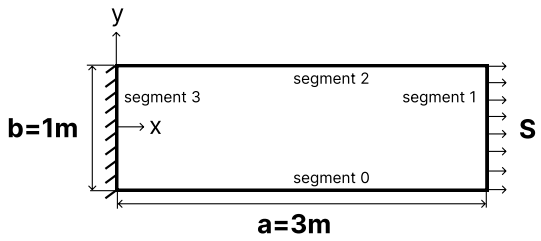

Figure 8: The restrained rectangular plate.

Fig. 9 presents values w_i (5) examined at 11 points of each segment, taking into account the collocation points ($M = 10$) placed at the roots of the Chebyshev polynomials and the Greville-inspired knots (3). As can be seen, the singularity occurs in the upper and lower left corners of the plate, on a fixed segment.

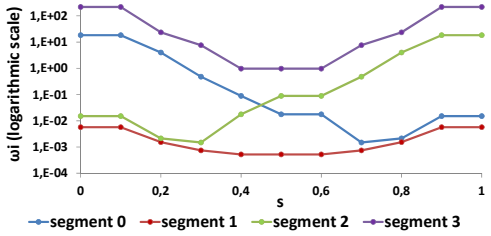

Figure 9: Distribution of w_i.

Once again, both adaptive approaches are applied and presented in Fig. 10. The parameters used are $\alpha = 0.6$, $\alpha_p = 0.5$ and $q = 2$.

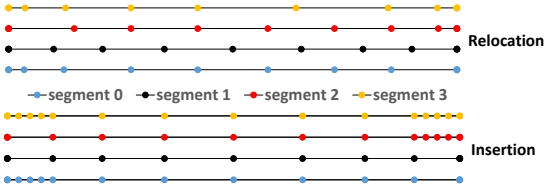

Figure 10: The knots after adaptive refinement.

As shown in Fig. 10, knots automatically densify in singular regions in both approaches.
At the fixed left segment, stresses σ_x peak at the corners and rapidly decay into a smooth, oscillation-free profile. The stress distributions obtained using PIES with B-spline functions and adapted knots are shown in Fig. 11. As seen in Fig. 11, the line representing the non-adaptive PIES (80 DOFs) solutions exhibits pronounced numerical oscillations, a typical symptom of poorly chosen knots. The results obtained using the classical BEM (28 quadratic elements, 120 DOFs, BEASY Mechanical Design) exhibit the same or even worse behavior. The lines representing PIES solutions after adaptation

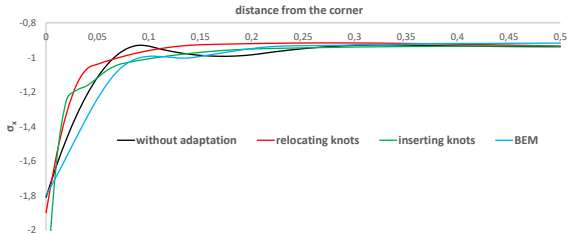

Figure 11: Stress values σ_x at a distance from the corner.

are significantly flatter towards the center of the segment and reach lower values, which better capture the physical nature of the stress concentration at the corner. The relocation version gives slightly better results, as it converges fastest to a stable stress value away from the corner and shows no oscillations. Moreover, it does not increase degrees of freedom (like in inserting strategy to 104 DOFs), as it is based solely on the existing knots.

CONCLUSIONS

In this paper, an approach was developed that integrates adaptive refinement into the PIES method. The core of the research focuses on a knot adaptation strategy driven by the problem's physical nature, specifically by analyzing the second derivatives of the solutions. Two distinct refinement strategies, knot relocation and knot insertion, were implemented and evaluated. It should be emphasized that the proposed strategy in PIES is efficient, as it involves only modifications to the approximation series and does not affect the modeled boundary geometry.
Two examples are analyzed: a plate with a hole under tension and a restrained rectangular plate, both of which exhibit corner stress concentrations due to singularities. Based on the solved problems, the proposed adaptive refinement strategy significantly improves the accuracy of the obtained results. They are characterized by much greater accuracy and the ability to capture singularities, also in comparison to classical BEM.
An essential advantage of the proposed approach is that in PIES, refinement is performed exclusively at the solution-approximation level, without the need to remodel or modify the geometry. Moreover, the singularity detection procedure is straightforward to implement. Since the solution is approximated globally over segments, the identification of regions requiring refinement can be performed using series with derivatives of B-spline functions.
The study determined the parameter values necessary for adaptive refinement. However, further investigation using additional examples would allow for even more effective selection. Using indicators of singularity other than the second derivative of the solutions can also be considered.

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A HYBRID PIES–PINN APPROACH FOR SOLVING TWO-DIMENSIONAL ORTHOTROPIC POTENTIAL PROBLEMS

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KEYWORDS
Parametric Integral Equation System (PIES), Physics-Informmed Neural Network (PINN), PIES-PINN coupling, 2D boundary value problems

ABSTRACT
This paper presents a hybrid computational framework that integrates the Parametric Integral Equation System (PIES) method with Physics-Informed Neural Networks (PINNs) for the analysis of two-dimensional orthotropic potential problems. The proposed approach enables the decomposition of the computational domain into multiple subdomains, where the solution is obtained independently using either PIES or PINN, depending on the characteristics of the analyzed region. Coupling between adjacent subdomains is achieved through the enforcement of compatibility and equilibrium conditions at their interfaces. The proposed framework is validated through a representative numerical example, which confirms its accuracy and applicability.

1. INTRODUCTION

In recent years, machine learning techniques have attracted significant attention owing to their ability to address problems governed by partial differential equations (PDEs). Among these approaches, Physics-Informed Neural Networks (PINNs) [1] have emerged as a particularly promising framework and have been successfully applied to a wide range of forward and inverse problems [2]. The central idea of PINNs is to reformulate a PDE-constrained boundary value problem as an optimization problem, in which the governing equations, boundary conditions, and available data are incorporated directly into the loss function through automatic differentiation.

Despite their considerable potential, PINNs continue to face several important challenges, including the high computational cost of network training and difficulties associated with complex geometries and large-scale problems. To address these limitations, considerable research efforts have been devoted to the development of more efficient and scalable PINN frameworks. Among the proposed approaches, domain decomposition has emerged as a particularly promising strategy. In this framework, the computational domain is partitioned into a collection of subdomains, each represented by an independent neural network. This concept has led to the development of several domain-decomposition-based PINN formulations, including Conservative Physics-Informed Neural Networks (CPINNs) [3] and Extended Physics-Informed Neural Networks (XPINNs) [4]. Beyond improving scalability, these methods naturally facilitate distributed computing and have been shown to substantially reduce training times through the parallel execution of neural network training on multiple graphics processing units (GPUs) [5].

In this study, a hybrid computational framework integrating PINNs with the Parametric Integral Equation System (PIES) [6,7] is proposed for solving two-dimensional potential boundary value problems in orthotropic media. The proposed framework leverages the complementary strengths of both methodologies to provide a more flexible solution strategy.

The PIES method can be regarded as an analytical extension of classical Boundary Integral Equations (BIEs), founded on the formal separation of boundary geometry representation from the approximation of boundary variables. As with conventional BIE-based formulations, PIES reduces the dimensionality of the governing problem by one, allowing the field solution within the domain to be determined solely from information prescribed on its boundary. A distinctive feature of PIES is that it eliminates the need for discretization of both the problem domain and the boundary. This property arises naturally from the parametric representation of the boundary geometry and the decoupling of geometric modelling from boundary-function approximation. As a result, PIES avoids the mesh generation process required by the Finite Element Method (FEM) and the boundary discretization inherent to the Boundary Element Method (BEM).

In the proposed framework, the computational domain is decomposed into a set of subdomains, within which the solution is independently obtained using either the PIES or PINN formulation. The global solution is established by enforcing compatibility and equilibrium conditions at the interfaces between neighbouring subdomains. Owing to the parametric nature of the PIES method, the interfaces between PIES- and PINN-based subdomains can be described by parametrically defined segments, while the geometric representation remains independent of the approximation scheme adopted for the boundary physical quantities. In the present study, the field variables and the corresponding fluxes along the subdomain interfaces and the external boundary are approximated within the PIES framework using Chebyshev series expansions.

The proposed framework extends the hybrid strategy introduced in [8], where potential boundary value problems governed by the Laplace equation were investigated. In the present study, this approach is further generalized to a broader class of orthotropic potential problems. The paper presents the theoretical foundations of the PINN and PIES formulations, details the procedure for their coupling within the proposed hybrid framework and assesses the accuracy of the resulting solutions.

2. PROBLEM FORMULATION

In this study, an orthotropic potential problem defined in a two-dimensional domain Ω bounded by the boundary Γ is

considered. The governing equation of the problem is expressed as

$$k_1 \frac{\partial^2 u}{\partial x_1^2} + k_2 \frac{\partial^2 u}{\partial x_2^2} = 0. \quad (1)$$

The above equation is supplemented by prescribed Dirichlet boundary conditions g on Γ_g and Neumann boundary conditions h on Γ_h, where $\Gamma = \Gamma_g \cup \Gamma_h$. The analysis assumes the absence of internal heat sources and constant material properties, characterized by the coefficients k_1 and k_2, which represent the conductivities along the principal orthotropic directions x_1 and x_2, respectively.

As schematically illustrated in Fig. 1, the proposed framework employs a domain decomposition strategy, in which the domain Ω is partitioned into two subdomains: Ω_{PINN}, in which the solution is approximated using the PINN formulation, and Ω_{PIES}, where the solution is obtained within the PIES framework.

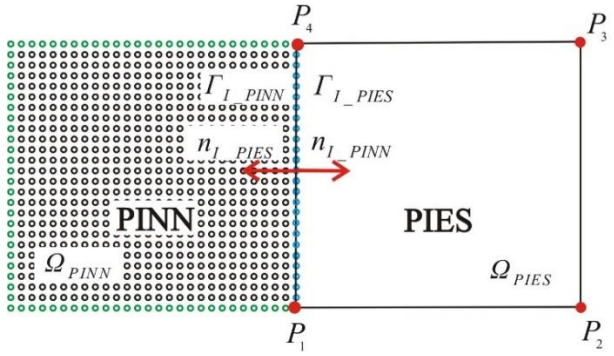

Fig. 1. Schematic illustration of the domain decomposition into the PINN subdomain Ω_{PINN} and the PIES subdomain Ω_{PIES}, with the common interface defined by $\Gamma_{I\,PINN}$ and $\Gamma_{I\,PIES}$.

It should be emphasized that the representation of the computational domain differs fundamentally between the two methods. In the PINN approach, the domain is represented by a set of collocation points distributed within the domain interior and along its boundary. In contrast, in the PIES method, the geometry of the problem is defined exclusively by the boundary, which, in the present case, is described by four parametric linear segments $\boldsymbol{\Gamma}_j(s)$ determined by four corner points $\boldsymbol{P}_j$.

A brief overview of the PIES and PINN formulations is provided below. The procedure adopted for coupling these methods at the interface between the subdomains is described in Section 3.

2.1 Parametric Integral Equation System

The PIES method is a computational approach for solving boundary-value problems, developed based on an analytical modification of the BIE formulation. As a result of this modification, similarly to the classical BIE, the analysis of a boundary-value problem in PIES can be reduced exclusively to the analysis performed on the boundary of the considered domain, which leads to a mathematical reduction of the problem from the domain to its boundary. However, the introduced modification results in a formal decoupling of the boundary geometry representation from the approximation of boundary solutions in PIES, enabling independent strategies for geometric modelling and solution approximation. This feature distinguishes PIES from the conventional boundary element method (BEM), which is widely established as the standard numerical approach for the solution of BIEs.

An elementary example of such a representation is shown in Fig. 1. For the considered PIES subdomain Ω_{PINN}, it is sufficient to define only the boundary of the domain using four linear parametric segments $\boldsymbol{\Gamma}_j(s)$, without the need for additional discretization procedures. As a result, the PIES approach avoids the domain discretization inherent to the finite element method (FEM) and eliminates the necessity of direct boundary discretization required in BEM. Consequently, the amount of geometric information required for numerical modelling is significantly reduced. As demonstrated in Fig. 1, the boundary geometry in PIES can be represented by only four corner points $\boldsymbol{P}_j$, whereas conventional BEM formulations require a substantially larger number of boundary elements and associated nodes for boundary representation.

The PIES formulation for the orthotropic potential problem considered in this study is expressed as follows

$$0.5u_l(\bar{s}) = \sum_{j=0}^{n} \int_{s_{j-1}}^{s_j} \{\overline{U}_{lj}^*(\bar{s},s)p_j(s) - \overline{P}_{lj}^*(\bar{s},s)u_j(s)\} J_j(s)ds, \quad (2)$$

where

$$l = 1,2,\dots,n, s_{l-1} \le \bar{s} \le s_l, s_{j-1} \le s \le s_j.$$

The parametric representation of the boundary segment $\boldsymbol{\Gamma}_j(s)$, defined by its components in the x_1 and x_2 directions, $\Gamma_j^{(1)}(s)$ and $\Gamma_j^{(2)}(s)$, respectively, is analytically embedded into Eq. (2) through the fundamentalintegral kernels $\overline{U}_{lj}^*(\bar{s},s)$ and $\overline{P}_{lj}^*(\bar{s},s)$, defined as follows

$$\overline{U}_{lj}^*(\bar{s},s) = \frac{1}{2\pi\sqrt{k_1 k_2}} ln \frac{1}{\left(\eta_1^2/k_1 + \eta_2^2/k_2\right)^{0.5}}, \quad (3)$$

$$\overline{P}_{lj}^*(\bar{s},s) = \frac{1}{2\pi\sqrt{k_1 k_2}} \frac{\eta_1 n_j^{(1)}(s) + \eta_2 n_j^{(2)}(s)}{\eta_1^2/k_1 + \eta_2^2/k_2}. \quad (4)$$

where

$$\eta_1 = \bar{\Gamma}_l^{(1)}(\bar{s}) - \Gamma_j^{(1)}(s), \eta_2 = \bar{\Gamma}_l^{(2)}(\bar{s}) - \Gamma_j^{(2)}(s). \quad (5)$$

In Eq. (4), $n_j^{(1)}(s)$ and $n_j^{(2)}(s)$ denote the components of the outward unit normal vector to the boundary, while $J_j(s)$ represents the Jacobian of the parametric transformation. Furthermore, $\bar{\Gamma}_l^{(1)}(\bar{s})$ and $\bar{\Gamma}_l^{(2)}(\bar{s})$ represent the scalar components of the l-th boundary segment that define the position of the source point $\bar{s}$ in the parametric domain. Within the PIES framework, the solution of the boundary-value problem is reduced to determining the boundary distributions of the potential $u_j(s)$ and the corresponding flux $p_j(s)$ along each parametric boundary segment $\boldsymbol{\Gamma}_j(s)$.

Owing to the separation between boundary geometry approximation and boundary function approximation inherent in PIES, different approximation schemes can be independently adopted for these quantities. In the present study, the boundary functions are approximated using Chebyshev polynomial expansions in the form

$$u_j(s) = \sum_{k=0}^{K-1} u_j^{(k)} T_j^{(k)}(s), \quad (6)$$

$$p_j(s) = \sum_{k=0}^{K-1} p_j^{(k)} T_j^{(k)}(s), \quad (7)$$

where $u_j^{(k)}$ and $p_j^{(k)}$ represent the corresponding expansion coefficients and $T_j^{(k)}(s)$ denotes the Chebyshev polynomials of the first kind. The boundary variables on individual segments $\boldsymbol{\Gamma}_j(s)$ are approximated by Chebyshev series, allowing the accuracy of the numerical solution to be systematically controlled through the number of retained terms K, independently of the boundary geometry representation. This feature constitutes a key advantage of PIES over conventional BEM approaches, where increased accuracy is typically achieved through boundary mesh refinement and the introduction of additional boundary elements.

2.2 Physics-Informed Neural Network

The concept of PINNs is based on using a neural network $\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})$ as an approximation of the solution field $u(\boldsymbol{x})$, where $\boldsymbol{x}$ represents the spatial coordinates and $\boldsymbol{\theta}$ denotes the set of trainable network parameters. The approximation is constructed for the considered PDE, defined herein by Eq. (1). For the neural network approximation to be admissible, i.e., $u(\boldsymbol{x}) \approx \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})$, the following conditions must be satisfied

$$\boldsymbol{D}[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = f(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega, \tag{8}$$

$$\boldsymbol{B}_g[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = g, \quad \boldsymbol{x} \in \Gamma_g, \tag{9}$$

$$\boldsymbol{B}_h[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = h, \quad \boldsymbol{x} \in \Gamma_h, \tag{10}$$

where $\boldsymbol{D}[\cdot]$ represents the governing differential operator, whereas $\boldsymbol{B}_g[\cdot]$ and $\boldsymbol{B}_h[\cdot]$ denote the boundary operators associated with the Dirichlet and Neumann boundary conditions, respectively. These operators are defined on the corresponding boundary segments Γ_g and Γ_h, with the entire boundary satisfying $\Gamma = \Gamma_g \cup \Gamma_h$. In addition, $f(\boldsymbol{x})$ is a prescribed source function defined within the domain i.e., for $\boldsymbol{x} \in \Omega$. For the orthotropic potential problem considered in this study, the governing differential operator and the corresponding boundary operators take the following form

$$\boldsymbol{D}[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = k_1 \frac{\partial^2 \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_1^2} + k_2 \frac{\partial^2 \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_2^2}, \tag{11}$$

$$\boldsymbol{B}_g[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta}), \tag{12}$$

$$\boldsymbol{B}_h[\boldsymbol{x};\mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})] = k_1 \frac{\partial \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_1} n_1(\boldsymbol{x}) + k_2 \frac{\partial \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_2} n_2(\boldsymbol{x}), \tag{13}$$

where $n_1(\boldsymbol{x})$ and $n_2(\boldsymbol{x})$ denote the components of the outward normal vector to the boundary. Moreover, the source term is assumed to be $f(\boldsymbol{x}) = 0$.
The training of the PINN model for solving the boundary-value problem consists of determining the optimal neural network parameters $\boldsymbol{\theta}^*$ such that conditions (11)–(13) are satisfied.

3. COUPLING OF PIES AND PINN

To couple PIES and PINN, the compatibility conditions at the common interface between the subdomains must be considered. The interface boundaries are defined by $\Gamma_{I\,PINN}$ and $\Gamma_{I\,PIES}$ corresponding to the sides of Ω_{PINN} and Ω_{PIES}, respectively, as illustrated in Fig. 1. These conditions can be expressed in the following form

$$u(\boldsymbol{x})|_{\Gamma_{I\,PINN}} = u(s)|_{\Gamma_{I\,PIES}}, \tag{14}$$

$$\nabla u(\boldsymbol{x})|_{\Gamma_{I\,PINN}} \boldsymbol{n}_{I\,PINN} = -\nabla u(s)|_{\Gamma_{I\,PIES}} \boldsymbol{n}_{I\,PIES}, \tag{15}$$

where $\boldsymbol{n}_{I\,PINN}$ and $\boldsymbol{n}_{I\,PIES}$ denote the outward unit normal vectors to the interface boundaries of the respective subdomains. For the differential equation (1), Eqs. (14) and (15) enforce continuity of the potential field and conservation of flux across the interface, thereby ensuring a physically consistent coupling between the PINN and PIES subdomains. It should be noted that the quantities $u(\boldsymbol{x})$ and $\nabla u(\boldsymbol{x})$ are represented differently in the PINN and PIES formulations. In the PINN subdomain, for $\boldsymbol{x} \in \Gamma_{I\,PINN}$, these quantities are approximated by a neural network and can be expressed as

$$u(\boldsymbol{x})|_{\Gamma_{IPINN}} \approx \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta}), \tag{16}$$

$$\nabla u(\boldsymbol{x})|_{\Gamma_{IPINN}} \boldsymbol{n}_{IA} \approx k_1 \frac{\partial \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_1} n_1(\boldsymbol{x}) + k_2 \frac{\partial \mathcal{N}(\boldsymbol{x};\boldsymbol{\theta})}{\partial x_2} n_2(\boldsymbol{x}). \tag{17}$$

In contrast, within the PIES formulation, the quantities $u(s)|_{\Gamma_{I\,PIES}}$ and $\nabla u(s)|_{\Gamma_{I\,PIES}} \boldsymbol{n}_{I\,PIES}$ are approximated by Chebyshev series expansions, as defined in Eqs. (6) and (7).
In the proposed hybrid PIES–PINN framework, the boundary value problem is solved iteratively to determine the optimal neural network parameters $\boldsymbol{\theta}^*$, while simultaneously satisfying the interface conditions defined by Eqs. (14) and (15) within the PIES formulation. A flowchart of the coupling algorithm is presented in Fig. 2.

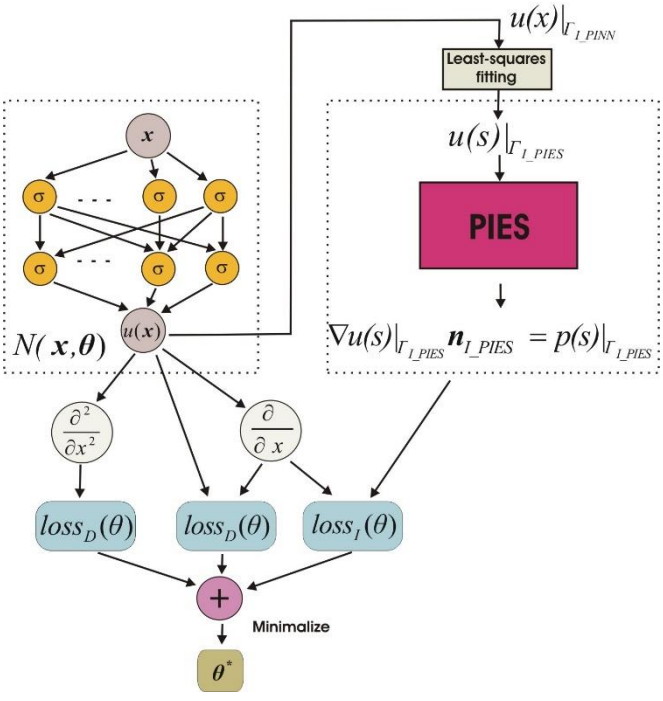

Fig. 2. Flowchart of the coupling algorithm between the PIES and PINN formulations.

The training of the neural network is carried out by minimizing the objective function

$$\boldsymbol{\theta}^* = \underset{\boldsymbol{\theta}^*}{argmin}(\boldsymbol{\mathcal{L}}_D(\boldsymbol{\theta}) + \boldsymbol{\mathcal{L}}_B(\boldsymbol{\theta}) + \boldsymbol{\mathcal{L}}_I(\boldsymbol{\theta})), \tag{18}$$

where $\boldsymbol{\mathcal{L}}_D(\boldsymbol{\theta})$ is the loss associated with the residual of the governing differential equation, $\boldsymbol{\mathcal{L}}_B(\boldsymbol{\theta})$ represents the

contribution of the boundary conditions and $\boldsymbol{\mathcal{L}}_I(\boldsymbol{\theta})$ accounts for the enforcement of the interface conditions at the PINN–PIES coupling boundary. The individual loss terms are defined as follows

$$\boldsymbol{\mathcal{L}}_D(\boldsymbol{\theta}) = \frac{1}{N_D}\sum_{i=1}^{N_D}\left|k_1\frac{\partial^2\mathcal{N}(\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial x_1^2} + k_2\frac{\partial^2\mathcal{N}(\boldsymbol{x}_i;\boldsymbol{\theta})}{\partial x_2^2}\right|^2, \tag{19}$$

$$\boldsymbol{\mathcal{L}}_B(\boldsymbol{\theta}) = \frac{1}{N_{B_g}}\sum_{i=1}^{N_{B_g}}\left|\mathcal{N}\left(\boldsymbol{x}_i^g;\boldsymbol{\theta}\right)g\left(\boldsymbol{x}_i^g\right)\right|^2 + \frac{1}{N_{B_h}}\sum_{i=1}^{N_{B_h}}\left|k_1\frac{\partial\mathcal{N}\left(\boldsymbol{x}_i^h;\boldsymbol{\theta}\right)}{\partial x_1}n_1\left(\boldsymbol{x}_i^h\right) + k_2\frac{\partial\mathcal{N}\left(\boldsymbol{x}_i^h;\boldsymbol{\theta}\right)}{\partial x_2}n_2\left(\boldsymbol{x}_i^h\right) - h\left(\boldsymbol{x}_i^h\right)\right|^2, \tag{20}$$

$$\boldsymbol{\mathcal{L}}_I(\boldsymbol{\theta})) = \frac{1}{N_I}\sum_{i=1}^{N_I}\left|k_1\frac{\partial\mathcal{N}(\boldsymbol{x}_i^I;\boldsymbol{\theta})}{\partial x_1}n_1(\boldsymbol{x}_i^I) + k_2\frac{\partial\mathcal{N}(\boldsymbol{x}_i^I;\boldsymbol{\theta})}{\partial x_2}n_2(\boldsymbol{x}_i^I) - \left(-\nabla u(s)|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}\right)\right|^2. \tag{21}$$

where N_D denotes the set of collocation points in the subdomain Ω_{PINN}, N_{B_g} and N_{B_h} denote the sets of collocation on the boundary and N_I is the number of collocation points located on the interface between the subdomains.

The last term in the loss function (21), $-\nabla u(s)|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$ represents the coupling with the PIES subdomain. Its evaluation requires solving the PIES problem at each training iteration. First, the current PINN approximation of the interface solution, $u(\boldsymbol{x})|_{\Gamma_{I\,PINN}}$ is projected onto the PIES interface and approximated by the Chebyshev expansion given in Eq. (6), using a least-squares fitting procedure. The obtained Chebyshev expiation (6) is then used to solve the PIES equations, yielding the corresponding interface flux $\nabla u(s)|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$ represented by the expansion defined in (7).

The Chebyshev expansions (6) and (7) involve the coefficients $u_j^{(k)}$ and $p_j^{(k)}$, respectively. The coefficients $u_j^{(k)}$ in (6), describing the interface potential $u(s)|_{\Gamma_{I\,PIES}}$ are updated at each training iteration through a least-squares approximation of the current PINN prediction $u(\boldsymbol{x})|_{\Gamma_{I\,PINN}}$. In contrast, the coefficients $p_j^{(k)}$ in (7) corresponding to the interface flux $\nabla u(s)|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$, are obtained from the solution of the PIES system.

The PIES equations are solved using the collocation approach described in [6]. Collocation points are distributed in the parametric domains of the line segments forming the boundary. By evaluating Eq. (2) at the collocation points associated with the parameter $\bar{s}$, a system of algebraic equations is obtained. The size of this system depends on the number of boundary segments and on the number K of terms retained in the Chebyshev expansions. This leads to the matrix equation

$$[H^*]\begin{Bmatrix}p\\p_I\end{Bmatrix} = [G^*]\begin{Bmatrix}u\\u_I\end{Bmatrix}. \tag{22}$$

The matrices $[H^*]$ and $[G^*]$ are constructed by evaluating both regular and singular integrals. Details of the numerical implementation can be found in [6,7]. The vector u contains the coefficients of the Chebyshev expansions representing the potential $u_j(s)$ in Eq. (6) on the external boundary, whereas u_I contains the corresponding coefficients on the interface $\Gamma_{I\,PIES}$. Similarly, the vectors p and p_I contain the coefficients of the Chebyshev expansions approximating the flux $p_j(s)$ in (7) on the external boundary and on the interface, respectively. It should be emphasized that the matrices $[H^*]$ and $[G^*]$ are assembled only once prior to the iterative coupling procedure.

4. ACCURACY ANALYSIS OF THE SOLUTIONS

This section presents the results of a preliminary accuracy assessment of the proposed approach for the two-dimensional orthotropic problem defined in Fig. 1. In the analysis, the boundary of the subdomain Ω_{PIES} was dwscribed using four linear parametric segments $\boldsymbol{\Gamma}_j(s)$, defined by four corner points, with one segment selected to represent the interface $\boldsymbol{\Gamma}_{I\,PIES}(s)$ between the subdomains. The square subdomain Ω_{PINN} was discretized using a uniform grid of 100×100 collocation points, including 100 points located on the interface $\Gamma_{I\,PINN}$. The interface quantities $u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}$ and $-\nabla u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$ were approximated within the PIES formulation using the first five terms of the Chebyshev expansions given in (6) and (7). The PINN model was implemented using a fully connected feedforward neural network consisting of six hidden layers with 100 neurons in each layer. The network takes the two-dimensional vector $\boldsymbol{x} = \{x_1, x_2\}$ as input and returns a scalar output u, corresponding to the approximated pointwise distribution of the field variable within the computational domain and along its boundary. The orthotropic material parameters were set to $k_1 = 1$ and $k_2 = 5$ for both subdomains. The external boundaries were subjected to Dirichlet boundary conditions of the form

$$u(x_1, x_2) = -10x_1^2 + 2x_2^2 + 3x_1x_2 + 4x_1 + 5x_2 + 6. \tag{23}$$

The function defined by Eq. (23) satisfies the governing equation, Eq. (2), and therefore constitutes the exact analytical solution to the problem under consideration.

The proposed hybrid PIES–PINN algorithm was implemented using the PyTorch framework. Figure 3 illustrates the solution fields predicted within the PINN domain and compares them with the corresponding analytical distributions obtained from Eq. (23). The presented results were generated after 10,000 training iterations using the ADAM optimization algorithm. The close agreement between the numerical and analytical solutions demonstrates the accuracy and effectiveness of the proposed hybrid approach.

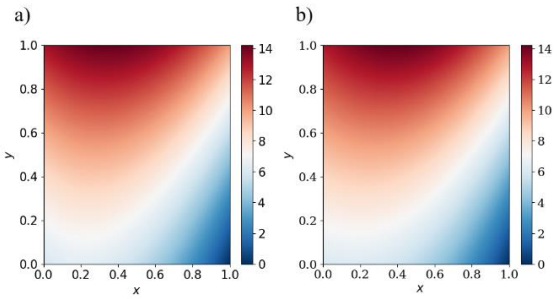

Fig. 3. Error distribution of the solutions in the PINN subdomain compared with the analytical solution given by Eq. (23).

Figure 4 compares the final PIES approximations of the interface potential, $u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}$ and the corresponding normal flux, $\nabla u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$, on $\Gamma_{I\,PIES}$ with the exact analytical solutions, shown as black lines.

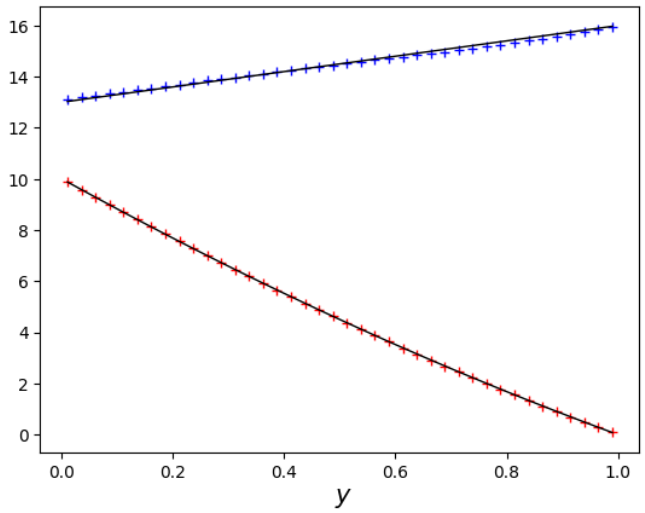

Fig. 4. Interface potential, $u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}$ (blue), and normal flux, $\nabla u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES}$ (red), obtained by PIES on $\Gamma_{I\,PIES}$ compared with the analytical solutions (black lines).

The analytical expression for the interface potential, $u(\boldsymbol{x})|_{\Gamma_{I\,PIES}}$ is provided by (23). The corresponding exact normal flux is obtained by taking the normal derivative of (23) and is given by

$$\nabla u(x_1, x_2)|_{\Gamma_{I\,PIES}}\boldsymbol{n}_{I\,PIES} = (-20x_1 + 3x_2 + +4)n_1 + (4x_2 + 3x_1 + 5)n_2. \quad (24)$$

Both quantities are represented by the first five terms of the Chebyshev series expansions introduced in Eqs. (6) and (7). Figure 5 shows the convergence histories of the individual components of the loss function defined in Eq. (18) over successive training iterations.

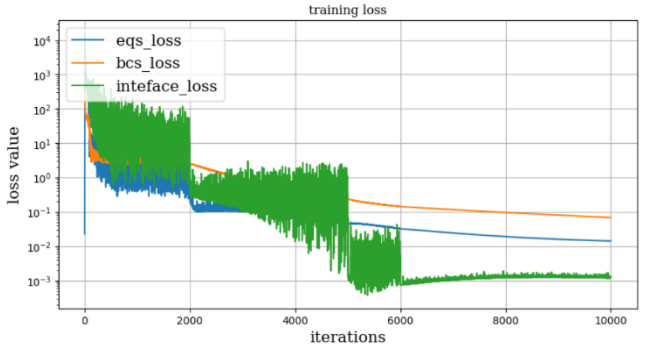

Fig. 5. Evolution of the individual components of the loss function defined in Eq. (18) during the first 10,000 training iterations.

The presented results demonstrate the potential for coupling PIES with PINNs under the proposed consistency conditions.

5. CONCLUSIONS

In this work, a coupling strategy between PIES and PINNs is proposed for solving two-dimensional boundary value problems, using the orthotropic potential equation as a representative example. The proposed approach is particularly suitable because PIES requires only the specification of the boundary of the considered domain, while allowing the geometric representation of the boundary to be decoupled from the approximation of boundary solutions expressed through Chebyshev series.

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INFLUENCE OF SELECTED IFPIES PARAMETERS ON SOLUTIONS IN THE DOMAIN

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KEYWORDS
IFPIES, directed interval arithmetic, uncertainty

ABSTRACT

In the paper, the authors present a search for the best parameters of the FMM tree and groups of solution points in the interval fast parametric integral equations system (IFPIES) from an accuracy and efficiency point of view. Previously, the IFPIES has been successfully applied to modelling uncertainly defined 2D potential BVPs to obtain solutions only on the boundary. The combination of the modified fast multipole tree with the interval parametric integral equations system (IPIES) reduced the method's computational time and RAM usage. A similar technique was applied to find solutions in the domain. However, some parameters associated with the FMM tree and the way of creating solution groups impact the speed of searching for solutions in the domain. The paper aims to determine the optimal parameter values in terms of CPU time and accuracy as a basis for further research.

INTRODUCTION

The interval fast parametric integral equations system (IFPIES) (Kużelewski et al. 2022) belongs to a family of modern computational methods derived from the parametric integral equation system (PIES) (Zieniuk 2003). The members of the family are in a constant state of development and are employed to solve a variety of two- and three-dimensional boundary value problems (BVPs) described by different types of differential equations (e.g., Laplace, Helmholtz, or Navier–Lame). Also, they offer the same remarkable advantages compared to the FEM (Zienkiewicz 1977, Zienkiewicz et al. 2013, Babuska et al. 2004), the BEM (Brebbia et al. 1984, Banerjee and Butterfield 1981, Katsikadelis 2002), or the still being developed modern FEM-BEM hybrids (Nedjar 2017), isogeometric analysis (IgA) (Hughes et al. 2005) and the Virtual Element Method (VEM) (Veiga et al. 2019)). First and foremost are the lack of discretization of the problem's boundary and domain (sometimes called mesh-free) and the ability to improve solution accuracy with only a slight modification of the input data (specifically, the number of collocation points). The PIES family methods utilize parametric functions widely used in computer graphics (among others, Bézier and B-spline curves, Coons and Bézier surface patches). Due to the direct inclusion of these functions into the mathematical formalism of the PIES family methods, the boundary is defined by a small number of control points. Also, using parametric functions reduces the problem's dimensionality by 1. Improving the accuracy of solutions requires only a slight adjustment to the number of collocation points, unlike in the BEM and the FEM, where the discretization process must be repeated. Such an approach differs from a finite difference scheme used in a wide family of mesh-free methods (such as particle methods (Seo et al. 2021), Galerkin methods (Jaśkowiec and Pluciński 2018) or cloud methods (Duarte and Oden 1996)). Also, the part of the PIES family, starting with the interval PIES (Zieniuk et al. 2018), considers the uncertainty in the boundary shape, boundary conditions, and material parameters in modelling and solving BVPs. It should be noted that existing interval element methods (such as the interval FEM (Muhanna et al. 2005) or the interval BEM (Zalewski et al. 2009)) are unable to address uncertainty in the boundary shape for 2D or 3D problems.

Our previous studies on the PIES family methods (including various aspects, i.e. the use of NURBS curves (Zieniuk et al. 2016) or inverse problems (Zieniuk et al. 2014)) have confirmed high accuracy compared to BEM or FEM. However, early implementations of classical and interval PIES required time-consuming calculations and RAM utilization (increased with the problem size squared). It is due to the use by the PIES, similarly to the BEM, dense non-symmetric coefficient matrices. Therefore, to compute the PIES matrices, we need $O(N^2)$ operations and another $O(N^3)$ operations to solve the obtained system using direct solvers (where N is the number of equations of the algebraic equations system). The authors of this paper applied the fast multipole method (FMM) (Greengard and Rokhlin 1985; 1987, Greengard 1988) to eliminate this disadvantage, which resulted in obtaining the fast PIES (FPIES) (Kużelewski and Zieniuk 2020) and the IFPIES (Kużelewski et al. 2022). Application of the FMM required some modifications in the binary tree

(Kużelewski et al. 2020) and resulted in obtaining the system of algebraic equations $\boldsymbol{A} \cdot \boldsymbol{x} = \boldsymbol{b}$ which is generated implicitly in the PIES (and the IFPIES) and solved by the iterative GMRES solver (Saad and Schultz 1986) (and its interval version prepared ourselves). Only the result of the multiplication of the matrix $\boldsymbol{A}$ by the vector of unknowns $\boldsymbol{x}$ is stored in memory, in contrast to the conventional PIES, which has to store the full dense non-symmetric matrix $\boldsymbol{A}$ and the vector $\boldsymbol{b}$ in RAM. Previous studies on the FPIES and IFPIES have focused on efficient and accurate solutions on the boundary (Kużelewski and Zieniuk 2020; 2022), as well as a comparison with a competing method, such as the fast multipole BEM (FMBEM) described in (Liu and Nishimura 2006). Also, the authors of this paper presented the first attempt to find solutions in the domain using the IFPIES (Kużelewski and Zieniuk 2026). However, some parameters related to the FMM, such as the number of tree levels or the number of solution points per group, should be predetermined to avoid searching for their values before solving each examined problem.
The main goal of this paper is to determine the optimal values of two parameters of the IFPIES: the number of tree levels and the number of solution points in groups, for solving any example. Optimal means the number of tree levels and the number of solution points in groups, which gives highly accurate results in the shortest CPU time. The efficiency and accuracy of the IFPIES for different parameter values are tested on the problem with a polygonal domain.

THE INTERVAL FAST PIES

The IFPIES for 2D potential problems, as a direct successor of the IPIES (Zieniuk et al. 2018) and the FPIES (Kużelewski and Zieniuk 2020), was clearly described in (Kużelewski et al. 2022). It should be noted that the method is based on the Taylor expansion and the modified FMM tree, and that some modifications to the IPIES kernels were necessary to describe the boundary shape in the parametric reference system adequately. Also, the modified directed interval arithmetic and interval numbers, in combination with the Taylor expansion, required defining some variables as complex intervals, i.e., the real or imaginary part of a complex number is treated as an interval.
The general IFPIES formula has the following form:

$$\frac{1}{2}\boldsymbol{u}_l(\widehat{\boldsymbol{\tau}}) = \sum_{j=1}^{n} \mathbb{R}\left\{ \int_{s_{j-1}}^{s_j} \widehat{\boldsymbol{U}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau}) \boldsymbol{p}_j(s) \boldsymbol{J}_j^{(c)}(s) ds - \int_{s_{j-1}}^{s_j} \widehat{\boldsymbol{P}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau}) \boldsymbol{u}_j(s) \boldsymbol{J}_j^{(c)}(s) ds \right\}, \quad (1)$$

$$l = 1, 2, ..., n,\ s_{l-1} \leq \widehat{s} \leq s_l,\ s_{j-1} \leq s \leq s_j,$$

$$\widehat{\boldsymbol{\tau}} = \boldsymbol{S}_l^{(1)}(\widehat{s}) + i\boldsymbol{S}_l^{(2)}(\widehat{s}), \boldsymbol{\tau} = \boldsymbol{S}_j^{(1)}(s) + i\boldsymbol{S}_j^{(2)}(s),$$

where: $\widehat{s}$ and s are defined in the parametric coordinate system, s_{j-1} (s_{l-1}) correspond to the beginning while s_j (s_l) to the end of complex interval segment $\boldsymbol{S}_j^{(c)}$ ($\boldsymbol{S}_l^{(c)}$), n is the number of parametric segments that creates a boundary of the domain in 2D, $\boldsymbol{J}_j^{(c)}(s)$ is the complex interval Jacobian, $\boldsymbol{u}_j(s)$ and $\boldsymbol{p}_j(s)$ are interval parametric boundary functions on individual complex segment $\boldsymbol{S}_j^{(c)}$ of the interval boundary, $\mathbb{R}$ is the real part of complex function, $\boldsymbol{S}_j^{(1)}$, $\boldsymbol{S}_j^{(2)}$, $\boldsymbol{S}_l^{(1)}$, $\boldsymbol{S}_l^{(2)}$ are the interval components connected with the direction of coordinates in a 2D Cartesian reference system. These components have the form of directed intervals (Markov 1995).
Expressions $\widehat{\boldsymbol{U}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau})$ and $\widehat{\boldsymbol{P}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau})$ are the following complex interval kernels (modified kernels) (Kużelewski and Zieniuk 2022):

$$\widehat{\boldsymbol{U}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau}) = -\frac{1}{2\pi} \ln(\widehat{\boldsymbol{\tau}} - \boldsymbol{\tau}),\ \widehat{\boldsymbol{P}}_{lj}^{*(c)}(\widehat{\boldsymbol{\tau}}, \boldsymbol{\tau}) = \frac{1}{2\pi} \frac{\boldsymbol{n}_j^{(c)}}{\widehat{\boldsymbol{\tau}} - \boldsymbol{\tau}},$$

where $\boldsymbol{n}_j^{(c)} = \boldsymbol{n}_j^{(1)} + i\boldsymbol{n}_j^{(2)}$ is the complex notation of normal vector to the interval curve, which creates segment j, $^{(1)}$ and $^{(2)}$ are directions of coordinates in a 2D Cartesian reference system.
The example of interval boundary shape is presented in Fig. 1, where $\boldsymbol{P}_i$ ($i = \{1, 2, ..., k, k+1, ...n\}$) means interval points defining the boundary.

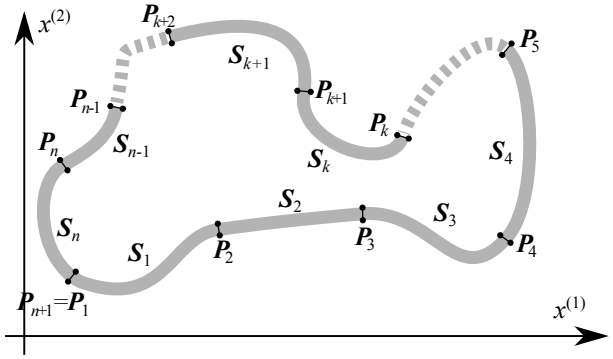

Figure 1: The Example of Interval Boundary Shape

At last, the following formula presents the final form of the IFPIES for solving uncertainly defined BVPs on the boundary Kużelewski et al. (2022):

$$\frac{1}{2}\boldsymbol{u}_l(\widehat{s}) = \sum_{j=1}^{n} \mathbb{R}\left\{ \frac{1}{2\pi} \sum_{l=0}^{N_T} (-1)^l \cdot \left[\sum_{k=0}^{N_T} \sum_{m=l}^{N_T} \frac{(k+m-1)! \cdot \boldsymbol{M}_k(\boldsymbol{\tau}_c')}{(\boldsymbol{\tau}_{el} - \boldsymbol{\tau}_c)^{k+m}} \cdot \frac{(\boldsymbol{\tau}_{el}' - \boldsymbol{\tau}_{el})^{m-l}}{(m-l)!} \right] \cdot \frac{(\widehat{\boldsymbol{\tau}} - \boldsymbol{\tau}_{el}')^l}{l!} \right\} - \sum_{j=1}^{n} \mathbb{R}\left\{ \frac{1}{2\pi} \sum_{l=0}^{N_T} (-1)^l \cdot \left[\sum_{k=1}^{N_T} \sum_{m=l}^{N_T} \frac{(k+m-1)! \cdot \boldsymbol{N}_k(\boldsymbol{\tau}_c')}{(\boldsymbol{\tau}_{el} - \boldsymbol{\tau}_c)^{k+m}} \cdot \frac{(\boldsymbol{\tau}_{el}' - \boldsymbol{\tau}_{el})^{m-l}}{(m-l)!} \right] \cdot \frac{(\widehat{\boldsymbol{\tau}} - \boldsymbol{\tau}_{el}')^l}{l!} \right\}, \quad (2)$$

where: $\boldsymbol{M}_k(\boldsymbol{\tau}_c') = \sum_{l=0}^{k} \frac{(\boldsymbol{\tau}_c - \boldsymbol{\tau}_c')^{k-l}}{(k-l)!} \boldsymbol{M}_l(\boldsymbol{\tau}_c)$ and $\boldsymbol{N}_k(\boldsymbol{\tau}_c') = \sum_{l=1}^{k} \frac{(\boldsymbol{\tau}_c - \boldsymbol{\tau}_c')^{k-l}}{(k-l)!} \boldsymbol{N}_l(\tau_c)$, interval moments $\boldsymbol{M}_l(\boldsymbol{\tau}_c)$ and $\boldsymbol{N}_l(\boldsymbol{\tau}_c)$ are elementary integrals computed twice only (require a short computation time), which have the form:

$$\boldsymbol{M}_l(\boldsymbol{\tau}_c) = \int_{\boldsymbol{s}_{j-1}}^{\boldsymbol{s}_j} \frac{(\boldsymbol{\tau} - \boldsymbol{\tau}_c)^l}{l!} \boldsymbol{p}_j(s)\, \boldsymbol{J}_j^{(c)}(s) ds,$$

$$\boldsymbol{N}_l(\boldsymbol{\tau}_c) = \int_{\boldsymbol{s}_{j-1}}^{\boldsymbol{s}_j} \frac{(\boldsymbol{\tau} - \boldsymbol{\tau}_c)^{l-1}}{(l-1)!} \boldsymbol{n}^{(c)} \boldsymbol{u}_j(s)\, \boldsymbol{J}_j^{(c)}(s)\, ds,$$

and N_T is the number of terms in the Taylor series. The interval points $\boldsymbol{\tau}_c'$ and $\boldsymbol{\tau}_{el}'$ are obtained by applying the FMM technique (tracing the tree structure up and down).

INTERVAL SOLUTIONS IN THE DOMAIN

The IFPIES (2) yields solutions only on the boundary. To find solutions in the domain, a modification of the IPIES integral identity is required, as presented in (Kużelewski and Zieniuk 2026).

Similarly to the concept presented in (Kużelewski and Zieniuk 2025), we obtained the complex interval integrands $\boldsymbol{U}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau})$ and $\boldsymbol{P}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau})$ in the following form:

$$\begin{aligned} \boldsymbol{U}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau}) &= -\frac{1}{2\pi} \ln\left(\boldsymbol{x}^{(c)} - \boldsymbol{\tau}\right), \\ \boldsymbol{P}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau}) &= \frac{1}{2\pi} \frac{\boldsymbol{n}^{(c)}(s)}{\boldsymbol{x}^{(c)} - \boldsymbol{\tau}}, \end{aligned} \tag{3}$$

where $\boldsymbol{x}^{(c)} = \boldsymbol{x}^{(1)} + i\boldsymbol{x}^{(2)}$, $\boldsymbol{U}_k^*(\boldsymbol{x}, s) = \mathbb{R}\{\boldsymbol{U}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau})\}$ and $\boldsymbol{P}_k^*(\boldsymbol{x}, s) = \mathbb{R}\{\boldsymbol{P}_k^{*(c)}(\boldsymbol{x}^{(c)}, \boldsymbol{\tau})\}$.

The same fast multipole tree and moments $\boldsymbol{M}_l(\boldsymbol{\tau}_c)$ and $\boldsymbol{N}_l(\boldsymbol{\tau}_c)$ were calculated previously during solving the IFPIES. However, to find solutions in the domain, we are moving only to the highest level of the tree.

We also expanded interval integral identity using Taylor series about any point $\boldsymbol{x}_c \in \{\boldsymbol{x}_{c1}, \boldsymbol{x}_{c2}, ..., \boldsymbol{x}_{ck}\}$ close to some groups points of solutions (presented in Fig.2) to reduce the number of computations.

At last, the integral identity in the IFPIES has the following form:

$$\boldsymbol{u}(\boldsymbol{x}) = \sum_{j=1}^{n} \mathbb{R} \sum_{l=0}^{N_T} \frac{(-1)^l}{2\pi} \cdot \sum_{k=0}^{N_T} \frac{(k-1)! \cdot \left(\boldsymbol{M}_k(\boldsymbol{\tau}_c) - \boldsymbol{N}_k(\boldsymbol{\tau}_c)\right)}{\left(\boldsymbol{x}_c - \boldsymbol{\tau}_c\right)^k} \Bigg) \frac{\left(\boldsymbol{x}^{(c)} - \boldsymbol{x}_c\right)^l}{l!}, \tag{4}$$

where $\boldsymbol{N}_0(\boldsymbol{\tau}_c) = 0$.

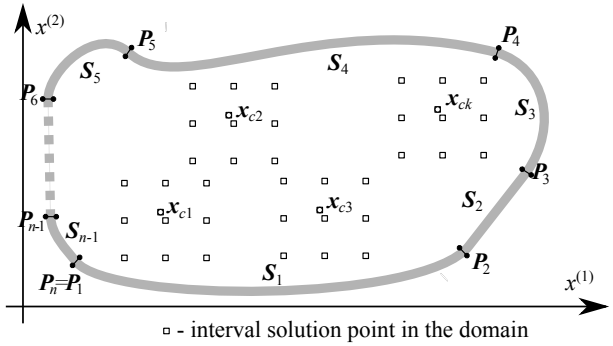

Figure 2: Groups of interval solution points approximated by corresponding point x_{ci} $(i = 1, 2.., k)$

NUMERICAL RESULTS

All presented tests are performed on a PC with an Intel Core i7-14700KF processor and 128 GB of RAM. The program is compiled with g++ (with `-O3 -march=native` optimization) on a 64-bit Ubuntu Linux operating system (kernel 6.14.0).

The example is the current flow through the square plate presented in Fig. 3. Boundary conditions presented in the figure mean interval potential $\boldsymbol{V}$ (red electrodes) and interval flux $\frac{\partial \boldsymbol{V}}{\partial \boldsymbol{n}}$ on the rest of the boundary (n means normal vector to the boundary segment).

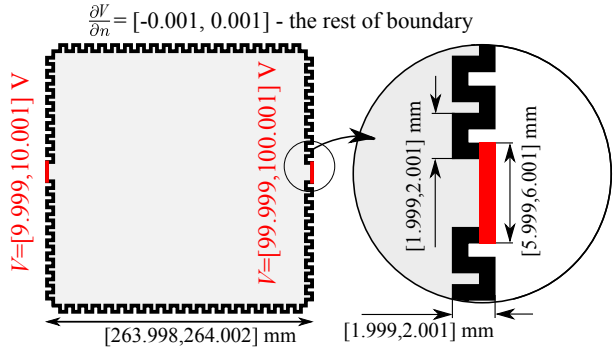

Figure 3: The Example of Current Flow Through the Square Plate

The Number of Solution Points in Group

The first research focused on determining the number of solution points in the group that yield high solution efficiency. Efficiency is presented as the CPU time required to compute solutions in the domain and the RAM utilization.

The IFPIES approximation of the modified IPIES kernels uses 20 terms in the Taylor series, and the GMRES tolerance is 10^{-8}. These parameters were determined based on findings in Kużelewski and Zieniuk (2023) and Kużelewski and Zieniuk (2024) regarding the search for optimal IFPIES parameters. The number of collocation

points is the same across all segments (from 2 to 8). Therefore, systems of 2 040 to 8 160 algebraic equations were solved. The same number of terms in the Taylor series was also used to approximate the integral identity.

The number of solution points in the domain was equal to 260 100, and the distance between solution points was equal to 0.5 mm, as shown in Fig. 4.

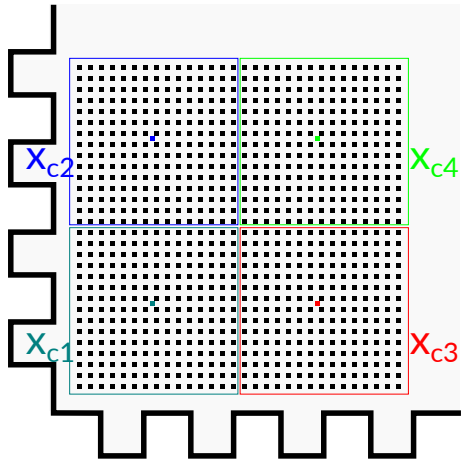

Figure 4: Location of Solution Points Approximated by x_c Points (Coloured Line Mark the Boundary of the Group Approximated by Particular x_c)

Table 1: CPU Time of Solutions in the domain (in s)

The number of solution groups x The number of points in group						
100 x 2601	225 x 1156	289 x 900	900 x 289	1156 x 225	2601 x 100	10404 x 25
2 collocation points						
83.3	37.84	29.56	10.11	8.29	**5.85**	11.3
3 collocation points						
151.61	60	45.55	15.02	11.9	**7.16**	11.7
4 collocation points						
205.09	80.35	63.96	20.41	15.72	**8.93**	11.83
5 collocation points						
320.07	102.25	79.7	25.46	19.55	**10.16**	12.05
6 collocation points						
322.45	127.47	97.88	30.62	23.6	**11.86**	12.17
7 collocation points						
385.66	151.18	118.4	36.71	28.83	13.65	**12.62**
8 collocation points						
452.56	177.76	137.88	43.14	33.11	15.66	**12.88**

As shown in Tab. 1, the lowest CPU time is observed for the smallest solution groups. Therefore, in future studies, we should apply a large number of solution groups. Also, the size of utilized RAM depends only on the number of collocation points and uses respectively (starting from 2 collocation points) about 15, 21, 29, 39, 50, 62 and 77 MB. Similarly, the time to solve the IFPIES is about 0.87, 1.67, 2.78, 4.24, 6.16, 8.66, and 11.54 s (starting from 2 collocation points).

The Number of Tree Levels

In this research, we considered different tree sizes in the fast multipole procedure. The main parameters of the IFPIES are the same as in the previous test. However, taking into account the results of the previous test, the search for the best number of tree levels uses 2601 solution groups, with 100 points per group.

Table 2: Influence of Tree Levels Number on CPU Time of Solutions in the Domain (in s)

The number of tree levels						
4	5	6	7	8	9	10
2 collocation points						
3.59	3.9	3.9	5.7	7.61	12.87	22.59
3 collocation points						
5.01	5.35	5.33	7.16	8.79	14.21	24.26
4 collocation points						
6.53	6.8	6.81	8.61	9.9	15.68	25.2
5 collocation points						
8.11	8.41	8.37	10.16	11.12	17.04	26.76
6 collocation points						
9.87	10.11	10.15	11.87	12.36	18.58	28.12
7 collocation points						
11.71	12.14	12.11	13.56	13.7	19.72	29.61
8 collocation points						
12.38	14.19	14.24	15.62	15.13	21.69	31.6

As shown in Tab. 2, as in the previous example, the lowest CPU time is observed for the fewest tree levels. However, we should also consider the IFPIES modelling process across different numbers of tree levels to strike a balance between the CPU time required to solve IFPIES and the time spent searching for solutions in the domain.

Table 3: Total CPU Time of Solving the IFPIES and Searching Solutions in the Domain (in s)

The number of tree levels						
4	5	6	7	8	9	10
2 collocation points						
7.41	5.86	**5.85**	6.56	8.44	13.9	24.28
3 collocation points						
13.93	9.75	**9.7**	8.82	10.28	15.74	26.43
4 collocation points						
22.75	14.67	14.73	**11.39**	12.3	17.85	27.97
5 collocation points						
34.39	21.18	21.11	**14.46**	14.8	20	30.2
6 collocation points						
49.53	29.15	29.11	18.01	**17.66**	22.61	32.39
7 collocation points						
67.43	38.83	38.84	22.01	**20.83**	24.82	34.77
8 collocation points						
89.34	50.95	51.27	27.15	**24.68**	28.24	37.88

As shown in Tab. 3, the time to solve and search for domain solutions in the IFPIES depends on the number of collocation points. However, as a compromise, we should use 7 tree levels in the presented example. It means that the tree is composed of 255 cells (4 levels means 31 cells, while 10 levels means 2039 cells).

CONCLUSIONS

The paper presents a method for searching for the values of two parameters of the IFPIES: the number of tree levels and the number of solution points in groups, which are optimal when searching for solutions in the domain. Optimal means that the number of tree levels and the number of solution points in groups that give solutions in the shortest CPU time. The results obtained in this study lead to the conclusion that, to compute a large number of solutions in the domain efficiently, we should use a large number of solution groups, each with a relatively small number of solution points. Also, the number of tree levels should be a compromise between the time required to solve the IFPIES and the time required to search for solutions in the domain. It should be noted that the optimal number of tree levels is determined by the average number of cells in the tree.

FUTURE RESEARCH

The results suggest that further research is needed to examine more examples with complicated boundary shapes and multi-connected boundary problems. Another area of research should also focus on solving BVPs described by different equations, such as Navier–Lamé equations.

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EFFICIENT 2D ELASTOPLASTIC ANALYSIS USING PIES WITH GLOBAL TELLES TRANSFORMATION FOR HANDLING WEAKLY SINGULAR DOMAIN INTEGRALS

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KEYWORDS
PIES, Elastoplastic analysis, Weakly singular domain integrals, Telles transformation, Subdivision technique

ABSTRACT

This paper presents an efficient 2D elastoplastic analysis using the PIES method, integrated with a global quadrature-point transformation technique based on Telles' method to compute weakly singular domain integrals. This strategy directly resolves the singularity across the entire, undivided global plastic zone, eliminating the need for local subdivision of the integration domain. By doing so, it fully exploits the unique advantage of PIES, in which the plastic region is represented as a single global parametric surface patch. Numerical experiments conducted on a classic elastoplastic benchmark validate the proposed strategy. The results demonstrate that incorporating the global Telles transformation into PIES integration yields high accuracy and numerical stability, while significantly reducing computational time.

INTRODUCTION

Due to its nonlinearity and dependence on load history, 2D elastoplastic analysis requires an incremental-iterative approach. It should be emphasized that at each iterative step, it is necessary to determine the domain integrals (responsible for the current state of plastic strain). In the classical finite element method (FEM) (Zienkiewicz 1977, Liu and Quek 2003, de Borst et al. 2012), this requires a dense mesh of finite elements covering the entire domain. In contrast, the boundary element method (BEM) (Aliabadi 2002, Gao and Davies 2002, Becker 1992) forces the introduction of internal cells over the expected plastic zone. Both procedures incur high data-preparation costs, especially since the extent of the plastic zone is not known a priori.

Mentioned imperfections prompted the authors to develop a version of the parametric integral equation system (PIES) (Zieniuk and Bołtuć 2006) for 2D elastoplastic problems (Bołtuć 2016a;b). Instead of dividing the plastic domain into multiple subdomains, it models it globally, typically with a single surface patch. This stems from the method's fundamental advantage of completely separating shape approximation from solution approximation. Just as the boundary in PIES is defined in a 1D parametric reference system, the plastic domain is spanned in a 2D parametric space. Any surface patches of any degree can be used to define the plastic zone. Most commonly, bilinear patches are employed to model polygonal regions, and bicubic patches to regions with curved boundaries (Salomon 2006, Farin 2002). Their definition is very simple and involves specifying only control points. Modifying such a region is also trivial, as it only requires changing the positions of the patch's selected control points. This is important given the lack of prior knowledge about the location and size of the plastic zone, which can lead to changes in its extent. PIES addresses this defect in yet another way: it often allows modeling of the entire domain under consideration with a single patch, not just the intended plastic zone, thereby eliminating the need for subsequent modifications in the event of incorrect assumptions.

The global formulation of the plastic zone as a single parametric patch significantly simplifies the geometric modeling stage, but also introduces specific conditions for the numerical evaluation of the resulting domain integrals - they can be calculated globally. The elastoplastic PIES equations contain domain integrals with varying degrees of singularity (Bołtuć 2016a;b): one exhibits a weak singularity of the type $1/r$, whereas the second contains a strongly singular kernel of the type $1/r^2$. The strongly singular function can be converted into parts: one weakly singular over the domain, and the second one (still strongly singular) that is analytically transformed into an easier to determine integral over the boundary of the plastic zone (Gao 2005). Therefore, finally, the most important part is computing integrals over a domain with a weak singularity. Due to the incremental-iterative nature of the solution algorithm, they must be calculated continuously at each load increment and iteration step to account for dynamically changing plastic strain fields. Therefore, the choice of a numerical integration technique is crucial to the method's overall efficiency.

Till now, the weakly singular domain integrals in PIES have been resolved globally using the surface subdivision technique (Aliabadi 2002, Gao and Davies 2002,

Becker 1992). The core concept of this approach is to partition the integration domain into a set of triangular subdomains that converge precisely at the current singular point. Although this strategy effectively handles the singularity by introducing a local coordinate transformation and enabling the use of the standard Gauss-Legendre quadrature, it does not fully exploit a fundamental principle of PIES - global representation of the domain. However, it should be emphasized that, in general, the integrals are calculated globally over the domain, and the division is performed only to handle a singularity. Because the subdivision pattern changes continuously as the singular points move, recalculating new subdomains and new integration points becomes a computational bottleneck. To eliminate this drawback and fully exploit the benefits of global plastic domain modeling, the authors proposed implementing a transformation technique developed by Telles (Telles 1987).
This paper aims to demonstrate that combining the elastoplastic PIES method with the Telles transformation enables the full utilization of the advantages of global plastic zone modeling in singular numerical integration. By eliminating the subdivision of it into subtriangles and retaining a single integration domain, it is possible to reduce computational time while maintaining high accuracy and stability in the iterative elastoplastic analysis process. To validate the proposed approach, numerical tests are conducted, analyzing the error values obtained by comparison with analytical solutions, processor times, and the number of iterations performed, using the subdivision approach and the Telles transformation to calculate weakly singular integrals over the plastic domain.

PIES FOR 2D ELASTOPLASTIC ANALYSIS

For 2D elastoplastic problems without body forces, PIES can be formulated as follows (Bołtuć 2016a;b)

$$0.5\dot{\boldsymbol{u}}_l(\bar{s}) = \sum_{j=1}^{n}\int_{s_{j-1}}^{s_j}\left\{\mathbf{U}^*_{lj}(\bar{s},s)\dot{\boldsymbol{p}}_{\boldsymbol{j}}(s) - \mathbf{P}^*_{lj}(\bar{s},s)\dot{\boldsymbol{u}}_{\boldsymbol{j}}(s)\right\}J_j(s)ds + \int_{\Omega}\boldsymbol{\sigma}^*_l(\bar{s},\boldsymbol{y})\dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y})d\Omega(\boldsymbol{y}), \tag{1}$$

where $l, j = 1, 2, ..., n$, $s_{l-1} \le \bar{s} \le s_l$, $s_{j-1} \le s \le s_j$ and n is the number of boundary segments. The variable s is a parameter defined in a 1D parametric reference system, $\bar{s}$ is the collocation point in that system, and $J_j(s)$ is the Jacobian for the transformation to that system. The boundary solutions on segment j are represented by $\dot{\boldsymbol{p}}_{\boldsymbol{j}}(s)$ and $\dot{\boldsymbol{u}}_{\boldsymbol{j}}(s)$.
$\dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y})$ denotes the plastic strain within the plastic domain Ω ($\boldsymbol{y} \in \Omega$), which acts as the initial strain field in the initial strain approach applied in this paper.
The complete forms for the fundamental solutions $\mathbf{U}^*_{lj}(\bar{s},s)$, $\mathbf{P}^*_{lj}(\bar{s},s)$ and $\boldsymbol{\sigma}^*_l(\bar{s},\boldsymbol{y})$ can be found in (Bołtuć 2016a;b).
Due to the nonlinear nature of elastoplastic problems and the history-dependent behavior of plastic deformations, the PIES equation (1) cannot be solved directly as in linear elasticity. Instead, an incremental-iterative procedure (Aliabadi 2002) must be employed to trace the loading path and to handle the coupling of the plastic strain fields. Consequently, an additional internal stress equation is required at each iteration to evaluate the actual stress state within the domain, thereby verifying the yield criterion and updating the plastic strains. The integral identity for stresses can be expressed as (Bołtuć 2016a;b)

$$\dot{\boldsymbol{\sigma}}(\boldsymbol{x}) = \sum_{j=1}^{n}\int_{s_{j-1}}^{s_j}\left\{\hat{\boldsymbol{D}}^*_j(\boldsymbol{x},s)\dot{\boldsymbol{p}}_j(s) - \hat{\boldsymbol{S}}^*_j(\boldsymbol{x},s)\dot{\boldsymbol{u}}_j(s)\right\}J_j(s)ds + \int_{\Omega}\hat{\boldsymbol{\Sigma}}^*(\boldsymbol{x},\boldsymbol{y})\dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y})d\Omega(\boldsymbol{y}) + \hat{\hat{\boldsymbol{f}}}\dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{x}), \tag{2}$$

where $\boldsymbol{x}, \boldsymbol{y} \in \Omega$. The expressions for $\hat{\boldsymbol{D}}^*_j(\boldsymbol{x},s)$, $\hat{\boldsymbol{S}}^*_j(\boldsymbol{x},s)$, $\hat{\boldsymbol{\Sigma}}^*(\boldsymbol{x},\boldsymbol{y})$ and the free term $\hat{\hat{\boldsymbol{f}}}$ are given in (Bołtuć 2016a;b).
The main feature of PIES is the separation of shape approximation from solution approximation. Therefore, completely different yet highly effective methods can be used to perform these two stages. The approaches applied in this paper are presented in the next section.

NUMERICAL IMPLEMENTATION

Parametric Boundary and Domain Modeling

In equations (1) and (2), the boundary geometry is represented by segments defined in the parametric reference system with parameter s (where $s_{j-1} \le s \le s_j$ for segment j). This allows the use of any curves known in computer graphics (Farin 2002, Salomon 2006) for boundary representation. For polygonal shapes, each segment is modeled globally using a first-degree curve. In curvilinear geometry, the number of curves and their degrees depend on the shape's complexity. Third-degree curves are most commonly used because they offer a well-balanced trade-off between computational efficiency and numerical stability, and they are usually sufficient for modeling any shape in practical problems. This approach is different than the strategy used in the so-called element methods (e.g. BEM), where classical discretization using elements of various types and degrees is typical. In the presented paper, Bézier curves are used to model the boundary.
The nonlinear nature of the elastoplastic analysis necessitates the introduction of the domain integrals, as is evident in both equations (1) and (2). In the PIES

method, the plastic zone Ω is not divided into a grid of cells like in BEM or finite elements like in FEM, but is described by parametric surface patches spanned in a two-dimensional parameter space $(v, w) \in [0, 1] \times [0, 1]$. The first-order (bilinear) rectangular surface patch is used to represent the polygonal plastic region, while the curvilinear plastic zones are defined with third-order (bicubic) patches. In this paper, rectangular Bézier surface patches are applied.

Fig. 1 shows, in a general way, how to define the geometry in the following methods: FEM - where finite elements are used in the entire domain, BEM - where the boundary is defined by boundary elements and the plastic region by the cells, and PIES - where the boundary segments are modeled entirely by means of curves and their control points, and the plastic region is also created globally by means of a single surface patch defined by its four corner points.

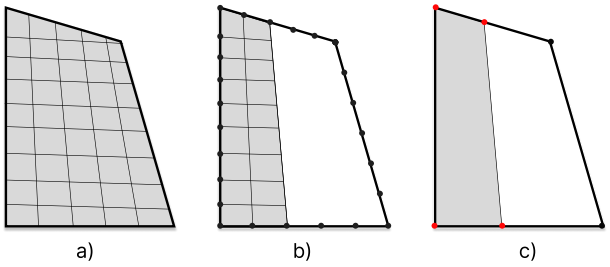

Figure 1: Modeling the Boundary Geometry and Plastic Zone in: a) FEM, b) BEM, c) PIES.

Numerical Treatment of Domain Integrals

The domain integrals are present in both PIES equations (1) and (2). From a numerical point of view, these integrals pose a major computational challenge because both are singular integrals, though to different degrees. The integral in the PIES equation (1) with kernel $\boldsymbol{\sigma}_l^*(\bar{s}, \boldsymbol{y})$, after substituting the function describing the rectangular Bézier patch (Farin 2002, Salomon 2006), takes the following form

$$\int_\Omega \boldsymbol{\sigma}_l^*(\bar{s}, \boldsymbol{y}) \dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y}) d\Omega(\boldsymbol{y}) = \int_{v_{d-1}}^{v_d} \int_{w_{d-1}}^{w_d} \boldsymbol{\sigma}_l^*(\bar{s}, v, w) \dot{\boldsymbol{\varepsilon}}^p(v, w) J_d(v, w) dv dw, \tag{3}$$

and exhibits a so-called weak singularity of the type $1/r$. The next integral over the domain from equation (2), after the same transformations as above, takes the form

$$\int_\Omega \hat{\hat{\boldsymbol{\Sigma}}}^*(\boldsymbol{x}, \boldsymbol{y}) \dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y}) d\Omega(\boldsymbol{y}) = \int_{v_{d-1}}^{v_d} \int_{w_{d-1}}^{w_d} \hat{\hat{\boldsymbol{\Sigma}}}^*(\boldsymbol{x}, v, w) \dot{\boldsymbol{\varepsilon}}^p(v, w) J_d(v, w)\, dv\, dw, \tag{4}$$

and its kernel exhibits a strong singularity of the type $1/r^2$.

Variables $v_{d-1}, v_d, w_{d-1}, w_d$ are respectively the beginning and the end of the domain surface patch (of index d) with parameters v, w. $J_d(v, w)$ is the Jacobian of the transformation to the parametric domain $0 \leq v, w \leq 1$. Special numerical treatment is required for both integrals. The integral (3) can be accurately resolved via surface subdivision (Aliabadi 2002, Gao and Davies 2002, Becker 1992). The core concept of this approach is to partition the plastic domain into a set of triangular subdomains converging at the singular point. By introducing a local coordinate transformation for each subdomain, the Jacobian of the coordinate system effectively cancels the weak singularity. This allows for the standard Gauss-Legendre quadrature to be applied with high precision.

The integral (4) needs another treatment. There is an approach developed for elastoplastic problems integrating in (Gao 2005), which decomposes the strongly singular integral into two components: one that is weakly singular (and can be resolved via the aforementioned subdivision), and the second one (still strongly singular) that is analytically transformed into an easier to determine integral over the boundary of the plastic zone.

This approach to integrating domain singular integrals has been used by the authors in previous studies (Bołtuć 2016a;b). However, the subdivision method's main drawback is its high computational cost. For each triangle-like subdomain (three if the singular point is on the boundary, four if it is within the domain), the coordinates must be independently transformed and the Gauss-Legendre quadrature points determined. This situation is worsened by the need to repeat the operation many times, as the division varies with the singular point's current position, and by the iterative nature of the entire procedure. As a result, it becomes the main bottleneck of the algorithm. For this reason, it was decided to test an alternative integration approach to eliminate the high computational cost of region partitioning and to leverage the global modeling of the plastic area even for singularity treatment.

The transformation technique has been developed in (Telles 1987). This method applies a special global cubic mapping directly to the entire domain, without subdivision. This transformation automatically shifts and densifies the Gauss-Legendre quadrature points in the immediate vicinity of the singularity, enabling precise isolation and smoothing. The procedure used is presented below using the example of integral (3).

Let's assume that the position of the singular point $\bar{s}$ in the normalized coordinate system of the plastic domain $[-1, 1] \times [-1, 1]$ is represented by two components: $\bar{\eta}_v$ (for the v direction) and $\bar{\eta}_w$ (for the w direction). For each axis, the coordinates of the singular point in the new quadrature space are independently calculated using the function (Aliabadi 2002)

$$\bar{\xi}_k = \sqrt[3]{\bar{\eta}_k(\bar{\eta}_k^2 - 1) + |\bar{\eta}_k^2 - 1|} + \sqrt[3]{\bar{\eta}_k(\bar{\eta}_k^2 - 1) - |\bar{\eta}_k^2 - 1|} + \bar{\eta}_k, \tag{5}$$

where the index $k \in \{v, w\}$ represents the direction of integration.
Then the standard Gauss-Legendre quadrature points ξ_k (ξ_v, ξ_w) are transformed from the range $[-1, 1]$ using Telles' polynomial to

$$\eta_k(\xi_k) = \frac{(\xi_k - \bar{\xi}_k)^3 + \bar{\xi}_k(\bar{\xi}_k^2 + 3)}{1 + 3\bar{\xi}_k^2}, \tag{6}$$

with the corresponding Jacobian of the transformation

$$\frac{d\eta_k}{d\xi_k} = \frac{3(\xi_k - \bar{\xi}_k)^2}{1 + 3\bar{\xi}_k^2}. \tag{7}$$

Substituting the transformed coordinates (6) and incorporating the respective Jacobians for both directions (7), the final form of the domain integral (3) becomes

$$\int_{-1}^{1}\int_{-1}^{1} \sigma_l^* \left(\bar{s}, \eta_v(\xi_v), \eta_w(\xi_w)\right) \dot{\varepsilon}^p \left(\eta_v(\xi_v), \eta_w(\xi_w)\right) J_d \left(\eta_v(\xi_v), \eta_w(\xi_w)\right) \frac{d\eta_v}{d\xi_v}\frac{d\eta_w}{d\xi_w} d\xi_v d\xi_w. \tag{8}$$

The same procedure is applied to integral (4) in the weakly singular part. The only difference lies in the location of the singular points. In integral (3), it lies on the boundary, whereas in integral (4), it is inside the domain. The new locations of the integration points for the 10-point standard Gauss-Legendre quadrature, compared with those from the previously used subdivision technique, are shown in Fig. 2.
Fig. 2 illustrates the adaptive behavior of the transformed Gauss-Legendre quadrature points within a plastic domain depending on the position of the singular point. The left figures (red) show quadrature points densely packed along the lower boundary, as the singularity is located at its center. For the right ones (blue), the transformation forces the points to cluster in a dense, concentric pattern immediately around the central singular point within the domain. As can be seen, in contrast to the subdivision technique, which structurally divides the integration domain into triangles to cluster points near the singular point, Telles' approach preserves the original domain layout. It achieves the necessary point concentration solely through a mathematical transformation of the coordinate space, thereby reducing the total number of integration points. However, the use of the Telles transformation has one significant drawback. Using an odd number of quadrature points causes one of the nodes to fall exactly at the midpoint of the interval, which can lead to the Jacobian becoming zero or changing sign, and consequently to errors.

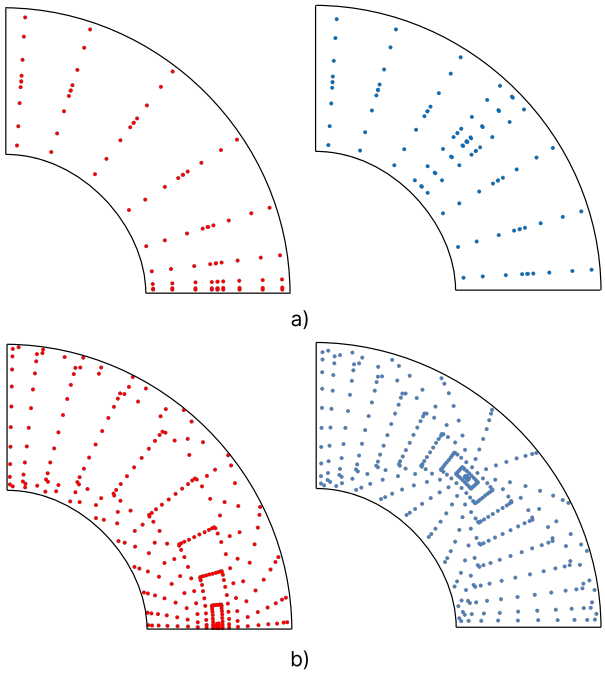

Figure 2: Distribution of Gauss-Legendre Quadrature Points within a Plastic Domain transformed by: a) Telles' Method, b) Subdivision Technique.

Incremental-iterative Solution Algorithm

The boundary functions $\dot{\boldsymbol{p}}_{\boldsymbol{j}}(s)$ and $\dot{\boldsymbol{u}}_{\boldsymbol{j}}(s)$ and plastic strains $\dot{\boldsymbol{\varepsilon}}^p(\boldsymbol{y})$ from equation (1) are approximated by series presented explicitly in (Bołtuć 2016a;b). While the boundary functions are approximated using one-dimensional series, the plastic strains are defined within the domain and require a two-dimensional series approximation over a set of internal points. These series can use any basis functions; in this paper, Lagrange polynomials are involved. Then they are substituted into equation (1) to replace the continuous, unknown functions with a combination of known basis functions and unknown coefficients that need to be calculated. Using the collocation method (Gottlieb and Orszag 1997), the PIES equation is transformed into a system of equations by imposing the condition that it is satisfied at predetermined boundary collocation points. For linear problems, solving the system of equations generated in this way completes the basic stage of solving the boundary value problem. In nonlinear problems, the system cannot be solved in a single step because the equations themselves depend on the unknown solution.
To solve the elastoplastic PIES (1), an incremental-iterative solution procedure based on the explicit scheme

presented in (Aliabadi 2002) is employed. The total loading path is divided into a sequence of small load increments. Within each increment, an iterative process is performed to satisfy both PIES and the material constitutive relations. The algorithm for a typical load increment can be summarized as follows:

1. The load at first yield can be calculated from the elastic solution obtained for the total value of external load without plastic strains. Then the remaining load is applied incrementally.

2. At each load increment:

   - assume plastic strain increment to 0 at the first iteration and the last determined value at the subsequent ones.
   - solve equations (1) and (2) calculating integrals (3) and (4).
   - compute the current increment of plastic strains using the plastic flow rule and the hardening law (Aliabadi 2002).
   - verify convergence comparing the change in plastic strain increments between successive iterations against a predefined tolerance.
   - depending on the convergence test, go to the first step for a new iteration or stop the process.

It should also be remembered that the distribution of both collocation points on the boundary and internal points in the domain (for the plastic strains approximation) is crucial to the accuracy of the solutions. For this purpose, the distributions of points corresponding to the roots of Chebyshev polynomials (Abramowitz and Stegun 1972) are used in this paper, as they have given the best results in previous research (Bołtuć 2016a;b, Zieniuk and Bołtuć 2006).

RESULTS

The numerical benchmark used to validate the proposed elastoplastic PIES formulation is the classic Lamé's problem, which considers a thick-walled cylinder subjected to internal pressure (Fig. 3). The analysis is carried out under the assumption of a plain strain condition. The material behavior is governed by the von Mises yield criterion combined with a perfectly plastic material model. The comprehensive set of material properties and model parameters is specified as follows: $\sigma_0 = 30$ MPa, $E = 21\,000$ MPa, $\nu = 0.3$ and $H' = 0$. Since the size and extent of the plastic zone are unknown in advance, a rectangular third-degree Bezier patch is used to model the entire problem domain. This allows avoiding subsequent changes if the initial guess proves incorrect.

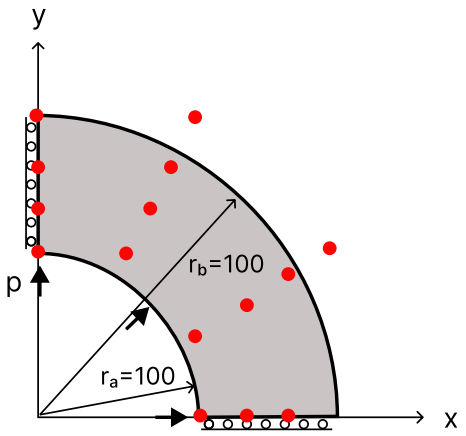

Figure 3: The Quarter of a thick Cylinder under internal Pressure.

The main goal of the study is to verify the integration procedure for weakly singular domain integrals using classical Gauss-Legendre quadratures and Telles' transformations. For this purpose, Lamé's problem is solved using PIES, with a model having 56 degrees of freedom (7 collocation points on each boundary segment) and 25 points in the plastic domain to approximate plastic strains. Calculations are performed in 6 increments with a tolerance 0.00001.

The approach is analyzed by comparing the results with the analytical solution (Prager and Hodge 1951) and with those obtained using PIES, with the subdivision method as the singularity treatment technique. In addition to the accuracy of the results, the computational time and the number of performed iterations are examined.

The first test concerns the accuracy of the final results obtained using both integration techniques. The displacement of the external boundary in response to the applied pressure $p = 22.5$ MPa is calculated for all increments. The radial and hoop stresses at the fifth increment at a load level near collapse ($p = 21.21$ MPa) are analyzed for seven values of radius r, where r is the distance of the tested point from the axis of symmetry of the cylinder. The tests are performed for different numbers of points (N) in Gaussian-Legendre quadrature. Based on the determined solutions, the average relative errors are calculated and presented in Table 1.

As shown in Table 1, for both integration methods analyzed, the tendency for relative errors to increase with decreasing quadrature point count is evident. At 60-100 points, both methods yield very similar, high accuracies. However, the approach using the Telles transformation is much more stable and accurate, maintaining a similar error rate across most tested values of N and converging even with the minimum value. In contrast, the subdivision method yields satisfactory results only for dense quadratures ($N \geq 60$), and for fewer points it produces

Table 1: Average relative errors [%] for stresses and displacements using different singularity handling techniques and number of points in the quadrature.

N	**Telles**			**Subdivision**		
	σ_r	σ_q	u_r	σ_r	σ_q	u_r
100	1.29	1.21	0.61	1.30	1.20	0.57
80	1.28	1.17	0.59	1.29	1.21	0.61
60	1.29	1.25	0.62	1.11	1.23	0.67
40	1.45	1.14	0.56	10.86	4.84	4.55
20	6.05	2.32	0.97	lack of convergence		

Table 2: Comparison of computation time and number of iterations for the analyzed singularity handling techniques.

N	**Telles**		**Subdivision**	
	Time [s]	**I**	**Time** [s]	**I**
100	207.05	46	819.49	47
80	138.11	48	517.62	47
60	75.93	46	303.42	48
40	36.39	48	245.45	90
20	13.08	58	lack of convergence	

very large errors or fails to converge ($N = 20$). The difference is particularly clear at $N = 40$, where the average relative errors for the subdivision method increase rapidly, whereas for the Telles method they are still below 1.5%.

To validate the errors generated by the PIES method, they can be compared with reference results obtained by FEM and BEM. While the errors reported in (Bołtuć 2016b) were evaluated across a larger set of points, they nevertheless provide a highly valid reference point. The average relative stress errors presented in that study are 1.639% and 2.225% for σ_r and σ_θ using BEM (Gao and Davies 2002), and 1.439% and 1.638% using FEM (ALGOR). It should be emphasized that, for the BEM model, 266 degrees of freedom were used, whereas for FEM, it was 2522 degrees of freedom. As can be seen, the results obtained using PIES are more accurate with much smaller degrees of freedom.

The Telles method demonstrates significantly higher computational efficiency for the analyzed problem because it requires several times fewer quadrature points while maintaining good accuracy and stability. It comes from the fact that in the subdivision method, the total number of points used in practice is larger because they are duplicated across subdomains. Thus, for singular points on the boundary, they are used three times, while for those within the domain, four. For this reason, the next step in this study is to compare the time required to perform calculations for various N values across both integration algorithms. The averaged results from three runs, along with the total number of iterations (I) across all load increments, are shown in Table 2.

The computational times presented in Table 2 were obtained on a standard desktop computer, running code without any optimizations or parallelization. Therefore, time values should not be considered definitive indicators of the algorithms' performance in production systems, because different software and hardware techniques can significantly reduce them. Instead, at this proof-of-concept stage of the research, the analysis focuses on examining the proportions, trends, and dynamics of the decrease in computational time when switching to the new technique for handling singularities.

As shown in Table 2, regardless of the chosen method, reducing the number of quadrature points results in a natural, rapid decrease in execution times. The key observation, however, is that the Telles method is at least approximately 4 times faster than the subdivision method across all variants. This demonstrates the enormous performance gain that results directly from eliminating the need to repeatedly duplicate quadrature points across subdomains.

In terms of iterations, the Telles approach demonstrates algorithmic stability, maintaining a constant number of iterations across almost all tested values of N, with only a slight increase at the lowest point count. In contrast, the subdivision method is highly sensitive to reducing the number of quadrature points. Its iterative process becomes unstable as N decreases, leading to a rapid increase in the number of iterations before the algorithm ultimately fails to converge.

The last numerical experiment is conducted with varying numbers of internal points to approximate plastic strains (which are singular points in the domain), thereby directly reflecting the system's complexity when calculating domain integrals. The main goal of the study is to determine how both approaches handle increasing task size and whether they scale to more computationally demanding models. The dependence of computational time on the number of internal points for the Telles method and the subdivision method is shown in Fig. 4. Tests are performed using 60 points in the Gauss-Legendre quadrature.

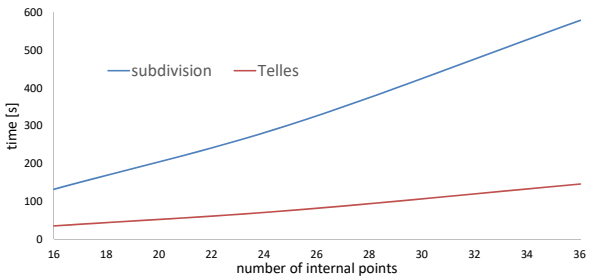

Figure 4: Dependence of the Calculation Time on the Number of internal Points.

As can be seen in Fig. 4, the increase in the vertical distance between the curves suggests that, as the number of internal points increases, the efficiency advantage of the Telles method (red line) over the subdivision method (blue line) will grow at an increasingly rapid rate, making it a more scalable solution.

CONCLUSIONS

In this paper, an efficient numerical strategy for evaluating weakly singular domain integrals within the 2D elastoplastic PIES method has been successfully implemented and validated. By replacing the surface subdivision technique with the global Telles coordinate transformation, the approach fully aligns its numerical integration phase of the singular integrals over the domain with its global geometric modeling philosophy. As in previous versions of the method, the plastic region is modeled with rectangular Bézier surface patches of varying degrees, and the solutions are approximated with Lagrange polynomials.

Based on the numerical analysis of the classic elastoplastic benchmark, the proposed approach yields a performance gain, executing approximately 4 times faster than the subdivision technique across all tested configurations. This advantage is achieved because the Telles method avoids duplicating quadrature points across multiple subdomains, which, in the subdivision technique, formed the primary computational bottleneck. Moreover, the Telles method demonstrates resilience to reductions in the number of integration points. While the subdivision technique fails to converge at certain levels of quadrature points, the Telles method ensures convergence and yields reliable solutions. Scalability tests with varying numbers of internal points confirm that the global transformation approach performs well even with increased task size.

Although the use of the Telles transform, like any method, is associated with certain limitations, such as the requirement for an even number of quadrature coefficients, this drawback is less significant given the results obtained. The studies conducted clearly demonstrate that this approach significantly reduces computational time compared to the subdivision method for identical N values. While maintaining a high solution accuracy, the overhead from the even-quadrature rigor becomes negligible.

The preliminary tests presented in this article, which should obviously be expanded, demonstrate the accuracy and effectiveness of this approach. The time benefits should be even greater after code optimization and parallelization on high-performance workstations, which the authors intend to implement in the final version of the proposed PIES method.

FUTURE RESEARCH

Building on the promising results of this study, future research should focus on optimizing and parallelizing the code to leverage the speed advantages of the global Telles transformation fully. Another key direction is to expand the proposed formulation to 3D elastoplastic problems using parametric volume patches. Additionally, the methodology can be extended to handle more complex material models, including isotropic and kinematic strain hardening.

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ADVANCED MODELING AND ALGORITHMIC TECHNIQUES FOR SELF-SIMILAR IP NETWORK TRAFFIC SIMULATION AND PERFORMANCE EVALUATION

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Modeling and Simulation, Communication Networks, Teletraffic, Stochastic Self-similar Processes, Synthetic Self-Similar Sequence Generators.

ABSTRACT

This paper investigates the self-similar (fractal) characteristics of real communication network traffic across a broad range of time scales. These characteristics differ significantly from those predicted by conventional traffic models based on Poisson and Markov-modulated Poisson processes. To accurately capture the observed self-similar behavior, advanced self-similar models for sequential and fixed-length sequence generators are developed, along with efficient algorithms for simulating self-similar IP network traffic. Numerical examples are presented, and the resulting simulation outcomes are analyzed to evaluate the effectiveness of the proposed models and algorithms.

INTRODUCTION

The rapid expansion of broadband networks and the Internet has led to exponential growth in the number of users, connected devices, and network traffic. Modern high-speed communication networks support a diverse range of multimedia applications, including voice, video, and data services, each with distinct traffic characteristics and service quality (QoS) requirements (Lokshina and Lanting 2026; Radev and Lokshina 2010). A major objective in telecommunications is the development of a unified high-speed network infrastructure capable of supporting heterogeneous traffic while providing different levels of QoS. Studies of real communication network traffic have demonstrated that teletraffic, i.e., the collection of information, transport, and control processes within telecommunication networks, exhibits self-similar (fractal) characteristics over a wide range of time scales. Self-similarity can be broadly classified into deterministic and stochastic forms (Radev and Lokshina 2010). This study focuses on stochastic self-similarity. Two forms of stochastic self-similarity are commonly distinguished. A continuous-time stochastic process $\mathbf{Y}_t$ is strictly self-similar with a self-similarity parameter H ($1/2<H<1$) if the processes $\mathbf{Y}_{ct}$ and $c^H\mathbf{Y}_t$ possess identical finite-dimensional probability distributions for every positive scaling factor c. This definition, based on probability distributions, differs from that of second-order self-similarity, which is characterized by the behavior of the mean, variance, and autocorrelation function. A stochastic process $\mathbf{X}$ is asymptotically second-order self-similar with $0.5<H<1$ if, for each k large enough, its autocorrelation converges according to $\rho_k^{(m)} \to \rho_k$, as $m\to\infty$, where $\rho_k=E[(X_i-\mu)(X_{i+k}-\mu)]/\sigma^2$. Throughout this work, the terms exact self-similarity and asymptotic self-similarity are used interchangeably to refer primarily to the long-range dependence reflected in the tail behavior of the autocorrelation function. The modeling and simulation of self-similar IP network traffic are performed using advanced synthetic traffic generators, which are broadly categorized into two classes: sequential generators and fixed-length sequence generators. Both classes are investigated and evaluated in this study.

PROPERTIES OF LONG-RANGE DEPENDENT SELF-SIMILAR PROCESSES

In recent communication networks, the most striking feature of some second-order self-similar processes is that the cumulative functions of the aggregated processes do not degenerate as the non-overlapping batch size m tends to infinity. Such processes are known as Long-Range Dependent (LRD) processes. This is in contrast to traditional processes used in modeling IP network traffic, all of which include the property that the cumulative functions of their aggregated processes degenerate as the non-overlapping batch size m tends to infinity, i.e., $\rho_k^{(m)}\to 0$ or $\rho_k^{(m)}=0(|k|>0)$, for $m>1$. The equivalent definition of long-range dependence is given as (1).

$$\sum_{k=-\infty}^{\infty} \rho_k = \infty \tag{1}$$

Another definition of LRD is presented as (2),

$$\rho_k \sim L(t)k^{-(2-2H)}, \quad \text{as} \quad k \to \infty \tag{2}$$

where $1/2<H<1$ and $L(\cdot)$ slowly varies at infinity, i.e. for all $x>0$ it could be determined as (3).

$$\lim_{t\to\infty} \frac{L(xt)}{L(t)} = 1 \tag{3}$$

Hurst parameter H characterizes the relation in (2), which defines the form of the tail of the cumulative function. One can show that it holds for $1/2<H<1$, as given in (4).

$$\rho_k = \frac{1}{2}\left[(k+1)^{2H} - 2k^{2H} + (k-1)^{2H}\right] \tag{4}$$

For 0<H<1/2, the process is Short Range Dependent (SRD) and could be presented as (5).

$$\sum_{k=-\infty}^{\infty} \rho_k = 0 \tag{5}$$

For H=1, all autocorrelation coefficients are equal to one, regardless of how far apart in time the sequences are. This case has no practical importance in real telecommunication network traffic modeling. If H>1, then (6) is true.

$$\rho_k = \begin{cases} 1 & \text{for} \quad k = 0 \\ \frac{1}{2} k^{2H} g(k^{-1}) & \text{for} \quad k > 0 \end{cases} \tag{6}$$

where

$$g(x) = (1+x)^{2H} - 2 + (1-x)^{2H} \tag{7}$$

One can see that $g(x)\to\infty$ is as H>1. If 0<H<1 and H≠1/2, then the first nonzero term in the Taylor expansion of g(x) is $2H(2H\text{-}1)x^2$. Therefore, (8) is true.

$$\rho_k / (H(2H-1)k^{2H-2}) \to 1, \quad \text{as} \quad k \to \infty \tag{8}$$

In the frequency domain, an essentially equivalent definition of LRD for a process X with given spectral density (9)

$$f(\lambda) = \frac{\sigma^2}{2\pi} \sum_{k=-\infty}^{\infty} \rho_k e^{ik\lambda} \tag{9}$$

is that, in the case of LRD processes, this function is required to satisfy the following property (10),

$$f(\lambda) \sim c_{f_1} \lambda^{-\gamma}, \quad \text{as} \quad \lambda \to 0 \tag{10}$$

where c_{f_1} is a positive constant and 0< γ<1, γ= 2H-1<1.

As a result, LRD manifests in the spectral density, which follows a power law near the origin. This implies that f(0)=$\Sigma_k\rho_k$=∞.

Consequently, it requires a spectral density that tends to +∞ as the frequency λ approaches 0. For a Fractional Gaussian Noise (FGN) process, the spectral density $f(\lambda,H)$ is given by (11),

$$f(\lambda, H) = 2c_f (1 - \cos(\lambda)) B(\lambda, H) \tag{11}$$

with 0<H<1 and -$\pi\le\lambda\le\pi$, where (12) is true,

$$c_f = \sigma^2 (2\pi)^{-1} \sin(\pi H)\Gamma(2H+1)$$

$$B(\lambda, H) = \sum_{k=-\infty}^{\infty} |2\pi k + \lambda|^{-2H-1} \tag{12}$$

and σ^2 =Var[X_k] and Γ(·) is the gamma function.

The spectral density $f(\lambda,H)$ in (11) complies with a power-law at the origin, as shown in (13),

$$f(\lambda, H) \to c_f \lambda^{1-2H}, \quad \text{as} \quad \lambda \to 0 \tag{13}$$

where 1/2< H <1.

SEQUENTIAL GENERATORS OF SELF-SIMILAR TELETRAFFIC

Although it is possible to construct a sequential Markovian model that reproduces self-similar traffic behavior, such an approach has a significant limitation. The relationship between the model parameters and the resulting self-similar characteristics is highly complex, making parameter selection and interpretation difficult. Markovian models for self-similar traffic typically require numerous control parameters spanning a wide range of values. Consequently, parameter calibration and control in sequential generators are considerably more complicated than in fixed-length sequence generators designed to produce self-similar processes with a specified Hurst parameter. To address these challenges, this study investigates several efficient sequential generators for modeling self-similar network traffic, as detailed below.

Fractal-Binomial-Noise-Driven Poisson Processes

For the standard fractal renewal process (FRP), inter-event times are independent random variables. The marginal probability density function (PDF) of such a fractal renewal process can be defined as (1),

$$f(t) = \begin{cases} 0, & t \le A, \\ \delta A^{\delta} t^{-(\delta+1)}, & t > A \end{cases} \tag{14}$$

where 0<δ<2. Selecting δ in this interval proves far superior to 0<δ<1 for the same required values, allowing the inter-event time PDF to be further improved. The improved PDF of the FRP decays as a power law, as shown in (15),

$$f(t) = \begin{cases} \delta A^{-1} e^{-\delta t/A}, & 0 < t \le A, \\ \delta e^{-\delta} A t^{-(\delta+1)}, & t > A, \end{cases} \tag{15}$$

which is continuous for all t, producing a smoother spectral density function. A method based on the fractal-binomial-noise-driven Poisson processes (FBNDP) adds M independent and identically distributed (iid) alternating FRPs to generate a fractal binomial noise process that serves as the rate function for a Poisson process. The FBNDP requires five input parameters to generate self-similar sequences: A, δ, R, Δt, and M. The resulting Hurst parameter H assumes the value (α+1)/2. The proposed algorithm advances with the intervals Δt. If S is a simulation clock that advances in time, and $S^{(j)}$ is the elapsed time of the j-th FRP sequence, then $S^{(j)}=\tau_0^{(j)} + \tau_1^{(j)} + \ldots + \tau_k^{(j)}$ for some k and j=1,2,…,M, where $\tau_k^{(j)}$ is the inter-arrival time. The sequence of self-similar pseudo-random numbers $\{X_0,X_1,\ldots\}$ is obtained using the steps shown in Figure 1.

Step 1: *For each j=1,2,…,M, generate $\tau_0^{(j)}$ from (16),*

$$\tau_0^{(j)} = \begin{cases} -\delta^{-1} A \log[U(\delta^{(V-1)})(\delta^{(V-U)})^{-1}, & V \ge 1, \\ A V^{1/(1-\delta)}, & V < 1, \end{cases} \tag{16}$$

where

$$V \equiv \frac{1 + (\delta - 1)e^{\delta}}{\delta} U \tag{17}$$

and U is an independent and identically distributed (iid), uniformly distributed random variable over the unit interval [0,1]; set $S^{(j)}= \tau_0^{(j)}$.

Step 2: *Find j^* and $S^{(j)}$ such that j^*=argmin$_j${$S^{(j)}$}.*

Step 3: *Calculate (18).*

$$x = \begin{cases} 0, & \text{if} \quad S^{(j^*)} < A, \\ 1, & \text{if} \quad S^{(j^*)} \ge A. \end{cases} \tag{18}$$

Step 4: *If x=1, then X_0 should be drawn from a Poisson probability distribution with λ=1. If x=0, then X_0=0.*

Step 5: *Set i=1 and y=0. Advance the simulation clock, i.e., $S\leftarrow S^{(j^*)}$.*

Step 6: *Construct a new inter-event time* $\tau_i^{(j)}$ *from (19),*

$$\tau_i^{(j)} = \begin{cases} -\delta^{-1} A \log[U], & U \geq e^{-\delta}, \\ e^{-1} A U^{-1/\delta}, & U < e^{-\delta}, \end{cases} \quad (19)$$

and set $S^{(j^*)} \leftarrow S^{(j^*)} + \tau_i^{(j)}$.

Step 7: *Find a new* j^* *such that* $j^* = \text{argmin}_j\{S^{(j)}\}$, *and compute* $S^{(j)} - S$.

Step 8: *Repeat Steps 6 through 8 to obtain x as in (18).*

Step 9: *Advance the simulation clock, i.e.,* $S \leftarrow S^{(j^*)}$, *and set* $y = y + x$.

Step 10: *Repeat Steps 6 through 10 within a time slot of length* Δt.

Step 11: *Compute* $X_i = POISS(y)$, *set* $y=0$, *and* $i=i+1$.

Step 12: *Repeat Steps 6 through 11 until i=n, where n is the number of sample points.*

Figure 1: FBNDP Algorithm

It took 1 min. 38 sec. to generate a traffic sample sequence of 1,048,576 data points. Table 1 reports the mean estimates of the Hurst parameter H, along with the corresponding 95% confidence intervals for the means (in parentheses), obtained using the wavelet-based H estimator for the FBNDP method with four input values: A=9.92, R=200, M=4 to 14, and H=0.6 to 0.9. The results confirm that the most appropriate aggregate level is M=10.

Table 1: Precision of Sequences Generated with FBNDP

M	Mean Values of Estimated *H*			
	0.6	0.7	0.8	0.9
4	0.6023 (0.575, 0.630)	0.6864 (0.659, 0.714)	0.7817 (0.754, 0.809)	0.8469 (0.819, 0.874)
6	0.6085 (0.581, 0.636)	0.6871 (0.660, 0.715)	0.7804 (0.753, 0.808)	0.8496 (0.822, 0.877)
8	0.6080 (0.581, 0.636)	0.6870 (0.660, 0.715)	0.7803 (0.753, 0.808)	0.8489 (0.821, 0.876)
10	0.6089 (0.581, 0.636)	0.6875 (0.660, 0.715)	0.7827 (0.755, 0.810)	0.8502 (0.823, 0.878)
12	0.6053 (0.578, 0.633)	0.6903 (0.663, 0.718)	0.7832 (0.756, 0.811)	0.8501 (0.823, 0.878)
14	0.6049 (0.577, 0.632)	0.6895 (0.662, 0.717)	0.7842 (0.757, 0.812)	0.8497 (0.822, 0.877)

Superposition of Fractal Renewal Processes

A method based on the superposition of fractal renewal processes (SFRP) combines a group of independent and identically distributed fractal renewal processes (FRP). This method requires three parameters: α and A from the individual FRPs, and M, which is the number of FRPs superimposed. The resulting Hurst parameter H, and the mean μ and the variance σ^2 of the marginal output distribution, can be defined according to (20):

$$H = (\alpha + 1)/2$$
$$\mu = \mathrm{E}[X_n] = \lambda \quad (20)$$
$$\sigma^2 = Var[X_n] = (1 + (1/t_0))^{\alpha} \lambda$$

where

$$\lambda = M\delta[1 + (\delta - 1)^{-1} e^{-\delta}]^{-1} A^{-1} \quad (21)$$

is the aggregate arrival rate of events per unit time interval. The sequence of self-similar pseudo-random numbers $\{X_0, X_1, \ldots\}$ is generated using the steps shown in Figure 2.

Step 1: *For each* $j=1,2,\ldots,M$, *and* $i=0$, *generate* $\tau_0^{(j)}$ *from equations (3) and (4); set* $S^{(j)} = \tau_0^{(j)}$.

Step 2: *Find* j^* *such that* $j^* = \text{argmin}_j\{S^{(j)}\}$ *and set* $X_0 = S^{(j^*)}$.

Step 3: *Advance the simulation clock, i.e.,* $S \leftarrow S^{(j^*)}$.

Step 4: *Set* $i=i+1$. *Construct a new inter-event time* $\tau_i^{(j)}$ *from equation (6) and set* $S^{(j^*)} \leftarrow S^{(j^*)} + \tau_i^{(j)}$.

Step 5: *Find a new* j^* *such that* $j^* = \text{argmin}_j\{S^{(j)}\}$, *and compute* $S^{(j)} - S$.

Step 6: *Advance the simulation clock, i.e.,* $S \leftarrow S^{(j^*)}$.

Step 7: *Repeat Steps 4 through 6 until i=n, where n is the number of sample points.*

Figure 2: SFRP Algorithm

The results from the simulation with M=10, A=3.8, and other conditions similar to those of the FBNDP method, along with the mean estimates H and 95% confidence intervals for the means (in parentheses), are shown in Table 2. The mean estimates for H=0.6 and 0.7, along with 95% confidence intervals for the means in parentheses, for the FBNDP algorithm are 0.6086 (0.581, 0.636) and 0.6875 (0.66, 0.715). Their ΔH(%) values are +1.44 and -1.789. For H=0.6, 0.7, and 0.8, the overall relative error is less than 5%; however, for H=0.9, it exceeds 5%. The same characteristics obtained with the SFRP algorithm are 0.6166 (0.589, 0.644) and 0.7091 (0.682, 0.737), and their ΔH(%) are +2.759 and -1.296. The SFRP approach yields more accurate results, with relative errors of +2.76%, +1.29%, -0.17%, and -3.49% for H = 0.6, 0.7, 0.8, and 0.9. Similar to the FBNDP algorithm, the H value estimates obtained using the SFRP algorithm exhibited a transition from positive to negative bias as H increased.

Table 2: Precision of Sequences Generated with SFRP

M	Mean Values of Estimated *H*			
	0.6	0.7	0.8	0.9
4	0.6076 (0.580, 0.635)	0.6986 (0.671, 0.726)	0.7929 (0.765, 0.820)	0.8603 (0.833, 0.888)
6	0.6076 (0.580, 0.635)	0.6986 (0.671, 0.726)	0.7929 (0.765, 0.820)	0.8655 (0.833, 0.888)
8	0.6192 (0.592, 0.647)	0. 7084 (0.681, 0.736)	0.7997 (0.772, 0.827)	0.8655 (0.838, 0.893)
10	0.6166 (0.589, 0.644)	0. 7091 (0.682, 0.737)	0.7986 (0.771, 0.826)	0.8686 (0.841, 0.896)
12	0.6134 (0.586, 0.641)	0. 7085 (0.681, 0.736)	0.7978 (0.770, 0.825)	0.8718 (0.844, 0.899)
14	0.6143 (0.587, 0.642)	0. 7063 (0.679, 0.734)	0.7983 (0.771, 0.826)	0.8708 (0.843, 0.898)

Superposition of Autoregressive Processes

A method based on the superposition of autoregressive processes (SAP) generates asymptotically self-similar sequences when aggregating several independent autoregressive processes. In the simplest case, this can be presented as the sum of two autoregressive processes of the first order, as shown in (22):

$$z_{1i} = A_{1i} z_{1,i-1} + y_{1i}$$
$$z_{2i} = A_{2i} z_{2,i-1} + y_{2i}, \quad i = 1,2,\ldots \quad (22)$$

where A_{1i} and A_{2i} are randomly chosen from a beta distribution $B(\alpha_1, \alpha_2)$ on [0,1] with the shape parameters α_1 and α_2, where $\alpha_1 > 0$, $\alpha_2 > 0$; y_{1i} and y_{2i} are a pair of

independent and identically distributed sequences of random variables with mean of zero and variance σ^2=1. Using least-squares fitting, we obtain the following empirical relationship: α_2=7.7929∗log(H)+4.9513. Then the Hurst parameter H is linearly dependent on the shape parameter 2 of a beta distribution, while 1 can be chosen arbitrarily. For instance, α_1 was set to 1 in all cases studied. The PDF $f(x)$ of the beta distribution is defined as (23),

$$f(x)=\begin{cases}\dfrac{x^{\alpha_1-1}(1-x)^{\alpha_2-1}}{\beta(\alpha_1,\alpha_2)} & 0<x<1\\ 0, & \text{otherwise}\end{cases} \qquad (23)$$

where $\beta(\alpha_1,\alpha_2)$ is given as (24).

$$\beta(\alpha_1,\alpha_2)=\int_0^1 x^{\alpha_1-1}(1-x)^{\alpha_2-1}dx=\frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)} \qquad (24)$$

This method, based on a superposition of the autoregressive processes, involves the steps presented in Figure 3.

Step 1: *Set i=i+1. Determine z_{1i} and z_{2i} using (9).*
Step 2: *Calculate the sum X_i= z_{1i} + z_{2i}, i=1,2,.....*
Step 3: *Repeat Steps 1 and 2 until i=n, where n is the number of sample points.*

Figure 3: SAP Algorithm

An asymptotically self-similar traffic sequence {X1,X2,…} of 1,048,576 data points was generated in 3.5 seconds. Unlike other sequential generators, the SAP generator does not require an aggregation level as an input parameter. However, it requires the shape parameter (set to 1). The mean estimates H obtained using algorithms for sequential generators are shown in Tables 3 and 4 for H=0.6, 0.7, 0.8, and 0.9. The cumulative relative error for the FBNDP method was less than 5%, except for H=0.9. For the SEFR method, the cumulative relative errors were +6.21%, +1.78%, -1.43%, and -5.37% for H=0.6, 0.7, 0.8, and 0.9. Like the FBNDP method, the H value estimates exhibited a transition from positive to negative bias as H increased.

Table 3: Precision of Generated Sequences (H = 0.6; 0.7)

Methods	Mean Values of Estimated H and ΔH			
	0.6		0.7	
	$\hat{H}$	ΔH	$\hat{H}$	ΔH
FBNDP	0.6086 (0.581, 0.636)	+1.440	0.6875 (0.660, 0.715)	-1.789
SFRP	0.6166 (0.589, 0.644)	+2.759	0.7091 (0.682, 0.737)	+1.296
SAP	0.5989 (0.595, 0.603)	-0.182	0.6852 (0.680, 0.690)	-2.112

Table 4: Precision of Generated Sequences (H = 0.8; 0.9)

Methods	Mean Values of Estimated H and ΔH			
	0.8		0.9	
	$\hat{H}$	ΔH	$\hat{H}$	ΔH
FBNDP	0.7827 (0.755, 0.810)	-2.157	0.8502 (0.823, 0.878)	-5.538
SFRP	0.7986 (0.771, 0.826)	-0.174	0.8686 (0.841, 0.896)	-3.491
SAP	0.7845 (0.781, 0.788)	-1.937	0.8971 (0.894, 0.900)	-0.325

For α_2=2.9 and 8.1, all Hurst parameter estimates obtained from the traffic sample sequences generated by the SAP method exceeded the corresponding required values. The relative errors were +24.14% and +10.45%; in this case, the results were overestimated. For α_2=21.3 and 71.5, the relative errors were -0.01% and -8.45%.

FIXED-LENGTH SEQUENCE GENERATORS OF SELF-SIMILAR TELETRAFFIC

The most frequently studied discrete-time models of self-similar IP network traffic are fractional autoregressive integrated moving average (F-ARIMA) and fractional Gaussian noise (FGN) processes. These models effectively capture long-range dependence and self-similar traffic characteristics and are used to generate self-similar traffic, as detailed below.

Fractional Autoregressive Integrated Moving-Average Processes

Let us consider the F-ARIMA(0,d,0) method for generating self-similar sequences, where d is the fractional differencing parameter, 0<d<1/2, and let us generate the process $\mathbf{X}$={X_i: i=0,1,2,…,n} with a normal marginal distribution, a mean of zero, and the variance. σ_0^2, and the autocorrelation function (ACF), {ρ_k} (k=0,±1,…) that defined as (25),

$$\rho_k=\gamma_k/\gamma_0=\frac{\Gamma(1-d)\Gamma(k+d)}{\Gamma(d)\Gamma(k+1-d)}, \qquad (25)$$

where

$$\gamma_k=\sigma_0^2\frac{(-1)^k\Gamma(1-2d)}{\Gamma(k-d+1)\Gamma(1-k-d)}$$

The fixed-length sequence of self-similar pseudo-random numbers is obtained by using the steps shown in Figure 4.

Step 0: *Set N_0=0 and D_0=1. X_0, the first pseudo-random element in the self-similar sequence, is generated from the normal distribution $N(0,\sigma_0^2)$, where σ_0^2 is the required variance of X_i.*
Step i: (i=1,…,n-1). *Compute $mean_i$ and var_i of X_i recursively, using the following equations:*

$$N_i=\rho_i-\sum_{j=1}^{i-1}\phi_{i-1,j}\rho_{i-j},$$
$$D_i=D_{i-1}-N_{i-1}^2/D_{i-1},$$
$$\phi_{ii}=N_i/D_i,$$
$$\phi_{ij}=\phi_{i-1,j}-\phi_{ii}\phi_{i-1,i-j},\qquad j=1,...,i-1,$$

where ϕ_{ij}, $i=0,\quad j=0,...,n-1,$ is given by

$$\phi_{ii}=-\binom{i}{j}\frac{(j-d-1)!(i-d-j)!}{(-d-1)!(i-d)!},$$
$$mean_i=\sum_{j=1}^{i}\phi_{ij}X_{i-j},\ \mathrm{var}_i=(1-\phi_{ii}^2)\,\mathrm{var}_{i-1},$$

Generate X_i from N($mean_i$, var_i), increase i by 1.
Step n: *If i=n, then stop.*

Figure 4: F-ARIMA Algorithm

It took 4 hours, 1 min. and 24 sec. to generate the traffic sample sequence of 1,048,576 data points. The algorithm based on the F-ARIMA method is too computationally intensive for long sample sequences.

Fractional Gaussian Noise and Daubechies Wavelets

In this work, we propose a new advanced generator of pseudo-random self-similar sequences based on Fractional Gaussian Noise (FGN) and Daubechies Wavelet (DW), called the FGN-DW method. The generators based on DW produce more accurate self-similar sequences than those based on Haar Wavelets (HW) (Radev and Lokshina 2010). In that case, not only are estimates of H obtained from DW closer to the true values than those from HW, but also variances obtained from DW are much lower. The reason is that DW produces smoother wavelet coefficients in the Discrete Wavelet Transform (DWT) than HW does. The HW functions are discontinuous and do not exhibit good time–frequency localization, since their Fourier transforms decay as $|\lambda|^{-1}$, $\lambda \to \infty$, resulting in a decomposition with poor scaling properties. Accordingly, DW yields more accurate coefficients than HW. The proposed approach for generating synthetic self-similar FGN sequences in the time domain is based on DWT. Wavelets can provide compact representations for a class of FGN processes, because their structure naturally matches the self-similar structure of the LRD processes. This paper demonstrates that the FGN-DW method is sufficiently fast for practical generation of synthetic self-similar sequences to serve as input data for simulations. Wavelet analysis transforms a sequence into a time-scale grid, where time and scale are used instead of frequency, because, unlike in the Fourier transform, the mapping is not directly related to frequency. The wavelet transform generates the wavelet coefficients $d_x(i,j)$ from a sequence of given numbers. For an LRD process, the variance of the wavelet coefficients at each level i is defined by (26),

$$E[d_x(j,\cdot)^2 = CP_f 2^{i(2H-1)} \tag{26}$$

where $C > 0$ and (18) is true,

$$P_f = 2(2\pi)^{1-2H} c_\gamma E(2H-1)\sin((1-H)\pi) \tag{27}$$

where $E(\cdot)$ is the Euler function and c_γ is a positive constant. The power parameter P_f, which plays a key role in determining the absolute scale of the LRD process and its effects in applications, is an independent quantitative parameter with the dimension of variance. A spectral estimator can be obtained by calculating the time average Q_i of the $d_x(i,\cdot)$ at a given scale, which can be described as (28),

$$Q_i = \frac{1}{n_i}\sum_{j=1}^{n_i} d_x^2(i,j), \tag{28}$$

where n_i is the number of wavelet coefficients at scale i, i.e., $n_i = 2^{-i}n$, where n is the number of data points. The estimator uses a weighted linear regression as the variance of $\log_2(Q_i)$ varies with $\log_2(2^i)$. An estimated Hurst parameter $\hat{H}$ can be obtained with the linear regression (29),

$$y_i = \log_2(Q_i) = (2\hat{H}-1)i + c + 1/(n_i \ln 2) \tag{29}$$

where c is a constant. The wavelet transform provides good resolution in both time and scale, whereas the Fourier transform provides high frequency resolution but lacks time localization. The fixed-length sequence of self-similar pseudo-random numbers is obtained using the steps shown in Figure 5. It took only 2 seconds to generate a sequence of 1,048,576 data points using this algorithm.

Step 1: *Given H, start with i=1 and continue until i=n. Calculate a sequence of values* $\{f_1, f_2, \ldots, f_n\}$ *using* (30),

$$f(\lambda, H) = c_f \,|\,\lambda\,|^{1-2H} + O(|\,\lambda\,|^{\min(3-2H,2)}), \tag{30}$$

where $c_f = \sigma^2 \cdot (2\pi)^{-1}\sin(\pi H)\Gamma(2H+1)$, $O(\cdot)$ *represents the residual error and* $f_i = \hat{f}(\pi i / n; H)$, *the value of the frequencies* f_i *corresponds to the spectral density of an FGN process for* f_i *ranging between* $\pi/n \div \pi$.

Step 2: *Multiply* $\{f_i\}$ *by realizations of the independent exponential random variable with the mean of one to obtain* $\{\hat{f}_i\}$, *because the spectral density estimated for a given frequency is asymptotically distributed as an independent exponential random variable with mean* $f(\lambda,H)$.

Step 3: *Generate a sequence* $\{Y_1, Y_2, \ldots, Y_n\}$ *of complex numbers such that* $|\,Y_i\,| = \sqrt{\hat{f}_i}$ *and the phase* Y_i *is uniformly distributed on the interval* $[0, 2\pi]$. *This random phase technique preserves the spectral density corresponding to* $\{\hat{f}_i\}$. *It also makes the marginal distribution of the final sequence normal and satisfies the requirements for FGN.*

Step 4: *Calculate two synthetic coefficients of orthonormal Daubechies wavelets. The output sequence* $\{X_1, X_2, \ldots, X_n\}$ *representing an approximately self-similar FGN process is obtained by applying the inverse Daubechies wavelet transformation of the sequence* $\{Y_1, Y_2, \ldots, Y_n\}$.

Figure 5: FGN-DW Algorithm

SIMULATION RESULTS AND PERFORMANCE EVALUATION

The Hurst parameter estimates for the wavelet-based H estimator are shown in Tables 5 (H=0.6 and 0.7) and 6 (H=0.8 and 0.9) with different wavelet coefficients. The results are averaged over 30 sequences and show that for all input H values, the F-ARIMA and FGN-DW methods produce sequences with less-biased H values than other methods. The mean estimates of H, obtained using the proposed wavelet-based algorithm, are shown in Table 7. Analysis of the mean times required to generate a sequence of a given length shows that sequential generators are more suitable for practical network simulations than the F-ARIMA-based generator because they are faster. However, these generators require more input parameters, and selecting appropriate values is a remaining problem. In addition, the problem of defining the relationship between the Hurst parameter and the two shape parameters of a beta distribution in the SAP case also remains.

Let us consider generating self-similar time series using Algorithm 5 with one million data points, a Hurst parameter H = 0.5, a time-scaling factor of 10, and 10 wavelet

coefficients. The results are presented in Figure 6. For the generated time series, the values $Q(j_1)$ were calculated as a function of j_1, as illustrated in Figure 7. The application of linear regression (29) produced a better representation of the analytical values, as shown in Figure 8. The results demonstrate self-similar and long-range dependence characteristics of the generated time series, shown in Figure 9. The Hurst parameter H was determined with high precision, yielding an exact value of 0.504 for the original H=0.5. The results were estimated based on correspondence criteria with parameter Q=0.16155, and a time series of 1,000,000 values was generated using a Pareto distribution with α=1.8. Then, they were multiplied by a self-similar time series generated using Algorithm 5. The wavelet criteria provided good results for $Q(j_1)$ as a function of j_1, consistent with results in Figure 7. Considering the complete time interval (j1,j2)=[1,16], the optimal values were recorded at (j1,j2)=[2,13], and the correspondence criteria parameter Q increased from 0.16155 to 0.36899. The linear regression helped yield the Hurst parameter with better precision (0.502 versus 0.504, for the exact value of 0.5). The logarithmic plot obtained by weighted linear regression is shown in Figure 10. The confidence intervals are shown with vertical lines. They are small for j_1<13 and are obtained with a probability of 0.95 after generating 2^{10} (1,024) samples.

Table 5: Precision of FGN-DW Sequences (H = 0.6; 0.7)

Wave-let coeff.	Mean Values of Estimated H and ΔH			
	0.6		0.7	
	$\hat{H}$	ΔH	$\hat{H}$	ΔH
2	0.6019 (0.574, .630)	+0.323	0.6984 (0.671, 0.726)	-0.229
4	0.6026 (0.575, 0.630)	+0.433	0.7039 (0.676, 0.731)	+0.554
8	0.6026 (0.575, 0.630)	+0.430	0.7031 (0.676, 0.731)	+0.445
16	0.6013 (0.574, 0.629)	+0.214	0.6987 (0.671, 0.726)	-0.185

Table 6: Precision of FGN-DW Sequences (H = 0.8; 0.9)

Wave-let coeff.	Mean Values of Estimated H and ΔH			
	0.8		0.9	
	$\hat{H}$	ΔH	$\hat{H}$	ΔH
2	0.7943 (0.767, 0.822)	-0.709	0.8898 (0.862, 0.917)	-1.137
4	0.8055 (0.778, 0.833)	+0.684	0.9074 (0.880, 0.935)	+0.821
8	0.8039 (0.776, 0.831)	+0.486	0.9049 (0.877, 0.932)	+0.545
16	0.7962 (0.769, 0.824)	-0.474	0.8938 (0.866, 0.921)	-0.694

Table 7: Variances of Hurst Parameter

Wave-let coeff.	Mean Values of Estimated H and ΔH			
	0.6	0.7	0.8	0.9
2	1.949e-04	2.154e-04	2.393e-04	2.670e-04
4	2.508e-04	2.540e-04	2.586e-04	2.645e-04
8	2.440e-04	2.432e-04	2.438e-04	2.456e-04
16	2.116e-04	2.055e-04	2.007e-04	1.973e-04

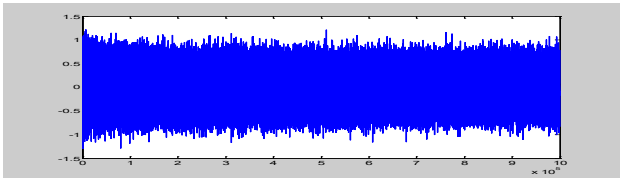

Figure 6: Self-similar Time Series with 1,000,000 Values

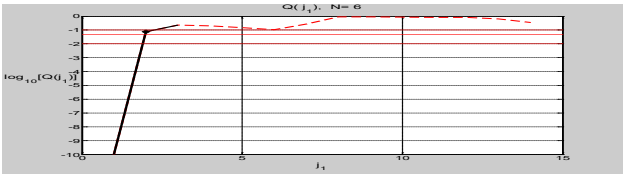

Figure 7: Evaluation of Self-similar Time Series for Different Scale Parameters

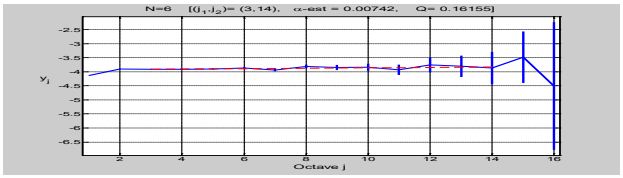

Figure 8: Comparison between Generated Time Series and Analytical Values

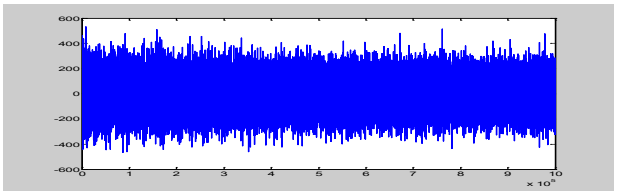

Figure 9: Product of Self-similar and Pareto Time Series

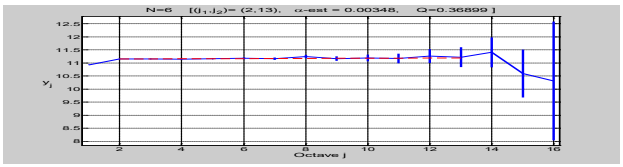

Figure 10: Comparison between Generated Time Series and Obtained Values with Weighted Linear Regression

CONCLUSION

Real communication network traffic exhibits self-similar characteristics across a broad range of time scales, distinguishing it fundamentally from the behavior predicted by traditional traffic models. Because self-similar models accurately capture the long-range dependence and burstiness observed in real IP network traffic, they provide a reliable foundation for network performance evaluation, resource allocation, capacity planning, and the provision of service quality (QoS). In this work, advanced self-similar models and efficient algorithms for simulating self-similar IP network traffic were developed and evaluated. Both sequential and fixed-length sequence generators were investigated to confirm their ability to reproduce statistical characteristics of real network traffic. Simulation results showed the effectiveness of the proposed models and algorithms for generating realistic self-similar traffic.

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AN SIS–TYPE CONTINUOUS–TIME MODEL OF TUBERCULOSIS SPREAD IN A HETEROGENEOUS POPULATION WITH A LINEAR AND A BILINEAR TRANSMISSION FUNCTION

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KEYWORDS

SIS model, transmission function, heterogeneous population, tuberculosis

ABSTRACT

In this paper we propose a model of infectious disease transmission in a heterogeneous population consisting of two subpopulations: individuals with low (LS) and high (HS) susceptibility to an infection. A pathogen can be transmitted among each subpopulation and from HS to LS. The novelty of our paper lies in considering two types of transmission function: based on the mass action law and linear, while preserving populations' heterogeneity and autonomous demographic processes in each subpopulation. We focus on the existence and local stability of stationary states appearing in the system. Theoretical results are supplemented with numerical simulations. We fit simulated values to epidemic data accounting for TB spread within and outside prisons in India. These data concern the tuberculosis epidemic in India in the years 2015–2023, where inmates are considered as HS.

INTRODUCTION

In a given population it is often possible to pinpoint individuals with a high susceptibility to an infection. Thanks to this distinction, we can identify two subpopulations that differ in their risk of getting infected (Noppert et al., 2023). A low–risk subpopulation (LS) is a subpopulation composed of individuals with low susceptibility to the infection. People with a high susceptibility constitute a high–risk subpopulation (HS). Naturally, the pathogen can be transmitted within each subpopulation and between subpopulations. However, in many cases individuals from HS form a group that wilfully separates from the whole society in the context of illness transmission, for example homeless people in the case of tuberculosis (Figueroa-Munoz and Ramon-Pardo, 2008) or people with high sexual activity in the case of HIV transmission. Furthermore, such separation can be imposed and HS can be constituted from prisoners Samal and Singh (2025) or isolated hospitalized patients. For this reason most new infections in HS arise from contacts among HS only. Hence, it is reasonable to assume that there is no transmission of illness from LS to HS. What is more, this assumption makes a mathematical analysis of epidemiological models simpler. However, we must still assume the possibility of transmission of TB from HS to LS since individuals can contract tuberculosis while interacting with released inmates, discharged patients and prison or hospital service, who interact with high–risk communities.

In each subpopulation we can naturally distinguish two groups – susceptible and infected people. Indicating more groups, for example dividing infected individuals into infectious and non–infectious ones, leads to more complicated computations. Moreover, the number of groups is determined by the structure of epidemic data – in most cases only the number of infections without classification is published. For many diseases an infected individual does not gain immunity after recovery and can be susceptible again. Models with this assumption are called SIS (*susceptible–infected–susceptible*) ones, which are widely analyzed in the literature. In (Kuhl, 2021) one can find an exemplary SIS model with mathematical analysis. The authors in (Kuehn and Widder, 2016) introduce an SIS model for a heterogeneous population and extend it with stochastic terms. However, they do not consider demographic processes. The issue of time persistence is investigated in (Clancy, 2018). The author shows that heterogeneity helps in eliminating a disease in a population. SIS models can also be applied to the analysis of $COVID$–19 spread – the authors in

(Gerasimov et al., 2020) emphasized the role of populations' heterogeneity in the appearance of early–stage and late–stage infection.

When describing illness transmission, one assumes that the whole population is well–mixed and that individuals encounter each other at random. A crucial thing is determining a function that describes such transmission. Basically, functions based on the mass action or the standard incidence function (Wei and Cui, 2012) are proposed. Both approaches rely on the assumption that the number of new infections is proportional to the multiplication of the number of healthy and infected people. However, in some cases a linear function can be used. This usage concerns bovine tuberculosis (BTB, caused by *Mycobacterium bovis*), which pathogen can be mainly transmitted with unpasteurized milk (Collins et al., 2022). This problem is mainly encountered in India (Ramanujam and Palaniyandi, 2023) and other undeveloped countries (Tulu, 2021). One can also address the problem of undetected or unreported cases by incorporating of a linear function Abadie et al. (2020). Applying the linear function as a transmission function is very rare. In (Mulkern and Nosrati, 2022) the authors used linear models of epidemic spread to investigate the effect of decreasing a constant in the linear function on flattening a curve showing the number of infected people over a given time horizon. As real data they used data from the $COVID$–19 pandemic. This pandemic is also considered in Tutsoy and Polat (2022), where one investigates linear models that better fit COVID–19 dynamics in Turkey than non–linear systems. Naturally, using a linear function instead of a bilinear one simplifies or even allows mathematical analysis.

In this paper we introduce and investigate a continuous–time SIS model of TB spread in a heterogeneous population, in which the spread is described by two types of functions. We use a bilinear function built on the mass action law for describing TB transmission between humans, which is caused by (caused by *Mycobacterium tuberculosis sensu stricto*). A linear one reflects spread of BTB. However, even for BTB there is human–to–human transmission (United States Department of Health and Human Services, 2011). Hence, the proposed concept can be applied to real cases of epidemics. Results presented herein are a continuation of research from our previous papers (Bodzioch et al., 2019) and (Choiński et al., 2019). The models in these papers also capture the dynamics of the epidemic in a heterogeneous population, but they do not include a linear term for disease transmission. The outcomes from these papers confirm the medical hypothesis that to investigate the dynamics of epidemic spread in a heterogeneous population, one must examine the spread not only in a particular subpopulation, but in the whole population. The novelty of our paper lies in a complex approach to modeling disease spread thanks to: assuming populations' heterogeneity, by considering different values of parameters for each subpopulation, considering transmission of infection with both airborne (human–to–human) and non–airborne ways, and assuming demographical processes occurring in each subpopulation independently. According to our knowledge, in the literature there is no paper that spans these issues and provides analytical results. Our aim is to fill this gap.

We want to relate our analytical results to a real scenario of TB epidemic, which can be reflected with both linear and non–linear terms. For this reason, in this paper we fit the parameters of our model to data on the spread of TB in India from 2015–2023. In this country TB incidence arises from contact between individuals and the consumption of unpasteurized milk and dairy products. Such possibilities of infection allows us to incorporate to, respectively, linear and non–linear components into our system. We choose prisoners as HS and the remaining individuals constitute LS. As stated earlier, we assume that inmates, because of their contained environment, do not contract TB from the outside community. Since new convicts are generally diagnosed before entering jails, we can assume autonomy of the prison community. Moreover, since one strictly supervises prison nutrition and food delivered to penitentiaries is tested, we neglect TB transmission in prison when consuming infected dairy products.

Although we validate our model for the specific case of TB spread, we should stress that the proposed system is universal and that the results of its analysis can be applied to other TB epidemic scenarios. Moreover, thanks to considering linearity during pathogen transmission, the model can constitute a basis for constructing models of the spread of any infectious disease in a heterogeneous population where unobserved incidence should be reported. Clearly, one can also use the results obtained herein to investigate stability in systems where transfer between variables is guaranteed with both linear and bilinear functions.

This paper is organized as follows. Firstly we propose the model with the constant migration into HS and discuss its basic properties. Then we indicate the stationary states of the system and investigate their local stability. Later we present the result of the numerical simulation. Finally we discuss the outcomes arising in the paper.

FORMULATION OF MODEL AND ITS BASIC PROPERTIES

Now we describe the components of the model. Every variable and most parameters are marked with the subscript i equal to 1 or 2. These values correspond to LS and HS, respectively. If in a given equation no value is given, then $i \in \{1, 2\}$. The variables S_1 and S_2 denote the sizes of healthy groups in the given subpopulation. The number of infected people is denoted by I_1 and I_2. We assume that each demographic and epidemiological process in each subpopulation is independent. New individuals migrate into the particular subpopulation with a constant inflow C_i. Individuals in LS and HS contract TB by human–to–human contact among their own subpopulations at, respectively, β_1 and β_2, rates. Moreover, HS–to–LS transmission is captured with

β coefficient. The linear term with ε constant reflects incidences in LS that are unobserved or emerged from consuming TB–infected products. Recovery, natural and TB–related death coefficients are denoted with γ_i, μ_i and α_i, respectively. The values of all parameters besides C_1 belong to the interval $(0, 1)$. The proposed model reads

$$\begin{aligned}
\dot{S}_1 =& C_1 - \varepsilon S_1 - \beta_1 S_1 I_1 - \beta S_1 I_2 \\
&+ \gamma_1 I_1 - \mu_1 S_1, && \text{(1a)}\\
\dot{I}_1 =& \varepsilon S_1 + \beta_1 S_1 I_1 + \beta S_1 I_2 \\
&- (\gamma_1 + \mu_1 + \alpha_1) I_1, && \text{(1b)}\\
\dot{S}_2 =& C_2 - \beta_2 S_2 I_2 + \gamma_2 I_2 - \mu_2 S_2, && \text{(1c)}\\
\dot{I}_2 =& \beta_2 S_2 I_2 - (\gamma_2 + \mu_2 + \alpha_2) I_2. && \text{(1d)}
\end{aligned}$$

For the sake of simplification we define

$$k_i := \gamma_i + \mu_i + \alpha_i.$$

Moreover, we introduce variables

$$N_i := S_i + I_i,$$

naturally meaning the size of each subpopulation.

See that Eqs. (1c)–(1d) constitute a two–dimension autonomous system. This observation is reasonable in the context of TB epidemics mentioned in the introduction – inmates (HS) are isolated from the rest of the population. The analysis of such a system is presented in our previous paper (Bodzioch et al., 2019).

Basic Properties Of The Model

It is clear that the right–hand side of Eqs. (1) fulfills the local Lipschitz condition. We establish the existence, uniqueness and positivity of solutions of the system for any positive initial conditions. Adding Eqs. (1a)–(1b) or Eqs. (1c)–(1d) by sides gives

$$\dot{N}_i = C_i - \mu_i N_i - \alpha_i N_i, \tag{2}$$

what can be estimated from above as

$$\dot{N}_i \leq C_i - \mu_i N_i. \tag{3}$$

Solving Eq. (3) yields

$$N_i(t) \leq \left(N_i(0) - \frac{C_i}{\mu_i} \right) \exp\left(-\mu_i t\right) + \frac{C_i}{\mu_i}.$$

For $N_i(0) > \frac{C_i}{\mu_i}$ the solution decreases and $S_i(t)$, $I_i(t)$ are bounded from above as $S_i(t), I_i(t) \leq N_i(t) \leq N_i(0)$. For $N(0) \leqslant \frac{C}{\mu}$ the solution can increase with a limitation:

$$S_i(t),\ I_i(t) \leq N_i(t) \leq -c \exp\left(-\mu_i t\right) + \frac{C_i}{\mu_i} \leq \frac{C_i}{\mu_i},$$

where c is a positive constant. Hence, $S_i(t)$ and $I_i(t)$ are defined for every $t > 0$. We can clearly also estimate Eq. (2) from below as $\dot{N}_i \geq C_i - (\mu_i + \alpha_i) N_i$, what together with Eq. (3) generates an invariant set

$$\left\{ S_i + I_i : S_i + I_i \in \left(\frac{C_i}{\mu_i + \alpha_i}, \frac{C_i}{\mu_i} \right) \right\}.$$

Stationary States

Let us indicate stationary states of the form (S_1, I_1, S_2, I_2) appearing in System (1). We obtain the theorem:

Theorem 1. *In System* (1) *there exist two stationary states: without infection in HS*

$$E_0 = \left(\widehat{S}_1, \widehat{I}_1, \widehat{S}_2, 0 \right),$$

where $\widehat{S} = \frac{k_2}{\beta_2}$ *and*

$$\begin{aligned}
\widehat{S}_1 =& \frac{\beta_1 C_1 + \mu_1 k_1 + (\mu_1 + \alpha_1)\varepsilon - \sqrt{\Delta}}{2\mu_1 \beta_1},\\
\widehat{I}_1 =& \frac{\beta_1 C_1 - \mu_1 k_1 - (\mu_1 + \alpha_1)\varepsilon + \sqrt{\Delta}}{2\beta_1(\mu_1 + \alpha_1)}, \qquad (4)\\
\Delta =& (\beta_1 C_1 + \alpha_1 \varepsilon - \mu_1 k_1 - \mu_1 \varepsilon)^2 \\
&+ 4\mu_1 \varepsilon (\beta_1 C_1 + \alpha_1 k_1 + \alpha_1 \varepsilon),
\end{aligned}$$

always existing, and postive (endemic) $E_e = \left(\bar{S}_1, \bar{I}_1, \bar{S}_2, \bar{I}_2 \right)$, *where*

$$\bar{S}_2 = \frac{k_2}{\beta_2}, \quad \bar{I}_2 = \frac{C_2 - \bar{S}_2}{\mu_2 + \alpha_2}$$

and

$$\begin{aligned}
\bar{S}_1 =& \frac{\theta + \beta_1 C_1 + \mu_1(\gamma_1 + \mu_1) - \sqrt{\bar{\Delta}}}{2\beta_1 \mu_1},\\
\bar{I}_1 =& \frac{\beta_1 C_1 - \theta - \mu_1(\gamma_1 + \mu_1) + \sqrt{\bar{\Delta}}}{2\beta_1(\mu_1 + \alpha_1)}, \qquad (5)\\
\bar{\Delta} =& \left(\theta - \beta_1 C_1 + \mu_1(\gamma_1 + \mu_1) \right)^2 + 4\beta_1 C_1,\\
&\textit{where} \quad \theta := (\mu_1 + \alpha_1)(\varepsilon + \beta \bar{I}_2).
\end{aligned}$$

E_e exists if

$$C_2 > \frac{k_2}{\beta_2}. \tag{6}$$

Proof. Eq. (1d) for any stationary state yields $0 = I_2(\beta_2 S_2 - k_2)$ giving two separate cases.

• Firstly assume that $I_2 = 0$. From Eq. (1c) we get $S_2 = \widehat{S}_2$. Eqs. (1a)–(1b) can be rewritten as

$$\begin{aligned}
0 =& C_1 - \varepsilon S_1 - \beta_1 S_1 I_1 + \gamma_1 I_1 - \mu_1 S_1, && \text{(7a)}\\
0 =& \varepsilon S_1 + \beta_1 S_1 I_1 - k_1 I_1. && \text{(7b)}
\end{aligned}$$

Eq. (7b) gives

$$I_1 = \frac{\varepsilon S_1}{k_1 - \beta_1 S_1}. \tag{8}$$

Positivity of I_1 implies

$$S_1 < \frac{k_1}{\beta_1}. \tag{9}$$

From Eq. (2) we get

$$0 = C_1 - \mu_1(S_1 + I_1) - \alpha_1 I_1. \tag{10}$$

Substituting Eq. (8) into Eq. (10) and further transformations yield

$$P(S_1) := \mu_1\beta_1 S_1^2 - \Big(\beta_1 C_1 + \mu_1 k_1 + \\ + (\mu_1+\alpha_1)\varepsilon\Big)S_1 - C_1 k_1 = 0. \tag{11}$$

The discriminator of $P(S_1)$ can be rephrased as Δ from (4). Clearly, $\Delta > 0$. Moreover, $P(0) > 0$ and $P''(0) > 0$. Hence, the zeros of $P(S_1)$ are positive real numbers and read

$$S_{a,b} = \frac{1}{2\mu_1\beta_1}\Big(\beta_1 C_1 + \mu_1 k_1 + \mu_1\varepsilon \mp \sqrt{\Delta}\Big).$$

Observe that from Eq. (10) one must hold

$$S_1 < \frac{C_1}{\mu_1}. \tag{12}$$

It is easy to check that $S_a < \frac{C_1}{\mu_1}$ and $S_b > \frac{C_1}{\mu_1}$. Hence, only S_a can be S_1 coordinate of the sought equlibirium. Moreover, such S_1 satisfies Ineq. (9). From (10) we get form of $\widehat{I}_1$ expressed in (4). Naturally, the obtained E_0 state always exists because of any conditions restricting its coordinates.

• Now assume that $S_2 = \bar{S}_2$. From Eq. (2) for any stationary state we have

$$0 = C_2 - \mu_2(S_2 + I_2) - \alpha_2 I_2,$$

what in our case transforms to $I_2 = \bar{I}_2$. Required positivity of $\bar{I}_2$ leads to Condition (6). We rephrase Eq. (10) as $I_1 = \frac{C_1}{\mu_1} - S_1$ and substitute the obtained depnedence into Eq. (1a) for another postulated state. After proper transformations we get

$$\bar{P}(S_1) := \mu_1\beta_1 S_1^2 - \Big(\theta + \beta_1 C_1 + \mu_1(\gamma_1+\mu_1)\Big) \\ + C_1(\alpha_1+\mu_1) = 0.$$

We treat the above analogously as Eq. (11) and therefore consider only $S_1 = \bar{S}_1$ from (5). We also easily get $I_1 = \bar{I}_1$. □

STABILITY OF STATIONARY STATES

Local Stability

In this section one investigates local stability of stationary states of System (1). We begin from the state without infection in HS.

Theorem 2. *E_0 is locally stable if*

$$C_2 < \frac{\mu_2 k_2}{\beta_2}. \tag{13}$$

Proof. The Jacobian matrix for E_0 can be written as a block matrix $\begin{pmatrix} M_1 & M^* \\ 0 & M_2 \end{pmatrix}$, where

$$M_1 = \begin{pmatrix} -(\varepsilon+\mu_1) - \beta_1\widehat{I}_1 & \gamma_1 - \beta_1\widehat{S}_1 \\ \varepsilon + \beta_1\widehat{I}_1 & -k_1 + \beta_1\widehat{S}_1 \end{pmatrix},$$

$$M_2 = \begin{pmatrix} -\mu_2 & -\beta_2\widehat{S}_2 + \gamma_2 \\ 0 & \beta_2\widehat{S}_2 - k_2 \end{pmatrix}.$$

The characteristic polynomial of M_1 reads

$$P_1(\lambda) := \lambda^2 + \lambda\big(\varepsilon + \mu_1 + k_1 + \beta_1(\widehat{I}_1 - \widehat{S}_1\big) \\ + (\varepsilon + \mu_1 + \beta_1\widehat{I}_1)(k_1 - \beta_1\widehat{S}_1) \\ - (\varepsilon + \beta_1\widehat{I}_1)(\gamma_1 - \beta_1\widehat{S}_1).$$

and its discriminant can be transformed to

$$\Delta_1 = \big(\varepsilon + \mu_1 + \beta_1(\widehat{I}_1 + \widehat{S}_1) - k_1\big)^2 \\ + 4(\varepsilon + \beta_1 I_1)(\gamma_1 - \beta_1\widehat{S}_1).$$

In order to produce two real negative zeros of $P_1(\lambda)$, one must assume that

$$P_1(0) > 0, \quad P_1'(0) > 0, \quad \Delta_1 > 0. \tag{14}$$

Assuming

$$\gamma_1 > \beta_1\widehat{S}_1 \tag{15}$$

implies fulfillment of the first and second third inequality from (14). It is easy to check that Ineq. (15) is always true. The second inequality from (14) holds if

$$k_1 > \beta_1\widehat{S}_1.$$

We neglect the above inequality since it is weaker than Ineq. (15). Hence, there is no restriction on desirable characters of zeros of P_1.

The eigenvalues of M_2 are $\lambda_1 = -\mu_2 < 0$ and $\lambda_2 = \beta_2\frac{C_2}{\mu_2} - k_2$. The second eigenvalue is negative if Ineq. (13) holds. This inequality becomes the condition for the E_0 local stability. □

Theorem 3. *E_e is locally stable whenever it exists.*

Proof. We write the Jacobian matrix for E_e as a block matrix $\begin{pmatrix} J_1 & J^* \\ 0 & J_2 \end{pmatrix}$, where

$$J_1 = \begin{pmatrix} -\beta_1\bar{I}_1 - \beta\bar{I}_2 - (\varepsilon+\mu_1) & \gamma_1 \\ \varepsilon + \beta_1\bar{I}_1 + \beta\bar{I}_2 & -k_1 \end{pmatrix},$$

$$J_2 = \begin{pmatrix} -\beta_2\bar{I}_2 - \mu_2 & \beta_2 k_2 + \gamma_2 \\ \beta_2\bar{I}_2 & -(1-\beta_2)k_2 \end{pmatrix}.$$

The matrices J_1 and J_2 are analogous to the matrix M_1 from the earlier part of the proof. Therefore, we state that each J_1 and J_2 unconditionally generates two negative real zeros under the E_e existence. □

Let us remind Theorems 1–3. Combining them, we conclude that

Corollary 1. *If E_e exists, then E_{df} loses its stability. Always exactly one stationary state is locally stable.*

From Theorem 2 and Corollary 1 we state that reducing an inflow of (even only healthy) individuals into HS enables eradication of infection in this subpopulation.

NUMERICAL SIMULATION

Here we depict the dynamics of System (1) for the parameters fitted to real data. We illustrate the spread of tuberculosis in India during the years 2015–2023. As stated in the introduction, as HS we choose prisoners, and the remaining citizens comprise LS. For simulation we used demographic and epidemiological data from annual statistics. The population size of India for each particular year is taken from Worldometer (2026). In Office of the Registrar General and Census Commissioner (2026) one can find the numbers of births and deaths. The Central Tuberculosis Division in India provides data on TB cases Central Tuberculosis Division (2026). Numbers of prisoners are obtained from the National Crime Records Bureau National Crime Records Bureau (2026). The collected numbers allow us to estimate the values of γ_i, μ_i and C_i. Table 1 comprises the obtained values.

Table 1: The Values of demographic Parameters for System (1).

Name	Described process	Value
C_1	Inflow into LS	26080290
C_2	Inflow into HS	1259
μ_1, μ_2	Natural death	0.0065
γ_1	Recovery in LS	0.85
γ_2	Recovery in HS	0.54
α_1	TB–related death in LS	0.116
α_2	TB–related death in HS	0.135

There is no medical definition of a transmission coefficient. For that reason there is a need to estimate these types of coefficients that appear in our models. In order to do that, we use the built–in *lsqcurvefit* function in the Matlab software. Thanks to this function, we find the values of transmission coefficients for which the matching error is minimal. The obtained values for System (1) are presented in Table 2. The order of magnitude of ε is higher than the orders of magnitude of β_1 and β_2. This is expected since ε is included in the linear term, whereas the remaining coefficients correspond to the bilinear terms, which are products of the numbers of healthy and infected people. Moreover, observe that $\beta_2 > \beta_1$. It is also anticipated because transmission of TB among prisoners is more effective than TB spread outside penitentiaries. Furthermore, β has the least value due to inmates' isolation.

Table 2: The Values of Transmission Coefficients for System (1).

Notation	Value
ε	$4.4666 \cdot 10^{-5}$
β_1	$5.1415 \cdot 10^{-9}$
β_2	$6.7901 \cdot 10^{-7}$
β	$6.6109 \cdot 10^{-10}$

In Figure 1 one can find the comparison between real and simulated values of variable I_1 (corresponding to infected individuals, excluding prisoners) for System (1).

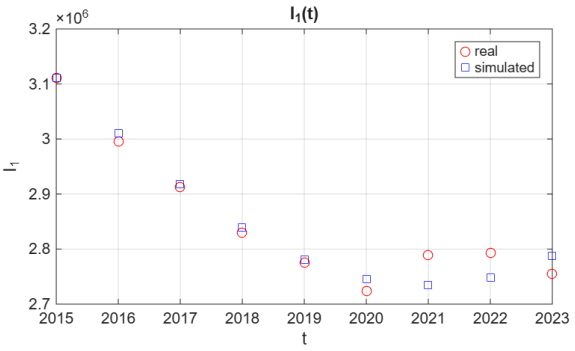

Figure 1: The real and simulated Values of I_1 for System (1) in Years 2015–2023.

DISCUSSION

In this paper we have introduced and investigated the model (1) which captures the dynamics of tuberculosis spread in a heterogeneous population. For such a population two subpopulations are indicated: people with low (LS) and high (HS) susceptibility to the infection. The work is a continuation of our research presented in (Bodzioch et al., 2019) and (Choiński et al., 2019). The novelty of the results presented herein lies in considering two types of illness transmission functions, the linear function and the function based on the mass action law, in the model. We were able to provide analytical results, preserving populations' heterogeneity and autonomous demographic processes in each subpopulation.

System (1) has two stationary states: without infection in HS (E_0) and endemic (E_e). The first state exists unconditionally, the second state disappears if the inflow into HS is sufficiently small. E_0 is locally stable under the fulfillment of one inequality, which is reversed to the inequality guaranteeing the existence of E_e. Stability of E_e in System (1) is always unrestricted unless E_e vanishes.

What is important, in System (1) there is no actual disease–free state in which there is no infection in both subpopulations. The absence of this state is uncommon in epidemiological modelling. However, in most cases complete eradication of the disease can occur when external measures, such as vaccinations or isolation, are implemented. In our model we assume there are no such actions and that individuals within their own subpopulations interact freely. For this reason we consider a lack of this disease–free state to be reliable. Furthermore, we should emphasize that the structure of the model is uncommon and we conducted only its initial analysis. The model can be extended with terms reflecting for example prevention campaigns, including vaccinations. If we consider such extensions, then the other stationary states can be reached. In (Choiński et al., 2019) we modified the four–dimensional model, including disease detection programs in HS. We ob-

tained stationary states that differed from those in the model without this modification. Our future work will focus on extending System (1) to include terms related to these programs.

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Optimization Methodology

DIRECT VERSUS INDIRECT METHODS IN SINGULAR OPTIMAL CONTROL: A STRUCTURAL BENCHMARK

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ABSTRACT

Direct transcription methods are standard tools for optimal control, but their behaviour on singular arcs is difficult to judge from the objective value alone. This paper presents a compact simulation benchmark showing how different direct transcriptions represent a bang-singular-bang optimal control. A structure-preserving indirect solution is used as a reference: it fixes the qualitative control pattern, enforces the singular feedback analytically, and optimises the switching times. We compare five direct formulations under mesh refinement and over structured sets of initial conditions. The results show that some methods reproduce a competitive objective value while replacing the singular arc by mesh-dependent switching. The comparison therefore separates objective accuracy from structural accuracy. The study supports a simple conclusion for simulation practice: singular optimal-control benchmarks should report structural diagnostics, such as switch count and total variation, together with objective values.

INTRODUCTION

Optimal control computations are often assessed by the value of the objective functional. In simulation studies this is not always enough. The computed control may represent a therapy, intervention, or operating policy, and therefore its qualitative structure matters. A control with almost the same cost but a different switching pattern can have a different practical meaning and may indicate a numerical artefact rather than a genuine continuous-time optimum.

This issue is especially important in control-affine problems with singular arcs. On a bang arc the control is at one of its bounds. On a singular arc the switching function vanishes over a nontrivial time interval, and the control is recovered from higher-order optimality conditions (Bonnard and Chyba, 2003; Schättler and Ledzewicz, 2015; Bodzioch, 2024). Direct transcription methods do not usually enforce this structure explicitly. They introduce control variables on a mesh and solve a nonlinear program, so a singular feedback may be represented by chattering, many small interior segments, or a mesh-dependent switching pattern. This is consistent with recent direct-collocation works on nonsmooth and singular optimal control, where structure detection and mesh adaptation are treated as separate numerical ingredients (Miller et al., 2021; Pager and Rao, 2022).

This also explains why the distinction between indirect and direct methods matters here. Indirect methods start from the Pontryagin Maximum Principle (PMP) and exploit the analytical switching structure. They can be very accurate when the structure is known, but require additional analysis and reliable initial guesses for switching times. Direct methods instead discretise the state and control and solve the resulting nonlinear program with general-purpose optimisation software (Biegler, 2010; Kelly, 2017; Wächter and Biegler, 2006). This makes them flexible and easy to apply, but the singular structure is recovered only from mesh variables. Thus the two approaches are compared not only as optimisation algorithms, but as different ways of representing the same continuous-time control structure.

The present paper builds on the single-input benchmark problem introduced by Bajger et al. (2019) and Bodzioch (2024). The cited works formulate an indirect switching-time algorithm and apply it to a population-control model with a non-standard objective functional. The benchmark is suitable for the present comparison because it admits locally optimal singular arcs and its optimal control structure depends on both the dynamics and the penalty term in the objective functional. Hence, it is a useful test case for studying whether direct transcriptions recover the qualitative structure of a singular solution. For the reference initial condition, the indirect solution has the structure: full control, singular feedback, and then full control again. The two transition times are optimised by the switching-time indirect method. This solution is used as a continuous-time structural reference, not as another direct mesh transcription.

The contribution of this paper is experimental. We do not propose a new direct transcription. Instead, we ask a focused question:

When a singular solution is known, do standard direct transcriptions recover its structure under mesh refinement, or do they only recover a similar objective value?

The answer is method-dependent. Some direct methods remain structurally stable, while others produce increasingly oscillatory controls as the mesh is refined. The main message is that objective-only benchmarking is unreliable for singular optimal control. Thus, the benchmark combines objective values with structural diagnostics: switch count, total variation of the control, and a residual measuring proximity to the singular manifold.

BENCHMARK MODEL AND SINGULAR STRUCTURE

We consider the single-input population model

$$\dot{n}_1 = \lambda_1 n_1 - n_1 u, \quad \dot{n}_2 = \lambda_2 n_2(1 - n_1 - n_2), \qquad (1)$$

with $u(t) \in [0,1]$. The variables n_1 and n_2 denote control-sensitive and control-resistant subpopulations. The Bolza objective is

$$J(u) = \omega_1 n_1(T) + \omega_2 n_2(T) + \int_0^T [\eta_1 n_1(t) + \eta_2 n_2(t) + \xi G\Big(\tfrac{n_2(t)-n_1(t)}{\varepsilon}\Big) + \theta u(t)\Big]\, \mathrm{d}t, \qquad (2)$$

with terminal time T, where $G(z)$ is an activation function (e.g. $G(z) = \frac{1}{2}(1 + \tanh z)$).

The benchmark parameters are those of the reference case in Bodzioch (2024): $T = 14$, $\lambda_1 = 0.2$, $\lambda_2 = 0.1$, $\eta_1 = 2$, $\eta_2 = 3$, $\varepsilon = 0.01$, $\xi = 1$, $\theta = 0.5$, $\omega_1 = 5$, $\omega_2 = 10$. The main (reference) initial condition is $n_1(0) = 0.44$, $n_2(0) = 0.05$.

This point lies in the regime $n_1(0) > n_2(0)$, where the singular mechanism associated with the smooth dominance penalty is active in the reference solution. The penalty weight ξ is structurally important: when $\xi = 0$, the objective no longer contains the dominance penalty that creates the singular-feedback mechanism considered here. The present benchmark intentionally uses $\xi = 1$ in order to test the singular case.

For this model the singular feedback used by the indirect reference is

$$u_S(n_1, n_2) = \frac{\lambda_1 n_1 - \lambda_2 n_2(1 - n_1 - n_2)}{n_1} + \frac{\varepsilon^2 \lambda_2(\eta_2 - \eta_1) n_2 + 2\varepsilon\xi\lambda_2 n_2 G'\big(\frac{n_2-n_1}{\varepsilon}\big)}{\xi n_1 G''\big(\frac{n_2-n_1}{\varepsilon}\big)}. \qquad (3)$$

The associated singular manifold in the state plane depends on the constant Hamiltonian value c and is described by

$$F_{\mathrm{arc}}(n_1, n_2; c) = \varepsilon\big(\eta_1 + \lambda_1\theta - c + (\eta_2 - \eta_1)n_2\big) + \varepsilon\xi G\big(\tfrac{n_2-n_1}{\varepsilon}\big) - \xi(1 - n_1 - n_2) G'\big(\tfrac{n_2-n_1}{\varepsilon}\big) = 0. \qquad (4)$$

Here $H \equiv c$ along an autonomous extremal. For the derivation of the singular feedback and the singular-arc equation, see Bodzioch (2024).

INDIRECT SWITCHING-TIME REFERENCE

We briefly recall the indirect construction used to obtain the reference; see Bodzioch (2024). Consider a scalar-control problem

$$\dot{n}(t) = f(n(t)) + g(n(t))u(t), \qquad u(t) \in [0,1],$$

with Hamiltonian

$$H(n, p, u) = p^\top(f(n) + g(n)u) + L(n, u).$$

The switching function is

$$\Phi(t) = \frac{\partial H}{\partial u}(n(t), p(t), u(t)).$$

When the Hamiltonian is affine in u, the sign of Φ determines the bang-bang control whenever $\Phi \neq 0$. If $\Phi \equiv 0$ on an interval, the control is singular and must be recovered by differentiating the switching function until the control appears explicitly. In the benchmark model this calculation gives the feedback (3).

The switching-time method assumes a symbolic control structure $\sigma = (\sigma_0, \sigma_1, \ldots, \sigma_k)$, $\sigma_j \in \{0, S, 1\}$, where 0 denotes the lower bound, 1 the upper bound, and S a singular-feedback interval. The associated switching times satisfy

$$0 = t_0 < t_1 < \cdots < t_k < t_{k+1} = T.$$

For a fixed structure σ, the control is constructed interval by interval: on intervals labelled 0 or 1 it is set to the corresponding bound, and on intervals labelled S it is set to the singular feedback $u_S(n(t))$. Hence the only optimisation variables are the switching times $\tau = (t_1, \ldots, t_k)$.

The infinite-dimensional optimal-control problem is thereby reduced to the finite-dimensional problem of minimising $J = J(t_1, \ldots, t_k)$ for the prescribed structure. The gradient used in the reference computation is approximated by finite differences,

$$\frac{\partial J}{\partial t_i} \approx \frac{J(t_1, \ldots, t_i + \Delta, \ldots, t_k) - J(t_1, \ldots, t_i, \ldots, t_k)}{\Delta},$$

followed by a projected descent step that preserves the ordering of the switching times. Each evaluation of J requires integration of the state equations under the constructed control. The important point for the present benchmark is that the singular arc is generated analytically, not by independent mesh-level control variables.

For the reference case, $k = 2$ and $\sigma = (1, S, 1)$, so the switching-time variables are only t_1 and t_2. This gives a low-dimensional structural reference against which the direct transcriptions can be compared.

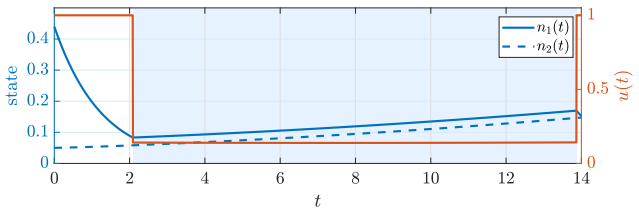

Figure 1: Indirect reference for $n(0) = (0.44, 0.05)$. The shaded interval is the singular arc on $[t_1, t_2]$.

The indirect solution (IND) to the problem (1)–(2) used as reference has the structure $1 \to S \to 1$ with

$$t_1 = 2.082214, \quad t_2 = 13.867390, \quad J_{\text{ref}} = 11.901121.$$

For this reference trajectory, the constant Hamiltonian value is $c_{\text{ref}} \approx 1.01124$. The same c_{ref} is used below when the direct trajectories are tested against the reference singular manifold (4). The structural switch count of this reference is 2, corresponding to the transitions $1 \to S$ and $S \to 1$. The sampled reference control has discrete total variation $\text{TV}(u_{\text{ref}}) = 1.7230$. This quantity is formally defined later; here it should be read simply as the accumulated variation of the plotted control profile. Figure 1 shows the reference state and control profiles. The shaded region marks the singular interval, on which the plotted control is neither an independent optimisation variable nor a bang-bang interpolation; it is the feedback (3).

DIRECT TRANSCRIPTIONS AND DIAGNOSTICS

We compare five direct transcriptions: forward Euler (FE), backward Euler (BE), trapezoidal (TR), multiple shooting (MS), and direct collocation (DC). The first three are low-order finite-difference or quadrature-based NLP transcriptions, while the last two are standard higher-level direct optimal-control formulations. In our implementation, the Pyomo-based transcriptions and the CasADi shooting/collocation formulations use IPOPT as the NLP solver (Hart et al., 2011; Andersson et al., 2019; Wächter and Biegler, 2006). All methods use the same continuous-time model, objective, and control bounds before transcription. The main mesh sweep uses

$$N \in \{50, 100, 200, 400, 800, 1600\}.$$

The main comparison is performed for $n_1(0) = 0.44$, $n_2(0) = 0.05$. Additional runs at $N = 400$ are used to test robustness over three initial-condition families:

(i) a triangular set with $n_1(0) > n_2(0)$, representing the singular-capable regime;

(ii) a near-diagonal set with $n_1(0) \approx n_2(0)$, probing the transition zone of the smooth penalty;

(iii) a full rectangular 20×20 grid (400 initial conditions), used to assess structural artefacts across the state plane.

The initial-condition sweeps are not interpreted through reference switching-time errors, because a separate indirect reference was not generated for every point. They are still informative: switch counts and total variation are absolute structural diagnostics of the computed controls and do not require a reference trajectory.
The reported diagnostics are intentionally few. The relative objective gap is

$$\delta J = \frac{|J - J_{\text{ref}}|}{1 + |J_{\text{ref}}|}.$$

The denominator avoids an artificial scaling problem if the reference objective is small. We use the absolute, not signed, gap because a direct transcription may occasionally produce a value below J_{ref} due to discretisation error or a structurally different control; such a value should not automatically be interpreted as a better continuous-time solution.
For a control profile $u_0, u_1, \ldots, u_N$ sampled at times $0 = t_0 < t_1 < \cdots < t_N = T$, the detected switch count is

$$U_\tau = \#\{j : \ |u_{j+1} - u_j| > \tau\}, \qquad \tau = 0.02.$$

This counts visible changes in the computed control profile. It is not meant to identify exact analytical switching times; rather, it detects excessive mesh-level switching.
The total variation is

$$\text{TV}(u) = \sum_{j=0}^{N-1} |u_{j+1} - u_j|.$$

This is the discrete analogue of the variation of a bounded-variation function. A bang-bang control with a small number of switches has small total variation. By contrast, a rapidly oscillating control has large total variation even if the state trajectory and objective value look reasonable. In this benchmark, U_τ and $\text{TV}(u)$ are therefore the main structural diagnostics.
Finally, for the main reference trajectory we use a state-space diagnostic, the singular-arc residual

$$D_{\text{arc}} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |F_{\text{arc}}(n_1(t), n_2(t); c_{\text{ref}})| \, \mathrm{d}t.$$

Here $F_{\text{arc}}(\cdot, \cdot; c_{\text{ref}}) = 0$ is the reference singular manifold from (4), with the Hamiltonian constant fixed at the indirect-reference value. Thus D_{arc} is the mean absolute residual of the correctly shifted singular-manifold equation on the reference singular interval. The same c_{ref} is used for every direct method, so that all trajectories are compared with one and the same reference manifold. In the numerical implementation, the integral is approximated by the corresponding quadrature sum on the time

grid of each method. It measures whether the computed state lies close to the singular manifold, not whether the computed control is singular in the PMP sense. This distinction is important: an oscillating bang-bang control may keep the state near the manifold while generating that motion in a numerically artificial way.

For the main reference case, exact structural recovery would give a switch count close to the structural value 2. This criterion is too strict for broad initial-condition sweeps, where a direct method may represent a smooth singular segment by several small numerical changes. Therefore, in the batch analysis we report the severe-over-switch rate

$$P_{10} = \Pr(U_\tau > 10),$$

computed over completed runs. The threshold is conservative: it does not classify mild segmentation as chattering, but it detects the large switch counts produced by unstable transcriptions. A lower threshold such as $U_\tau > 5$ would also flag many multiple-shooting and collocation runs, although their total variation remains close to the reference scale. Thus P_{10} is interpreted together with $\mathrm{TV}(u)$: the former detects excessive switching complexity, while the latter distinguishes small local discretisation effects from large-amplitude chattering.

RESULTS

Main Mesh-Sensitivity Experiment

Table 1 reports the diagnostics at the representative mesh size $N = 400$. The objective values alone would suggest that all methods except perhaps forward Euler are nearly equivalent. This is misleading. Forward Euler has a small objective gap, 0.296%, but its control has 106 switches and $\mathrm{TV}(u) = 92.64$. By contrast, multiple shooting and collocation have 9 switches and total variation about 2.4. The trapezoidal method has the smallest direct objective gap at this mesh, but still produces 25 detected switches. Backward Euler suppresses the amplitude of oscillations, as indicated by $\mathrm{TV}(u) = 2.06$, but its switch count remains elevated.

Table 1: Diagnostics for $n(0) = (0.44, 0.05)$ at $N = 400$.

Method	J	δJ [%]	U_τ	$\mathrm{TV}(u)$	D_{arc}
FE	11.8629	0.2960	106	92.64	0.002529
BE	11.9268	0.1990	18	2.06	0.000111
TR	11.9016	0.0038	25	3.13	0.0000065
MS	11.9031	0.0150	9	2.40	0.0000091
DC	11.9017	0.0046	9	2.39	0.0000081
IND	11.9011	0	2	1.72	$< 10^{-6}$

Figure 2 shows representative control profiles at $N = 400$. This visual comparison explains why the objective gap alone is not sufficient.

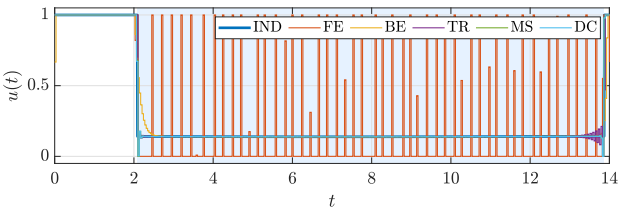

Figure 2: Control profiles at $N = 400$ for the indirect reference and all direct methods. The shaded interval is the singular interval $[t_1, t_2]$.

Forward Euler generates a rapidly alternating pattern over most of the singular interval. The trapezoidal method is less extreme in the objective table, but still contains local oscillations near the end of the singular interval. Direct collocation follows the reference much more closely at the structural level, although it is still a mesh representation rather than an analytically enforced singular feedback. The direct state trajectories corresponding to these runs are almost indistinguishable at the plot scale; hence the structural error is primarily visible in the controls, not in the states.

Figure 3 shows the same comparison over all mesh sizes. Forward Euler is the clearest example of objective convergence without structural convergence: its objective gap decreases from 1.43% to 0.081%, while its switch count increases from 35 to 381 and its total variation from 11.40 to 367.32. Mesh refinement therefore gives the discretisation more freedom to approximate the singular feedback by rapid switching. Trapezoidal transcription converges very well in objective value, but the switch count remains around 25–29. Multiple shooting and collocation are the most stable direct baselines: their objective gaps are small and their structural metrics stay close to the reference scale.

The singular-arc residual provides a complementary state-space diagnostic. At $N = 400$, forward Euler has $D_{\mathrm{arc}} \approx 2.53 \times 10^{-3}$ and backward Euler about 1.11×10^{-4}, whereas trapezoidal transcription, multiple shooting and direct collocation are at the 10^{-5}–10^{-6} scale and the indirect reference is numerically close to zero. Nevertheless, a small state-space residual does not imply correct control structure: the trapezoidal solution remains close to the singular manifold while still producing 25 detected control switches. Hence D_{arc} complements, rather than replaces, $(U_\tau, \mathrm{TV}(u))$.

Initial-Condition Robustness Checks

The robustness runs answer a different question from the main mesh sweep. They do not prove that each computed control matches a separately generated indirect optimum. Instead, they test whether the structural behaviour observed for the representative point persists over organised sets of initial states. This is useful for

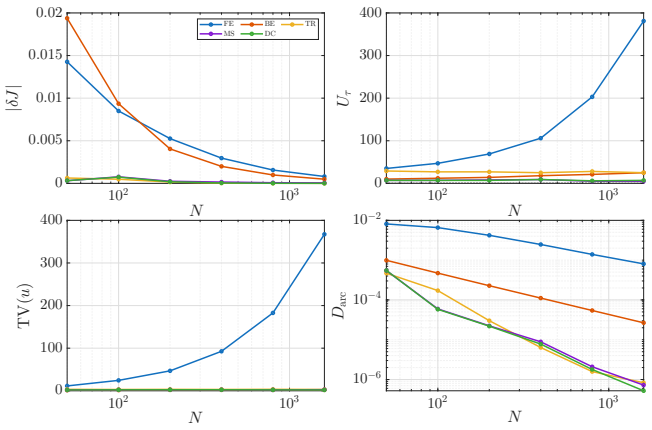

Figure 3: Mesh sensitivity for the main initial condition. The panels show objective gap, switch count, total variation, and the singular-arc residual D_{arc} computed with the reference Hamiltonian constant c_{ref}.

simulation benchmarking because it reveals whether an artefact is local to one trajectory or systematic across a region of the state plane.

Table 2 summarises the initial-condition sweeps. The triangular family, $n_1(0) > n_2(0)$, contains 210 requested points, the near-diagonal family, $n_1(0) \approx n_2(0)$, 108 requested points, and the full grid 400 requested points. To keep the table compact, the reported percentages and distributional summaries are computed over completed runs for each method. The table gives the severe-over-switch rate $P_{10} = \Pr(U_\tau > 10)$ and medians with 90th percentiles for U_τ and $\mathrm{TV}(u)$.

The robustness results strengthen the mesh-sensitivity conclusion. In the singular-capable triangular region, forward Euler and trapezoidal transcription over-switch in essentially all completed runs. Forward Euler has median $U_\tau = 210$, while trapezoidal transcription has median $U_\tau = 132$ and a 90th percentile above 300. Near the diagonal, where the activation term changes most rapidly, trapezoidal transcription becomes even more unstable structurally, with median $U_\tau = 226$. Direct collocation is the most robust baseline in this study: its severe-over-switch rate is at most 2% in all three initial-condition families. Multiple shooting has a similar median structure but fewer completed runs; across the three sweeps it completed 619/718 cases, compared with 2822/2872 for the other four methods combined. Even in these best cases, the median U_τ remains above the structural reference value 2, so the recovered controls are structurally close but not exact $1 \to S \to 1$ reconstructions.

Table 2: Initial-condition robustness at $N = 400$, computed over completed runs. Medians are followed by 90th percentiles in parentheses. P_{10} is the percentage of completed runs with $U_\tau > 10$.

Family	Method	P_{10}	U_τ	TV(u)
$n_1 > n_2$	FE	100.0	210 (244)	119.3 (136.1)
	BE	98.0	16 (40)	2.02 (2.85)
	TR	100.0	132 (325)	14.6 (81.3)
	MS	18.7	8 (13)	2.34 (3.33)
	DC	0.5	8 (9)	2.30 (3.09)
$n_1 \approx n_2$	FE	90.7	206 (243)	117.3 (135.7)
	BE	63.6	28 (41)	1.80 (3.10)
	TR	93.5	226 (337)	37.4 (129.1)
	MS	27.0	8 (17)	2.39 (3.59)
	DC	1.9	8 (10)	2.56 (3.23)
full grid	FE	72.0	177 (239)	104.0 (133.7)
	BE	90.1	25 (75)	2.05 (2.88)
	TR	72.0	88 (299)	8.16 (111.4)
	MS	14.5	7 (12)	2.18 (3.21)
	DC	2.0	8 (9)	2.16 (3.14)

DISCUSSION

The benchmark separates two notions of convergence. Objective convergence asks whether the computed cost approaches a reference value. Structural convergence asks whether the computed control has the same qualitative organisation as the reference. The forward Euler results show that these notions can diverge strongly: the objective gap decreases with mesh refinement, but the control becomes increasingly oscillatory.

The metrics have complementary roles. The objective gap is necessary but not sufficient. The singular-arc residual checks whether the state follows the expected reference manifold and clearly separates the forward-Euler state trajectory from the higher-quality state approximations. It still does not determine how the control generates that motion: trapezoidal transcription has a residual of the same small order as shooting and collocation, yet substantially more mesh-level switching. The switch count and total variation are therefore still needed to assess control structure.

The robustness sweeps are useful even without a separate indirect reference for every initial condition. They should not be used to report switching-time errors or objective gaps relative to unknown references. They can, however, reveal systematic discretisation artefacts. The triangular set tests the singular-capable regime, the near-diagonal set probes the transition where the smooth dominance penalty is most sensitive, and the full grid assesses structural artefacts across the initial-condition plane. This is why Table 2 reports structural complexity rather than reference errors.

The methods differ in characteristic ways. Forward Euler is the clearest chattering-prone method: it attains competitive objective values by creating rapidly switching controls. Backward Euler is more dissipative; its total variation is moderate, but its switch count remains elevated in many runs. The trapezoidal method is ac-

curate in objective value for the main trajectory, yet it is structurally fragile in the initial-condition sweeps, especially near the diagonal. Multiple shooting and direct collocation are the strongest direct baselines. They do not reproduce the two structural switches of the indirect solution exactly, but their switch counts and total variation remain close to the reference scale. Among them, collocation has the best completion and robustness profile in the present data.

The indirect method is not proposed as a universal replacement for direct transcription. It requires an assumed switching structure and an analytic singular feedback. Its role here is to provide a structural reference. In practical workflows, the most useful approach may be hybrid: a direct method identifies approximate transition regions, possibly using mesh or structure-detection ideas, and an indirect refinement then enforces the singular feedback and optimises switching times.

CONCLUSION

This paper presents a compact simulation benchmark for mesh sensitivity in singular optimal control. The benchmark uses a single-input population model with a known $1 \to S \to 1$ indirect reference and compares five standard direct transcriptions. The main finding is that small objective gaps can coexist with severe structural artefacts. Forward Euler is the strongest example: it approaches the reference objective as the mesh is refined, while its switch count and total variation grow rapidly. Trapezoidal transcription is objective-accurate but can over-switch across initial conditions. Multiple shooting and direct collocation are substantially more stable, with direct collocation giving the strongest overall baseline in the tested suite.

For short simulation studies involving singular arcs, we recommend reporting at least three quantities beyond the objective value: switch count, total variation, and a problem-specific singular-arc residual when available. This minimal diagnostic set is sufficient to detect the central failure mode: approximating the cost of a singular control without approximating its structure.

FUTURE WORK

A natural continuation is to generalise the structure-preserving switching-time framework beyond the scalar-control setting. The additional analytical and numerical issues arising in that setting are deliberately left outside the scope of the present single-input benchmark and will be treated separately. Further work will focus on automated structural diagnostics and on the effect of adaptive refinement on mesh-dependent switching.

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GRADIENT-BASED SWITCHING-TIME OPTIMIZATION FOR SINGULAR OPTIMAL CONTROL: LOCALITY, ADMISSIBILITY AND PMP DIAGNOSTICS

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Singular optimal control, switching-time optimisation, indirect methods, gradient descent, Adam, Armijo line search, Pontryagin Maximum Principle diagnostics

ABSTRACT

Switching-time optimisation is attractive for singular optimal-control problems when the control structure is known: the singular feedback can be imposed analytically and only the switching times are updated. This paper studies the optimisation layer of such a structure-preserving method for a single-input benchmark with a prescribed structure. We ask which update behaviours are reliable and which diagnostics are needed to distinguish numerically attractive but invalid candidates. A global benchmark with ten update rules shows that final objective value alone may give a false ranking: from remote starts some adaptive methods attain values below the reference value only by leaving the admissible singular-feedback domain and producing large residuals of the Pontryagin Maximum Principle (PMP). A separate local stress test removes this feasibility confound. The experiments support a diagnostic-gated workflow for this benchmark: preserve switching-time ordering and feedback admissibility, screen candidates by PMP residuals, and compare objective values only among diagnostically acceptable candidates. Near a valid reference, non-monotone updates may still be useful when these diagnostics remain satisfactory.

INTRODUCTION

Singular arcs remain a delicate part of numerical optimal control. They are classical in Pontryagin Maximum Principle (PMP) based control theory (Pontryagin et al., 1962; Bonnard and Chyba, 2003), but remain numerically demanding. In a control-affine system the Hamiltonian is linear in the control and bang-bang decisions are governed by the sign of the switching function. On a singular interval, however, the switching function and a number of its derivatives vanish, so the control must be recovered from higher-order optimality conditions. Consequently, a small discretised objective value is not by itself evidence of a valid singular candidate: the singular feedback may violate its bounds, Hamiltonian constancy may fail, or the switching function may have the wrong sign on nonsingular arcs.

The model considered here originates from the chemotherapy-resistance problem of Bajger et al. (2019). In that work the optimal structure contains a full-dose arc, a singular arc and another full-dose arc, and the switching times were updated by a gradient method with a fixed learning rate. The later structure-preserving switching-time algorithm of Bodzioch (2024) developed this idea into an indirect procedure: once the structure is fixed, the singular feedback is inserted analytically and the reduced objective is optimised only over the switching times. That approach makes the singular arc explicit, but leaves a numerical issue that is often hidden in direct transcriptions: which switching-time update behaviours are reliable, and which diagnostics are needed to decide whether a returned point is an acceptable singular-control candidate?

Recent numerical work addresses singular or nonsmooth optimal control through adaptive Radau collocation and structure identification (Pager and Rao, 2022a,b), mesh refinement for nonsmooth solutions (Miller et al., 2021), switched formulations (Andrés-Martínez and Ricardez-Sandoval, 2022), and finite elements with explicit switch detection (Nurkanović et al., 2024). Singular-specific alternatives also include continuation for known structures and uniform trigonometrisation of the control (Abadia and Rao, 2022; Mall et al., 2021). These approaches seek to expose or regularise difficult control structure before or during the direct solve. Our setting is complementary: the structure is assumed known, and we study only the reliability of the reduced switching-time optimisation layer.

The contribution is experimental rather than a new convergence theory. We test three issues: whether objective-only ranking can favour inadmissible remote solutions, whether local admissibility implies PMP consistency, and whether monotone objective decrease is necessary for successful local recovery. The results motivate a diagnostic-gated comparison in which admissibility and PMP checks precede ranking by the objective value. The aim is therefore not to identify a universally best optimiser, but to determine which numerical behaviours can be trusted for this prescribed sin-

gular structure and how returned candidates should be screened.

REDUCED SWITCHING-TIME PROBLEM

For the single-input benchmark, $u(t) \in [0,1]$, $n(0) = n_0$, and $n = (n_1, n_2)$ satisfies

$$\dot{n}_1 = \lambda_1 n_1 - n_1 u, \quad \dot{n}_2 = \lambda_2 n_2 (1 - n_1 - n_2),$$

and the objective is

$$J(u) = \omega_1 n_1(T) + \omega_2 n_2(T) + \int_0^T \Big(\eta_1 n_1(t) + \eta_2 n_2(t) + \theta u(t) + \tfrac{\xi}{2} \Big(1 + \tanh \tfrac{n_2(t) - n_1(t)}{\varepsilon} \Big) \Big) dt,$$

where the last term represents an activation penalty term. We use the parameters of Bodzioch (2024): $T = 14$, $n_0 = (0.44, 0.05)$, $\lambda_1 = 0.2$, $\lambda_2 = 0.1$, $\eta_1 = 2$, $\eta_2 = 3$, $\xi = 1$, $\varepsilon = 0.01$, $\theta = 0.5$, $\omega_1 = 5$ and $\omega_2 = 10$.
Writing the dynamics as $\dot{n} = f(n) + g(n)u$ and the integrand as $L(n,u)$, the Hamiltonian is $H(n,p,u) = p^\top (f(n) + g(n)u) + L(n,u)$, where p is the costate vector; the switching function is $\Phi = \partial H / \partial u$. The prescribed structure is $1 \to S \to 1$, where 1 denotes the upper-bound control $u = 1$ and S denotes the singular-feedback arc. Thus only two switching times are optimised: $\tau = (t_1, t_2)$, $0 < t_1 < t_2 < T$. For a fixed τ, the state is integrated forward with $u = 1$ on the outer arcs and with $u = u_S(n)$, the analytical singular feedback derived in Bodzioch (2024), on the middle arc. The reduced objective is denoted by $J(\tau)$. In the present paper the gradient is computed only with respect to the two switching-time variables,

$$\frac{\partial J}{\partial t_i}(\tau) \approx \frac{J(t_1, \ldots, t_i + \Delta, \ldots, t_2) - J(\tau)}{\Delta}, \qquad i = 1, 2,$$

where $\Delta > 0$ is the finite-difference increment. The notation extends naturally to more switches, but no such extension is used in the experiments. The fixed-step update has the form

$$\tau^{m+1} = \Pi \left(\tau^m - \alpha \nabla J(\tau^m) \right),$$

where $\alpha > 0$ is the learning rate and Π denotes the ordering safeguard for $0 < t_1 < t_2 < T$. Other tested methods replace this elementary step by a line-search, momentum, adaptive-scaling or limited-memory quasi-Newton update.
We denote the reference switching-time vector by

$$\tau^* = (t_1^*, t_2^*) = (2.082205, 13.867394),$$

and the corresponding objective value by

$$J_{\text{ref}} = J(\tau^*) = 11.90109731.$$

This admissible solution is used as a numerical reference for the prescribed $1 \to S \to 1$ structure, not as an independent proof of global optimality; PMP consistency is assessed by the residual criteria introduced below.

DIAGNOSTICS AND EXPERIMENTS

Let $u_{\min}$ and $u_{\max}$ be the minimum and maximum sampled singular-feedback values and let v_u be the integrated box-constraint violation. We define the scalar control excess

$$e_u = \max\{0,\, u_{\max} - 1,\, -u_{\min},\, v_u\}.$$

A run is called control-feasible if $e_u \le 10^{-6}$. The Hamiltonian residual is

$$r_H = \max_t |H(t) - \bar{H}|, \qquad \bar{H} = \frac{1}{T} \int_0^T H(t)\, dt,$$

which measures the deviation from Hamiltonian constancy for the fixed-terminal-time autonomous problem. The switching-function residual is structure-aware:

$$r_\Phi = \frac{1}{T} \int_0^T \begin{cases} \max\{0, \Phi(t)\}, & t < t_1, \\ |\Phi(t)|, & t_1 \le t \le t_2, \\ \max\{0, \Phi(t)\}, & t > t_2, \end{cases} dt.$$

Under the minimisation convention used here, the outer arcs with $u = 1$ require $\Phi \le 0$; hence r_Φ penalises a wrong sign on those arcs and a nonzero switching function on the singular arc. A run is called PMP-good in this numerical study if $r_H < 10^{-2}$ and $r_\Phi < 10^{-2}$. These tolerances are screening thresholds for the present benchmark, not general sufficient optimality conditions. To measure non-monotonicity of the reduced objective, we count accepted iterations with $J_{m+1} > J_m$, where m is the accepted-iteration index. This simple diagnostic is sufficient here because all fixed-step and Armijo local runs are monotone, whereas Adam accepts many small local increases even when the overall trace decreases.
The global benchmark compares ten update rules: fixed, diminishing and projected gradient descent (GD), Armijo backtracking, momentum, adaptive moment estimation (Adam), RMSProp (root mean square propagation), AdaGrad (adaptive gradient), limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) after switching-interval reparametrisation, and an Adam-L-BFGS hybrid (Armijo, 1966; Kingma and Ba, 2015). Five global starts s_i are used:

$$s_1 = (2.082205, 13.867394), \quad s_2 = (1.5, 13),$$
$$s_3 = (0.75, 8), \quad s_4 = (3.5, 12), \quad s_5 = (5, 13.5).$$

The local stress test uses eight starts ℓ_i near τ^*, with $\ell_1 = \tau^*$, and compares fixed-step GD with $\alpha \in \{8 \times 10^{-5}, 1.5 \times 10^{-4}, 3 \times 10^{-4}\}$ against Armijo and Adam. The underlying optimal-control problem is unchanged.

RESULTS

Global Benchmark: Locality and False Objective Improvements

Figure 1 gives the first, objective-only view. All methods recover the same value from s_1. From the remaining

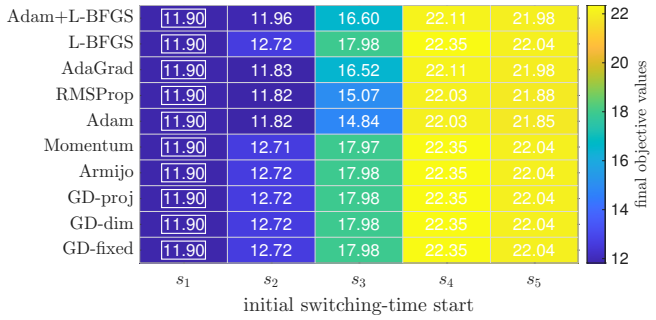

Figure 1: Global benchmark: final objective values for ten update rules and five initial switching-time starts. Black frames mark runs that are both control-feasible and PMP-good.

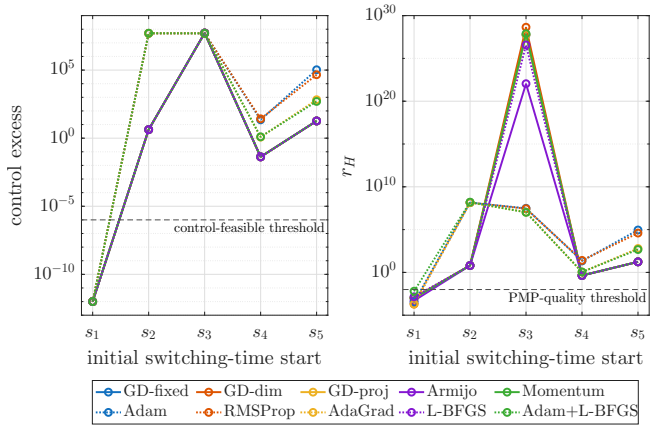

Figure 2: Global benchmark: diagnostics by start. Left: control excess e_u. Right: Hamiltonian residual r_H. Dashed lines mark the control-feasibility threshold (left) and the r_H threshold used in the PMP-good criterion (right).

starts, however, the final objective depends strongly on the initial switching times. In particular, Adam, RMSProp and AdaGrad give objective values below J_{ref} from s_2. If J were the only reported metric, these runs would appear to improve the reference solution.

The diagnostic view in Figure 2 explains why these apparent improvements are false. Except for the reference start s_1, the runs violate either the control-feasibility threshold, the Hamiltonian-residual threshold, or both. The three objective-improving remote runs are listed in Table 1. Their objective values are lower by about 7–8×10^{-2}, but this is accompanied by $u_{\max} \approx 5 \times 10^{7}$ and $r_H \approx 10^{8}$. Hence they are not admissible singular-control candidates.

Local Stress Test: Recovery and Residuals Across Variants

The local stress test asks a different question. We start close to the reference solution and remove the severe

Table 1: False objective improvements from the global benchmark. All three runs have $J < J_{\mathrm{ref}}$ but fail both admissibility and PMP diagnostics.

Method	Start	$J - J_{\mathrm{ref}}$	e_u	r_H
Adam	s_2	-7.96×10^{-2}	5×10^{7}	1.43×10^{8}
RMSProp	s_2	-7.96×10^{-2}	5×10^{7}	1.44×10^{8}
AdaGrad	s_2	-7.12×10^{-2}	5×10^{7}	1.42×10^{8}

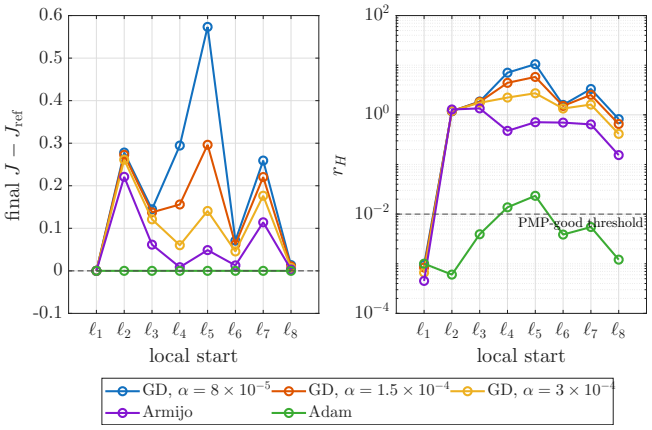

Figure 3: Local stress test for all tested local variants. Left: final objective offset relative to J_{ref}. Right: Hamiltonian residual r_H. The dashed line in the right panel is the r_H threshold used in the PMP-good criterion.

global feasibility confound: all 40 local runs are control-feasible. The remaining issue is whether the method returns to the reference objective while satisfying the PMP screening criteria.

Figure 3 compares the local variants. Increasing the fixed learning rate improves some objective offsets, but the Hamiltonian residual is more selective. Table 2 shows that Adam meets the defined recovery criterion for six of eight starts, whereas the fixed-step variants and Armijo meet it only at the reference start. Thus even with admissible controls, J must be read together with PMP residuals.

Table 2 summarises the local test. Larger fixed steps improve some residuals but not the recovery count; Armijo remains monotone and local, whereas Adam recovers most perturbations and accepts frequent objective increases.

The nonzero "Inc." value for Adam in Table 2 shows that successful local recovery can coexist with a non-monotone objective trace; no causal claim is implied.

DISCUSSION

The experiments separate two failure modes: remote objective descent can produce inadmissible, PMP-poor candidates, while near the reference all controls may re-

Table 2: Local stress test. Recovery means control-feasible, PMP-good and $|J - J_{\text{ref}}| < 10^{-4}$. "Inc." is the mean number of accepted objective increases.

Variant	Recovered	Mean r_H	Inc.
GD, $8{\cdot}10^{-5}$	1/8	3.29	0.0
GD, $1.5{\cdot}10^{-4}$	1/8	2.24	0.0
GD, $3{\cdot}10^{-4}$	1/8	1.41	0.0
Armijo	1/8	0.664	0.0
Adam	6/8	0.00665	42.9

main admissible but PMP residuals still distinguish the updates. Neither low J nor monotone decrease is therefore sufficient on its own.

The practical protocol suggested by this benchmark is: preserve $0 < t_1 < t_2 < T$; exclude candidates with singular-feedback bound violations; screen the remaining candidates with r_H and r_Φ; and only then compare objective values and trace behaviour. Near a valid candidate, a non-monotone update may be retained when these diagnostics remain satisfactory. Wall time is secondary: faster refinement matters only after admissibility and PMP checks have been passed. This is an empirical recommendation, not a convergence theorem.

CONCLUSIONS

For the prescribed $1 \to S \to 1$ benchmark, objective-only switching-time optimisation is unreliable from remote starts, and local admissibility alone does not guarantee PMP consistency. Fixed-step and Armijo updates recover only the reference local start, whereas Adam recovers six of eight perturbations while accepting small objective increases. The reduced problem should therefore be handled diagnostically: enforce switching-time ordering and feedback admissibility, screen candidates with r_H and r_Φ, and compare J only within the accepted set.

FUTURE WORK

The next step is to incorporate admissibility and PMP residuals directly into the switching-time update as acceptance filters, and to combine such local refinement with structure estimates supplied by direct or collocation methods (Pager and Rao, 2022a; Andrés-Martínez and Ricardez-Sandoval, 2022; Nurkanović et al., 2024).

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Robust Set-Based Design by Monte-Carlo Simulation and Decision Tree

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Computer Aided Engineering (CAE), Robust design, Set-Based Design, Sampling, Decision tree.

ABSTRACT

Robust Set-Based Design aims at identifying regions of the design space that satisfy functional requirements under uncertainty. This paper proposes a simulation-driven exploration strategy combining Monte Carlo simulation and supervised machine learning. Candidate solutions are generated through random sampling of the design characteristics and evaluated under multiple realizations of external and modelling uncertainties. Each solution is then classified as infeasible, conditionally feasible, or robust according to its compliance with the required properties. A decision-tree classifier is subsequently trained to learn the structure of the design space and extract explicit design rules defining robust solution regions. The method is applied to the preliminary sizing of an external gear pump. Results show that the proposed approach efficiently identifies robust design regions while providing an interpretable representation of the design space. Unlike QCSP-based methods, the proposed framework can directly integrate simulation models and black-box behavioural relations, making it particularly suitable for complex engineering design problems.

INTRODUCTION

The increasing complexity of modern engineered systems, combined with growing requirements related to performance, sustainability, cost, reliability, and lifecycle adaptability, challenges the effectiveness of traditional design approaches. Conventional point-based design typically relies on the early selection of a single solution concept, followed by iterative refinements. While this strategy may reduce the apparent design space, it often leads to premature decisions, costly redesign loops, and limited exploration of alternative solutions .

To overcome these limitations, Set-Based Design (SBD) was originally developed and popularized through Toyota's product development system. Rather than committing early to a single design alternative, SBD promotes the simultaneous exploration of multiple feasible solution sets and progressively narrows these sets as knowledge is acquired and constraints become better understood (Ward et al., 1995; Sobek et al., 1999).

The fundamental principle of SBD is to delay irreversible decisions until sufficient evidence is available to eliminate infeasible or suboptimal alternatives (Al-Ashaab et al, 2021).

Despite its recognized benefits, the practical implementation of SBD remains challenging. Exploring large sets of candidate solutions significantly increases the computational effort required for performance evaluation and feasibility assessment. As system complexity grows, the size of the design space expands exponentially, making exhaustive evaluation impractical (Toche et al, 2020). Furthermore, uncertainties related to manufacturing variability, operating conditions, and environmental factors must be incorporated to ensure robust decision-making. Consequently, efficient methods capable of evaluating large populations of design alternatives and rapidly identifying promising solution regions are required.

Beyond design exploration, another major challenge concerns the management of uncertainty (Dantan et al, 2021). Engineering systems are continuously exposed to variability originating from manufacturing tolerances, material properties, assembly deviations, environmental conditions, operational disturbances, and degradation phenomena. A design solution that satisfies requirements under nominal conditions may become unacceptable when such variations occur. Robust Design addresses this challenge by seeking solutions whose performances remain acceptable despite the presence of uncertainty (Bras et al, 1995) . Originating from the work of Taguchi (Taguchi, 1987), Robust Design aims to minimize performance variability while maintaining compliance with functional requirements (Chen et al, 1996), (Verma et al, 2022). Consequently, robustness has become a fundamental criterion in modern product development, particularly in safety-critical and high-value engineering applications where design failures can generate substantial economic and operational consequences.

The combination of Set-Based Design and Robust Design represents a natural evolution of system design methodologies. While SBD explores multiple candidate solutions, Robust Design evaluates their sensitivity to uncertainty. Their integration therefore enables the identification of design regions that are not only feasible but also resilient to variability (Bird et al, 2023). Instead of converging toward a single optimal point, the objective becomes the identification of robust design sets that maintain acceptable performance over a range of operating and manufacturing conditions. Such an approach significantly reduces redesign activities, facilitates decision-making under uncertainty, and improves product reliability throughout the lifecycle. However, the simultaneous exploration of large design spaces and uncertainty domains presents important computational challenges, especially when nonlinear constraints and multiple interacting variables are involved (Shallcross et al, 2022).

To overcome these limitations, (Qureshi et al, 2010 and 2014) proposed a robust Set-Based Design method based on the

Quantified Constraint Satisfaction Problem (QCSP) framework (Gent et al, 2005). In this approach, design variables are represented through existential quantifiers, whereas uncertainty sources are modeled using universal quantifiers. This formulation allows robustness requirements to be explicitly incorporated into the design model by verifying that all admissible uncertainty realizations satisfy the design constraints. Combined with interval arithmetic and domain-splitting mechanisms, the QCSP framework systematically explores the design space and progressively identifies robust solution regions while eliminating infeasible areas. The resulting method provides a rigorous mathematical foundation for robust Set-Based Design and has demonstrated its capability on several mechanical design problems involving dimensional, material, and manufacturing uncertainties.

Despite its theoretical rigor, the QCSP-based approach may suffer from scalability issues when dealing with high-dimensional design spaces, strongly nonlinear models, or computationally expensive simulations. The repeated subdivision of domains and robustness verification procedures can rapidly generate a combinatorial explosion in the number of subdomains to evaluate. Consequently, the industrial adoption of QCSP frameworks remains limited for complex engineering systems requiring thousands or millions of evaluations.

To address these limitations, this paper proposes an alternative Robust Set-Based Design method that combines Monte Carlo Simulation (MCS) and Supervised Machine Learning. Monte Carlo simulations are first employed to generate a large population of design alternatives and evaluate their robustness under uncertainty through stochastic sampling. The generated dataset is then used to train supervised learning models capable of approximating the feasibility and robustness boundaries within the design space. In particular, Decision Trees provide an interpretable representation of design rules, enabling the identification of robust design regions and the extraction of explicit engineering knowledge. By replacing exhaustive domain exploration with data-driven learning, the proposed approach significantly reduces computational effort while preserving the ability to identify robust solution sets. The resulting framework combines the exploratory power of Set-Based Design, the uncertainty management capabilities of Robust Design, and the predictive efficiency of machine learning, thereby offering a scalable alternative for robust engineering design under uncertainty.

MATHEMATICAL FORMULATION OF ROBUST SET-BASED DESIGN

To support the identification of robust design regions under uncertainty, the proposed framework relies on two complementary mathematical representations. The first representation formulates product design as a parametric design problem, where a set of design characteristics is searched within a multidimensional design space while satisfying performance requirements. The second representation employs First-Order Logic (FOL) to formally describe the interactions between design variables, uncertainty sources, behavior relations, and required properties. This logical formalization provides a rigorous foundation for robust Set-Based Design by enabling the explicit representation of uncertainty and the definition of robust solution sets. The framework derives from the integration of the Characteristics-Properties Model (CPM) and First-Order Logic developed by (Dantan et al, 2013).

Parametric Formulation of a Design Problem

The design problem is represented using the Characteristics-Properties Model (CPM), where the product is described through five fundamental entities:

$$M=\{C,P,R,D,EC,MC,RP\} \qquad (1)$$

where:

- $C=\{C_1,C_2,\ldots,C_n\}$ denotes the set of characteristics (design variables),
- $P=\{P_1,P_2,\ldots,P_m\}$ denotes the set of properties (performance indicators),
- RP denotes the set of required properties,
- D denotes the design space,
- R denotes the set of relations linking characteristics and properties,
- EC denotes the set of external conditions,
- MC denotes the set of modeling conditions.

The objective of design is to determine values of the characteristics C such that the resulting properties satisfy the required properties RP (Malmiry et al, 2016).

The design space is defined as the Cartesian product of the admissible domains of all design variables: $D=D_1\times D_2\times\cdots\times D_n$ with $C_i \in D_i$ where D_i is the admissible interval or domain associated with characteristic C_i.

The product properties result from physical, behavioral or empirical relations: $P=R(C,EC,MC)$. More generally, $P_j=f_j(C,EC,MC)$ for $j=1,\ldots,m$ where f_j represents a physical model, surrogate model, simulation model, empirical law or analytical equation.

Set-Based Design Formulation

A candidate design solution is represented as s, $s=(C'_1,C'_2,\ldots,C'_n)$ such that $s \in D$. C'_1 : *a value of* C_1.

A design solution is considered feasible if all required properties are satisfied: $P_j \in RP_j \ \forall j=1,\ldots,m$ or equivalently $g_k(C,EC,MC)\leq 0$ for all performance constraints g_k.

Instead of searching for a single solution point, Set-Based Design searches for a family of admissible solutions: S. Each solution is represented as a subset (box, interval region, hypervolume) of the design space. The objective becomes the identification of robust regions rather than isolated optimal points.

The feasible design set is therefore:

$$S=\{s\in D \mid R(C,P,RP,EC,MC)\} \qquad (2)$$

First-Order Logic Formulation

To formally characterize robustness in a Set-Based Design context, a mathematical language capable of explicitly representing design variables, uncertainty sources, behavioral relations, and performance requirements is required. First-Order Logic (FOL) provides such a framework through the

use of predicates and quantifiers, allowing the rigorous expression of feasibility and robustness conditions. Within the proposed Characteristics–Properties Model (CPM), FOL enables the representation of the relationships linking product characteristics, generated properties, external conditions, and required properties. The resulting formulation transforms the design problem into a quantified reasoning problem, where existential quantifiers are used to identify admissible design solutions and universal quantifiers are employed to guarantee the satisfaction of requirements over entire ranges of variability and uncertainty. Consequently, the robust Set-Based Design problem can be formulated as the search for subsets of the design space that remain feasible for all admissible realizations of external and modeling uncertainties, thereby providing a formal basis for QCSP-based exploration.

Existence of a Feasible Design.
The existence of at least one feasible solution is expressed by the existential quantifier: $\exists si \mid si \subseteq D$ such that $si \vDash R(C,P,RP)$ meaning that at least one assignment of design characteristics satisfies all requirements.

First Robust Design Formulation
The article proposes a first Robust Design as follows:

$$S=\{\cup si \in S \mid si \vDash \forall C \ \forall EC \ R(C,P,RP,EC)\} \quad (3)$$

This formulation states that: *There exists a subset of the design space such that for all admissible values of the characteristics and for all admissible external conditions, the required properties remain satisfied.* The use of universal quantifiers guarantees robustness over the entire domain of variation.

General Robust Design Formulation
The robust solution set can be generalized as:

$$S_R=\{s \subseteq D \mid \forall EC \ \forall MC \ R(C,P,RP,EC,MC)\} \quad (4)$$

which means that every realization of the uncertainty variables preserves the required behavior.
The objective is to identify all subsets of the design space whose associated performances satisfy the required properties despite the presence of variability, uncertainty and modeling imperfections. This formulation provides the theoretical basis for QCSP-based exploration methods.

Interpretation for Set-Based Design
The proposed formalization provides a unified representation of quantitative design problems by integrating design variables, performance requirements, physical relations, and uncertainty sources within a single mathematical framework. The design process is expressed as a structured mapping between characteristics, representing controllable design decisions, and properties, representing the resulting system performances. This representation establishes a rigorous foundation for describing design feasibility and evaluating the influence of uncertainty on product behavior.
The introduction of First-Order Logic significantly extends the expressive power of the formulation. Unlike conventional optimization approaches that focus on identifying a single feasible or optimal solution, the logical framework enables the explicit representation of existence, admissibility, and robustness using quantified expressions. Existential quantifiers allow the search for admissible design solutions within the design space, while universal quantifiers ensure that these solutions remain valid over the complete range of variability associated with design parameters, environmental conditions, and modeling assumptions. Consequently, uncertainty is not treated as a post-design verification problem but becomes an intrinsic element of the design formulation itself.
An important outcome of this formalization is the natural transition from point-based design to Set-Based Design. Instead of determining a unique design configuration, the objective becomes the identification of subsets of the design space that satisfy the required properties. These subsets represent feasible design regions that preserve flexibility during the decision-making process. When uncertainty is incorporated through universal quantification, the retained subsets correspond to robust design regions, i.e., regions in which all admissible solutions satisfy the required performances despite the presence of variability and incomplete knowledge. This interpretation is fully consistent with the principles of Set-Based Concurrent Engineering, where knowledge acquisition progressively reduces the design space until only robust and feasible solution regions remain.
The logical formulation directly supports Set-Based Design:

- classical design: $\exists C$ searches a single design point;
- set-based design: $\exists S \subseteq D$ searches a solution region;
- robust set-based design: $\exists S \subseteq D$ s.t. $\forall EC, \forall MC$ all requirements remain satisfied.

From a computational perspective, the resulting formulation can be viewed as a Quantified Constraint Satisfaction Problem (QCSP), where robustness verification consists of evaluating whether a candidate design set satisfies the required properties for all admissible uncertainty realizations. The QCSP framework provides a rigorous technique for exploring the design space.

PROPOSED EXPLORATION TECHNIQUES

QCSP-Based Design Space Exploration

The Quantified Constraint Satisfaction Problem (QCSP) approach provides a rigorous mechanism for exploring design spaces under uncertainty by extending classical Constraint Satisfaction Problems (CSP) with existential and universal quantifiers. In this framework, design variables are represented by existentially quantified variables, whereas uncertainty sources such as manufacturing variations, material variability, operating conditions, or modeling imperfections are represented by universally quantified variables. The design problem is formulated as a set of characteristics, domains, and constraints, and the objective is to identify subsets of the design space that satisfy all performance requirements despite uncertainty.
To perform these evaluations efficiently, the QCSP framework employs interval arithmetic, which enables the propagation of variable intervals through the design equations without requiring exhaustive point-by-point evaluations. This interval-based reasoning allows large infeasible regions to be eliminated early while progressively refining promising areas of the design space. The final outcome is a classification of

the design space into infeasible regions, feasible-but-non-robust regions, and robust solution regions. Although mathematically rigorous and capable of explicitly handling uncertainty, the computational cost of QCSP increases rapidly with the number of variables and constraints, motivating the development of alternative exploration methods based on stochastic simulation and machine learning.

Proposed Technique based on Monte Carlo Simulation and Decision Tree

To overcome QCSP limitations, an alternative exploration strategy is proposed. The approach combines Monte Carlo simulation and supervised machine learning to approximate the robust design space while preserving the fundamental principles of Robust Set-Based Design presented in previous section.

Unlike QCSP, which performs a systematic exploration of the design space through quantified reasoning and domain subdivision, the proposed method transforms the robust design problem into a supervised classification problem. The objective is to learn the boundaries separating infeasible, feasible, and robust regions directly from simulation data (Homri et al, 2023), (Dantan et al, 2026). The resulting decision tree provides an explicit and interpretable representation of the robust design space in the form of a set of decision rules that can be directly translated into candidate design regions.

The algorithme is organized into four successive stages: (i) generation of candidate design solutions through random sampling of the design characteristics, (ii) generation of uncertainty scenarios by sampling the external and modeling conditions, (iii) simulation-based evaluation and robustness labelling of each candidate solution, and (iv) learning and extraction of robust design regions using a decision tree classifier. The figure 1 summarize the algorithm.

Step 1: Generation of Candidate Design Solutions

The first stage consists of generating a large set of candidate design solutions through random sampling of the design characteristics: $C=(C_1,C_2,\dots,C_n)$ from their admissible domains: $C_i \in D_i$

A Monte Carlo sampling procedure is applied to create a representative population: $S=\{s_1,s_2,\dots,s_N\}$ where each candidate solution corresponds to a realization of the design characteristics: $s_i=(c_{i,1},c_{i,2},\dots,c_{i,n})$

The generated population constitutes an initial approximation of the design space. Since no assumptions are made regarding the existence or location of feasible solutions, the exploration remains unbiased and suitable for highly nonlinear design problems.

Step 2: Sampling of External and Modeling Conditions

For each candidate design solution sis_isi, a secondary Monte Carlo sampling is performed on the uncertainty space.

The uncertainty space is composed of: $EC=\{EC_1,\dots,EC_r\}$ representing external conditions and $MC=\{MC_1,\dots,MC_q\}$ representing modeling conditions.

A relatively small number M of uncertainty realizations is generated: $U_i=\{u_{i,1},u_{i,2},\dots,u_{i,M}\}$ where $u_{i,j}=(EC_{i,j},MC_{i,j})$.

This secondary sampling aims to approximate the variability associated with environmental conditions, manufacturing variability, user behavior, model uncertainty, or simplification errors without requiring an exhaustive robustness verification.

Step 3: Simulation-Based Evaluation and Labelling

Each pair composed of a candidate solution and an uncertainty realization is evaluated through the set of behavioral relations: $P=R(C,EC,MC)$. The computed properties are then compared with the required properties RP.

For a given candidate solution s_i, the compliance function is defined as: $y_{ij}=1$ if RP is satisfied or 0 otherwise for every uncertainty realization u_{ij}. The set $Y_i=\{y_{i,1},y_{i,2},\dots,y_{i,M}\}$ summarizes the behavior of solution s_i under uncertainty.

The final label associated with sis_isi is determined according to the following robustness criteria:

– Class 0: Infeasible Solution

The required properties are never satisfied: $\forall j,\ y_{ij}=0$

which leads to: $Label(si)=0$. This class corresponds to infeasible regions of the design space.

– Class 1: Conditionally Feasible Solution

The required properties are satisfied for at least one uncertainty realization but not for all: $\exists j,\ y_{ij}=1$ and $\exists k,\ y_{ik}=0$ leading to: $Label(si)=1$. These solutions correspond to feasible but non-robust regions.

– Class 2: Robust Solution

The required properties are satisfied for all sampled uncertainty realizations: $\forall j,\ y_{ij}=1$ leading to: $Label(si)=2$

This class corresponds to robust solution sets consistent with the formal robust design definition introduced in previous section.

Step 4: Decision Tree Learning and Extraction of Robust Regions

The labelled dataset is then used to train a decision tree classifier. The objective of the classifier is to learn the mapping: $f: C \rightarrow Label$ where the input variables are the design characteristics and the output is the robustness label.

During training, the decision tree recursively partitions the design space by identifying characteristic thresholds maximizing class discrimination. Each path from the root node to a terminal leaf defines a conjunction of constraints:

$$\begin{aligned} &C_1 \in [a_1,b_1] \\ &C_2 \in [a_2,b_2] \\ &\vdots \\ &C_n \in [a_n,b_n] \end{aligned} \qquad (5)$$

These constraints define a multidimensional hyper-rectangle (or design box) within the original design space. The extraction of all such branches provides a collection of robust design sets: $B_R=\{B_1,B_2,\dots,B_K\}$ which approximate the robust solution space originally sought through QCSP.

The proposed exploration framework combines the formal rigor of Robust Set-Based Design with the computational efficiency of data-driven techniques. The three-level labelling strategy further distinguishes infeasible, conditionally feasible, and robust solutions, providing a richer characterization of the design space than traditional binary classifications. The extracted decision rules constitute interpretable design knowledge that can be directly exploited during engineering decision-making and design-space reduction.

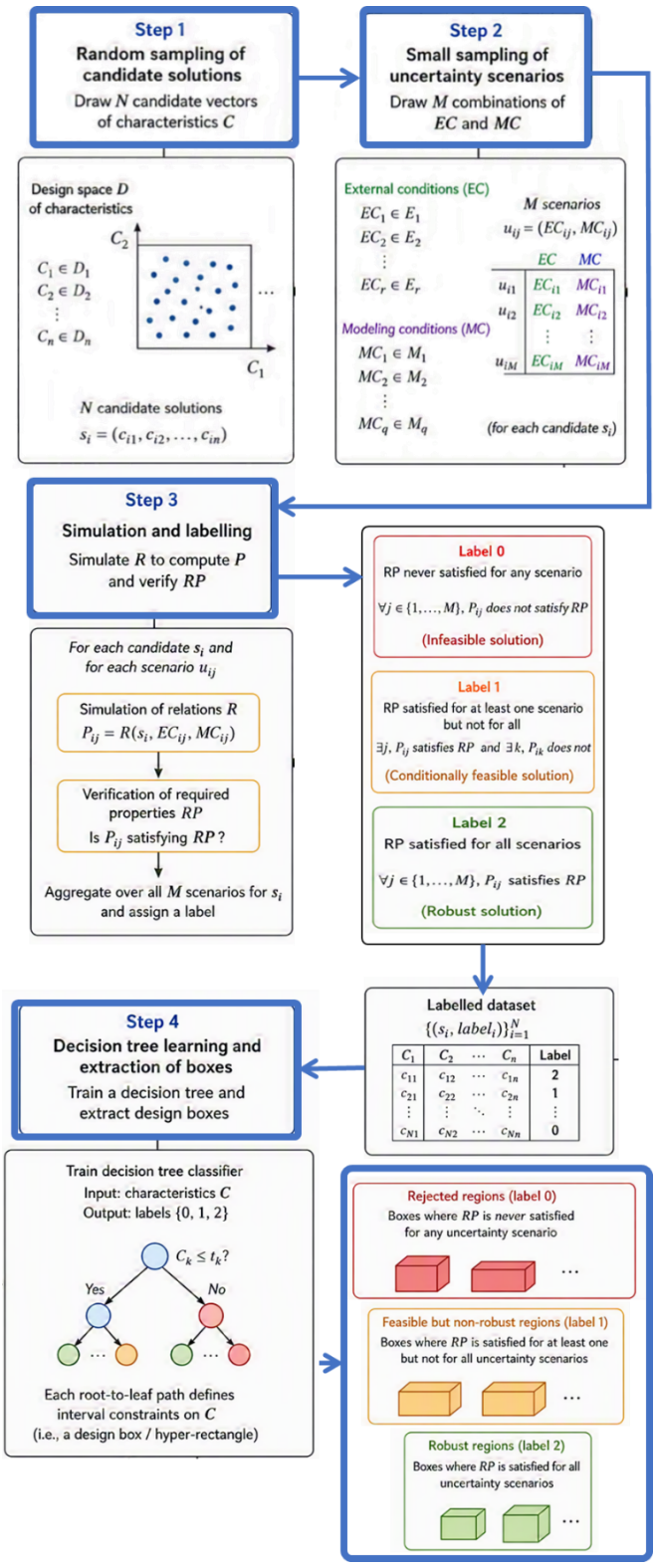

Figure 1: Algorithm.

To assess the effectiveness of the proposed method, the next section applies the framework to the preliminary sizing of an external gear pump. The objective is to identify robust regions of the design space defined by the pump characteristics while accounting for uncertainties associated with operating conditions and behavioral models. The resulting robust design sets are then compared with those obtained using the QCSP-based exploration approach in order to evaluate the capability of the proposed Monte Carlo and decision-tree framework to efficiently approximate the robust solution space.

APPLICATION: PRELIMINARY SIZING OF AN EXTERNAL GEAR PUMP

Case Study Overview

The proposed exploration methodology is applied to the preliminary sizing of an external gear pump, using the case study introduced in (Dantan et al, 2013). The objective of the case study is to identify admissible and robust regions of the design space for selecting a pump configuration able to satisfy functional requirements under several sources of uncertainty. More specifically, the design task consists in determining suitable values of the main pump characteristics, such as displacement and drive speed, so that the resulting delivery rate and required power remain compatible with the expected performance specifications.

The external gear pump is characterized by manufacturer performance curves involving the displacement V, the delivery rate Q, the pressure difference p, the drive torque M, the drive speed nnn, and the required power P. The behavioral model also involves volumetric efficiency η_v and hydraulic-mechanical efficiency η_{hm}, which introduce additional uncertainty into the relation between design characteristics and product properties.

The main design objective is therefore to identify robust configurations satisfying two required properties:

$$Q \geq 23 \text{ l/min} \qquad (6)$$
$$P \leq 5 \text{ kW}$$

where Q denotes the pump delivery rate and P denotes the required drive power. The design characteristics considered in the case study include the pump displacement V, defined in the interval [8,16] cm³/rev, and the drive speed nnn, defined in the interval [1500,3000] rpm.

CPM Representation of the Gear Pump

The case study is formulated using the Characteristics–Properties Model, which distinguishes between product characteristics directly controlled by the designer and product properties resulting from physical or empirical relations. In this framework, the designer acts on the characteristics, while the properties are indirectly obtained through the behavioral relations of the system.

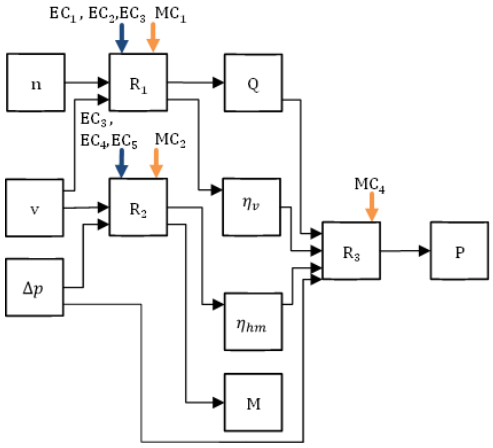

Figure 2: CPM Diagram of an External Gear Pump.

For the external gear pump, the main characteristics are defined as: C = {V,n,p} where:

– V is the pump displacement;
– n is the drive speed;
– p is the pressure difference.

The main properties are: P={Q,M,P,η_v,η_{hm}} where:

– Q is the delivery rate;
– M is the drive torque;
– P is the required power;
– η_v is the volumetric efficiency;
– η_{hm} is the hydraulic-mechanical efficiency.

The behavioral relations R link the characteristics to the properties through analytical, empirical, or manufacturer-based models. A simplified example of such a relation is:

$$Q=V\cdot n\cdot\eta_v\cdot 10^{-5} \quad (7)$$

This relation shows that the delivery rate depends not only on controllable characteristics, such as displacement and speed, but also on efficiency terms affected by uncertainty.
The case study also includes several external and modelling conditions. The external conditions account for uncertainty sources such as drive speed fluctuation, displacement variation, leakage from joints, power transmission loss, and back-pressure fluctuation. The modelling conditions represent uncertainty due to metamodel errors or simplifying assumptions related to efficiency models.

QCSP-based Exploration Results

The QCSP-based exploration was applied to the preliminary sizing problem of the external gear pump in order to identify feasible and robust regions of the design space under uncertainty. Following the principles of Quantified Constraint Satisfaction Problems, the design space defined by the pump displacement V and the rotational speed nnn was partitioned into elementary boxes. Each box was then evaluated through interval reasoning by considering the admissible variations of the external conditions EC and modelling conditions MC. The resulting classification distinguishes three categories of regions: infeasible regions (Label 0), uncertain or partially feasible regions (Label 1), and robust regions (Label 2).

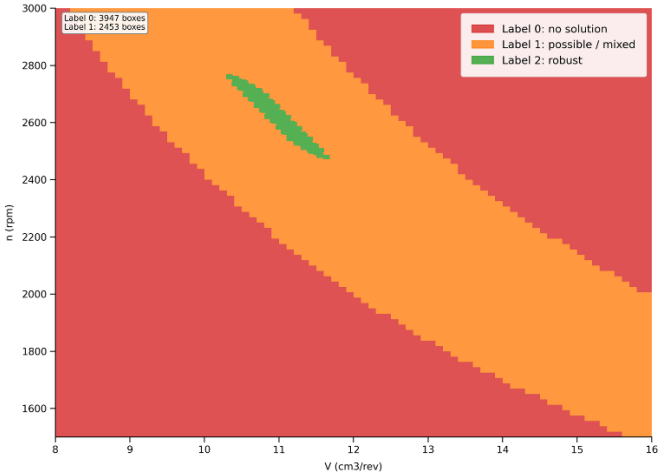

Figure 3: General Robust Design Results.

The QCSP exploration was performed using a discretization of 80×80 design boxes. The corresponding map reveals a clear separation between infeasible regions and potentially feasible regions. Large portions of the design space are classified as infeasible because the required flow rate cannot be achieved or because the required power exceeds the admissible limit under at least one uncertainty realization.
A finer exploration was subsequently performed using a resolution of 200×200, corresponding to 40,000 elementary design boxes. This increased resolution significantly improves the description of the frontier separating feasible and infeasible regions.
The results indicate that 25,136 boxes are classified as infeasible, 14,421 boxes are classified as uncertain or partially feasible, and 423 boxes are directly classified as robust at this stage of the interval analysis. This result highlights the conservative nature of QCSP reasoning when large uncertainty intervals are considered. Indeed, robust classification requires that all possible realizations of uncertainty satisfy the required properties simultaneously. Consequently, many regions that appear satisfactory under nominal conditions are transferred to the uncertain category because at least one extreme uncertainty configuration violates a requirement.
From an engineering perspective, the QCSP map provides valuable information regarding the structure of the design space. The red region identifies configurations that can be definitively excluded from further investigation, while the orange region corresponds to candidate areas requiring additional subdivision or numerical verification.

Application of the Proposed Monte Carlo and Decision-Tree Approach

The proposed method was applied to the same external gear pump sizing problem using the four-step workflow presented in Section 3. First, 10,000 candidate solutions were randomly generated within the design space defined by the displacement V∈[8,16] cm³/rev and the rotational speed n∈[1500,3000] rpm. For each candidate solution, a set of uncertainty scenarios was generated by sampling the external conditions and modelling conditions. The behavioral relations were then simulated to evaluate the resulting flow rate and required power, and each candidate solution was assigned a robustness label according to the proposed three-level classification strategy.
The resulting labelled design space is shown in the two-dimensional representation of the sampled solutions. The distribution of labels reveals a clear structure within the design space, with large infeasible regions, intermediate conditional regions, and a smaller set of robust solutions. The classification obtained from the Monte Carlo simulations indicates that 7,207 candidate solutions were assigned to Label 0 (infeasible), 2,456 solutions were assigned to Label 1 (conditionally feasible), and 337 solutions were assigned to Label 2 (robust). These results confirm that only a limited portion of the initial design space satisfies all performance requirements under all sampled uncertainty realizations.
The labelled design space demonstrates that robustness is not uniformly distributed. Robust solutions mainly appear in a restricted region corresponding to larger pump displacements combined with moderate rotational speeds. This behaviour is physically consistent with the pump model. Increasing the

displacement improves the achievable flow rate, while excessively increasing the rotational speed may lead to higher power consumption and therefore violate the power constraint. Consequently, only a specific combination of displacement and speed simultaneously satisfies both the flow-rate and power requirements under uncertainty.

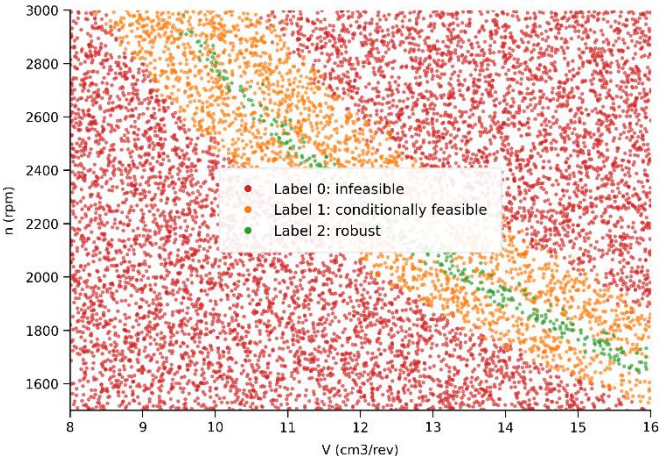

Figure 4: Results of Steps 1,2 and 3

The labelled dataset was subsequently used to train a decision-tree classifier whose objective was to learn the relationship between the design characteristics (V,n) and the robustness labels. The complete decision tree contains 865 nodes, 433 leaves, and reaches a maximum depth of 31. The resulting classifier achieves an accuracy of approximately 94.0 % on the test dataset, demonstrating its ability to reproduce the robustness classification generated by Monte Carlo simulation.

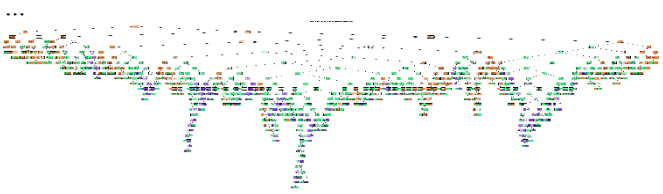

Figure 5: Decision Tree of Steps 4

A major advantage of the proposed approach is the interpretability of the learned model. Unlike black-box surrogate models, the decision tree explicitly identifies the decision boundaries separating infeasible, feasible, and robust regions. The first levels of the tree indicate that the displacement VVV is the dominant design parameter. Several of the first splits occur around values close to V≈12V \approx 12V≈12 cm³/rev, highlighting the strong influence of the pump displacement on the ability to satisfy the flow-rate requirement. The rotational speed nnn appears as a secondary but still significant discriminant variable.

The extraction of all terminal branches of the decision tree yields a collection of multidimensional boxes, or pavés, representing the partition of the design space learned by the classifier. A total of 433 boxes were extracted from the complete tree. Among them, 121 boxes correspond to infeasible regions, 221 boxes correspond to conditionally feasible regions, and 91 boxes correspond to robust regions. The visualization of these pavés clearly highlights the structure of the robust solution space and provides a direct interpretation of the decision boundaries learned from the simulation data.

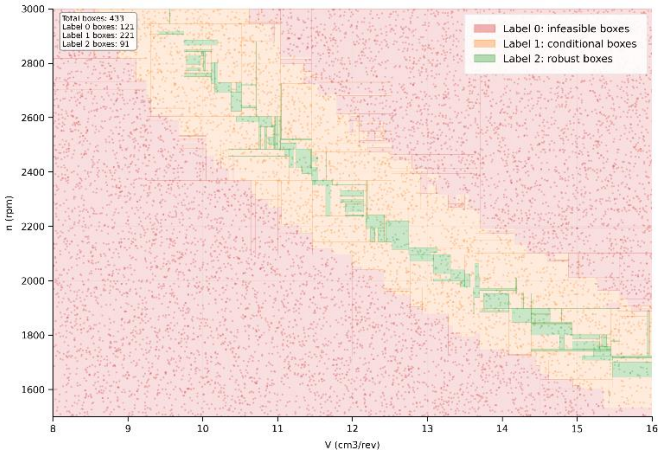

Figure 6: Boxes of Design Space

Compared with the QCSP-based exploration, the proposed method provides a fundamentally different representation of robustness. QCSP relies on interval reasoning and guaranteed classifications, which often leads to large uncertain regions when uncertainty intervals are broad. In contrast, the Monte Carlo and decision-tree approach statistically estimates robustness and directly learns the boundaries of robust regions from simulation results.

Overall, the results demonstrate that the combination of Monte Carlo simulation and decision-tree learning constitutes an effective alternative for exploring robust design spaces. The method successfully transforms a complex robustness-verification problem into an interpretable classification problem. It generates explicit design rules, identifies candidate robust regions, and provides a compact representation of the design space while maintaining a substantially lower computational complexity than exhaustive QCSP exploration. These characteristics make the proposed approach particularly attractive for preliminary design studies involving uncertainty and large design spaces.

From a computational perspective, the proposed Monte Carlo and decision-tree framework is significantly more efficient than QCSP-based exploration. While QCSP requires recursive domain subdivision and exhaustive consistency checks on thousands of design boxes, the proposed approach relies on a fixed number of simulations and a single supervised learning stage. In the considered case study, robust regions can be identified with only 10,000 candidate solutions and 20 uncertainty scenarios per solution, whereas QCSP requires the evaluation of thousands of subdomains and additional refinement iterations to progressively isolate reliable solution sets.

CONCLUSION

This paper proposed a novel exploration strategy for Robust Set-Based Design that combines Monte Carlo simulation and supervised machine learning to identify robust regions of a design space under uncertainty. Building upon the formal Characteristics–Properties Model (CPM) and First-Order Logic framework, the method transforms the robust design problem into a classification problem in which candidate

solutions are automatically labelled according to their level of compliance with the required properties.

The proposed approach first generates candidate design solutions through random sampling, then evaluates their behaviour under multiple realizations of external and modelling uncertainties. A three-level labelling scheme distinguishes infeasible, conditionally feasible, and robust solutions. A decision tree is subsequently trained to learn the structure of the design space and to extract explicit design rules in the form of multidimensional design boxes.

Compared with QCSP-based exploration, the proposed method offers several important advantages. First, robustness assessment is performed through simulation and can therefore incorporate complex behavioural models, numerical simulations, black-box models, metamodels, and multiphysics analyses without requiring explicit analytical formulations of the constraints. In contrast, QCSP approaches fundamentally rely on analytical or interval-based constraint representations and become increasingly difficult to apply when the behavioural relations are available only through simulation models.

Second, the proposed approach significantly improves scalability. Rather than recursively subdividing the design space and evaluating quantified constraints, robustness is estimated through Monte Carlo sampling and generalized through machine-learning-based rule extraction. This considerably reduces the computational burden associated with exhaustive quantified exploration while preserving the ability to identify robust solution regions.

Third, the methodo produces directly interpretable results. The extracted decision-tree branches constitute explicit design knowledge that can be readily used during engineering design activities, facilitating communication between designers, experts, and decision makers. Unlike many optimization or surrogate-based approaches, the resulting robust regions remain fully traceable and understandable.

Overall, the results demonstrate that Monte Carlo simulation combined with supervised machine learning constitutes a promising alternative to classical QCSP exploration for robust Set-Based Design. The proposed framework preserves the fundamental objective of identifying robust solution sets under uncertainty while extending the applicability of the approach to simulation-driven engineering problems for which analytical constraint formulations are difficult or impossible to obtain. Future work will focus on higher-dimensional design problems, adaptive sampling strategies, active-learning mechanisms, and the integration of more advanced machine-learning models for the exploration of large-scale engineering design spaces.

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BIOGRAPHY

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Simulation Tools

QUANTUM KERNELS WITHOUT THE QUANTUM COMPUTER: DEMONSTRATING A SIMULATION-BASED STYLOMETRY PIPELINE FOR ATTRIBUTING AI-GENERATED TEXT

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KEYWORDS

Quantum kernel methods, statevector simulation, language model attribution, stylometry, kernel concentration

ABSTRACT

Quantum kernel methods are usually discussed as if a quantum computer were required. We present a working tool that decides which of two language models, Gemma 3 or Qwen 2.5, wrote a text. Its quantum kernels are computed by exact simulation on conventional processors; the simulation is the measuring instrument, not a substitute for hardware. Each text is reduced to fourteen principal components of its stylometric features, encoded one per qubit in a small quantum circuit. A classifier separates the two models' texts by how much the texts' quantum states overlap. Running 83 configurations (three days of computing) located the method's limit, and the similarity matrices showed why: as qubits are added, the similarities shrink towards zero until almost all text pairs score nearly the same. On an independent test set, a classical kernel given identical inputs is about eight percentage points more accurate, and each model's stylometric fingerprint, which we hoped to credit to the quantum kernel, comes from the shared feature pipeline. Three lessons generalise: map a method's limits in simulation before paying for quantum hardware, compare against a classical baseline on identical inputs, and test once on data generated after the design is fixed.

INTRODUCTION

Given a text, it is increasingly useful to know not only whether a machine wrote it but which machine did. The question matters to publishers, examination boards and regulators, and emerging transparency rules for generated content make it more urgent. The literature on detecting machine-generated text is substantial (Wu et al. 2025), but it mostly addresses the easier question of human versus machine rather than the harder question of which model wrote a text (Huang et al. 2025).

Stylometry offers one route, long established for human authors (Stamatatos 2009): writing style leaves measurable traces, and those traces identify an author. Language models are thought to leave such traces too, because training may push each model towards particular sentence lengths, rates of punctuation and levels of formality.

Quantum kernel methods (Havlíček et al. 2019; Schuld and Killoran 2019) offer another way to compare such traces. The features measured from each text set the parameters of a small quantum circuit, and the similarity of two texts is the overlap of the resulting quantum states, a comparison that might capture structure that a conventional distance measure misses.

We built a tool to test that promise on one practical task: deciding whether a text was written by Gemma 3 (Gemma Team 2025) or by Qwen 2.5 (Qwen Team 2024). This paper is about the tool and what building it taught us; the full accuracy study is published separately (Kopanov and Atanasova 2026). We concentrate on what developers of simulation tools can reuse: how the system is assembled, what exact simulation contributed, and which of our expectations did not survive a comparison with a classical baseline.

No quantum computer was involved at any stage: every similarity score was obtained by simulating the circuit exactly on conventional processors. We do not present this as a limitation awaiting better hardware. The simulation was the measuring instrument, and it is what makes our result precise enough for others to build on. That result is negative in two respects: given identical inputs, a classical kernel is more accurate than the quantum kernel, and the writing-style profile we had hoped to credit to the quantum kernel comes from the feature pipeline that both methods share.

SYSTEM ARCHITECTURE

The tool answers one question: which of two known language models wrote a given text (Figure 1). The data are 5800 short texts. We wrote 2900 prompts (short writing tasks) spread over ten topics, and had each prompt answered once by Gemma 3 (27B, instruction-tuned) and once by Qwen 2.5 (32B, instruction-tuned), so both models write about exactly the same topics and only style separates them. The two

classes are therefore the same size, and the chance-level accuracy is 50%.

The prompts were split 80/20 before anything was fitted: 2320 prompts (4640 texts) for training and 580 prompts (1160 texts) for validation, that is, for comparing configurations and choosing settings. No prompt contributes to both. A separate test set of 1000 texts, generated only after all design decisions had been fixed, is described in the Results section. Unless stated otherwise, the accuracies reported below are measured on the validation set.

Feature Extraction

Two kinds of feature are extracted from each text. The first is a word-frequency profile weighted by term frequency–inverse document frequency (TF-IDF), so that widely used terms count for less. Common function words and any term found in more than 70% of texts are removed first, and the 3000 most frequent remaining words and word pairs are kept. The second is a set of eighteen interpretable linguistic features, such as average word and sentence length, vocabulary richness, part-of-speech proportions, punctuation density, formality, and the rates of questions and exclamations. Together they form a 3018-dimensional feature vector for each text.

That is far too many values for a small circuit, which needs one qubit (quantum bit) per value. The tool therefore standardises the features, reduces them with principal component analysis (PCA) to fourteen components, and rescales each component into $[0, 2\pi]$ so that it can serve directly as a rotation angle. Every step that is fitted to data, from the vocabulary to the rescaling, is fitted on the training data only. The fourteen components retain less than 7% of the variance of the original 3018 features; the comparison with the classical baseline in the Results section shows that the information needed to tell the two models apart survives nonetheless.

Quantum Kernel and Classifier

The fourteen angles are encoded into fourteen qubits with the ZZFeatureMap circuit of the Qiskit library: each angle rotates one qubit, neighbouring qubits are then entangled in a chain (linear entanglement), and this pattern is applied twice (two repetitions). Such an encoding circuit is called a feature map. The similarity of two texts, called the kernel, is the fidelity (squared overlap) of their quantum states, a number between zero and one: $K(x, x') = |\langle\phi(x)|\phi(x')\rangle|^2$, where $\phi(x)$ is the state that the circuit prepares from the angles x of one text. The matrix of all pairwise similarities, the kernel matrix, is passed to a standard support vector machine, which learns the boundary between the texts of the two language models. The classifier sees the data only through this matrix, which is why the properties of the kernel matter so much.

The Classical Baseline

What lets the tool measure the effect of the quantum kernel, rather than merely classify texts, is the second branch in Figure 1. The same input recipe (the same features, compression and rescaling, refitted by the same code on the same data split and random seed) also feeds a classical support vector machine with a radial basis function (RBF) kernel, which serves as the baseline. Because the inputs are produced identically, a difference in accuracy can be attributed to the kernel rather than to the input representation.

This branch was not part of our original design. Our first round of experiments compared the quantum classifier only with a classical model trained on the full 3018-dimensional feature vector, which left open the objection that the compression, not the kernel, limited the accuracy. Adding the baseline on identical inputs was the most valuable change we made: it turns "our quantum classifier reaches 88%" into a statement about the quantum kernel itself. Its code and results are released with the rest of the tool.

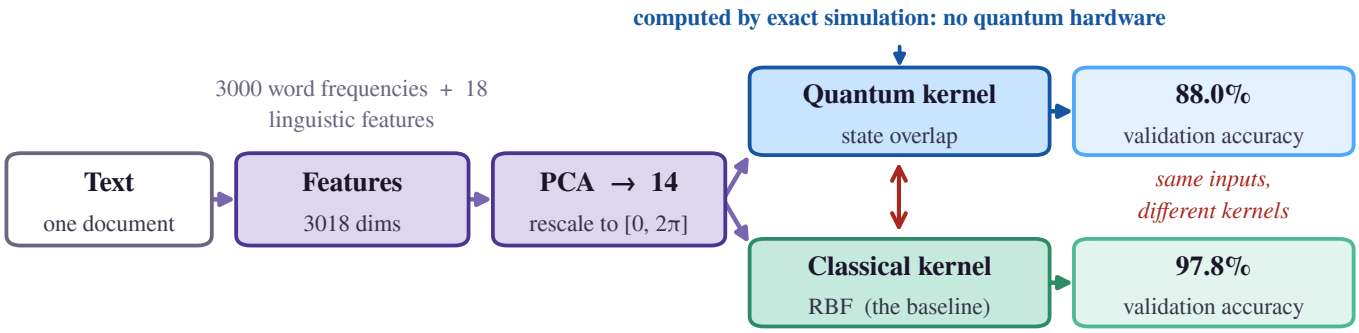

Figure 1: The Pipeline: One Shared Feature-Extraction Stage Feeds the Same Fourteen Principal Components to a Quantum and a Classical Kernel (Accuracies on the 1160-Text Validation Set; 87.3% and 95.6% on the Independent Test Set)

The pipeline is written in Python, using scikit-learn and NLTK for feature extraction, scikit-learn for compression and the classical classifiers, and Qiskit 1.4.3 with Qiskit Machine Learning 0.8.3 for the quantum kernel. The benchmark configurations are run from one training script, whose settings (the number of qubits, the training-set size and the simulator) are passed as environment variables; a second script runs the classical baseline and the kernel measurements, and a third evaluates the classifiers on the independent test set. The eight-qubit configuration trained on 600 texts builds its kernel matrix in under a minute on one 64-core node, which is why we use it in the live demonstration; it is chosen for speed, not accuracy, as the Results section shows. The code, the corpus, the results of all configurations and records of both software environments are available at https://github.com/Kopanov/QSVM-text-attribution.

SIMULATION SETUP

The quantum kernel is evaluated with Qiskit's FidelityStatevectorKernel, which simulates the circuit exactly. There is no measurement sampling (a real quantum processor would estimate each similarity from many repeated runs of the circuit) and no hardware noise: the similarity returned is the value defined by the ideal circuit, to machine precision, and repeating a run with the same software versions returns the same value.

This determinism is what makes each result traceable to its cause. On a quantum processor, sampling error and decoherence (loss of the quantum state through interaction with its surroundings) would have been added to the effect we wanted to isolate. Whenever an accuracy figure changed, we could not then have said whether the cause was the feature map, the training-set size or drift in the device. In simulation, every difference follows from a design choice or, as one small configuration later showed, from the version of a numerical library.

Running all 83 configurations took about 72 hours on 64-core nodes of a high-performance computing cluster; about 61 of these hours were spent building kernel matrices. The most expensive single run, with sixteen qubits and 4640 training texts, took just under 18 hours. The cost of building a kernel matrix grows quadratically with the number of texts and exponentially with the number of qubits.

With this computing time we mapped where the method works and where it fails. We varied four design parameters: circuit width (the number of qubits, from 4 to 16, which equals the number of principal components kept), training-set size (600 to 4640 texts), the feature map, and circuit depth (the number of repetitions of the encoding pattern); the regularisation strength of the support vector machine was also varied and made little difference. Doing this on quantum hardware first would have cost more and revealed less, because hardware noise would have affected every measurement.

RESULTS

Comparison on Identical Inputs

The fourteen-qubit quantum classifier reached 88.02% on the validation set, and no configuration we tried did better. The obvious suspect was the compression to fourteen components, so we gave the classical baseline the same fourteen components, computed by the same code from the same data split and random seed. It reached 97.76%. Its regularisation parameter was chosen on the validation set, but that does not explain the gap: with the same setting as the quantum classifier it still reached 96.98%, and even a linear support vector machine on the same fourteen components reached 96.64%. The compression was not the bottleneck; the kernel was. On the independent test set, the baseline reached 95.60% and the quantum classifier 87.30%, a gap of 8.30 percentage points, far larger than the statistical uncertainty of an accuracy measured on 1000 texts (a 95% confidence interval of about ±2 percentage points).

The comparison also overturned a second expectation. Our pipeline produces an interpretable writing-style profile, or fingerprint, for each model: Gemma 3 writes with richer vocabulary, denser punctuation and more questions and exclamations, while Qwen 2.5 writes longer, more complex and more formal sentences. We expected to credit this fingerprint to the quantum kernel, and to argue that, because the evidence is spread across many features, an adversary could not defeat the method by altering any single feature.

When we computed the fingerprint in the same way for the classical baseline, we obtained essentially the same profile, led by the same few features, and this removed the basis for both claims. The importance score behind the fingerprint is computed only from the principal-component loadings and the share of variance that each component explains. It never uses the kernel or the trained classifier, so any classifier trained on these components inherits the same fingerprint; the small differences we saw arise only because the baseline refits the PCA in a later software environment. The fingerprint describes what the representation retained, not how either classifier used it. We therefore make no interpretability claim and no robustness claim for the quantum kernel. This is an instance of a pattern documented across quantum machine learning, in which an apparent quantum advantage disappears once a properly matched classical baseline is included (Bowles et al. 2024).

Kernel Concentration

If the kernel is the bottleneck, the kernel matrix itself should show why. Figure 2 plots the average similarity that the quantum kernel assigns to two different texts as the number of qubits grows. It falls two- to threefold for every two qubits

added; at fourteen qubits it is 413 times smaller than the value of the classical kernel on the same inputs.

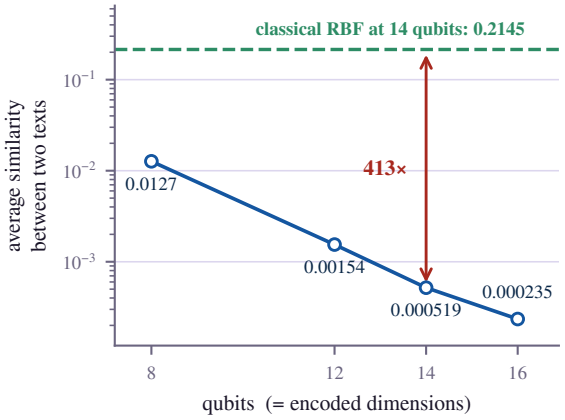

Figure 2: Average Similarity Between Two Different Texts under the Quantum Kernel versus Number of Qubits, with the Classical Kernel's Value at Fourteen Qubits on the Same Inputs (Logarithmic Scale; Data from Kopanov and Atanasova 2026)

As qubits are added, the encoded quantum states become nearly orthogonal (their overlap approaches zero). The kernel then gives almost every pair of texts a similarity close to zero, and the classifier must draw its boundary from values that barely differ from one another. This effect, known as exponential kernel concentration, was predicted theoretically (Thanasilp et al. 2024); here it is measured directly on a working text-classification pipeline.

Two practical consequences follow. First, more training data helps only up to a point. At fourteen to sixteen qubits, accuracy stops improving at roughly 220–250 training texts per qubit, and at sixteen qubits, increasing the training set from 3480 to all 4640 texts *reduced* accuracy by 4.56 percentage points. Second, the obvious ways of making the circuit more expressive (able to represent more complicated patterns) reduce accuracy. The depth and entanglement tests used smaller runs, each scored on its own validation set of 200 to 800 texts. Increasing the circuit depth from two to three repetitions of the encoding pattern lowered accuracy by 5.00 to 9.25 percentage points in every configuration where we tested it (three in all). The feature map with the richest entanglement that we tested, in which every qubit interacts with every other, reached only 65–66% accuracy, against 86–88% for the ZZFeatureMap in the same eight-qubit runs, and its kernel took about five times longer to compute.

We therefore suggest that studies of this kind report the average similarity between different items alongside accuracy: accuracy shows that something is wrong, whereas this quantity shows what is wrong.

Independent Test-Set Evaluation

Three configurations reached the 88% ceiling. The small one used eight qubits and 600 training texts; it was the best of several dozen small runs, each scored on its own small validation set of 200 to 400 texts rather than on the 1160-text set. The two large ones used twelve and fourteen qubits with 3480 training texts and were scored on the full 1160-text validation set. The small configuration was far cheaper and looked like the better engineering choice.

Independent testing showed otherwise. After every design decision had been fixed, we wrote 500 new prompts, generated 1000 new texts from them, and evaluated each configuration on these texts exactly once. The large configurations kept their accuracy: 87.30% at fourteen qubits (against 88.02% on validation) and 88.40% at twelve, a configuration that also builds its kernel matrix in about a quarter of the time needed at fourteen qubits. The small configuration fell from 88.00% to 79.80%. It was also numerically fragile: when the run was repeated with NumPy, the numerical library underlying the pipeline, upgraded from version 1.26.4 to 2.2.6 and nothing else changed, its validation accuracy fell from 88.00% to 85.75%. This suggests that small configurations of this kind are sensitive to low-level numerical changes; the large configurations, evaluated under the new version on the independent test set, stayed within one percentage point of their validation accuracy. When a configuration is chosen as the best of many on a small validation set, its accuracy estimate is optimistically biased, and only data that played no part in the choice can reveal this.

OPEN-SET EXTENSION: A PROPOSAL

The tool described above chooses between two known models. In practice the question is often open-set: the text may come from a known model or from one the tool has never seen. We have designed a second mode for this question and present it as a proposal rather than a result; it is not part of the public repository. In this mode, a model is enrolled once by storing the quantum states of 500–1000 of its texts, and a new text is scored by its average similarity to this stored set. No texts from other models are needed, so each model can be enrolled on its own.

Its limitations must be stated. The score is the same kind of average similarity that Figure 2 shows falling towards zero as qubits are added, so this mode inherits the concentration problem. The scores are not calibrated probabilities: turning them into decisions requires held-out texts from known and unknown models, a decision threshold and a calibration step, none of which exists yet. We make no robustness claim, and nothing in the scheme requires a quantum kernel; a classical kernel whose values lie in a fixed range and have been checked not to shrink towards zero could be used instead.

CONCLUSION

We built a tool that identifies which of two language models wrote a given text, with every quantum computation carried out by exact simulation on conventional processors. Its best quantum configurations reach 87–88% accuracy on new texts, whereas the classical baseline given the same inputs reaches about 96%. The cause is exponential concentration of the quantum similarity measure, which we measured directly and could not overcome with more data, deeper circuits or richer entanglement.

For developers of simulation tools, the method that produced this result is more useful than the result itself. Systematic simulation mapped the working range of the method in about three days of computing time on a cluster, with every result traceable to a design choice or a software version rather than to hardware noise. A classical baseline on identical inputs ruled out the plausible explanation that the compression limited the accuracy, showed that the kernel did, and prevented an interpretability claim that we would otherwise have made. Testing once on data generated after the design was fixed exposed a configuration whose validation accuracy was overstated (88.0% on validation, 79.8% on new texts). None of this is specific to quantum computing: characterise a method in simulation before committing to expensive experiments, compare methods on identical inputs, and generate the test data only after the design is fixed. Each of these practices costs little compared with publishing a result that does not hold.

FUTURE WORK

Three directions follow. The first is an encoding that resists concentration: rescaling the input angles to a narrower range, then trainable feature maps and projected quantum kernels, which compare only parts of the quantum states (Kopanov and Atanasova 2026), with progress measured by the similarity statistic reported here rather than by accuracy alone. The second is an evaluation of the open-set proposal: how many texts an enrolment needs, where to set the decision threshold, how to calibrate the score, and whether paraphrasing can defeat it. The third is execution on real quantum processors, at the circuit widths and training-set sizes where our measurements show the method starting to fail, with the exact noiseless simulation as the reference.

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A PROBABILISTIC RISK-RANKING FRAMEWORK FOR ASSESSING FOOD CHAIN AND ENVIRONMENTAL RISKS OF NITRIFICATION INHIBITORS IN GRAZED PASTORAL SYSTEMS

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KEYWORDS

Environmental Exposure Assessment, Food Chain Modelling, Food Safety, Nitrification Inhibitors, Probabilistic Risk Assessment, Risk Ranking

ABSTRACT

Nitrification inhibitors (NIs), particularly dicyandiamide (DCD), 2-chloro-6-(trichloromethyl) pyridine (nitrapyrin), 3,4-dimethylpyrazole phosphate (DMPP), and 3,4-dimethylpyrazole succinic acid (DMPSA), are applied, either as coatings on granular nitrogen fertilisers, in combination with liquid fertilisers, slurries and digestate and can also be potentially applied alone in grazed pastoral systems targeting urine patches to reduce nitrous oxide emissions and nitrate leaching. Their persistence, mobility and potential transfer into the food chain are important for the continued safe use of these compounds. Nevertheless, no simulation framework currently integrates their residues in crops, pastures and soils assesses their exposure and risk ranking. This paper outlines the development of a modular, probabilistic risk-ranking framework for evaluating the food-chain and environmental risks of DCD and DMPP that could be extended to other nitrification inhibitors. The framework applies a two-stage approach: an initial semi-quantitative weighted scoring of persistence, mobility and exposure-pathway relevance, followed by a probabilistic layer that propagates variability and uncertainty in measured residue concentrations using Monte Carlo simulation. With further development, the framework has the potential to be implemented as a practical decision-support tool for food safety and environmental risk management stakeholders.

INTRODUCTION

DCD, Nitrapyrin, DMPP and DMPSA are applied to grazed pasture to slow the microbial conversion of ammonium to nitrate, reducing nitrate leaching and nitrous oxide emissions from urine patches (Adhikari et al., 2021a; Di and Cameron, 2002; Zerulla et al., 2001). These environmental benefits are well established, but the same properties that make NIs effective, persistence in soil and mobility within the soil-plant system, raise the possibility of transfer into pasture, grazing animals and, ultimately, milk and meat (Saggar et al., 2026a, b; Pal et al., 2015). This concern was realised in New Zealand in 2013, when DCD use was voluntarily suspended after traces of DCD residues were detected in exported milk. Although the amounts of DCD detected were trace amounts this came in the context of an absence of a formal acceptable daily intake (ADI) or maximum residue limit (MRL) for the compound. By the time the traces were detected DCD commercially released in New Zealand in February 2004 by Ravensdown under the trade name eco-n to mitigate nitrogen leaching and N_2O emissions from livestock urine, had already actively applied on roughly 4% to 5% of farms nationwide, illustrating the potential exposure pathway before this regulatory gap was identified.

A recent review by Adhikari et al. (2022) reported that the United States Environmental Protection Agency (USEPA) has registered nitrapyrin as an NI. However, neither DCD nor DMPP are included on USEPA Toxic Substance Control Act (TSCA) inventory. These reviewers also reported that in the European Union (EU), the use of NIs falls under Regulation (EC) No 2003/2003 (EUR-Lex 2003). This regulation states that N process inhibitors can be added to N fertilisers such as ammonium nitrate and urea. All three NIs are registered in another recent EU regulation that ensures the proper registration, evaluation, authorization, and restriction of chemicals (REACH) to protect human health and the environment. However, minimum and maximum allowance level (% inhibitor presented by mass of the total N in the fertiliser) have been established for only DCD and DMPP. The minimum and maximum values for DCD are 2.3 and 4.5 % (w/w) and for DMPP are 0.8 and 1.6% (w/w), respectively (European Commission 2014). No integrated simulation framework currently exists that combines residue monitoring, exposure-pathway modelling and comparative risk ranking for NIs used as plant growth enhancing compounds. Existing chemical risk-ranking methods (scoring systems, hazard quotients, risk matrices, probabilistic approaches and multi-criteria decision analysis) are well developed for pesticides and industrial contaminants but have not been developed to the specific data limitations of NIs, where toxicological reference values are largely not available. This paper describes the design (Figure 1) and current implementation of a probabilistic risk-ranking framework intended to fill this gap, and outlines how the framework is being developed, tested and validated using published residue and environmental fate data.

FRAMEWORK ARCHITECTURE AND METHODOLOGY

The framework is organised as a modular pipeline (Figure 1). A structured database first consolidates three categories of input: (i) NI concentrations in soil and pasture reported in the published literature, (ii) physicochemical and environmental fate parameters (persistence, sorption/mobility, degradation rate) drawn from independent, peer-reviewed sources rather than manufacturer technical data sheets, and (iii) literature-derived toxicological information, which remains partial for both compounds. This database feeds two linked assessment stages.

Stage 1 and Stage 2: How They relate

Stage 1 is a semi-quantitative weighted scoring step. Each compound is scored against a small set of criteria, persistence, mobility, frequency and magnitude of residue detection, and relevance of each exposure pathway (soil-pasture-animal transfer and soil-water transport), to produce a preliminary relative ranking. Stage 2 does not operate on the Stage 1 score itself; but it relates back to the same underlying residue concentration measurements and fits probability distributions to them by pathway and matrix (soil, pasture, water), propagating this variability and the associated parameter uncertainty through Monte Carlo simulation. This generates a distribution of exposure estimates per pathway rather than a single point value, from which probabilistic comparisons between two NIs, including the probability that their relative ranking would reverse under resampling, can be derived. Thus, Stage 1 therefore an interpretable, low-data-demand screening output, while Stage 2 provides a statistically refined output based on the availability of sufficient residue data; both stages are reported alongside one another rather than one replacing the other.

Weighting Factor Selection

Stage 1 weights are currently assigned using literature-informed relative importance of each criterion (e.g. weighting persistence and pathway relevance more heavily than detection frequency alone, reflecting their greater influence on long-term exposure), reviewed by the project team. This is an interim, expert-informed approach; a planned sensitivity analysis will test how the final ranking responds to plausible alternative weighting schemes, and pairwise comparison methods are being considered as a more formal elicitation route.

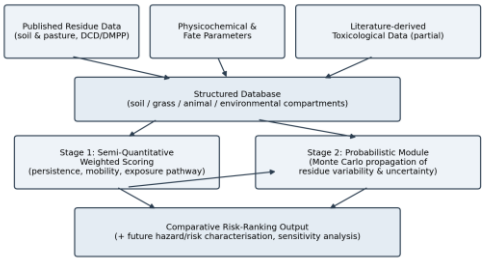

Figure 1: Schematic Architecture of the Risk-Ranking Framework

DATA BASIS

The structured database underlying the framework is designed to draw on published measurements of DCD and DMPP concentrations in soil and pasture following application, together with literature-reported physicochemical and environmental fate parameters for both compounds. This literature basis is intended to capture how NI concentrations vary over time after application and across soil type and climatic conditions, both of which are required inputs for the Stage 2 probabilistic layer. As further field studies are published, the framework has been designed so that updated residue datasets can be incorporated without requiring changes to its underlying structure.

IMPLEMENTATION STATUS

To date, the structured database and the Stage 1 weighted-scoring module have been implemented and tested using illustrative, literature-derived values to confirm the scoring logic. The exposure-pathway structure (soil-pasture-animal and soil-water transport) has been defined and mapped onto the database schema. The Stage 2 Monte Carlo module is under development. The distribution-fitting and simulation logic has been prototyped, but propagation across all pathways is not yet complete. Hazard characterisation, and the quantitative risk characterisation (risk quotients) it would support, is deferred pending the compilation of sufficient toxicological reference data for DCD and DMPP, and is treated in the current framework as an optional module to be activated once such data become available.

ILLUSTRATIVE EXAMPLE

Table 1 illustrates the form of the Stage 1 output using representative, literature-consistent qualitative fate characteristics of DCD and DMPP (DCD is generally reported as more persistent and more mobile in soil than DMPP; Di and Cameron, 2002; Zerulla et al., 2001). This example is illustrative of the ranking structure only and does not represent a finalised quantitative analysis; a full quantitative ranking will be reported once suitable published residue data have been compiled and Stage 2 development is complete.

Table 1: Illustrative Stage 1 Ranking Structure (Not Final Results)

Criterion	DCD	DMPP
Persistence	High	Medium
Mobility	High	Low-Medium
Detection frequency	Medium	Low
Composite Stage 1 score	Higher	Lower

VALIDATION AND ADAPTABILITY

Validation is planned in three complementary ways. First, Stage 1 rankings will be cross-checked against an independent set of published residue values not used in the initial scoring. Second, a sensitivity analysis will quantify how rankings respond to changes in the weighting scheme

described above. Third, for Stage 2, the probability of rank reversal between DCD and DMPP under resampling will be reported directly, giving a quantitative, decision-relevant measure of how robust the comparative ranking is to residue variability, addressing a specific need for evidence that a probabilistic ranking is sound.

The framework's modular structure, a shared database schema for physicochemical, fate, residue and (when available) toxicological data, is intended to generalise beyond DCD and DMPP to other emerging agricultural chemicals with similar data limitations. At present this generalisability is a design goal rather than a tested capability: applying the framework to another compound would currently require populating the same data structure with compound-specific residue and fate data, and re-deriving pathway weights, rather than any change to the underlying framework logic. Testing this adaptation on a second compound is planned as future work.

CONCLUSION AND FUTURE WORK

This paper has outlined the architecture and current implementation status of a probabilistic risk-ranking framework for NIs in grazed pastoral systems, developed using published residue and environmental fate data for DCD and DMPP. The demonstration will walk through the database structure, the Stage 1 scoring workflow and early Stage 2 prototyping, and will discuss the practical challenges of building a risk-ranking framework on compound class with no established toxicological reference values. Immediate next steps are completion of the Monte Carlo propagation across all exposure pathways, sensitivity analysis of the weighting scheme, and reporting of rank-reversal probabilities as a validation metric; longer-term work will incorporate hazard characterisation once toxicological data become available and test the framework's adaptability to a second agrochemical.

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HOW A SMALL FINE-TUNED MODEL CAN RIVAL A LARGE GENERAL-PURPOSE MODEL IN CULINARY TASKS

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KEYWORDS

Small language models, Fine-tuning, LoRA, Evaluation metrics, Culinary NLP

ABSTRACT

It was tested whether language models with at most one billion parameters can rival a large general-purpose model in a narrow, structured domain. Culinary tasks were used as a controlled benchmark covering recipe classification, nutritional-value prediction, instruction generation, and description generation. Three public small models, Qwen3-0.6B, GPT-2-Large, and Llama-3.2-1B, were compared with GPT-4.1 before and after LoRA fine-tuning on 30 000 recipes. Fine-tuned Llama-3.2-1B approached GPT-4.1 in generation and classification, and exceeded it in structured nutritional prediction. The results show that domain-specialised small models can be a practical alternative to frontier models in well-defined tasks.

INTRODUCTION

Large language models owe much of their success to scale, but this scale also brings high computational, financial, and infrastructural costs. For many practical applications the central question is therefore not whether the largest available model performs well, but whether a much smaller model can become competitive after domain adaptation.

We examine this question in culinary tasks, a domain combining structured labels, numerical nutritional values, procedural text, and natural-language descriptions. The hypothesis is that a properly fine-tuned small language model ($\leq 1\,B$ parameters) can approach or exceed a large general-purpose model in such tasks. The study compares pretrained and LoRA-fine-tuned SLMs with GPT-4.1 under identical prompts and validation data. It is based on work reported in (Krajewski 2026).

STATE OF THE ART

Transformers (Vaswani et al. 2017) and subsequent pretrained models such as BERT (Devlin et al. 2019) and GPT-type systems (Yenduri et al. 2024) established large-scale pretraining as the dominant NLP paradigm. Meanwhile, the cost of training, fine-tuning, and deploying very large models has stimulated research into Small Language Models (SLMs) (Kim et al. 2025). We treat models up to one billion parameters as small.

Parameter-Efficient Fine-Tuning, especially Low-Rank Adaptation (LoRA) (Hu et al. 2022), makes such models easier to specialise, as only small adapter matrices are trained while the base model remains frozen. Previous studies (Łępicki and Kurek 2025) indicate that SLMs can be competitive in specialised tasks, but systematic sub-1B versus frontier-model comparisons across several structured tasks remain limited. Culinary NLP has also been explored for recipe-related tasks (Rita et al. 2024), (Sugioka et al. 2024), making it a suitable benchmark for testing scale against specialisation.

MODELS AND TASKS

The evaluated SLMs were Qwen3-0.6B (Yang et al. 2025), GPT-2-Large (Radford et al. 2019), and Llama-3.2-1B (Touvron et al. 2023). GPT-4.1 (OpenAI, Kumar et al. 2025) served as the large general-purpose reference model and was not fine-tuned.

Four tasks were considered: recipe classification, nutritional-value prediction for six nutrients, generation of cooking instructions from dish name and ingredients, and generation of short dish descriptions from instructions. Together they cover classification, regression-like numerical similarity, and free-form generation.

MEASURES OF QUALITY

The evaluation framework was designed for comparability across heterogeneous tasks: classification, structured numerical prediction, and free-form generation. Because

the tasks differ fundamentally in output structure, a unified metric is neither feasible nor desirable. Instead, each task category is evaluated with task-appropriate criteria under a shared experimental protocol.

Quantitative Measures and their Limitations

For multi-class classification, standard accuracy was used. Given the balanced distribution of categories in the validation set, accuracy provides a transparent and interpretable measure.

Text generation evaluation presents greater difficulty. Classical n-gram–based metrics such as BLEU, ROUGE, or METEOR penalize lexical and syntactic variation even when semantic equivalence is preserved (Hu and Zhou 2024). Culinary instructions frequently admit multiple equivalent formulations (e.g., reordering steps that are causally independent, using synonyms for ingredients or actions). In such cases, surface-level overlap underestimates true semantic similarity.

For this reason, semantic-level metrics were preferred. BERTScore (Zhang et al. 2020) was employed to compare contextual token embeddings between generated and reference texts. Additionally, cosine similarity between sentence embeddings (Yin and Zhang 2024) was computed to capture global semantic alignment. These metrics do not rely strictly on token overlap and better reflect functional equivalence in procedural instructions.

Two additional reference-free metrics were included. Perplexity (Chen et al. 2018) was measured to assess fluency and syntactic plausibility. Readability (Eleyan et al. 2020) was computed using established readability indices to detect excessive structural complexity. While neither directly measures culinary correctness, both provide useful constraints on linguistic quality.

Nutritional value prediction required a different approach. Exact numerical matching is inappropriate because minor deviations may be nutritionally negligible. Therefore, similarity was defined using a logarithmic-relative formulation:

$$p = e^{-|\ln((a+\varepsilon)/(b+\varepsilon))|} \tag{1}$$

where a denotes the predicted value, b the ground truth, and ε a stabilizing constant. The logarithmic term ensures scale invariance, penalizing proportional rather than absolute error. The exponential transformation bounds the similarity score within $(0, 1]$. Values above a threshold of 0.7 were considered sufficiently similar; this value was chosen to balance tolerance to minor deviations with practical nutritional relevance; pilot checks indicated stable model ranking under moderate threshold variations. The parameter ε was chosen empirically per nutrient ($\approx 10 - 15\%$ of the median nutrient value) to avoid instability for small magnitudes. This formulation provides three advantages: proportional fairness across nutrients with different scales, tolerance for small nutritional deviations, and interpretability of similarity as a continuous confidence measure.

Human-based Evaluation and its Potential Automation

Automatic metrics alone cannot guarantee culinary validity. A generated instruction may achieve high semantic similarity while containing procedural flaws (e.g., incorrect cooking order, omission of essential steps), so a structured human-evaluation protocol was introduced.

For instruction generation, six binary conditions were defined: linguistic correctness, culinary validity, absence of procedural errors, appropriate ingredient handling, logical completeness, and presence of time specification. Each condition was evaluated independently, enabling fine-grained analysis rather than a single aggregated score. The criteria are a result of observation of incorrect answers returned by the trained networks.

For description generation, analogous binary criteria were defined: semantic validity, linguistic correctness, proper length compression, lack of hallucinations, sensory informativeness, and inclusion of key dish features.

The protocol was constructed inductively by analyzing recurring failure modes in baseline outputs. Importantly, the criteria avoid subjective notions such as taste preference and instead focus on structural and procedural adequacy.

Although currently implemented as human annotation, the structure of the protocol suggests partial automatization potential. Several criteria (e.g., time specification, length constraints, lexical overlap with ingredients) can be programmatically detected. Others, such as logical consistency, may be approximated through rule-based or model-based validators. The protocol thus provides a foundation for future reference-free automatic culinary evaluation.

DATASETS USED

The experimental design relied on clearly separated data sets to prevent leakage and to ensure fair comparison between models.

Three distinct data sets were used: (i) pretrained corpora implicitly embedded in each model, (ii) a supervised fine-tuning dataset, and (iii) an independently constructed validation dataset.

THE TESTING ENVIRONMENT

Experiments were conducted in Python using PyTorch and Hugging Face Transformers (Transformers 2019). Fine-tuning was performed on a local system with RTX 4070 GPU, 12 GB VRAM. LoRA adapters were applied to attention projection matrices. Training times ranged from 1 to 10 hours depending on task.

Pretraining

All evaluated small models were used in their publicly available pretrained versions. Their original pretraining corpora are not identical and remain partially undisclosed in detail. Importantly, none of the models were originally specialized for structured culinary reasoning tasks such as nutrient prediction.
GPT-4.1 was evaluated in its publicly available inference configuration without domain-specific fine-tuning. This ensures that any performance gains observed in small models after specialization are attributable to controlled adaptation rather than hidden domain training.

Fine-tuning

Fine-tuning was performed using a dataset `recipe` (Shengtao 2019) of over 30 000 English-language recipes sourced from AllRecipes (Kappel et al. 2026). It contains fields including dish name, ingredient list, preparation instructions, nutritional values per serving, category labels, and metadata.
The dataset was selected for the following properties: balanced distribution across culinary categories, wide nutritional range (low- and high-calorie dishes), natural, non-synthetic language, minimal missing values, and consistent formatting. Records were transformed into instruction–target pairs tailored to each of the four defined tasks. Prompts were standardized to reduce variance induced by formatting differences. For generative tasks, instruction templates constrained output format (e.g., short answer requirements) to avoid uncontrolled verbosity. No record from this dataset was included in the validation set.
During the fine-tuning, the biggest challenges were the high risk of overfitting and the irregularity of the training data. The former was mitigated by using such LoRA techniques like relaxed warm-up and cool-down of the learning rate, as well as monitoring the loss function and recording several points of training progress. The latter was addressed by using a specific structure for each command and expected response, with specific tokens defining the start and stop of the target value.
By way of example of overtraining let us mention that one of the systems proposed to bake the scrambled eggs in an oven, like pancakes.

Validation and Experimental Control

A separate validation dataset with 150 manually curated recipes was constructed. Recipes were collected from multiple public culinary sources and manually verified. Each record contains: dish name, ingredient list with proportions, full preparation instructions, nutritional values per serving (calories, fat, cholesterol, carbohydrates, sugar, protein), assigned category label, and descriptive summary.

Table 1: Classification Accuracy [%]

Metric \ Model	GPT-4.1	Qwen3-0.6B	GPT-2-Large	Llama-3.2-1B
		Basic / Fine-tuned		
1	2	3 / 4	5 / 6	7 / 8
Classification accuracy	68.5	52 / 54	37 / 65	11 / 67

Table 2: Instruction Generation: automatic Metrics

Metric \ Model	GPT-4.1	Qwen3-0.6B	GPT-2-Large	Llama-3.2-1B
		Basic / Fine-tuned		
1	2	3 / 4	5 / 6	7 / 8
BERTScore	0.45	0.26 / 0.26	0.25 / 0.24	0.22 / 0.34
Cosine embedding similarity	0.87	0.77 / 0.78	0.75 / 0.77	0.74 / 0.83
Perplexity	13.30	6.11 / 6.51	6.30 / 7.37	6.24 / 6.30
Readability	74.44	75.18 / 78.92	77.00 / 77.30	72.71 / 78.71

Nutritional values were computed proportionally using standardized USDA tables, for internal consistency.
The validation dataset was designed to maximize heterogeneity: multiple cuisines and cultural origins, varied preparation times, simple and multi-step dishes, both low- and high-calorie meals. This heterogeneity reduces the risk of biased evaluation toward specific recipe types. All models, including GPT-4.1, were evaluated on the identical validation dataset under the same prompt formulations. No gradient updates were performed on this dataset. This guarantees comparability and eliminates data leakage between training and evaluation phases.

RESULTS

This section presents experimental outcomes in three stages: (i) evaluation of the large reference model without domain adaptation, (ii) evaluation of pretrained small models, and (iii) evaluation after LoRA-based fine-tuning. All models were tested on the same validation dataset using identical prompt formulations.

Large Model – No Fine-tuning

GPT-4.1 was evaluated as a strong general-purpose baseline which establishes an upper reference bound for generative fluency and structured reasoning without domain-specific specialization. It achieved 68.5% accuracy on the curated validation dataset (Tab. 1, col. 2).
Automatic metrics computed on generated instructions have been collected in Tab. 2, col. 2. Results of human evaluation of instruction generation have been shown in Tab. 3, col. 2. For description generation, the human-based evaluation yielded the percentages of outputs satisfying the assumed criteria listed in Tab. 4, col. 2.
Structured nutritional prediction revealed substantial weaknesses. The percentage of dishes whose predicted nutrient value reached at least 0.7 similarity with ground truth was: calories (69%), fat (44%), cholesterol (30%), carbohydrates (19%), sugar (18%), and protein (10%). As it can be seen, only the results for calories exceeded 50%. These results were collected in Tab. 5,

Table 3: Instruction generation [%]: human-based evaluation. Narrow grey bars: reference value (from col. 2, respectively in each row)

Metric \ Model	GPT-4.1	Qwen3-0.6B	GPT-2-Large	Llama-3.2-1B
		Basic / Fine-tuned		
1	2	3 / 4	5 / 6	7 / 8
Linguistic correctness	99	83 / 80	67 / 83	73 / 96
Culinary validity	94	80 / 74	64 / 77	71 / 89
Absence of procedural errors	88	56 / 62	50 / 67	67 / 86
Appropriate ingredient handling	96	64 / 76	71 / 83	62 / 92
Logical completeness	92	57 / 74	59 / 70	67 / 81
Presence of time specification	98	55 / 98	79 / 97	84 / 98

Table 4: Description generation [%]: human-based evaluation. Narrow grey bars: reference value (from col. 2); bright bars: larger than reference

Metric \ Model	GPT-4.1	Qwen3-0.6B	GPT-2-Large	Llama-3.2-1B
		Basic / Fine-tuned		
1	2	3 / 4	5 / 6	7 / 8
Semantic validity	100	42 / 91	63 / 82	- / 86
Linguistic correctness	92	84 / 75	74 / 76	- / 79
Appropriate compression	100	80 / 92	68 / 94	- / 88
Relevance to the recipe	99	89 / 74	91 / 78	- / 85
Absence of hallucinations	92	98 / 83	68 / 83	- / 88
Sensory informativeness	94	41 / 73	72 / 65	- / 83
Inclusion of key dish features	100	84 / 71	66 / 68	- / 79

col. 2. These results indicate that while GPT-4.1 performs strongly in linguistic and structural generation tasks, its performance in precise quantitative nutrient estimation is limited.

Pretrained Small Models

Base versions of Qwen3-0.6B, GPT-2-Large, and Llama-3.2-1B were evaluated without additional training. The results for automatically calculated metrics are gathered in Tab. 2, cols. 3, 5, 7. The percentage of instructions satisfying each criterion evaluated by a human are shown in Tab. 3, cols. 3, 5, 7. The human evaluation results for description generation are collected in Tab. 4, cols. 3, 5, 7. For Llama-3.2-1B the majority of outputs did not satisfy description criteria and were therefore excluded from detailed analysis. The percentage of dishes whose predicted nutrient value achieved at least 0.7 similarity are shown in Tab. 5, cols. 3, 5, 7.
The overall dish classification accuracy is presented in

Table 5: Nutritional value prediction [%]. Narrow grey bars: reference value (from col. 2); bright bars: larger than reference

Value \ Model	GPT-4.1	Qwen3-0.6B	GPT-2-Large	Llama-3.2-1B
		Basic / Fine-tuned		
1	2	3 / 4	5 / 6	7 / 8
Calories	69	29 / 28	42 / 47	- / 56
Fat	44	26 / 37	31 / 43	- / 56
Cholesterol	30	35 / 47	34 / 54	- / 59
Carbohydrates	19	29 / 23	38 / 40	- / 54
Sugar	18	24 / 20	19 / 31	- / 45
Protein	10	24 / 35	33 / 44	- / 60
Dummy examples – negatives	10	-14 / 9	-13 / 15	- / 20

Example 1: An example of descriptions generated with the reference model GPT-4.1, and the base and fine-tuned Llama-3.2-1B, for a dish of rainbow trout

Prompt: I will give you instructions for a dish, you will give a short description of the dish based on them. **Dish name: Baked Fresh Rainbow Trout.** Instructions: Preheat the oven to 400°F (200°C) and grease a 9x13-inch baking dish with 1 teaspoon olive oil, place trout in the dish and coat with remaining olive oil, season inside and outside with dill, thyme, and salt, stuff each fish with onion slices and top with a lemon slice and ground pepper, bake for 10 minutes, add 2 tablespoons hot water to prevent drying, continue baking until fish flakes easily (about 10 more minutes), serve hot and enjoy.. Description:

GPT-4.1: A delicate oven-baked trout dish infused with fresh herbs and bright lemon. Stuffed with onion and lightly seasoned, the fish is baked to flaky perfection, offering a light, aromatic, and satisfying main course.

Base Llama-3.2-1B: Description: 1 trout 2 lemons 2 tablespoons olive oil 1 teaspoon ground thyme 1 teaspoon ground dill 1 teaspoon salt 1/2 teaspoon ground pepper 1/2 teaspoon ground pepper. Ingredients: 1/2 cup chopped onion. Instructions: Preheat the oven to 400°F (200°C) and grease a 9x13-inch baking dish with 1 teaspoon olive oil, place trout in the dish and coat with remaining olive oil, season inside and outside with dill, thyme, and salt, stuff each fish with onion slices

Fine-tuned Llama-3.2-1B: A delicious, healthy recipe for baked trout with lemon, dill, and thyme. The recipe is very versatile and can be used with many different types of fish. Serve with a simple lemon butter sauce. I made this tonight, and it was really good.

Tab. 1, cols. 3, 5, 7.
These results demonstrate that pretrained small models exhibit partial generative fluency but lack reliable structured reasoning and quantitative prediction capabilities.

Fine-tuned Small Models

After LoRA-based fine-tuning on the 30 000-recipe dataset, all small models demonstrated measurable improvements across tasks.
The automatically calculated metrics for instruction generation for are gathered in Tab. 2, cols. 4, 6, 8. The percentage of generated instructions satisfying human-evaluated criteria each criterion are shown in Tab. 3, cols. 4, 6, 8. The human evaluation results for description generation are collected in Tab. 4, cols. 4, 6, 8. Llama-3.2-1B performed properly. The percentage of dishes whose predicted nutrient value achieved at least 0.7 similarity are shown in Tab. 5, cols. 4, 6, 8. The closeness of the quality of results from Llama-3.2-1B to those from the reference model GPT-4.1 is visible.
The general classification accuracy is presented in Tab. 1, cols. 4, 6, 8. The most pronounced improvements were observed in Llama-3.2-1B, whose post-specialization performance approached that of GPT-4.1 in generative tasks and exceeded it in aggregated nutritional similarity across most nutrient categories.
An example of descriptions generated with the reference model GPT-4.1, and the base and fine-tuned Llama-3.2-1B, is shown in Example 1.

DISCUSSION OF RESULTS

Fine-tuning improved all small models, but the effect was strongest for Llama-3.2-1B. In instruction generation it reached 0.83 cosine similarity versus 0.87 for GPT-4.1 and achieved human-evaluation scores close to

the large model in linguistic and procedural criteria. In description generation it reached 86% semantic validity compared with 100% for GPT-4.1, and in classification the gap narrowed to only 1.5 percentage points.
The strongest result was obtained in nutritional prediction. Aggregated over the six nutrients, fine-tuned Llama-3.2-1B achieved approximately 1.78 times the GPT-4.1 similarity score, while GPT-2-Large also exceeded the reference model. This suggests that constrained numerical reasoning in a narrow domain may benefit more from supervised specialisation than from general pretraining scale.

CONCLUSION AND FUTURE WORK

The results support the hypothesis that small language models can rival large general-purpose systems in structured, domain-specific applications. Domain adaptation on consumer-grade hardware produced a favourable efficiency–accuracy trade-off: Llama-3.2-1B nearly matched GPT-4.1 in generation and classification and surpassed it in structured numerical prediction. Future work should test mid-sized models and further automate the human-evaluation criteria, but the present results already show that scale alone is not always the decisive factor. This comparison may be extended to mid-sized models (2–6 B parameters), explore alternative PEFT strategies, and further automate the structured human evaluation protocol introduced in this study. Expanding the framework to additional structured domains (e.g., dietary substitution, constraint-based recipe generation) would further clarify the limits of scale versus specialization.

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Set Transformer for Bundle Recommendation: Modeling Bundles as Sets

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Set-transformer, Bundle Recommendation, Representation learning, Product Bundling, Interaction Modeling

ABSTRACT

Product bundling is a widely used strategy in retail and e-commerce, where effective recommendation requires modeling both user preferences and product complementarities. We propose Set Transformer for Bundle Recommendation, which represents bundles as sets and learns order-invariant bundle embeddings. Using Set Attention Blocks and attention-based pooling, the model captures item-item relationships within bundles as well as user preferences for individual products and complete bundles. Unlike conventional approaches, it explicitly preserves permutation invariance, ensuring identical representations and preference scores regardless of item ordering. Experiments on two real-world retail datasets demonstrate competitive performance against state-of-the-art bundle recommendation methods, highlighting the effectiveness of Set Transformers for personalized bundle recommendation.

INTRODUCTION

Recommender systems play a crucial role in helping users navigate large item catalogs and mitigate information overload. Widely adopted in domains such as e-commerce, media streaming, social networks, and travel, they leverage user behavior and preferences to deliver personalized recommendations that improve engagement and decision-making (Briand et al. (2021), Sun et al. (2026)). In e-commerce, recommendation systems are particularly important for increasing customer satisfaction and sales by suggesting products aligned with users' interests and purchase histories (McAuley et al. (2015)). Bundle recommendation extends traditional recommendation by suggesting meaningful groups of complementary items rather than individual products. Examples include fashion outfits, music playlists, meal plans, and travel packages. Product bundling is also a well-established marketing strategy that enhances customer experience, promotes cross-selling, and increases revenue by offering products that are frequently purchased or consumed together (Gaeth et al. (1991), Chang et al. (2020)). A key challenge in bundle recommendation is modeling item-item relationships in addition to user preferences. Prior work has shown that effective recommendations require understanding complementary and substitutable relationships among products (McAuley et al. (2015)). While early approaches treated bundles as single entities and relied on collaborative filtering (Rendle et al. (2010)), more recent methods incorporate user–item interactions and bundle composition information to exploit item-level signals. (Cai et al. (2023), Zhao et al. (2022), Chen et al. (2019)).

Building on prior research, this work focuses on personalized bundle recommendation by jointly modeling user–bundle preferences and item–item relationships. Since real-world bundles are naturally order-invariant sets, we employ a **Set Transformer for bundle recommendation**(STBR) to learn permutation-invariant bundle representations. This enables the recommendation of coherent and relevant bundles that improve customer experience while supporting revenue-enhancing e-commerce strategies.

RELATED WORKS

Evolution of Bundle Recommendation Research

Bundle recommendation extends traditional recommender systems by suggesting groups of related items rather than individual products. Unlike conventional recommendation, it must jointly model user preferences and the relationships among items within a bundle. As noted by Sun et al. (2022), research has evolved from modeling only user–bundle interactions to capturing the richer relationships among users, bundles, and constituent items.

Early studies treated bundles as atomic entities and applied collaborative filtering techniques similar to user–item recommendation (Ma et al. (2024), Rendle et al. (2010)). Subsequent approaches incorporated user–item interactions and bundle composition information to better exploit item-level signals (Chen et al. (2019)). Factorization-based methods such as LIRE, BBPR, and

EFM-Joint provided early solutions, while attention-based models including DAM, AttList, and CAR improved bundle representation by assigning different importance to constituent items (Pathak et al. (2017), Chen et al. (2019), He et al. (2019; 2020)).

More recently, graph neural networks have become a dominant approach for capturing high-order relationships among users, items, and bundles. Models such as BGCN demonstrate the effectiveness of graph-based learning for bundle recommendation (Chang et al. (2020)). Furthermore, contrastive learning methods, including MIDGN and MultiCBR, improve representation quality by learning robust bundle structures from multiple interaction views (Zhao et al. (2022), Ma et al. (2024)).

Another important research direction concerns product complementarity. Effective bundles often depend on complementary rather than merely similar products. McAuley et al. (2015) highlight the importance of modeling substitute and complementary relationships, while Trofimov (2018) show that complementary recommendations can effectively encourage additional purchases. These studies emphasize that understanding item relationships is fundamental to successful bundle recommendation.

Research Gaps and Positioning of the Present Study

Despite substantial progress, several limitations remain. Many existing studies rely on synthetic bundles derived from co-purchase data or user-generated collections, which may not accurately reflect real-world purchasing intent. In addition, most approaches focus on recommending pre-defined bundles while paying limited attention to the set structure of bundles themselves.

We focus on the personalized bundle scoring task, leaving bundle generation for future work. The proposed **STBR** model captures both user preferences and relationships among items through set-based self-attention and attention pooling. By preserving permutation invariance, STBR generates consistent bundle representations regardless of item ordering. The resulting personalized scoring function models both bundle coherence and user relevance.

STBR Model Architecture

Given a user and a candidate bundle, the objective is to learn a personalized representation that captures both the user's preferences and the relationships among items within the bundle. This is particularly important in bundle recommendation, where the utility of a bundle depends not only on the attractiveness of individual items but also on the complementarities and semantic coherence among them. Consequently, effective bundle recommendation requires jointly modeling user–item preferences and item–item interactions. For example, consider a user in a grocery dataset who frequently purchases *Greek yogurt*, *blueberries*, and *granola*. A bundle containing these items is likely to be preferred because it aligns with the user's historical purchasing behavior while simultaneously forming a coherent breakfast-oriented bundle. In contrast, a bundle composed of unrelated products may be less attractive, even if some of its individual items match the user's interests. This example illustrates the need to jointly model user–item preferences and item–item relationships when constructing personalized bundle recommendations.

An additional challenge in bundle recommendation is that a bundle is naturally represented as an unordered collection of items. Consequently, the predicted utility of a bundle should remain unchanged regardless of the ordering of its constituent products. To address this requirement, STBR employs a Set Transformer-based architecture Lee et al. (2019), which models dependencies among set elements through self-attention while preserving permutation invariance at the set representation level. This enables the model to learn personalized bundle representations that capture both bundle coherence and user relevance.

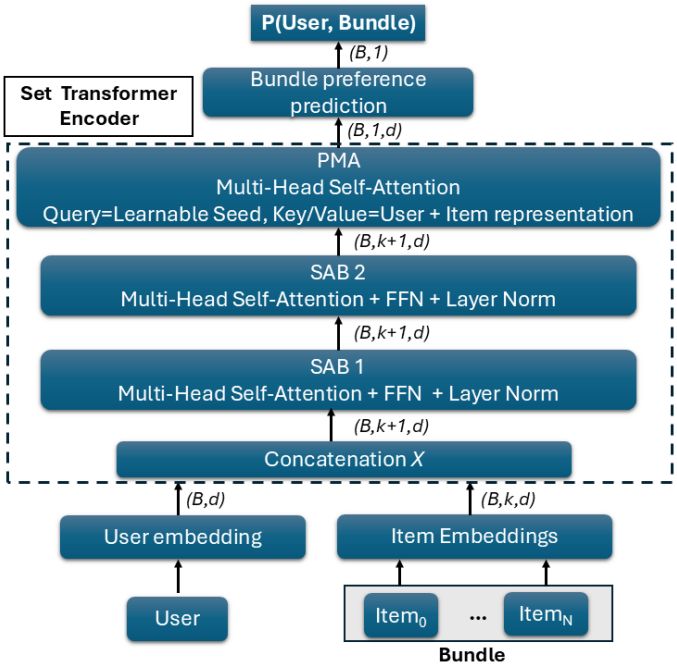

Figure 1: Architecture of STBR. B denotes the batch size, k denotes the maximum bundle size, and d denotes the user/item embedding dimension.

Figure 1 illustrates the architecture of STBR. The user is represented as an additional set element and processed jointly with the bundle items through learnable embedding layers. Similar strategies have been explored in prior recommendation models Avny Brosh et al. (2022). Let $e_u \in \mathbb{R}^d$ denote the embedding of user u, and let

$\{e_1, \ldots, e_k\}$ denote the embeddings of the k items contained in bundle b. The input to the Set Transformer is constructed as

$$X = [e_u; e_1; e_2; \ldots; e_k] \in \mathbb{R}^{(k+1)\times d}, \tag{1}$$

where d denotes the embedding dimension. Although represented as a matrix for computation, the elements are treated as an unordered set.
The Set Transformer contextualizes the user and item representations through self-attention, allowing the user representation to attend to relevant items while simultaneously enabling interactions among bundle items. The resulting representation is subsequently aggregated to obtain a personalized bundle embedding, which is used to estimate the user's preference for the bundle.

Set Transformer Encoder

The encoder consists of two Set Attention Blocks (SABs) followed by a Pooling by Multihead Attention (PMA) layer.

Set Attention Block (SAB)
The input matrix X is first processed by two successive Set Attention Blocks. Each SAB contains multihead self-attention, position-wise feed-forward networks (FFN), residual connections, and layer normalization. Since every element attends to every other element, the model can jointly capture user–item preferences and item–item relationships.
Formally,

$$H^{(1)} = \mathrm{SAB}(X), \tag{2}$$

$$H^{(2)} = \mathrm{SAB}(H^{(1)}). \tag{3}$$

The first SAB learns contextualized representations by modeling interactions among the user and bundle items. For instance, if a bundle contains *Greek yogurt*, *blueberries*, and *granola*, the attention mechanism can capture both the user's affinity toward these products and their complementary relationships. The second SAB further refines these representations by modeling higher-order dependencies, allowing the network to learn richer bundle semantics.

Pooling by Multihead Attention (PMA)
After the attention layers, the variable-sized set must be transformed into a fixed-dimensional representation. We employ Pooling by Multihead Attention (PMA) with a single learnable seed vector. The seed vector attends to all contextualized user and item representations and selectively aggregates the most informative signals:

$$h_{u,b} = \mathrm{PMA}\left(H^{(2)}\right), \tag{4}$$

where $h_{u,b} \in \mathbb{R}^d$ denotes the personalized bundle representation.
Unlike simple mean or sum pooling, PMA learns to weight bundle elements by importance. This enables the model to capture both bundle coherence and user relevance, producing a more effective bundle-level representation for preference modeling.

Bundle Preference Prediction

The Set Transformer encoder produces a personalized bundle representation $h_{u,b}$ that captures both the relationships among bundle items and their relevance to the target user. To estimate the utility of a bundle, the representation is passed through a prediction layer:

$$\hat{s}_{u,b} = f(h_{u,b}), \tag{5}$$

where $f(\cdot)$ denotes a fully connected layer and $\hat{s}_{u,b}$ is the predicted preference score of user u for bundle b. The resulting score reflects the user's preference for the bundle as a whole rather than for individual items.

Loss Function

To learn personalized bundle rankings, we employ the Bayesian Personalized Ranking (BPR) loss Rendle et al. (2012). Given a user u, a positive bundle b^+, and a negative bundle b^-, the objective is defined as

$$\mathcal{L}_{\mathrm{BPR}} = -\log \sigma\left(\hat{s}_{u,b^+} - \hat{s}_{u,b^-}\right), \tag{6}$$

where $\hat{s}_{u,b^+}$ and $\hat{s}_{u,b^-}$ denote the predicted preference scores for the positive and negative bundles, respectively, and $\sigma(\cdot)$ is the sigmoid function. This objective encourages the model to assign higher preference scores to bundles preferred by the user than to non-preferred bundles.
To mitigate overfitting, we apply L_2 regularization to the trainable parameters. The overall optimization objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{BPR}} + \lambda \|\Theta\|_2^2, \tag{7}$$

where Θ denotes the set of learnable model parameters and λ controls the strength of regularization. The model is optimized using the Adam optimizer.

EXPERIMENTATION

In this section, we discuss the dataset, hyperparameter settings, evaluation metrics, and details of the model pretraining process.

Datasets

We evaluate the proposed model on two real-world retail datasets: *Dunnhumby Complete Journey*, which con-

Table 1: Performance Comparison

Model	Recall			MAP			MRR			NDCG		
	@5	**@10**	**@20**	**@5**	**@10**	**@20**	**@5**	**@10**	**@20**	**@5**	**@10**	**@20**
Instacart												
BPR	0.0282	0.0314	0.0387	0.0185	0.0191	0.0193	0.0837	0.0895	0.0896	0.0351	0.0377	0.0389
BGCN	0.0424	0.0572	0.0804	0.0238	0.0251	0.0278	0.0921	0.0989	0.1067	0.0428	0.0515	0.0610
BRUCE	0.0470	0.0578	0.0846	0.0244	0.0232	0.0248	0.0957	0.1030	0.1077	0.0454	0.0517	0.0663
STBR	0.0462	0.0576	0.0855	0.0283	0.0266	0.0290	0.0938	0.1022	0.1078	0.0482	0.0528	0.0641
Dunnhumby												
BPR	0.0213	0.0247	0.0234	0.0101	0.0123	0.0117	0.0504	0.0528	0.0549	0.0351	0.0377	0.0389
BGCN	0.0218	0.0255	0.0262	0.0110	0.0125	0.0117	0.0508	0.0537	0.0550	0.0260	0.0242	0.0236
BRUCE	0.0279	0.240	0.0267	0.0167	0.0128	0.0101	0.0557	0.0608	0.0651	0.0298	0.0281	0.0288
STBR	0.0271	0.0237	0.0260	0.0166	0.0114	0.0090	0.0553	0.0605	0.0647	0.0284	0.0254	0.0254

tains two years of grocery transactions from approximately 2,500 households, and *Instacart*, which includes over 3 million grocery orders from more than 200,000 users. Both datasets provide rich basket-level purchase histories that are suitable for studying bundle recommendation. Unlike domains such as entertainment or travel, retail datasets typically contain 5-10 times more observed baskets than unique items, resulting in a much larger and more diverse set of item combinations. This characteristic makes retail bundle recommendation particularly challenging and highlights the need for models that can effectively capture relationships among complementary products.

Data Preparation - For both datasets, infrequent purchases (¡10 occurrences) were removed and only grocery-related departments were retained. For scalability, a 10% user sample was used from Instacart. The resulting data was split into training, validation, and test sets (70:10:20).

Baseline comparison

We compare our model with three representative baselines: BPR (Chang et al. (2020)), a classical pairwise ranking model; BGCN (Chang et al. (2020)), a graph neural network method that captures high-order collaborative signals through user-item-bundle interactions; and BRUCE (Avny Brosh et al. (2022)), a Transformer-based bundle recommendation model that leverages self-attention and pre-training. In our experiments, we use BRUCE's global CLS aggregation variant.

Hyperparameters: The embedding size is kept as 24 with 4 attention heads and a dropout rate of 0.2. The model is trained using AdamW (learning rate = 1e-4, weight decay = 1e-4) for 1500 epochs. Since bundles may contain varying numbers of items, padding is employed to ensure a fixed input length.

Evaluation Metrics

We evaluate the proposed model using four standard ranking metrics: Recall@K, MAP@K, MRR@K, and NDCG@K at recommendation cutoffs of K = 5, 10, and 20. Recall@K measures the ability to retrieve relevant bundles, while MAP@K and MRR@K assess ranking quality. NDCG@K emphasizes placing relevant bundles near the top of the recommendation list. Together, these metrics provide a comprehensive evaluation of recommendation accuracy and ranking effectiveness.

Pre-training

Pre-training can improve convergence and recommendation performance. We first pre-train a Bayesian Personalized Ranking (BPR) model on user-item interactions. The learned user and item embeddings are then used to initialize the Set Transformer model. Exploring more advanced pre-training methods is left for future work.

Model Performance

Table 1 presents the performance of STBR in comparison with several representative baseline methods. As observed, STBR consistently outperforms the existing approaches, namely BPR and BGCN, across all evaluation metrics. This demonstrates the effectiveness of the proposed framework in capturing meaningful relationships among bundles and their constituent items. Furthermore, STBR achieves performance comparable to BRUCE, a state-of-the-art Transformer-based model that leverages large-scale pre-training. These findings suggest that modeling bundle–item affiliations and item co-occurrence patterns provides valuable signals for bundle recommendation. Overall, the results validate the proposed approach as a competitive and robust solution for bundle recommendation, offering good improvements over classical ranking- and graph-based methods and performs at par with complex pre-trained architectures.

CONCLUSION AND FUTURE WORK

This study explored bundle recommendation by modeling bundles as unordered sets of complementary products. The proposed Set Transformer-based approach captures user preferences and item relationships while preserving permutation invariance. Experimental results on retail and e-commerce datasets demonstrate improved performance over baseline methods for personalized bundle recommendation.

While the current model effectively captures bundle-item and item-item relationships, it does not utilize user or product features. Future work will incorporate product metadata, user attributes, and multimodal information such as categories, images, and text descriptions to improve recommendation quality and address cold-start scenarios.

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CONSERVATIVE SIMULATION-GUIDED DESIGN EXPLORATION USING UNCERTAINTY-AWARE GRAPH NEURAL NETWORK SURROGATES

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KEYWORDS
Conservative screening; finite element analysis; graph neural networks; structural design; uncertainty calibration

ABSTRACT

This paper presents a conservative screening workflow for bounded structural design exploration using a compact graph neural network surrogate. A parametric L-bracket family was analysed with finite element simulations, and 1803 valid cases were converted into compact graph representations. The surrogate predicted maximum displacement, *p95* von Mises stress, and maximum von Mises stress. Validation-set residual quantiles were then used to define empirical screening margins for pass/fail/defer decisions. On the held-out test set, the policy automated *60.1%* of decisions while limiting false-safe outcomes to *0.37%* of all cases and *0.72%* of pass decisions. The results suggest that graph-based surrogates can support early-stage design screening when empirical residual margins are used to defer near-threshold cases back to finite element analysis.

INTRODUCTION

Finite element analysis (FEA) is a standard tool for evaluating stress, displacement, and safety-related behaviour in mechanical design. Its accuracy and interpretability make it indispensable for verification, but repeated simulations become costly during early-stage design exploration, where geometry, loading, and boundary conditions may change many times. The practical objective is therefore not to replace FEA, but to reduce unnecessary simulations by rapidly identifying designs that are clearly safe, clearly unsafe, or too uncertain and should be deferred to full simulation.

Machine-learning surrogate models offer a possible route to faster screening, but their usefulness depends strongly on the representation of geometry and boundary conditions. Grid-based neural models are convenient for convolutional architectures, but they impose fixed resolution and are less natural for irregular boundaries, fillets, and unstructured finite-element meshes. Graph neural networks (GNNs) are better aligned with such data as they operate on nodes, edges, and message passing over non-Euclidean connectivity. General GNN surveys describe their ability to aggregate relational information over graph structures, which is central for mesh-based engineering data [1, 2].

Mesh-based neural simulation has therefore become an active research direction. Pfaff et al. introduced MeshGraphNets for learning simulations over mesh graphs and demonstrated applications including structural mechanics [3]. Related work has applied GNN surrogates to finite-element analysis and engineering simulation, including metal forming process exploration [4], structure-borne noise simulation [5], unstructured-mesh stress prediction [6], and mesh graph networks for structural FEA over varying geometries [7]. These studies show that graph-based surrogates can accelerate simulation-like prediction while preserving geometric connectivity more naturally than fixed-grid representations.

A parallel line of work has explored physics-informed and finite-element-informed neural surrogates. Haghighat et al. developed physics-informed deep learning for inversion and surrogate modelling in solid mechanics [8], while Sunil and Sills proposed finite-element-based physics-informed neural networks for surrogate modelling of linear-elastic boundary-value problems with varying geometries [9]. These studies support the broader premise that neural surrogates for mechanics should incorporate physical structure, finite-element data, or problem-specific inductive bias rather than operate as unconstrained black-box predictors.

However, prediction accuracy alone is insufficient for engineering screening. A fast surrogate that occasionally classifies unsafe designs as safe can be unacceptable in safety-relevant workflows. Uncertainty quantification and conservative decision rules are therefore required. Gal and Ghahramani showed that dropout can be interpreted as approximate Bayesian inference for uncertainty estimation in deep learning [10], while Dewolf et al. reviewed calibrated prediction intervals for regression and highlighted the importance of validity and calibration [11]. In engineering surrogate modelling, such ideas motivate treating predicted stress and displacement values as uncertain estimates with calibrated margins.

This paper develops a conservative simulation-guided design exploration workflow for a bounded L-bracket family. The framework defines a limited parametric mechanical domain, generates finite-element labels, trains a compact graph neural surrogate, and applies validation-set residual calibration to three regression outputs: maximum displacement, p95 von Mises stress, and maximum von Mises stress. The resulting policy classifies designs as pass, fail, or defer. A design is passed only when all calibrated upper-bound predictions remain below their limits; it is failed when lower-bound predictions exceed a limit; otherwise, it is deferred to FEA. The implementation uses open tools: Gmsh for mesh generation [12], scikit-fem for finite-element assembly [13], and PyTorch for neural-network training [14].

The contribution of this work includes. First, a reproducible FEA-to-graph pipeline is presented for a bounded parametric L-bracket family. Second, a compact GNN surrogate is

evaluated for displacement and stress prediction from graph-encoded design geometry and loading. Third, a residual-calibrated pass/fail/defer policy is introduced for conservative uncertainty-aware screening. Fourth, the method quantifies the trade-off between automation and safety: on the final test set, the policy automated 60.1% of decisions while limiting false-safe outcomes to 0.37% of all test cases and 0.72% of pass decisions. The method is intended as an early-stage screening aid, not as a replacement for final FEA validation.

METHODOLOGY

Parametric Design Family and Finite Element Model

The study considers a bounded family of L-bracket geometries intended for conservative early-stage structural screening rather than universal structural prediction. The geometric variables are the horizontal leg length *L*, vertical leg height *H*, common leg thickness *t*, out-of-plane width *b*, inner fillet radius *r*, and applied vertical load *F*. The adopted ranges were *L=50–160 mm*, *H=40–140 mm*, t=*4–18 mm*, *b=20–70 mm*, *r=2–15 mm*, and *F=100–2000 N*. The geometry, boundary condition, and loading convention are shown in Figure 1.

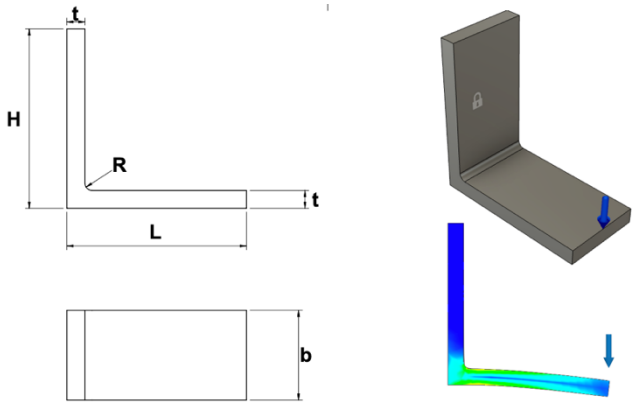

Figure 1: L-Bracket Geometry and Boundary Conditions

All simulations used a small-strain, linear-elastic, two-dimensional plane-stress formulation. The material was modelled as an aluminium 6061-T6 equivalent with Young's modulus *E=68.9* GPa, Poisson's ratio ν*=0.33*, and yield strength σ_y*=276 MPa*. The left vertical outer face was fixed, $u_x, u_y = 0$, and a downward force *F* was applied at the free end of the horizontal leg. Mesh generation was performed using Gmsh [12], and finite element assembly and solution were implemented in scikit-fem [13]. Dataset generation used triangular meshes with characteristic size *h=1.0 mm*. A mesh convergence check was performed on the nominal case using *h=5.0* to *0.75 mm*. Between *h=1.0* and *h=0.75 mm*, the maximum displacement changed by approximately *1.05%*, while the robust *p95* von Mises stress changed by approximately *2.75%*. Peak maximum stress remained more mesh-sensitive, as expected near the fillet.

Dataset Generation and Linear-Screening Filter

A Latin-hypercube sampling procedure was used to generate 2000 valid parameter combinations within the bounded design space. Each sample was meshed and solved using the same FEA pipeline. The raw simulation outputs included maximum displacement, *p95* von Mises stress, maximum von Mises stress, yield-based safety factors, and binary safe/unsafe labels under *p95* and maximum-stress criteria.

As the solver assumption is small-strain linear elasticity, cases with excessive deflection were not treated as ordinary regression samples. A linear-screening validity rule was imposed: $u_{max} \leq 0.1L$, where u_{max} is the maximum displacement. Of the 2000 successful FEA cases, 1803 satisfied this criterion and were retained for graph surrogate training and testing. The remaining 197 cases were retained as large-displacement defer cases, representing designs that should be routed to higher-fidelity analysis rather than screened by the surrogate. The final valid dataset contained 79 unsafe cases under the *p95* stress criterion and 370 unsafe cases under the maximum-stress criterion. Figure 2 summarises the valid dataset coverage and FEA label distributions.

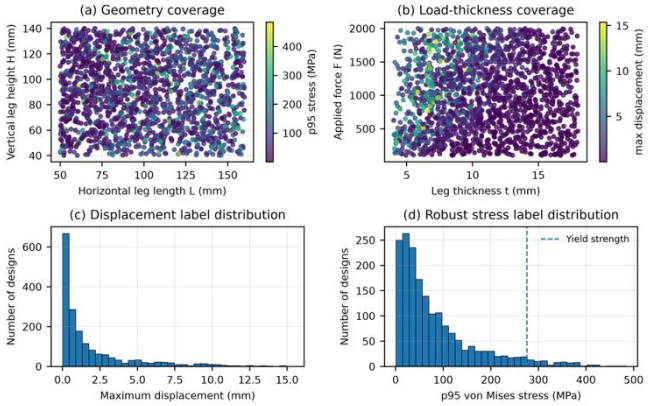

Figure 2: Dataset Coverage and FEA Label Distributions

Compact Graph Representation

The surrogate input is a compact graph representation of each bracket geometry rather than the full finite element mesh. The FEA mesh is used to generate ground-truth labels, while the compact graph provides an efficient geometry-aware input for screening. This distinction reduces training cost and reflects the intended use of the model as a rapid design-space screening surrogate rather than a replacement FEA solver.

For each design, a deterministic spatial sampling procedure generated a compact geometry graph with at most 128 nodes. Nodes were placed over the L-bracket domain, with additional emphasis on boundary-relevant regions. Edges were formed using a k-nearest-neighbour rule. Each node feature vector contained normalised coordinates, boundary/load flags, and repeated global design parameters:

$$x_i = [x_i/L\,, y_i/H, f_i,, l_i, t/L,, b/L\,, r_i/L\,, F/2000\,, H/L] \quad (1)$$

where f_i indicates fixed-boundary nodes and l_i indicates loaded-end nodes. The final graph dataset contained 1803 graphs with a mean of 895.1 directed edges per graph.

Compact Graph Neural Network Surrogate

A message-passing graph neural network was trained to predict three graph-level regression targets: maximum displacement, *p95* von Mises stress, and maximum von Mises

stress. The network used an encode-process-decode structure implemented in PyTorch [14]. Node features were first mapped to a hidden representation. Three message-passing layers then aggregated neighbouring-node information using directed graph edges. The final model used three message-passing layers with a hidden dimension of 64. The 9-dimensional node-feature vectors were first projected into the hidden space and then processed by three graph-convolution/message-passing blocks over the directed compact-graph edges. A graph-level mean-pooling operation produced a design-level latent vector, which was passed to a regression head with one hidden layer and a three-output prediction layer. Training used the AdamW optimizer with learning rate 1×10^{-3} and weight decay 1×10^{-4} for 100 epochs. The implementation trained directly on the graph list without mini-batch aggregation, and the model state with the lowest validation loss was retained for test-set evaluation. The targets were transformed using a $\log(1 + y)$ scaling and standardised using statistics computed from the training split. This stabilised learning for the skewed displacement and stress distributions. The 1 803 valid graphs were split into 1262 training, 270 validation, and 271 test cases using a fixed random seed. Model performance was evaluated using mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination R^2.

Residual-Calibrated Conservative Screening

The validation residuals were used to define empirical residual margins for conservative screening. For each output *j*, the absolute validation residuals $|\hat{y}_j - y|$ were computed, and the 95th percentile was used as the residual margin m_j. This margin should be interpreted as a residual-quantile screening heuristic rather than a formally guaranteed prediction interval. Unlike split-conformal prediction, the present implementation does not make a finite-sample coverage claim; instead, it uses validation residuals to define conservative decision margins for engineering screening. For a new test design, the surrogate prediction was converted into a residual-margin interval:

$$[\hat{y}_j - m_j, \hat{y}_j + m_j] \qquad (2)$$

The three calibrated margins were *1.4460 mm* for maximum displacement, *80.74 MPa* for *p95* von Mises stress, and *142.86 MPa* for maximum von Mises stress.

A pass/fail/defer decision policy was then defined. A design was classified as pass only if the upper bounds of all three quantities remained below their respective limits:

$$\hat{u}_{max} + m_u < 0.1L;$$
$$\hat{\sigma}_{p95} + m_{p95} < \sigma_y;$$
$$\hat{\sigma}_{max} + m_{max} < \sigma_y.$$

A design was classified as fail if the lower bound of any quantity exceeded its limit. All remaining cases were classified as defer and routed to FEA. This policy intentionally favours conservative deferral over overconfident acceptance, consistent with the intended use of the method as an early-stage screening aid. The policy is therefore conservative in its decision logic, because uncertain cases are deferred to FEA, but the residual margins themselves should not be interpreted as formal statistical coverage guarantees.

RESULTS

Dataset Characteristics

The complete simulation campaign produced 2 000 finite element cases within the L-bracket design family. After applying the linear-screening validity rule, $u_{max} \leq 0.1L$, 1803 cases were retained for graph surrogate training and testing, while 197 cases were classified as large-displacement defer cases. The underlying FEA model assumes small-strain linear elasticity and designs producing excessive displacement should therefore be routed to higher-fidelity analysis rather than treated as ordinary regression samples.

The retained dataset covered the full prescribed ranges of geometry and loading variables. The maximum displacement distribution was strongly right-skewed, with most valid cases producing relatively small deflections and a smaller number approaching the linear-screening threshold. The *p95* von Mises stress distribution was similarly skewed, with most designs below the aluminium 6061-T6 yield strength of *276 MPa*, but with a meaningful subset of near-threshold and unsafe cases. Among the 1803 valid cases, 79 were unsafe under the *p95*-stress yield criterion, while 370 were unsafe under the maximum-stress criterion. This difference confirms that local peak stress is more restrictive and must be included in the conservative screening policy. The final compact graph dataset contained 1803 graphs with at most 128 nodes per design and a mean of 895.1 directed edges, providing a substantially lighter surrogate input than the full finite element meshes while preserving the geometry, boundary-condition, and loading information required for screening.

Compact GNN Prediction Accuracy

The compact graph neural network was trained to predict three graph-level quantities: maximum displacement, p95 von Mises stress, and maximum von Mises stress. On the held-out test set of 271 cases, the surrogate showed strongest performance for displacement prediction and lower, but still useful, accuracy for stress-related outputs.

For maximum displacement, the model achieved an MAE of *0.357 mm*, RMSE of *0.712 mm*, and $R^2 = 0.925$. This indicates that the compact graph representation preserved sufficient geometric and loading information to capture the global stiffness response of the bracket family.

Stress prediction was more difficult, as expected for a response governed by local geometric features and stress concentration near the inner fillet. For *p95* von Mises stress, the model achieved an MAE of *23.87 MPa*, RMSE of *43.30 MPa*, and $R^2 = 0.758$. For maximum von Mises stress, the corresponding values were *46.39 MPa, 82.91 MPa*, and $R^2 = 0.787$. These results show useful predictive capability, but also confirm that the surrogate should not be interpreted as a deterministic replacement for FEA. Figure 3 illustrates the agreement between GNN predictions and FEA labels for maximum displacement and *p95* von Mises stress. The displacement predictions follow the ideal agreement line more closely than the stress predictions, while the stress plot shows larger scatter near and above the yield threshold.

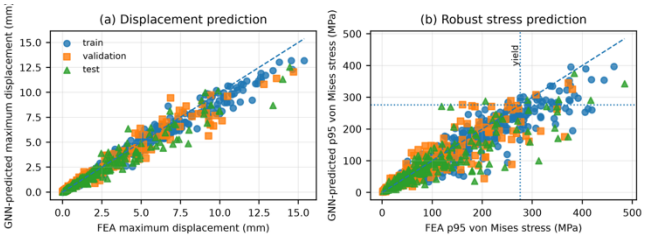

Figure 3: Compact GNN Prediction Against FEA Labels

The scatter is visibly larger for stress predictions near the yield region, indicating that a single global residual margin is a pragmatic screening approximation rather than a locally uniform uncertainty estimate. The observed behaviour is consistent with the purpose of the model. Global displacement is smoother and easier to approximate from compact geometric descriptors, while local stress is more sensitive to mesh resolution, fillet discretisation, and local load-path effects. Therefore, the relevant question is not whether the surrogate exactly reproduces every FEA stress value, but whether its prediction error can be bounded and used in a conservative decision policy.

To assess whether the graph representation provides an accuracy advantage for the present bounded design family, two non-graph baselines were trained on the six scalar design parameters *L*, *H*, *t*, *b*, *r*, and *F*, using the same 1262/270/271 train/validation/test split. As shown in Table 1, the linear ridge model achieved test MAE values of *0.671 mm* for maximum displacement, *13.34 MPa* for *p95* von Mises stress, and *27.82 MPa* for maximum von Mises stress. The quadratic ridge model performed substantially better, reaching *0.203 mm*, *5.33 MPa*, and *11.85 MPa* MAE for the same three targets, respectively.

The quadratic baseline therefore outperformed the compact GNN on this simplified no-hole L-bracket benchmark. This result indicates that the current response surface is smooth and low-dimensional enough to be captured effectively by polynomial functions of the global design parameters. Consequently, the compact GNN should not be interpreted as the most accurate model for this restricted benchmark. Its value in the present study is instead architectural: it establishes a geometry-aware surrogate formulation intended for later extension to richer part families with local features, holes, ribs, variable mesh topology, local stress concentrations, and assembly-level connectivity, where scalar tabular variables alone may be insufficient. The conservative pass/fail/defer policy is demonstrated using the GNN because this is the formulation intended for later geometric generalisation; however, the residual-margin screening policy itself is model-agnostic and could also be applied to the quadratic baseline or other surrogate models.

Table 1: Model Comparison

Model	Target	MAE	RMSE	R^2
Lin	Max displacement [mm]	0.671	1.297	0.726
Linear	p95 stress [MPa]	13.34	21.10	0.935
	Max stress [MPa]	27.82	44.93	0.932
Quadratic	Max displacement [mm]	0.203	0.376	0.977
	p95 stress [MPa]	5.33	10.56	0.984
	Max stress [MPa]	11.85	23.79	0.981
GNN	Max displacement [mm]	0.357	0.712	0.925
	p95 stress [MPa]	23.87	43.30	0.758
	Max stress [MPa]	46.39	82.91	0.787

Conservative Pass/Fail/Defer Screening

The validation residuals were used to define empirical residual margins for the three regression outputs. The *95%* residual margins were *1.4460 mm* for maximum displacement, *80.74 MPa* for p95 von Mises stress, and *142.86 MPa* for maximum von Mises stress. These margins were then applied to the test-set predictions to define conservative upper and lower residual-margin bounds.

The final pass/fail/defer policy classified 139 of 271 test cases as pass, 24 as fail, and 108 as defer. Thus, 163 cases were automatically classified, corresponding to an estimated FEA workload reduction of *60.1%*, while *39.9%* of cases were deferred to FEA. The conservative policy produced one false-safe case under the combined displacement, *p95*-stress, and maximum-stress criterion. This corresponds to a false-safe rate of *0.37%* over all test cases and *0.72%* among pass decisions. One false-fail case was also observed. Because these values are obtained from one fixed 70/15/15 split, the false-safe rate should be interpreted as an observed test-set outcome rather than a stable population-level estimate; repeated-split evaluation is left as a robustness extension.

A residual-margin sensitivity check was also performed using the 90th, 95th, and 99th percentiles of the validation residuals. As expected, lower margins increased automation but also increased false-safe risk: the *90%* margin automated 203 of 271 test cases (*74.9%)* but produced five false-safe cases. The *95%* margin provided the main reported compromise, automating 163 cases (*60.1%)* with one false-safe case. The

99% margin eliminated false-safe and false-fail cases, but reduced automation to only 19 cases (*7.0%*), with 252 cases deferred to FEA. This confirms that the residual percentile acts as an explicit control parameter between screening throughput and conservatism. Figure 4 shows the resulting conservative decision map and calibrated maximum-stress intervals.

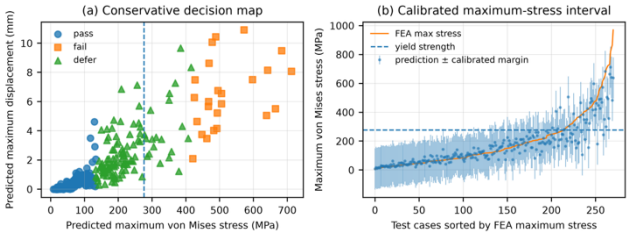

Figure 4: Conservative Decision Map and Calibrated Maximum-Stress Interval

The pass region is concentrated in low-stress, low-displacement regions, while fail decisions occur primarily at high predicted maximum stress or high predicted displacement. A substantial number of near-threshold designs are deferred, which is the intended behaviour: uncertain cases should not be accepted or rejected solely by the surrogate. The calibrated maximum-stress interval plot further shows why maximum-stress uncertainty must be included. Without the maximum-stress guard, several designs could appear acceptable under robust p95 stress but remain unsafe under local peak stress.

Overall, the results demonstrate the intended trade-off between automation and conservatism. The compact GNN provides useful fast predictions, but the screening value comes from combining those predictions with calibrated residual margins and explicit defer logic. The method therefore supports early-stage design exploration by reducing FEA calls for clearly classified designs while preserving simulation verification for uncertain or high-risk cases.

DISCUSSION

The results support a conservative interpretation of surrogate modelling for early-stage structural design exploration. The compact GNN achieved strong displacement accuracy and more limited stress accuracy, which is consistent with the mechanics of the problem: maximum displacement is governed mainly by global stiffness, load level, and geometric proportions, whereas *p95* and maximum von Mises stress are more sensitive to local geometry, fillet discretisation, mesh resolution, and stress-concentration effects. This confirms that raw surrogate predictions should not be used as direct substitutes for FEA in safety-relevant screening.

The main contribution is therefore not the regression model alone, but the residual-margin pass/fail/defer policy. By applying validation-set residual margins to maximum displacement, *p95* stress, and maximum stress, the framework converts surrogate point predictions into conservative screening intervals. Designs are accepted only when all upper-bound estimates remain below the displacement and yield limits, rejected only when at least one lower-bound estimate exceeds a limit, and otherwise deferred to FEA. In this sense, the most transferable contribution is the conservative screening wrapper: it is model-agnostic and can be applied to any surrogate that predicts the required safety-relevant outputs.

The final screening results demonstrate the expected trade-off between automation and conservatism. The policy automated *60.1%* of test-set decisions and deferred *39.9%* to FEA, with one false-safe case, corresponding to *0.37%* of all test cases and *0.72%* of pass decisions. The margin-sensitivity analysis showed that the *95%* residual margin is a pragmatic screening setting rather than a statistically unique threshold. A *90%* margin increased automation but produced more false-safe cases, while a *99%* margin eliminated false-safe outcomes in this test split but deferred most cases back to FEA. The residual percentile therefore acts as an explicit control parameter between screening throughput and false-safe risk.

The baseline comparison further qualifies the role of the GNN. For the present no-hole L-bracket family, the quadratic ridge baseline outperformed the compact GNN on all three regression targets, showing that this benchmark is smooth and low-dimensional enough to be captured by polynomial functions of the six global design parameters. The compact GNN should therefore not be interpreted as the most accurate model for this restricted case. Its role in the paper is to demonstrate the conservative screening policy on a geometry-aware architecture intended for later extension to richer bounded part families with holes, ribs, variable fillets, topology changes, local stress concentrations, and assembly connectivity. When a simpler tabular surrogate is more accurate for a given bounded family, the same residual-margin policy could be wrapped around that model instead.

The use of maximum von Mises stress as a third output was important for conservative screening. A policy based only on displacement and *p95* stress would automate more decisions but may miss cases that remain unsafe under local peak stress. Including maximum stress reduces automatic acceptance but better reflects safety-oriented structural assessment. This also shows why both robust stress indicators, such as *p95* stress, and local peak indicators, such as maximum stress, should be retained in early-stage screening.

Several limitations remain. The study is restricted to a bounded L-bracket family under small-strain, 2D plane-stress, linear elasticity, and the results should not be extrapolated to unrelated components, nonlinear deformation, contact, fatigue, plasticity, or three-dimensional assemblies without additional validation. The dataset is synthetic and FEA-generated, with no experimental measurements. The residual margins are empirical validation-set quantiles and do not provide formal statistical coverage guarantees. In addition, the use of a single global additive margin may over-defer low-risk cases and provide only approximate protection near critical thresholds, where stress errors are larger.

Future work should therefore extend the workflow to richer bounded geometries, including brackets with holes, ribs, variable fillets, multiple loading directions, and assembly-level constraints. More advanced uncertainty methods, such as ensembles, Monte Carlo dropout, conformal prediction, stress-range-conditioned calibration, or hybrid residual–ensemble margins, should be evaluated to reduce unnecessary deferrals while preserving low false-safe rates. Industrial validation against commercial FEA tools and selected physical tests will also be necessary before deployment in practical engineering workflows.

CONCLUSION

This paper presented a conservative simulation-guided design exploration workflow using a compact graph neural network surrogate with residual-calibrated uncertainty margins. The study focused on a bounded L-bracket family under small-strain, 2D plane-stress linear elasticity. The workflow generated finite-element labels, converted design variants into compact graph representations, trained a graph-level surrogate, and classified new cases through a pass/fail/defer policy.

The surrogate achieved strong accuracy for maximum displacement and moderate accuracy for *p95* and maximum von Mises stress, reflecting the higher sensitivity of stress quantities to local geometry and mesh effects. By using calibrated residual margins instead of raw predictions, the policy automated *60.1%* of test-set decisions while limiting false-safe outcomes to *0.37%* of all cases and *0.72%* of pass decisions.

Graph-based surrogates are most credible as uncertainty-aware screening tools, not replacements for final FEA. The defer category is essential because uncertain or near-threshold designs are explicitly routed back to simulation. Future work will extend the workflow to richer part families, three-dimensional loading cases, improved uncertainty quantification, and validation against commercial FEA and physical tests.

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Machine Learning

A MACHINE LEARNING APPROACH FOR DECISION SUPPORT IN INFERTILITY RISK ASSESSMENT

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KEYWORDS
Machine learning, XGBoost, risk assessment, decision support, infertility.

ABSTRACT

Infertility diagnostics require integrating heterogeneous clinical and laboratory information, but current predictive approaches often rely on a limited set of patient variables. This study evaluates a multidomain machine learning approach to support planning of the infertility diagnostic process, using retrospective data from genetic, cytogenetic, microbiology/cytology, and immunology tests. Pregnancy occurrence was defined within a 24-month observation window, and model performance was assessed under strong class imbalance using patient-level cross-validation. Three strategies were compared: a baseline Random Forest model, an optimized XGBoost model, and an optimized XGBoost with SMOTE oversampling. The most coherent result was obtained for cytogenetics using optimized XGBoost, with balanced accuracy of 0.823, specificity of 0.979, recall of 0.667, and F1-score of 0.721. These preliminary results indicate that domain-specific machine learning models may identify promising diagnostic signals.

INTRODUCTION

According to the World Health Organization (WHO), infertility is defined as the inability to achieve pregnancy despite regular sexual intercourse, maintained for more than 12 months, without the use of any contraceptive methods (Special Programme of Research, Development, and Research Training in Human Reproduction World Health Organization 2025). Infertility may be considered across many aspects of reproductive medicine and usually concerns both partners (Szamatowicz and Szamatowicz 2020). Assisted reproductive technology (ART) has already helped many infertile couples fulfil the desire to have a child. Since the birth of the first child conceived by *in vitro* fertilization (IVF) in 1978, ART has evolved substantially over the last four decades (Fauser 2019). It should be emphasized that IVF procedures are complex and require intensive patient monitoring, while clinicians and embryologists are responsible for making several key decisions. In the context of risk assessment in the multifaceted process of infertility treatment, and due to different levels of specialist experience, some of these decisions may be highly subjective and/or associated with a high risk of error, which may irreversibly affect clinical outcomes and results (Koperwas and Głowacka 2017).

In recent years, biomedical data describing the diagnosis and treatment of infertility, including medical questionnaires and diagnostic and genetic test results, have been increasingly collected and stored. As a result, researchers worldwide have been developing novel treatment-support methods based on artificial intelligence (AI), whose main purpose is to facilitate a more individualized approach to infertility treatment, for example through intelligent drug dosing or clinical decision support systems (Güell 2024). Considering the complexity of infertility treatment, the analysis and classification of acquired data have become key challenges. The symbiosis between clinical expertise and personalized AI models may potentially improve clinical outcomes. Machine learning (ML), deep learning (DL), and natural language processing (NLP) represent important subfields of AI that are increasingly relevant to the processing of medical data analysis (Dimitriadis et al. 2022). Supervised learning methods used in ART include, among others, decision trees, linear or logistic regression, k-nearest neighbors, support vector machines, and random forests (Dimitriadis et al. 2022; Riegler et al. 2021).

Risk assessment in infertility diagnosis and treatment is a dynamically developing area in which AI algorithms are increasingly applied. A noticeable increase in interest in the use of ML algorithms in infertility diagnosis and treatment has been observed after 2017. Along with the development of risk assessment algorithms, two main methodological groups can be distinguished: methods based on the analysis of numerical data carrying information about the patient and diagnostic results, and methods based on the analysis of images obtained from specialized diagnostic examinations (Güell 2024). The analysis and classification of large amounts of diagnostic data, especially when infertility treatment is considered across many aspects of reproductive

medicine, has therefore become an important problem. The complexity of this field has attracted researchers working on machine learning, particularly deep neural networks, which may support more accurate clinical decisions based on the results of multiple examinations performed during infertility treatment (Hanassab et al. 2024).

Nevertheless, current solutions remain strongly limited because they are often based on a narrow range of clinical features, such as female age, the number of ovarian follicles, or hormone levels. No solutions were identified that would jointly consider factors of genetic, cytogenetic, immunological, and microbiological origin. This confirms the need to develop a comprehensive risk assessment system for infertility treatment based on the multidirectional use of machine learning methods for processing, analyzing, and classifying a broad spectrum of diagnostic test results.

In this study, we present preliminary models to make a step towards a multidomain machine learning-based risk assessment system in infertility diagnosis. The proposed work focuses on the integration of heterogeneous diagnostic data from several infertility-related domains, including genetics, cytogenetics, microbiology/cytology and immunology, and on the evaluation of predictive performance across these domains. The initial modelling pipeline is based on a Random Forest and XGBoost classifiers, patient-level stratified group cross-validation, and metrics appropriate for imbalanced classification, including balanced accuracy, precision, recall, and F1-score. The main attention is paid to the problem of class imbalance and a relatively small feature set, which are typical limitations of real-world clinical datasets and may substantially affect model generalization. This paper aims to understand infertility-related clinical patterns from data and develop a multidomain ML pipeline for infertility risk assessment, with future work comparing baseline models against oversampling and regularization strategies to improve stability and clinical utility. The present study should therefore be regarded as an initial proof-of-concept focused primarily on diagnostic laboratory data, with future extensions planned to incorporate broader clinical and demographic information.

MATERIALS AND METHODS

Ethical Approval, Consent and Data Protection

The research protocol, including the retrospective use of clinical and laboratory data for machine learning analysis, was submitted to the Scientific Research Ethics Committee of the Academy of Silesia and approved by Resolution No. 7/KEBN/2025, adopted on 15 December 2025. The study used authentic retrospective records generated during routine diagnostic and clinical activity at the Gyncentrum Sp. z o.o. Fertility Clinic in Katowice. Data were handled in accordance with applicable data-protection requirements, including the General Data Protection Regulation (GDPR; in Poland: RODO). Direct personal identifiers were not used in the analytical datasets. Access to source data was restricted to authorized personnel, and the analyses were performed on secured, pseudonymized research datasets.

Clinical Data Source and Diagnostic Domains

The retrospective dataset was extracted from the database used at the Gyncentrum Sp. z o.o. Fertility Clinic. The analysis was organized into domain-specific datasets corresponding to the results of genetic testing, cytogenetic testing, microbiological/cytological testing, and immunological testing. Genetics has been divided into ‘package’ and ‘remaining’ as dictated by the clinic’s diagnostic protocol. Additionally, for the auxiliary clinical-reference analysis in the genetics package domain for strategy 2, patient age and sex were added as complementary demographic covariates. Missing values indicated that the diagnostic test had not been performed.

According to Table 1, the raw data matrix included 6,641 records in the genetics package and remaining genetics domain, 3,309 records in cytogenetics, 7,386 records in microbiology/cytology, and 2,614 records in immunology. Diagnostic results were encoded as three-level variables: abnormal test result = 0, inconclusive result = 1, and normal result = 2. This encoding allowed the use of tabular machine learning models while retaining the ordinal-like structure of laboratory outcomes. The original diagnostic measurements were not pooled across tests as raw numerical values. Reference ranges and interpretative criteria were test-specific and were applied at the routine laboratory-reporting stage. Each retained diagnostic test group was represented by a harmonized laboratory interpretation assigned by clinical laboratory specialists at the clinic. The input matrix was prepared separately for each diagnostic domain. Genetic tests represented infertility-related genetic variants or diagnostic panels; cytogenetic tests represented chromosomal number and structure abnormalities; microbiological/cytological tests represented infection-, inflammation-, microbiome-, and cytology-related findings; and immunological tests represented immune-cell, cytokine, NK-cell, KIR/HLA-C, and antibody-related markers.

Table 1: Domain-Specific Data Summary Before and After Preprocessing

Domain	Raw records	Features before	Records after	Features after
Genetics total	6,641	18	6,131	9
Cytogen.	3,309	8	3,022	2
Micro./cyto.	7,386	6	7,163	4
Immunol.	2,614	7	731	3

Outcome Definition and Time Horizon

The binary target variable was pregnancy occurrence. A patient was assigned to the positive class if at least one pregnancy-related record was identified in predefined groups, including prenatal testing, non-invasive prenatal testing, selected post-mortem pregnancy-related testing after miscarriage, beta-hCG-related information, or other pregnancy-confirming entries available in the database. The negative class was defined more restrictively. A patient was labelled as no pregnancy only if the patient met infertility diagnostic or treatment criteria, had at least 24 months of

observable history from the first to the last relevant database appearance, and had no pregnancy-related evidence within that time window. The final modelling cohorts used for classification are shown in Table 2.

Table 2: Final Modelling Cohorts Used for Classification

Domain	n	Positives	Negatives	Pos. class
Gen. pkg.	719	79	640	10.99 %
Gen. rem.	792	20	772	2.53 %
Cytogen.	1,999	229	1,770	11.46 %
Micro./cyto.	5,019	143	4,876	2.85 %
Immunol.	287	25	262	8.71 %

Feature Filtering and Missing Data

Tests with fewer than 100 observations, patients with exclusively missing values in the corresponding diagnostic-result columns, and two tests for which an unambiguous numerical interpretation could not be assigned were removed. In modeling strategies requiring explicit imputation, missing values were handled by a KNN-based procedure using Manhattan distance, 10 neighbors, and distance weighting. After imputation, values were rounded and clipped to the valid interval from 0 to 2, so that no impossible diagnostic codes were introduced.

For exploratory assessment of structure after preprocessing, UMAP projections were generated for each domain and colored by test date, first-visit date, consultation number, physician, prediction label, and test-result categories (McInnes et al. 2018). These visualizations suggested that the domains were not equally structured and that positive cases were sparse, frequently embedded within the dominant negative class. The projections were therefore used as qualitative support for data understanding, not as formal evidence of predictive separability.

Modelling Strategies

Three modelling strategies were evaluated. Strategy 1, used as a baseline, was based on a Random Forest classifier. It used one input sheet per diagnostic domain, available numeric predictors, the pregnancy label, and patient-level grouping by patient identifier. Stratified group cross-validation was used to prevent the same patient from appearing in both training and validation folds. Five folds were used for most domains, whereas two folds were used for domains with very small numbers of positive cases. The Random Forest classifier used 50 trees, square-root feature selection, bootstrapping, no maximum tree depth, minimum samples per leaf equal to one, and a fixed classification threshold of 0.5. Class imbalance was addressed through manually defined class weights.

Strategy 2 used an XGBoost classifier with hyperparameter optimization. The input matrix was analyzed without explicit imputation because missing values were passed directly to XGBoost. For each patient-level cross-validation fold, the training and validation sets were defined using the patient identifier so that records from the same patient could not appear in both parts of the split. Class weights were calculated only from the training set according to class frequencies, with an additional positive-class boost. Candidate XGBoost configurations were evaluated using a grid over tree depth and learning rate: max_depth in {2, 3, 4, 5, 6, 7} and learning_rate in {0.0005, 0.001, 0.005, 0.01, 0.05, 0.1, 0.5}. The classifier used 50 estimators, a binary logistic objective, a log-loss evaluation metric, histogram-based tree construction, and a fixed random state with fold-specific seed offset. For every candidate configuration, predicted probabilities were converted into binary labels using thresholds from 0.05 to 0.95 in steps of 0.01. The best fold-specific model and threshold were selected using balanced accuracy as the primary criterion.

Strategy 3 extended the XGBoost workflow by adding SMOTE with fold-wise preprocessing. In each patient-level cross-validation split, KNN imputation was fitted within the training partition before SMOTE, while the validation fold remained unseen during imputation and oversampling. Class weights were also computed from the training partition only. The hyperparameter grid included the same max_depth and learning_rate values as in Strategy 2, with an additional oversampling_rate parameter in {1.0, 1.25, 1.5, 1.75, 2.0}, where 1.0 represented no oversampling. Since diagnostic variables were encoded as 0/1/2 categories, synthetic SMOTE values were rounded and clipped to the valid 0–2 range. Classification thresholds from 0.05 to 0.95 were evaluated in 0.01 increments, using balanced accuracy as the primary selection metric.

Evaluation Metrics

Because the final cohorts were highly imbalanced, accuracy was not used as the main criterion. Model performance was assessed using sensitivity, specificity, precision, F1-score, and balanced accuracy. Recall is also known as sensitivity. For five-fold settings, results were reported with 95% confidence intervals based on the t-distribution. Confidence intervals were not reported for two-fold settings because such estimates are unreliable.

RESULTS

Baseline Random Forest - Strategy 1

The baseline Random Forest pipeline confirmed that the task is highly sensitive to class imbalance and thresholding. In the genetics package domain, balanced accuracy reached 0.581, with precision of 0.141, recall of 0.717, and F1-score of 0.233. The aggregated confusion matrix contained 57 true positives and 22 false negatives, but also 356 false positives and 284 true negatives, indicating a sensitivity-oriented model with limited positive predictive value.

The remaining domains showed weaker or degenerate behavior. According to Table 3, genetics remaining achieved balanced accuracy of 0.398 and detected only five true positives. Cytogenetics produced an all-positive classifier: recall was 1, but specificity was 0 and balanced accuracy was therefore 0.5. Microbiology/cytology showed near-all-positive behavior, with recall of 0.959 but precision of only

0.028. Immunology reached balanced accuracy of 0.523 but was based on only 25 positive cases. Overall, Strategy 1 served primarily as a diagnostic baseline rather than as a clinically interpretable model.

Table 3: Cross-Validation Results for Strategy 1

Domain	BA	Prec.	Rec.	F1
Gen. pkg.	0.581	0.141	0.717	0.233
Gen. rem.	0.398	0.014	0.229	0.026
Cytogen.	0.500	0.115	1.000	0.205
Micro./cyto.	0.493	0.028	0.959	0.055
Immunol.	0.523	0.103	0.612	0.164

Optimized XGBoost Without Explicit Imputation - Strategy 2

Strategy 2 provided the most coherent overall results. In genetics package, balanced accuracy increased to 0.775, with specificity of 0.668, recall of 0.882, and F1-score of 0.248. This model detected most positive cases, but precision remained low because of low pregnancy prevalence and a substantial number of false positives. As a simple clinical point of reference, an additional Strategy 2 analysis was performed for the genetics package domain after adding patient age and sex to the input matrix. The enriched dataset improved balanced accuracy to 0.812 and specificity to 0.752, while maintaining high recall of 0.873. Genetics remaining achieved balanced accuracy of 0.702 and recall of 0.910, but F1-score remained limited at 0.184, indicating a sensitivity-oriented pattern.

The highest performance was obtained for cytogenetics (see Table 4). The optimized XGBoost model achieved balanced accuracy of 0.823, specificity of 0.979, recall of 0.667, and F1-score of 0.721. The aggregated confusion matrix contained 1,732 true negatives, 38 false positives, 150 true positives, and 79 false negatives. This was the only domain in which high specificity was combined with meaningful sensitivity and a high F1-score. Microbiology/cytology showed only marginal signal, with balanced accuracy of 0.523 and F1-score of 0.069. Immunology reached balanced accuracy of 0.653 and F1-score of 0.237, but the estimate remains unstable because of the small positive class.

Table 4: Cross-Validation Results for Strategy 2

Domain	BA	Spec.	Rec.	F1
Gen. pkg.	0.775	0.668	0.882	0.248
Gen. pkg. enrich.	0.812	0.752	0.873	0.330
Gen. rem.	0.702	0.494	0.910	0.184
Cytogen.	0.823	0.979	0.667	0.721
Micro./cyto.	0.523	0.451	0.596	0.069
Immunol.	0.653	0.584	0.721	0.237

XGBoost with SMOTE - Strategy 3

This fold-wise preprocessing produced results broadly comparable to Strategy 2 (see Table 5). In genetics package, balanced accuracy was 0.783, with specificity of 0.684, recall of 0.882, and F1-score of 0.258. Genetics remaining achieved balanced accuracy of 0.702, but the estimate remains unstable because this domain contains very few positive cases.

The cytogenetic domain remained the strongest result, with balanced accuracy of 0.823, specificity of 0.979, recall of 0.667, and F1-score of 0.721. Microbiology/cytology showed only marginal performance, whereas immunology reached balanced accuracy of 0.662. Overall, fold-wise imputation confirmed that SMOTE did not provide a consistent advantage over optimized XGBoost alone, although it reduced the preprocessing leakage risk in this strategy.

Table 5: Cross-Validation Results for Strategy 3

Domain	BA	Spec.	Rec.	F1
Gen. pkg.	0.783	0.684	0.882	0.258
Gen. rem.	0.702	0.440	0.964	0.179
Cytogen.	0.823	0.979	0.667	0.721
Micro./cyto.	0.528	0.446	0.609	0.070
Immunol.	0.662	0.603	0.721	0.245

DISCUSSION

Pregnancy prediction from infertility-related diagnostic results is strongly dependent on the modelling pipeline and on the diagnostic domain. The baseline Random Forest strategy demonstrated that class weighting with a fixed 0.5 threshold may lead to excessive positive prediction under severe imbalance. The baseline Random Forest should be interpreted in the context of known limitations of tree ensembles under severe class imbalance. In skewed binary classification, standard bootstrap samples may contain only a few minority-class cases, producing majority-biased or unstable minority-class performance. Recommended adaptations include class weighting, balanced bootstrap sampling and under- or over-sampling within the ensemble workflow (Brownlee 2021). Recent work shows that combining Random Forest pruning with re-sampling can further improve imbalanced classification (Faiz et al. 2026). Therefore, our Random Forest results should be treated as a simple baseline, not as evidence that RF-based methods are unsuitable for infertility data. In contrast, the optimized XGBoost strategy improved model behavior by using model-native missing-value handling, fold-specific class weights, and threshold optimization. This was particularly evident in cytogenetics, where the model achieved the best balance between sensitivity and specificity.

The cytogenetic findings should be interpreted with caution, but they currently represent the most promising group. One possible explanation is that the retained cytogenetic features identify a more clinically coherent patient subgroup, which may reflect a direct relationship between chromosomal abnormalities and reproductive outcomes (Zhao et al. 2025). By contrast, microbiological/cytological features may be too sparse, too coarse, or temporally misaligned with the pregnancy outcome. Low performance in this domain should

therefore not be interpreted as evidence that the microbiome or cytological status is clinically irrelevant.

The 24-month threshold was introduced to reduce false negative labelling caused by short or incomplete follow-up. This choice improves the credibility of the retrospective label, but it also reduces the number of usable observations and strengthens class imbalance. Therefore, all reported results should be interpreted as performance estimates for a filtered retrospective cohort rather than for the entire clinic population.

The SMOTE experiment shows that imbalance correction is not automatically beneficial. In the optimal Strategy 3 configurations, oversampling_rate was 1.0 for genetics package, genetics remaining, and cytogenetics, indicating that no additional synthetic oversampling was selected as beneficial in these domains. Higher oversampling rates were selected only for microbiology/cytology (oversampling_rate = 1.75) and immunology (oversampling_rate = 1.5), but these domains still showed weak or unstable performance. This suggests that the effect of SMOTE was domain-dependent and that synthetic oversampling alone was not sufficient to overcome sparse, discrete, and partly imputed diagnostic feature spaces.

Feature importance was inspected for the enriched genetics package to assess whether the prediction was driven by clinically interpretable demographic and laboratory variables rather than by potential artefacts of data collection. These results suggest that age provides complementary clinical context to genetic test results.

From a practical implementation perspective, predictive models should also use larger, multicenter datasets that include heterogeneous patient populations and reflect the specific nature of clinical practice worldwide. Retrospective data have already been shown to be useful for enabling decision-making in the context of ART (Curchoe 2023). Achieving this goal requires cooperation between stakeholders representing multiple disciplines in artificial intelligence and healthcare (Wiens et al. 2019). Published machine-learning studies in infertility and ART usually address narrower tasks, such as semen-quality or male-fertility prediction, embryo ploidy or viability assessment, and IVF pregnancy or live-birth prediction using age, ovarian-reserve, hormonal, embryo or treatment-related variables (Ory et al. 2022; Zhou et al. 2022; Barnes et al. 2023; Wang et al. 2022). The literature summary indicates that existing approaches rarely integrate broader diagnostic domains such as genetic, immunological, and microbiological factors. Direct numerical benchmarking against IVF outcome-prediction models is therefore limited, and the present results should be interpreted as preliminary domain-specific diagnostic signals from a multidomain infertility work-up dataset. Moreover, future systems should also include developing tools for processing electronic health records (EHRs), because infertility treatment begins and ends with medical interviews in which diagnostic results and other relevant clinical information are recorded (Meskó and Topol 2023; Thirunavukarasu et al. 2023).

The study has several limitations. First, there is no external or prospective validation cohort. Patient-level cross-validation reduces leakage, but it does not prove generalizability to another clinic or future clinical workflow. Second, the feature sets are small and domain-specific. In addition, the number of positive cases was limited in several domains, particularly in genetics remaining and immunology. For this reason, a conventional independent data split was not used, as it would have produced very small and statistically unstable positive subsets for model tuning and testing. Instead, patient-level stratified group cross-validation was used as an internal validation strategy to maximize data use while preventing patient-level leakage. Therefore, the reported metrics should be interpreted as internally optimized cross-validation estimates rather than performance measured on a fully independent external test set. Third, the outcome is pregnancy, not live birth. It is clinically meaningful as an intermediate endpoint, but it does not represent the full reproductive outcome. Finally, missingness is clinically informative in retrospective diagnostic data because a missing test may mean that the test was not indicated, not ordered, or performed outside the observation window.

CONCLUSION

This work established a multidomain machine learning pipeline for integrating complex infertility-related diagnostic data and evaluating predictive performance across distinct research domains. The strongest preliminary result was obtained for cytogenetic testing using the optimized XGBoost strategy, which achieved balanced accuracy of 0.823, specificity of 0.979, recall of 0.667, and F1-score of 0.721. The results support the feasibility of domain-specific machine learning for infertility diagnosis planning.

The broader conclusion is that machine learning may support infertility diagnostics only when embedded in a rigorous clinical-data pipeline. Such a pipeline must respect patient-level grouping, temporal follow-up, class imbalance, missing-data mechanisms, and the clinical meaning of the target label. In its present form, the system should be understood as a research framework for identifying promising diagnostic signals and supporting future development of personalized infertility diagnostics.

FUTURE RESEARCH

In the present study, fold-wise KNN imputation was implemented for the SMOTE-based Strategy 3, and patient age and sex were added as complementary covariates in an additional genetics package analysis. Further work will extend the dataset and apply this enrichment to other diagnostic domains by incorporating additional clinically relevant variables. The system should also be developed toward clinically useful risk scores, and more detailed domain-specific explanations, followed by external or prospective validation before practical decision-support deployment.

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MULTI-AGENT SIMULATION OF PHISHING USING A METAMODEL INTEGRATING MACHINE LEARNING FOR DETECTION

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Phishing, multi-agent modeling and simulation, HiLLS, Cybersecurity, Supervised learning, Behavioral detection.

ABSTRACT

Phishing is an evolving threat that exploits advanced social engineering and evasion techniques, while undermining traditional detection techniques. This article presents a hybrid framework combining multi-agent simulation (MAS) and supervised machine learning (ML) supported by a simulation metamodel to improve detection techniques for resilient cybersecurity.

The system models the dynamic interactions between various agents: attackers, users, attack, defense, and ML module within an AnyLogic environment. PhishingAttaque agents rely on strategies inspired by Lockheed Martin's Cyber Kill Chain (CKC), with the ability to select or change the type of attack. Formal behavior modeling is based on the HiLLS language, while a generic metamodel ensures a consistent representation of entities, events and information flows.

ML integration is achieved through a Java-Python architecture, in which Java drives the evolution of agents and Python, via the pypline API, executes the classification models in real time. This interoperability enables adaptive detection of suspicious behavior.

The results show that the proposed framework offers a robust and realistic platform for analyzing phishing campaigns, testing defensive strategies and supporting cybersecurity training.

1 INTRODUCTION

Phishing attacks represent one of the most persistent and evolving threats in the cybersecurity landscape today. Indeed, according to [1], 71% of organizations experienced at least one successful phishing attack in 2023. These attacks fundamentally exploit human weaknesses through techniques, the main one being social engineering [2]. They have evolved considerably in sophistication, now incorporating advanced evasion strategies, personalized content, and real-time adaptation mechanisms [3].

Phishing generally relies on a discrete event process involving several key steps as illustrated in the generic diagram see Figure 1. It takes several forms including email phishing, the most widespread form using mass, non-targeted attacks; spear phishing, targeting individuals or small groups; whaling, targeting senior officials; clone phishing reproducing legitimate messages; and pharming, redirecting users to fraudulent website.

In this context, traditional detection systems based on static signatures or predefined rules show significant limitations when faced with multifaceted and dynamic attacks [5]. These approaches struggle, in particular, to capture the complex interactions between the various actors in a sociotechnical system.

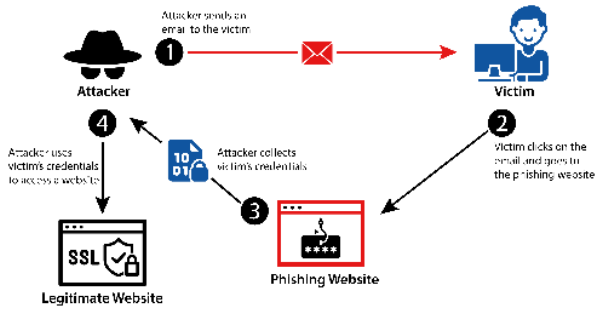

Figure 1 : Phishing Process. Adapted from [4]

Despite advances in cybersecurity, several major challenges persist, including: (i) the lack of clear methodological foundations to support the modeling and assessment of phishing attacks in dynamic environments; (ii) the inability of classical systems to represent the evolving interactions between attackers and users; (iii) the lack of realistic platforms to experiment with and test phishing strategies; (iv) the lack of seamless integration between behavioral simulation and Artificial Intelligence; and (v) the difficulty in generalizing detection models in varied contexts.

In response to the situation described above, the proposed platform enables the modeling of complex human behaviors, the simulation of realistic cyberattack and defense scenarios, and the integration of adaptative detection mechanisms based on supervised ML techniques.

With this in mind, the main objective of this article is to design an intelligent simulation framework that allows: (1) modeling phishing campaigns in a dynamic environment; (2) formally representing behaviors using a metamodel; (3) integrating ML algorithms for adaptive detection; and (4) evaluating different defense strategies.

To address this issue, this paper presents an innovative framework based on multi-agent simulation, combined with supervised learning techniques supported by a generic simulation metamodel, in order to better understand, detect, and anticipate phishing attacks.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 presents the proposed methodology, including the conceptual tools employed, namely HiLLS, the CKC, the generic HiLLS-based phishing metamodel, and the integration of ML into MAS framework.

Section 4 focuses on the experimental framework, scenarios studied, and results obtained to validate the proposed approach using email phishing as the case study. Finally, the last section concludes the paper and outlines directions for future research.

2 RELATED WORKS

In the fight against mass phishing attacks, several approaches have been used, according to the scientific literature. These approaches can be categorized into two types: traditional methods and modern methods that take into account technological advancements and the limitations of traditional approaches.

Traditional approaches: Several researchers have proposed traditional approaches based on heuristic filters, signatures, and blacklisting mechanisms to detect these attacks. Indeed, according to [6], phishing detection can be achieved through the analysis of the structural characteristics of suspicious emails and URLs. This approach is also explored in [7], where the authors propose a method based on analyzing the content of web pages to identify phishing sites. However, in [8], the authors emphasize that mass phishing attacks heavily exploit social engineering techniques and the human factor to increase their effectiveness.

Current approaches: With the evolution of cyber threats, several studies have focused on using modeling, simulation, and ML techniques to improve the detection of mass phishing campaigns. According to the authors of [9], supervised learning models can automatically identify malicious behavior from large volumes of data. However, beyond simple detection, understanding the dynamic interactions between the different actors in a cyberattack requires more sophisticated modeling approaches. In this regard, the work presented in [10] shows that multi-agent systems are a suitable solution for representing the interactions between users, attackers, and security mechanisms in distributed environments. Other studies, such as those in [11], propose hybrid approaches combining simulation and ML to improve detection accuracy and adaptation to new forms of phishing.

This work shows that the integration of metamodels and artificial intelligence algorithms constitutes a relevant approach for the simulation and detection of massive phishing attacks in complex cybersecurity environments.

3 METHODOLOGICAL FRAMEWORKS FOR MULTI-AGENT PHISHING DETECTION

The proposed framework is based on a hybrid architecture combining three fundamental pillars:

3.1 Conceptual foundations

3.1.1 HiLLS Formalism

HiLLS is a graphical formalism for specifying discrete event system (DES) models for analysis using methodologies such as simulation, formal analysis, and implementation [12, 13]. HiLLS was designed to simulate a wider range of DESs with greater expressiveness and communicability, through concrete graphical notations from software engineering, following DEVS concepts to make it accessible to a diverse user community [12, 13]. HiLLS is a graphical language whose abstract syntax is derived from the integration of concepts from systems theory and software engineering. HiLLS provides a unified abstract syntax for describing DES, a concrete graphical-textual syntax, and three major semantic domains: a unified modeling language (UML) for implementation, a discrete event system specification (DEVS) for simulation, and a formal specification [18] (Figure 2).

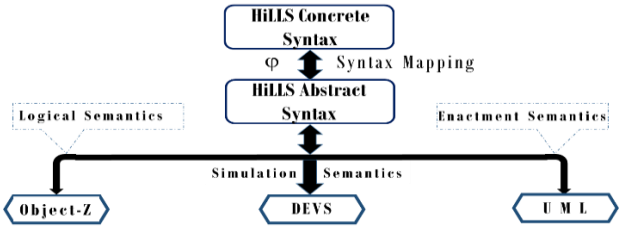

Figure 2 : Overview of the HiLLS Formalism [12]

The HiLLS formalism provides an improvement for representing the notion of entities by introducing two categories of entities: (i) The HiLLS system (HSystem) which represents an entity with temporal behavior (equivalent to an atomic or coupled DEVS model); (ii) The HiLLS class (HClass) which represents an entity without any temporal behavior, and which is used to model the passive resources of the system.

3.1.2 Cyber Kill Chain-Based Behavioral Modeling

The CKC provides a methodological framework for breaking down the different stages of a cyberattack, from initial reconnaissance to the attacker's final objective.

This model offers a structured view of how an attack unfolds, highlighting potential entry points and critical phases where defensive actions can be deployed to interrupt or neutralize the threat. It consists of seven generic stages: (1) Reconnaissance (i.e., target identification); (2) Arming (i.e., preparation of the operation); (3) Delivery (i.e., launch of the operation); (4) Exploitation (i.e., access to the victim); (5) Installation (i.e., establishing a foothold within the victim's organization); (6) Command and Control (C&C or C2) (i.e., remote control of implants); (7) Actions on the objective (i.e., achieving the mission objective).

3.1.3 Multi-Agent Simulation

MAS model autonomous agents capable of perceiving, reasoning, interacting and adapting to their environment. In cybersecurity, they enable the simulation of interactions among users, attackers, infrastructures, and defense mechanisms, providing a flexible framework for evaluating cyberattack and defense scenarios [15]. Combined with ML, MAS further support adaptative behavior modeling, vulnerability detection, defense optimization, and anticipation of attacker actions in dynamic environments.

3.2 Generic HiLLS-Based Phishing Metamodel

3.2.1 Generic Metamodel

The metamodel shown in Figure 3 is based on the HilLS principle and proposes a MAS architecture for detecting phishing attacks that integrates ML. In this model, Attacker, Security, User, Detection, Attack, Phishing Attack, and Link are defined as HSystems, representing autonomous systems capable of interacting dynamically within the simulation

environment. Conversely, PhishingScenario, Message, and Phishing Message are modeled as HClasses, as they represent structures manipulated by the different systems. The metamodel highlights interaction relationships between Attacker and User, association relationships between Attack, PhishingAttack, and link, as well as a specialization relationship between Message and PhishingMessage. The Detection system monitors and analyzes suspicious behavior using ML, while Security coordinates protection and learning mechanisms.

Thus, the metamodel offers a modular and hierarchical representation of the structural and behavioral relationships involved in the simulation of phishing attacks.

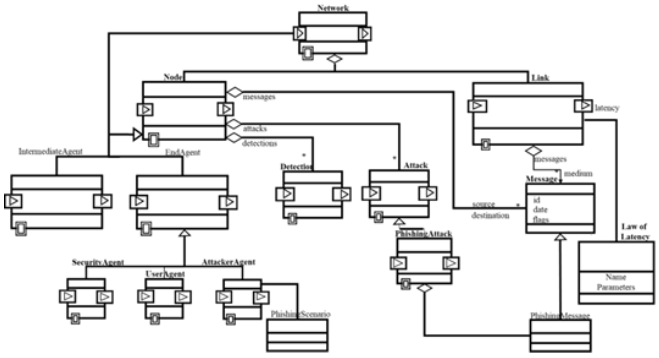

Figure 3: A Simulation Metamodel for detecting Phishing Attacks

3.2.2 Phishing Attack Model

The HSystem PhishingAttack is modeled in Figure 4 using the HiLLS formalism, with its behavior represented by a state-transition diagram structured accordin to the CKC phases. The HSystem receives input messages (In?m) that trigger external transitions and generates output messages (Out!m) associated with internal transitions during attack execution.

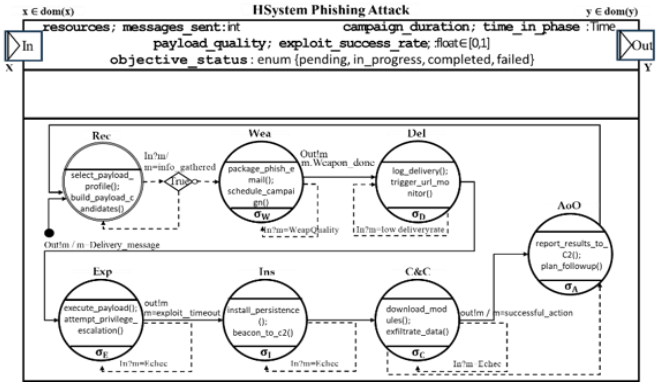

Figure 4: HiLLS Phishing Attack Model

3.2.3 Phishing Detection Model

Figure 5 presents the HiLLS model of HSystem Detection, dedicated to detecting phishing attacks in a MAS environment. The agent's dynamic behavior is described in the third compartment through a state-transition diagram structured around four phases: Idle, Collect, Train, and Detect. Each phase evolves under the influence of input and output messages, generating either external or internal green transitions to other states. Thus: (1) the Idle phase receives activation events or analysis requests; (2) the Collect phase receives traffic data, suspicious messages, URLs, or user behavior. The information recorded includes: (i) role ∈ {role_CEO, role_CIO, role_Employee, role_Finance, role_HR}; (ii) has_received_training; (iii) has_sensitive_access; (iv) vigilance_level; (v) emails_received_count, (vi) email_received_at, (vii) has_sensitive_access, (vii) clicked. (3) The Training phase receives the datasets and parameters necessary for training the model. This step includes preprocessing, feature selection, and data splitting. Preprocessing cleans and normalizes the data to improve processing efficiency. Static and dynamic packet features are then selected to distinguish normal traffic from attacks using ML algorithms.

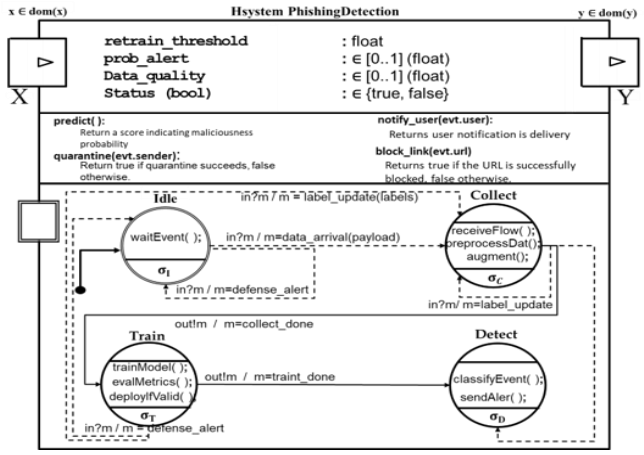

Figure 5: HiLLS Model for Phishing Attack Detection

The proposed framework uses several classification algorithms capable of distinguishing normal traffic from malicious traffic, including KNN, SVM, RF, and LR [16]. Finally, the data is split into training and test sets. Finally, (4) the Detect phase receives the data to be classified in order to identify phishing activities.

3.3 Machine Learning Integration

3.3.1. Hybrid Architecture

The framework presented is based on a hybrid approach combining modeling-simulation and ML. At the formal level, the HiLLS is used to define the simulation metamodel, while the ML algorithms are integrated. At the operational level, the simulation is implemented using Anylogic, and the learning models are developed with scikit-learn.

The framework combines two approaches in a nested pattern, as illustrated in Figure 6; (1) Modeling and simulation integrating ML, where agents adapt their behavior through learning, and (2) ML integrated with the simulation, where the data generated by the simulation is used to train the learning models.

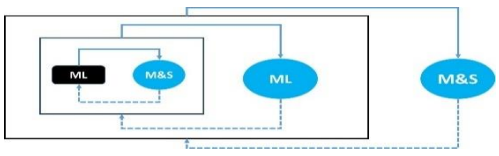

Figure 6: Hybridization Simulation-ML

A simulation metamodel integrating ML is first designed, then a training scheme is implemented using the simulated data. The learned model is then fed back into the simulation to evaluate the performance of the trained system compared to the untrained model.

3.3.2. Operational Interoperability

The simulation metamodel incorporates a learning function via the ML-based detection system. Operational interoperability is achieved between the AnyLogic software (written in Java), used as the simulator, and the Scikit-Learn package (written in Python), used as the solver. This Java-Python interoperability is implemented using the Pypline API, allowing AnyLogic to connect and communicate with a local Python installation, which can be exported as JAR files. The data generated by the simulation model is stored in an AnyLogic database, which the ML algorithm uses for training and decision-making, as shown in Figure 7.

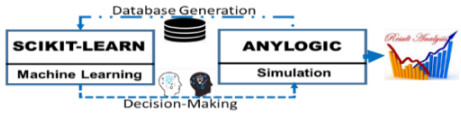

Figure 7: Integration of Simulation and ML [17]

4 EXPERIMENTAL FRAMEWORKS

4.1 Experimental Framework, scenarios

The experimental results presented in this paper consider a large-scale, non-targeted phishing attack scenario.

4.1.1. Experimental context

This document analyzes the results obtained from a simulated phishing attack developed using AnyLogic. The four study scenarios, S0 to S3, are distinguished by the activation of two defense mechanisms: the SecuritySystem and the ML module.

4.1.2. Scenario Definition

Table 1 presents different scenarios defined to evaluate the impact of security mechanisms and ML on system performance.

Table 1: Experimental Simulation Scenarios

Scenario	Designation	Security-Activated	ML-Activated	Scientific Purpose
S0	Basic Model	False	False	Defenseless reference
S1	Security only	True	False	Security Contribution
S2	ML only	False	True	Contribution of ML
S3	Complete	True	True	Complete Model

4.1.3. Learning Model

The dataset under consideration contains 257,413 entries with 11 columns, including 10 individual characteristics and 1 binary target. Each instance represents an event: "a massive phishing email arrives at a user's address within the organization." The distribution has 92.1% of instances with "clicked (clk)" = False and 7.9% with (clk) = True. Note that the concepts of (clk) = False and (clk) = True refer to the user's reaction to a phishing email, representing the user's behavior in the face of a phishing attack and serving as a target variable for ML. This distribution is said to be highly skewed. The correlation matrix between the features and the target allows for the identification of bivariate linear associations. A strong correlation between two features can signal collinearity, which is particularly detrimental to linear regression. Therefore, this correlation matrix, shown in Figure 8, was used to eliminate highly correlated variables.

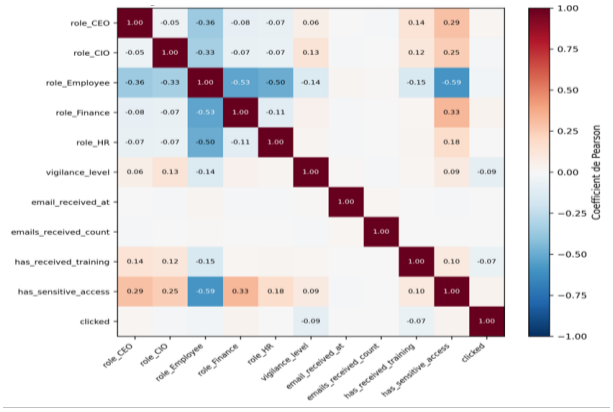

Figure 8: Correlation Matrix including the Target Variable

Ultimately, only three variables were selected as a discriminative feature for distinguishing phishing from legitimate emails: role_Employee, role_Finance, and role_CEO. While their correlations with the target variable (clk) are relatively weak, they remain more informative than the other candidate variables.

In the context of phishing attack detection, it is crucial to implement effective evaluation measures to test the performance of detection systems. Among the most common measures are Accuracy, the ROC curve, and the Curve AUC. The superimposed ROC curves in Figure 9 visually summarize the discriminatory capabilities of the four algorithms. The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) at different decision thresholds. The blue curve, which rises rapidly towards the upper left corner, corresponds to the most efficient classifier; a curve aligned along the diagonal (dotted line) corresponds to pure chance (AUC=0.5).

Logistic regression (in blue) is clearly above all other curves across most of the FPR range, confirming its superior AUC of 0.615. SVM (in green) follows in second place with 0.557. Random Forest (in red) and KNN (in orange) are very close to random chance. This figure provides the most compelling visual summary to support the choice of Logistic Regression as the selected algorithm.

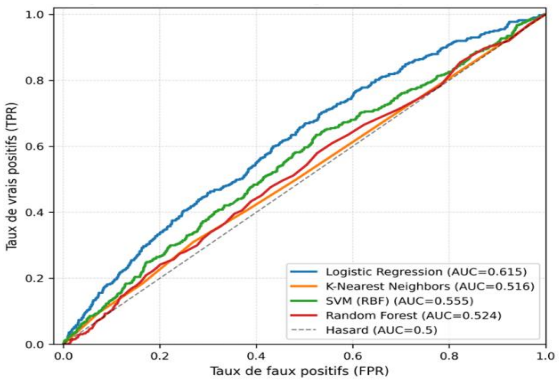

Figure 9: ROC Curves of the 4 Algorithms

Following this comparative study, Logistic Regression emerged as the most suitable algorithm for classifying massive phishing attacks, achieving an AUC-ROC of 0.615, higher than the other algorithms. It should be noted that a higher AUC indicates a better ability to rank instances by risk, which is precisely the objective of a behavioral detection system.

4.2 Results

The simulation provides a detailed temporal trace of the model's behavior under fixed nominal parameters. It allows observation of the propagation dynamics and the defense system's reaction during a campaign.

To analyze these behaviors in detail and comparatively, each scenario S0, S1, S2, and S3 will be studied using four types of curves, denoted respectively a); b); c); and d).

a) Real-Time State Distribution (RTSD): This corresponds to the proportion or distribution of the different states at each instant during the evolution of a phishing campaign. This distribution allows us to: (1) analyze the propagation of the attack; (2) evaluate the effectiveness of the detection mechanisms; and (3) dynamically adjust the countermeasures.
b) Cumulative Attack Progression (CAP): This allows observation of the accumulation of impacts over time, unlike a snapshot, thus enabling assessment of the overall scope and dynamics of the attack.
c) Mean Time To Detect (MTTD): This is a global performance indicator for the detection system. It allows: (1) evaluation of the responsiveness of detection mechanisms; (2) reduction of the overall impact of attacks (the faster the detection, the fewer victims); (3) comparison of different strategies (static rules vs. ML); (4) use as an indicator in multi-agent security.
d) Vigilance Level Distribution (VLD): This is the proportion of users in each of the following states: (1) 10%; (2) 20%; (3) 30%; (4) 40%; (5) 50%; (6) 60%; (7) 70%; (8) 80%; (9) 90%; (10) 100%.

4.2.1. Scenario S0: Base Model

S0 illustrates the uncontrolled spread of a phishing campaign in the absence of any defense mechanisms. At the end of the simulation, a compromise rate of 51% is observed. The statistical variables (stat_detectedCount, stat_recoveredCount) indicate that the security and recovery mechanisms are inactive. Indeed, the counters associated with detection and recovery remain at zero (Figure 10), reflecting the absence of any monitoring, recovery, or automatic response processes to the phishing attack. Similarly, some derived metrics, such as the final rate of remedied compromises or the MTTD, also remain at zero, since no detection is performed during the simulation. These results are therefore consistent with the configuration.

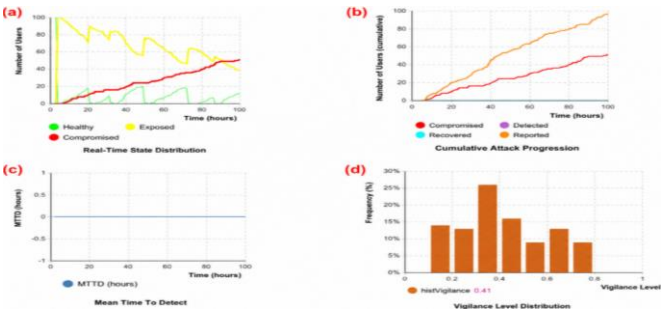

Figure 10: Simulation Experiment Table Scenario S0

The RTSD curve shows a rapid transition from the Healthy state to the Exposed state, then to the Compromised state, without any regression. This indicates spontaneous reporting activity (active Exposed→Healthy transition at the given vigilance level) that remains functional in all scenarios.

4.2.2. Scenario S1: Security alone

With the securitySystem enabled and ML disabled, the dynamics change qualitatively. A total of 52 compromises were observed, of which 40 were detected and 29 users recovered at the end of the campaign Figure 11.
The MTTD reached 21 hours and 30 minutes, consistent with responseDelay=21h. The counter stat_falsePositiveCount=32 reflects the activity of the periodic scan on the Healthy population (falsePositiveRate).
The effect of the defense is visible on the CAP curve: the compromised trajectory stabilizes while the detected and recovered curves rise progressively, indicating an active remediation cycle.

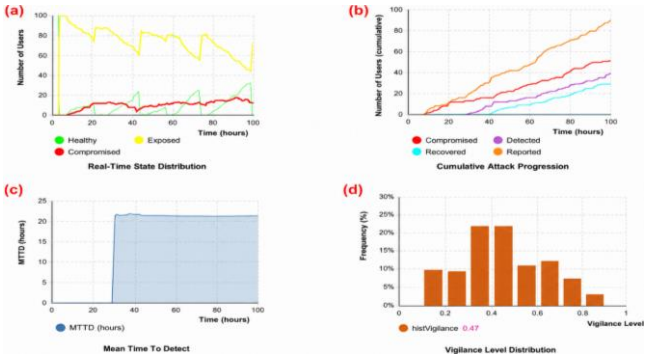

Figure 11: Simulation Experiment Table Scenario S1

4.2.3. Scenario S2: ML alone

S2 activates ML with CSV coupling operational and SecuritySystem disabled. The stat_mlCycleCount counter reaches 50 export cycles over the 100 simulated hours. At the end of the simulation, 54 compromised users are observed (slightly higher than in S0), with no detection or recovery (consistent with activateSecurity = false), and a remarkably high stat_reportedCount counter of 100. This signal saturation is the most notable effect of ML in the absence of remediation: the model encourages users to report more suspicious emails, which disrupts the propagation chains by returning to the Healthy state.

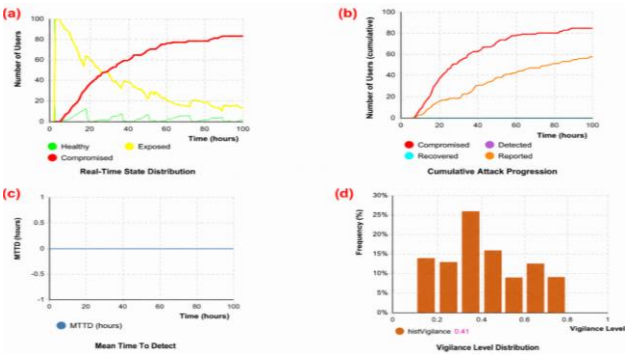

Figure 12: Simulation Experiment Table Scenario S2

4.2.4. Scenario S3: Complete

In S3, both the SecuritySystem and the ML are activated. The Python integration runs continuously, and the security system ensures the complete detection and remediation cycle. The results are the best of the four scenarios: 65 compromised users, of which 52 were detected and 45 recovered, representing a recovery rate of 69%. The average MTTD is 21.70 hours. The average vigilance level reaches 0.500, higher than those of S0 and S1, thanks to the combined effect of the ML and the recovery mechanisms.

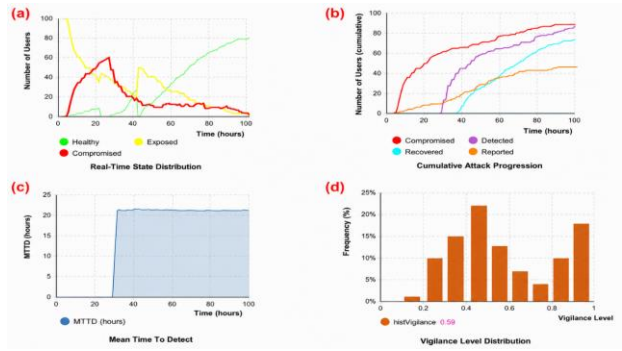

Figure 13: Simulation Experiment Table Scenario S3

These results confirm the synergy between the ML, which acts preventively, and the SecuritySystem, which ensures detection and recovery.

CONCLUSION AND PERSPECTIVES

This work presents a hybrid framework for modeling, simulating, and detecting mass phishing attacks. By combining the HiLLS formalism, Multi-Analysis of Systems (MAS), and real-time machine learning (ML), this approach links formal modeling with AI applied to cybersecurity.

Scenario evaluation (S0 to S3) demonstrates the value of this synergy: (1) Isolated ML (S2): increases vigilance and reporting, but remains insufficient without operational remediation. (2) ML-Security coupling (S3): offers the best results with 69% of accounts recovered and an average vigilance level raised to 0.50. (3) Evaluation metrics: the joint analysis of indicators (RTSD, CAP, VLD, and MTTD) quantifies the impact of human factors and response times.

This framework constitutes a rigorous virtual testbed for evaluating defensive strategies without exposing real infrastructure.

From a future perspective, we propose three priority areas for extending this work: (1) Continuous learning and advanced AI through the integration of reinforcement learning to adapt detection to variations in attacks, particularly those generated by AI; (2) Targeted scenarios and human factors: Modeling spear-phishing via the organization's social graph and enriching user agents with psychological variables; (3) Variation and real-world environment: Connecting the platform to operational tools or real data streams to evolve towards a cybersecurity digital twin..

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Sequential Linearization for Physics-Informed Extreme Learning Machines: A Benchmark-Motivated Fix for Nonlinear PDE Failures

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KEYWORDS

physics-informed extreme learning machine, Burgers equation, sequential linearization, Picard iteration, nonlinear PDEs, benchmark

ABSTRACT

Physics-informed extreme learning machines (PIELMs) turn the PDE training problem into a one-shot linear solve. That property is exactly what makes them fast on linear problems, but it is also the point where the method starts to lose its edge on genuinely nonlinear equations. This paper addresses that gap with a simple sequential linearization strategy for the one-dimensional viscous Burgers equation. The nonlinear convection term is frozen from the previous iterate, the resulting linearized problem is solved again with the existing PIELM machinery, and the process is repeated until the update norm is small enough.

The result is a minimal fixed-point wrapper around the existing PIELM solver that keeps the closed-form spirit of the method while recovering a large part of the lost accuracy. On a sweep across four viscosities ($\nu = 0.10, 0.05, 0.02, 0.01$), the sequential method consistently improves over the one-shot linear baseline, reducing the average RMSE to about 51% of the baseline level while keeping the runtime in the sub-second range. The accompanying comparison suite also shows how the method sits relative to a small set of PINN baselines in terms of accuracy, wall-clock time, and memory.

Introduction

Physics-informed extreme learning machines were attractive from the beginning because they offer a different computational trade-off from gradient-based physics-informed neural networks (PINNs). Instead of training all network parameters by backpropagation, PIELMs fix the hidden-layer representation and solve for the output weights in a regularized linear system. In linear PDE settings this can be extremely efficient (Dwivedi and Srinivasan 2020, Huang et al. 2006).

The difficulty appears when the PDE itself becomes nonlinear. The closed-form least-squares step is then no longer enough to capture the nonlinear coupling, and the one-shot solve starts to behave like a linear surrogate of a nonlinear target. That is the core motivation for this paper. The benchmark-oriented literature around PIELMs already indicates that the method is competitive on many smooth linear problems, but it also leaves a clear gap on nonlinear regimes and on systematic comparisons against alternative families (Hao et al. 2023, Takamoto et al. 2022, Struniawski 2026). The present work is a direct answer to that gap.

The manuscript is written in a practical spirit, emphasizing what can be executed efficiently, where the original method fails, how much of that failure can be recovered, and at what computational cost. The proposed modification is intentionally minimal. Rather than redesigning the PIELM formulation, we retain the existing linear solver and wrap it inside a classical Picard (fixed-point) iteration. At each iteration, the nonlinear convection coefficient is evaluated using the solution obtained at the previous step, converting the nonlinear Burgers equation into a linear problem that can again be solved by the standard PIELM least-squares procedure. This process is repeated until the relative change between successive iterates falls below a prescribed tolerance. Picard's iteration is one of the simplest and most widely used linearization strategies for nonlinear partial differential equations because it replaces a difficult nonlinear solve with a sequence of linear problems while avoiding the need to evaluate Jacobian matrices, as required by Newton-type methods (Bathe 1996, Allgower and Georg 1990, Ferziger and Perić 2002). Implementing the approach as an outer fixed-point loop allows the original one-shot PIELM algorithm to remain unchanged, ensuring that the comparison isolates the effect of the nonlinear treatment rather than differences in the underlying solver implementation.

Background

For a PDE of the form

$$\mathcal{N}[u](x,t) = 0, \tag{1}$$

PIELMs approximate the solution with a fixed-feature model and determine the output weights from a linear system. The method is especially natural when $\mathcal{N}$ is linear. In that case the residual operator can be written directly in the least-squares objective and the solve remains compact and fast (Dwivedi and Srinivasan 2020). The general idea is consistent with the ELM literature: random hidden features, deterministic output solve, and a strong emphasis on conditioning and regularization (Huang et al. 2006, Struniawski et al. 2023). For nonlinear PDEs, however, the residual contains products of unknown solution terms and their derivatives. In Burgers' equation, the term uu_x is exactly the part that breaks the one-shot linearization. One can still fit a PIELM to a linearized residual, but that requires updating the nonlinear coefficient from the previous iterate.

This is where the present paper fits. The method is not presented as a replacement for all nonlinear solvers. It is a practical fix for a specific failure mode: when the one-shot PIELM is too crude on a nonlinear PDE, a sequential linearization wrapper may recover accuracy without abandoning the original solver design.

The broader literature place this choice in context. PINNs remain the most common baseline for nonlinear PDEs, but their training cost is high and their optimization can be brittle (Raissi et al. 2019, Karniadakis et al. 2021). At the same time, the benchmark papers on PIELM-family methods show that the model family is broad enough to deserve a nonlinear extension strategy rather than being judged only on linear PDEs (Struniawski 2026, Hao et al. 2023, Takamoto et al. 2022). The current manuscript reuses that benchmark mindset but narrows the question to one equation and one algorithmic change.

Method

The target problem is the one-dimensional viscous Burgers equation

$$\partial_t u + u\,\partial_x u - \nu \partial_{xx} u = 0, \tag{2}$$

on the space-time domain $x \in [-1, 1]$, $t \in [0, 1]$, with homogeneous Dirichlet boundary conditions

$$u(-1, t) = u(1, t) = 0, \tag{3}$$

and the standard sinusoidal initial condition

$$u(x, 0) = -\sin(\pi x). \tag{4}$$

The experiment evaluates four viscosities, $\nu \in \{0.10, 0.05, 0.02, 0.01\}$, where the smaller values make the problem increasingly nonlinear and therefore more demanding for the solver.

The sequential linearization scheme used in the code is a Picard-style fixed-point iteration. Starting from a one-shot linear PIELM baseline, the nonlinear convection coefficient is frozen from the previous prediction and the PDE is solved again. In the implementation, the linearized residual is

$$\partial_t u^{(k+1)} + u^{(k)} \partial_x u^{(k+1)} - \nu \partial_{xx} u^{(k+1)} = 0, \tag{5}$$

where $u^{(k)}$ is the solution from the previous iterate. The loop stops when the relative update norm falls below a tolerance threshold or when the maximum number of Picard steps is reached.

The solver is intentionally simple. The existing PIELM machinery is reused without changing the model family, the hidden-layer representation, or the direct least-squares solve. That keeps the method close to the original PIELM philosophy. The only new ingredient is the outer fixed-point loop.

Algorithmically, the method is:

1. Solve the linearized Burgers problem with the existing linear PIELM solver.
2. Evaluate the current iterate on the collocation set.
3. Freeze the nonlinear convection coefficient from that iterate.
4. Resolve the new linearized PDE with the same PIELM solver.
5. Check the relative update $\|u^{(k+1)} - u^{(k)}\|_2 / \|u^{(k+1)}\|_2$.
6. Stop when the update is below the tolerance or the iteration cap is reached.

The code uses a maximum of five Picard iterations in the paper profile and a stopping tolerance of 10^{-3}. The comparison suite keeps the same scheme but also reports the full Picard history so convergence can be inspected directly. The nonlinear solve is not presented as a theorem with a formal convergence proof. That would be a different paper. Here the claim is narrower: on the tested Burgers regime, sequential linearization materially improves the baseline while preserving the speed advantage of PIELM.

Experimental Setup

The experiments are implemented in the local PyPIELM-based Burgers scripts published alongside in this paper. The paper run uses the same solver structure throughout, with a larger collocation set and hidden layer than the smoke-test mode. In the full profile, the paper configuration uses 1,800 collocation points, 180 boundary points, 280 initial-condition points, hidden width 280, and three main viscosities in the simple

run. The comparison suite extends that to four viscosities and multiple random seeds and hidden sizes.
The reference solution is generated by a fine finite-difference solver on the final-time slice. This makes the comparison self-contained and avoids importing an external benchmark artifact just to make the plot. The experimental outputs include three types of figures: the Picard convergence history, the accuracy-versus-cost scatter plots, and a representative solution comparison against the reference curve.
The extended comparison suite was used to make the evaluation harder to dismiss. It includes the one-shot linear PIELM baseline, the sequential linearized version, and a small set of PINN baselines from PyPIELM: VanillaPINN, AdaptivePINN, FourierPINN, MuonPINN, and ResidualAdaptivePINN. The suite records RMSE, relative L^2, MAE, R^2, maximum absolute error, wall-clock time, peak memory, boundary-condition error, initial-condition error, iteration count, and the final Picard update norm. This combination is enough to judge both accuracy and computational cost without hiding behind a single metric.

Results

The main numerical result is straightforward: the sequential method consistently improves over the one-shot linear baseline. Across the four viscosities, the Picard-iterated PIELM reduces the average RMSE to roughly half of the baseline value. More precisely, the mean ratio of Picard RMSE to linear-baseline RMSE across the four tested viscosities is about 0.515, i.e. an average reduction of about 48.5%.
The gain is strongest in the easier, more diffusive regimes. At $\nu = 0.10$, the sequential method cuts the baseline RMSE by about 72%, and at $\nu = 0.05$ it cuts it by about 63%. At $\nu = 0.02$ and $\nu = 0.01$ the improvement is still present but smaller, which is what one would expect as the nonlinear regime becomes harder and the fixed-point wrapper has to work against stronger convection effects. The same trend is visible in the convergence plots, where the Picard update norms decrease in a few iterations but do not collapse uniformly across all viscosities.
Runtime remains one of the main strengths of the approach. Even after sequential updates, the PIELM-based method stays in the sub-second range on the tested cases, while the best PINN baselines generally need around 10–13 seconds for the same setups. On average, the best-performing baseline in each viscosity band is roughly 300 times slower than the Picard-iterated PIELM in this benchmark. That ratio is large enough that it cannot be ignored even when a PINN wins on RMSE for some individual viscosity settings.
The comparison suite also shows that the sequential method occupies a distinct middle ground. The one-shot linear baseline is very fast but too inaccurate on

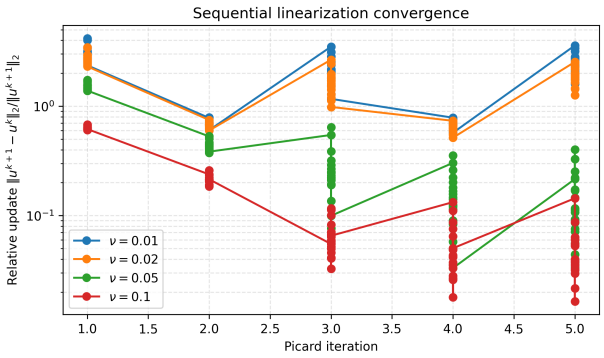

Figure 1: Sequential linearization convergence history. The update norm decreases across Picard iterations, with smaller viscosities remaining harder to stabilize.

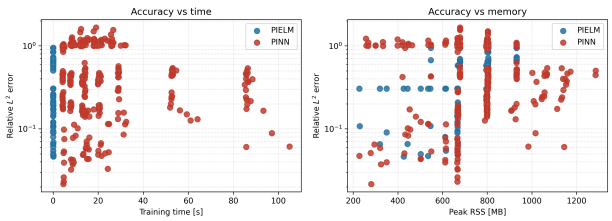

Figure 2: Accuracy versus time and memory for the Burgers comparison suite. The sequential PIELM stays much faster than the PINN baselines while improving on the one-shot linear baseline.

the nonlinear problem. The PINNs are more flexible and can win on accuracy in the smoother viscosity cases, but they pay for it in time and memory. The sequential PIELM sits in between: not always the most accurate method on the table, but usually the most attractive option when the goal is to preserve PIELM-like speed while recovering nonlinear accuracy.
This is visible in the saved assets. Fig. 1 shows the update norms over Picard iterations, Fig. 2 shows the accuracy-versus-cost trade-off, and Fig. 3 compares the final-time profile against the reference solution for the representative case. The figures are meant to be read together rather than in isolation.

Discussion and Limitations

The main takeaway is not that sequential linearization makes PIELMs universally superior. It does not. The takeaway is that the failure mode of one-shot PIELMs on nonlinear PDEs is not a dead end. A very small fixed-point wrapper already recovers a large share of the lost accuracy while keeping the direct-solve character of the method. That is useful because it preserves the practical reason people reach for PIELMs in the first place: speed and reproducibility.
The price is iteration cost, and the cost grows as the equation becomes harder. That is visible in the smaller-

Table 1: Compact Burgers summary for burgers_comparison_results.csv. Each cell shows median [q1, q3] over successful runs grouped by viscosity and method.

ν	Method	n	RMSE	rel-L^2	Time [s]	Peak RSS [MB]
0.01	Linear PIELM	20	0.3851 [0.3851, 0.3851]	0.9472 [0.9472, 0.9473]	0.005734 [0.004189, 0.007871]	804.4 [677.8, 931.1]
0.01	Picard PIELM	20	0.272 [0.2616, 0.2789]	0.6691 [0.6435, 0.686]	0.03475 [0.02499, 0.04838]	804.4 [677.8, 931.1]
0.01	Vanilla PINN	20	0.1809 [0.1738, 0.1847]	0.445 [0.4276, 0.4544]	10.43 [6.753, 15.12]	804.4 [677.8, 931.1]
0.01	Adaptive PINN	20	0.1776 [0.1718, 0.1852]	0.4369 [0.4227, 0.4555]	10.57 [6.903, 15.56]	741.1 [677.7, 931]
0.01	Fourier PINN	20	0.4197 [0.4109, 0.4502]	1.032 [1.011, 1.107]	15.32 [11.14, 20.78]	741.1 [677.7, 931]
0.01	Muon PINN	20	0.4223 [0.4137, 0.436]	1.039 [1.018, 1.072]	10.76 [6.884, 15.62]	741.1 [677.7, 931.1]
0.01	Residual-adaptive PINN	20	0.2064 [0.1897, 0.2188]	0.5077 [0.4666, 0.5383]	40.93 [25.28, 62.4]	931.8 [802.1, 1032]
0.02	Linear PIELM	20	0.3246 [0.3246, 0.3246]	0.8427 [0.8427, 0.8427]	0.00534 [0.003896, 0.00724]	802.5 [802, 803.2]
0.02	Picard PIELM	20	0.2328 [0.227, 0.2385]	0.6044 [0.5893, 0.6191]	0.03378 [0.0244, 0.04382]	802.5 [802, 803.2]
0.02	Vanilla PINN	20	0.1447 [0.1391, 0.1487]	0.3757 [0.3611, 0.3861]	10.23 [6.758, 14.7]	802.5 [802, 803.2]
0.02	Adaptive PINN	20	0.1428 [0.1385, 0.1497]	0.3707 [0.3595, 0.3886]	10.35 [6.784, 14.95]	802.6 [802.1, 803.3]
0.02	Fourier PINN	20	0.3932 [0.3895, 0.4185]	1.021 [1.011, 1.086]	15.29 [11.11, 20.36]	802.6 [802.1, 803.3]
0.02	Muon PINN	20	0.4014 [0.3924, 0.4155]	1.042 [1.019, 1.079]	10.62 [6.885, 15.3]	802.6 [802.1, 803.3]
0.02	Residual-adaptive PINN	20	0.1716 [0.1506, 0.1818]	0.4455 [0.3909, 0.4719]	40.82 [25.13, 61.22]	929.8 [802.6, 1083]
0.05	Linear PIELM	20	0.1852 [0.1852, 0.1852]	0.5817 [0.5817, 0.5817]	0.005416 [0.004551, 0.007537]	798.7 [796.9, 799.5]
0.05	Picard PIELM	20	0.06588 [0.06126, 0.0789]	0.2069 [0.1924, 0.2478]	0.03515 [0.02631, 0.05022]	798.7 [796.9, 799.5]
0.05	Vanilla PINN	20	0.05878 [0.05485, 0.06138]	0.1846 [0.1723, 0.1928]	10.4 [6.758, 14.94]	798.7 [796.9, 799.5]
0.05	Adaptive PINN	20	0.05828 [0.05487, 0.06165]	0.183 [0.1723, 0.1936]	10.44 [6.813, 15.01]	798.8 [797, 799.5]
0.05	Fourier PINN	20	0.3236 [0.3217, 0.3477]	1.016 [1.01, 1.092]	15.08 [11.11, 20.22]	798.8 [797, 799.5]
0.05	Muon PINN	20	0.336 [0.3256, 0.3512]	1.055 [1.023, 1.103]	10.7 [7.036, 15.45]	798.8 [797.1, 799.6]
0.05	Residual-adaptive PINN	20	0.07715 [0.0615, 0.08943]	0.2423 [0.1932, 0.2809]	40.73 [25.01, 60.79]	865.8 [798.7, 1071]
0.10	Linear PIELM	20	0.06654 [0.06654, 0.06654]	0.305 [0.305, 0.305]	0.006375 [0.004728, 0.00827]	640.2 [487.4, 666.9]
0.10	Picard PIELM	20	0.01848 [0.0129, 0.02356]	0.08471 [0.05915, 0.108]	0.04313 [0.03032, 0.05431]	640.2 [487.4, 666.9]
0.10	Vanilla PINN	20	0.01138 [0.009439, 0.01566]	0.05216 [0.04327, 0.07178]	12.14 [7.18, 16.34]	640.3 [487.6, 667]
0.10	Adaptive PINN	20	0.01162 [0.008594, 0.01552]	0.05326 [0.0394, 0.07115]	11.6 [7.135, 16.87]	640.4 [452.8, 667]
0.10	Fourier PINN	20	0.2232 [0.2214, 0.2346]	1.023 [1.015, 1.075]	16.33 [11.57, 22.52]	640.4 [434.6, 667]
0.10	Muon PINN	20	0.2384 [0.2248, 0.2541]	1.093 [1.03, 1.165]	11.84 [7.301, 17.15]	640.5 [423.8, 666.8]
0.10	Residual-adaptive PINN	20	0.03162 [0.02608, 0.03638]	0.1449 [0.1196, 0.1668]	42.15 [26.05, 69.57]	668.7 [613.9, 923.5]

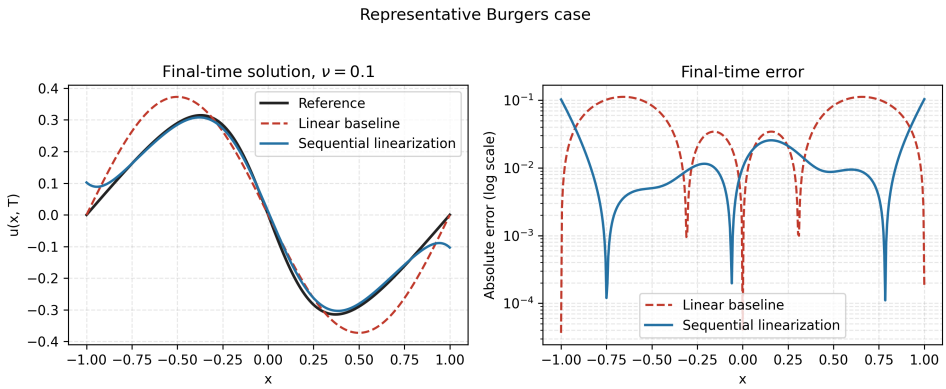

Figure 3: Representative Burgers solution comparison for the smallest viscosity case. The sequential method tracks the finite-difference reference more closely than the one-shot linear baseline.

viscosity cases, where the Picard updates remain larger and the improvement over the linear baseline is less dramatic. There is also no formal guarantee that the method will converge on strongly nonlinear, discontinuous, or shock-dominated problems. The method should therefore be presented honestly as a pragmatic extension, not as a universal solver.

Conclusion

This paper proposed a sequential linearization scheme for PIELMs and tested it on the one-dimensional viscous Burgers equation. The method is deliberately simple: use the existing linear PIELM solver, freeze the nonlinear convection term from the previous iterate, and repeat until the update norm is small. In the benchmarked cases, that was enough to improve the baseline materially while retaining the speed advantage of the PIELM formulation.

The results support a practical conclusion. If a one-shot PIELM is too crude on a nonlinear PDE, a Picard-style wrapper is a low-cost next step before moving to a heavier solver class. The code, outputs, and summary tables in this repository are sufficient to reproduce the experiments and to build the final manuscript around them.

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Physics-Informed Extreme Learning Machines for Steady-State Amplified Spontaneous Emission: A Feasibility Study

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KEYWORDS

physics-informed extreme learning machine, amplified spontaneous emission, rate equations, sequential linearization, Picard iteration, benchmark

ABSTRACT

Physics-informed extreme learning machines (PIELMs) replace iterative neural network training with a direct least-squares solve, enabling fast and reproducible solutions for linear physics-informed problems. However, their performance degrades in nonlinear settings, where the one-shot formulation behaves as a linear approximation of a nonlinear system. This work proposes a simple extension based on outer Picard-style sequential linearization and evaluates it on a steady-state amplified spontaneous emission (ASE) rate-equation model.

The problem is formulated as a zero-dimensional coupled nonlinear system for the excited-state population and photon density. Nonlinearity is handled by iteratively updating a linearized PIELM solve, while the inner solver remains unchanged. Reference solutions are obtained via continuation combined with `fsolve` and bounded least-squares refinement. The method is evaluated across three synthetic regimes (soft, paper, and stiff), demonstrating consistent convergence and improved accuracy under increased iteration depth. In the main configuration, the method achieves a threshold relative error of approximately 18.6%, with RMSE values of 1.30×10^{-2} for the excited-state population and 1.34×10^{-1} for photon density. The results highlight a clear trade-off between computational cost, stability, and accuracy in the proposed sequential PIELM formulation.

INTRODUCTION

Physics-informed extreme learning machines (PIELMs) were introduced as an alternative computational trade-off to gradient-based physics-informed neural networks (PINNs). By keeping the hidden representation fixed and solving only for output weights via a linear system, they enable fast and reproducible training (Huang et al. 2006, Dwivedi and Srinivasan 2020). This framework has proven effective on smooth PDE benchmarks and in broader studies of physics-informed learning, although most applications remain limited to linear or mildly nonlinear problems (Raissi et al. 2019, Karniadakis et al. 2021, Hao et al. 2023, Takamoto et al. 2022, Struniawski 2026).

This paper provides a feasibility study in a less explored direction by considering a steady-state amplified spontaneous emission (ASE) rate-equation model instead of standard fluid or diffusion benchmarks. ASE and laser physics are naturally formulated in terms of coupled rate equations describing pumping, inversion, and photon-density growth (Siegman 1986, Giles and Desurvire 1991, Coldren et al. 2012), making them a representative class of nonlinear systems for mesh-free solvers when kept in a compact form.

The manuscript follows a minimal extension of prior PIELM implementations. The aim is not to propose a general solver for ASE systems, but to demonstrate that a simple sequential linearization wrapper can extend the applicability of the existing PIELM framework to a steady-state ASE model without introducing a full nonlinear optimization procedure.

BACKGROUND AND LITERATURE OVERVIEW

PIELM and related work

The ELM family is based on frozen random hidden features and a deterministic output solve (Huang et al. 2006). PIELM adapts that idea to physics-informed problems by putting the PDE residual directly into the linear system and solving the output weights from collocation and boundary/initial constraints (Dwivedi and Srinivasan 2020). In practice, that gives a simple and fast alternative to gradient-based PINNs, which are more flexible but usually require more optimization effort and more tuning (Raissi et al. 2019, Karniadakis et al. 2021).

The benchmark literature makes the same point from

a different angle. PINNacle and PDEBench showed that scientific machine learning models should be judged across accuracy, speed, memory, and robustness, not just on a single loss value (Hao et al. 2023, Takamoto et al. 2022). Other PIELM-related work also showed that the method is sensitive to feature scaling, activation choice, and the quality of the linear solve (Struniawski et al. 2023, Struniawski 2026). Those observations matter here because a nonlinear fixed-point wrapper is only useful if the inner linear solve remains stable.
Several extensions have since been proposed, including local and domain-decomposition variants that improve scalability and nonlinear approximation capabilities (Dong and Li 2021, Liu et al. 2023, Wang et al. 2025).

ASE rate-equation physics

Amplified spontaneous emission (ASE) originates from spontaneous emission events that are subsequently amplified while propagating through an active gain medium. In laser cavities the same physical mechanisms eventually lead to stimulated emission and oscillation, whereas in optical amplifiers the amplified spontaneous emission forms a broadband background that influences gain, saturation, and noise performance. Despite the diversity of physical implementations, including semiconductor optical amplifiers and rare-earth-doped fiber amplifiers, their steady-state behavior is commonly described using coupled rate equations for the excited-state population and the photon density (or optical power) (Siegman 1986, Coldren et al. 2012, Giles and Desurvire 1991).
The rate-equation formalism expresses the competition between several physical processes. Pump absorption transfers population into the excited state, spontaneous relaxation depletes that population, stimulated emission couples the population inversion to the optical field, and optical losses remove photons from the system. These mechanisms are inherently nonlinear because the stimulated emission rate depends on the product of the excited-state population and the photon density. Consequently, even highly simplified steady-state models exhibit threshold behavior, gain saturation, and multiple operating regimes that cannot be reproduced by purely linear models.
From a numerical perspective, these nonlinear coupling terms are precisely what make ASE models attractive benchmarks for physics-informed solvers. Classical numerical approaches typically solve the resulting nonlinear algebraic systems using continuation methods, Newton-type iterations, or other fixed-point techniques. In realistic fiber-amplifier simulations, the rate equations are additionally coupled with propagation equations along the fiber, producing boundary-value problems that require repeated nonlinear solves (Giles and Desurvire 1991, Desurvire 1994, Becker et al. 1999). Although the present work intentionally avoids these additional spatial complexities, the underlying nonlinear structure remains representative of the challenges encountered in practical ASE modeling.
The present feasibility study therefore adopts the smallest model that still preserves the essential mathematical characteristics of ASE dynamics. We consider a zero-dimensional steady-state system consisting of coupled equations for the upper-state population and photon density. A synthetic quadratic saturation term is included solely to regularize the high-pump regime and maintain bounded solutions; it is introduced as a numerical stabilization device rather than as a claim of physical completeness. This deliberate simplification allows the study to isolate the behavior of the proposed sequentially linearized PIELM without introducing additional complications arising from spatial propagation, wavelength dependence, polarization, or detailed cavity dynamics.
Importantly, the computational framework is independent of this simplified f ormulation. T he sequential PIELM algorithm operates only on the residual equations and their linearization, meaning that more realistic ASE models—including longitudinal propagation, multi-level rare-earth kinetics, wavelength-resolved gain spectra, or experimentally calibrated parameters—can be incorporated without changing the overall solution strategy. The present model should therefore be viewed as a controlled benchmark that evaluates the numerical feasibility of the proposed approach before extending it to experimentally relevant ASE systems.

METHOD

Steady-state ASE model

The simplified ASE model consists of two coupled steady-state algebraic equations describing the interaction between the excited-state population and the photon density. The unknown variables are the excited-state population density N_2 and the normalized pho-ton density ρ_{ph}. The prescribed model parameters com-prise the pump rate W_p, the total active-ion concentra-tion N_{tot}, the upper-state lifetime τ_{21}, the photon life-time τ_{ph}, the stimulated-emission cross section σ_{21}, the spontaneous-emission coupling factor β, and the saturation coefficient s.
The governing equations are

$$0 = W_p(N_{tot} - N_2) - \frac{N_2}{\tau_{21}} - \sigma_{21} N_2 \rho_{ph}, \tag{1}$$

$$0 = \beta \frac{N_2}{\tau_{21}} + \sigma_{21} N_2 \rho_{ph} - \frac{\rho_{ph}}{\tau_{ph}} - s\rho_{ph}^2. \tag{2}$$

The first equation describes conservation of the excited-state population. Pumping transfers atoms into the excited state, spontaneous decay removes population

through radiative and non-radiative relaxation, while stimulated emission transfers energy from the excited population into the optical field. The second equation describes photon generation and depletion. Photons are created through spontaneous emission coupled into the optical mode and stimulated emission, whereas cavity losses and the saturation term remove photons from the system.

Physics-informed Extreme Learning Machine

A single-hidden-layer extreme learning machine approximates the solution by

$$u(\mathbf{x}) = \sum_{i=1}^{L} \beta_i g(\mathbf{w}_i^T \mathbf{x} + b_i), \tag{3}$$

where $g(\cdot)$ is the activation function, L denotes the number of hidden neurons, $\mathbf{w}_i$ and b_i are randomly initialized input weights and biases, and β_i are the output weights. Unlike conventional neural networks, only the output weights are optimized. The hidden-layer parameters remain fixed throughout the computation, allowing the solution to be obtained from a single linear least-squares problem rather than iterative gradient-based optimization.

Instead of minimizing the discrepancy between predictions and labelled data, PIELM minimizes the residuals of the governing equations evaluated at collocation points.

For a collection of collocation points

$$\Omega_c = \{x_i\}_{i=1}^{N_c},$$

the residual operator is assembled into a linear system

$$A\beta \approx b,$$

where matrix A contains evaluations of the governing operators acting on the hidden-layer basis functions and β contains the unknown output weights.

The output weights are computed by solving

$$\min_{\beta} \|A\beta - b\|_2^2 + \lambda \|\beta\|_2^2,$$

where λ denotes a small ridge regularization parameter.

Sequential linearization

The nonlinearity originates from the products

$$N_2 \rho_{ph},$$

which prevent the governing equations from being written as a single linear least-squares system.

To retain the direct PIELM solver, these products are linearized through a Picard fixed-point iteration.

At iteration k

Algorithm 1 Sequential PIELM with Picard Linearization

1: Initialize $k \leftarrow 0$
2: Initialize $(N_2^{(0)}, \rho_{ph}^{(0)})$
3: **repeat**
4: Assemble residual matrix using $(N_2^{(k)}, \rho_{ph}^{(k)})$
5: Solve least-squares PIELM system
6: Update solution $(N_2^{(k+1)}, \rho_{ph}^{(k+1)})$
7: $k \leftarrow k + 1$
8: **until** convergence criterion is satisfied

$$N_2 \rho_{ph} \approx N_2^{(k)} \rho_{ph}^{(k-1)},$$

or equivalently

$$N_2^{(k-1)} \rho_{ph}^{(k)}.$$

The quantities from the previous iteration are treated as constants, transforming the nonlinear equations into a linear system for the current iterate.

Reference solution

To evaluate the proposed approach, a numerical reference solution is computed independently using continuation along the pump parameter.

For each pump value, the previous solution is used as the initial guess for SciPy (Virtanen et al. 2020). If convergence is not achieved, a bounded nonlinear least-squares solver is employed as a fallback. This continuation strategy avoids branch switching near threshold and provides a stable reference across the entire pump sweep.

Experimental Setup

The codebase uses three synthetic presets: *soft*, *paper*, and *stiff*. The *paper* configuration serves as the main balanced feasibility case, while the other two demonstrate robustness with respect to parameter selection. All experiments use the same zero-dimensional model, pump sweep, and PIELM structure.

The pump is swept from $0.1\times$ to $5\times$ the threshold value. In the paper setup, the solver uses 41 collocation points for the sequential solve and 401 points for evaluation. The reference branch is obtained via continuation over the pump range. Default parameters include a hidden width of 192, ridge regularization of 10^{-9}, and a Picard limit of 10 iterations.

Performance is evaluated using RMSE for N_2 and ρ_{ph}, relative L^2 error, threshold and slope errors, training and reference times, iteration count, and the final Picard update norm, providing a compact assessment of accuracy and efficiency.

RESULTS

Table 1 summarizes results for the paper preset. The PIELM solution closely follows the reference for the upper-state population and provides a reasonable approximation of the photon-density branch. While the threshold estimate is not exact, it remains within a feasible error range for a first s ynthetic benchmark. The small slope error indicates that the above-threshold trend is captured correctly, despite minor deviations in curvature.

Table 1: Main Metrics for the Paper Preset.

Metric	Value
RMSE N_2	1.302485×10^{-2}
RMSE ρ_{ph}	1.342373×10^{-1}
Relative L^2 N_2	1.441667×10^{-2}
Relative L^2 ρ_{ph}	2.063854×10^{-1}
Reference threshold pump	1.550503×10^{1}
PIELM threshold pump	1.262030×10^{1}
Threshold relative error	1.860512×10^{-1}
Reference slope	2.290421×10^{-2}
PIELM slope	2.415239×10^{-2}
Slope absolute error	1.248187×10^{-3}
Training time	7.23×10^{-1} s
Reference time	3.00×10^{-2} s
Picard iterations	10

Fig. 1 presents the photon-density response over the full pump sweep. The PIELM solution reproduces the threshold transition and overall growth trend but slightly overestimates the reference in the high-pump regime, consistent with the reported error metrics. This confirms that the method is suitable for feasibility analysis, though not yet a calibrated physical model.

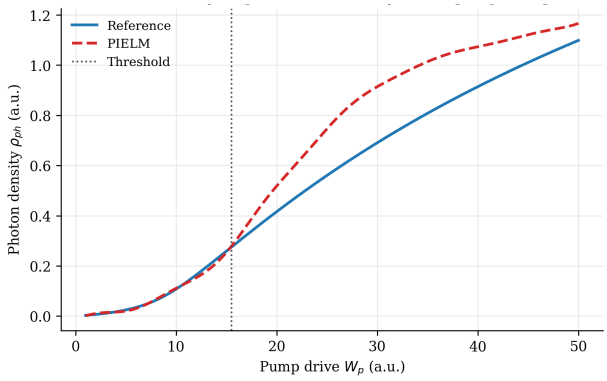

Figure 1: Photon density versus pump drive for the paper preset. The vertical line marks the threshold pump.

Fig. 2 illustrates the coupled threshold dynamics more explicitly. The upper-state population agrees well near threshold, where the system is most sensitive, while deviations increase away from threshold but remain bounded within the expected range for this synthetic setup.

Fig. 3 shows the Picard iteration history, demonstrating that the sequential correction actively reduces the residuals. The convergence is not strictly monotone, which further supports interpreting the approach as a practical fixed-point w rapper r ather t han a f ully converged nonlinear solver.

The regime comparison in Table 2 highlights the trade-off b etween a ccuracy a nd c ost. T he s oft r egime is computationally cheapest but least accurate, the paper regime provides a balanced compromise, and the stiff regime achieves the best accuracy at increased iteration cost.

DISCUSSIONS AND LIMITATIONS

The main limitation of this study is that it represents a synthetic feasibility investigation rather than a physically calibrated ASE model. The governing equations are expressed in normalized form, and the parameter values are chosen to provide a controlled numerical benchmark rather than to reproduce a specific experimental system. However, the formulation is modular, so the parameter block can be directly replaced by experimentally derived values without modifying the solver structure.

A second limitation concerns the scope of the physical model, which is restricted to a zero-dimensional steady-state description. Consequently, spatial propagation, spectral dependence, and cavity-specific e ffects ar e not included, limiting direct applicability to realistic ASE devices. The present work should therefore be interpreted as a numerical feasibility study, with a natural extension toward inverse identification u sing measured gain or spectral data.

Finally, the nonlinear solution strategy is based on a Picard fixed-point i teration. T his c hoice p reserves the simplicity of the PIELM framework but inherits the well-known limitations of fixed-point s chemes, including slower convergence and reduced robustness in strongly coupled regimes. Although the stiff c ase demonstrates good performance, more advanced nonlinear strategies may be required for larger-scale or highly nonlinear ASE models.

CONCLUSION

This work uses the existing PyPIELM codebase to evaluate a sequential linearization strategy for a steady-state ASE rate-equation model. The approach is straightforward: nonlinear coupling terms are frozen, the resulting linearized system is solved using the standard PIELM least-squares formulation, and the process is iterated until convergence. On the main paper preset, this yields a consistent feasibility result with an identifiable threshold branch, satisfactory slope accuracy, and computational cost compatible with short conference requirements.

The regime comparison further clarifies t he b ehavior of

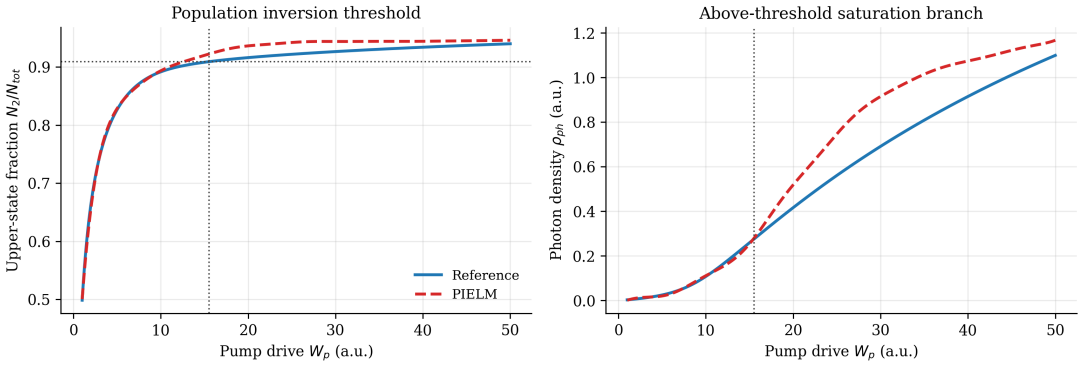

Figure 2: Threshold behavior for the paper preset. The left panel shows the upper-state fraction and the right panel shows the photon-density branch.

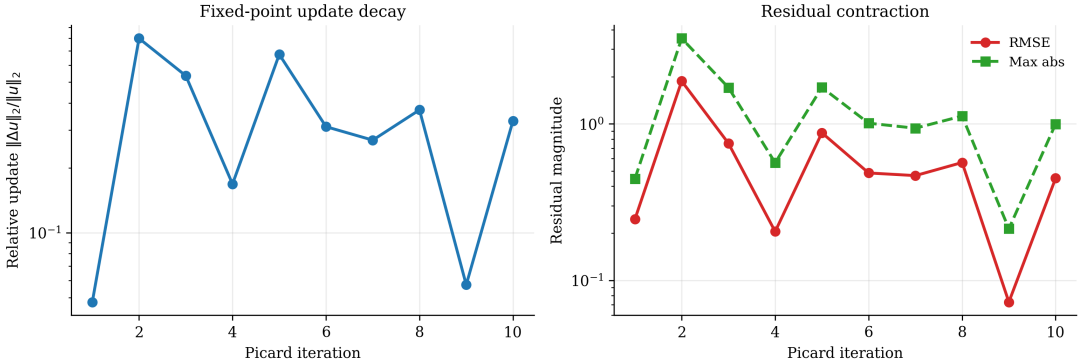

Figure 3: Picard history for the paper preset. The update norm and residuals decrease overall, but the nonlinear regime remains nontrivial.

Table 2: Compact Comparison across the Synthetic Regimes.

Preset	Ref. time (s)	Train time (s)	Total (s)	Iters	Thr. rel. error	RMSE ρ_{ph}
soft	0.011	0.038	0.049	10	2.29×10^{-1}	3.51×10^{-1}
paper	0.030	0.723	0.753	10	1.86×10^{-1}	1.34×10^{-1}
stiff	0.012	0.304	0.316	12	2.16×10^{-3}	1.53×10^{-2}

the method. The balanced configuration provides sufficient accuracy to support the feasibility claim, while the stiff configuration demonstrates that the same framework can achieve significantly improved accuracy under more favorable parameter settings. Future work will replace synthetic parameters with experimentally informed ASE models and extend the approach toward inverse parameter identification rather than purely forward simulation.

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Large Language Models

Dynamic Super-Resolution Strategy for Efficient Scene Text Legibility Classification

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ABSTRACT

Scene text analysis remains highly challenging due to busy visual environments (e.g., complex backgrounds or partial occlusions) and low-resolution inputs. To streamline this process, we investigate the impact of a lightweight Single Image Super-Resolution (SISR) model on binary scene text classification into legible and illegible categories. We propose an integrated pipeline leveraging a Residual Feature Distillation Block (RFDB) network and a compact EfficientNetV2B0 classifier. To handle variable image resolutions, we introduce a dynamic, resolution-aware strategy that automatically directs image crops to the optimal pathway (×4 RFDB, ×2 RFDB, or bicubic resizing) based on their native height. The proposed approach consistently outperforms all fixed baselines, achieving a test PSNR of 31.534 dB and an SSIM of 0.878. Furthermore, thanks to the super-resolution step, it was possible to expand the backbone's final feature map from 1×5 to 2×10, reducing feature variance without increasing the parameter count. Combining this architectural enhancement with hyperparameter optimization within Optuna fundamentally stabilizes the training process, suppressing validation loss divergence. The final optimized pipeline achieves a competitive test balanced accuracy of 83.60% and a ROC-AUC of 0.948 on the Media-Text dataset. These results confirm that the synergy between lightweight super-resolution strategy and robust classifier design offers an exceptional trade-off between reconstruction quality, training stability, and classification accuracy for resource-constrained deployments.

I. INTRODUCTION

Thanks to the development of computer vision technologies, the detection and analysis of text in natural scenes has grown as a crucial research area. Detection systems, however, must contend with numerous challenges and image distortions, making this process exceedingly difficult. These primarily include a vast diversity of fonts, variable scales, colors, layouts (e.g., arbitrary and extreme orientations), and fluctuating illumination conditions. A significant problem and barrier for algorithms remains the highly complex background—natural objects (such as grass, leaves, or mountains) and urbanized artificial structures (e.g., windows, fences, or guardrails) frequently exhibit textural properties closely resembling the text itself, leading to many false detections (Bai et al., 2017; Ghai et al., 2024). The entire task is further complicated by physical image degradation in the form of noise, camera blur, occlusion (partial covering of the text), and overall low sensor resolution (Ghai et al., 2024). Initially, traditional localization methods dominated in solving these problems, including connected component-based algorithms (e.g., MSER) that grouped pixels with uniform properties to separate characters from the background. Unfortunately, in complicated natural environments, these methods generated many false positives and required complex, computationally expensive filtering rules (Zhu et al., 2017; Ghai et al., 2024). Deep learning and Convolutional Neural Networks (CNNs) enabled the construction of breakthrough integrated end-to-end systems. Modern architectures easily handle extreme text shapes through precise, pixel-level localization. By combining Recurrent Neural Networks (RNNs) and attention mechanisms, they efficiently translate regions into character sequences, significantly reducing error accumulation between detection and recognition stages (Ghai et al., 2024; Huang et al., 2025).

Going beyond mere detection and recognition, the classification of text images has long been an essential component of reading systems. Early contributions within document analysis emphasized the necessity of distinguishing between machine-printed text, handwriting, and noise; identifying these types is vital because printed and handwritten texts require significantly different segmentation and recognition techniques (Yefeng Zheng et al., 2004). As research shifted from scanned documents to real-world photography, dealing with large volumes of visual data became a primary challenge. Since only a small fraction of natural images (estimated at 10–15%) contains text, applying complex text reading algorithms directly to all images is highly inefficient. Therefore, systems used classification primarily as a rapid pre-processing step to discard non-text images and save substantial computational power (Bai et al., 2017). Over time, this task evolved chronologically to address new complexities in scene images, leading to the crucial distinction between natural text, which is a physical part of the environment (e.g., street signs), and artificial text that was digitally added to the image during the editing stage (e.g., movie captions). This also encompasses the crucial division into legible and illegible text, where the latter—degraded by blur or extreme perspective—carries no useful information and can disrupt the results of the systems. As recent studies show, compact deep learning models are successfully utilized for this type of fine-grained classification problems. Networks from the EfficientNet family have proven their superiority in these tasks, offering an excellent compromise between high

accuracy and low computational demands. Thanks to their optimized design and minimized classification heads, these models outperform much heavier, traditional architectures like ResNet50, making them an ideal solution for efficient complex systems (Kalisz et al., 2025).

Despite such immense progress in the areas of text detection and classification, low resolution and significant image degradation still pose a barrier; therefore, Scene Text Image Super-Resolution (STISR) algorithms are increasingly applied as an essential pre-processing step. Unlike classic Single Image Super-Resolution (SISR) methods, which focus solely on improving overall image sharpness, STISR algorithms aim to simultaneously enhance visual quality and the legibility of the characters themselves (Ma et al., 2023; Park and Ko, 2025). Early approaches in this trend introduced the so-called categorical Text Prior (TP) into the process, generated by pre-trained recognition models, to guide the networks toward the correct letter shapes during the reconstruction stage (Ma et al., 2023). However, because explicit categorical information could lead to error propagation and the creation of visual hallucinations in the event of a mistake by the base recognizer, engineers had to refine this process. The Non-Categorical Prior (NCAP) was introduced, utilizing hidden representations from the penultimate layers of the network, which stabilizes the reconstruction and prevents the model's overconfidence problem (Park and Ko, 2025). In parallel with the development of NCAP, new architectures such as StyleSRN began embedding unique text style information within the super-resolution process. This approach guarantees that the finally reconstructed text maintains not only its structural integrity but also its stylistic consistency, preserving original fonts and 3D effects even in the most challenging real-world conditions (Yuan et al., 2025).

Adaptive computation has also been explored in computer vision. V-MoE routes image-patch representations to selected experts (Riquelme et al., 2021), while CAMixerSR selectively applies additional attention to complex textures during super-resolution (Wang et al., 2024). These approaches illustrate how computational processing can be adapted to the characteristics of the input.

Although STISR methods can improve text readability, they often rely on computationally demanding architectures. Furthermore, approaches guided by explicit categorical text priors may propagate recognition errors, potentially introducing visual hallucinations or distorted character strokes when the underlying recognizer misinterprets degraded inputs (Ma et al., 2023; Park and Ko, 2025). Motivated by the need for efficient preprocessing, this research investigates the impact of lightweight SISR on scene text legibility classification. The proposed pipeline combines RFDB-based reconstruction with a compact EfficientNetV2B0 classifier and uses a resolution-aware strategy to select an appropriate preprocessing path for each text crop.

II. MATERIALS AND METHODS

Illegible text refers to any image-based writing whose visual quality is too compromised to be successfully interpreted by human annotators or machine recognition algorithms. Such unreadable instances are typically the result of physical or environmental distortions, including severe camera blur, poor lighting and contrast, extreme angles, or objects partially blocking the text (Kalisz et al., 2025).

A. Data

The publicly available Media-Text dataset comprises 400 highly diverse images, encompassing 193 unique resolutions to enhance research within the media industry. In total, the dataset contains 7,744 text instances, each annotated with a four-vertex bounding box. A significant emphasis during the annotation process was placed on identifying and isolating illegible text. Specifically, 973 instances were categorized as illegible. These represent textual content that lacks sufficient visual clarity or resolution to be reliably interpreted by human annotators or automated recognition systems, typically degraded by factors such as motion blur, low contrast, partial occlusion, or extreme perspective (Kalisz et al., 2024).

Publicly-Text collection, a comprehensive dataset comprising 1,393,560 text instances, was compiled by extracting data from several popular and widely adopted public datasets: ICDAR2013 (Karatzas et al., 2013), ICDAR2015 (Karatzas et al., 2015), ICDAR2019 (Chng et al., 2019), COCO-Text (Veit et al., 2016), Total-text (Ch'ng and Chan, 2017), SCUT-CTW1500 (Yuliang et al., 2017) and TextOCR (Singh et al., 2021). The aggregated data was subsequently analyzed and categorized based on text clarity and original labels. As a result of this process, 895,624 text instances were classified and annotated as legible, whereas 497,936 instances were designated as illegible.

All image crops were evaluated based on their height and width dimensions. To maintain a consistent horizontal orientation, any image in which the height exceeded the width was rotated by 90 degrees. Furthermore, since the Latin alphabet was only taken into account, all text instances written in other alphabets were also categorized as illegible across the collections.

B. Model Architectures

The EfficientNet family revolutionized CNN design by introducing a compound scaling method that uniformly balances network depth, width, and resolution. Its baseline model, EfficientNet-B0, developed via Neural Architecture Search (NAS), uses MBConv blocks to optimize both accuracy and computational cost. This results in highly efficient models requiring significantly fewer parameters than traditional architectures while excelling in transfer learning (Tan and Le, 2019). EfficientNetV2 evolves this architecture to address V1's training bottlenecks, such as slow training at high resolutions. It replaces early-layer MBConv blocks with Fused-MBConv to better utilize hardware accelerators and uses training-aware NAS for non-uniform scaling, which caps maximum resolution and selectively deepens later stages. Its key innovation is progressive learning with adaptive regularization—training starts with small images and weak regularization, gradually increasing both. Consequently, EfficientNetV2 offers a highly compact architecture that trains and operates significantly faster without sacrificing classification accuracy (Tan and Le, 2021).

Residual Feature Distillation Block (RFDB) is a key module of the RFDN network, designed for lightweight and fast single image super-resolution (SISR). It represents an optimized evolution of the IMDB block, where inefficient channel splitting operations were replaced by flexible Feature

Distillation Connections (FDC). In the RFDB, the classic 3x3 convolutions previously used for channel compression and reduction are replaced with much lighter 1x1 convolutions. The 3x3 filters are retained only in the main body of the block, which is essential to effectively process the spatial context of the image. An additional innovation in this module is the use of Shallow Residual Blocks (SRB), which integrate a single 3x3 convolutional layer with an identity connection and an activation unit. This allows the model to better utilize the benefits of fine-grained residual learning without introducing any extra parameters. As a result, RFDB creates an architecture that is computationally much lighter than its predecessors, while ensuring higher reconstruction quality (Liu et al., 2020).

C. Evaluation

To rigorously evaluate the proposed system, the experimental process was divided into two integrated processing stages. Based on the analysis of the dataset's characteristics, including the median shape of the extracted text instances, and guidelines from previous studies, we opted to initially normalize the dimensions of the input images to a fixed aspect ratio of 2:5, which resulted in a baseline resolution of 32 × 160 pixels. The height of 32 pixels was specifically selected as it represents the minimal acceptable input size for most popular Convolutional Neural Network (CNN) architectures (Kalisz et al., 2025). To mitigate the visual degradation of the images and investigate the impact of super-resolution methods on subsequent analysis, we trained and employed a model based on the RFDB architecture as an advanced preprocessing algorithm. Its primary objective was to increase the resolution of the text image to a target size of 64 × 320 pixels, while simultaneously restoring lost details and character edges. These reconstructed, enhanced images were then fed into a classifier driven by a compact network from the EfficientNet family, which was designed to ultimately categorize the text instances into legible or illegible classes. Detailed procedures, training configurations, and the obtained results for each of these modules are presented in the following subsections.

All experiments were conducted using a server with the following specifications: 2x RTX A5500 24GB GPU, 128 GB RAM, AMD EPYC 7232P 8-Core Processor.

III. EXPERIMENTS AND RESULTS

A. Data Description

Figure 1 illustrates the distribution of the aspect ratio (defined as the ratio of the longer side to the shorter side) across the dataset categories. As observed in the plot, the median aspect ratio for illegible text instances (2.50) is slightly higher than that of the legible instances (2.25). Furthermore, the distribution of illegible texts exhibits a significantly longer tail, reaching an aspect ratio of up to 20.0, which indicates that extreme, highly elongated text bounding boxes are more frequently associated with illegibility.

Figure 2 presents randomly selected pairs of legible and illegible text samples from the utilized datasets. Notably, the ICDAR2013 dataset serves as an exception in this comparison, as it did not contain any text instances explicitly annotated as illegible.

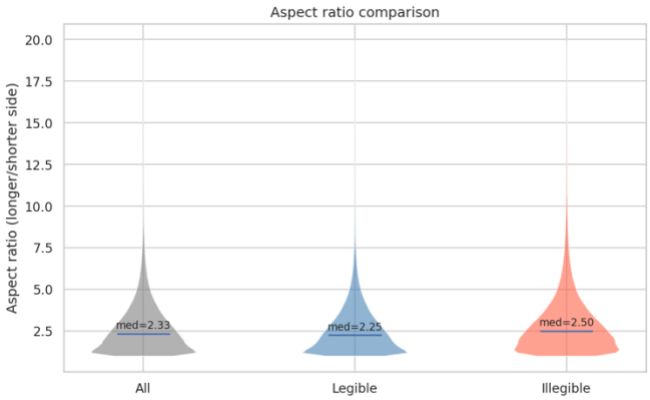

Figure 1. Violin plot comparing the aspect ratios of all, legible, and illegible text instances.

Figure 2. Qualitative comparison of randomly selected legible and illegible samples from public databases.

B. RFDB for Text Image Super-Resolution

To evaluate the effectiveness of the RFDB-based architecture for scene text image restoration, a series of experiments was conducted. From the previously described Publicly-Text collection, we selected 40,196 images with a native crop height of at least 64 pixels. A synthetic degradation pipeline was employed. Each high-resolution crop was bicubically resized to a canonical target of 64×320 pixels and subsequently downsampled to generate degraded inputs for two distinct scale factors: a ×2 model (32×160 input) and a ×4 model (16×80 input). The dataset was randomly partitioned into training, validation, and test sets using an 80:10:10 split. To optimize the network's capacity and learning dynamics, an independent grid search was performed for each upscaling factor. The evaluated search space encompassed: (i) learning rates {1e-3, 5e-4, 1e-4}, (ii) filter widths {48, 64, 128}, and (iii) network depths {4, 6} blocks. Batch sizes were evaluated across {16, 32, 64, 128} for the ×2 model (with 64 proving optimal) and fixed at 256 for the ×4 model. All models were trained using the Adam optimizer to minimize the Mean

Absolute Error (L1) loss function to better preserve sharp text edges.

$$\mathcal{L}_{SR}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \| H_{RFDN}(I_i^{LR}) - I_i^{HR} \|_1$$

where $\mathcal{L}_{SR}(\theta)$ is the super-resolution training loss (the mean absolute error) as a function of the network parameters θ. N is the number of training crops; i indexes the individual samples. $H_{RFDN}(\cdot)$ denotes the RFDN reconstruction network. I_i^{LR} and I_i^{HR} are, respectively, the ii-th low-resolution input crop and its corresponding ground-truth high-resolution target. $\|\cdot\|_1$ is the ℓ_1 norm, so that the loss equals the average reconstruction error over the training crops.

The experimental results revealed that the optimal hyperparameter configuration was scale-invariant: the combination of a 1e-3 learning rate, 128 filters, and 6 blocks ranked first in both scenarios. The best ×2 model achieved a validation PSNR of 36.563 dB, while the ×4 model reached 27.038 dB, which directly reflects the greater challenge of hallucinating four times more spatial information. When evaluated on the test set, the fixed ×2 and ×4 pipelines yielded an overall PSNR of 36.596 dB (SSIM: 0.962) and 27.047 dB (SSIM: 0.84), respectively. Furthermore, the grid search demonstrated the high parameter efficiency of the RFDB architecture; even a heavily constrained variant utilizing only 48 filters and 4 blocks (~0.39M to ~0.40M parameters) performed within a margin of 0.811 dB (for ×2) and 1.340 dB (for ×4) on the test set compared to the optimal ~4 million parameter models. This confirms that the network provides an excellent balance between high reconstruction quality and the low computational inference costs required for an efficient preprocessing stage.

As shown in Table 1, the validation and test metrics align almost perfectly across all configurations (with PSNR differences of ≤0.05 dB), indicating a complete absence of selection bias and confirming the robust generalization of the trained models to unseen data.

Table 1: Comparison results of hyperparameter tuning for the best and worst configuration for each scale. LR – Learning Rate, PSNR – Peak Signal-to-Noise Ratio [dB], SSIM – Structural Similarity Index Measure. Both the validation and test datasets contain 4,020 samples.

Hyperparameters tuning					**Validation Dataset**		**Test Dataset**	
Scale	Filters	Blocks	LR	Params	PSNR	SSIM	PSNR	SSIM
X2	128	6	0.001	3.99M	36.563	0.962	36.596	0.962
X2	48	4	0.0001	0.39M	35.736	0.955	35.785	0.956
X4	128	6	0.001	4.03M	27.038	0.837	27.047	0.840
X4	48	4	0.0001	0.40M	25.724	0.800	25.707	0.803

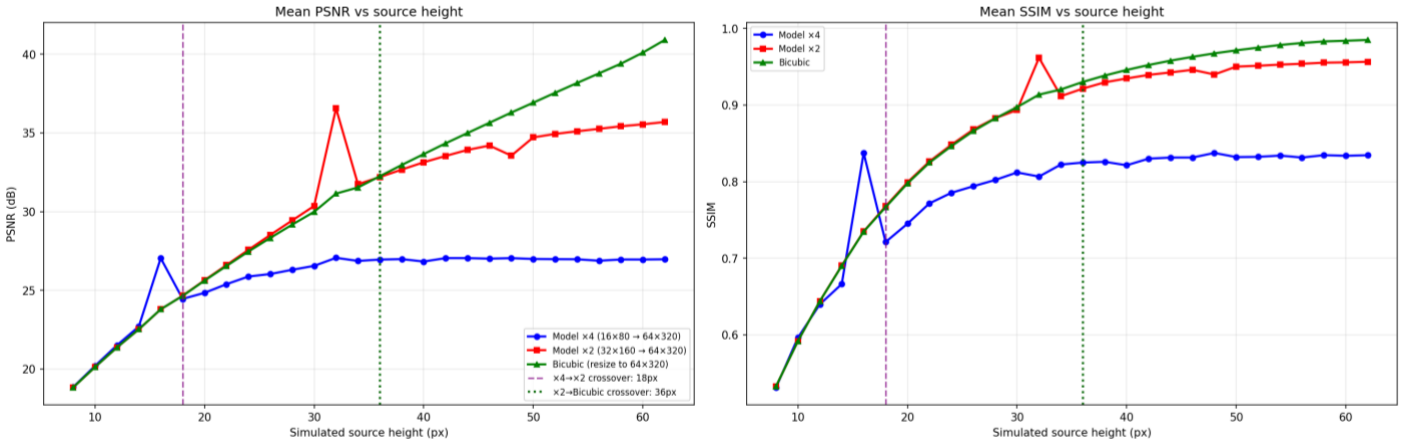

Figure 3. Performance comparison of the fixed ×2, ×4, and bicubic pipelines across varying simulated source heights in terms of PSNR and SSIM metrics

The analysis highlights the high parameter efficiency of the RFDB architecture. For the simpler ×2 upscaling task, reducing the model capacity by approximately 10 times (from ~3.99M down to ~0.39M parameters) results in a modest test PSNR penalty of just 0.827 dB. However, this performance gap becomes substantial for the ×4 scale (a drop of 1.314 dB), demonstrating that hyperparameter sensitivity and the need for greater model capacity scale directly with the difficulty of the reconstruction task. Overall, even the most constrained configurations maintain competitive baseline performance, proving that the RFDB framework provides an excellent trade-off between image fidelity and the low inference latency required for a scalable preprocessing stage.

C. Multi-Model Comparison and Dynamic Strategy

In real-world scenarios, scene text crops arrive at highly variable resolutions, often falling well below the canonical target of 64×320 pixels. To investigate the optimal restoration approach across different input sizes, we conducted a head-to-head comparison of three distinct pre-processing pipelines: a fixed ×4 model, a fixed ×2 model, and a plain bicubic resizing baseline that bypasses learned models entirely. The neural networks were evaluated using their respective optimal configurations (128 filters, 6 blocks). The performance of these pipelines was analyzed across 28 simulated source heights, ranging from 8 to 62 pixels. The results of the experiment are shown in Figure 3.

A detailed crossover analysis revealed that the reconstruction quality is non-monotonic when applying a single, fixed strategy to all resolutions. Specifically, the ×4 model demonstrated superior performance for severely degraded inputs with a native height below 18 pixels. However, for intermediate heights between 18 and 35 pixels, the ×2 model effectively capitalized on the additional input detail, outperforming both the ×4 model and the bicubic method. Notably, once the source height reached 36 pixels or higher, the effectiveness of the neural networks leveled off. At this point, standard bicubic resizing became the optimal strategy, given that the original inputs already retained sufficient high-frequency information.

Based on these results, we introduced a dynamic, resolution-aware routing strategy that directs each input crop to the most appropriate pipeline based on its native height (height < 18 px routed to ×4; 18–35 px routed to ×2; ≥ 36 px routed to bicubic).

$$P(h) = \begin{cases} \text{RFDN}_{\times 4}, & h < 18 \\ \text{RFDN}_{\times 2}, & 18 \le h < 36 \\ \text{bicubic}, & h \ge 36 \end{cases}$$

As presented in Table 2, this proposed dynamic strategy consistently outperformed every fixed pipeline, including the strong bicubic baseline, across all metrics on both datasets. Figure 4 illustrates the routing rule on a representative test crop degraded to three source heights, one per branch. In every regime the path the router selects is also the highest-scoring one: RFDN ×4 at 16 px (28.44 dB / 0.922, against 24.87 / 0.858 for bicubic), RFDN ×2 at 32 px (37.77 / 0.982 against 33.13 / 0.970), and plain bicubic interpolation at 48 px (39.33 / 0.992 against 36.25 / 0.979 for the ×2 model). Two details deserve comment. At 16 px the ×2 branch is indistinguishable from bicubic (24.86 dB / 0.858): its input must first be interpolated from 16 × 80 up to 32 × 160 and therefore carries no information the interpolation did not already have, so a learned model applied below its native input resolution degenerates into the very operation it was meant to replace. Conversely, the two metrics disagree in how sharply they separate the paths once reconstruction is already good — at 48 px PSNR still gives bicubic a 3.08 dB margin over the ×2 model, while their SSIM differs by only 0.013 (0.992 against 0.979).

Table 2: Performance comparison of fixed super-resolution pipelines versus the proposed dynamic strategy.

Strategy	Validation Dataset		Test Dataset	
	PSNR	SSIM	PSNR	SSIM
X4	25.755	0.785	25.746	0.787
X2	30.269	0.864	30.268	0.865
Bicubic	31.178	0.872	31.158	0.873
Dynamic (proposed)	**31.546**	**0.877**	**31.534**	**0.878**

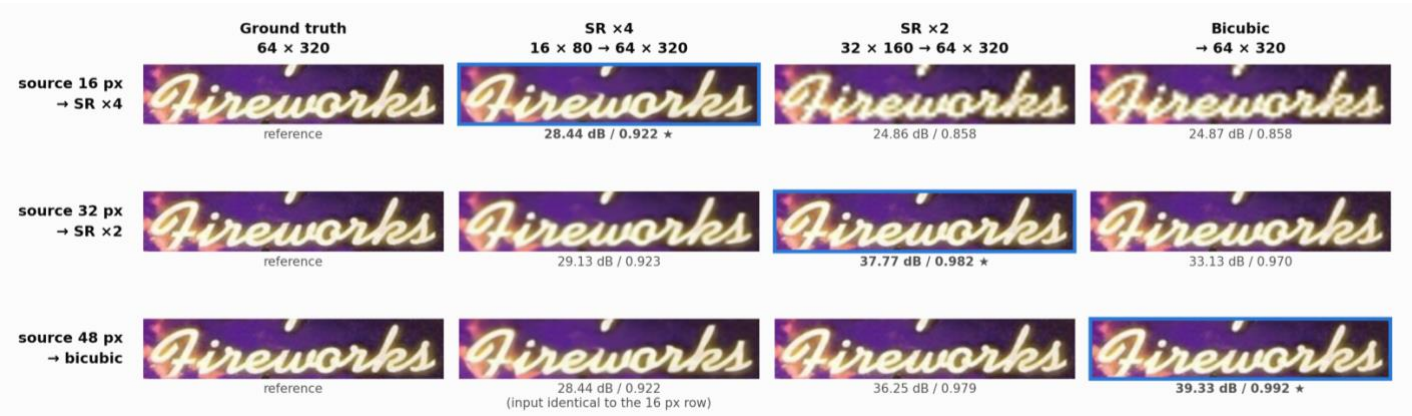

Figure 4. Reconstruction paths across the three routing regimes; outlined panel is the router's choice, ★ = best PSNR.

D. Illegible Text Classifier

To filter out illegible text regions and thereby streamline the operation of comprehensive scene text analysis systems, we formulated a binary classification task to distinguish between legible and illegible text regions. To determine a suitable configuration, hyperparameter tuning was conducted using the Optuna framework (Akiba et al., 2019). The optimization was driven by the Tree-structured Parzen Estimator (TPE) algorithm, a sequential Bayesian strategy that models the distributions of good and bad outcomes to sample new parameter candidates in regions with the highest probability of improvement. Two workers ran in parallel, one per GPU, sharing a single study database. The search space comprised nine hyperparameters: (i) backbone architecture type {EfficientNetB0, EfficientNetV2B0}, (ii) fine-tuning depth (number of unfrozen backbone stages), (iii) spatial dropout rate, (iv) learning rate, (v) weight decay, (vi) batch size, and

(vii–ix) data augmentation magnitudes for rotation, brightness and contrast. The study consumed approximately 674 GPU-hours in total. Based on its results, we selected the compact EfficientNetV2B0 backbone, equipped with a minimal classification head regularized via SpatialDropout2D.

In every trial the classification head was trained by minimizing the categorical cross-entropy between the predicted class probability p_i and the ground-truth y_i:

$$L_{\mathrm{CE}} = -\frac{1}{m}\sum_{i=1}^{m} \log \hat{p}_{i,y_i}$$

The selected configuration uses a learning rate of 6.7×10^{-4}, a spatial dropout rate of 0.407, six of the seven backbone stages left trainable, a batch size of 512, and augmentation magnitudes of 0.238 (rotation), 0.097 (brightness) and 0.083 (contrast); weight decay was driven to 1.0×10^{-6} and is therefore effectively inactive. Adding the dual-SR preprocessing step (described in section B) and then tuning the classifier both act on the learning curve, and their effects are visible on measures that do not depend on how long each run happened to last. Figure 5 shows the training and validation loss of the three configurations on shared axes, with the vertical distance between a configuration's two curves being its generalization gap. Compared over the 52 epochs the three runs share, the validation minimum improves monotonically from 0.3056 in the baseline to 0.2883 with SR pre-processing and 0.2865 after tuning. The curve also becomes progressively quieter, the mean epoch-to-epoch change in validation loss falling from 0.011 to 0.008 to 0.006. The practical consequence is visible in the width of the near-optimal region: the baseline and the untuned SR run hold four and three epochs respectively within 0.005 of their best validation loss, whereas the tuned run holds eleven, so the retained checkpoint no longer depends on which single epoch happens to be lowest.

Part of the SR step's contribution is architectural. At 64×320 the backbone's final feature map is 2×10 rather than the baseline's 1×5, so each of the 1,280 pooled features is averaged over 20 spatial positions instead of 5, and the vertical dimension is no longer collapsed to a single cell before pooling. The vector reaching the classification head has the same 1,280 dimensions and the model the same parameter count in both cases, only the estimate of each feature becomes lower-variance. Because this step changes the SR model and the input resolution together, their contributions cannot be separated from these runs.

Only a few of the nine hyperparameters matter. Learning rate is the only one that strongly affects the optimum itself: every leading trial lies in the $2 \times 10^{-4} – 1 \times 10^{-3}$ decade, while rates below 5×10^{-5} give the study's worst results (median 0.3006 against 0.2862). EfficientNetV2B0's median loss falls below EfficientNetB0's best, though it was sampled five times more often. Spatial dropout must be at least ≈0.35, above which the exact value barely matters; brightness and contrast correlate with no outcome ($|\rho| \leq 0.24$) and rotation matters only as a threshold the forty trials sampled below 0.1 hold all ten of the largest post-minimum divergences and reach their minimum at a median epoch of 8.5 against 22 above it (Mann–Whitney $p < 10^{-7}$), while beyond 0.1 further rotation brings no benefit. The response surface is flat, validation loss spans 0.046 across the 96 completed trials but only 0.006 among the leading forty, so the final selection was made on learning-curve stability rather than on the tuning objective alone.

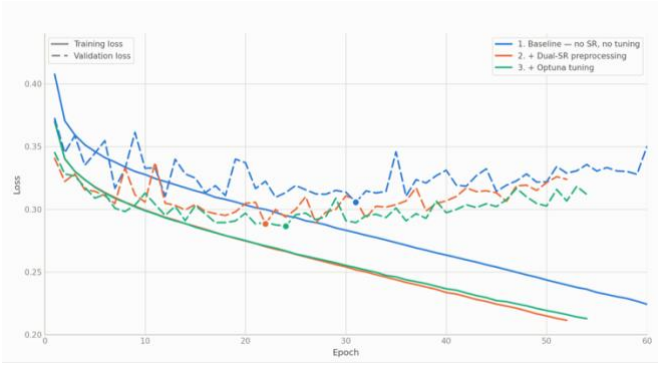

Figure 5. Training (solid) and validation (dashed) loss for the baseline, the dual-SR, and the tuned dual-SR configuration over the first 60 epochs; ringed markers mark each run's selected (lowest-validation-loss) epoch.

Table 3 reports the resulting classification performance. Since the production split carries a markedly different class prior (12.6% illegible, against 35.7% in training and validation), the fixed argmax rule is not a meaningful operating point; each model is therefore evaluated at the threshold that maximizes balanced accuracy on the validation split, and that same threshold is transferred to the test split. Both pipeline steps improve every validation metric monotonically. On the production split, balanced accuracy rises from 82.8% to 83% to 83.6% and ROC-AUC from 0.940 to 0.948, so the added stability does not come at the cost of discriminative power; illegible F1 improves from 0.674 to 0.695, corresponding to ten additional illegible regions correctly rejected out of 973.

Table 3: Validation and production-split performance of the three configurations, at thresholds selected on validation. BACC – Balanced Accuracy, F1 – F1 score for illegible class.

Variant	Validation Dataset		MediaText	
	BACC	F1	BACC	F1
Baseline	0.865	0.817	0.828	0.674
Dual-SR	0.871	0.826	0.830	0.689
Tuned Dual-SR	**0.875**	**0.831**	**0.836**	**0.695**

IV. CONCLUSIONS AND FUTURE WORK

The results clearly demonstrate that naively applying a single "best" trained super-resolution model to all incoming images is suboptimal. The proposed strategy successfully combines the specific regimes of local strength for each pipeline, achieving the highest overall test PSNR of 31.534 dB and an SSIM of 0.878. Notably, the dynamic resolution strategy selected the empirically best-performing pipeline for 100% of the evaluated heights on the unseen test set, confirming its exceptional accuracy and reliability for practical deployment in a pre-processing stage.

An architectural evaluation of the Super-Resolution stage further reveals that even shallower RFDB-based variants with significantly reduced parameter counts achieved highly satisfactory reconstruction performance. The evaluation metrics of these lightweight models remained remarkably close to those of the deeper target architectures, demonstrating that compact RFDB designs offer an optimal trade-off between computational efficiency and image enhancement

quality, making them particularly suitable for resource-constrained inference environments.

Furthermore, integrating the super-resolution module into the pre-processing pipeline substantially enhanced the quality and spatial resolution of the feature representation (increasing feature map dimensions from 1×5 to 2×10) when paired with the EfficientNet-based classifier. This spatial enhancement provided effective regularization during training, flattening the learning curves and yielding noticeable improvements in overall classification accuracy and stability.

Finally, the developed model holds significant potential to serve as a foundational backbone architecture for downstream scene text analysis tasks. Future research will explore leveraging this backbone model for fine-grained text categorization, particularly in distinguishing between natural and artificial scene text.

A further direction concerns disentangling the two effects that the super-resolution stage currently couples: the learned reconstruction of high-frequency detail and the mere increase in spatial resolution. As noted above, these cannot be separated in the present runs, since the SR model and the input resolution change together. Future work will address this through a controlled ablation that holds the classifier input fixed at 64×320 while varying only how it is produced by the RFDB model versus plain bicubic upscaling. Comparing classification performance under identical resolution would isolate the net contribution of the learned super-resolution from that of the resolution increase, and repeating the protocol across the router's source-height regimes would indicate at which resolutions learned reconstruction is genuinely worthwhile.

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CROSS-LINGUAL AGREEMENT BETWEEN TEXT AND SPEECH EMOTION RECOGNITION: TOWARDS ROBUST MULTIMODAL PERCEPTION IN SOCIAL ROBOTS

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ABSTRACT

This study analyzes cross-modal consistency and language differences in multimodal emotion recognition systems used in social robotics. Maintaining this consistency is crucial for natural human-robot communication, yet the aspect remains underexplored in the literature. The methodology combines a review of multimodal datasets and emotion recognition models, the development of a universal tool for analyzing cross-modal consistency, and an assessment of how language differences affect model performance.
The evaluation covers four datasets (approximately 59,000 utterances) and 10 models across three modalities: speech, text, and facial expression. It examines model performance, cross-modal consistency measured with Cohen's and Fleiss' κ coefficients, and the influence of language. The results indicate that text-speech consistency depends primarily on the emotionality of the text and the characteristics of the corpus, not solely on language. Both recognition performance and text-speech agreement are among the lowest for the Polish corpora, but the high variability across corpora within the same language shows that dataset difficulty and quality play a key role. The analysis also reveals a shortage of natural corpora in which all modalities simultaneously convey a significant emotional signal. The developed tool enables effective validation of existing solutions and analysis of new datasets and models.

INTRODUCTION

The development of applied artificial intelligence (AI), and particularly large language models (LLM), has made AI-based tools increasingly a part of everyday life. This trend also aligns with the assumptions of social robotics and human-robot interaction (HRI), according to which AI should support humans in a variety of situations. Algorithms are increasingly being used to analyze utterances, determine their sentiment, and identify the emotions contained within them. Therefore, it is crucial to further develop these methods toward a more natural interpretation of emotions.
In the case of social robotics, multimodality, i.e., the use of information from various sources, is a key element enabling natural human-machine communication. However, this also presents a number of challenges. Research indicates the need for better modeling of the relationships between input channels. Determining the impact of individual signals on the data fusion process and developing uniform methods for assessing the effectiveness of such solutions remains a particular challenge (Pillalamarri and Shanmugam 2025).
Just as the biblical Tower of Babel symbolized barriers resulting from language differences, the limited availability of multilingual emotional resources can lead to problems with the correct interpretation of emotions in both interpersonal communication and HRI. Another limitation is the heavy focus of emotion recognition research on high-resource languages while many other languages remain underrepresented. The scarcity of appropriate datasets hinders the development of models capable of accounting for cultural and linguistic differences (Muhammad et al. 2025).
The main goal of the research is to develop and evaluate a universal framework for the analysis of multimodal emotion recognition, enabling the study of consistency between modalities and the impact of language differences, with particular emphasis on the Polish language in the context of social robotics.

RELATED WORK

Multimodal Emotion Recognition

The complexity of human emotions is reflected in both how they are expressed and perceived. Humans communicate their emotional state through multiple channels simultaneously – through facial expressions, the content of their utterances, intonation, tempo and tone of voice,

as well as physiological signals, which are often beyond their control. Therefore, contemporary research on emotional artificial intelligence is increasingly focusing on a multimodal approach which increases classification efficiency and improves the system's resilience to interference and missing data. An example of a solution utilizing this concept is the EmbraceNet+ architecture, which enables maintaining high-quality emotion recognition even with incomplete input data (Heredia et al. 2022). The integration of verbal and nonverbal communication allows for more accurate determination of the user's emotional state. This allows the system to adapt its communication accordingly, building trust and improving the quality of HRI.

Cross-modality Consistency

One of the greatest advantages of a multimodal approach is the ability of individual information channels to complement each other. If data from one modality is unavailable, noisy, or ambiguous, the system can utilize information from other sources. In practice, however, an additional challenge arises: individual modalities may indicate different emotional states, or one may not convey any clear signal. In such situations, examining cross-modal consistency becomes crucial. In HRI, the consistency of expression conveyed by facial expressions and the robot's behavior influences the correct interpretation of the user's intentions and increases the naturalness of the interaction. Inconsistency between modalities or the use of rigid behavioral patterns can lead to reduced trust and the perception of the robot as unnatural (Stock-Homburg 2022).

Linguistic Context and Resource Constraints

An additional challenge in multimodal emotion recognition is the dependence of some modalities, particularly speech and text, on language and cultural context. Current research on automatic emotion recognition in conversations is heavily focused on English, which accounts for the majority of analyzed cases, while other languages remain significantly underrepresented. The limited number of available emotional speech datasets in less common languages makes model development and evaluation difficult. The lack of sufficiently diverse data leads to limited model generalization and the risk of biased results, which is a significant obstacle to the application of social robots in multilingual environments. A solution to this problem is the creation of new multilingual datasets and conducting cross-linguistic comparative studies, which still constitute a small portion of published work (Alhussein et al. 2025). Examples of initiatives aimed at reducing this gap include evaluation campaigns such as PolEval, which cover tasks related to emotion recognition in Polish speech (`https://poleval.pl/tasks/task4`).

RESEARCH ASSUMPTIONS

The above literature review highlights significant research challenges related to improving the naturalness of communication between humans and social robots. Due to the need to verify cross-modal consistency and the limited availability of multilingual datasets for emotion recognition, the following research objectives were formulated:

- conduct a review of existing multimodal datasets for emotion recognition, with particular emphasis on Polish-language and multilingual datasets,
- conduct a review of modern models for emotion recognition from text, speech, and facial expressions, with emphasis on their effectiveness,
- develop a tool (pipeline) that will easily integrate various multimodal datasets and classification models, enabling predictions, cross-modal consistency analysis, and assessment of language differences for individual modalities,
- conduct research to evaluate the effectiveness of models on various datasets and analyze cross-modal consistency using Cohen's and Fleiss' coefficients,
- conduct research on the behavior of models on multilingual datasets and compare the results obtained for Polish-language datasets with other languages.

The research focused primarily on two modalities: speech and text, with emphasis on the Polish language, which is still relatively underrepresented. Facial expression and its consistency with the other modalities were also analyzed.

MULTIMODAL EMOTION DATASETS

Emotion recognition datasets, particularly language-specific and multilingual corpora, remain limited resources. The datasets used for multimodal consistency analysis in this project are presented below:

- Dataset of Emotional Speech in Polish (nEMO) (Christop 2024) - a corpus developed for the Polish language, comprising over three hours of emotionally charged speech. The texts used reflect the full phonetic range of the Polish language, while recording the material in a home environment allowed for a more natural-sounding voice.
- Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) (Livingstone and Russo 2018) - a verified collection of speech and vocal recordings performed by 24 professional actors with North American accents. Data is available in audiovisual, video, and audio formats and covers two levels of emotional intensity.
- Multimodal EmotionLines Dataset (MELD) (Poria et al. 2019) - the collection contains approximately 13,000 utterances from dialogues of the TV series Friends. It also includes audio tracks and video footage, enabling multimodal emotion analysis and

considering interactions between participants.

- Collection of Multilingual Emotional Speech Corpora (CAMEO) (Christop and Czajka 2025) - this is a unified reference dataset consisting of 13 emotional speech corpora in eight different languages, including the aforementioned nEMO and RAVDESS. The dataset was prepared to ensure consistent conditions for evaluating and comparing emotion recognition models.

Table 1 presents a comparison of the datasets in terms of the most important parameters in multimodal analysis: the language, data modality, descriptive categories, the number of samples, and other characteristic features. All corpora are publicly available for scientific research.

Table 1: Comparison of Multimodal Datasets for Emotion Recognition Problems

Dataset	Language	Modality	Classes	Samples	Other
nEMO	Polish	text, speech	anger, fear, happiness, neutral, sadness, surprise	4 481	acted and amateur speech, 9 professional actors, semantically neutral sentences
RAVDESS	English	text, speech, facial expression	angry, calm, disgust, fearful, happy, neutral, sad, surprised	1 440 speech, 1 012 song	acted speech and song, 24 actors, 2 fixed neutral sentences, two intensity levels
MELD	English	text, speech, facial expression	anger, disgust, fear, joy, neutral, sadness, surprise	13 706	multi-party dialogues from the series "Friends", natural emotional text, noisy audio
CAMEO	Bengali, English, French, German, Italian, Polish, Russian, Spanish	text, speech	17 emotions (93% are anger, disgust, fear, happiness, neutral, sadness, surprise)	41 265	acted and read speech, unified set of 13 standard corpora in 8 languages

EMOTION RECOGNITION MODELS

The model selection focused primarily on two modalities: text and speech. Due to the importance of the Polish language in this study, models representing different linguistic capabilities were considered, with Polish as the main reference point. Facial expression models were additionally included as a linguistically independent modality:

- Chatbot emotion recognition model (Skowroński 2023) - a proprietary model for analyzing emotions in Polish texts, utilizing an LSTM architecture and word representations (word embeddings) trained on the National Corpus of Polish (NKJP) and the Polish Wikipedia. It was trained on an artificially generated dataset from a chatbot and achieved an accuracy of 96% for general data and 87% for a dataset manually tagged by humans.
- Fine-tuned RoBERTa Model for Emotion Classification in Polish `https://huggingface.co/visegradmedia-emotion/Emotion_RoBERTa_polish6` - the RoBERTa-base model tuned for classifying emotions in texts written in Polish. It was developed specifically for analyzing user opinions and content published on social media, achieving high performance, particularly in recognizing fear and disgust.
- Emotion English DistilRoBERTa-base (Hartmann 2022) - a version of the DistilRoBERT model tuned for emotion recognition in English texts. It classifies texts into Ekman's six basic emotions and a neutral class, using data from several diverse text sources from social media to TV dialogues.
- Multilingual Emotion Classification Model (Borisov 2026) - a multilingual model based on the Transformer architecture, trained on a synthetic dataset of over one million multi-labeled examples in 23 languages, enables emotion classification in a multilingual environment, achieving performance comparable to models dedicated to English.
- xlm-roberta-large-pooled-emotions10 `https://huggingface.co/poltextlab/xlm-roberta-large-pooled-emotions10-v3` - model designed to classify emotions in texts in seven languages (each language has its own specific model). It uses SentencePiece tokenization and predicts one of ten emotional categories, including enthusiasm, pride, and hope.
- Parallel 2D CNN-Transformer Encoder (Skowroński et al. 2024) - a speech analysis architecture combining parallel convolutional layers for feature extraction with a Transformer encoder using self-attention. The best results were obtained for the Polish nEMO corpus (84.60% accuracy), while maintaining performance on English recordings (63.00%).
- SpeechBrain Emotion Recognition with wav2vec2 trained on the IEMOCAP dataset (Ravanelli et al. 2021) - an open-source speech processing platform offering a set of tools and pre-built models. It enables the implementation of a wide range of speech analysis tasks and provides solutions that achieve competitive results in numerous benchmarks.
- Qwen2-Audio-7B-Instruct (Chu et al. 2024) - a large-scale audio-language model that analyzes diverse audio signals and performs tasks based on natural language instructions. By handling complex acoustic scenes, it can interpret recordings containing multiple speakers and background sounds.

- Vision Transformer (ViT) Model (Pakov 2024) - facial expression recognition model based on the Vision Transformer architecture. It was fine-tuned on the FER2013 dataset and classifies processed facial images into seven basic emotional states.

Table 2 presents a comparison of the emotion recognition models in terms of the most important parameters: analysis language, modality, classes and accuracy of these models provided by the authors.

Table 2: Comparison of Emotion Recognition Models for Multimodal Research

Classification Model	Language	Modality	Classes	Accuracy
Chatbot	Polish	text	anger, disgust, fear, happiness, neutral, sadness, surprise	0.87
Fine-tuned RoBERTa	Polish	text	anger, disgust, fear, joy, none of them, sadness	0.82
Emotion English DistilRoBERTa-base	English	text	anger, disgust, fear, joy, neutral, sadness, surprise	0.66
Multilingual Emotion Classification	multilingual (23 languages)	text	anger, contempt, disgust, fear, frustration, gratitude, joy, love, neutral, sadness, surprise	0.84
xlm-roberta-large-pooled-emotions10	multilingual	text	anger, disgust, enthusiasm, fear, hope, joy, neutral, none, pride, sadness	0.82
Parallel 2D CNN-Transformer Encoder based on nEMO	Polish	speech	anger, fear, happiness, neutral, sadness, surprise	0.85
Parallel 2D CNN-Transformer Encoder based on RAVDESS	English	speech	anger, disgust, fear, happiness, neutral, sadness, surprise	0.63
SpeechBrain wav2vec2 on IEMOCAP	English	speech	anger, happiness, neutral, sadness	0.79
Qwen2-Audio-7B-Instruct	multilingual	speech	Audio LLM model (prompted)	not reported
ViT for facial expression recognition	—	facial expression	angry, disgust, fear, happy, neutral, sad, surprise	0.71

EXPERIMENTAL EVALUATION

Pipeline

One of the project's main goals was to create a universal tool for examining cross-modal consistency in emotion recognition for social robotics. Its key features include the unification and harmonization of model classes and datasets according to Paul Ekman's widely adopted theory of six basic emotions (anger, disgust, fear, joy, sadness, and surprise) with an additional neutral category, while allowing researchers to map or discard other classes. The pipeline also enables easy integration of new datasets and models, efficient prediction for all configured models through a caching mechanism, comprehensive analyses including confidence correlation, cross-modal consistency, Cohen's and Fleiss' coefficients, and statistical tests, as well as automatic generation of reports, tables, and figures. The pipeline consists of four core modules that define the entire process - INGESTOR, PREDICTOR, ANALYZER and REPORT GENERATOR. The tool's architecture, including modules, is shown in Figure 1.

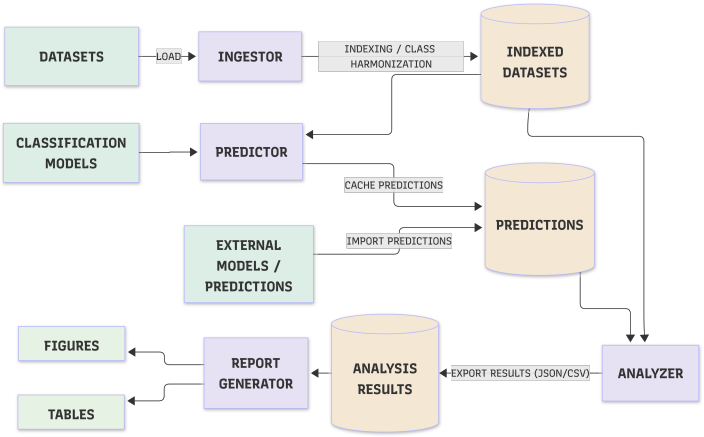

Figure 1: Architecture of the proposed Evaluation Pipeline

The pipeline is publicly available at `https://github.com/SkowRon96/Cross_Lingual_Agreement_TSER`.

Experimental Setup

Study 1 – Cross-modal consistency analysis. The first s tudy a imed t o a ssess c ross-modal consistency in emotion recognition. Analysis was conducted on the RAVDESS, nEMO, and MELD datasets, using all modalities. All available models for individual languages, as well as a multilingual model, were used. The models' prediction performance for each dataset was assessed using accuracy and Macro-F1, taking into account the problem of imbalanced classes. Subsequently, the agreement between modalities (focusing on text and speech, with facial expression as an additional modality) was analyzed using Cohen's κ (Eq. 1) and Fleiss' κ (Eq. 2) coefficients al ong wi th 95 % co nfidence intervals. Cohen's κ is used when comparing two classification sources. This index measures the extent to which the agreement between classifications exceeds the agreement expected if the classes were assigned randomly.

$$\kappa_{\text{Cohen}} = \frac{P_o - P_e}{1 - P_e}, \tag{1}$$

where P_o denotes the observed agreement between two classifiers and P_e the expected agreement by chance. Fleiss' κ is an extension of Cohen's coefficient to the situation where the agreement of a larger number of classifiers is analyzed.

$$\kappa_{\text{Fleiss}} = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \tag{2}$$

where $\bar{P}$ denotes the average observed agreement between all classifiers, and $\bar{P}_e$ the average expected agreement resulting from the class distribution.

The coefficients take values $-1 \leq \kappa \leq 1$, where $\kappa = 1$ corresponds to perfect agreement, $\kappa = 0$ to agreement at the random level, and $\kappa < 0$ to agreement worse than expected by chance.

Study 2 – Language impact on cross-modal consistency. The second study focused on the influence of language on cross-modal consistency. Analysis was conducted on the CAMEO and MELD datasets, which together cover eight languages. Only two of the models operate across all of them, and both were used: Qwen2-Audio-7B-Instruct for the speech modality and the Multilingual Emotion Classification Model for the text modality. The audio model was applied in a zero-shot setting, i.e. the emotion label was requested through a natural-language prompt, without fine-tuning on the evaluated corpora. MELD served as a reference set, being the only corpus among those considered with natural, non-acted utterances. The study assessed the prediction performance across languages and the applicability of the zero-shot approach to emotion recognition. Additionally, text and speech consistency was analyzed using Cohen's κ coefficient.

Results

The experiments utilized four datasets totaling approximately 59,000 utterances and 10 models operating in three modalities: text, speech, and facial expression.
In Study 1, the performance was first assessed for all models used, per dataset and modality. The results are presented in Table 3. Based on the obtained results, several important observations can be made. First and foremost, the nEMO and RAVDESS datasets contain utterances with little semantic content, which means the text modality does not convey sufficient emotional content. This results in low performance of the text models, particularly noticeable for the RAVDESS dataset. It follows from the design of these corpora, where actors utter a fixed set of semantically neutral sentences, and is confirmed by all text models converging to the same accuracy regardless of architecture.
For the nEMO dataset, the dedicated Polish-language model achieves significantly better results than the zero-shot approach. For the English-language RAVDESS and MELD datasets, the differences between the models are much smaller, and in some experiments, the zero-shot approach achieves comparable or even better results. The exception is the model trained on the IEMOCAP dataset, which typically achieves lower performance, although for the speech modality in the MELD dataset, it achieved the best result (0.41), outperforming the zero-shot and CNN-RAVDESS models.

Table 3: Performance comparison of emotion recognition models across multimodal datasets

Classification Model	Dataset	Modality	Accuracy	Macro-F1
Multilingual Emotion Classification	nEMO	text	0.17	0.11
Fine-tuned RoBERTa	nEMO	text	0.17	0.13
xlm-roberta-large-pooled-emotions10	nEMO	text	0.17	0.13
Chatbot	nEMO	text	0.17	0.10
Qwen2-Audio-7B-Instruct	nEMO	speech	0.26	0.19
Parallel CNN-Transformer [nEMO]	nEMO	speech	0.88	0.88
Emotion English DistilRoBERTa-base	RAVDESS	text	0.20	0.05
Multilingual Emotion Classification	RAVDESS	text	0.20	0.05
Qwen2-Audio-7B-Instruct	RAVDESS	speech	0.85	0.85
Parallel CNN-Transformer [RAVDESS]	RAVDESS	speech	0.78	0.77
SpeechBrain wav2vec2 on IEMOCAP	RAVDESS	speech	0.18	0.09
ViT for facial expression recognition	RAVDESS	facial expression	0.47	0.44
Emotion English DistilRoBERTa-base	MELD	text	0.54	0.42
Multilingual Emotion Classification	MELD	text	0.46	0.33
Qwen2-Audio-7B-Instruct	MELD	speech	0.38	0.33
SpeechBrain wav2vec2 on IEMOCAP	MELD	speech	0.41	0.20
Parallel CNN-Transformer [RAVDESS]	MELD	speech	0.22	0.14
ViT for facial expression recognition	MELD	facial expression	0.23	0.15

The facial expression modality did not exceed an accuracy of 0.47 (RAVDESS) and dropped to 0.23 (MELD), where it was the weakest of the three modalities. Despite using different methods for extracting representative frames, the models achieved relatively low results, which may be attributed to the presence of noise in the images, multiple participants, and the limited generalizability of models trained on more homogeneous datasets.
An interesting observation is the MELD dataset, for which the text modality achieved higher performance than the speech modality (0.54 vs. 0.38). This confirms that text carries significant emotional information in dialogues. This is also evident when comparing the results for the RAVDESS and MELD datasets, where accuracy and Macro-F1 indicate a significantly higher emotional content in the text layer of dialogues. Because the analyzed corpora contain both acted and natural utterances, the direction of conflict between the text and speech modalities was additionally analyzed. The results are presented in Figure 2. The analysis indicates that the speech modality plays a dominant role for the RAVDESS and nEMO corpora, as the textual content does not contain sufficient emotional signals. MELD exhibits a different pattern, with text derived from natural dialogues and constituting a more significant source of emotional information than speech. The results confirm

that the modality carrying the most important emotional signal depends on the characteristics of the given datasets.

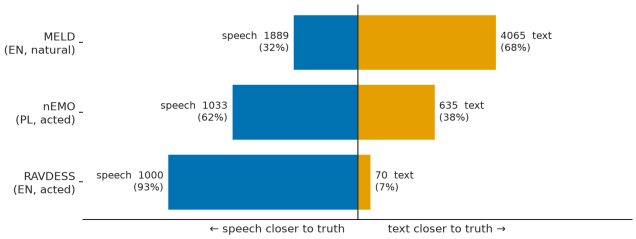

Figure 2: Direction of Text–speech Conflicts across Emotion Recognition Datasets

Next, the cross-modal consistency was analyzed using Cohen's and Fleiss' κ coefficients, co mputed between one reference model per modality: Emotion English DistilRoBERTa-base for English text, the Multilingual Emotion Classification M odel f or P olish t ext, Qwen2-Audio-7B-Instruct for speech, and the ViT model for facial expression. Models fine-tuned o n t he analyzed corpora were excluded here, as their agreement would reflect training on the same data rather than cross-modal correspondence. The results are presented in Table 4.

Table 4: Cross-modal agreement comparison.

Dataset	Coef	Modalities	κ	95% CI
nEMO	Cohen	text–speech	0.015	[0.01, 0.02]
RAVDESS	Cohen	text–speech	0.000	[0.00, 0.00]
RAVDESS	Cohen	text–facial expr.	0.000	[0.00, 0.00]
RAVDESS	Cohen	speech–facial expr.	0.297	[0.27, 0.33]
RAVDESS	Fleiss	text–speech–facial expr.	0.015	[0.00, 0.03]
MELD	Cohen	text–speech	0.202	[0.19, 0.21]
MELD	Cohen	text–facial expr.	0.038	[0.03, 0.04]
MELD	Cohen	speech–facial expr.	0.041	[0.03, 0.05]
MELD	Fleiss	text–speech–facial expr.	0.065	[0.06, 0.07]

For the nEMO dataset, a lack of consistency was observed between the text and speech modalities, which is due to the neutral nature of the transcripts. For RAVDESS, there is moderate consistency between speech and facial expression (0.297), while text shows no consistency with either speech or facial expression. For MELD, however, moderate consistency between text and speech is observed (0.202), while the values for pairs containing the facial expression modality are close to zero, primarily due to noise in the data.

Analysis of all three modalities simultaneously using Fleiss' κ also indicates a lack of global consistency, as a single inconsistent modality lowers the overall score. Excluding the neutral class did not significantly change the κ value, confirming that the low agreement is not solely due to the dominance of this class. Analysis of 95% confidence intervals confirms that the main contrast between datasets is not a coincidence.

Ultimately, low κ values for some modality pairs should be interpreted in light of the corpus characteristics. In the case of RAVDESS, the zero text to speech correspondence results primarily from the lack of emotional information in the text layer, rather than from a true lack of cross-modal dependence, as confirmed by the MELD results and the analysis of conflict direction.

The second study analyzed the impact of language on emotion recognition performance and consistency between the text and speech modalities, using the Multilingual Emotion Classification Model for text and Qwen2-Audio-7B-Instruct for speech. The results are presented in Table 5.

Table 5: Comparison of model performance and cross-modal consistency across multilingual datasets

Language	CAMEO Dataset	Text		Speech		κ	95% CI
		Acc	F1	Acc	F1		
fr	cafe	0.09	0.04	0.40	0.31	0.002	[0.00, 0.00]
en	crema_d	0.15	0.04	0.79	0.80	0.000	[0.00, 0.00]
en	emns	0.14	0.06	0.25	0.22	0.039	[0.03, 0.05]
it	emozional	0.15	0.08	0.28	0.21	0.012	[0.01, 0.01]
en	enterface	0.75	0.75	0.63	0.63	0.428	[0.40, 0.46]
en	jl_corpus	0.20	0.13	0.30	0.30	0.107	[0.09, 0.13]
es	mesd	0.23	0.19	0.27	0.22	0.043	[0.03, 0.06]
pl	nemo	0.17	0.09	0.26	0.19	0.014	[0.01, 0.02]
fr	oreau	0.45	0.44	0.50	0.49	0.247	[0.20, 0.29]
de	pavoque	0.60	0.29	0.77	0.64	0.181	[0.16, 0.20]
en	ravdess	0.20	0.05	0.85	0.85	0.000	[0.00, 0.00]
ru	resd	0.25	0.24	0.27	0.25	0.127	[0.11, 0.15]
bn	subesco	0.14	0.09	0.31	0.20	-0.019	[-0.03, -0.01]
en	MELD*	0.46	0.33	0.38	0.33	0.183	[0.17, 0.19]

* dataset is not a part of the CAMEO collection

The results indicate that the Qwen2-Audio approach outperforms the multilingual text model in most cases. Among the CAMEO corpora, the only exception is ENTERFACE (0.75 vs. 0.63); text also outperforms speech for the MELD reference set (0.46 vs. 0.38). However, the advantage of speech models is partially due to the characteristics of the corpora, where text often lacks sufficient emotional information. The highest scores were obtained for Germanic datasets, but the high variance within English indicates that the characteristics of the corpus are more important than the language itself.

Text and speech consistency analysis using Cohen's κ revealed low values for most languages. The highest consistency was obtained for enterface ($\kappa = 0.428$), while for the Polish nEMO dataset, the value was close to zero ($\kappa = 0.014$). This confirms previous observations that cross-modal consistency depends primarily on the presence of emotional information in the text, not the language itself. Additionally, differences between Accuracy and Macro-F1 indicate the problem of imbalanced classes. For example, for PAVOQUE the results suggest uneven recognition of individual emotions.

Additionally, the performance of the Qwen2-Audio model using a zero-shot approach on various language sets was analyzed. Performance results are presented in Figure 3, while the distribution of predictions for individual emotions is presented in Figure 4.

The obtained results indicate significant variability in

model performance across corpora. These differences are not solely due to language—for example, the German PAVOQUE corpus achieves high scores, while some English corpora exhibit significantly lower performance. This indicates that the characteristics and difficulty of the corpus also have a significant impact on the results.

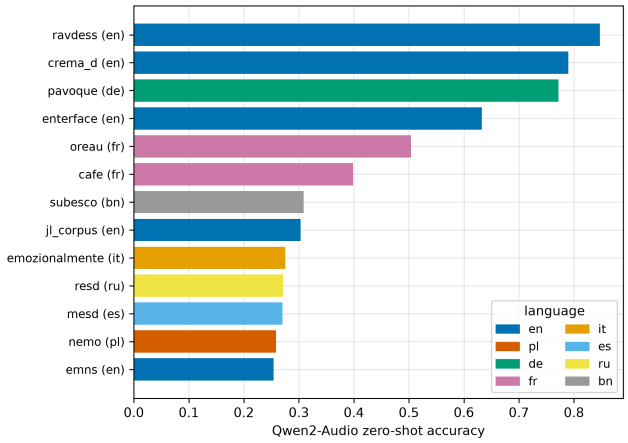

Figure 3: Qwen2-Audio Accuracy Distribution across CAMEO Datasets

Analysis of the prediction distribution shows that for more difficult corpora, the model tends to limit predictions to a few dominant emotion classes, primarily fear, sadness, and anger, while sparsely selecting the classes happiness, surprise, and disgust. This suggests a partial collapse of prediction for less represented or more difficult linguistic domains.

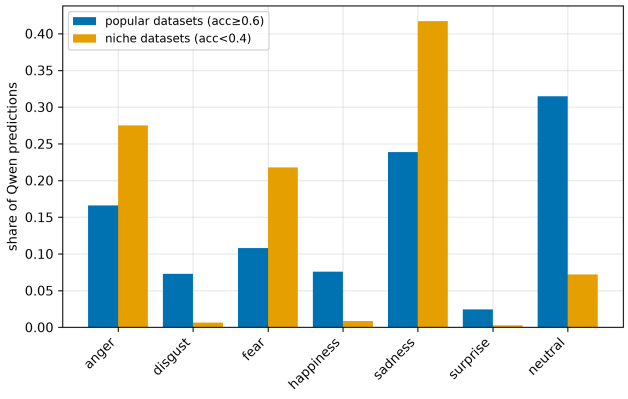

Figure 4: Emotion Category Distribution for popular and Niche Datasets

CONCLUSION AND FUTURE WORK

In summary, the conducted research indicates that assessing cross-modal consistency is a key step in the validation of emotion recognition systems. Correspondence between text and speech is not a property of the modalities themselves, but depends on whether a given channel actually contains emotional information. In the RAVDESS and nEMO actor corpora, the text proved to be emotionally empty, so the lack of consistency does not indicate a modality conflict, but rather the absence of an emotional signal in the textual layer. Only the analysis of natural MELD dialogues demonstrated measurable consistency ($\kappa = 0.202$), showing that before using a single modality, its suitability must be validated on data representing real-world conditions.

The research also confirmed a gap related to the limited availability of multimodal datasets for niche languages. Analysis of eight languages revealed a predominance of English-language resources and high variability in the results of the zero-shot Qwen2-Audio model. This indicates that current audio-LLMs do not yet represent a reliable solution for cross-linguistic emotion recognition. In the case of Polish, due to the lack of natural emotional text-speech corpora, speech currently remains the most stable source of information, while text should be used depending on the emotional context of the utterance.

A significant contribution of this work is a review of datasets and emotion recognition models, together with a research pipeline covering the entire process of cross-modal consistency analysis, from data harmonization to result reporting, which allows new datasets and models to be evaluated efficiently.

The limited number of natural, multilingual emotional corpora, particularly for Polish, remains a limitation of this research. Future work should focus on creating such resources and extending multimodal systems with additional sources of information, such as biomedical signals. However, their acquisition and integration remain demanding and time-consuming due to the need to capture data reflecting actual emotional communication.

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Exploring OODA Loops through Categorical Formalisation and Executable Aspect Semantics

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KEYWORDS
OODA, Attestation, Trusted Computing, Rewriting, Simulation

ABSTRACT

We present a category-theoretic formalisation of the Observe–Orient–Decide–Act (OODA) loop in which each stage is modelled as an object in a category OODA, and cross-cutting concerns are captured as aspects: functors from OODA into endofunctor categories carrying their own rewrite-based operational semantics. This construction gives OODA an executable structure whose properties including fixed points, termination, timing and the relationship between an aspect's internal model and evolving reality, can be explored through simulation rather than only stated formally. We instantiate the approach using computational trust, established through remote attestation, as a worked aspect, building on a prior category-theoretic model of trust. Using a small scenario, we trace the execution of an OODA-loop cycle utilising the rewrite rules from initial observation through to a set of possible actions, and show how this executable structure exposes properties of OODA itself including the divergence between an OODA loop's internal model of reality and reality as it actually evolves, and the temporal cost of reaching a fixed point.

INTRODUCTION

The use of Observe, Orientate, Decide and Action Loops (OODA) developed by Boyd (Boyd et al. 2018) and further extended by others (Angerman 2004) is well established in military applications as a mechanism for analysing a situation and formulating a response (Brehmer 2005). As systems become progressively more complex and OODA is employed in a more extensive role it becomes clear that managing various aspects of systems being observed needs to be better understood (Brehmer 2007).
A second area of investigation is the computational trust of computing systems, through the use of devices such as the Trusted Platform Module (TPM), trusted execution environments (TEEs) (Shepherd and Markantonakis 2024), enclaves for confidential computing (Anasuri 2023) and remote attestation (Oliver 2021). We focus here on the process of attestation and the establishment of trust.
As the latter is less well known but now becoming an integral part of defence technologies (Ryu et al. 2024, Sahu et al. 2025), it is necessary to understand how this aspect works within the OODA scheme itself. This is especially true when we are utilising multiple OODA loops and OODA loops in real-time processing contexts. We explore the nature of observations and how they are constructed with respect to their aspects. Given this we continue the paper with a walk through of an OODA-loop demonstrating how these principles apply to orientation, decisions and actions through their executional semantics.
The primary contribution of this paper is not a trust model, but an executable categorical foundation of OODA in which aspects are represented as functors into endofunctor categories and rewrite semantics. This enables systematic exploration (simulation) of the behaviour, timing and divergence between an OODA's loop's internal model and evolving reality.

FORMALISATION

In this section we describe the formalisation of OODA and of the trust aspect. We then show how trust (and other) aspects are mapped into OODA.

OODA

We construct a category **OODA** containing the objects $\mathtt{Obs}, \mathtt{Ori}, \mathtt{Dec}$ and $\mathtt{Act}$, and morphisms $\mathtt{ori}, \mathtt{dec}$ and $\mathtt{act}$ according to the diagram shown in figure 1. This represents the simplest formalism of an OODA loop - we ignore feedback loops and the idea of a *next* morphism from an action to a new observation for good reason, firstly being simplicity, and the latter that feedback loops can be modelled as instantiations of new OODA loops with new observations, and that *next* might not exist 'inside' an OODA loop: time is linear and loops unfold, hence *next* plays a very different role which we explore in our simulation later.
$\mathtt{Obs}$ denotes observations of some system — the totality of information gathered during an observation cycle,

$$\mathsf{Obs} \xrightarrow{\mathsf{ori}} \mathsf{Ori} \xrightarrow{\mathsf{dec}} \mathsf{Dec} \xrightarrow{\mathsf{act}} \mathsf{Act}$$

Figure 1: Category OODA

subject to the inherent limits of observability, access and time. No assumption is made as to whether such observations are structured, directed or complete, nor that they constitute a full or faithful representation of the system being observed. An observation also contains information about *how* the observation was conducted

Ori is generated through the process ori and contains the analysis, inferences and conclusions drawn from the observation. Orientation may reveal hidden structures, perform abstractions and incorporate contextual knowledge beyond what the observation itself contained. From a single observation multiple orientations can be constructed.

Decisions Dec are produced from Ori by dec and encode a policy-driven response to the gaps between observed and expected states identified by the orientation — determining which gaps are addressed, deferred or accepted; the null decision is always admissible. Actions Act are produced from Dec by act and represent the concrete execution of those decisions. Every action, including the null action, constitutes a determinate outcome of the pipeline.

Trust and Attestation

In (Oliver and Kuure 2026) we introduced a category **TRUST** (specifically a topos) for the expression of [computational] trust and described how the operations of attestation, verification and ultimately decision of an element's level of trust is established:

$$\begin{aligned} \text{Objects} &: \mathsf{E}, \mathsf{C}, \mathsf{R}, \mathsf{D} \\ \text{Pullback} &: \mathsf{C} \cong_{\mathsf{E}} \mathsf{M} \times \mathsf{S} \\ \text{Morphisms} &: \mathsf{attest} : \mathsf{E} \to \mathsf{C}, \mathsf{verify} : \mathsf{C} \to \mathsf{R}, \\ &\quad \mathsf{decide} : \mathsf{R} \to \mathsf{D} \\ \text{Classifier} &: \mathcal{A} : \mathsf{E} \to \mathsf{D} \\ \text{Evaluation} &: \mathsf{eval} : \mathsf{E} \times (\mathsf{A}, \Upsilon, \Delta) \end{aligned} \tag{1}$$

The classifier $\mathcal{A}$ captures the entire attestation process: $\mathsf{decide} \circ \mathsf{verify} \circ \mathsf{attest}$ and is used as a shorthand for the composition of these three operations.

Elements and claims are modelled as a pullback shown in figure 2 relating the integrity measurements and signature (identity) of the element with the contents of a claim; which is further uniquely identified by some identifier.

Verification is the process of checking the validity and veracity of claims against known norms or expected values into some result space. This result space is modelled as a co-product as in equation 2.

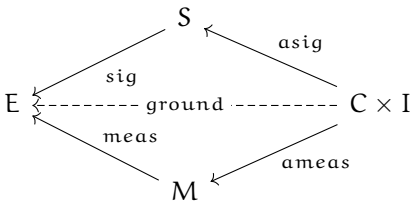

Figure 2: Pullback of Claims

$$\mathsf{R} = \coprod_{1\ldots n} \{\mathsf{R}_1 \ldots \mathsf{R}_n\} \amalg \{\mathsf{R}_{\mathsf{ERR}}\} \tag{2}$$

Finally a decision is made mapping the result spaces to a trust decision as shown in figure 3.

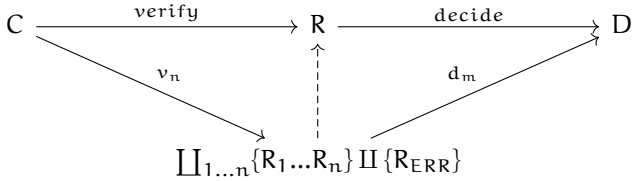

Figure 3: Mapping to Decisions

Trust is treated as an intuitionistic concept(Amoore 2014); lack of evidence does not imply that a system is trusted, or that something not being not trusted is not the same as being trusted (Becker et al. 2012). We utilise a Heyting Algebra to encode trust levels (Ren et al. 2025) in our decision space: this structure is a lattice with least and greatest elements $\bot$ and $\top$, relation $\leq$, and equipped with operators *meet* $\wedge$, *join* $\vee$ and *implication* $\to$ where $(\mathsf{c} \wedge \mathsf{a}) \leq \mathsf{b} \equiv \mathsf{c} \leq (\mathsf{a} \to \mathsf{b})$. The lack of excluded middle allows us to better reason about the idea of not trusting something: $\neg\neg\mathsf{a} \neq \mathsf{a}$. For example, in earlier work we utilise six levels of trust:

- $\top$ Successfully attested identity and integrity
- $\mathsf{D}_{\mathsf{NEW}}$ Successfully attested identity and integrity, but one which is 'new' to our knowledge and need to be monitored in some sense.
- $\mathsf{D}_{\mathsf{AUTH}}$ Validated identity without any integrity measurement present, eg: secure boot
- D_{M} Validated integrity measurements without any signature present
- D_{S} Validated identity (signed) but invalid integrity measurements
- $\bot$ Failure (including error situations)

The ordering and other properties are:

$$\begin{aligned}&\bot \leq \mathsf{D_S} \leq \mathsf{D_{AUTH}} \leq \mathsf{D_{NEW}} \leq \top\\&\bot \leq \mathsf{D_M} \leq \mathsf{D_{NEW}} \leq \top\\&\mathsf{D_{AUTH}} \| \mathsf{D_M} \quad \text{(incomparable) and } \mathsf{D_{AUTH}} \wedge \mathsf{D_M} = \bot\end{aligned} \tag{3}$$

Finally, the actual operations of the attestation pipeline are expressed as functions over elements, verifications and decisions: $(\mathsf{A}, \Upsilon, \Delta) : (\mathsf{C^E}, \mathsf{R^C}, \mathsf{D^R})$. This structure we will discuss when mapping our formalism and simulation to a real attestation environment, and also to the nature of actions within OODA.

Aspects of Observations (Trust)

To bridge between **OODA** and **TRUST**, we construct a functor **OBST**, which actually maps **OODA** to the category of End[**TRUST**, **TRUST**]. The codomain is chosen as an endofunctor category because aspects are *processes acting on trust models* rather than just trust models.

To further demonstrate the properties of this we begin by constructing the observation endofunctor $\mathsf{OBST}(\mathsf{Obs}) = \mathsf{B}$. The corresponding orientation, decision and action endofunctors are introduced later when discussing execution and simulation. Thus B should be understood as one element of a larger family of aspect-processing transformations.

It should be emphasised that trust is used here as a concrete aspect. An aspect is represented by a functor from **OODA** into an appropriate endofunctor category whose objects provide the operational semantics of that aspect. Trust and attestation merely provide the first worked example.

B is the machinery that processes the 'how' of an observation, ensuring that both the resultant observation is consistent with the structural rules of **TRUST** and the resultant observation is well-formed, either by successful attestation or application of default rules.

B is defined by a set of rewrite rules acting on the state of each element in the observation (Barbierato et al. 2026). Each element is in a state of being unattested (initial), attested, verified or decided.

$$\frac{e \vdash \mathsf{Init} \quad \text{attest}(e) = c}{e \vdash \mathsf{Attested}(c)} \text{ Attest} \tag{4}$$

$$\frac{e \vdash \mathsf{Attested}(c) \quad \text{verify}(c) = r}{e \vdash \mathsf{Verified}(r)} \text{ Verify} \tag{5}$$

$$\frac{e \vdash \mathsf{Verified}(r) \quad \text{decide}(r) = d}{e \vdash \mathsf{Decided}(d)} \text{ Decide} \tag{6}$$

$$\frac{e \vdash s \quad s \neq \mathsf{Decided}(d)}{e \vdash \mathsf{Decided}(\bot)} \text{ Default} \tag{7}$$

We say that a fixed point of B exists when no rule above can alter the element's state, ie: $\mathsf{B}(\mathsf{B}(X)) = \mathsf{B}(X)$, which occurs when every element satisfies $e :\vdash \mathsf{Decided}(d)$; this is then the output of B.

For example, if we make an observation $\mathtt{obs}_1$ with elements e_1 and e_2 existing in the trust domain as an input to that observation, then we can record that we attest, verify and decide the trust of those elements. For example (simplifying the 3 operations above) we can define rules to run the attestation pipeline as below and graphically as in 4. Note: we write $\mathcal{A}(e) = d$ as shorthand for $e \vdash \mathtt{Decided}(d)$.

$$\frac{e1}{\mathcal{A}(e1) = \top} \text{ Attest Pipeline} \tag{8}$$

And as, in this example e_2 was never attested, then a default value is applied:

$$\frac{e2}{\mathcal{A}(e2) = \bot} \text{ Trust Default} \tag{9}$$

$$\{e1, e2\} \xrightarrow{\mathsf{B}} \begin{array}{l}\{e1, e2,\\ \mathcal{A}(e1) = \top,\\ \mathcal{A}(e2) = \bot\}\end{array}$$

Figure 4: Endofunctor B Applying Rules

The observation $\mathtt{obs}_1$ is then the output of B and the above $\{e1, e2, \mathcal{A}(e1) = \top, \mathcal{A}(e2) = \bot\}$ is a fixed point of B.

Now that the mechanisms of B have been introduced, **OBST** similarly functions on the other objects in **OODA** giving the following $\mathbf{OBST}(\mathsf{Ori}) = \mathsf{O}$, $\mathbf{OBST}(\mathsf{Dec}) = \mathsf{D}$ and $\mathbf{OBST}(\mathsf{Act}) = \mathsf{A}$. Again O, D and A are endofunctors in the category End[TRUST, TRUST]. These will be better explained in the following section as they follow the same form as B.

OBST by definition also operates on the morphisms in **OODA**: $\mathbf{OBST}(\mathsf{ori}) = \mathtt{bo} : \mathsf{B} \to \mathsf{O}$, $\mathbf{OBST}(\mathsf{dec}) = \mathtt{od} : \mathsf{O} \to \mathsf{D}$ and $\mathbf{OBST}(\mathsf{act}) = \mathtt{da} : \mathsf{D} \to \mathsf{A}$. Categorically these are natural transformations as they are operating between functors and therefore allows us to specify further properties of the workings on the aspects. We shall briefly discuss these later.

SIMULATION

The purpose of the simulation is not to evaluate attestation technology itself, but to demonstrate the executional semantics of the trust aspect (Shu et al. 2016). Specifically, we show how observations containing trust-relevant entities are transformed through the rewrite system B until a fixed point is reached.

Use Case

Consider a tactical unit consisting of a mobile communications hub and the users (field operatives) of that system as in figure 5. Each of these devices operated utilising a TEE and can provide its identity and integrity for verification through the attestation process (Kuure 2024).

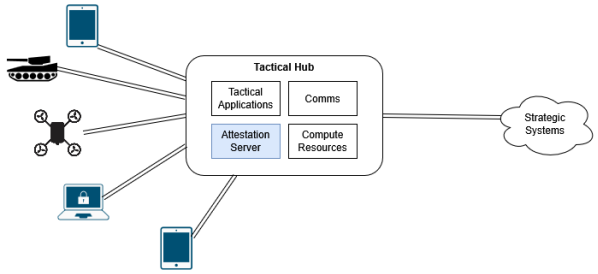

Figure 5: Example System

Observation of this system is required to be made periodically as well as under certain situational changes: attachment of mobile device, change of communications provider and so on. In each situation we wish to ensure, both at the tactical communications hub as well as the strategic, centralised hub that the system's integrity has not been compromised. If a compromise in the system has been detected then application of suitable mitigation actions needs to be decided upon and performed (Fusano et al. 2011).
We introduced B formally in the earlier section. The functors O, D and A to be introduced in the following sections are explained illustratively but follow a similar, formal structure, semantics and pattern to B.
The prototype implementation currently under development derives the event traces from the Jane Attestation Server environment (`https://github.com/iolivergithub/jane`) and digital forensics tooling (Oliver 2020). The simulator wraps the OODA formalism around the attested systems allowing execution and replay of the rewrite rules as necessary. During execution of the simulation, an operator may actively modify the underlying computing environment to demonstrate real-time changes and the gap between reality and any divergences observed.

Observation

The system in figure 5 is observed at a particular moment to contain 3 elements: $\mathtt{obs}_1 = \{\mathtt{drone}_1, \mathtt{laptop}_1, \mathtt{mobiledevice}_1\}$ one drone, one laptop, one mobile device. The observation therefore represents only those elements visible at the time of measurement and not necessarily the complete state of reality, for example, due to network latencies, real-time constraints of communications etc.

These are mapped to elements $\{\mathtt{d1}, \mathtt{l1}, \mathtt{m1}\}$ in the trust domain. It can be that things observed are actually composite or decomposed/structured elements in the domain of the aspects. The initial state and applications of B is shown in figure 6.

$$\begin{array}{c} \{\mathtt{m1} : \mathtt{Init} \\ \mathtt{l1} : \mathtt{Init} \\ \mathtt{d1} : \mathtt{Init}\} \\ \downarrow \mathsf{B} \\ \{\mathtt{m1} : \mathtt{Attested(c1)} \\ \mathtt{l1} : \mathtt{Attested(c2)} \\ \mathtt{d1} : \mathtt{Attested(c3)}\} \end{array} \xrightarrow{\mathsf{B}} \begin{array}{c} \{\mathtt{m1} : \mathtt{Verified(r1)} \\ \mathtt{l1} : \mathtt{Verified(r2)} \\ \mathtt{d1} : \mathtt{Verified(r3)}\} \end{array}$$

Figure 6: Internal Operations of B

Figure 7 shows the final decision step and the fixed point of B. Here we see that the mobile device has been fully attested, the laptop's signature has been verified, but not its integrity, and finally due to some error the drone fails all of its verifications. Termination is guaranteed if the rewrite system is finite and strongly normalising. This is a necessary property that must be ensured in the semantics for any aspect. Investigation into systems that do not terminate is left for future work.

$$\begin{array}{c} \{\mathtt{m1} : \mathtt{Verified(r1)} \\ \mathtt{l1} : \mathtt{Verified(r2)} \\ \mathtt{d1} : \mathtt{Verified(r3)} \\ \Xi, \ldots\} \end{array} \longrightarrow \begin{array}{c} \{\mathtt{m1} : \top \\ \mathtt{l1} : \mathsf{D}_{\mathsf{AUTH}} \\ \mathtt{d1} : \bot \\ \Xi, \ldots\} \end{array}$$

Figure 7: Resultant Fixpoint of B

In all of the above it is necessary to understand the inputs and outputs to B can contain other structures from **TRUST**, such as claims, Ξ, the specific attestation/verification functions: A, Υ and Δ.
A discussion of the temporal and ordering aspects of observation (and other OODA concepts) is made later in this paper. It is noted that reaching a conclusion, the fixed-point of B, and any other similar mechanism, incurs measurable computational cost.

Orientation

The observation stage produces a fixed point of the trust aspect through repeated application of the endofunctor B which contains a model enriched with all the attestation information gained, including the claims, results, decisions, context etc (Johnson 2023). Orientation is concerned with revealing additional structure and relationships implicit in the trust-enriched observations B(X).
It identifies the laptop and drone as the sources of deviation from the desired configuration. More generally, orientation may derive aggregate properties. Orientation may also calculate metrics over these partitions

such as the number or proportion of trusted devices, the presence of critical failures, or the connectivity of trusted elements within the observed system. The purpose of orientation is therefore not to establish trust but to transform trust decisions into information suitable for decision making. The output of the orientation stage identifies where the system differs from an acceptable or expected state and forms the basis for constructing alternative future system states during the decision stage.

Similarly to B, $\mathbf{OBST}(\mathsf{Ori}) = \mathsf{O}$, where again O is an endofunctor in [**TRUST**, **TRUST**]. Unlike B, O accumulates semantic structures over those classifications. These structures become part of the oriented model and may subsequently be used later in the OODA process when constructing candidate future states (Fusano et al. 2012). For example, structures such as the sets below might be utilised. Note, some systems having 'secure boot' (eg: UEFI) can be considered trusted despite not providing evidence from 'measured boot' (full integrity check with TPM).

$$\begin{aligned}\mathsf{Trusted}(X) &= \{e \mid A(e) = \top \vee A(e) = \mathsf{D}_{\mathsf{AUTH}}\} \\ &\quad \setminus \mathsf{Suspicious}(X) \\ \mathsf{Suspicious}(X) &= \{e \mid \Xi(e) \subseteq \{...\}\} \\ \mathsf{Untrusted}(X) &= \{e \mid A(e) \neq \top \wedge A(e)! = \mathsf{D}_{\mathsf{AUTH}}\} \\ &\quad \setminus \mathsf{Suspicious}(X)\end{aligned}$$

The sets above are not additions to the TRUST category itself but derived structures constructed by O over the fixed-point output of B. They therefore preserve the underlying trust model while providing additional information suitable for decision making.

Figure 8 illustrates the action of O. The fixed-point observation produced by B is retained, while additional structures such as those above are constructed over it. Thus O enriches rather than replaces the output of B.

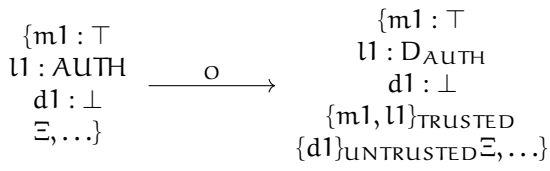

Figure 8: Input to and Output of an Orientation Through O

The models allow for multiple orientations for a given observation. In the examples here we do not show this as in a simulation we wish to follow a single path at this stage. Also, to demonstrate multiple orientations requires richer set of aspects which is not the point of the current work.

Decision

Decision is concerned with constructing candidate future states that resolve, mitigate or otherwise respond to those discrepancies. The purpose of decision is therefore not to interpret the system but to determine what acceptable states of the system might be achieved.

The morphism $\mathsf{dec} : \mathsf{Ori} \to \mathsf{Dec}$ admits many possible decision states, always including the null-decision where nothing changes.

Similarly $\mathbf{OBST}(\mathsf{Dec}) = \mathsf{D}$, where again D is an endofunctor in [**TRUST**, **TRUST**]. The input to D is the oriented model produced by O, including both the trust-enriched observation and any derived structures (Taillandier and Therond 2011).

The model identifies the laptop and drone as sources of deviation from the desired state. Several candidate future states may therefore be constructed. Figure 9 illustrates three such possibilities:

1. Attempt to restore trust in the drone whilst accepting state of the devices: $\mathsf{dec}_{\mathsf{T}}$
2. Remove the drone from the system and continue operation with the remaining trusted and partially trusted elements: $\mathsf{dec}_{\mathsf{X}}$
3. Null Decision: $\mathsf{dec}_{\mathsf{NULL}}$

Thus $\mathsf{dec}(\mathsf{ori}_1) \to \{\mathsf{dec}_{\mathsf{T}}, \mathsf{dec}_{\mathsf{X}}, \mathsf{dec}_{\mathsf{NULL}}\}$ and the Dec objects are the ones carrying the goals; D enriches these goals carrying decision objects into structures in **TRUST** in this case. Figure 9 shows a possible configuration of our model for D.

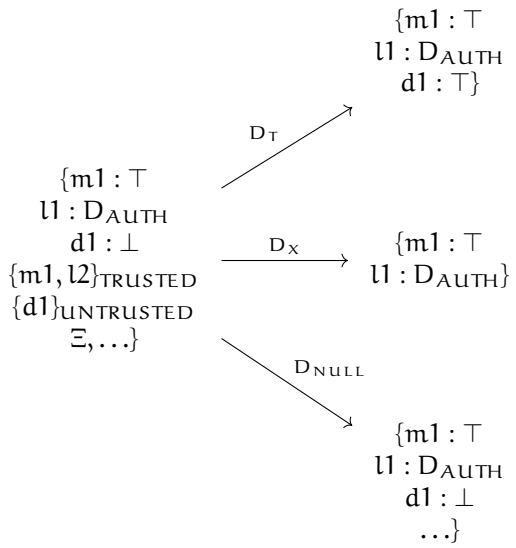

Figure 9: Possible Decisions Generated by dec and Enriched by D

The goal states themselves are not additions to **TRUST**, but derived structures constructed over the output of O. They therefore preserve the underlying

Table 1: Action Sequences

Decision	Action Sequence
$\mathtt{dec}_{\mathsf{T}}$	updateDevice(d1);rebootDevice(d1); attest(d1);verify(d1); decide(d1)
$\mathtt{dec}_{\mathsf{X}}$	dropDevice(d1)
$\mathtt{dec}_{\mathsf{NULL}}$	doNothing()

trust model whilst introducing policy and mission objectives.

Action

Again, following the established pattern the morphism $\mathtt{act} : \mathsf{Dec} \rightarrow \mathsf{Act}$ picks decisions and maps these to, again, potentially one or more actions. $\mathbf{OBST}(\mathsf{Act}) = \mathsf{A}$, where again A is an endofunctor in $[\mathbf{TRUST}, \mathbf{TRUST}]$.
The actions themselves in Act are operations or functions that operate upon the system being observed by the OODA-loop. In many respects this has a similar semantics to B in that the A carries an internal operational semantics and further operations as rewrite rules, again almost mirroring B: updateDevice, rebootDevice, dropDevice, attest, verify, decide etc.
Given the decisions we may then employ the following actions assuming that we operate within the confines of circumscription, ie: nothing else happens and we only perform actions within a specific context (McCarthy 1980). Table 1 shows some possible action sequences.
Of course, we may formulate the actions such that other elements do change resulting in some form of branching where other elements may change due to other actions also being included.

Next

In most descriptions of OODA, there is invariably an arrow leading from the action back to observation; this is where the notion of an OODA-loop comes from. However this idea of next is misleading in that while conceptually we keep repeating OODA *loops*, each loop is effectively independent of any previous over time.
This leads us to one aspect that is not covered in simulations of OODA and that is how reality and the output of the OODA loop relate. The system acted upon by the actions may not admit those actions or have other responses, meaning the expected (ideal) results of an action never occur. This of course is then discovered during the next OODA loop. Figure 10 shows this graphically:
An observation is made of some reality Σ, the output of which is a new reality Σ'_n, however at this point in time reality is now Σ'_R resulting in a gap between our expected reality and reality itself.

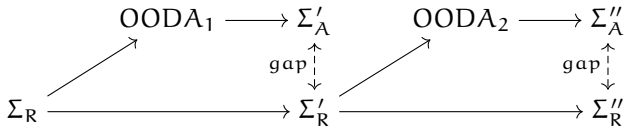

Figure 10: OODA Loops Through Time

One aspect of simulation is that we can ignore this and use that to fine tune our responses, decisions etc and how we operate within a given OODA loop without worrying about this: $0 \leq \mathtt{gap}(\Sigma_R, \Sigma_n) < \epsilon$. We are also allowed to explore situations where reality diverges, possibly rapidly: $\mathtt{gap}(\Sigma_R, \Sigma_n) \geq \epsilon$. This remains the focus of future work but its incorporation into the simulation is critical for proper understanding of what happens next.

TEMPORAL ASPECTS

In many idealised situations it is assumed that OODA operates efficiently and sufficiently fast that the gap between any reality and the final actions are negligible. This leads to two related temporal aspects: the time it takes for an OODA loop to operate, and the amount of gap between the observation and reality (Miller et al. 2021).
Using Computational Tree Logic (CTL), an OODA loop's execution must progress through observation, orientation, decision to action. Actions eventually give rise to systems that can be observed by other OODA loops. The idea of a loop emerges from repeated executions over reality rather than any intrinsic cyclic structure of OODA itself. Within the OODA loop itself, execution progresses as follows:

$$\mathbf{AG}(\mathsf{Obs} \rightarrow \mathbf{AF}(\mathsf{Ori}))$$
$$\mathbf{AG}(\mathsf{Ori} \rightarrow \mathbf{AF}(\mathsf{Dec}))$$
$$\mathbf{AG}(\mathsf{Dec} \rightarrow \mathbf{AF}(\mathsf{Act}))$$

Our argument is that no intra-loop relation connects Act back with Obs. Instead, successive instantiations of OODA are related only at the level of the simulation trace as below. This is a property of loop instantiations over time and not a morphism of the loop. The *idea* of a loop is emergent from repeated execution against reality.

$$\mathbf{AG}(\mathsf{Act}_i \rightarrow \mathbf{AF}(\mathsf{Obs}_{i+1}))$$

OODA admits early termination of a loop and we model this by admitting null decisions and actions as described earlier. We assume that observation and orientation

complete, but only then can nothing else happen, that is, the loop 'terminates'.
The next issue is the amount of time a loop takes which can be expressed as the time it takes for the operational mechanism **OBST**(Obs) plus some operational overhead τ. This follows similarly for the other objects resulting in T_{OODA}

$$\begin{aligned}\mathsf{T}_{Obs}(X) &= \mathsf{T}_{B}(X) + \tau_{Obs}\\ \mathsf{T}_{Ori}(X) &= \mathsf{T}_{O}(X) + \tau_{Ori}\\ \mathsf{T}_{Dec}(X) &= \mathsf{T}_{D}(X) + \tau_{Dec}\\ \mathsf{T}_{Act}(X) &= \mathsf{T}_{A}(X) + \tau_{Act}\\ \mathsf{T}_{OODA}(X) &= \mathsf{T}_{Obs} + \mathsf{T}_{Ori} + \mathsf{T}_{Dec} + \mathsf{T}_{Act}\end{aligned}$$

Each aspect can be similarly calculated (and similarly for O,D and A) assuming sequential execution for each element in X:

$$\mathsf{T}_{B}(X) = \sum_{e \in X} (\mathsf{t}_{attest}(e) + \mathsf{t}_{verify}(e) + \mathsf{t}_{decide}(e)) \quad (10)$$

The time complexity of an OODA cycle becomes a function of both the number of observed entities and the computational complexity of aspect processing.
The amount of time it takes to perform the attestation pipeline can be derived from the exponentials in **TRUST**. Certain attestation operations and extensive verification may then affect the choice of decision procedures and even the sufficiency of the orientation.
These temporal observations have a direct effect upon the OODA loop and reality itself. A stable system is one where $\mathtt{gap}(\Sigma_R, \Sigma_n) < \epsilon$ where epsilon is directly related to the amount of time available for the processing of OODA. This remains for further study.

CONCLUSION

This work has developed from remote attestation with multiple devices in highly configurable and volatile field situations based on critical systems (Oliver and Müller 2026). Utilising these environments we can perform many of the attacks and other scenarios against this system and perform OODA type analyses. As remote attestation and computational trust are an integral part of this system these become the primary aspect to be investigated - as described in this paper.
We have demonstrated the formalisation of OODA categorically and shown that aspects such as computational trust can be expressed as functors into endofunctor categories of these aspects. This gives a natural structure for rewrite systems and therefore an executable semantics. Further to this we expose many properties of OODA such as the termination of observations of aspects, temporal properties and the divergence of the observed model from reality.
A prototype based on existing tooling such as the Jane Remote Attestation Server and other system security and performance tools and formalisation using the Lean 4 thorem prover are a work in progress. Jane has specifically been constructed around the structures in **TRUST** and is amenable to scaffolding that allows us to implement the OODA-loop in this context.
Future work is now focussed on the generalisation of the framework to multiple and interacting aspects. Further categorical machinery such as natural transformations also provide interesting properties that need to be investigated as discussed below.
The natural transformations $\mathsf{OBST}(\mathtt{ori}) = \mathtt{bo}$, $\mathsf{OBST}(\mathtt{dec}) = \mathtt{od}$, $\mathsf{OBST}(\mathtt{act}) = \mathtt{da}$ preserve aspect consistency between stages and guarantee that structures generated by one endofunctor are admissible inputs to the next. For example the functor bo effectively states *'the observation phase has completed and produced a well-formed, enriched observation of trust, then we can proceed'* and that $\mathsf{O}(\mathsf{B}(X))$ can be evaluated.
Also the relationship between bo and the temporal calculation T_B has interesting properties and could be interpreted as some form of timeout or other temporal constraint on the system. This has obvious implications when working with OODA in highly real-time oriented systems.
Extension to multiple aspects is relatively trivial in the categorical model with a generalisation of **OBST** and additional categories for those other aspects: $\mathbf{OODA} \xrightarrow{\mathsf{OBST}_X} [X, X], X \in \{\mathbf{TRUST}, \mathsf{PERF}, \mathsf{NET}, \ldots\}$. For example, network latency and availability is one area of investigation with the demonstrator (Blaha 2018). One challenge that is under active investigation is the relationship between aspects, for example, if a device is unavailable then attestation is not possible, suggesting that there are relationships between, say an aspect PERF and **TRUST**.
The property of divergence between the model and reality has been discussed; we would continue with not only a proper metricisation of this, but also how divergence can be modelled between concurrently operating OODA loops. This would give not only a measure of the validity of our OODA analyses, but also of the stability of the system being observed.
Overall, the exercise of placing OODA-loops (and in this specific case, computational trust) on a firm mathematical basis has allowed us to explore many of the facets of OODA and how it actually operates in reality with the safety of being able to simulate many unexpected situations.

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Parallel Computing

A DESIGN-TIME ARCHETYPE LENS FOR HLA OBJECT MODELS

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KEYWORDS
Distributed simulation, High Level Architecture, Federation Object Model, Object Model Template, RPR FOM, design-time analysis, model archetype, code generation, SimGe.

ABSTRACT

High Level Architecture (HLA) object models define the shared data contracts of distributed simulations. Although standard schema validation and recent structural and semantic metrics diagnose local complexity, neither characterizes an object model's overall architectural orientation. This paper introduces a design-time archetype lens that provides a model-level architectural interpretation. By evaluating the balance between persistent object state and transient event exchange, the lens classifies schemas as state-centric, event-driven, or hybrid. This classification is operationalized via the Semantic Mass Ratio and a derived domain-dominance score. The resulting profile identifies key federate-side implementation concerns prior to code generation and execution. A case study on the modular RPR FOM v3.0 suite demonstrates a predominantly state-centric baseline while revealing distinct hybrid and event-driven functional modules.

INTRODUCTION

The High Level Architecture (HLA) is a widely used standard for distributed simulation, defined by its framework rules (IEEE 2025a), federate interface specification (IEEE 2025b), and the Object Model Template (OMT) (IEEE 2025c, Dahmann et al. 1997). HLA object models define the shared data contract: the Federation Object Model (FOM) establishes the shared federation-wide contract, while the Simulation Object Model (SOM) specifies the subset relevant to a particular federate. Because modern HLA federations rely on modular FOM development (Möller et al. 2007, Wenguang et al. 2009), architectural analysis must be performed on the fully resolved model to capture inherited and merged elements accurately. Beyond documentation, an object model is an implementation contract that shapes discovery, attribute updates, interaction delivery, subscription filtering, and code generation, directly impacting federate software structure.

Most object model evaluation still focuses on syntactic validation, which is necessary but not sufficient. Recent work extends this analysis on two fronts. Structural and topological analysis measures the shape of the object and interaction class hierarchies, identifying conditions such as deep inheritance, broad concrete populations, and fragmented depth distributions (Topçu 2026). Semantic and kinetic analysis measures the resolved payload carried by object and interaction classes, together with update-related metadata and payload concentration (Topçu 2026a). Although these frameworks provide detailed domain-specific diagnostics, they do not directly reveal the overall architectural character of the object model. In particular, they do not provide a model-level interpretation of the balance between persistent object state and transient interaction exchange.

This paper addresses this missing interpretation layer by introducing a design-time archetype lens for HLA object models. The lens asks a simple architectural question: is the model mainly state-centric, event-driven, or a balanced hybrid? This distinction matters because it indicates where implementation and code-generation effort should be concentrated: persistent-state management and attribute reflection for state-centric models, interaction dispatch and event handling for event-driven models, or coordinated support for both in hybrid models.

The paper evaluates the proposed lens using the Real-time Platform Reference Federation Object Model (RPR FOM) v3.0, standardized under SISO-STD-001.1-2025 (SISO 2025). As a mature reference suite encompassing both extensive persistent entity representations and discrete operational control interfaces, RPR FOM exhibits rich architectural diversity across its functional modules, making it an ideal evaluation benchmark.

The contributions of this paper are: (i) a design-time archetype lens that adds a model-level cross-domain interpretation above structural and semantic diagnostics; (ii) a Semantic Mass Ratio and derived domain-dominance score for identifying and qualifying state-centric, event-driven, and hybrid HLA object models; and (iii) a module-level RPR FOM case study that demonstrates how different functional modules exhibit

different archetype profiles within the same reference suite. The remainder of this paper first reviews structural and semantic foundations. It then introduces the archetype lens along with the governing Semantic Mass Ratio and domain-dominance formulation. Subsequently, the modular RPR FOM case study is presented, followed by a discussion of federate engineering and code-generation implications. The paper concludes by addressing current limitations and outlining future research directions.

BACKGROUND AND MOTIVATION

Structural and Topological Analysis

Structural and topological analysis evaluates the geometry of the object model hierarchy and focuses on shape rather than payload. Prior work introduced metrics that quantify horizontal sprawl, vertical depth, depth fragmentation, and abstraction imbalance (Topçu 2026). Hierarchy directly affects implementation: a broad interaction hierarchy enlarges the dispatch surface, a deep object hierarchy increases traversal and aggregation cost, and a concrete-heavy structure inflates generated-code volume. Topology alone does not capture payload; structural analysis must be complemented by semantic analysis.

Semantic and Kinetic Analysis

Semantic and kinetic analysis evaluates the resolved data carried by object and interaction classes. Prior work defined a class-level Semantic Weight based on recursively resolved OMT data types (Topçu 2026a); for object classes, this weight may be adjusted by update-related metadata, since static, conditional, and periodic attributes do not impose the same design-time update exposure.

This framework yields indicators of semantic saturation (mean payload normalized by depth and population) and payload dispersion (local concentration around hotspots), which together identify whether a domain is light, balanced, saturated, or concentrated (Topçu 2026a). The framework treats object and interaction class domains separately, since they represent persistent state and transient events respectively; this distinction is preserved here.

ARCHETYPE LENS

This section defines the archetype lens independently of any particular object model. The subsequent section applies this method to the RPR FOM module set, followed by a discussion of practical engineering implications.

Table 1: Position of the archetype lens in the analytical pipeline

Layer	Main question	Example indicators	Output
Structural topology	What shape does the hierarchy have?	Shape, depth, breadth metrics (Topçu 2026)	Broad, deep, fractured, abstract, concrete
Semantic payload	How much data does the model carry?	Class-level weight and dispersion (Topçu 2026a)	Light, balanced, saturated, concentrated
Archetype lens	Which domain dominates the model?	M_{OC}, M_{IC}, R, A (this paper)	State-centric, event-driven, hybrid

Motivation and Analytical Position

Metric-level diagnostics are useful but hard to interpret as a whole. A model may exhibit acceptable hierarchy depth, moderate payload saturation, and high interaction dispersion, patterns that together require a higher-level interpretation layer. In this paper, an archetype lens is a design-time interpretive abstraction that maps the relative semantic mass of the Object Class and Interaction Class domains to a model-level architectural category. The Semantic Mass Ratio provides the quantitative mechanism for this interpretation.

The lens complements, rather than replaces, prior structural and semantic metrics. Topological metrics characterize hierarchy shape (Topçu 2026), while semantic metrics evaluate payload saturation and dispersion (Topçu 2026a). The archetype lens then explains the balance between the two HLA domains. This separation defines a three-level analytical pipeline (Table 1) and prevents over-interpretation: a state-centric model is not automatically inefficient, and a broad interaction hierarchy is not automatically problematic. The purpose of the lens is to expose the dominant design orientation so that implementation strategies can match it.

Semantic Mass

Let $\mathscr{P}_{OC}$ denote the set of concrete object classes with active sharing semantics in the resolved model and let $\mathscr{P}_{IC}$ denote the corresponding set of concrete interaction classes. For each class i, W_i denotes its class-level Semantic Weight, obtained by aggregating the recursively resolved structural weights of all inherited and modularly merged attributes or parameters. Primitive data types contribute weights based on their declared

representation size, records sum the weights of their fields, and arrays combine element weight with effective cardinality. For object classes, attribute contributions are adjusted according to their OMT update category, such as static or periodic; for interaction classes, parameter contributions are aggregated without an update-volatility adjustment. The resulting value is a design-time approximation of resolved payload rather than an exact encoded wire size. The complete weighting rules and calibration rationale are presented in (Topçu 2026a).

The Total Object Mass and Total Interaction Mass are defined as

$$M_{OC} = \sum_{i \in \mathscr{P}_{OC}} W_i, \qquad M_{IC} = \sum_{i \in \mathscr{P}_{IC}} W_i. \tag{1}$$

M_{OC} represents the cumulative semantic payload of persistent state. M_{IC} represents the cumulative semantic payload of transient interactions. These quantities are not runtime traffic measurements; they describe the payload surface implied by the static object model.

Semantic Mass Ratio

The Semantic Mass Ratio R is defined as

$$R = \frac{M_{OC}}{M_{OC} + M_{IC}}, \qquad M_{OC} + M_{IC} > 0. \tag{2}$$

The ratio lies in the interval $[0, 1]$ whenever the denominator is positive. Values near 1 indicate object-domain dominance; values near 0 indicate interaction-domain dominance; intermediate values indicate a balanced model. When $M_{OC} + M_{IC} = 0$, the model carries no semantic payload and R is undefined; such modules are reported as non-semantic support modules (see Table 3).

The interpretation bands in Table 2 are heuristic defaults rather than empirically fitted or theoretically universal thresholds. They implement an approximately two-to-one dominance rule (i.e., one domain carries approximately twice the semantic mass of the other): $R > 0.65$ indicates object-domain dominance, $R < 0.35$ indicates interaction-domain dominance, and the intermediate interval represents a hybrid model. These values provide an interpretable baseline that can be calibrated across broader object-model corpora. Implementation guidance aligned with each archetype is detailed in Table 4.

Domain Dominance

The archetype label can be complemented by a domain-dominance score A, which indicates how far the model lies from the balanced center:

$$A = 2 \, | \, R - 0.5 \, | \, , \qquad A \in [0, 1]. \tag{3}$$

Table 2: Object model archetype classification

Archetype	Criterion	Interpretation	Primary Implementation Focus
State-Centric	$R > 0.65$	Persistent object state dominates the semantic mass	Efficient management of persistent object state and federate-specific state exposure
Event-Driven	$R < 0.35$	Transient interaction payload dominates the semantic mass	Efficient routing, decoding, and handling of transient interactions
Hybrid	$0.35 \leq R \leq 0.65$	Persistent state and transient events are both substantial	Balanced support for both persistent state and transient event handling

$A = 0$ indicates no dominance by either domain and therefore represents a maximally balanced hybrid profile. In contrast, $A = 1$ indicates complete dominance by one domain. The score measures the magnitude of domain dominance, not the strength or confidence of the assigned archetype; the direction of dominance is determined from R.

For reporting purposes, domain dominance is qualified as *weak* ($A < 0.30$), *moderate* ($0.30 \leq A < 0.70$), or *strong* ($A \geq 0.70$). Hybrid models therefore inherently exhibit weak domain dominance by construction ($A \leq 0.30$), reflecting that neither domain substantially outweighs the other.

FOM-level and SOM-level Archetypes

The archetype lens operates at two scopes. At the FOM level, it captures the global federation contract. At the SOM level, it characterizes federate-specific projections. Because individual federates (e.g., physical platforms vs. simulation managers) publish and subscribe to distinct subsets of the FOM, their local SOM archetypes often diverge from the global contract. Pre-implementation code generation must therefore align with the local SOM archetype rather than assuming a uniform federation-wide profile.

RPR FOM CASE STUDY

We evaluated the archetype lens across the fifteen modular definitions of S ISO-STD-001.1-2025 (RPR FOM v3.0) (SISO 2025). Metrics were computed using SimGe (Topçu 2026b) under standard HLA dependency resolution to assess the fully merged schemas. Table 3 reports the resulting module-level classifications.

Table 3: SISO-STD-001.1-2025 module archetype classification

Module	M_{OC}	M_{IC}	R	A	Dom.	Archetype
Aggregate	5 180	0	1.000	1.000	strong	State-Centric
Base	480	0	1.000	1.000	strong	State-Centric
Communication	1 578	174	0.901	0.801	strong	State-Centric
DER	12 583	731	0.945	0.890	strong	State-Centric
IO	480	839	0.364	0.272	weak	Hybrid
Logistics	480	664	0.420	0.161	weak	Hybrid
Minefield	20 624	7 548	0.732	0.464	moderate	State-Centric
Physical	49 442	246	0.995	0.990	strong	State-Centric
SE	15 980	5 252	0.753	0.505	moderate	State-Centric
SIMAN	480	21 825	0.022	0.957	strong	Event-Driven
UA	1 146	40	0.966	0.933	strong	State-Centric
Warfare	15 020	2 332	0.866	0.731	strong	State-Centric

*Non-semantic support modules ($M_{OC} = M_{IC} = 0$, R, A undefined): Enumerations, Foundation, Switches.

The results reveal all three archetypes within the reference suite, reflecting functional design intent. A large group (Physical, Warfare, DER, Aggregate, Communication, UA, and Base) exhibits strong state-centric dominance ($R > 0.85$, $A \geq 0.70$). Within this group, Physical carries the largest individual semantic mass ($M_{OC} = 49{,}442$), accounting for over 70% of the suite's persistent state due to extensive platform enumerations, coordinate kinematics, and articulated parts. For these modules, software generation must prioritize efficient instance storage, dirty-state tracking, and reflection scaffolding. In contrast, Minefield and SE exhibit moderate dominance ($R \approx 0.73$–0.75); while persistent state dominates, their interaction domains carry substantial payload for sensor queries, breach attempts, and detonation effects, requiring concurrent dispatch support.

The hybrid archetype emerges in IO and Logistics ($A < 0.30$), indicating a balanced dual-domain role. Functionally, Logistics pairs continuous entity status with discrete replenishment requests (e.g., `ServiceRequest`, weight 173), while IO couples baseline platform state with operational action messages (`IOaction`, weight 426). Notably, $M_{OC} = 480$ is identical across Base, IO, Logistics, and SIMAN because the latter three declare no new concrete object classes; under modular resolution, each inherits only the foundational baseline state (`BaseEntity`) from Base.

The event-driven archetype is decisively represented by SIMAN ($R = 0.022$, $A = 0.957$). Spanning 45 interaction classes and 115 parameters, it exhibits extreme event-payload concentration around operations such as `TransferOwnership` (weight 1,765). SIMAN demonstrates that the lens readily isolates specialized control modules within a predominantly state-oriented federation suite.

ENGINEERING AND TOOLCHAIN IMPLICATIONS

The R ratio guides software design pattern selection before code generation. Although HLA 4 introduces new runtime delivery mechanisms (Möller et al. 2022; 2026), the archetype lens operates at the schema level and does not model these directly. The archetypes classify the information-exchange orientation of object models and federate-side implementations; they do not classify or prescribe RTI products. At design time, the lens identifies whether the generated federate code should emphasize persistent-state handling, interaction dispatch, or both. Tool-supported HLA development frameworks have already shown the value of reducing implementation burden through reusable infrastructure and automated support (Falcone et al. 2017).

Table 4: Archetype-driven implementation guidance

Archetype	Generator Focus	Federate Support & Actions
State-Centric	State accessors, reflection scaffolding, selective subscription	State caches, update filtering; verify if SOM projections can reduce state exposure
Event-Driven	Dispatch tables, parameter decoders, handler registration	Constant-time routing, callback queues; inspect payload hotspots and tree breadth
Hybrid	Dual-pipeline: synchronized state accessors and typed dispatch	Integrated caches with prioritized event queues; isolate state and event threading

Table 4 summarizes the dominant implementation concerns implied by each archetype. The guidance is design-time: it indicates which generator strategies and federate-side support structures should be prioritized, not which RTI parameters to tune.

The archetype lens fits into the broader dashboard layering introduced in Table 1, extended at the front by a volume layer (class, attribute, and parameter counts) and at the back by the implementation guidance summarized in Table 4. SimGe computes the required metrics after model resolution and presents the result as a dashboard-level diagnostic (Topçu 2026b), taking the simulation engineer from raw model data to actionable implementation guidance without requiring manual metric computation.

LIMITATIONS

The archetype lens is a static design-time diagnostic. It captures the structural and semantic potential of a resolved schema rather than measuring observed network traffic, latency, or memory consumption. Key limitations include: (i) **Static basis:** Runtime traffic depends heavily on dynamic scenario rates, DDM regions, and filtering; (ii) **Heuristic thresholds:** The R boundaries (0.35, 0.65) and dominance bands provide an interpretable baseline that requires ongoing calibration against broader corpora; (iii) **Weight calibration:** Metric values depend on the underlying semantic weight model (Topçu 2026a), though relative module rankings

remain stable; and (iv) **Diagnostic intent:** Archetypes guide software scaffolding rather than mandating model refactoring.

CONCLUSION AND FUTURE WORK

This work introduces a design-time abstraction layer that supports pre-implementation reasoning about the architectural intent expressed in HLA object models. Implemented as an archetype lens, this layer complements structural and topological (Topçu 2026) and semantic and kinetic (Topçu 2026a) analyses by adding a cross-domain architectural view. It distinguishes whether an object model is oriented primarily toward persistent object state, transient interaction exchange, or a balance between the two. The Semantic Mass Ratio provides the quantitative basis for this comparison, while the domain-dominance score expresses how far the resulting profile lies from the balanced center.

The RPR FOM case study showed that the reference suite is predominantly state-centric, but this overall result does not apply uniformly to every module. Several modules contain a meaningful interaction component, and individual modules exhibit meaningful architectural diversity across all three archetypes. The lens therefore captures both the dominant architectural character of a resolved object model and the functional variation within a modular FOM suite.

The resulting archetype characterizes static schema orientation to inform early design choices. Its engineering utility lies in directing pre-implementation focus toward persistent-state management, interaction dispatch, or balanced dual-pipeline scaffolding before federate deployment.

Future work will proceed along three directions: (i) extending module-level analysis to role-specific SOM projections; (ii) calibrating the R and A thresholds against a wider set of reference FOMs and operational federation models; and (iii) linking archetype profiles more directly to generated-code strategies in SimGe, including state caching, selective reflection, event routing, and hybrid generation profiles.

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DECLARATION OF GENERATIVE AI USE

During manuscript preparation, OpenAI's ChatGPT was utilized for language refinement and readability improvements under authorial oversight. After using this tool, the author carefully reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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BEYOND SINGLE-STREAM TESTING: A LARGE-SCALE STATISTICAL EVALUATION OF MODERN PSEUDO-RANDOM NUMBER GENERATORS

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ABSTRACT

Pseudo-random number generators (PRNGs) are critical components of modern high-performance computing and artificial intelligence applications, where both statistical robustness and reproducibility are essential. This study provides a large-scale comparative evaluation of representative PRNGs from several major families, including Xoshiro, Philox, PCG, and MRG32k3a. Unlike conventional assessments that typically examine a single output stream per generator, we perform a systematic campaign involving more than 10,000 independent streams, each evaluated with the BigCrush battery of the TestU01 library. The resulting dataset, corresponding to over 4.5 cumulative years of computation, offers one of the most extensive empirical investigations of modern PRNGs to date. Our analysis confronts the observed behavior of the generators with the claims made by their designers and reveals substantial variability across streams: the highest observed success rate is 72%, and nearly all generators fail at least one BigCrush test under some conditions. All detected failures are documented in detail. To ensure a fair comparison, all generators are tested under a unified experimental protocol designed to closely reflect practical deployment scenarios. Finally, the complete set of results, scripts, and experimental artifacts is released through a public Git repository, enabling full reproducibility and reuse by the community.

INTRODUCTION

Pseudorandom number generators (PRNGs) are widely used in various fields of modern computer science, including the numerical simulation of physical models, Monte Carlo simulations, artificial intelligence (AI), and machine learning. To obtain high-quality results, the generators on which our simulations rely must be of excellent quality. The quality of a generator depends on various aspects: whether the numbers are generated uniformly, whether the generator is portable across different platforms, whether the sequence can be changed (with a different initialization state) and is repeatable (Autunes and Hill, 2024), whether the period is as large as possible, whether the generator uses little memory and delivers numbers quickly, and whether it produces uncorrelated sequences which pass common statistical tests. The ability to be easily parallelized is also important for intensive computing (Hill et al., 2013).

In this paper, propose an update of a preprint (Wartel and Hill 2026) focusing on the statistical quality of the generators, i.e., their ability to generate a sequence of numbers with the lowest possible correlation between them. This has to be achieved with a perfectly deterministic approach to ensure debugging as well as the repeatability of numerical experiments. The fact that we only evaluate the statistical quality of PRNGs means that our results do not reflect their overall performance. A generator that may seem inadequate according to this criterion could nonetheless be relevant for other uses.

In our study, we investigate and compare the statistical performance of the main modern generators found in machine learning applications and frameworks. The use of generators, their initialization, and their parallelization are frequent sources of non-reproducibility in machine learning and artificial intelligence studies (Hill et al., 2024).

For a given sequence of numbers, there are several methods for checking whether these numbers correctly simulate randomness. Historically, Donald Knuth introduced a first series of statistical tests, published in the second volume of "The Art of Computer Programming". Despite their age, these tests remain relevant both for academic pursuit and as a reference constantly republished (Knuth, 2014). In 1995, G. Marsaglia published the first series of tests called *Die Hard*—the source code of which is no longer available. These tests were extended in 2005 under the name *Die Harder*. The NIST Statistical Test Suite (STS) is also a famous test suite, particularly for assessing the ability of random numbers to be cryptographically secure.

A more advanced library of test suites was proposed by Pierre L'Ecuyer and Richard Simard. They released the first version of the TestU01 library in 2007 and they plan to develop a version for 64-bit generators. This library offers a wide range of tests, as well as test batteries. Among the proposed batteries, the BigCrush battery is still currently the most stringent battery for generators. It is, to the best of our knowledge, the most capable test battery for evaluating the quality of PRNGs. This library also allows us to modify tests already implemented (with our own parameters) and to add new tests. We do not use such possibilities in our study; we simply use the BigCrush battery in TestU01 version 1.2.3, which contains over 100 tests (L'Ecuyer et al., 2007).

In 1997, Makoto Matsumoto presented the Mersenne Twister (MT) generator (Matsumoto and Nishimura, 1998). This generator shook up the world of PRNGs and possesses some excellent qualities. Its outstanding features include a period

of 2^{19937}-1, and it is claimed to be equidistributed in up to 623 dimensions. This generator can also be easily parallelized and delivers good quality streams (Autunes and Hill, 2023). MT is known to be weak for cryptographic applications and has a few other limitations; this, however, does not prevent its use in many programs or simulations. MT is still proposed as the default generator for major languages (C++, Python…) and libraries.

However, new generators have emerged to further improve our ability to model randomness. MRG32k3a was proposed in 1999 (L'Ecuyer, 1999), Xorshift in 2003 (Marsaglia, 2003), Philox in 2011 (Salmon et al., 2011), PCG in 2014 (O'Neill, 2014), and the Xoshiro generators, which are improvements on the Xorshift generators, were released recently (Blackman and Vigna, 2022). The latest generator to be tested is Xoshiro1024**, released in 2019. This latter family of generators is presented as a viable alternative to MT, sometimes faster and more statistically robust (Blackman, 2021). The Philox generator is used as the default generator in TensorFlow and PyTorch.

The aim of this study is to evaluate the statistical quality of the above-mentioned generators. The study begins with a presentation of the methods used to test these generators. The method section will also present the resources used and the compilation methods. Next, each generator or family of generators will have a section dedicated to the presentation of its principles and the results of the batteries of tests on the various streams. The generators are then compared with each other, once again on their statistical capabilities. A discussion will be provided to understand the differences between the various results obtained. The study concludes with a summary of the main results and important points. All the source code used in this study is available on a public Git repository at the following url:
https://gitlab.isima.fr/thwartel/testu01_variousgenerators

MATERIAL & METHOD

The evaluation of the statistical quality of the selected generators involves carrying out a series of tests and then evaluating the results obtained. The generators considered are: Philox, PCG, MRG32k3a, and some generators from the Xoshiro family (Xoshiro256++, Xoshiro256** and Xoshiro1024**). The Mersenne Twister has been intensively studied in the past, and still quite recently (Antunes et al., 2023).

The tests we retain for our study are those of the TestU01 library (L'Ecuyer et al., 2007), and more specifically all the tests of the BigCrush battery. Instead of running this battery on a single initialization, we want to test and compare our generators intensively with many streams, corresponding to their use in machine learning or high-performance computing. We thus use several initial states to test the generators on different streams, and we attempt to select the different initialization states as an end user would do. Each of these streams provides a result file from the BigCrush battery. The results are analyzed for potential weaknesses, and the success rates are compared with other generators and with the claims made by the PRNG creators.

In order to conduct fair research, all generators are tested under similar conditions. For each generator, we use its official implementation, which can be found on the web page of the PRNG creator. To use the generators correctly, we follow the recommendations given by their designers. These conditions are often linked to the initialization of the generator. Thus, the approach followed for each generator initialization is as follows: if an initialization method is recommended, we follow it. If there is no constraint, or if randomness is required to fill the generator, we use the Mersenne Twister generator to provide this randomness. We chose to use MT because this generator is widespread and available in most modern languages. This generator is still likely to be used by scientists for many random use cases.

To test different initial states, we use integers as seeds in the generator or in MT. The old term seed can be misleading. It is important to realize that what is called a seed is just an index on the internal initialization state of a generator. Such a state is often much larger than a single integer (6 double numbers for MRG32k3a, more than 2 KB for MT, etc.). Like many end users accustomed to the counter generators proposed at the 2011 Supercomputing conference ("Parallel Random Numbers: As Easy as 1, 2, 3" was the best paper of this conference), we use sequential indexes (seeds), the first being 0, the next seed being 1, and so on until 1000. This way, at least 1,000 streams are tested from at least 1,000 initial states.

For the generators of the Xoshiro family, like Xoshiro256++, Sebastiano Vigna recommends filling in the initial state 's' (of 32 bytes: 256 bits) using the splitmix64 generator (Blackman and Vigna, 2022). There are no other constraints. So, we use the so-called seed, which we can vary from 0 to 1000, to obtain a thousand different initial states. This seed is placed in splitmix64, then s is filled.

To keep track of the initializations, we write the seed (index) used and the corresponding initial state (the real seed) in the output file of each battery. These files are named BigCrush_n.txt where n represents the seed used. They contain the result of the BigCrush battery on this seed. The file names may also include *High* or *Low* to indicate the tested bits (MSB or LSB). For each initial state, we run the BigCrush battery, necessitating between 3 to 4 hours of computing depending on the processor generation.

It is important to test all the bits generated, but BigCrush only tests 32 bits. To overcome this problem for all 64-bit generators, a common practice is to test each stream twice: first on the low-order 32 bits, and then the second time on the high-order 32 bits. There may be several ways of using all the bits, but one method proposed by Blackman is to return the low and high parts of each number alternately. Thus, for testing a hundred 32-bit numbers requested, we only require 50 drawings of 64 bits. We preferred to use a method that is probably closer to the real use of the generator: if the user only needs 32 bits, they will take the low or high part and

won't keep the remaining part for the next use. This is a more time-consuming method, but less cumbersome in terms of storage and implementation.

The batteries are run independently, so we rely on the repeatability of the generators' initialization method to guarantee that each seed gives us exactly the same initial state, and that each initial state provides exactly the same sequence of numbers. In theory, all generators should provide the same sequence of numbers from the same seed. As revealed by Antunes and Hill (Antunes and Hill, 2024), this is not always the case, and in fact depends mainly on the implementation of the initialization method and also on the platform and/or framework used. Here, we stick to the C language, with identical implementations, and we did not experience a lack of repeatability.

Some of the generators we are testing only generate 32 bits per drawing. Therefore, in order to have the same sample sizes, we are testing 2002 initial states for 32-bit generators. Each test we run produces an output file. We therefore obtain 2002 output files for 32-bit generators. The results of these tests are retrieved automatically thanks to Bash scripts. We obtain the following information: the success rate of the streams, the number of initial states that pass or fail the battery on all bits, and the distribution of failures across the various tests. In this way, we are able to distinguish the initial states that result in streams with very good statistical qualities.

HPC Hardware Used

We have used a part of our laboratory computing cluster running Ubuntu 20.04.5 LTS, slurm-wlm 19.05.5, and GCC 9.4.0. We have been using 3 types of nodes:

- Two AMD EPYC 7452 processors with 32 physical cores and 2 GB of RAM per core;
- Two Intel® Xeon® CPU E5-2670 v2 @ 2.50 GHz, 10 physical cores and 2 GB of RAM per core;
- Two Intel® Xeon® CPU E5-2670 0 @ 2.60 GHz, 8 physical cores and 2 GB of RAM per core.

The compilations were done with GCC on a 64-bit Intel Xeon Silver 4310 CPU @ 2.10GHz. Hyper-Threading is not activated on the cluster we used. To match real-world usage, programs are compiled using the -O2 optimization option. All the nodes feature a 64-bit x86 architecture.

RESULTS

The Xoshiro Family

In the Xoshiro generator family, we test the Xoshiro256++, Xoshiro256** and Xoshiro1024** generators. The name comes from the sequence of operations “Xor, shift, rotation”. The number following the name indicates the size in bits of the generator state. This size has an impact on the generation cycle (Blackman and Vigna, 2022) but it is also suggested that it has an impact on generation quality (Vigna, 2016).

These generators come with the symbol of an operator `+` or `*` that represents a “scrambler”. This makes modifications to the returned value. The aim of these modifications is to improve statistical quality. However, `+` and `*` still show weaknesses in the lower bits and can be considered for floating-point use, since the lowest bits of floating-point numbers have a smaller impact on the final double precision value. The authors advise using the `**` scrambler for general purposes since it allows a shuffling of all bits of the Linear Feedback Shift Register. Among other things, scramblers prevent the systematic failure of certain BigCrush tests. Each scrambler is different and brings different features. They are fully explained in (Blackman and Vigma 2021) paper on the subject.

The Xoshiro generators are in fact an improvement on the Xorshift and Xoroshiro families, which presented some statistical problems and recurring issues if the bits are processed in reverse order (Lemire et al., 2019). In this paper, we do not analyze bits in reverse order. This family of generators generates numbers with 64-bit resolution. As explained in the methods section, each stream has been tested twice (once for the 32 least significant bits and once for the 32 most significant bits). Therefore, we have dealt twice with the same 1001 initialization states & streams.

Figure 1 shows some statistical results obtained with these generators. It shows the number of times each BigCrush battery test failed (at least one test). A distinction is made between generators, but not between high and low bits. We can see that the generators have the same weaknesses and the same strengths. They face difficulties on the latest `CollisionOver` tests, on `ClosePairs`, and `RandomWalks`. They have relatively similar success rates, between 71.1% for Xoshiro1024** and 68.2% for Xoshiro256**. Some streams are BigCrush resistant, but not all of them, so it is important to choose the right streams.

`MatrixRank` and `LinearComp` tests are rarely triggered, showing that scramblers have a significant impact on the generation quality. There's a significant difference between Xoshiro256++ and the other two: on `LinearComp` tests, the failure rate is almost two times greater for Xoshiro256++, showing that scramblers therefore have different efficiencies.

These results are for high and low bits combined. If we distinguish between them, high bits are more robust, with the difference between success rates being on the order of a few percent—at most 2.20% in favor of high bits for Xoshiro256**. The distribution of test failures is relatively similar for both high and low bits, across all 3 generators.

For each of the generators, we can therefore obtain high-quality streams (i.e., streams that pass BigCrush on all bits). Following the order of the generators on the graph, they show respectively 515, 467, and 511 high-quality streams for 1001 initial states. The probabilities of obtaining such quality streams are therefore respectively 51.4%, 46.7%% and 51.0%.

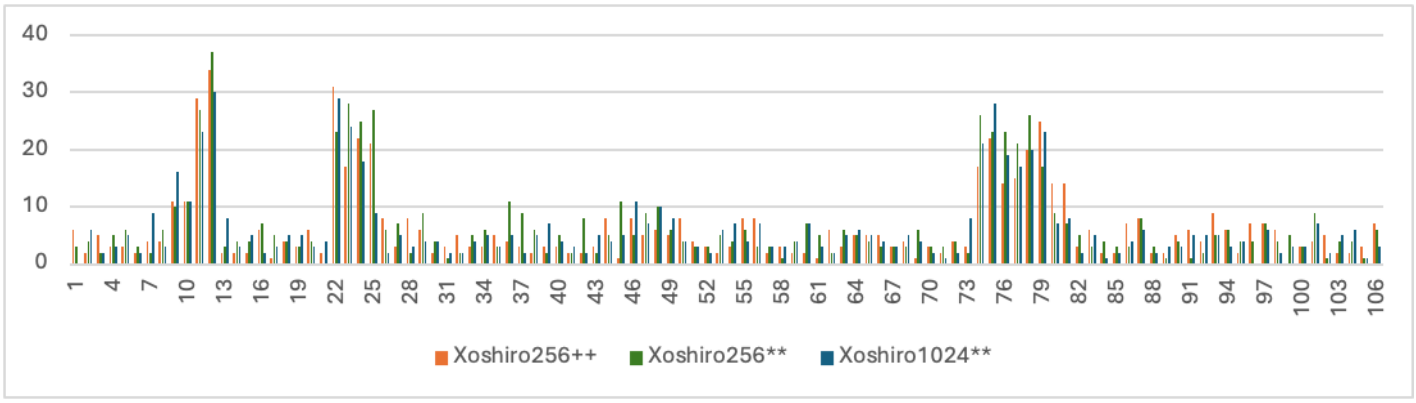

Figure 1: Distribution of Failed Tests for Xoshiro256++ - Xoshiro256** - Xoshiro1024** using BigCrush Test Suite

In the same way, there are streams that don't pass the battery on either the high or low bits. These streams are therefore completely unusable. They account for 82, 101, and 88 unusable streams respectively for 1001 initial states. The distribution of test failures is relatively similar for both high and low bits across all three generators.

MRG32k3a

MRG32k3a is a generator designed by Pierre L'Ecuyer and presented in 1999. It is a multiple recursive generator (MRG). What makes its quality so good is the choice of parameters (L'Ecuyer, 1999). This generator has a 32-bit resolution, although it returns 64 bits. There is no indication of which bits to choose, so it was tested in the same way as presented by Pierre L'Ecuyer in his TestU01 user guide (L'Ecuyer et al., 2007). Given that this generator was designed by the creator of the test battery, we can suppose that the parameters were chosen with Crush resistance in mind. Even if this is the case, passing the full test battery remains a guarantee of quality. This is the oldest generator tested in our study.

The generator was tested with 2002 different initial states, due to its 32-bit resolution. For its initialization, a 384-bit initial state must be filled in, corresponding to 6 double-precision numbers of 8 bytes. This is broken down into two groups of 3 double-precision numbers. The combination of 6 values at 0 must be avoided. In addition, each G_i group must be less than a m_i value. Having no further instructions, we take the numbers a user could use (0, 1, 2, 3, ...) and place them as an index in MT, and then use 6 MT double values to fill in the states of MRG32k3a.

Figure 2 shows the distribution of MRG32k3a failures on the BigCrush battery. They were obtained using 2002 initial states. The first notable point is that there are 3 main peaks of failure. These correspond to the last `CollisionOver` tests, `ClosePairs`, and `RandomWalks`. The maximum number of test failures is less than 40 for 2002 tests, i.e., less than 2% of occurrences. The overall success rate of the generator (success on all tests in a battery) is 69.73%. Due to the 32-bit resolution, there is no success rate on high or low bits. Of the 2002 streams tested, 1365 are Crush resistant, and therefore are of good quality.

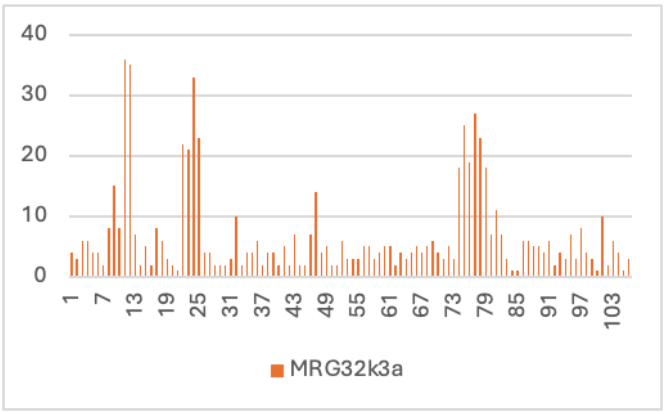

Figure 2: Distribution of Failed Tests for MRG32k3a using BigCrush Test Suite

This generator, although old, has excellent statistical qualities. Only a small number of tests in BigCrush show some problems. Because of its rather good success rate, it can be a generator of choice if you only need 32-bit randomness and if the generation time is not a constraint, since it is many times slower than generators from the Xoshiro family. A similar version, MRG63k5a, generates numbers with a 63-bit resolution. The quality of this generator is not assessed here.

Philox

Philox is a generator introduced in 2011 by John Salmon. It's a counter-based generator. It has an initial state producing initial values. To obtain other values, a counter must be incremented. This generator is advertised as not being usable for cryptography (as are all the generators in this study), but it is claimed to be the fastest Crush-resistant generator on GPUs (Salmon, 2011). It is the default generator used by the TensorFlow and PyTorch libraries. Its counter aspect makes it very practical for parallelization: if you know the number of values required, you can easily advance through the generation cycle.

This generator is available in several languages and with different resolutions. Implementations of these generators are given in the Random123 library (Salmon and Moraes, 2016).

For the study, we chose to use Philox4x32, which means a 32-bit output resolution and 4 values generated in parallel by the generator. A 32-bit resolution is what TestU01 requires. Initialization of this generator consists in filling an array with 2 elements of 32 bits. We use the MT generator, in which we place the user's seed (from 0 to 2002) and fill the initial state with MT. Since the generator gives us a result with a 32-bit resolution, we use 2002 initial states to have as many output files as the 64-bit generators.
To use the generator, you need to increment the counter for each generation. The counter is represented by four 32-bit numbers. Theoretically, you can generate $4 * 2^{4*32}$ numbers on the same initial state. In our case, BigCrush uses around 2^{38} values, so we only use the first 2 numbers of the increment as they allow us to go up to 2^{66} values.

Figure 3 summarizes the distribution of Philox failures across the 106 BigCrush tests. Once again, we can see that failures are mainly clustered on the `CollisionOver`, `ClosePairs`, and `RandomWalks` tests. The test with the highest failure rate (test number 11) presents 43 occurrences, representing a failure rate of 2.1%. The other tests are below 30 occurrences. The global success rate for this generator with the BigCrush battery is 69.78%. Of the 2002 initial states tested, 1397 were of high stochastic quality. Note that these states were only tested on the first 2^{38} values. This should be sufficient for many applications. Since Philox is sometimes available by default or implemented in a library, it can be considered a very good generator for 32-bit randomness.

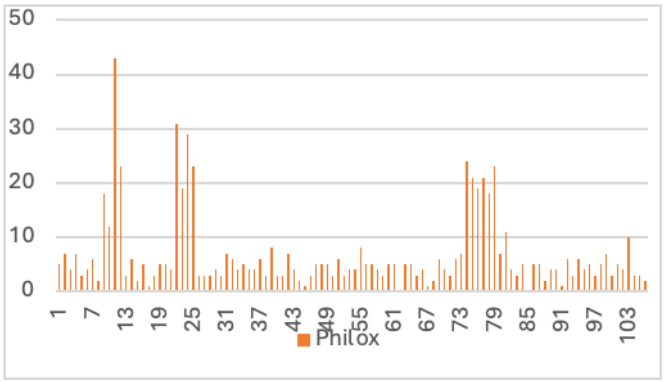

Figure 3: Distribution of Failed Tests for Philox4x32 using BigCrush Test Suite

PCG

Permuted congruential generators (PCGs) were proposed in 2014 by Melissa O'Neill. This is a family claiming excellent stochastic quality as the generators are advertised as Crush resistant. This family is an evolution of the Linear Congruential Generator (LCG) family. The quality of these generators relies on the skillful choice of coefficients, chosen by Melissa O'Neill with information from Pierre L'Ecuyer (O'Neill, 2014). As with the MRG32k3a generator, the coefficients announced as the most interesting by Pierre L'Ecuyer were perhaps chosen to pass the BigCrush tests. Passing the test battery is nevertheless a guarantee of high quality. This new family claims to be better than LCG in every respect (O'Neill, 2014); knowing the structural weaknesses of LCG, this quality is highly expected.

PCGs are easy to use, and implementations can be found on the `https://pcg-random.org` page with a user manual. This generator was claimed to meet the needs of parallel applications because, once the initial state has been set, an increment can be used to produce a completely random stream. In this way, from the same initial state, we can produce many streams of equal quality, and the authors expected no correlation; this is why PCG is used in machine learning and as the default generator for the NumPy library.
Unfortunately, as we mentioned in (Hill et al., 2024), experience showed that PCG is weak when you use it intensively (e.g., in High-Performance Computing, Machine Learning, …). Thus, an extension has been proposed recently: PCG64DXSM. The bad news is that Vigna has recently shown that both are weak, not only PCG but also PCG64DXSM. In addition, we have shown that they were slower than other older known PRNGs.
In our case, we have taken the basic implementation of PCG in its minimal version. We chose a 32-bit version because this is what TestU01 requires. Higher-resolution versions are available, but they are not studied here. To use it, we need to fill in an initial state consisting of a 64-bit integer. We then select an increment and obtain a stream. To properly initialize the generator, we place our seed value (0, 1, 2, ..., 2002) in the state, and set the increment to 1. Following this, due to the fact that our seed uses only a few bits, the first generated value is highly likely to be 0. To counter this, we generate one value that we discard. This generation changes the generator state and places it in a configuration that uses more bits.

To obtain as many output files as for the other 32-bit generators, we use 2002 initial states on the generator. Figure 4 shows the distribution of PCG32 failures in the BigCrush battery. Like all the other generators, the failure peaks are placed on the same tests, which are: the latest `CollisionOver`, `ClosePairs`, and `RandomWalks` tests. The tests with the highest failure rate show 33 and 32 occurrences respectively for tests 12 and 25, representing an occurrence rate of less than 1.7%. The other tests are below 30 failures. Out of the 2002 initial states tested, 1398 streams passed BigCrush. Thanks to the 32-bit generation, each stream is tested only once.

The generator has a BigCrush success rate of 69.8%. The first tested stream is not Crush-resistant. To obtain this stream, we set the generator's state to 0 and then perform a warm-up to reach a state with better entropy. However, the resulting increment is 1, meaning it only uses 1 bit. The failure of the battery may be due to improper initialization.
The quality of this generator in BigCrush tests makes it very interesting. However, it is important to note the issues raised by NumPy and Vigna. Despite its good results, it should not be preferred. Its implementation in different languages makes it versatile. It is the default generator in NumPy, at least its version PCG64DXSM.

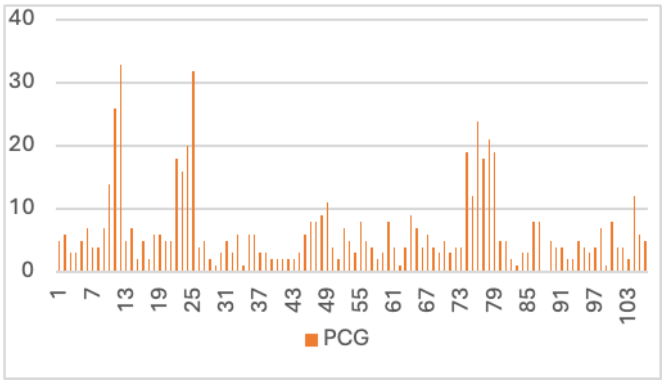

Figure 4: Distribution of Failed Tests for PCG32 using BigCrush Test Suite.

DISCUSSION

Each generator presents interesting results in the protocol we applied. However, there are a few points to note from Table 1: the 64-bit generators have a lower BigCrush battery success rate. This may be due to the fact that each stream is tested twice, once on the upper bits and then on the lower bits. This method seems more rigorous. We also note that the Xoshiro1024** generator has a larger initial state than other generators in the same family, but this doesn't give it better quality. The Xoshiro256++ generator is better than Xoshiro256**, as announced (Blackman et al., 2021).
If the user needs a 64-bit generator, Xoshiro256++ could be the best choice. It is the one with the best statistics. In the case of a 32-bit generator, high bits are the most robust. However, this type of generator has a higher cost, uses a larger initial state than Philox and PCG, and its use will produce 64 bits where only 32 are needed, so there may be some loss. This is why the Philox generator is a perfectly coherent alternative—PCG could also be considered purely based on our results. So is MRG32k3a, but the initial state is very large (384 bits), compared with 64 bits for Philox4x32 and PCG32. In terms of ease of use, Philox should be prioritized, as the implementation is available in default libraries and user functions are provided with them.

It's worth noting that a 64-bit generator stream passes BigCrush only if both the lower bits and the higher bits pass BigCrush. This makes the task more complicated. Perhaps this is why there is such a gap between the Xoshiro family and other generators.

Table 1: Some Generator Statistics

Generator name	Failed batteries	Overall success rate	Success rate of MSBs	Success rate of LSBs
Xoshiro256++	675	51.45%	72.70%	70.50%
Xoshiro256**	733	46.65%	68.80%	67.70%
Xoshiro1024**	675	51.05%	71.00%	71.20%
Philox	605	69.78%	/	/
MRG32k3a	606	69.73%	/	/
PCG32	603	69.83%	/	/

Comparison between Philox and PCG

Philox and PCG are both 32-bit generators, implemented in libraries and used for parallelization. It therefore makes sense to compare these generators directly. Table 1 already shows battery success rates in favor of PCG (a difference between the two of 0.005% over 2002 tests). PCG has passed only 2 more tests.

Figure 5 shows the distribution of Philox4x32 and PCG32 generator failures on the BigCrush test battery. The same 3 failure peaks can be seen, as well as a striking similarity between the two generators. Each test has a similar number of failures. A difference can be made on test numbers 11 and12. Both are `CollisionOver` tests, but with different parameters (L'Ecuyer and Simard, 2007). Test number 11 was more successful for PCG, with 26 failures versus 43 for Philox. On test number 12, the opposite is true, with Philox doing better with 23 failures versus 33 for PCG. So, the same test but with different parameters is better passed by one or the other of the generators.
These two generators produce extremely similar results. They should be tested on more initial states so as to have more samples to analyze. It would also be necessary to test other characteristics expected of good PRNGs to differentiate them. We can mention the generation speed, the cycle length, the ability to parallelize properly, and the memory footprint.

Conditions of our Reproducibility Testing

Partial tests were rerun using the TestU01 library on a Mac running macOS with GCC 15, and on the same cluster using a different version of GCC (GCC 13). TestU01 can be compiled with GCC 12, but additional options are required for more recent compilers (GCC 13 and later). The suite has been slightly modified to support newer hardware (configuration files) and to allow compilation with recent compilers.

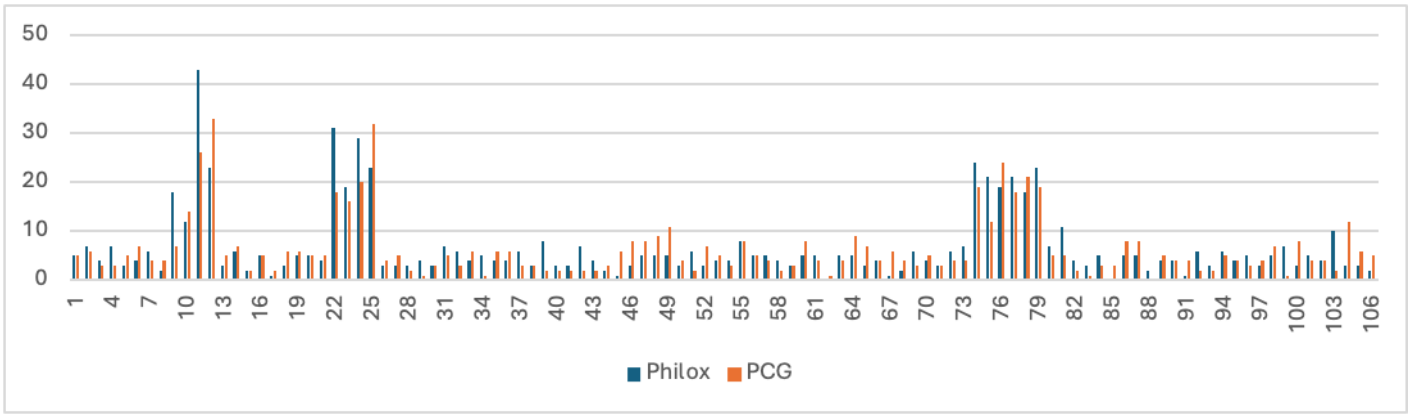

Figure 5: Distribution of Failed Tests for Philox4x32 and PCG32 using BigCrush Test Suite

The compilation process may still produce some warnings, but no errors. Here are the 2 URLs we use as repositories:

- https://gitlab.limos.fr/testu01/benchmark
- https://gitlab.limos.fr/testu01/library

The modified version for the final testing library is no longer backward compatible. The following modifications were made:

- Functions within the same file were reordered (a function declared before another function was moved after it).
- Functions with an implicit void return type were made explicit (otherwise, recent compilers assume an int return type).

The testing result files produced by the student developing the first results were compared with the new results to ensure the reproducibility of what we presented in this paper.

CONCLUSION

Good-quality randomness is a fundamental requirement for a wide range of scientific computing applications. In this work, we performed a large-scale statistical evaluation of six widely used PRNGs that are commonly regarded as capable of passing the most demanding statistical test suites. Although all of these generators are advertised as being resistant to the TestU01 Crush batteries and are indeed able to pass at least one complete BigCrush campaign, our results demonstrate that this property is highly dependent on the selected stream. For every generator considered, we identified initial states that fail the BigCrush battery, highlighting that passing BigCrush is not an intrinsic property of a generator but rather of specific initialization conditions. Moreover, because PRNG designers rarely disclose the initial states used to validate their generators, the reproducibility of published statistical evaluations remains limited.

A major contribution of this work is the introduction of a BigCrush success rate as a quantitative indicator of PRNG robustness across streams. By evaluating more than 10,000 independent streams per generator, we show that statistical reliability should be assessed probabilistically rather than from a single successful execution. Despite relying on substantially different design principles developed over nearly two decades, the generators exhibit remarkably similar statistical behavior: the gap between the best and worst generators is only 2.16% when considering 32-bit outputs. This observation suggests that recent algorithmic advances have not translated into a measurable improvement in BigCrush success rates.

Perhaps the most striking observation is that none of the evaluated generators consistently passes the BigCrush battery, though most of them claim to be BigCrush resistant. Even the best-performing generators succeed for only about 70% of the tested streams. This raises an important question for the community: have current PRNG designs reached the practical limits imposed by existing statistical constructions, or has the BigCrush battery become sufficiently stringent that occasional failures should be expected even for high-quality generators? Answering this question would require evaluating a source of true randomness against the same testing protocol, providing an empirical upper bound for achievable success rates.

From a practical perspective, our results indicate that the evaluated generators remain suitable for most sequential applications. However, for large-scale parallel simulations or applications requiring exceptionally high statistical confidence, stream initialization deserves considerably more attention than is generally assumed. For modern PRNGs, the choice of the initial state is no longer equivalent to selecting a simple seed, and poorly chosen streams may significantly degrade statistical quality. Whenever possible, practitioners should therefore rely on streams whose statistical behavior has been experimentally validated.

Several directions naturally emerge from this work. Extending the study to additional 64-bit generators would help determine whether their lower success rates stem from intrinsic architectural differences or from the specific algorithms currently available. Including widely used reference generators such as Mersenne Twister would also provide a valuable baseline. Further comparisons between Philox and recent PCG variants, including PCG64DXSM, would clarify the relative merits of generators designed for similar applications. Finally, complementing statistical quality with performance metrics—including throughput, memory footprint, and the impact of compiler optimizations

such as -O3—would provide a more comprehensive evaluation framework for selecting PRNGs in scientific computing.

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DECLARATION OF GENERATIVE AI USE

While preparing this work the authors used ChatGPT, Gemini and Google Translate to assist with grammar, spelling, language refinement for readability and improvements of its impact. After using AI tools, the authors carefully reviewed and edited the paper and take full responsibility for its content.

Autonomous Systems

Pathplanning in Dynamic Environments

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KEYWORDS
dynamic pathplanning

ABSTRACT

Automation in logistics requires efficient path-planning algorithms for autonomous robots that can operate reliably in dynamic environments with unpredictable obstacles. In this work, several established approaches—including Multi-Agent A*, Conflict-Based Search (CBS), Push and Wait (PAW), and Windowed Hierarchical Cooperative A* (WHCA*)—were implemented and evaluated within a custom-developed simulation environment. The simulation models a warehouse as a discrete grid-based environment and incorporates both autonomous robots and randomly moving pedestrians as dynamic disturbances.

The experimental results demonstrate that classical Multi-Agent A* is only practical for small-scale scenarios due to its exponentially increasing computational complexity. CBS provides a significant improvement but exhibits considerable runtime variability with occasional performance outliers. WHCA* proves to be robust, adapts flexibly to environmental changes, and requires only moderate computational resources. The newly implemented PAW approach achieves the highest overall efficiency and is particularly well suited for real-world logistics applications because of its reactive strategy.

Overall, the findings indicate that reactive and time-window-based algorithms offer the greatest practical applicability and provide a promising foundation for deployment in dynamic warehouse environments.

INTRODUCTION

Autonomous mobile robots (AMRs) are increasingly used for material handling, order fulfillment, and internal transportation in warehouses. Their operation depends on path-planning algorithms that compute collision-free routes while supporting efficient and scalable coordination. This becomes particularly challenging when many robots share narrow aisles and common loading or storage areas.

The coordination of multiple robots is commonly formulated as Multi-Agent Path Finding (MAPF). MAPF algorithms must prevent collisions and resolve competition for shared space while optimizing objectives such as travel time, throughput, or total path length. Existing approaches range from optimal joint search to hierarchical, time-window-based, and reactive methods. They therefore offer different trade-offs between solution quality, computational cost, scalability, and response time. Classical MAPF often assumes a static environment with known agent trajectories. Real warehouses, however, contain human workers, manually operated vehicles, and other obstacles whose movements cannot be predicted reliably. These dynamic obstacles may invalidate previously computed routes at any time. A practical planner must consequently detect changes, revise its decisions quickly, and maintain safe robot motion without repeatedly incurring excessive replanning costs. Responsiveness to unpredictable movement is therefore as important as the quality of the initial plan.

In this setting, planning and execution cannot be separated completely. Offline methods can optimize routes before movement begins, but even a high-quality solution loses value when an unmodeled obstacle blocks a critical cell. Online and reactive methods instead incorporate current observations, although frequent updates may reduce global coordination quality. Comparing these strategies therefore requires measuring both computational effort and their effect on completing warehouse tasks.

This work investigates the suitability of Multi-Agent A*, Conflict-Based Search (CBS), Windowed Hierarchical Cooperative A* (WHCA*), and Push and Wait (PAW) for such environments. We developed a custom grid-based warehouse simulation containing autonomous robots and randomly moving human agents. Stochastic human behavior provides a repeatable source of uncertainty and allows us to examine how planning strategies respond when obstacle motion differs from their assumptions. The algorithms are compared with respect to planning time, scalability, runtime stability, task-completion time, robustness, and adaptability. This evaluation highlights the strengths and limitations of optimal, windowed, and reactive MAPF strategies and assesses their suitability for logistics systems in which environmental changes and unpredictable moving obstacles are unavoidable.

LITERATURE SURVEY

Autonomous mobile robots (AMRs) provide greater flexibility than systems tied to fixed infrastructure, but large fleets require effective task allocation, traffic control, and coordination. The warehouse systems presented by Wurman et al. (2008) and Guizzo (2008) demonstrate both the industrial relevance and the complexity of multi-robot operation.

Path planning computes collision-free routes while optimizing criteria such as distance or travel time. Grid-based graph search is well suited to warehouse layouts, and A* provides complete and optimal paths under appropriate heuristic conditions (Hart et al. 1968). Its cost, however, grows rapidly when a state represents several robots. This coordination problem is formalized as Multi-Agent Path Finding (MAPF), for which Stern et al. (2019) provide standard definitions and benchmarks. MAPF research therefore focuses on balancing solution quality and computational feasibility.

MAPF methods range from optimal joint searches to prioritized, compilation-based, learning-assisted, and reactive approaches (Surynek 2022, Chung et al. 2024). Optimal methods provide strong solution guarantees but may scale poorly, whereas heuristic and reactive methods trade some path quality for faster decisions. The appropriate balance depends on fleet size, congestion, and how frequently the environment changes.

A direct joint-state A* search can find optimal multi-agent solutions but scales exponentially. Silver (2005) addressed this limitation with Cooperative A* and Windowed Hierarchical Cooperative A* (WHCA*). Both plan agents sequentially using space-time reservations, while WHCA* limits planning to a rolling window. This substantially reduces computation but can produce suboptimal behavior or deadlocks in congested environments.

Conflict-Based Search (CBS), introduced by Sharon et al. (2015), separates individual path planning from high-level conflict resolution. When a conflict occurs, constraints are added and the search branches on the involved agents. CBS often preserves optimality without exploring the complete joint state space, although interacting conflicts may cause costly replanning and substantial runtime variation. Improved, bounded-suboptimal, anytime, and learning-guided variants seek to reduce these limitations (Huang et al. 2022).

These variants modify conflict selection, merge strongly interacting agents, relax optimality bounds, or return progressively better solutions. Although they improve scalability in many instances, their dependence on conflict resolution can still produce computational peaks after unexpected environmental changes.

Dynamic warehouses make responsiveness as important as optimality. When humans or other unpredictable obstacles invalidate a plan, reactive approaches use current observations and local coordination instead of repeatedly computing complete global solutions. PAW follows this principle and is related to the Push-and-Swap method of Luna and Bekris (2011). Robots yield, wait, or temporarily displace other robots when conflicts occur. These actions may lengthen paths but support rapid adaptation.

This local response is consistent with behavior-based navigation, where simple behaviors produce robust reactions to changing surroundings (Arkin 1998). Such methods are particularly useful when obstacle trajectories cannot be predicted far enough ahead for reliable global planning.

Most MAPF formulations assume static environments and coordinated agents. Human workers cannot be assigned reservations or expected to follow a protocol, making their future movement uncertain. Human-aware navigation consequently combines pedestrian prediction, safety, social behavior, and dynamic replanning, as reviewed by Kruse et al. (2013). Reactive strategies and limited planning horizons are valuable because decisions can be revised during execution.

Grid-based simulation enables reproducible comparisons of planning time, scalability, path quality, and robustness without a large physical fleet. The continuous transportation tasks considered here are also related to lifelong MAPF and multi-agent pickup-and-delivery problems (Ma et al. 2017). Our experiments compare optimal, windowed, and reactive methods under identical stochastic human movement to assess computational efficiency, runtime stability, adaptability, and task completion.

OUR APPROACH

Our approach is a Push-and-Wait algorithm, analogous to Push-and-Swap strategies Luna and Bekris (2011). Since the human obstacles will eventually move away, waiting is a valid strategy. Both MA* and CBS are **offline planning approaches**, meaning that the complete paths for all robots are computed before execution begins. However, path planning can also be performed **online**, allowing robots to react to the current state of the environment during operation. One well-known online multi-agent pathfinding algorithm is **Push-and-Swap (PAS)**, introduced by Luna and Bekris (2011).

In PAS, each robot initially computes a shortest path to its goal and subsequently attempts to follow this path during execution. Whenever a robot encounters another robot blocking its intended movement, one of two operations is performed based on a predefined priority assignment. If the blocking robot has a lower priority, it is pushed aside to free the desired position. This operation may recursively trigger additional push operations if further robots obstruct the relocation. Conversely, if the blocking robot has a higher priority, the algorithm performs a **swap operation**, exchanging the positions of the two robots. This is achieved by exploiting a ver-

tex with a minimum degree of three and two unoccupied neighboring vertices, enabling a conflict-free exchange of positions.

PAS assumes a scenario in which each robot is assigned a single goal location that must be reached exactly once. Furthermore, all robots are required to remain at their goal locations until every robot in the system has successfully reached its destination. The logistics scenario considered in this work differs significantly from these assumptions. Robots continuously receive new transportation tasks and are not required to remain at their destination after task completion. Instead, they occupy the goal location for only a single time step, representing the loading or unloading of a package, before either receiving a new destination or becoming idle.

As a consequence, a simplified variant of PAS can be employed that omits the swap operation entirely. In the remainder of this work, this modified approach is referred to as **Push-and-Wait (PAW)**. In PAW, robots that are unable to clear the path immediately remain stationary but can still be displaced by other robots if necessary. Additionally, robots that have reached their destination cease blocking the environment after a single time step, since they either receive a new task or become inactive. These characteristics significantly reduce the likelihood of deadlocks and eliminate many of the situations that would otherwise require explicit swap operations.

Under these assumptions, the primary remaining source of deadlocks arises from **dead-end structures**, i.e., vertices with degree one. In such locations, robots may become trapped because no alternative movement options exist that would allow conflicts to be resolved through local push operations alone. Consequently, the performance and completeness of PAW depend strongly on the topological properties of the underlying environment, particularly the presence and distribution of dead-end corridors.

Most of the time, humans are treated simply as immovable obstacles. In other words, the algorithm initially includes all current human positions in the blocked set. This works largely without issues. As a result, robots only need to wait until humans move out of the way on their own. The only problem arises when two robots approach a human from opposite sides, and the human is unable to move in any other direction. In this case, a deadlock occurs. This type of deadlock can be avoided by checking, before the actual PAW execution, whether a human is completely enclosed by robots. If this is the case, the path of the blocked robot with the longest path is recomputed. This automatically creates space for the human.

EXPERIMENTS

In order to evaluate our algorithm, show that it works on our problem setting, and compare the performance to other algorithms in the literature, we developed a software that can generate warehouse instances of various size and with several parametrized characteristics and measure different algorithms' performance. Figure 1 shows the main view.

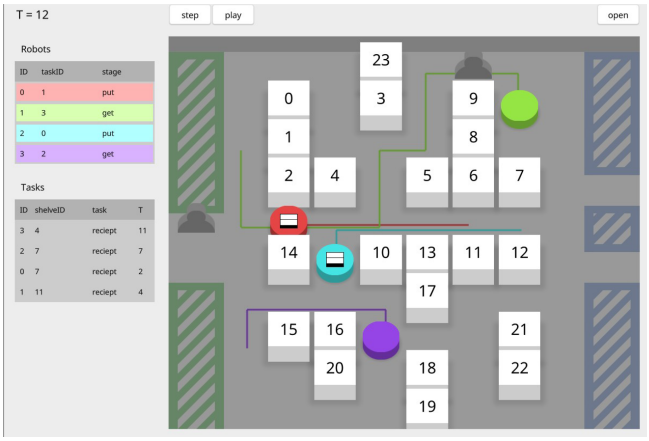

Figure 1: App Screenshot of generated warehouse instance

Warehouse Generation

To minimize the influence of a specific warehouse layout, we developed an algorithm that generates random but realistic (based on subjective judgment) warehouse environments that contain no dead ends and in which all shelves are reachable. First, an empty warehouse with a grid layout is generated. Then, alternating horizontal and vertical shelf rows with random gaps are placed until 2/5 of the warehouse area is filled with shelves. Subsequently, all shelves that are completely enclosed by other shelves are removed. Dead ends are eliminated by connecting nodes with one or zero outgoing edges in the navigation graph to the nearest connected component by removing shelves. Finally, it is checked whether the entire graph is connected; if not, a new warehouse is generated.

The Simulation

At the beginning of each simulation run, a predefined number of robots and human agents are placed at randomly selected starting positions within the warehouse environment. The robots are controlled by one of the path-planning algorithms under investigation, whereas the simulated human agents follow a stochastic movement model. Human movement is random but incorporates a directional persistence bias, making agents more likely to continue moving in their current direction than to change direction abruptly.

The simulation operates in discrete time steps. During each time step, a robot may either move to an adjacent

Table 1: Average planning time in a 15 × 15 warehouse with three robots

Algorithm:	A*	WHCA*	CBS	PAW
Mean time (ms):	263.203	0.203	0.437	0.085

grid cell or perform a loading or unloading operation associated with a package handling task.

Customer orders are generated at random intervals, with at least one future order always scheduled in advance. Each order corresponds either to the storage of an incoming package or the retrieval of an outgoing package. An order specifies both a storage location, represented by a shelf identifier, and a corresponding pickup or delivery location within the inbound or outbound logistics area of the warehouse. Orders are processed in a first-in–first-out (FIFO) manner, while the assignment of tasks to individual robots is determined by the respective planning strategy.

As long as pending orders exist, the simulation advances one time step at a time. When no active orders are available, the simulation clock advances directly to the arrival time of the next scheduled order, thereby avoiding unnecessary computation during idle periods.

Experimental Procedure

Two experiments are conducted. First, the average planning time of all algorithms is measured on a 16×16 warehouse with three robots. This experiment is intended to evaluate the improvement of the other algorithms compared to the basic Multi-Agent A* search. The main experiment evaluates all algorithms except the basic A* search on a 50×50 warehouse with five human agents and four to ten robots (cf. Fig. **??** for a large problem instance.), and 10 indipendent replications. (The A* search requires too much computation time for the given warehouse size and number of robots, and its poor performance is already clearly evident from the first experiment.)

RESULTS

The analysis of the first experiment shows that in a 15×15 warehouse with 3 robots, the basic A* search requires on average 263.203 ms for planning. WHCA* requires 0.203 ms, CBS 0.437 ms, and PAW only 0.085 ms (cf. table 1). It can be observed that A* already performs significantly worse than all other tested algorithms even in this small scenario.

For the large-scale experiment in 50×50 warehouses, both the average and maximum planning times were recorded. It can be observed that CBS performs significantly worse on average than WHCA* and PAW, and in particular its runtime increases more rapidly with the number of robots, cf. fig. 2.

Furthermore, the difference in maximum planning duration is even more pronounced, e.g., 1280 ms for 8 robots in CBS compared to only 29 ms and 127 ms for the other algorithms, cf. fig.3. This can be explained by the fact that CBS mostly only executes an existing plan, but when replanning is required it can become very time-consuming if conflicts require many iterations to resolve. For WHCA* and PAW, the variance in planning time is significantly lower. This is likely because PAW resolves conflicts immediately when they occur, rather than focusing on a future planning horizon. WHCA*, through its time-windowed planning approach, also distributes computation more evenly. Due to the lower variance in planning time and overall significantly better performance, both WHCA* and PAW are well suited for the simulated scenario. However, PAW is likely the most suitable for real-world deployment, as it not only performs slightly better on average, but—more importantly—its reactive nature makes it more adaptable to real-world situations with imprecise, continuous (non-grid-based) movement. The other algorithms, which rely more heavily on preplanning, would need to replan more frequently than in the simulation, and would therefore perform even worse. As can be seen in fig.4, PAW finds the best solution measured in number of steps to fulfill the task.

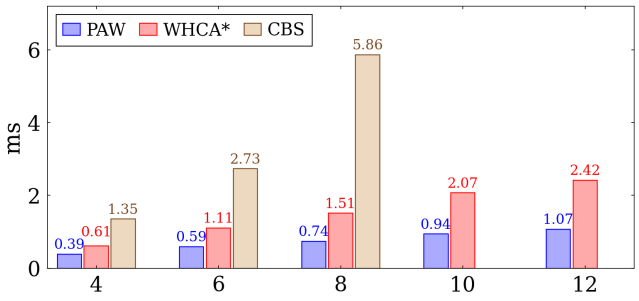

Figure 2: Mean planning duration in ms shown per number of robots

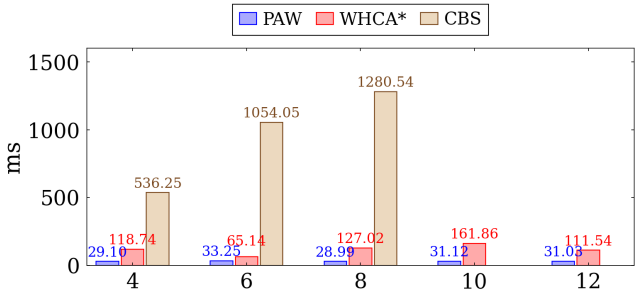

Figure 3: Maximum planning duration in ms shown per number of robots

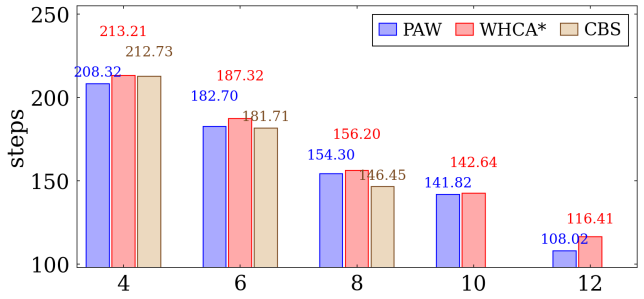

Figure 4: Average task completion time measured in steps per algorithm and number of robots

CONCLUSION

The experimental results show that reactive and time-window-based path-planning strategies are particularly well suited for dynamic warehouse environments. While classical Multi-Agent A* becomes impractical even for comparatively small scenarios, CBS improves scalability but exhibits considerable runtime variability when conflicts require extensive replanning. In contrast, WHCA* and especially PAW achieve consistently low planning times while maintaining competitive task completion times.

Among the evaluated approaches, PAW provides the most promising balance between computational efficiency, robustness, and practical applicability. Its reactive conflict-resolution strategy enables fast adaptation to moving obstacles and changing environmental conditions, which is especially relevant for warehouses shared by robots and human workers. This makes PAW a suitable candidate for real-world logistics systems in which precise long-term prediction of all agents is not feasible. Future work will focus on transferring the approach to more realistic and continuous warehouse settings. In particular, we are investigating its deployment in an innovative logistics concept in which the floor moves pallets instead of relying exclusively on autonomous mobile robots. Since the planning and movement of individual pallets can be mapped to the multi-agent path-planning problem studied in this work, we expect the presented approach to provide a strong foundation for such systems as well.

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MODEL FOR VISUAL TERRAIN RELATIVE NAVIGATION

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KEYWORDS
Neural network, Pattern recognition, Real-time, Software engineering, Model evaluation

ABSTRACT

Unmanned aerial vehicles (UAVs) are a rapidly developing and synergistic technology that complements other new technologies. The paper in question focuses on the use of artificial intelligence for visual Terrain-Relative Navigation (TRN) in UAVs, specifically utilizing computer vision based on deep learning and EfficientNet architecture, which is suitable for deployment on edge computing. To evaluate the effectiveness of such a navigation system in GNSS-denied environments, a simulation study was conducted. The main objective of this work is to evaluate the impact of input image resolution on the performance of a neural network-based multi-class classification model and to assess the model's robustness under varying input conditions simulating different UAV flight altitudes.

INTRODUCTION

In the current international geopolitical situation, the use of UAV has become increasingly important for both military and civilian applications. UAVs are required to operate in various environments, and traditional navigation methods, such as GNSS, can be unreliable due to intentional and unintentional interference. Visual TRN is a promising alternative that utilizes the onboard camera to extract features from the environment and determine the UAV's position and orientation.
The use of Convolutional Neural Networks (CNNs) in TRN has several advantages over traditional navigation methods. Firstly, CNNs are highly effective in extracting features from images, particularly in 3-channel RGB images. This allows the TRN system to operate effectively in a wide range of environments, including urban and natural landscapes. Secondly, CNNs are robust to noise and interference, making them suitable for operation in contested environments. Finally, the use of CNNs in TRN can provide a higher level of security and resilience, as they do not rely on external signals, such as GNSS, for navigation. The simulation is based on the assumption that the drone is flying at a maximum altitude of 120 meters from the nearest point of the earth's surface. This is in accordance with the regulations in force in the European Union (EU) for drones in the open category. Flight altitude influences the CNN-based TRN performance in a different way - although lower altitudes increase the visibility of fine terrain details, the spatial extent covered by a given image patch decreases, potentially resulting in an insufficient number of terrain features for robust classification. Therefore, it is important to consider the operating altitude when designing and implementing a TRN system using CNNs. Despite this limitation, the use of CNNs in TRN for UAVs has several advantages over traditional navigation methods. By optimizing the classified patch size, it is possible to achieve state-of-the-art performance. Additionally, the robustness of CNNs to noise and interference makes them suitable for operation in contested environments.

RELATED WORKS

A comprehensive learning-based approaches offer a promising solution by using deep learning models to directly regress the UAV's position from aerial imagery (Harvey et al., 2021). These models eliminate the need for explicit mapping and matching to a reference map, providing a more efficient and flexible navigation system. However, these systems can be sensitive to perception errors, leading to significant drift and inaccuracies in position estimation. To address this challenge, hybrid approaches, combine deep learning with classical optimization methods, have been proposed (Yao et al., 2024). These methods use neural networks to provide observations or features, while geometric models and optimization constraints are used to refine the final estimation of position. In the realm of classical image retrieval techniques, a notable methodology has been proposed as a means of navigation in GNSS-denied environments (Kinnari et al., 2022). This approach diverges from traditional techniques by utilizing CNNs as feature extractors for images and determining the location through the matching of image representations to orthophotos or reference databases. Previous TRN systems relied on manually designed features and optical flow, which were then combined with probabilistic filtering frameworks like particle filters and the Extended Kalman Filter (Lee et al., 2014). These methods explicitly modeled terrain geometry using digital elevation maps and homography constraints. However, their accuracy was limited by the reliability of the manually engineered feature representations. This realization led to a shift towards CNN-based methods, which learn distinguishable visual features directly from data while maintaining the fundamental principles of the Bayesian state estimation framework. Lidar-Inertial Odometry (LIO) systems, such as FAST-LIO2 (Xu et al., 2022), estimate trajectory in a purely relative manner without referring to an external global map. FAST-LIO2 propagates the UAV position based on a tightly coupled fusion of LiDAR and IMU measurements within an iterative Kalman filter, providing high local accuracy and robustness to dynamic maneuvers. However, such systems remain a form of odometry and not a full global Simultaneous Localization and Mapping (SLAM), as estimation errors accumulate over time leading to increasing trajectory drift in long missions

without additional global correction mechanisms or loop closing. In contrast, TRN approaches with CNNs introduce an additional source of information by referencing an external, global terrain representation (e.g., satellite maps or digital elevation models). TRN systems periodically perform image alignment to the reference map, allowing for estimation error correction. As a result, TRN does not eliminate local sensor errors, but introduces a global "re-anchoring" mechanism, limiting drift accumulation through periodic reference to an external coordinate system.

MOTIVATION AND CONTRIBUTION

One of the major challenges in operating UAVs is navigation, particularly over long distances. The ongoing conflict between Russia and Ukraine has highlighted the importance of long-distance UAV navigation, as optical fibers have been used as a means of guidance over extended periods. This article aims to address this challenge by proposing a novel concept for UAV navigation that utilizes visual TRN and edge computing. Specifically, we will demonstrate the feasibility of using CNNs deployed on an NVIDIA Jetson platform to navigate over long distances by leveraging terrain features. Unlike purely local state estimation methods, such as visual or LiDAR-inertial odometry, which estimate motion incrementally and are therefore susceptible to drift accumulation over time in the absence of global constraints or loop closures, TRN approaches incorporate external map information to provide absolute or quasi-absolute localization updates. These updates can be used to constrain and periodically correct accumulated error, thereby improving long-term navigation robustness in GPS-denied environments.

CONCEPT OF SIMULATED SYSTEM

The system is a classifier designed for UAV navigation over terrain. The assumption is that the altitude of the drone's flight is translated into the size of a single patch, which in turn defines the resolution of the navigation. This means that the size of the patch will determine the level of detail that the classifier can distinguish and recognize in the terrain. A significant advantage of this system is the use of transfer learning, which allows for the easy adaptation classification layer of the CNN to suit the specific terrain. This means that the classifier can be fine-tuned to recognize and classify features specific to the terrain that the UAV will be operating over. Furthermore, the use of multiple maps during the training process allows the classifier to be agnostic to the time of year and lighting conditions. This means that the classifier will be able to accurately classify and navigate the terrain regardless of the time of day, season or weather conditions. This is particularly useful for UAVs operating in outdoor environments where these factors can vary significantly.

Figure 1 presents the block diagram of the experimental simulation pipeline employed in this study. The workflow begins with the acquisition of aerial photographs, which constitute the primary input data. From each aerial image, multiple regions of interest (ROIs) are extracted.

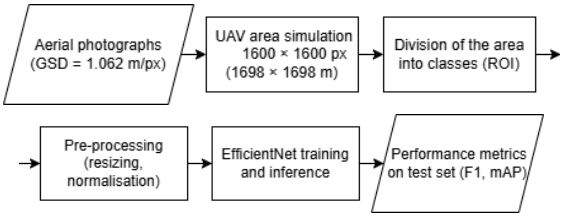

Figure 1: Block Diagram of the Experimental Simulation Pipeline.

Each ROI is therefore treated as an individual class within the formulated multi-class classification problem. Following ROI extraction, all image samples undergo a preprocessing stage to ensure consistency of the input data. This stage includes image resizing, normalization, and other transformations required to match the input specifications of the CNN. The preprocessed dataset is subsequently used for training an EfficientNet model. The performance of the trained classifier is assessed on an independent validation dataset using several complementary evaluation metrics. The primary metric is the F1-score, defined as the harmonic mean of precision and recall, providing a balanced measure of classification performance, particularly in the presence of class imbalance. Additionally, the model is evaluated using the mean Average Precision (mAP), computed for the multi-class classification task by averaging the AP values across all classes. Together, these metrics provide a comprehensive assessment of both the classification accuracy and the discriminative capability of the proposed approach.

Overall, the use of visual TRN with CNN on edge computing provides a robust and flexible solution for UAV navigation over known terrain. The ability to adapt the classifier to specific terrains and conditions through transfer learning, as well as the ability to operate regardless of season and lighting conditions, makes this system a powerful tool for a wide range of UAV applications.

Algorithms

Simulation was implemented using the EfficientNet architecture, as it provides an optimal trade-off between accuracy and computational complexity. Unlike heavier backbones, such as ResNet-101 or networks used in Mask R-CNN architectures, EfficientNet achieves comparable feature extraction performance with significantly fewer parameters and lower computational requirements. This is particularly important for autonomous navigation systems where available computational resources are limited.

The choice of EfficientNet is further justified by the previous success of CNNs in TRN tasks. CNN-based "LunaNet" for detecting lunar craters in TRN, achieving approximately 60% reduction in average position estimation error compared to traditional image processing methods (Downes et al., 2020) . The effectiveness of CNNs in terrain navigation, developing the FederNet system that uses a Mask R-CNN architecture to detect and align craters with reference maps. The proposed system achieved navigation accuracy below 400 m during autonomous lunar orbital navigation (Prete et al., 2022). Considering the proven effectiveness of CNNs in TRN tasks and the real-time requirements for operation on resource-constrained platforms, the use of EfficientNet offers a cost-effective solution for efficient feature extraction while maintaining low computational costs.

The EfficientNet architecture is based on a compound scaling method that uniformly scales the depth, width and resolution of the network using a set of hyperparameters (Tan and Le, 2019). This approach allows for a trade-off between computational resources and model performance. In this study, we focus on the problem of patch classification as a subtask of TRN. The goal is to classify a given patch of terrain into one of several predefined classes based on visual features. The evaluation was conducted using EfficientNet architectures, including EfficientNetB0, EfficientNetB1, EfficientNetB2, EfficientNetB3, to determine the best-performing model for the task. The EfficientNet architecture consists of an initial stem convolution, a sequence of MBConv blocks for hierarchical feature extraction, and a classification head that produces the final class predictions. EfficientNet-B0 begins with a stem convolutional layer followed by several stages of MBConv blocks. Each MBConv block consists of an optional expansion layer (omitted in MBConv1), a depthwise convolution, Batch Normalization, SiLU activation, a squeeze-and-excitation module, and a projection layer. When applicable, a residual skip connection is also employed. The overall architecture of EfficientNet-B0 is shown in Figure 2.

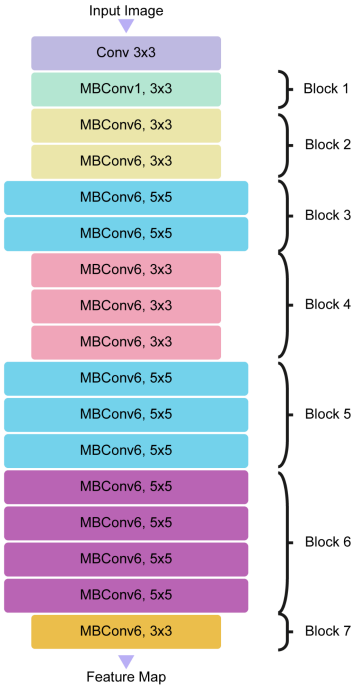

Figure 2: Architecture of EfficientNet-B0 with MBConv as Basic Building Blocks.

Datasets

The datasets consist of aerial images, which are classified as medium-resolution satellite images with a Ground Sampling Distance (GSD) of 1.062 m/px. The operational area of the UAV was simulated as a square with dimensions of 1600 x 1600 px, which corresponds to a real-world size of 1698,62 x 1698,62 m, examples are presented in Figure 3. The initial experiments focused on selecting the optimal number of model parameters, assuming that the size of each class is 300 m × 300 m.

a) 5th of May 2019 b) 28th of May 2023

Figure 3: Analyzed Area: a) Sample Training Map; b) Test Map

In the next phase of the experiments, the impact of reducing the size of the classified sub-area while simultaneously increasing the number of classes present in the analysis region was investigated. The study area represents a partially urbanized environment, through which a highway and critical power infrastructure pass. Specifically, the effects of decreasing the class size to 250 m × 250 m, 200 m × 200 m, 150 m × 150 m and 100 m × 100 m were analyzed, while maintaining a similar overall area for analysis. Representative samples from the training dataset are presented in Figure 4.

Figure 4: 100 m x 100 m Samples from Training Set: a) random Classes; b) same Class in different Seasons

The study aimed to find a balance between the level of detail and the computational efficiency of the model. By reducing the size of the classified sub-areas, we were able to increase the number of classes present in the analysis region, thereby improving the spatial resolution of the model. However, it is important to note that reducing the class size also has implications for the level of detail that can be captured in the model. Specifically, as the class size decreases, the number of pixels within each class also decreases, which can lead to a loss of information and a reduction in the accuracy of the model. Generally, small objects represented by a limited number of pixels may provide insufficient visual and semantic information for effective feature extraction (Sun et al., 2022), (Liu and Juang, 2026). By selecting the optimal number of model parameters for each reduced object class size, a model was developed that accurately captured the spatial patterns. Overall, these experiments highlight the importance of considering both the size and number of classes when developing a model for the analysis of spatial data. By carefully balancing these factors, we can develop models that accurately capture the spatial patterns of the variables of interest while minimizing the computational cost of the analysis.

The training dataset consisted of 6 tiles of aerial images taken between 2015 and 2022 at various times of the day and during different seasons ("National Geoportal" 2026). The test dataset, on the other hand, contained images taken in the spring of 2023. The training and testing of the TRN model involved the use of aerial images, where the goal was to estimate the position of the UAV based on the features extracted from the images. The model was trained on the training dataset, and the performance was evaluated on the test dataset. The results of the experiment were then analyzed to determine the accuracy and robustness of the TRN model. This experiment demonstrates the potential of TRN for UAV navigation and highlights the importance of selecting the optimal number of model parameters for accurate and efficient position and orientation estimates.

Neural Network Training

From a wide range of optimization techniques, the following methods were selected and applied: global average pooling, batch normalization, Adam optimizer and cosine learning rate scheduler. First, global average pooling is a layer in that is used after the last convolutional layer of a CNN to perform average pooling on the input tensor (Lin et al., 2013). This layer reduces the spatial dimensions of the input tensor while keeping the number of filters the same. This is useful for image classification tasks where the exact location of the object in the image is not important. Batch normalization is a technique that normalizes the activations of the layers in a neural network. It is applied to the output of a layer before applying the activation function. Batch normalization helps to speed up the training of the model, reduce the sensitivity to the initialization of the weights, and improve the generalization of the model (Ioffe and Szegedy 2015).
The Adam optimizer is a popular optimization algorithm for training neural networks. It is an extension of the Stochastic Gradient Descent (SGD) algorithm that uses the Adaptive Moment Estimation (ADAM) to update the weights of the model. The Adam optimizer is known for its ability to converge quickly and robustly to the optimal solution (Kingma and Ba 2014). The learning rate of 1e-4 was chosen to ensure a stable training process and avoid the possibility of overshooting the optimal solution. The cosine learning rate scheduler is used to adjust the learning rate during the training process (Xie et al., 2018). The scheduler used in this case has three parameters: initial learning rate, decay steps, and alpha. The initial learning rate is set to 1e-4, which is a relatively high value for the head of the network. The decay steps are calculated using the following formula:

$$decay\ steps = \left(\frac{training\ samples}{batch\ size}\right) \cdot epochs \quad (1)$$

This ensures that the learning rate is adjusted appropriately during the training process. The alpha parameter is set to 1e-2, which is used to control the rate of decay of the learning rate. Finally, the batch size was set to the maximum value that can be accommodated by a graphics card with a VRAM of 24 GB. The batch size is an important parameter in training neural networks as it affects the stability and speed of the training process. As illustrated in Table 1, the study was conducted using a fixed area of 300m x 300m.

Table 1: Fixed area of 300 m x 300 m, results are subject to a high level of uncertainty due to only 25 classes presence*

	params [mln]	F1* [%]	acc* [%]	mAP* [%]	batch	epoch
B0	5.3	100	100	100	8	56
B1	7.8	100	100	100	4	73
B2	9.2	100	1000	100	3	29
B3	12.0	100	100	100	2	57

During this study, the number of classes has been set to 25, which directly results from the chosen patch size. Transfer learning approach has been utilized, by setting up a model with parameter weights="imagenet". The last column in the table shows the number of epochs it took for the F1 metric to stabilize, which we chose for its high stability and tolerance on an imbalanced data set. Model input size was set to 192 pixels x 192 pixels, which is lesser than actual input size in pixels. The image was downsampled to presumed input size. The chosen model for further experimentation due to the earliest stabilization of results on the test set and the lowest epoch loss value is the EfficientNet-B2 model. This model exhibits the least signs of overfitting.
Possible applications of such a large coverage area as a single class, require either exceeding the permitted flight altitude for civil operations or using image stitching from multiple images captured during flights therefore further experiments aimed to simulate a reduced flight altitude. The input image size was proportionally reduced, which resulted from the decreasing size of individual objects. The number of classes increased proportionally, as shown in Table 2.

Table 2: Results for EfficientNetB2 with changing size of input area.

patch [m]	classes [-]	F1 [%]	acc [%]	mAP [%]	batch [-]	epoch [-]
300	25	100	100	100	3	29
250	36	100	100	100	5	29
200	64	95,8	96,9	98,4	7	30
150	121	93,5	95,1	98,2	7	71
100	289	75,7	79,2	86,7	16	80

The experimental results summarized in the table clearly indicate that the input image size has a significant effect on the performance of the neural network. A reduction in patch size leads to a measurable degradation in classification accuracy, accompanied by an increase in false positive detections. This behavior suggests that smaller input dimensions limit the amount of discriminative spatial information available to the model, which negatively impacts its ability to correctly differentiate between classes. This was expected, as the model found it more challenging to distinguish between different classes with less number of discriminative features in classified area.
A patch size corresponding to a 150 m × 150 m ground area was selected to satisfy the required accuracy criteria, as

demonstrated in Table 2. Using 96 × 96 × 3 input size and a batch size of 1, the EfficientNet-B3 model achieved a mean inference time of 15.7 ms per patch, corresponding to an effective processing rate of 63.7 FPS. The inference was performed on an NVIDIA RTX 3090 GPU, with the model loading onto the GPU taking approximately 870 ms. This processing speed is well above typical UAV frame rates, indicating that the model is suitable for near-real-time inference on UAV imagery.

CONCLUSIONS

The main objective of this work was to evaluate the impact of input image resolution on the performance of a neural network-based multi-class classification model and to assess the model's robustness under varying input conditions simulating different UAV flight altitudes. The task was formulated as a multi-class image classification problem, where each input patch is assigned to one of several predefined object classes. This objective was achieved through a series of controlled experiments in which different input patch sizes were analyzed and compared in terms of classification accuracy, F1-score and mAP. Additionally, different configurations and sizes of the EfficientNet architecture were evaluated to assess their influence on model performance.

The results of these experiments demonstrated that reducing the input image size had a significant impact on the model's performance. Overall, these experiments highlighted the importance of considering the impact of input image size on the performance of object classification models. The observed drastic decrease in classification quality between object sizes of 150×150 and 100×100 meters was considered a strong indication for adopting 100×100 meters as the minimum acceptable patch (object) size. This result also emphasizes the necessity of applying image stitching techniques if further reduction of the acquisition altitude, and consequently object size below 100×100 meters, is desired. By simulating the effects of reduced flight altitude, we were able to identify potential weaknesses in our model and take steps to address them. In future works, the presented simulation results will be validated on edge computing hardware, specifically the NVIDIA Jetson platform, to assess real-time performance under operational constraints. In addition, a dataset comprising RGB images will be collected during real UAV flights over the analyzed area, together with synchronized IMU sensor data, to enable validation of the simulation results under real-world operational conditions.

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A PARTICLE SWARM OPTIMIZATION ALGORITHM FOR MOBILE ROBOT PATH PLANNING WITH HERMITE SMOOTHING

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KEYWORDS
Mobile Robot, PSO algorithm, Optimization, Trajectory, Hermite Polynomials

ABSTRACT

This paper presents a hybrid path planning method that combines Particle Swarm Optimization (PSO) with Hermite spline smoothing for generating collision-free and smooth trajectories in two-dimensional environments. PSO is employed to optimize intermediate waypoints between a start and goal position while minimizing path length and avoiding obstacles. The resulting piecewise linear path is then refined using cubic Hermite interpolation to ensure smoothness and continuity suitable for robotic navigation. Simulation results demonstrate that the proposed method produces shorter and smoother paths compared to unsmoothed PSO solutions, making it suitable for real-time applications in mobile robotics and autonomous systems.

INTRODUCTION

Path planning is a fundamental problem in robotics and autonomous systems, requiring the generation of a collision-free trajectory between a start and a goal position. Classical approaches such as graph-based and sampling-based algorithms provide feasible solutions but often produce non-smooth paths that are not suitable for real-world motion execution (LaValle 2006). Smoothness is a critical requirement in practical applications, as abrupt changes in direction may lead to increased energy consumption, mechanical stress, and infeasible motion for nonholonomic systems.

Particle Swarm Optimization (PSO), originally introduced by Kennedy and Eberhart (Kennedy and Eberhart 1995), has emerged as an effective metaheuristic for solving nonlinear optimization problems, including robot path planning. PSO-based methods represent candidate solutions as a set of waypoints and iteratively refine them to minimize a cost function. Several studies have demonstrated the effectiveness of PSO in generating collision-free paths in static environments (Alam et al. 2020). However, due to the discrete nature of waypoint representation, PSO-generated paths often contain sharp turns and irregular geometries.

To address this limitation, various smoothing techniques have been proposed. Spline-based approaches, including cubic splines, Bézier curves, and NURBS, are widely used to generate continuous and smooth trajectories (Piegl and Tiller 1997). For instance, PSO combined with cubic spline interpolation has been shown to improve path continuity and reduce trajectory length (Zhang et al. 2020). Similarly, Bézier curve-based smoothing methods have been applied to reduce excessive turning and improve motion feasibility in multi-robot systems (Chen et al. 2024, Sun et al. 2021). Despite these advancements, many existing approaches rely on post-processing techniques without explicitly incorporating smoothness into the optimization objective.

In this paper, we propose a hybrid path planning framework that combines PSO-based waypoint optimization with an explicit turning angle penalty and Hermite spline smoothing. Unlike conventional approaches, the proposed method incorporates a curvature-related term directly into the fitness function, enabling the generation of smoother paths prior to post-processing. Subsequently, cubic Hermite splines are applied to ensure C^1 continuity and further refine the trajectory. The effectiveness of the proposed method is demonstrated through simulation experiments, showing improved smoothness and competitive path length compared to baseline approaches.

Unlike existing works that rely on implicit smoothness through spline interpolation, this study explicitly incorporates a turning angle penalty into the optimization objective, leading to improved geometric smoothness prior to post-processing. The main contributions of this work are summarized as follows:

- integration of a turning angle penalty into the PSO fitness function for explicit curvature control,
- application of Hermite spline interpolation for smooth trajectory generation,
- quantitative evaluation of path quality in terms of length, curvature, and smoothness.

MODEL OF MOBILE ROBOT

We assume that the input signals are represented by the left and the right wheel speeds ω_L and ω_R but output signals are expressed by linear and angular velocity v, and ω, respectively (Khan et al. 2023). The kinematic scheme is shown on Figure 1, where R is a radius of wheel, L is a distance between left and right wheel.

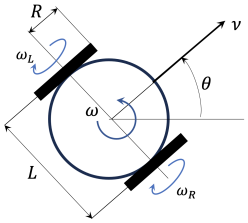

Figure 1: The Kinematic Scheme of Mobile Robot

Then, the forward kinematics can be defined as follows

$$v = \frac{R}{2}\left(\omega_R + \omega_L\right), \quad \omega = \frac{R}{L}\left(\omega_R - \omega_L\right). \tag{1}$$

and the inverse kinematics is expressed in the following form

$$\omega_L = \frac{1}{R}\left(v - \frac{\omega L}{2}\right), \quad \omega_R = \frac{1}{R}\left(v + \frac{\omega L}{2}\right). \tag{2}$$

PROPOSAL OF A HYBRID METHOD BASED ON PSO AND HERMITE CURVES

PSO algorithm based on evolutionary populations such as schools of fish and bird flocks. It was presented in (Kennedy and Eberhart 1995) where investigated abovementioned swarm behavior. The PSO contains a swarm of particles moving in a $\mathcal{S}$ dimensional space where a certain quality measure are being optimized. A position and velocity of each particle is represented by the vectors $X_i = [x_{i1}\ x_{i2}\ \ldots\ x_{i\mathcal{S}}]$ and $V_i = [v_{i1}\ v_{i2}\ \ldots\ v_{i\mathcal{S}}]$, respectively. Moreover, the velocity vector is bounded by the maximum and minimum values vector. The modified velocity and position of each particle can be manipulated according to the following equations

$$v_{ij}(k+1) = wv_{ij}(k) + c_1 r_1\left(p_{ij} - x_{ij}(k)\right) + \\ + c_2 r_2\left(p_{gj} - x_{ij}(k)\right),$$

$$x_{ij}(k+1) = x_{ij}(k) + v_{ij}(k+1), \qquad j = 1, \ldots, \mathcal{S},$$

where w is a inertia weight, c_1, c_2 are the acceleration constants, r_1 and r_2 are random numbers in the range of [0, 1].

Hermite Spline Smoothing

The use of Hermit curves to represent the path of a mobile robot stems primarily from their geometric properties. Hermit curves are characterized by reduced susceptibility to oscillations between control points while maintaining smoothness. Recently, there has been increased interest in their use in motion planning problems (Pan et al. 2025, Othman et al. 2021, Chen et al. 2023).
For consecutive waypoints $(\mathbf{p}_i, \mathbf{p}_{i+1})$, a cubic Hermite segment is defined as:

$$\mathbf{H}(t) = h_{00}(t)\mathbf{p}_i + h_{10}(t)\mathbf{m}_i + h_{01}(t)\mathbf{p}_{i+1} + h_{11}(t)\mathbf{m}_{i+1} \tag{3}$$

where basis functions are:

$$h_{00}(t) = 2t^3 - 3t^2 + 1 \tag{4}$$
$$h_{10}(t) = t^3 - 2t^2 + t \tag{5}$$
$$h_{01}(t) = -2t^3 + 3t^2 \tag{6}$$
$$h_{11}(t) = t^3 - t^2 \tag{7}$$

Tangents are approximated using finite differences:

$$\mathbf{m}_i = \frac{\mathbf{p}_{i+1} - \mathbf{p}_{i-1}}{2} \tag{8}$$

This guarantees C^1 continuity.

Fitness Function

Let the robot operate in a bounded 2D workspace:

$$\mathcal{W} \subset \mathbb{R}^2 \tag{9}$$

Given:

- Start position $\mathbf{x}_s$
- Goal position $\mathbf{x}_g$
- Set of obstacles $\mathcal{O}$

We seek a smooth trajectory:

$$\mathbf{x}(t) : [0, 1] \to \mathcal{W} \tag{10}$$

such that:

$$\mathbf{x}(0) = \mathbf{x}_s, \quad \mathbf{x}(1) = \mathbf{x}_g \tag{11}$$

while minimizing a composite cost function.
The cost function is defined as:

$$J = L + \lambda_1 P_{collision} + \lambda_2 P_{curvature} \tag{12}$$

Path length:

$$L = \sum_i \|\mathbf{x}_{i+1} - \mathbf{x}_i\| \tag{13}$$

Collision penalty:

$$P_{collision} = \sum_i \mathbb{I}(\mathbf{x}_i \in \mathcal{O}) \cdot \gamma \tag{14}$$

Curvature penalty:

$$P_{curvature} = \sum_i \theta_i^2 \tag{15}$$

Turning angle:

$$\theta_i = \cos^{-1}\left(\frac{\mathbf{v}_i \cdot \mathbf{v}_{i+1}}{\|\mathbf{v}_i\|\|\mathbf{v}_{i+1}\|}\right) \tag{16}$$

The vectors $\mathbf{v}_i$ and $\mathbf{v}_{i+1}$ represent consecutive path segments, and the angle between them quantifies the local turning behavior at each waypoint.
The Algorithm 1 presents the proposed hybrid path planning method with Hermit curve smoothing in pseudocode.

Algorithm 1 PSO-Based Path Planning with Hermite Smoothing

Require: Start S, Goal G, Obstacles O, number of waypoints N
Ensure: Smooth collision-free path P_{smooth}
Initialize swarm positions X_i and velocities V_i
Initialize personal bests $P_i \leftarrow X_i$
Initialize global best G^*
for iteration $t = 1$ to T **do**
  for each particle i **do**
    Construct full path $P_i = \{S, X_i, G\}$
    Evaluate fitness $f(P_i)$
    if $f(P_i) < f(P_i^{best})$ **then**
      $P_i^{best} \leftarrow X_i$
    end if
    if $f(P_i) < f(G^*)$ **then**
      $G^* \leftarrow X_i$
    end if
  end for
  Update velocities:

$$V_i \leftarrow wV_i + c_1 r_1 (P_i^{best} - X_i) + c_2 r_2 (G^* - X_i)$$

  Update positions:

$$X_i \leftarrow X_i + V_i$$

  Apply boundary constraints to X_i
end for
Construct optimal path $P^* = \{S, G^*, G\}$
Apply Hermite smoothing: $P_{smooth} \leftarrow \text{Smooth}(P^*)$
return P_{smooth}

RESULTS

All simulations were performed in Python environment. The following PSO algorithm parameters were assumed for the simulation experiments:

(i) a number of iterations is equal 150, a population size is 40, $c_1 = c_2 = 1.5$ and $w = 0.7$,

(ii) a number of iterations is equal 250, a population size is 100, $c_1 = c_2 = 1.5$ and $w = 0.7$,

(iii) a number of iterations is equal 150, a population size is 40, $c_1 = 1.5$, $c_2 = 2.0$ and $w = 0.7$.

(iv) a number of iterations is equal 250, a population size is 100, $c_1 = 1.5$, $c_2 = 2.0$ and $w = 0.7$.

For all simulations, the number of intermediate trajectory points was set to 5 and the number of Hermite curve segments was set to 20. The starting point was set to (2; 2) and the ending point to (8; 10). Finally, six simulation experiments were performed: one for (i) parameter set, two for (ii) set, one for (iii) set and two for (iv) set. The cost function values for each simulation were as follows:

- 84.357 for (i) parameter set,
- 84.345 for the first trial with parameters (ii) and 85.672 for the second trial with parameters (ii),
- 84.345 for (iii) parameter set,
- 84.330 for the first trial with parameters (iv) and 88.385 for the second trial with parameters (iv).

The simulation results demonstrate that the proposed hybrid path planning framework successfully combines the global search capability of Particle Swarm Optimization with the geometric refinement of Hermite spline interpolation. By representing candidate solutions as a finite set of intermediate waypoints, PSO effectively explores the feasible search space and identifies collision-free trajectories under obstacle constraints (see Figures 2-7).

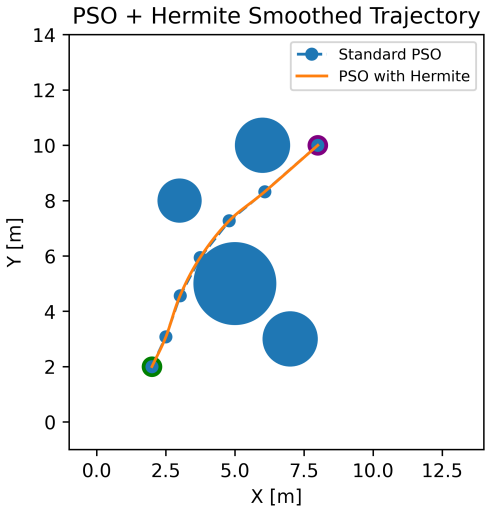

Figure 2: A Trajectory for (i) Parameter Set

A key observation from the experiments is that conventional PSO optimization based solely on path length

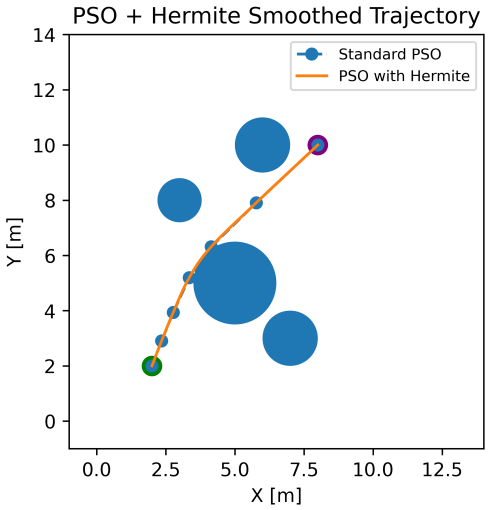

Figure 3: A Trajectory for the first Trial with (ii) Parameters

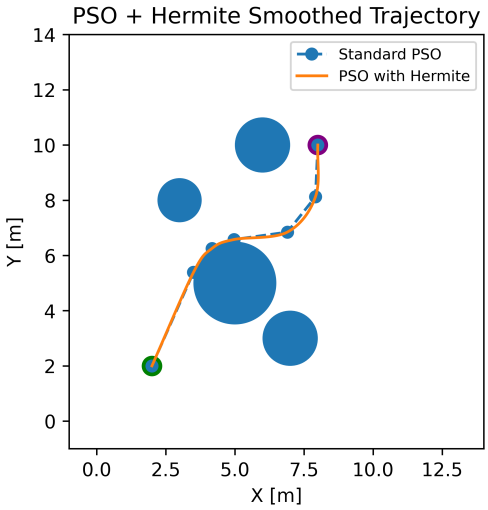

Figure 4: A Trajectory for the second Trial with (ii) Parameters

and obstacle avoidance often yields waypoint paths with abrupt directional changes. Such irregularities lead to high curvature values and poor motion feasibility. Such trajectories are shown in the Figures 4 and 7. To address this issue, the present work introduced an explicit turning angle penalty into the fitness formulation, thereby discouraging excessive directional deviations during the optimization process itself.

Quantitative analysis confirmed that the turning-angle-enhanced fitness function reduced the cumulative curvature of the generated raw path, while preserving near-optimal path length. Furthermore, the subsequent application of cubic Hermite spline smoothing significantly improved trajectory continuity by eliminating sharp corners and producing a continuously differentiable path. Although a slight increase in total trajectory length was observed after smoothing, this increase remained marginal compared to the substantial gain in smoothness and motion practicality.

Overall, the obtained results verify that explicit curvature penalization combined with Hermite post-smoothing provides a more balanced solution than conventional shortest-path-oriented PSO formulations.

CONCLUSION

This paper presented a hybrid trajectory planning method that integrates Particle Swarm Optimization, turning angle-based fitness penalization, and Hermite spline smoothing for mobile robot navigation in static two-dimensional environments. The proposed approach was motivated by the observation that standard PSO-based waypoint optimization, although effective in obstacle avoidance and path length minimization, frequently generates geometrically non-smooth trajectories with excessive turning behavior.

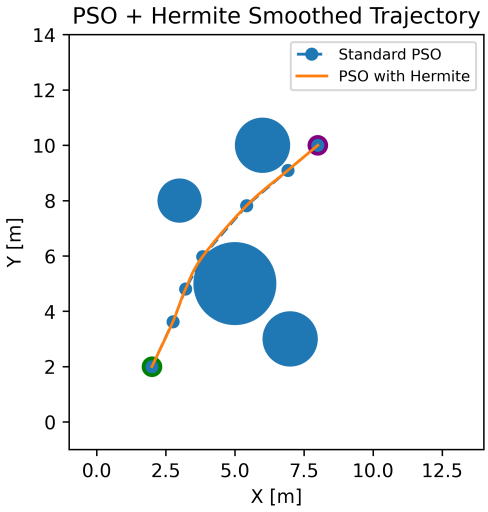

Figure 5: A Trajectory for (iii) Parameter Set

To overcome this limitation, a discrete turning angle functional was incorporated into the objective function to explicitly control local curvature during the optimization phase. This modification enabled the swarm to favor paths with smoother directional transitions even before post-processing. Subsequently, cubic Hermite spline interpolation was employed to transform the optimized waypoint sequence into a continuously differentiable trajectory suitable for practical robotic execution. Simulation results demonstrated that the proposed method achieves three simultaneous objectives: collision avoidance, competitive path length, and substantial smoothness improvement. Compared with raw PSO-generated trajectories, the hybrid method produced lower curvature, reduced directional discontinuities, and improved geometric feasibility with only minor compu-

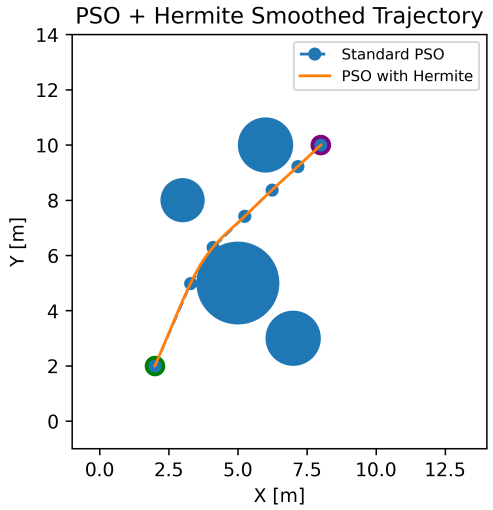

Figure 6: A Trajectory for the first Trial with (iv) Parameters

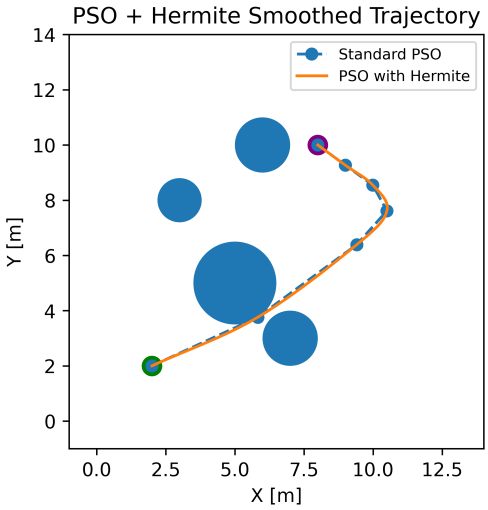

Figure 7: A Trajectory for the second Trial with (iv) Parameters

tational overhead.

The principal contribution of this study lies in the combination of explicit turning angle minimization with Hermite spline refinement, which provides a simple yet effective framework for generating robot-friendly trajectories without requiring complex dynamic models.

Future work will focus on extending the proposed formulation to dynamic obstacle environments, nonholonomic vehicle constraints, and three-dimensional autonomous navigation scenarios.

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Development of a Low-Cost Open-Hardware Throwable Mobile Robot for Remote Reconnaissance

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KEYWORDS
Throwable robot, mobile robot, reconnaissance robot, Raspberry Pi, 3D printing, open hardware, remote operation

ABSTRACT

This paper presents the design, implementation and validation of a low-cost open-hardware throwable mobile robot manufactured almost entirely using FDM (fused deposition modeling) 3D printing technology. The robot employs a Raspberry Pi Zero 2 W as the main controller together with a camera module, an inertial measurement unit (IMU), differential drive motors and a browser-based control interface.

INTRODUCTION

Modern military, police and rescue operations increasingly require rapid reconnaissance of hazardous environments before human personnel enter the area. Examples include hostage situations, bomb threats, collapsed buildings, urban combat and industrial accidents (Tadakuma 2009, Homeland 2016, Weissmann 2023). In these scenarios, remotely operated mobile robots significantly reduce the risk to human operators by providing visual information before intervention.

Among ground robotic platforms, throwable robots constitute a unique category designed to survive strong mechanical impacts resulting from being thrown through windows, doors or roof opening (Figure 1. After landing, the robot immediately begins transmitting video while remaining fully controllable by an operator (Czupryniak 2010). Modern rescue and law enforcement operations require the rapid acquisition of environmental data in high-risk zones. Throwable mobile platforms enable immediate visual reconnaissance before human personnel enter.

Despite their high operational performance, commercial tactical systems such as TRM (Lukasiewicz 2025) and Throwbot 2 (Recon 2026) (Figure 2) are characterized by prohibitive acquisition costs and proprietary hardware architectures, which limits their widespread deployment.

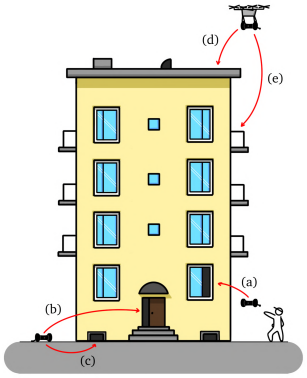

Figure 1: Examples of throwing robot use (based on (Tadakuma 2009))

The primary objective of this study is to design a low-cost, open-hardware inspection robot. Utilizing additive manufacturing (FDM) and readily available electronic components, the platform is engineered to withstand mechanical shock from a 2-meter drop, combining structural resilience with modularity, ease of repair, and accessibility for a broader range of research and operational applications.

Figure 2: Łukasiewicz Tactical Throwable Robot - PIAP (Lukasiewicz 2025) and Throwbot 2 by ReconRobotics (Recon 2026)

MECHANICAL DESIGN

The primary objective of the mechanical design was to develop a lightweight yet impact-resistant structure

capable of surviving repeated throws and drops while maintaining full operational capability. To achieve this goal, a modular architecture was adopted, allowing individual components to be easily replaced without redesigning the entire robot.
The complete mechanical structure was designed using Autodesk Fusion CAD software. All structural components were manufactured using FDM, significantly reducing production costs while enabling rapid prototyping and iterative development.
The robot employs a differential-drive configuration consisting of two independently driven wheels and a rear stabilizing tail (Figure 3). This arrangement provides excellent maneuverability while minimizing mechanical complexity. Differential drive also enables zero-radius turning, which is advantageous when operating in confined indoor environments. The kinematics of differential drive robots and their equations are widely discussed in the literature, and the most important formulas for the angular and rotational speed of the robot we can find in (Ibrahim 2023). We can see that if the wheel speeds are equal in magnitude but opposite in direction, the two-wheeled mobile robot will rotate in place around its own axis.
Material selection played a crucial role in achieving impact resistance. PLA Tough provides higher impact resistance than conventional PLA while preserving good dimensional accuracy (Naveed 2021). TPU was incorporated into the outer protective components to absorb kinetic energy during landing and reduce stress transmitted to the internal electronics (Iftekar 2023). The modular body consists of the following major subsystems: internal supporting frame, protective outer shell, wheel assemblies, rear stabilizing support, battery compartment,electronic mounting brackets. This architecture simplifies maintenance while improving the survivability of the robot after repeated mechanical impacts. Figure 3 shows the finished robot and his interior with the electronics installed. The vision system camera is visible on the hatch.

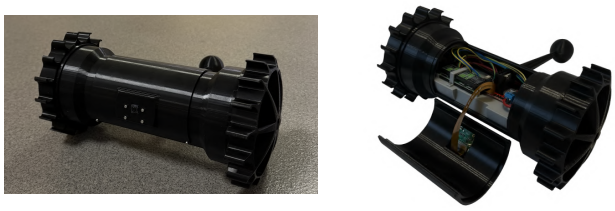

Figure 3: Finished robot exterior and internal electronics

ELECTRONIC ARCHITECTURE

The electronic system was designed using commercially available components selected according to four principal criteria: compact dimensions, low power consumption, sufficient computing performance, low overall cost. The Raspberry Pi Zero 2 W was selected as the central computing platform. Its quad-core ARM Cortex-A53 processor provides adequate computational capability for simultaneous video streaming, sensor acquisition and motor control while maintaining a compact footprint suitable for the robot enclosure. The hardware architecture (Figure 4) consists of: Raspberry Pi Zero 2W, built-in WiFi 2.4GHz and Bluetooth 4.2, Raspberry Pi Camera HD v2 8MPx, DC gear motors N20-BT43 micro, motor driver L298N, lithium battery pack 2200mAh, 2S, 7.4V, IMU Adafruit ICM-20948, 9 DoF. The camera continuously transmits live video to the operator through the wireless network (Figure 4). Motor control is achieved using pulse-width modulation (PWM), enabling smooth speed regulation of both drive wheels. An

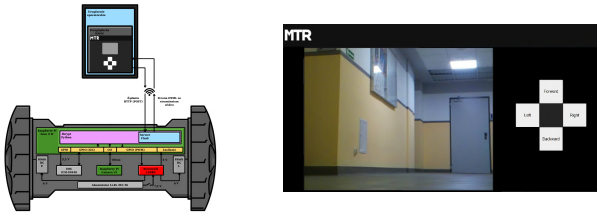

Figure 4: System operation diagram and a preview from the robot-mounted camera

inertial measurement unit (IMU) continuously measures robot orientation (Jaskot 2010, Norhafizan 2013). The acquired data are used to compensate camera orientation and improve operator situational awareness after landing, regardless of the robot's final orientation.

SOFTWARE ARCHITECTURE

The software was implemented using a client-server architecture. All computational tasks are executed directly on the Raspberry Pi, while the operator interacts with the robot through a standard web browser. The server application was developed in Python and is responsible for: motor control, sensor acquisition, camera management, wireless communication, video streaming. The graphical user interface was implemented using HTML, CSS and JavaScript. Consequently, no dedicated application installation is required on the operator device. The browser interface provides: real-time video transmission, robot driving controls, battery monitoring, connection status, automatic image orientation correction.
Sensor readings from the IMU are continuously processed to determine the robot orientation after landing (Bieda 2016). When necessary, the displayed video is automatically rotated and driving commands are adjusted accordingly. This feature eliminates operator confusion resulting from arbitrary landing orientations and significantly improves usability during reconnais-

sance missions. The adopted architecture provides a flexible and platform-independent control solution while maintaining low hardware requirements and straightforward deployment.

EXPERIMENTAL VALIDATION

The experiments focused on validating the structural durability, driving capability, wireless communication and video transmission following repeated impacts.
The functional evaluation verified all major subsystems before and after mechanical impact tests. The following functions were examined: wireless connection establishment, browser-based remote control, live video streaming, differential drive control, IMU data acquisition, automatic image orientation correction, battery-powered autonomous operation.
The robot successfully completed all functional tests. No communication failures or software instability were observed during normal operation. Video latency remained sufficiently low for manual navigation inside buildings, allowing real-time reconnaissance.
Mechanical validation consisted of repeated free-fall experiments from progressively increasing heights up to 2 m. The robot was dropped onto hard surfaces in multiple orientations to simulate realistic deployment conditions encountered during tactical operations. Each test was followed by a complete functional inspection. The following criteria were evaluated: structural integrity, wheel alignment, drivetrain functionality, camera operation, wireless communication, system boot reliability.

CONCLUSIONS

This paper presented the development of a low-cost open-hardware throwable mobile robot intended for remote reconnaissance in hazardous environments. The proposed platform combines additive manufacturing, commercially available electronics and web-based control software into an affordable reconnaissance system. Experimental evaluation demonstrated that the robot successfully fulfills all principal design objectives. The use of PLA Tough together with TPU proved to be an effective solution for balancing rigidity and impact resistance. Compared with commercial tactical robots, the proposed platform offers significantly lower manufacturing cost while retaining the essential functionality required for reconnaissance missions. Furthermore, the open architecture facilitates future modifications and educational applications. The total cost of the project is approximately 150 dollars.

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Deep Reinforcement Learning vs. PID Control: A Comparison of Autonomous Navigation of a Blimp in 3D Environment

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KEYWORDS

Blimp navigation, Deep Reinforcement Learning, RL, Obstacle avoidance, autonomous airship, sim-to-real, PID control.

ABSTRACT

This paper presents a comparative study of two control strategies for autonomous waypoint navigation of an outdoor blimp: a hand-tuned PID controller with reactive obstacle avoidance and a deep reinforcement learning (DRL) agent trained using Proximal Policy Optimization (PPO). Both methods achieve a 100% success rate in a 3D environment populated with axis-aligned obstacles and monitored via horizontal and vertical lidars. However, the DRL agent learns smoother, significantly shorter trajectories by proactively exploiting altitude to fly over sub-ceiling obstacles (243 vs. 427 steps, 25.45 m vs. 31.91 m path length). We explicitly analyze model limitations, sensor-hardware mismatches, and baseline design choices to contextualize these findings for future sim-to-real transfer.

INTRODUCTION

Unmanned Aerial Vehicles (UAVs) are deployed across environmental monitoring, disaster management, and inspection tasks. While rotary- and fixed-wing platforms suffer from limited endurance, Lighter-Than-Air (LTA) platforms such as autonomous blimps offer unique advantages for long-duration missions due to low power consumption and inherent physical safety (Bhat et al. 2024; Liu, Price, et al. 2021; Wang et al. 2026). However, blimp navigation is complicated by non-linear dynamics, buoyancy variations, and underactuation (Boase, Miah, and Gueaieb 2026; Fan et al. 2025). Efficient 3D obstacle avoidance requires handling high-dimensional sensor data under strict motion constraints (Li et al. 2023).

LITERATURE REVIEW

Classical PID controllers are simple and widely used (Fan et al. 2025), but manual gain tuning for non-linear LTA dynamics is challenging and often leads to suboptimal planar detours in dense 3D spaces (Wang et al. 2026). Recent reviews document the paradigm shift toward learning-based control for dirigible airships (Bhat et al. 2024; Boase, Miah, and Gueaieb 2026). Deep Reinforcement Learning (DRL) algorithms such as PPO (Schulman et al. 2017) and SAC (Fan et al. 2025) enable continuous adaptive trajectory generation. Advanced hybrid frameworks combining MPC, DRL, and RRT* (Sarathi et al. 2025), as well as bi-level RL architectures (Wang et al. 2026), have further improved tracking accuracy. Key challenges include bridging the sim-to-real gap via domain randomization and sensor fusion (Li et al. 2023; Liu, Singh, and Ahmad 2024; Neelesh 2025).

METHODOLOGY

Simulation Environment & Kinematic Model

We model a 3D workspace of dimensions $L_x \times L_y \times L_z = 10 \times 10 \times 5$ m containing static axis-aligned obstacles. The underactuated blimp state vector $\mathbf{s} = (x, y, z, \theta)^\top$ evolves according to continuous-time kinematic dynamics:

$$\dot{x} = v_f \cos\theta, \quad \dot{y} = v_f \sin\theta, \quad \dot{z} = v_v, \quad \dot{\theta} = \omega, \qquad (1)$$

where $v_f \geq 0$ is forward speed, v_v is vertical speed, and ω is yaw rate. Discrete-time integration uses Euler steps with $\Delta t = 0.1$ s.

Sensor Model & Mission Definition

The blimp carries two virtual lidar systems:

- **Horizontal lidar**: 24 rays covering 360° (15° resolution).
- **Vertical lidar**: 2 rays pointing straight down and up.

Readings are normalized to $[0, 1]$ by maximum range ($\sqrt{L_x^2 + L_y^2}$ and L_z). The mission requires visiting $K = 4$ waypoints in sequence ($r_{\text{wp}} = 0.5$ m acceptance radius). Episodes terminate upon goal completion, collision ($d_{\min} < 0.5$ m), or timeout ($T_{\max} = 2000$). For DRL, rewards include an alive bonus (+0.1), target progress ($d_{\text{prev}} - d_{\text{curr}}$), intermediate target bonus (+20), final completion (+100), and collision penalty (−10).

Classical PID Controller Baseline

The hand-tuned PID baseline comprises decoupled loops for heading ($\omega = K_p^\theta e_\theta + K_d^\theta \frac{de_\theta}{dt}, K_p^\theta = 0.8, K_d^\theta = 0.2$), altitude ($v_v = 0.5 e_z$), and forward speed ($v_f = \min(0.5 d_{xy}, 1.0)$). Reactive horizontal avoidance adds a repulsive yaw term $\omega \leftarrow \omega + 2.0(1 - d/1.5)^2 e_{\text{avoid}}$ when $d_{\min}^H < 1.5$ m.

Baseline Design Note: The PID obstacle avoidance operates purely in the horizontal plane (xy), reflecting standard 2D

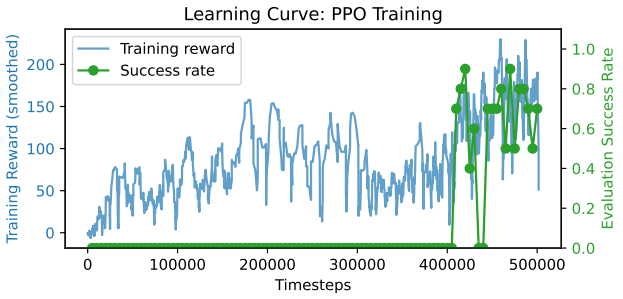

Figure 1: Episode reward and success rate during PPO agent training.

reactive industrial baselines to evaluate how unconstrained DRL exploits 3D space.

Deep Reinforcement Learning Controller

Formulated as an MDP with a 29D observation vector $\mathbf{o} = [\mathbf{h}, \mathbf{v}, \mathbf{g}]^\top$ (lidar scans and body-frame target vector $\mathbf{g}$). Actions $\mathbf{a} = [a_f, a_\omega, a_v]^\top$ map directly to scaled control inputs. An MLP policy network (2 layers of 64 units, tanh activations) was trained for 5×10^5 steps using PPO Schulman et al. 2017.

EXPERIMENTAL SETUP AND SCOPE ANALYSIS

The environment contains 3 vertical pillars of varying heights and 1 floating box obstacle. DRL training uses randomized initial poses across the map to ensure starting robustness (Fig. 1). Evaluation executes $N_{\text{eval}} = 50$ runs per controller from a fixed pose $(1, 1, 2.5, 0)$ to confirm execution determinism, integration stability, and system reliability under a deterministic policy.

Methodological Scope & Sim-to-Real Considerations

To ensure transparency regarding reviewers' observations:

1. *Kinematic vs. Dynamic Modeling*: The benchmark uses continuous kinematics without aerodynamic drag or wind turbulence; realistic flight requires dynamic physics simulators (AirSim/Isaac Gym).
2. *Generalization to Unseen Layouts*: Experiments were conducted on a single 3D course; multi-layout generalization remains a key future step.
3. *Hardware & Sensor Mismatch*: As shown in Fig. 2, our physical blimp prototype features a 2D horizontal lidar and IMU, lacking vertical lidar and 3D positioning.

RESULTS AND COMPARATIVE ANALYSIS

Performance metrics across 50 evaluation episodes are summarized in Table 1. Representative trajectories for PID and DRL are shown in Figures 3 and 4.

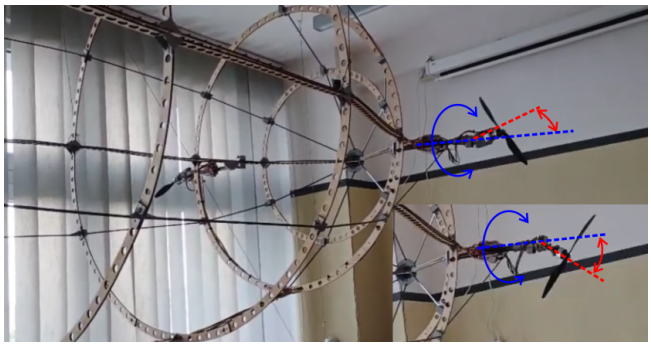

Figure 2: The real outdoor blimp platform during assembly.

Table 1: Performance comparison between PID and DRL (mean over 50 runs).

Metric	PID	DRL (PPO)
Success rate (%)	100.0	100.0
Avg. waypoints reached	4.0	4.0
Mean steps to success	427.0	243.0
Mean path length (m)	31.91	25.45
Optimal straight-line path (m)	23.25	
Mean min lidar clearance (m)	0.59	0.59

Both controllers achieve a 100% success rate. However, DRL completes the mission in nearly half the steps (243 vs. 427) and achieves a path length of 25.45 m (9.5% over the straight-line optimum of 23.25 m), compared to 31.91 m (37% over optimum) for PID.

Qualitatively, PID maintains a rigid altitude profile, making planar detours around pillars (Fig. 3). In contrast, DRL (Fig. 4) climbs directly over low pillars to minimize total 3D distance while keeping a safe 0.59 m obstacle clearance.

Computational & Power Trade-offs: PID requires zero training and $< 1\%$ CPU on Raspberry Pi 4, serving as a reliable safety fallback. DRL demands upfront training (5×10^5 steps, ≈ 1 h), but inference takes < 0.5 ms on embedded hardware while delivering superior 3D path optimization.

Quantitative Comparison and Qualitative Behaviour

Quantitatively, both controllers achieved a 100% success rate with an identical minimum obstacle clearance of 0.59 m, but the DRL agent completed the mission in almost half the steps (243 vs. 427) and achieved a path length of 25.45 m—only 9.5% over the straight-line optimum, compared to 31.91 m (37% over optimum) for PID. Qualitatively, while the PID controller maintained a rigid planar altitude profile leading to wide detours and heading oscillations near waypoints, the DRL policy proactively varied its altitude to fly directly over sub-ceiling obstacles. This proactive 3D reasoning eliminated overshoot near target acceptance radii and yielded significantly smoother trajectory transitions.

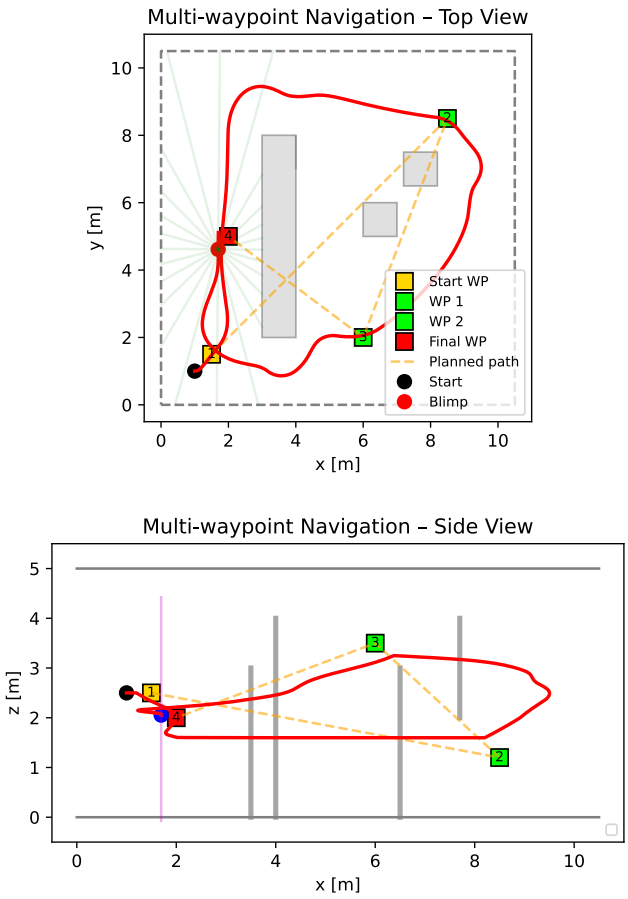

Figure 3: PID trajectory: planar detours around pillars in 2D.

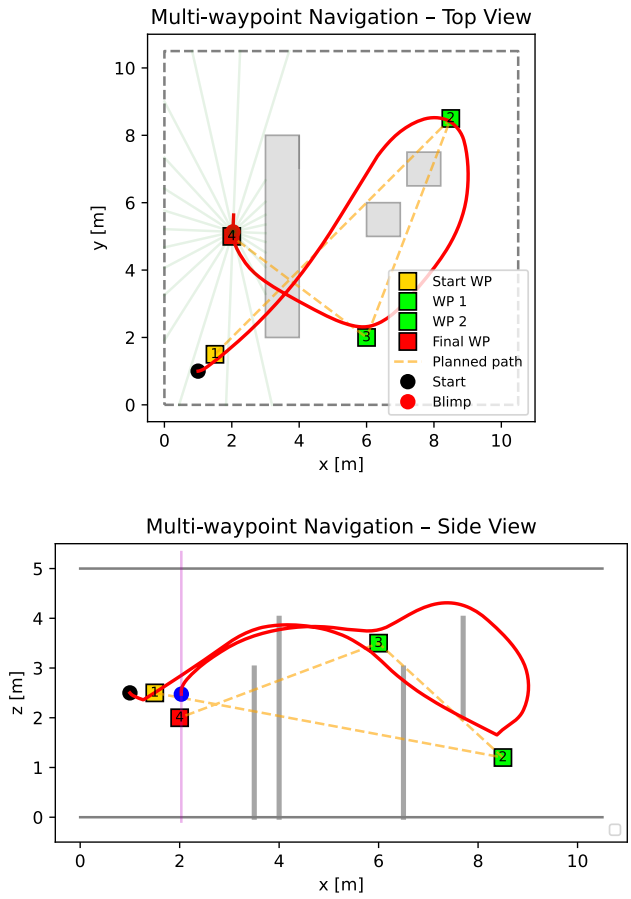

Figure 4: DRL trajectory: proactive 3D overflight of sub-ceiling pillars.

CONCLUSION AND FUTURE WORK

This study demonstrated that while PID and DRL achieve 100% navigation success, DRL yields a 43% step reduction and 20% path length reduction by leveraging vertical obstacle overflight. Future work will address policy transfer to physics-based dynamic simulators, evaluation on unseen obstacle configurations, and flight testing on the physical blimp platform.

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Engineering Simulation

Learning Strategy for XGBOOST based Energy Consumption Forecasting of an Injection Molding Machine

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KEYWORDS

Energy consumption forecasting, injection molding, XGBoost, learning curve, training-set size, prediction horizon, rolling-origin evaluation.

ABSTRACT

Energy-efficient monitoring of cyclic processes such as injection molding relies on accurate forecasting of electrical power consumption. This paper investigates the trade-offs between training-set size and prediction horizon for a recursive XGBoost model forecasting the instantaneous apparent power of an injection molding machine, using a run of 410 production cycles. Under a rolling-origin protocol, accuracy on this dataset saturates beyond approximately 125 training cycles, within 10% of the best achievable error over the tested training-set sizes, while the recursive error grows gradually rather than divergently, exceeding a 20% inflation threshold after about ten cycles and defining a practical recalibration cadence. A split-plot ANOVA shows that the two factors contribute approximately additively: their interaction explains only $\eta^2_G = 0.12$ of the variance, against 0.55 and 0.39 for the main effects, and becomes undetectable once under-trained models ($n < 25$) are excluded ($\eta^2_G = 0.05$, $p = 0.38$). A larger training window thus cannot compensate for long-horizon degradation, which is governed by recalibration rather than by the amount of training data. While these thresholds are specific to the machine studied, the proposed protocol is general and provides a reproducible basis for cost-effective, real-time energy monitoring in cyclic industrial settings.

INTRODUCTION

Energy efficiency and real-time process monitoring have become central concerns in modern manufacturing, where industrial activities account for a large share of global energy consumption. Within the polymer sector, injection molding is one of the most widely used shaping processes. Its electrical power demand dynamic carries rich information about the state and stability of the machine throughout production (Mianehrow and Abbasian 2017 [2]; Pan et al. 2025 [1]). Forecasting the instantaneous apparent power consumed by such a machine therefore provides a non-intrusive basis for real-time monitoring, anomaly detection and decision support, relying only on a three-phase power sensor rather than on a costly instrumentation of every subsystem. The injection molding process is intrinsically cyclic: each molded part results from a fixed sequence of stages (mold closing, injection, holding, plastification, cooling, ejection).

The power signal repeats a characteristic intra-cycle profile from one cycle to the next while still exhibiting inter-cycle variability. This structure makes the process well suited to supervised forecasting from lagged observations, and tree-based regressors such as XGBoost (Chen and Guestrin 2016 [3]) are attractive because they are fast to train, robust to noise and naturally compatible with the online retraining envisaged for an industrial deployment. In practice, a model embedded in a monitoring system is retrained as new production cycles accumulate. This raises an operational question that is rarely studied explicitly: how much historical data is needed to train the model? Collecting and storing data, and repeatedly retraining, all carry a cost, so it is valuable to know whether the accuracy keeps improving with the number of available cycles or whether it saturates beyond a certain point. A closely related question concerns the forecasting horizon: once a model is trained, how many cycles ahead can be predicted recursively before the prediction error justifies recalibration?

To the best of our knowledge, prior studies have used learning curves to size the training set for energy forecasting, but we found no prior study that explicitly quantifies whether training-set size and forecasting horizon interact under recursive multi-step forecasting in cyclic industrial processes.

This paper addresses both questions for the apparent power consumption of an injection molding machine. Our working hypothesis is that, for a cyclic process, predictive accuracy improves with the amount of historical training data only up to a learning threshold, beyond which additional training cycles yield diminishing returns. The contributions of this work are threefold. First, we formulate a reproducible, leakage-free protocol that separates the effect of the training-set size from that of the prediction horizon and averages each accuracy estimate over several rolling test origins, ensuring that the reported curves do not depend on any single test segment. Second, we introduce a criterion for selecting the optimal training size based on a tolerance to the best achievable error. Third, we derive a practical recalibration rule from the growth of the recursive error with the horizon, and, using a split-plot ANOVA, show that training-set size and horizon contribute approximately additively, so that data sizing and recalibration scheduling can, to a first approximation, be treated as separate operational decisions.

LITERATURE REVIEW

Machine learning forecasters are sensitive to the amount of training data: a small amount of training samples yields unstable models [4], whereas collecting, storing and repeatedly retraining on excess data inflate cost without improving accuracy [5, 6]. Identifying the smallest training set that attains near-optimal performance is therefore an operational concern. Existing research work formalizes the dependence of learning curves on training-set size: Viering and Loog [7] review the typically saturating shapes such learning-curves, and Mohr and van Rijn [8] survey the use of learning-curves for decision making on the data acquisition process and early stopping. Beyond forecasting, these curves have been used to quantify the amount of data a model actually needs: Ning et al. [9] train SegNet and U-Net on increasing portions of a high-resolution land-cover map and find that the benefit of additional data is uneven, some classes keep improving while others saturate early.

Closest to our work, Giola et al. [10] use learning curves to estimate the appropriate amount of training data for energy load forecasting, evaluating XGBoost among other models. In the specific setting of injection molding, Huang et al. [11] forecast process energy consumption with a Rolling-Learning Informer that is continuously retrained as new data arrive, outperforming recurrent and convolutional baselines; this confirms the relevance of energy forecasting for polymer processing, but relies on a heavy deep model and frequent updating rather than characterising the underlying data-size and horizon trade-offs.

Once a model is trained, a second question is how reliably its performance can be estimated and considering the forecasting horizon. In time series modeling, several strategies have been proposed to estimate forecasting performance. Among them, the *out-of-sample* (OOS), or holdout, approach reserves the most recent part of the series for testing and trains the model on the preceding observations, thereby preserving the temporal order of the data. Its simplest form uses a single fixed origin, but a single test block can yield a misleading estimate. Tashman [13] therefore recommends anchoring the test at several rolling origins and averaging over multiple test periods. Cerqueira et al. [14] report that blocked cross-validation is suitable for stationary series, whereas for non-stationary series the most accurate estimates are produced by out-of-sample methods, in particular holdout repeated over multiple testing periods. The trustworthy horizon is itself limited: Ben Taieb et al. [15] show that, under recursive multi-step forecasting, one-step errors accumulate and propagate as the horizon grows, largely independently of the base learner.

Deciding how often to retrain in response to this degradation is a cost-sensitive trade-off: Žliobaitė et al. [16] frame model updating as weighing the accuracy gained by retraining against its operational cost, and Zanotti [5] shows in a forecasting setting that less frequent retraining can preserve accuracy while cutting cost. The recalibration criterion we propose in this work makes this trade-off explicit for a cyclic industrial process.

MATERIALS AND METHODS

This section describes the materials and methods used to forecast the apparent power of an injection molding machine (Figure 1) using a machine learning model with a focus on the training and the recalibration strategy.

Figure 1: Injection Molding Machine

Process and Data

The data consist of the instantaneous apparent power S of an injection molding machine, sampled at 1 Hz during continuous production. The apparent power is computed from the three-phase active and reactive (a, b, c) components as:

$$S = \sqrt{P^2 + Q^2}. \tag{1}$$

With $P = P_a + P_b + P_c$ and $Q = Q_a + Q_b + Q_c$.

Each cycle comprises about 37 one-second samples. The analysed run yields several hundred consecutive cycles. The resulting series is a single univariate time series annotated with a cycle index. The dataset used in this study comprises 410 production cycles (Figure 2).

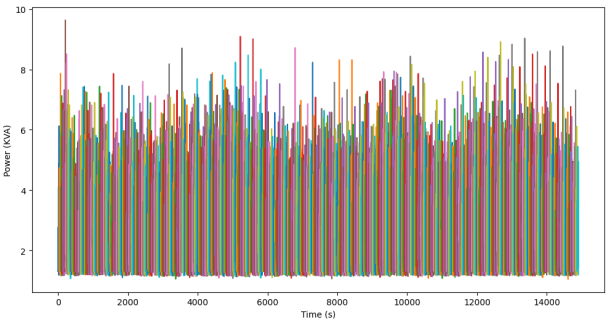

Figure 2: Instantaneous apparent Power Consumption during Production for each produced Part.

Problem Formulation

The base predictor is a one-step-ahead regression that is rolled out recursively. Let y_i denote the apparent power at sample i.

As expressed in the equation 2, the model forecasts the next sample from a window of past observations, where

L is the maximum lag and φ_i a set of statistical components (rolling mean, standard deviation, minimum, maximum, first and second differences, and an exponentially weighted mean) computed up to sample i.

$$\hat{y}_{i+1} = f(y_i, y_{i-1}, \ldots, y_{i-L+1};\ \varphi_i) \tag{2}$$

We set $L = 74$, i.e. two full cycles, so that the window spans the longest periodicity of interest. Inputs are scaled with a robust scaler fitted on the training window. At inference the model is applied recursively: each prediction is fed back as an input for the next step, and the model never observes the true test values. The objective of this work is to characterise how the forecasting accuracy depends on two operational quantities: the amount of historical data used for training, expressed as a number of production cycles n, and the length of the forecast, expressed as a prediction horizon h in cycles. Let f_n be the model trained on the n most recent cycles preceding a test segment, and let $e(n,h) = \mathrm{MAE}(f_n, h)$ be its instantaneous error when forecasting h cycles ahead recursively. The study addresses two questions:

Question 1 For a fixed reference horizon h_0, we search for the smallest training size beyond which accuracy no longer improves appreciably, i.e. a *learning threshold* n^* such that $e(n,h_0)$ has effectively plateaued for $n \geq n^*$.

Question 2 For a model trained at that threshold, we examine how the error grows with the horizon, $h \mapsto e(n^*,h)$, to determine how far ahead the model can forecast before recalibration becomes necessary.

Forecasting Model and Baselines

The forecast model considered in this work is XGBoost, a scalable gradient-boosting implementation of regression trees (Chen and Guestrin 2016 [3]). The model is an additive ensemble of K trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i) \tag{3}$$

It is built sequentially so that each new tree corrects the residual errors of the current ensemble while a regularisation term penalises tree complexity to limit overfitting. We use 1500 trees, a maximum depth of 6, a learning rate of 0.05, and column/row subsampling of 0.8; features are robustly scaled on the training window only, and an optional cycle-wise early stopping fixes the number of trees before refitting on the full window. A baseline is evaluated on the same test segment to provide a fair lower bound. The *seasonal naive* baseline repeats the last complete training cycle, so the forecast at each phase of the cycle equals the value observed one cycle ($m = 37$ samples) earlier:

$$\hat{y}^{\mathrm{Naive}}_{i+h} = y_{i+h-m} \tag{4}$$

where i is the last training sample, $h \geq 1$ the forecast step and $m = 37$ the cycle length.

Factor Interaction Analysis

To assess whether the training-set size and the prediction horizon interact, their joint effect on the instantaneous error is analysed with a split-plot analysis of variance (ANOVA) whose experimental unit is the test segment. At a given rolling origin all training sizes are scored against the same segment, so their errors are correlated, whereas each horizon is evaluated on its own segments; the training size is therefore a within-segment factor and the horizon a between-segment factor. The four random seeds, which only reflect subsampling randomness, are averaged within each segment, leaving four segments per horizon, and the error is modelled as

$$e_{ijk} = \mu + \alpha_i + \pi_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk} \tag{5}$$

where e_{ijk} is the seed-averaged error at horizon level i, training-size level j and segment k; μ is the overall mean; α_i and β_j are the horizon and training-size main effects; $\pi_{k(i)}$ is the effect of segment k nested within horizon i; $(\alpha\beta)_{ij}$ is their interaction; and ε_{ijk} is the training-size $\times$ segment residual. The horizon is tested against $\pi_{k(i)}$ and the training size and the interaction against ε_{ijk}, with a Greenhouse–Geisser correction for departures from sphericity. As statistical significance alone can be driven by the number of replicates, the practical importance of each term is assessed through the generalised effect size $\eta^2_G = \mathrm{SS}_{\mathrm{effect}}/(\mathrm{SS}_{\mathrm{effect}} + \mathrm{SS}_{\pi} + \mathrm{SS}_{\varepsilon})$, appropriate for designs with repeated measures [12], rather than through its p-value, where each sum of squares SS_X measures the variability attributed to source X. Approximate additivity between the training size and the prediction horizon is supported when the interaction accounts for a small fraction of the variance relative to the two main effects.

Experimental Protocol

The two questions of interest are deliberately separated into two experiments, which rely on two distinct evaluation schemes.

Training-size axis (learning curve)

The number of training cycles n is varied over $\{5, 10, 25, 50, 75, 100, \ldots, 300\}$. For every n, the test set is the *same* block of cycles, and the training window of n cycles ends immediately before it. Because all training sizes are evaluated against identical targets, any change in error reflects the amount of training data alone, isolating its effect. To remove the dependence of this comparison on one particular test block, the evaluation is performed under a *rolling-origin* [13] scheme (Figure 3): the test segment of the reference horizon is anchored at four test origins across the production run, and at each origin the model is retrained on the n cycles immediately preceding it. The error reported for a given n is the mean over these four origins and four random seeds (16 runs per training size).

Horizon axis (recalibration)

The horizon analysis follows a "train once, forecast further" logic that mirrors a conventional deployment: a single model is trained on one contiguous block of the threshold size (the first n^* cycles of the run) and then

rolled out recursively over increasing contiguous horizons $\{5, 10, \ldots, 40\}$ cycles, without any further retraining. This quantifies how the recursive error accumulates with the forecasting horizon. Because these two schemes differ in nature, their absolute error levels are *not directly comparable*: the rolling-origin learning curve reports an error averaged over 16 runs and short reference horizons, and therefore estimates the expected forecasting performance, whereas the train-once roll-out characterises a single deployment trajectory over a long horizon and yields systematically higher errors. We consequently use the rolling-origin curves to size the training set and the train-once roll-out to study horizon degradation, and we report the two families of figures separately. The joint response surface $e(n,h)$ is summarised as a heatmap whose cells are obtained with the same rolling-origin averaging, so that, within a row (a fixed horizon), all training sizes are scored against a shared set of test targets and the columns are directly comparable.

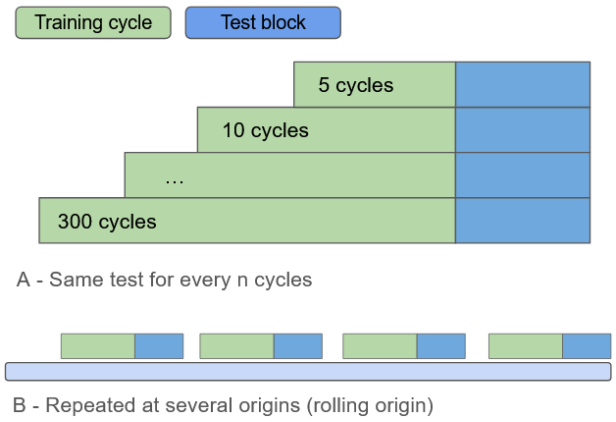

Figure 3: Rolling-origin Protocol for the Training-size.

Optimal Training Size

In this study, we define the optimal training size as the smallest number of cycles whose error is within a relative tolerance τ of the best error observed on the rolling-origin curve,

$$n^*(\tau) = \min\left\{ n : \mathrm{MAE}(n) \le (1+\tau)\,\mathrm{MAE}_{\mathrm{best}} \right\} \qquad (6)$$

with $\mathrm{MAE}_{\mathrm{best}} = \min_n \mathrm{MAE}(n)$. This criterion is less sensitive to local plateaus and irregular training-size spacing. The primary tolerance threshold adopted in this study is $\tau = 10\%$. For the forecast horizon axis, we define the recalibration threshold as the first horizon at which the error exceeds its minimum value by more than a relative tolerance ρ,

$$h_r(\rho) = \min\left\{ h : \mathrm{MAE}(h) > (1+\rho) \cdot \min_{h'} \mathrm{MAE}(h') \right\} \quad (7)$$

which translates directly into a practical retraining cadence. The reference tolerance adopted in this study is $\rho = 20\%$. A sensitivity analysis is performed to evaluate the robustness of h_r to the choice of ρ over the range $\rho \in [10\%, 50\%]$.

RESULTS AND DISCUSSION

Effect of Training Size and Horizon

Figure 4 reports the rolling-origin instantaneous error across the complete grid of training-set sizes and prediction horizons. Two monotonic trends emerge: along each row the error decreases with the number of training cycles and saturates towards the right, in agreement with the learning curve, while along each column it increases with the horizon, reflecting the accumulation of recursive prediction errors. The optimal operating region is therefore the lower-right corner, where a large training set and a short horizon yield approximately 0.29 kVA, against approximately 0.90 kVA in the least favourable cells that combine few training cycles with long horizons. Enlarging the training set thus shifts the entire horizon curve downward but does not suppress the intrinsic growth of the error with the horizon, indicating that the two factors contribute approximately additively. This is corroborated by a split-plot ANOVA with interaction (Table 1).

Over the full grid, both main effects dominate the response surface, the prediction horizon ($\eta^2_G = 0.55$) and the training-set size ($\eta^2_G = 0.39$), whereas their interaction, although statistically detectable ($p = 6.3 \times 10^{-3}$), accounts for only $\eta^2_G = 0.12$ of the variance, three to five times less than either main effect. Restricting the analysis to the saturated regime ($n \ge 25$) reinforces this conclusion: the interaction is no longer detectable ($p = 0.38$, $\eta^2_G = 0.05$), showing that the small residual coupling over the full grid originates almost entirely from the under-trained models ($n < 25$).

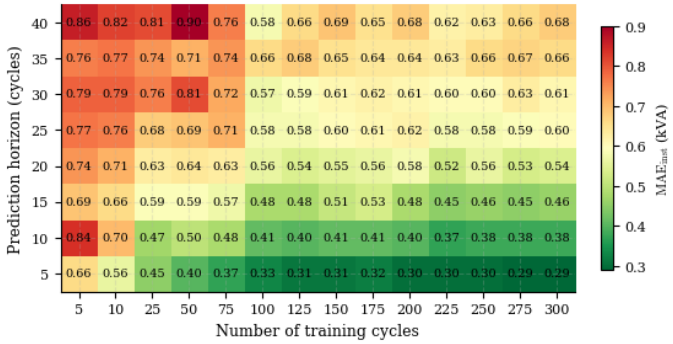

Figure 4: Instantaneous MAE (kVA) over the Grid of Training-set Sizes (columns) and Prediction Horizons (rows).

Table 1: Split-plot ANOVA of $\mathrm{MAE}_{\mathrm{inst}}$ over the training-size × horizon grid. Top: full grid; bottom: saturated regime ($n \ge 25$).

Source	η^2_G	F	df_1	df_2	ε_{GG}	p
Full grid (all n)						
Horizon	0.55	5.5	7	24	–	< 0.001
Training size	0.39	63.6	13	312	0.35	< 0.001
Interaction	0.12	1.93	91	312	0.35	0.006
Saturated regime ($n \ge 25$)						
Horizon	0.59	6.1	7	24	–	< 0.001
Training size	0.19	31.8	11	264	0.34	< 0.001
Interaction	0.05	1.08	77	264	0.34	0.38

Learning Curve and Training Threshold

Evaluated at the reference horizon of $h = 10$ cycles, the rolling-origin learning curve (Figure 5) exhibits the expected saturating shape, with the shaded band reporting the standard error over the 16 runs. At the smallest training size of five cycles the recursive forecast is poor ($\text{MAE}_{\text{inst}} \approx 0.95$ kVA); accuracy then improves sharply, reaching approximately 0.47 kVA by 25 cycles and 0.40 kVA near 100 cycles. Beyond 100–125 cycles the gains become marginal: the error decreases only slowly to a minimum of approximately 0.33 kVA near 250 cycles, the final hundred cycles yielding essentially no further improvement. Across the entire range, the model remains substantially below the seasonal-naive baseline (≈ 1.48 kVA). Applying the tolerance criterion defined in

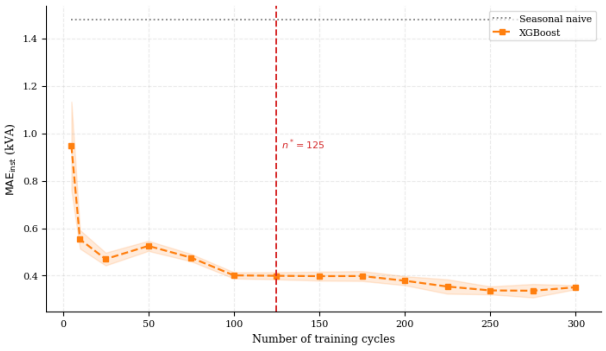

Figure 5: Instantaneous MAE versus number of training cycles at the reference horizon of $h = 10$. Shaded band: standard error over runs; dotted line; seasonal-naive baseline; the dashed line; learning threshold $n^* = 125$.

Equation (6) yields a learning threshold of $n^* = 125$ cycles at $\tau = 10\%$. The sensitivity of this threshold to the tolerance is reported in Table 2 and illustrated in Figure 6. A stringent tolerance ($\tau = 5\%$) raises the threshold to 225 cycles, the reference tolerance ($\tau = 10\%$) yields 125 cycles, and more permissive tolerances ($\tau = 15$–20%) lower it to 100 cycles. Over the entire tolerance range, n^* remains confined to the interval $[100, 225]$ cycles, all values lying on the saturated portion of the learning curve.

Table 2: Optimal training size as a function of the tolerance τ

τ	n^* (cycles)	MAE_{inst} (kVA)	gap vs. best
5%	225	0.34	+3.0%
10%	125	0.36	+9.1%
15%	100	0.37	+12.1%
20%	100	0.37	+12.1%

Forecasting Horizon and Recalibration

Once the training size has been fixed through the rolling-origin procedure, the horizon is analysed under a conventional time-series scheme that mirrors deployment: a single model is trained on the first 125 contiguous cycles of the production run and then applied recursively over increasing horizons, without further retraining (Figure 7).

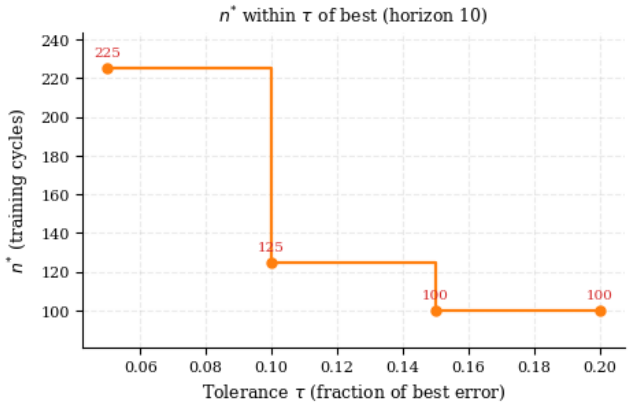

Figure 6: Training size n^* as a function of the tolerance τ.

The error starts at approximately 0.43 kVA at a five-cycle horizon, crosses the recalibration threshold of Equation (7) at the reference tolerance $\rho = 20\%$ near fifteen cycles, and rises smoothly to approximately 0.69 kVA at 40 cycles. This growth is gradual rather than divergent, a desirable property for a recursive forecaster, and the model remains substantially below the seasonal-naive baseline (≈ 1.8 kVA) across the entire horizon range.

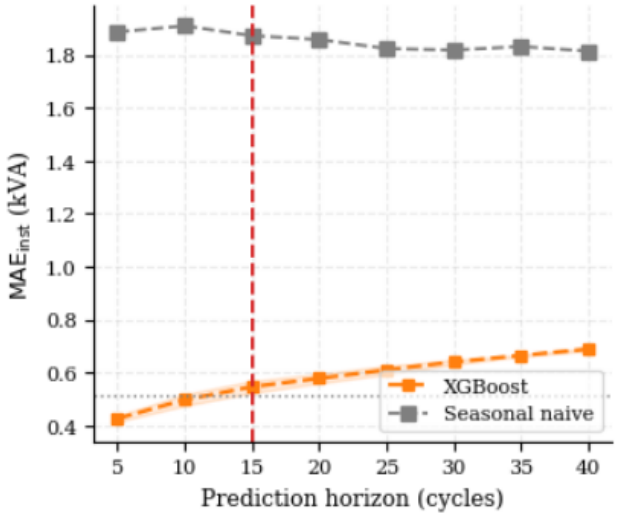

Figure 7: Instantaneous MAE versus prediction horizon for a model trained on the 125 cycle threshold, compared with the seasonal-naive baseline.

The sensitivity of the recalibration threshold to the tolerance ρ is reported in Table 3. It is computed from the rolling-origin errors of Figure 4 rather than from the single trajectory of Figure 7, so that the threshold is averaged over four test segments; the two estimates consequently differ by one grid step. The threshold is remarkably stable: $h_r = 10$ cycles for any tolerance between 10% and 25%, and 15 cycles between 30% and 40%; only at $\rho = 50\%$ does the smallest training size depart from this pattern, with $h_r = 20$ cycles. As horizons are evaluated in steps of five cycles, h_r is resolved to within five cycles. The recommended recalibration cadence is therefore ten to fifteen cycles, a range that is robust both to the choice of tolerance and to the evaluation scheme.

Table 3: Recalibration threshold h_r (cycles) as a function of the tolerance ρ and the training size n (rolling-origin errors).

ρ	$n = 100$	$n = 125$	$n = 225$
10%	10	10	10
15%	10	10	10
20%	10	10	10
25%	10	10	10
30%	15	15	15
40%	15	15	15
50%	20	15	15

Figure 8 illustrates a representative recursive forecast over the test cycles for a model trained on the threshold data. The model reproduces the repetitive intra-cycle profile and tracks the alternation of peaks and troughs, with the largest residuals occurring at the sharp transient peaks of the injection stage, which are inherently the hardest to predict. For this configuration the instantaneous error is $\mathrm{MAE}_{\mathrm{inst}} = 0.55$ kVA.

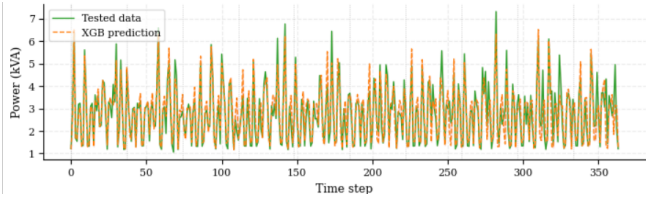

Figure 8: Recursive forecast on the test cycles for a model trained on the threshold data: measured power (green) versus prediction (orange).

Table 4 compares the three candidate operating points returned by the tolerance analysis. Accuracy improves monotonically with the training size, but the spread is small: moving from 100 to 225 cycles lowers the instantaneous MAE by only 0.06 kVA (0.58 $\rightarrow$ 0.52) and the RMSE by 0.08 kVA. This confirms that the 125-cycle threshold is a favourable compromise, retaining most of the accuracy of the largest window at roughly half the data-acquisition cost.

Table 4: Forecasting results for the three optimal training sizes n^* (train-once evaluation on the test cycles)

n^* (cycles)	$\mathrm{MAE}_{\mathrm{inst}}$ (kVA)	$\mathrm{RMSE}_{\mathrm{inst}}$ (kVA)
225	0.52	0.72
125	0.55	0.76
100	0.58	0.80

Discussion

The complete experimental campaign was carried out in approximately 313 minutes of computation time. The learning-curve analysis indicates that a training window of approximately 100–125 production cycles is sufficient to attain near-optimal forecasting accuracy, lying within the 10% tolerance of the best error observed on the grid. The lowest error is in fact reached with a somewhat larger window of about 225–250 cycles, yet the additional gain is marginal and operationally negligible, so that the two operating points differ only in a trade-off between data-acquisition effort and a residual accuracy improvement. Beyond this range the learning curve saturates, and the inclusion of further historical cycles provides no significant benefit, the split-plot ANOVA further shows that the training-set size and the prediction horizon contribute approximately additively to the forecasting error: their interaction accounts for only $\eta^2_G = 0.12$ of the variance, against 0.55 and 0.39 for the main effects, and becomes statistically undetectable ($\eta^2_G = 0.05$, $p = 0.38$) once the undertrained models ($n < 25$) are excluded.

Because the two factors barely interact, enlarging the training set lowers the overall error level but does not attenuate its growth with the horizon: adding more historical data therefore cannot improve long-horizon accuracy, which is instead governed by the recalibration strategy. This separation allows the training window to be sized and the recalibration schedule to be set as two distinct operational decisions rather than jointly optimised. This conclusion rests on four partly overlapping test segments per horizon from a single production run, so a weak interaction that the present design cannot resolve cannot be excluded. The re-calibration cadence of ten to fifteen cycles identified in the previous section, which is robust to the choice of tolerance, corresponds to roughly six to nine minutes of production at the nominal cycle time of about 37 s. Recalibrating at this rate preserves accuracy while keeping the overhead of repeated retraining acceptable, and it directly informs the scheduling of online model updates in a monitoring system. The reported numerical thresholds should therefore be interpreted as machine- and run-specific operating values rather than generally transferable thresholds.

The accuracy degradation at longer horizons is primarily a consequence of the recursive roll-out itself: each prediction is fed back as an input, so one-step errors accumulate and propagate as the horizon grows, largely independently of the chosen regressor [15]. A secondary factor is the limited temporal context available to the model. Here the lag window was deliberately set to two production cycles ($L = 74$ samples); this is a modelling choice rather than an intrinsic limitation of XGBoost, and a wider window could in principle give the model access to longer-range dependencies. More generally, tree-based regressors carry no explicit memory mechanism, so the retention of long-term temporal structure rests entirely on the engineered lag features.

In contrast, recurrent architectures such as Long Short-Term Memory (LSTM) networks maintain an internal state across time steps and may therefore be better suited to capturing long-range dependencies at longer horizons on this type of signal. However, this potential gain comes at a markedly higher computational cost: LSTM networks require gradient-based training over many epochs and are substantially more expensive to train and retrain than XGBoost, which is a decisive consideration for a monitoring system that is retrained online as production cycles ac-

cumulate. Overall, XGBoost is well suited to short-term forecasting with periodic recalibration, providing a computationally efficient and interpretable solution for the intelligent energy monitoring of the injection molding machine.

CONCLUSION AND FUTURE WORK

In this paper, we investigated how the training-set size and the forecast horizon affect a recursive XGBoost model forecasting the apparent power of a single injection molding machine, using a run of 410 production cycles. Two decoupled experiments, evaluated under a rolling-origin scheme, show that on this dataset accuracy saturates after about 125 training cycles, reaching within 10% of the best observed error, and that the recursive error grows gradually rather than divergently, crossing a 20% inflation threshold after about ten cycles. For this machine, this translates into a practical recalibration cadence of ten to fifteen cycles, robust to the choice of tolerance.

A split-plot ANOVA further shows that the two factors contribute approximately additively, and the learning threshold recomputed at each horizon remains confined to a narrow band (100–125 cycles) without growing appreciably with the horizon, so that data sizing and recalibration scheduling can, to a first approximation, be treated as two separate operational decisions. While these specific thresholds are tied to the machine and production run studied here, the proposed protocol decoupling the two factors and quantifying their interaction through a split-plot ANOVA is general and can be reapplied to size the training window and set the recalibration cadence of other cyclic industrial processes.

Future work will therefore aim to validate these findings across additional machines, materials and operating regimes, and to incorporate the thermal evolution of the machine as an additional model input. As thermal dynamics evolve slowly across successive cycles, they are expected to encode the long-range information that lagged power features alone cannot capture, thereby improving forecasting accuracy at extended horizons.

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A THREE-DEGREE-OF-FREEDOM MATHEMATICAL MODEL REPRESENTING THE APPARENT MASS AND FORE-AND-AFT CROSS AXIS APPARENT MASS OF THE HUMAN BODY EXPOSED TO VERTICAL VIBRATION

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KEYWORDS
3 DoF Model, whole-body vibration, posture, apparent mass, cross-axis apparent mass.

ABSTRACT

A model to represent both the apparent mass (AM) and fore-and-aft cross-axis apparent mass (CAM) of the seated human body under whole-body vertical vibration has been developed. The model is a three degree-of-freedom model with translational and rotational capabilities. The parameters of the model were optimised using the experimentally measured median and individual AM and CAM of 12 male subjects sitting with two different postures. In the first posture, the feet of the subjects were not supported while in the second posture the feet were supported. For both postures, the model gave a close fit to both the median and individual AM and CAM of the 12 subjects. The analysis indicated that the vertical degree of freedom was the main contributor to the AM. It also revealed that all three degrees of freedom contributed to the fore-and-aft CAM. The parameter optimisation demonstrated that some parameters are constrained by the measured biomechanical responses, whereas others exhibit considerable variability among subjects. This implies that multiple parameter combinations can provide comparable representations of the experimental apparent masses and cross-axis apparent mass.

INTRODUCTION

Human exposure to whole-body vibration (WBV) occurs in many occupational and transportation environments, including road and off-road vehicles, rail vehicles, and marine craft. Prolonged exposure to vibration can influence ride comfort, performance of activities, and, in some circumstances, health (Bovenzi and Betta 1994; Guruguntla 2026). Hence, investigating the biomechanical responses of the seated human body to vibration has received considerable attention over the past decades. Mathematical models provide an efficient means of representing the biomechanical responses to vibration over a wide range of conditions without the need for extensive time- consuming experimental testing.

Lumped-parameter models have been used to represent the biomechanical responses of the seated human body to vibration. These models idealise the body as a system of concentrated masses interconnected by springs and dampers, providing a practical compromise between physical realism and computational simplicity (Ghods et al. 2026). Several lumped-parameter models have been developed to predict vertical biomechanical responses, including apparent mass and seat-to-head transmissibility, and have been successfully applied in the analysis of vehicle ride dynamics, seat design, and vibration isolation systems (Fairley and Griffin 1989; Boileau and Rakheja 1998; Wei and Griffin 1998).

Experimental investigations have shown that the biomechanical response of seated occupants to vibration is not limited to the direction of vibration excitation. During vertical whole-body vibration, the human body also shows fore-and-aft dynamic forces at the seat interface owing to the coupled translational and rotational motion of the body (Nawayseh and Griffin 2005). This is described as the fore-and-aft cross-axis apparent mass and provides additional insight into the dynamic behaviour of seated human body when exposed to vertical vibration. To represent this coupled response, Nawayseh and Griffin (2009) developed a lumped-parameter model capable of simultaneously predicting both the vertical apparent mass and the fore-and-aft cross-axis apparent mass. Subsequent studies have further extended biomechanics modelling to different seating conditions and excitation directions, demonstrating the continued importance of lumped-parameter representations in the study of human vibration (Qiu and Griffin 2011; Nawayseh 2016). The development of alternative mathematical models remains valuable because lumped-parameter systems are not unique. Different model configurations may provide equally acceptable representations of experimentally measured biomechanical characteristics while offering different physical interpretations. So, there continues to be interest in developing and examining mathematical models that accurately reproduce measured apparent masses using relatively simple mechanical systems suitable for engineering analysis and simulation.

This study proposes a three-degree-of-freedom lumped-parameter model representing the vertical, fore-and-aft, and rotational motions of the seated human body during vertical whole-body vibration. The model is formulated to predict both the vertical apparent mass and the fore-and-aft cross-axis apparent mass. The parameters of the model are optimised using experimentally measured biomechanical responses. The proposed model provides an additional mathematical representation of the coupled biomechanical response of the seated human body and contributes to the continuing development of simplified models for vibration analysis and vehicle-seat applications.

METHODS

Experiment

The parameters of the model were calibrated using experiment data reported in (Nawayseh and Griffin 2003).

The objective of the experiment was to investigate the effect of the sitting posture on the apparent mass and the fore-and-aft cross-axis apparent mass during whole-body vertical vibration. In the present study, two postures were chosen: one with the feet of the subjects not supported on a footrest and the other with the feet supported on a footrest (Figure 1). Twelve male subjects took part in the experiment. A whole-body vibration machine that produces a peak-to-peak vertical displacement of 1 m was used to produce the vibration. The input vibration was random in the frequency range 0.25-25 Hz and at 4 different vibration magnitudes. In the present study the highest vibration magnitude (1.25 m/s^2) was used. For each condition, the duration of vibration exposure was 60 seconds.

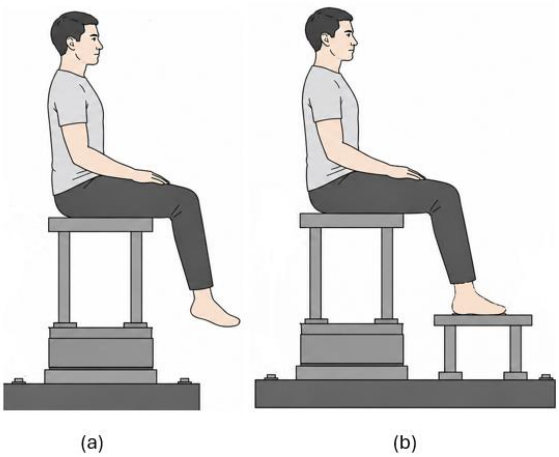

Figure 1 Postures used in this Work. (a) Feet Unsupported, (b) Feet Supported.

During the experiment, the subjects sat on triaxial force platform fixed on a rigid seat mounted on the vibration machine. The vertical acceleration and the vertical and fore-and-aft forces at the seat were measured during the exposure to vibration to calculate the apparent mass and the cross-axis apparent mass. The apparent mass was calculated using the cross-spectral density method between the vertical force and the vertical acceleration while the fore-and-aft cross-axis apparent mass was calculated using the cross-spectral density method between the fore-and-aft force and the vertical acceleration as shown in Equation 1.

$$M(\omega) = \frac{S_{af}(\omega)}{S_{aa}(\omega)} \qquad (1)$$

where, $M(\omega)$ is the apparent mass (or the cross-axis apparent mass), $S_{af}(\omega)$ is the cross-spectral density between the vertical (or fore-and-aft) force and the acceleration, and $S_{aa}(\omega)$ is the power spectral density of the acceleration.

THE PROPOSED MODEL

The model used in this study is a quantitative three-degree-of-freedom lumped parameter linear model (Figure 2). The model has 3 masses with mass 1 and mass 2 have translational motion while mass 3 has rotational motion in addition to the translational motion it gains from moving with the base. Since the proposed model is not a mechanistic model, none of the parameters of the model is intended to represent any specific part of the human body.

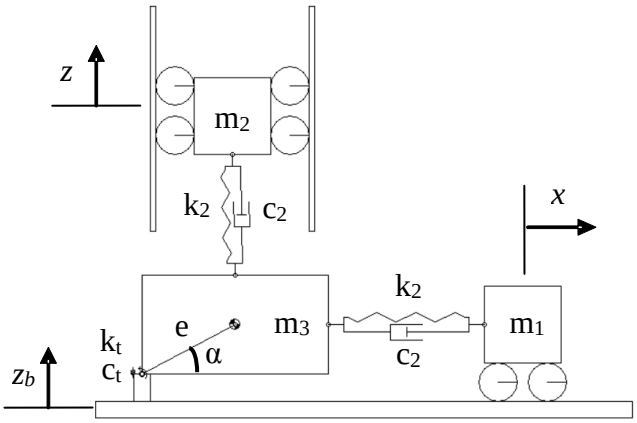

Figure 2 The Proposed 3 Degree-of-Freedom Model.

Mathematical Equations of the Model

The following equations of motion of the model were derived using Lagrange's method.

$$m_1\ddot{x} + c_1(\dot{x} - e\ \sin\alpha\dot{\theta}) + k_1(x - e\theta\sin\alpha) = 0 \qquad (2)$$

$$m_2\ddot{z} + c_2(\dot{z} - \dot{z}_b + e\cos\alpha\dot{\theta}) + k_2(z - z_b + e\theta\cos\alpha) = 0 \qquad (3)$$

$$m_3 e\cos\alpha\,\ddot{z}_b + m_3 e^2\ddot{\theta} + J_3\ddot{\theta} - c_1 e\sin\alpha\,(\dot{x} - e\sin\alpha\,\dot{\theta}) - c_2 e\cos\alpha\,(\dot{z} - \dot{z}_b - e\cos\alpha\,\dot{\theta}) + c_t\dot{\theta} - k_1 e\sin\alpha(x - e\sin\alpha\,\theta) - k_2 e\cos\alpha(z - z_b - e\cos\alpha\,\theta) + k_t\theta = 0 \qquad (4)$$

Where,
m_1, m_2, and m_3 are the masses of mass 1, mass 2, and mass 3 respectively.
J_3 is the mass moment of inertia of mass 3 about its centre of mass.
k_1 and c_1 are the horizontal stiffness and damping coefficient attached to mass 1.
k_2 and c_2 are the vertical stiffness and damping coefficient between mass 2 and mass 3.
k_t and c_t are the rotational stiffness and damping coefficient at the pivot between the body and the seat.
e represents the distance between the pivot and the centre of mass of mass 3.
α is the angle between e and the horizontal during equilibrium.
x represents the motion of mass 1 from its equilibrium position.
z represents the motion of mass 2 from its equilibrium position.
z_b represents the vertical motion of the seat where the force and acceleration used to calculate the apparent mass and cross-axis apparent mass were measured.
Represents the angular displacement of mass 3 from the equilibrium position

Since the vibration is in the vertical direction, the motion of the 3 masses contributes to the vertical force while only mass 1 and mass 3 contribute to the fore-and-aft force at the base (mass 2 can move only vertically). The equations for the vertical force ($f_z(t)$) and the fore-and-aft force ($f_x(t)$) at the base can be obtained from the inertia forces of the masses as follows:

$$f_z(t) = m_1\ddot{z}_b + m_2\ddot{z} + m_3\left(\ddot{z}_b + e\ddot{\theta}\cos\alpha\right) \quad (5)$$

$$f_x(t) = m_1\ddot{x} + m_3\left(e\ddot{\theta}\sin\alpha\right) \quad (6)$$

The apparent mass and fore-and-aft cross-axis apparent mass can be calculated by finding the Laplace transform of the above equations assuming zero initial conditions and solve for the complex ratio between $F_z(s)$ or $F_x(s)$ and $s^2Z_b(s)$ where $F_z(s)$ is the Laplace transform of the vertical force on the seat, $f_z(t)$, $F_x(s)$ is the Laplace transform of the fore-and-aft force at the seat, $f_x(t)$, $s^2Z_b(s)$ is the Laplace transform of the acceleration of the seat surface, $\ddot{z}_b(t)$, and s is $j\omega$ where j is $\sqrt{-1}$ and ω is the angular frequency in rad/s. The apparent mass (AM) and the fore-and-aft cross-axis apparent mass (CAM) can be found from the following equations:

$$AM = m_1 + m_2\left[\frac{D}{C} + \frac{E}{C}\frac{F-\frac{ED}{C}}{\frac{B^2}{A}+\frac{E^2}{C}-G}\right] + m_3 + m_3 e\cos\alpha\left[\frac{F-\frac{ED}{C}}{\frac{B^2}{A}+\frac{E^2}{C}-G}\right] \quad (7)$$

$$CAM = m_1\left[\frac{B}{A}\frac{F-\frac{ED}{C}}{\frac{B^2}{A}+\frac{E^2}{C}-G}\right] + m_3 e\sin\alpha\left[\frac{F-\frac{ED}{C}}{\frac{B^2}{A}+\frac{E^2}{C}-G}\right] \quad (8)$$

Where,
$A=m_1s^2+c_1s+k_1$
$B=c_1e\sin\alpha+k_1e\sin\alpha$
$C=m_2s^2+c_2s+k_2$
$D=c_2s+k_2$
$E=c_2e\cos\alpha+k_2e\cos\alpha$
$F=m_3es^2\cos\alpha+E$
$G=m_3e^2s^2+J_3s^2+k_1e^2\sin^2\alpha+k_2e^2\cos^2\alpha+k_t+c_1e^2s\sin^2\alpha+c_2e^2s\cos^2\alpha+c_ts$

Since the apparent mass and the cross-axis apparent mass are complex functions, the moduli of AM and CAM can be obtained from the real and imaginary parts of the complex functions, AM and CAM as shown in Equation 9 and Equation10:

$$AM_{\text{mod}}(\omega) = \sqrt{(ReAM(\omega))^2 + (ImAM(\omega))^2} \quad (9)$$

$$CAM_{\text{mod}}(\omega) = \sqrt{(ReCAM(\omega))^2 + (ImCAM(\omega))^2} \quad (10)$$

Optimisation of the Parameters of the Model

A total of twelve model parameters were optimized (m_1, m_2, m_3, J_3, c_1, c_2, c_3, k_1, k_2, k_3, e and α) using the interior-point algorithm within MATLAB. The function *fmincon* was used for this purpose as it allows for constraining the parameters to have values within a certain range. This is useful to avoid having a negative value for the optimized parameters. The optimised parameters were obtained by reducing the error between the experimentally measured moduli of the apparent masses and cross-axis apparent mass (Nawayseh and Griffin 2003) and those calculated from the model. So, the following objective function was used for that purpose:

$$err = \sum_{i=1}^{N}(AM_{mod_c} - AM_{mod_m})_i^{\,2} + w\sum_{i=1}^{N}(CAM_{mod_c} - CAM_{mod_m})_i^{\,2} \quad (11)$$

where,
$AM_{\text{mod_c}}$ and $AM_{\text{mod_m}}$ are the moduli of the calculated and measured apparent masses, respectively.
$CAM_{\text{mod_c}}$ and $CAM_{\text{mod_m}}$ are the moduli of the calculated and measured fore-and-aft cross-axis apparent masses respectively.
w is an arbitrary weighting factor employed to improve the prediction. This factor was needed as the order of magnitude of the apparent mass is greater than that of the cross-axis apparent mass.

RESULTS AND DISCUSSION

The Need for 3 Degree-of-Freedom Model

Previous studies have used a single-degree-of-freedom (SDOF) mass-spring-damper model to represent the vertical apparent mass measured at the seat surface (Toward and Griffin 2010). It has also been shown that two-degree-of-freedom (2 DOF) models provides a more accurate representation of the apparent mass measured on the seat. In the present study, however, the model must represent two responses (apparent mass and cross-axis apparent mass) simultaneously and hence, neither an SDOF nor a 2 DOF model can represent both responses simultaneously. This motivated the increase in the degree-of-freedom. Moreover, since the vibration in the present study is purely vertical, a rotational degree of freedom was needed to produce the cross-axis forces used to calculate the cross-axis apparent mass. In fact, Equations 7 and 8 show that the rotational degree of freedom contributed to both the apparent mass and the cross-axis-apparent mass. This is not surprising as the pitching motion of the seated human body during vertical vibration contributes to both the vertical force and the fore-and-aft cross-axis force.

Prediction of Median Data

The seat-person system is a coupled system in which the dynamic response of the seat affects the biomechanical response of its occupant and the biomechanical response of the occupant of the seat influences the dynamic response of the seat. The sitting posture has been reported to significantly influence the biomechanical responses to vibration (Nawayseh and Griffin 2003; Singh et al. 2025). Hence, it is important to model the responses of the human body to vibration under different postures. In this work, the model was used to represent the modulus of the median apparent mass and median cross-axis apparent mass of 12 subjects measured on the seat surface while the subjects sitting in two different postures (Figure 3). The figure shows that the model can provide a close fit to the moduli of the median apparent mass and cross-axis apparent mass with both the feet unsupported and the feet supported postures. The parameters of the model obtained by fitting the median data are given in the last row of Table 1 and Table 2.

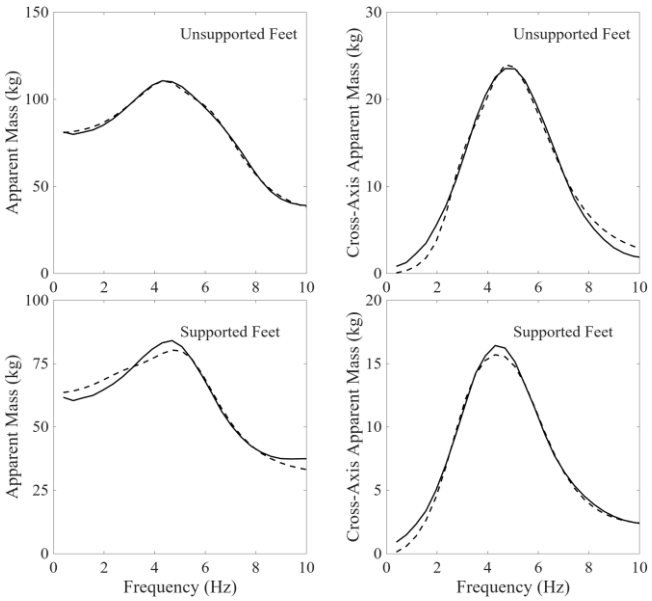

Figure 3 Median Apparent Mass and Cross-Axis Apparent Mass with 2 Sitting Postures: ———, Measured; - - - - - , Calculated.

The optimised parameters that were obtained from fitting the median data can then be very useful in evaluating the transmissibility of vehicle seats during vibration. This can be done by coupling the proposed mathematical model of the human body with a mathematical model of the seat. The obtained system can then be used to evaluate the transmissibility of the seat under vibration without the need for experimental procedure (Wei 2000; Qiu and Griffin 2011). Alternatively, an anthropodynamic mechanical dummy can be built using the optimised parameters of the model. This mechanical system can then be supported on the seat surface to evaluate the performance of seats experimentally by exposing the seat and dummy to vibration instead of using human subjects. This has two advantages, namely obtaining repeatable measurements and testing the performance of seats under extreme vibration conditions.

Prediction of Individual Data

The model was employed to represent the moduli of the apparent masses and fore-and-aft cross-axis apparent mass of the 12 subjects who took part in the experiment. The calculated (i.e. from the model) and measured (i.e. from the experiment) apparent masses and cross-axis apparent masses were in good agreement with both feet unsupported posture and the feet supported posture (Figures 4-7). The parameters of the model obtained for each subject are shown in Tables 1 and 2. The tables show high variability in the parameters among subjects. This is not surprising given the high inter-subject variability shown in the experimental data which was due to using subjects with wide range of mass, stature, body mass index (BMI) and age.

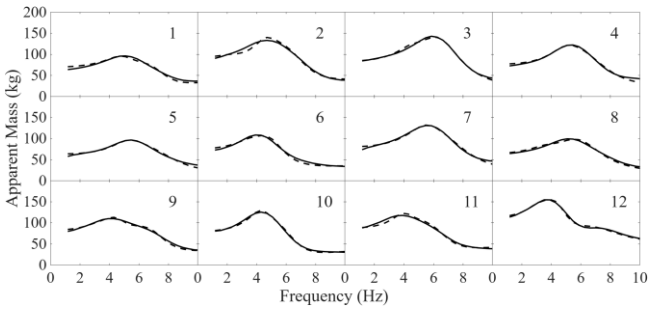

Figure 4 Individual Apparent Masses of 12 Subjects with The Unsupported Feet Posture. ——, Measured; - - - - -, Calculated.

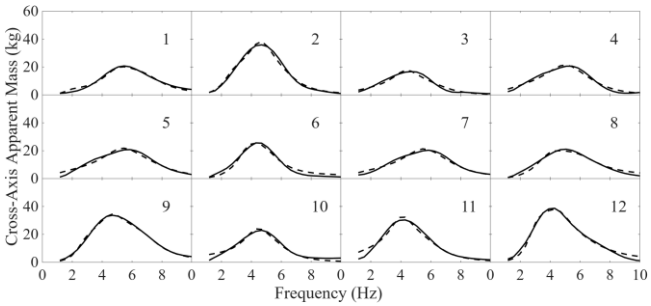

Figure 5 Individual Cross-axis Apparent Masses of 12 Subjects with The Unsupported Feet Posture. ——, Measured; - - - - -, Calculated.

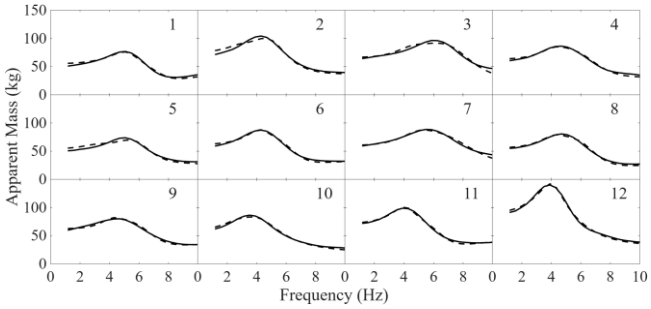

Figure 6 Individual Apparent Masses of 12 Subjects with The Supported Feet Posture. ——, Measured; - - - - -, Calculated.

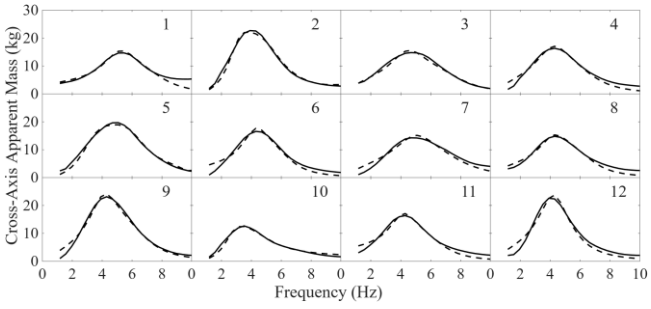

Figure 7 Individual Cross-axis Apparent Masses of 12 Subjects with The Supported Feet Posture. ——, Measured; - - - - -, Calculated.

Table 1 Optimised Parameters for Median (Last Row) and Individual Responses of 12 Subjects with The Feet Unsupported Posture

Sub	m_1 kg	m_2 kg	m_3 kg	k_1 N/m	k_2 N/m	k_t Nm	c_1 Ns/m	c_2 Ns/m	c_t Nsm	J_3 kgm^2	e m	α rad
1	43.91	25.69	0.00	6088.10	41166.00	20055.00	478.65	262.73	0.00	16.33	0.83	1.22
2	51.24	42.14	0.00	27932.00	61432.00	14492.00	0.00	572.14	787.84	27.83	0.66	1.16
3	28.56	53.98	1.60	10605.00	91297.00	1.18	0.02	917.44	326.86	11.82	0.85	1.43
4	33.13	41.92	1.63	14067.00	70718.00	0.13	0.00	646.41	402.79	8.18	0.87	1.36
5	33.63	29.99	0.00	13428.00	59575.00	0.00	0.00	377.14	1.48	0.03	0.06	1.30
6	22.45	35.61	17.83	33464.00	35363.00	20017.00	4897.60	531.42	378.03	0.02	0.93	1.42
7	34.12	46.49	0.00	13067.00	85519.00	0.00	0.00	761.65	6.56	0.17	0.13	1.41
8	0.00	43.29	21.56	33112.00	63598.00	12345.00	14024.00	878.02	285.11	0.00	0.97	1.45
9	45.16	37.59	0.00	15098.00	54497.00	20067.00	728.43	430.39	0.00	18.87	0.63	1.11
10	43.33	37.13	0.00	6684.30	42376.00	0.00	0.00	313.73	143.95	6.26	0.91	1.40
11	52.58	35.83	0.00	12933.00	44690.00	0.00	0.00	370.00	3.44	0.13	0.09	1.26
12	39.41	65.85	8.37	16230.00	88778.00	20098.00	567.40	1612.00	80.35	16.49	0.52	1.12
Med.	25.05	43.07	12.54	8582.20	62315.00	19152.00	412.10	851.85	0.01	12.96	0.72	1.35

Table 2 Optimised Parameters for Median (Last Row) and Individual Responses of 12 Subjects with The Feet Supported Posture

Sub	m_1 kg	m_2 kg	m_3 kg	k_1 N/m	k_2 N/m	k_t Nm	c_1 Ns/m	c_2 Ns/m	c_t Nsm	J_3 kgm^2	e m	α rad
1	42.51	13.18	0.00	6174.10	21570.00	0.00	0.00	0.00	2.24	0.05	0.12	1.23
2	15.87	35.71	24.56	30563.00	37690.00	17415.00	201.19	661.17	564.47	0.05	1.00	1.40
3	31.94	34.25	0.00	11942.00	69768.00	29.25	0.00	641.83	11.93	0.35	0.15	1.34
4	36.51	27.00	0.00	10598.00	39568.00	0.00	0.00	412.42	11.54	0.33	0.16	1.29
5	8.02	21.40	24.71	23261.00	30254.00	14287.00	37.36	374.96	478.32	0.05	0.92	1.35
6	39.21	23.17	0.00	6534.30	28175.00	0.00	0.00	218.22	155.88	6.25	0.92	1.34
7	25.78	34.62	0.00	10356.00	71191.00	0.00	0.00	716.29	13.25	0.31	0.19	1.37
8	32.34	23.91	0.00	6627.60	32833.00	0.00	0.00	252.59	82.28	2.93	0.61	1.34
9	42.15	20.22	0.00	15429.00	29581.00	0.00	0.00	190.25	620.29	14.45	0.94	1.08
10	1.90	43.41	16.43	33471.00	32787.00	6075.50	1617.30	929.67	209.78	0.05	1.00	1.48
11	48.23	24.44	0.00	4385.00	28904.00	0.00	0.00	143.48	104.70	3.63	0.91	1.35
12	31.67	60.14	0.00	12053.00	51776.00	0.00	0.00	1119.90	469.21	15.10	0.99	1.42
Med	19.30	30.94	13.17	25599	38346	14863	258.49	732.17	814.87	0.05	1.00	1.32

The optimized parameters exhibited the noticeable inter-subject variability despite the use of identical optimization settings for all subjects, including the same initial parameter guesses, parameter bounds, objective function, and optimization algorithm. As mentioned before, the parameters of the model were identified by minimizing the combined errors between the measured and calculated vertical apparent mass and fore-and-aft cross-axis apparent mass. Consequently, the identified parameters represent the best compromise for simultaneously reproducing two independent experimental biomechanical responses rather than fitting the vertical apparent mass alone as was done in previous studies. The relatively large variability observed for some parameters, particularly m_3, J_3, k_t, and c_t, suggests that these parameters are less uniquely identifiable than the principal translational parameters. This behaviour is characteristic of nonlinear constrained optimization problems, where several parameter combinations may produce similarly small objective-function values while providing comparable agreement with the experimental data.

Therefore, although the optimised parameters primarily reflect differences in the biomechanical responses of the subjects, part of the observed variability may also arise from the inherent non-uniqueness of the parameter identification process.

Supporting the feet produced systematic changes in only a limited number of the optimized parameters. Statistical analysis showed that mass m_2 and stiffness k_2 were significantly lower when the feet were supported than when they were unsupported (Wilcoxon signed-rank test, $p < 0.05$) whereas the remaining parameters did not differ significantly between the two postures (Wilcoxon signed-rank test, $p > 0.05$). Because identical optimisation settings were used for both postures, these systematic reductions are unlikely to be attributed only to the optimisation procedure but indicate genuine changes in the biomechanical behaviour associated with foot support. The reduction in m_2 and k_2 suggests that supporting the feet modifies the dynamic load transmission through the lower extremities, thereby altering the dynamic behaviour represented by the vertical degree of freedom of the model. In contrast, the absence of significant changes in the remaining parameters, including the geometric coupling parameters e and α, indicates that foot support has a relatively localized influence on the model structure and does not substantially alter the overall coupling mechanism between the translational and rotational motions. When the feet are unsupported, a greater proportion of body mass is transferred to the seat compared with the supported condition, which is reflected as an increase in mass 2 in the model. The associated increase in k_2 may indicate greater engagement of the upper-body muscles as subjects attempt to stabilise themselves on the seat.

An important feature of the present parameter identification procedure is that the optimisation was performed simultaneously using both the vertical apparent mass and the fore-and-aft cross-axis apparent mass. Unlike optimisation procedures based only on the vertical apparent mass, the inclusion of the cross-axis apparent mass provides an additional independent experimental constraint on the optimisation problem, reducing the freedom available to the optimiser and requiring the identified parameter set to reproduce both the translational and rotational biomechanical responses simultaneously. So, the optimised parameters are expected to possess greater physical credibility, as they satisfy two experimentally measured dynamic characteristics rather than a single response. This capability is particularly important for biomechanical models intended to represent the coupled translational and rotational behaviour of the seated human body under vertical vibration.

Sensitivity Analysis

Sensitivity analysis was performed to investigate the influence of the model stiffness and damping parameters on the predicted vertical apparent mass and fore-and-aft cross-axis apparent mass. The analysis was conducted by independently varying each stiffness and damping parameter by ±40% about the optimised parameter values obtained by fitting the median experimental apparent mass and cross-axis apparent mass responses of the 12 subjects.

The results demonstrate that the model predictions are most sensitive to the stiffness of the upper-body connection, k_2.

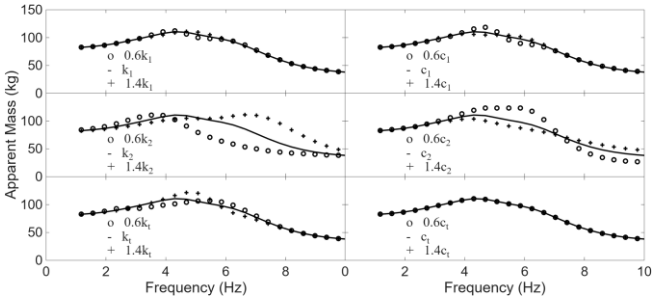

Figure 8: Changes in the Apparent Mass Resulting from Changes in the Optimised Model Parameters with The Feet Unsupported Posture

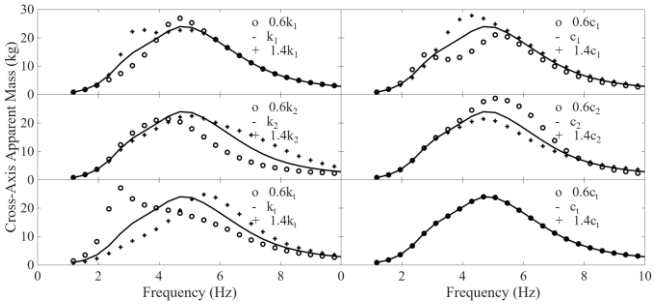

Figure 9: Changes in the Cross-Axis Apparent Mass Resulting from Changes in the Optimised Model Parameters with The Feet Unsupported Posture.

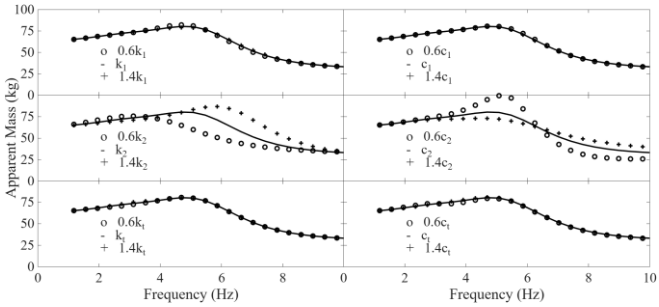

Figure 10: Changes in the Apparent Mass Resulting from Changes in the Optimised Model Parameters With The Feet Supported Posture

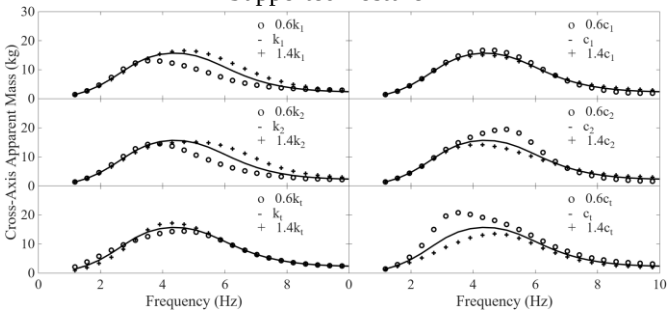

Figure 11: Changes in the Cross-Axis Apparent Mass Resulting from Changes in the Optimised Model Parameters with The Feet Supported Posture

Variations in k_2 produced the largest changes in both the vertical apparent mass and the fore-and-aft cross-axis apparent mass over the frequency range of interest. This behaviour is expected because k_2 governs the dynamic coupling between the upper body and the remainder of the

model, thereby influencing both the principal resonance observed in the vertical apparent mass and the fore-and-aft motion associated with body rotation.

The damping coefficient c_2 also produced noticeable changes in both responses, although its influence was generally smaller than that of k_2. Changes in c_2 affected the magnitude of the apparent mass at resonance but slightly shifted the resonance frequency. In contrast, variations in the lower-body stiffness and damping parameters k_1 and c_1 produced minor changes in the model responses, especially in the apparent mass, indicating that these parameters contribute to the overall dynamic behaviour of the model but have less influence on the principal resonance than the upper-body (vertical) system.

Changes in the rotational stiffness and damping coefficient k_t and c_t had little influence on the vertical apparent mass regardless of the posture. However, their effect on the fore-and-aft cross-axis apparent mass depended on posture: changes in k_t affected the fore-and-aft cross-axis apparent mass in the feet-unsupported posture, whereas changes in c_t had the greater effect in the feet-supported posture. These findings may support the call for having models that can predict the response in different sitting postures.

CONCLUSIONS

A three degree-of-freedom mass-spring-damper lumped parameter linear model was able to represent the moduli of the vertical apparent mass and fore-and-aft cross-axis apparent mass measured on a rigid seat during whole-body vertical vibration. The model was a coupled system with translational (both vertical and horizontal) degree of freedom as well as rotational degree of freedom. The model gave close fit to both the median and individual apparent masses and cross-axis apparent masses of 12 male subjects. The analysis showed that the vertical degree of freedom was the main contributor to the apparent mass. The analysis also showed contribution to the fore-and-aft cross-axis apparent mass from the 3 degrees of freedom. The parameter optimisation demonstrated that some parameters are constrained by the measured biomechanical responses, whereas others exhibit considerable variability among subjects. This implies that multiple parameter combinations can provide comparable representations of the experimental apparent masses and cross-axis apparent mass.

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Man-In-The-Loop Simulation

TOWARDS SOCIAL AUTOMATION: MATHEMATICAL MODELLING, SIMULATION AND CONTROL OF HUMAN-CENTRIC SYSTEMS

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social automation, human-in-the-loop systems, human–robot interaction, human-centric control, agent-based simulation, adaptive control, multi-agent systems, cyber-social systems

ABSTRACT

Nominal automation models may be insufficient when humans are active elements of feedback loops. This paper uses social automation as a control-oriented framework integrating technical dynamics, human states, and social interaction. A discrete-time cyber-human-social model is coupled with human-aware objectives and evaluated in a reproducible agent-based human-robot guidance scenario. Fifty paired Monte Carlo runs compare classical, human-in-the-loop (HITL), and Social Automation policies. Social Automation does not dominate all baselines: classical guidance is faster and HITL yields the lowest cognitive load. Relative to classical guidance, however, it reduces normalised control energy, mean cognitive load, and maximum individual load.

1. INTRODUCTION

Contemporary cyber-physical systems increasingly place humans inside sensing, decision, supervision, and intervention loops (Goodrich and Schultz 2007; Parasuraman and Riley 1997; Munir et al. 2013). Human-in-the-loop research addresses trust, workload, levels of automation, adaptive automation, and supervisory control (Parasuraman et al. 2000; Kaber and Endsley 2004; Lee and See 2004), while multi-agent and computational-social models address coordination and social influence (Olfati-Saber et al. 2007; Castellano et al. 2009). These streams are only partly integrated in feedback-control models.

In particular, human factors are often treated as disturbances or external constraints, and social influence is rarely represented as an explicit state-coupling mechanism. Recent cyberphysical-human, cyber-physical-social, and human-centric AI research motivates a common representation of technical, human, and social dynamics (Yildiz 2020; Yilma et al. 2021; Rožanec et al. 2023).

The research gap is therefore not the absence of human-in-the-loop modelling, but the limited integration of multiple socially connected human agents, human-factor states, and feedback-control objectives within one model. Trust, cognitive load, acceptance, delay, and perceived risk require explicit operational definitions in control objectives or constraints, while socially aware navigation highlights the need to connect interaction models to decision and control mechanisms (Akash et al. 2020; Guillén-Ruiz et al. 2023; Firmino de Souza et al. 2025).

This paper formulates social automation as a human-centric control framework coupling technical dynamics, individual human states, and social-interaction networks. The contribution is a discrete-time state-space formulation, boundedness conditions for the social layer, human-aware objectives, and a reproducible simulation procedure. The human-robot guidance case evaluates the internal behaviour and comparative implications of the proposed model.

2. FROM CLASSICAL AUTOMATION TO SOCIAL AUTOMATION

A nominal continuous-time technical model is represented in state-space form:

$$\dot{x}(t) = f\big(x(t), u(t), d(t)\big) \qquad (1)$$

In equation (1), x(t) is the technical state vector, u(t) is the control-input vector, d(t) represents exogenous disturbances, and f(·) is the plant vector field. In words, the instantaneous change of the technical state is determined by the current state, the applied control action, and external disturbances. A simple reference-tracking feedback law is:

$$u(t) = K[r(t) - y(t)] = Ke(t) \qquad (2)$$

In equation (2), r(t) is the desired reference output, y(t) is the measured output, e(t)=r(t)-y(t) is the tracking error, and K is the feedback-gain matrix. Thus the controller acts on the difference between the desired and observed outputs. Equations (1)-(2) are only a nominal illustration; robust, adaptive, and stochastic control already address uncertainty, noise, and disturbances. The issue considered here is the explicit representation of human and social states inside the feedback model.

In human-centric systems, low-order nominal models may omit intentional behaviour, heterogeneous preferences, bounded rationality, delayed responses, and group effects. Social automation extends rather than replaces established control methods by representing selected human and social variables explicitly.

For this paper, social automation is a control-oriented framework for modelling, simulating, and influencing systems in which humans, technical components, and social interactions form coupled feedback loops.

The objective is not to automate humans, but to design technical interventions that account for human limitations, support decision-making, and jointly consider safety, performance, cognitive burden, trust, and social acceptability.

3. HUMAN-CENTRIC SYSTEM MODEL

For simulation, a common sampling interval $\Delta t>0$ is introduced and the continuous vector fields are discretised by forward Euler. For compactness in equations (3)-(12), a sampled variable v[k] is written as v_k. The technical state, human state, and measured output are updated as follows:

$$x_{s,k+1} = x_{s,k} + \Delta t\, f_s\big(x_{s,k}, u_k, x_{h,k}, w_{s,k}\big) \qquad (3)$$

The corresponding human-state update is:

$$x_{h,k+1} = x_{h,k} + \Delta t\, f_h\big(x_{h,k}, y_{k-\tau}, c_k, w_{h,k}\big) \qquad (4)$$

The observed output is:

$$y_k = g\big(x_{s,k}, x_{h,k}\big) \qquad (5)$$

In equations (3)-(5), k is the discrete sample index; $x_{s,k}$ is the technical state, $x_{h,k}$ the stacked human-state vector, u_k the control input, y_k the measured output, and c_k the social-context vector. The disturbance terms $w_{s,k}$ and $w_{h,k}$ represent technical and human variability, respectively, while τ denotes an observation or response delay. Equation (3) states that the next technical state equals the current state plus the sampled technical dynamics; equation (4) applies the same convention to the human state; and equation (5) maps the joint technical-human state to the observable output. Every simulated transition uses this single k-to-k+1 convention.

For agent i, the complete human state contains perceived risk r_i, stress s_i, intention/compliance z_i, cognitive load l_i, and decision delay d_i. To avoid updating intention twice, the model separates the complete state from the four-dimensional non-intention state:

$$x_{h,i,k} = \big[r_{i,k}, s_{i,k}, z_{i,k}, l_{i,k}, d_{i,k}\big]^T$$

$$\tilde{x}_{h,i,k} = \big[r_{i,k}, s_{i,k}, l_{i,k}, d_{i,k}\big]^T$$

$$\tilde{x}_{h,i,k+1} = A_h \tilde{x}_{h,i,k} + B_h y_{i,k-\tau_i} + E_h c_{i,k} + \varepsilon_{h,i,k} \qquad (6)$$

Equation (6) updates only the non-intention variables. The matrices A_h, B_h, and E_h contain the discrete-time response coefficients; $y_{i,k-\tau_i}$ is the delayed local observation, $c_{i,k}$ is social context, and $\varepsilon_{h,i,k}$ is human-state noise. The intention variable z_i is updated exclusively by equation (7). Psychological variables are projected to admissible ranges and decision delay to a finite sample interval.

The social context for agent i is defined from differences between its intention and the intentions of neighbouring agents:

$$c_{i,k} = \sum_{j \in N_i} a_{ij}\,\big(z_{j,k} - z_{i,k}\big)$$

Here N_i denotes the neighbourhood of agent i and a_{ij} is the influence weight from agent j to i. This context enters the projected intention update:

$$z_{i,k+1} = \Pi_{[0,1]}\{\alpha_i z_{i,k} + c_{i,k} + b_i u_{g,k}\big(1 - z_{i,k}\big)\} \qquad (7)$$

In equation (7), α_i is the persistence coefficient, b_i is susceptibility to the guidance channel $u_{g,k}$, and $\Pi[0,1]$ projects intention onto its admissible interval. For the zero-input case $u_{g,k}=0$, if $\alpha_i=1$, $a_{ij}\geq 0$, and $\sum_j a_{ij}\leq 1$, the unprojected update has non-negative neighbour weights and self-weight $1-\sum_j a_{ij}$, and is therefore row-stochastic in the DeGroot sense (DeGroot 1974). Projection preserves $z_i\in[0,1]$. This boundedness statement concerns only the social layer, not stability of the complete coupled system.

4. CONTROL OBJECTIVES IN SOCIAL AUTOMATION

For comparison, a conventional finite-horizon quadratic control objective penalises technical tracking error and control effort:

$$J_c = \sum_{k=0}^{T-1}\big(\| x_{s,k} - x_{ref,k} \|_Q^2 + \| u_k \|_R^2\big) + \phi\big(x_{s,T}\big) \qquad (8)$$

In equation (8), $x_{ref,k}$ is the reference trajectory, Q is a positive-semidefinite state-weighting matrix, R is a positive-definite control-weighting matrix, and φ is the terminal cost. The human-centric controller augments this objective with human-aware and safety-related costs:

$$J_h^\star = \min_{\{u_k\}} \sum_{k=0}^{T-1}\big(\alpha L_{E,k} + \beta L_{S,k} + \gamma C_k + \delta L_{H,k}\big) \qquad (9)$$

In equation (9), L_E[k] is the effectiveness loss, L_S[k] the normalised safety-risk loss, C[k] the normalised control-effort cost, and L_H[k] the scalar human loss. All are defined so that larger values are undesirable. The human component is written explicitly as:

$$L_{H,k} = \sum_{q\in\{r,s,d,t,a\}} \omega_q\, \ell_{q,k}$$

In this compact expression, $\ell_{r,k}$ is mean perceived risk, $\ell_{s,k}$ mean stress, $\ell_{d,k}$ normalised mean decision delay, $\ell_{t,k}=1-\bar{t}_{r,k}$ is lack of trust, and $\ell_{a,k}=1-\bar{a}_k$ is lack of acceptance. The non-negative weights ω_r, ω_s, ω_d, ω_t, and ω_a sum to one; the outer weights α, β, γ, and δ in equation (9) are likewise non-negative. In the reported experiment the human-loss weights are 0.30, 0.25, 0.15, 0.15, and 0.15, and the outer weights are α=0.35, β=0.30, γ=0.10, and δ=0.25.

Technical constraints bound admissible control actions and safe technical states; human constraints bound cognitive load, stress, confusion, or fatigue and can impose a minimum trust level; social constraints concern transparency, non-manipulation, and fairness. Formally, u_k belongs to the admissible control set U, $x_{s,k}$ to the safe technical-state set X_s, and $x_{h,k}$ to the admissible human-state set X_h. These sets turn human-aware safety requirements into explicit constraints rather than qualitative recommendations.

5. SIMULATION FRAMEWORK

The simulation is a synchronous discrete-time agent-based model with technical, human, social-network, control, and evaluation layers. At each sample k, the current technical and human states are observed, the policy computes $u_{k,}$ and the social, human, queue, agent, and technical states are then advanced to k+1. Fifty paired Monte Carlo runs use common random numbers, which reduces comparison noise between policies.

At each transition the simulator (1) observes the current state, (2) computes the selected control policy, (3) evaluates the social context $c_{i,k}$ and updates intention through equation (7), (4) updates the remaining human variables through equation (6), (5) advances queues, robot guidance, and the technical state through equation (3), and (6) accumulates evacuation time, trajectory error, control energy, cognitive load, and maximum individual load.

This causal order prevents a human or social update from using a control signal that has not yet been computed and keeps the implementation consistent with equations (3), (6), and (7).

For policy p and Monte Carlo run r, trajectory error over the active evaluation horizon $T_{p,r}$ is measured by the discrete integral absolute error:

$$IAE_d^{p,r} = \Delta t \sum_{k=0}^{T_{p,r}-1} \| e_{p,r,k} \|_1 \qquad (10)$$

For vector-valued control, raw resource expenditure is the sampled squared Euclidean norm, and the dimensionless control-energy index is normalised against the mean classical-baseline energy across the paired replications:

$$E_U^{p,r} = \Delta t \sum_{k=0}^{T_{p,r}-1} \| u_{p,r,k} \|_2^2$$

$$U_C^{p,r} = \frac{E_U^{p,r}}{M^{-1} \sum_{r'=1}^{M} E_U^{classic,r'}} \qquad (11)$$

Mean cognitive load over the same active horizon is:

$$H_L^{p,r} = \frac{1}{T_{p,r}} \sum_{k=0}^{T_{p,r}-1} \bar{l}_{p,r,k} \qquad (12)$$

In equation (10), $e_{p,r,k}$ is the desired-minus-observed output vector. In equation (11), E_U[p,r] is raw control energy and M is the number of paired replications; normalisation makes the classical-baseline mean exactly 1.00. Equation (12) averages population mean load over the same active horizon, with exited agents assigned zero load. H_L reports cognitive load alone and is distinct from the composite human loss L_H in equation (9).

6. HUMAN-ROBOT ASSISTANCE SCENARIO

6.1 Scenario and Simulation Setup

The scenario contains N=100 agents in a directed 12-node facility with two exits, dynamic signage, an alarm channel, and node-level robot guidance. Agents differ in desired speed, reaction delay, compliance, stress, trust, and susceptibility to social influence. The robot provides local route guidance and supports, rather than replaces, human decisions.

The graph contains 22 directed edges. Edge length and width are sampled once using seed 2026000, with FIFO admission and width-dependent capacity. At each decision node, agent i assigns each feasible outgoing edge e=(v,w) the following cost:

$$C_{i,e,k} = \frac{L_e}{v_i} + D\left(w, x_{e,i}\right) + \frac{8\rho_{e,k}}{q_e} + 4s_{i,k} - 6g_{i,e,k}$$

Here $C_{i,e,k}$ is the route-choice cost; L_e/v_i is nominal traversal time, $D(w,x_{e,i})$ is remaining shortest-path distance, $\rho_{e,k}$ is the current queue, q_e is corridor capacity, $s_{i,k}$ is stress, and $g_{i,e,k}$ is edge guidance. With probability $1-p_{err,i}$ the minimum-cost edge is selected; otherwise a feasible edge is selected uniformly.

The local human-state observation contains hazard, normalised density, and guidance intensity. Desired speed is sampled from U(0.9,1.5) m/s, initial reaction delay from {1,...,5} samples, decision-error probability from U(0.05,0.15), trust from Beta(6,2), and initial stress and load from Beta(2,5). Initial intention/compliance is sampled from Beta(8,2) and clipped to [0.70,0.95].

The social graph is Watts-Strogatz with mean degree 4 and rewiring probability 0.10, fixed within each paired-seed run. Raw influence weights are sampled from U(0.10,0.40) and row-normalised so that the row sum is at most 0.40. Trust and acceptance are trajectory-derived observables used in the human-aware objective; they are not additional state components in equation (6).

Accordingly, each term entering the technical, safety, and human-aware objectives can be computed from the simulated trajectory without introducing an unobserved post-processing variable.

Three policies are compared. Classical uses static shortest-path signage. HITL redirects signage toward the least-congested exit every 10 s. Social Automation additionally uses state-dependent alarm, signage, and verbal guidance together with the explicit social-state update.

The horizon is 500 one-second transitions. Each policy is evaluated in 50 paired runs (seeds 2026001-2026050), terminating when all agents exit or the horizon is reached. The principal settings are listed in Table 1.

Table 1. Principal simulation and reproducibility settings

Number of agents	N = 100
Simulation horizon	T = 500 transitions
Time step	Δt = 1 s
Monte Carlo runs	50; seeds 2026001–2026050
Network type	Watts-Strogatz k=4, p=0.10; fixed per run
Compared strategies	classical, HITL, social automation

Building graph	12 nodes, 22 directed edges, 2 exits
Robot parameters	node-level replanning every 10 s; local guidance
Software	NumPy 2.3.5; deterministic adjacency lists

The stated equations, graph construction, policy logic, update order, seeds, and termination rule support methodological replication. The complete source code, model matrices, and policy gains are required for exact numerical replication. The reported runs used NumPy 2.3.5.

6.2 Results

Table 2 reports means ± sample standard deviations from 50 paired Monte Carlo replications. The statistics quantify variation across the paired runs.

Table 2. Simulation Results

Strategy	T_{ev} [s]	U_C [−]	H_L [−]	L_{max} [−]
Classical	112.7 ± 12.4	1.000 ± 0.110	0.392 ± 0.039	0.974 ± 0.022
HITL	118.5 ± 12.5	0.526 ± 0.056	0.026 ± 0.003	0.583 ± 0.044
Social Auto.	118.4 ± 11.3	0.656 ± 0.053	0.229 ± 0.027	0.876 ± 0.044

T_{ev} is evacuation time, U_C the normalised active-horizon control-energy index, H_L mean cognitive load, and L_{max} maximum observed individual load.

Classical guidance is fastest (112.7 s). Relative to classical guidance, Social Automation reduces normalised control energy from 1.000 to 0.656, mean cognitive load from 0.392 to 0.229, and maximum individual load from 0.974 to 0.876, corresponding to reductions of about 34.4%, 41.6%, and 10.1%, respectively, while increasing mean evacuation time to 118.4 s. HITL has a nearly identical evacuation time (118.5 s) and lower mean cognitive load than Social Automation. The proposed policy therefore does not dominate the baselines; it demonstrates a trade-off between technical speed, control expenditure, and selected human-aware indicators.

The 95% confidence intervals for evacuation time are [109.1,116.2] s for Classical, [115.0,122.1] s for HITL, and [115.2,121.6] s for Social Automation. For UC they are [0.969,1.031], [0.511,0.542], and [0.641,0.671], respectively; for HL they are [0.381,0.403], [0.025,0.027], and [0.222,0.237].

Sensitivity and ablation tests use the same paired seeds. Increasing reaction delay by 50% raises evacuation time to 126.0 s (95% CI [122.0,129.9]), whereas homogeneous agents reduce it to 106.9 s ([104.5,109.3]). Scaling social influence by ±50% leaves the aggregate results close to the baseline. Removing social coupling, the robot/verbal channel, or both changes mean human loss only from 0.229 to approximately 0.230 and leaves evacuation time virtually unchanged; the overlapping confidence intervals make the component-level contribution statistically non-conclusive. Reaction delay and population heterogeneity therefore have a stronger effect in this configuration than the tested social-weight range.

7. DISCUSSION

Social Automation reduces control energy and selected human-aware measures relative to classical static guidance, at the cost of longer evacuation time, while HITL attains the lowest cognitive-load measure. The sensitivity analysis shows that reaction delay and population heterogeneity affect the results more strongly than the tested range of social-coupling strength. The small ablation effects further indicate that the added social and robot/verbal mechanisms contribute only marginally under the reported configuration.

The principal contribution is a common discrete-time feedback representation of the technical state, individual human states, and network-mediated intention. The objective function makes trade-offs among effectiveness, control effort, safety, cognitive load, trust, acceptance, and delay explicit. The projected DeGroot-type update provides a bounded marginal consensus mechanism for the zero-input social layer (DeGroot 1974), but it is not a stability proof for the complete coupled system. The framework is therefore useful for testing how alternative policies redistribute performance across technical and human dimensions and for making human-aware costs and constraints explicit.

The policy ranking also depends on which cost is treated as primary. Classical guidance minimises evacuation time, whereas HITL and Social Automation alter the distribution of control effort and human load. The difference between HITL and Social Automation is especially informative. HITL reacts to congestion through periodic signage updates, while Social Automation combines route guidance with alarm, verbal, and social-state channels. Under the reported settings, these additional channels do not shorten evacuation time, but they change how effort and cognitive load are allocated. Policy selection should therefore follow the operating objective rather than a single aggregate ranking. Speed may be decisive in a time-critical evacuation, while control expenditure and peak individual load may receive greater weight in routine guidance or prolonged supervision. This is consistent with the principle that the appropriate level of automation depends on the task and on the human consequences of allocating control functions (Parasuraman et al. 2000; Kaber and Endsley 2004).

The comparison also clarifies the role of the weights in equation (9). The coefficients make design priorities explicit, but one fixed weight vector cannot represent every operating condition. A practical controller can first impose admissible bounds on safety, delay, and maximum individual load, and then optimise the remaining performance measures within those bounds. Reporting a weight-sensitivity study would show when the preferred policy changes and whether small adjustments produce abrupt changes in behaviour. A Pareto analysis of evacuation time, control energy, mean

load, and maximum load would be more informative than collapsing all outcomes into one score. Population averages should also be accompanied by distributions across agents. Because the social network is directed, agents with different network positions may receive different levels of guidance or indirect influence even when the population mean is favourable. The maximum-load measure partly addresses this issue, but quantiles and node-level summaries would connect the fairness constraint to observable outcomes.

These extensions require no change to the model architecture. Because the simulator logs transitions on a common time index, each objective term can be retained for every run and agent, allowing score changes to be traced to a cost component, policy action, or network position.

8. LIMITATIONS AND FUTURE WORK

The model currently uses a fixed social graph and simplified queue and robot dynamics. Future studies should evaluate whether the conclusions transfer to independent human-subject data and validated crowd or HRI benchmarks.

A defensible validation programme should separate calibration from evaluation: estimate human-state coefficients and distributions from observed reaction delay, compliance, stress, cognitive-load proxies, trust, and route choice; test the calibrated model on independent scenarios using trajectory-level and aggregate outcomes; and repeat the comparison under changes in topology, population composition, and disturbances. Further work should also examine time-varying social networks, nonlinear delayed-system stability, richer robot-human interaction, and adaptive control.

9. CONCLUSION

This paper presents social automation as a control-oriented framework integrating technical, human, and social dynamics. The discrete-time formulation separates socially updated intention from other human states and supports explicit human-aware objectives and constraints. In the reported experiment, Social Automation reduces control energy and selected load indicators relative to classical guidance but does not improve evacuation time or outperform HITL on cognitive load. The small ablation effects reinforce the methodological contribution and identify calibration and external validation as the next steps.

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MIXED-MODEL SIMULATION TO ENABLE HUMAN-AWARE DECISION-MAKING IN RECONFIGURABLE MANUFACTURING SYSTEMS

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Agent-based Simulation, Discrete Event Simulation, Human-centric, Reconfigurable Manufacturing System, Simulation Modeling.

ABSTRACT

Being human-centric is one of the key criteria for Industry 5.0. For such a purpose, manufacturing decisions must take human factors into consideration during operation. Yet, for a system paradigm such as a reconfigurable manufacturing system, high complexities during system analysis are inevitable, especially when human factors are taken into account. This is caused by the essence of the system: the varying reconfiguration actions that introduce uncertainties constantly. Therefore, this paper aims at enabling manufacturing decisions in a human-aware approach for reconfigurable manufacturing systems. The reconfiguration actions are linked with human fatigue, recovery and skill. Accordingly, a discrete event simulation is applied to model the manufacturing actions, with agent-based modeling used to model the individual human states, making it a mixed-model simulation that captures all the elements needed to be modeled. The paper first proposes a methodology for the human-aware decision-making in reconfigurable manufacturing systems and then models it using mixed-model simulation. The results are presented in a baseline model and three strategy-based models. The baseline model does not take decisions based on human factors. The strategy-based models prioritize human states to make reconfiguration decisions. The simulation result shows that prioritizing humans doesn't necessarily sacrifice production performance.

INTRODUCTION

Human factors is of high importance in Industry 5.0. For years, Industry 4.0 prioritized a seamless, data-driven inter-connectivity where the Internet of Things (IoT) prioritize maximizing throughput of the line. But a pivot is occurring where Industry 5.0 brings the human back from the periphery, re-centering the "Human-in-the-loop" not as a mere cost factor or a passive biological resource, but as the very backbone of industrial resilience (Ghobakhloo et al. 2024). Currently, the operator is increasingly recognized as a critical agent in industrial systems (Keshvarparast et al. 2024; Yao et al. 2023; Baratta et al. 2023).

Among the manufacturing system paradigms that align with Industry 5.0 principles, Reconfigurable Manufacturing Systems (RMS) emerged as an effective approach to volatile markets, providing capacity exactly as needed (Gu et al. 2025). However, the increasing flexibility of such system paradigms can put additional demands on human operators. As the system configurations evolve dynamically, the operators are required to cope with operational complexity, which increases cognitive and physical load. Unlike the rigid automation of the previous decade, Industry 5.0 leverages human cognitive flexibility to navigate the non-linear unpredictability of these reconfigurable layouts. In this perspective, the human operators are not replaced by automation but are acting as an essential partner in maintaining reliability and resilience. At the same time, human performance in a production system is inherently dynamic rather than static. Studies have shown that a massive percentage of industrial disruptions are tied directly to human variability, such as fatigue level, skill level, resilience, etc. Sometimes, these factors are interrelated rather than independent, evolving continuously with time, thus increasing the problem complexity (Yasue et al. 2024). Yet, a system with socio-technical harmony shall not treat operators as unfeeling, static parameters. Treating operators as autonomous agents with dynamic variables enables an RMS to detect early signs of exhaustion. Then, configuration logic can reassign the tasks in response to operators' dynamic states.

Much research has been done to integrate human factors into manufacturing systems, few of them studies

specifically on human factors integrated into an RMS. Very commonly, the traditional systemic approaches use human performance factors as productivity factors, rather than the most important criteria in manufacturing decisions. Kolus et al. (2018) conducted a systematic literature review of 73 empirical studies and showed that fatigue is a central mechanism of performance degradation, with defect rate reductions of up to 86% when human factors are properly addressed in system design. Endo and Haraguchi (2024) incorporated a fatigue model into a skill-based operator allocation method for cell production systems, showing that fatigue slows down the learning progression and that classical learning curves alone are not sufficient to capture operator behavior. Hajri-Gabouj and Darmoul (2015) presented that configuration selection in RMS should not rely on merely technical performance because the working conditions are going to impact the human factors, thus reduce productivity or cause absence. And therefore proposed a methodology to integrate human factors and ergonomics indicators with Analytic Hierarchy Process (AHP) to choose configurations that balance both technical and human performance.

The system paradigm such as RMS introduces complexity for two reasons when human factors are integrated: First, the availability of various configurations places direct demands on the operator. The frequent reconfiguration actions require a high level of human adaptability that is often non-linear. Thus, the associated learning curve can be irregular, causing the system to behave in a way that classical models fail to anticipate. Then, the tension between RMS varying demands and operator recovery creates a cognitive bottleneck that challenges the researchers to account for. Despite their high level of automation, RMS still require numerous manual interventions that expose operators to significant physical constraints. Bortolini et al. (2019) proposed a methodological framework for integrating safety, ergonomics and human factors into RMS design, identifying fatigue and stress as critical risk areas. Napoleone et al. (2023) showed through a systematic review that operator-related reconfigurability enablers are largely understudied, with only 7 out of 59 articles addressing this topic. Esmaeili-Qeshlaqi et al. (2024) integrated operator fatigue and skill level into a dynamic scheduling model for RMS, confirming that ignoring these factors leads to significantly suboptimal solutions. However, none of these studies considers operator fatigue as an active trigger for reconfiguration decisions. The physiological state of the operator is never used as the main criterion to determine when or how the system should reconfigure. This gap is precisely what the present work aims to address.

This research expects to treat human factors as an active, real-time trigger for logical reconfiguration, rather than a passive consequence of production. We aim to build a framework to integrate human factors into reconfigurable manufacturing analysis in a novel approach: not on the design level, but on the operation level, by making reconfiguration decisions mainly based on human factors, therefore enabling a "human-centric" RMS. The human-related factors are all assumed to evolve with time, and therefore the model is composed of time-dependent variables. By monitoring the operator's state for an entire day, for example, working or resting, the system can continuously update the human conditions. The objective is to develop a fatigue-aware reconfiguration strategy capable of dynamically adapting production routing mainly based on the operator's fatigue level, thereby maintaining system performance while preserving the well-being of the human operator.

Modeling of the proposed framework requires more than a simple flow analysis but a hybrid simulation architecture. Therefore, the software AnyLogic is applied for modeling of this human-centric RMS in a mixed-model simulation approach. By integrating Agent-Based Modeling (ABM) with Discrete Event Simulation (DES), the operator's states can be explicitly incorporated into the system logic, and the events during the operations can be captured. One baseline model and three strategy-based models were built to compare the performance of the production line when human factors are integrated in different ways.

This paper is organized in the following structure. Section *Methodology and Model* presents the methodology of the mixed-model simulation to enable human-aware decision-making in RMS. The detailed modeling of the human factors and operational factors in the RMS is presented, using a mixed model by combining DES and ABM. The performance index of operators and the system are then added into the model. Section *Reconfiguration Strategy and Results* presents the three decision strategies that are human-aware in different approaches, but all take human factors as the main decision criteria. Comparisons among the strategy-based models and baseline model are made to analyze the potentials of the proposed human-centric strategies. Section *Conclusion* discuss on this paper's contributions, limitations, and future perspectives.

ACRONYMS

The acronyms used throughout this paper are summarized in Table 1.

METHODOLOGY AND MODEL

This section introduces the methodology framework proposed for the research and then elaborates on the modeling process of the proposed system.

Table 1: Acronyms

Acronym	Definition
ABM	Agent-Based Modeling
AHP	Analytic Hierarchy Process
DES	Discrete Event Simulation
HEP	Human Error Probability
HF	Human Factors
IoT	Internet of Things
KPI	Key Performance Indicator
MLT	Manufacturing Lead Time
NFS	Normalized Fatigue Score
NLS	Normalized Learning Score
RMS	Reconfigurable Manufacturing System
WIP	Work-in-Process

Methodology Framework

The methodology (Fig.1) developed for this research employs a dynamic, integrated approach to capture the interdependency between system reconfigurability and human states. The research object is a human-aware RMS. The RMS is defined with its reconfigurability characteristic, its structure, and its functioning mode. It accords with the following characteristic:

- *Layout Reconfigurability:* The system is designed with modular cells that allow for rapid adjustments in capacity through the addition or removal of modules.
- *Sequential Process Line:* The nominal production flow follows a set sequence of operations, but the methodology accounts for convertibility, allowing for changes in the operating sequence or routing in response to system status.
- *Human-Machine Collaboration:* The model moves beyond purely technological design by placing human operators at the heart of the system, where their dynamic performance directly impacts decision-making and overall system output.

Then, the human factors include two approaches, the simple fatigue model that only concerns fatigue and skill and an advanced model that concerns fatigue, skill and recovery. In the following sections of this article, only the *fatigue recovery model* is presented, as it's the more accurate and advanced way to present the human states. The fatigue will decrease during micro-pauses or regular breaks, simulating realistic work-rest cycles.
The performance indicators are the commonly used indicators in production line management. The baseline model including all the information above is then built in mixed-model simulation environment, combining DES and ABM. DES captures all the events happening during the operations, and ABM captures the human agent

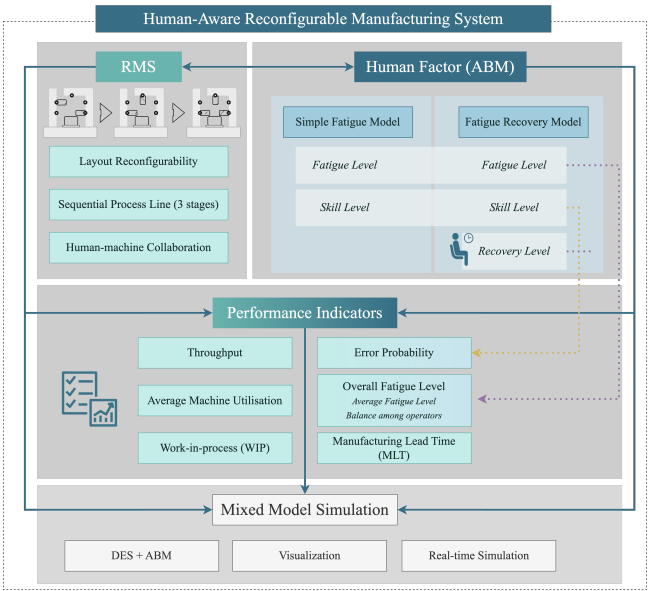

Figure 1: Methodology for Human-Aware Decision-Making in an RMS

behavior and states.
Although this paper do not take productivity as the main criterion, it is still interesting to observe its value for an analysis. By leveraging the mixed model within the AnyLogic simulation environment, the proposed framework facilitates real-time visualization and analysis of how human factors influence, and are influenced by, the operational structure of a Reconfigurable Manufacturing System (RMS).

Baseline Model

The baseline model is first built, with human factors integrated in the RMS, but reconfiguration decisions are not made based on human factors. Yet, human fatigue and skill do impact the production line in a conventional way. The following is the detailed baseline model with its system definition and assumptions, integrated human factors, and performance index.

1- System Definition and Assumptions

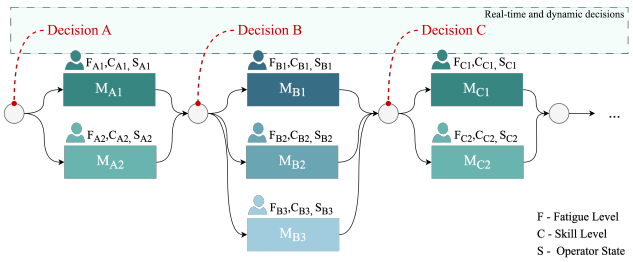

Figure 2: Sequential Reconfigurable Manufacturing System with Operators Integrated

The manufacturing system studied in this article is a typical sequential RMS (Fig.2) (Gu et al. 2025). Each process can have multiple choices of machines, mutually interchangeable, and each machine has an operator accordingly. Therefore, ideally, the process in this illustration has 12 configuration choices (with process A having 2 machines M_{A1}, M_{A2}, process B having 3 machines M_{B1}, M_{B2}, M_{B3}, and process C having 2 machines M_{C1}, M_{C2}). At the beginning of each process, a decision is made to decide which path the task will go to. Decisions are made in real-time and dynamically during the process.
The following assumptions are made:

- The fatigue level and skill level can be captured and presented based on functions evolving with time.
- The operator can have two states: {*Work, Rest*}. The working time will accumulate skill and fatigue; the resting time will reduce fatigue.
- The operators in each phase share the same skill category, making them interchangeable when needed.
- The tasks arrive at a constant rate and must go through each phase to be finished.
- Operators are assumed to be identical in every respect, with the same susceptibility to fatigue and learning.
- Any time when there is no workpiece on the machine is defined as a break, thereby triggering fatigue recovery.
- Each machine has its own dedicated queue; workpieces waiting for a busy machine accumulate independently in that machine's queue, which is particularly relevant for Strategy III (bottleneck-based operator reallocation)

2- Integration of Human-Aware Model
To present the dynamic influence of operators on the RMS, the DES is coupled with ABM for this subsection. The operators are seen as dynamic agents in the system. Specifically, an operator's fatigue level accumulates during the production process based on the duration of the task. When not working, the operator recovers, thereby reducing the fatigue level. Operators learn during the machining process, thereby improving their skill level. Fatigue and skill levels are updated in real time after each machining task is completed, and a function combining both factors influences the operator's probability of error in real time. For an operator, the key dynamic variables are: Skill ($skill \in [0,1]$) and Fatigue ($fatigue \in [0,1]$). The adopted formulation is developed based on combining the learning-fatigue model proposed by Digiesi et al. (2009) and Givi et al. (2015), in which the processing time of operator i is simultaneously affected by accumulated experience and current fatigue:

$$t_{x_i} = T_1 + \xi T_1 (x_i + u_i)^{-b} + (T_m - T_1) F_i, \qquad (1)$$

where T_1 is the basic processing time in the absence of fatigue and skill impact, T_m is the upper reference processing time, x_i is the number of units produced during the current work cycle, u_i denotes the experience accumulated during previous cycles, b is the learning index, and $F_i \in [0,1]$ is the operator's current fatigue level. ξ is the coefficient of the learning's impact on the processing time. Thus, accumulated production experience progressively reduces the learning-related component of processing time, whereas fatigue introduces an additional time penalty. Regarding the original formulations, please refer to the learning-forgetting research of Digiesi et al. (2009) and Givi et al. (2015).
Fatigue follows a nonlinear work–recovery process. During a working period of duration t_i, the fatigue level is updated as

$$F(t_i) = R(t_{i-1}) + [1 - R(t_{i-1})]\left(1 - e^{-\lambda t_i}\right), \qquad (2)$$

where $R(t_{i-1})$ is the residual fatigue inherited from the previous cycle and λ is the fatigue accumulation rate. During an idle or resting period of duration τ_i, the residual fatigue decreases according to

$$R(t_i) = F(t_{i-1}) e^{-\mu \tau_i}, \qquad (3)$$

where μ denotes the recovery rate, τ_i denotes recovery duration during cycle i. In the simulation, any period during which no part is processed by the operator is considered a recovery period.
The resulting human state also affects production quality. The normalized fatigue score (NFS) and normalized learning score (NLS) are defined as

$$NFS = \frac{F(t)}{F_{\max}}, \qquad NLS = \frac{t_x}{T_1}, \qquad (4)$$

and are converted into fatigue and learning utilities, $u_f = \beta NFS$ and $u_l = \alpha NLS$, where β converts the fatigue indicator into fatigue utility, and α converts the learning indicator into learning utility. Their weighted combination is expressed as

$$U(l, f) = w_l u_l + w_f u_f, \qquad (5)$$

from which the human error probability is obtained using the logarithmic mapping proposed by Givi et al. (2015):

$$\log\big(HEP\big) = 6 \log\big(U(l, f)\big). \qquad (6)$$

Equivalently, the human error probability can be expressed in power form as

$$HEP = U(l, f)^6. \quad (7)$$

This equivalent formulation is directly implemented in the AnyLogic simulation environment to compute the human error probability from the combined utility. The exponent is introduced to represent the nonlinear relationship between the combined utility and the human error probability. The operator states are updated continuously and synchronised with the production-flow model. In the baseline scenario, fatigue, learning, processing time and error probability influence system performance, but they are not used to trigger human-factor-based reconfiguration decisions. They subsequently serve as decision variables for the reconfiguration strategies introduced in the following section.

3- Performance Indicators and Feedback Logic

The methodology treats the outputs of the RMS-HF interactions as a set of multidimensional Performance Indicators (KPIs). These indicators are extracted in real-time to provide a comprehensive view of system health:

- *Operational KPIs:* These include Throughput (conform pieces/hour), Lead Time, Work-in-Process (WIP) per cell, and Average Machine Utilization.
- *Human-Centric KPIs:* The methodology tracks overall fatigue levels and workload balance among operators – quantified as the *fatigue dispersion*, the standard deviation of the operators' fatigue levels across the workforce at the end of the simulated period – as well as the frequency of required micro-pauses.
- *Quality KPIs:* These measure Error Probability through defect percentages, rework rates, and first-pass yield.

The final stage of the methodology involves a closed-loop feedback mechanism where the defined KPIs serve as triggers for system reconfiguration. By monitoring variables such as queue saturation or operator exhaustion, the system can trigger layout changes to leverage human factors and increase overall productivity. This entire architecture is simulated in real-time using AnyLogic, providing a robust visualization tool to reconcile technical efficiency with operator well-being.

RECONFIGURATION STRATEGIES AND RESULTS

Based on the methodology proposed in Section *METHODOLOGY AND MODEL*, different strategies can be proposed. The purpose is to provide different modeling and integration approaches for human factors in RMS, and at the same time, to make a comparison among the performance results. The following paragraphs explain the three strategies proposed in this research.

Strategy I: Fatigue Threshold-Based Reconfiguration

The first strategy (Fig.3) is based on the continuous monitoring of each operator's fatigue level. During production, the fatigue value of every operator is compared with a predefined critical threshold. When an operator reaches this threshold, the corresponding workstation is temporarily placed on hold and the operator enters a recovery period.

Figure 3: Illustration of Strategy I: Fatigue Threshold-Based Reconfiguration

During this interruption, parts are redirected toward the other available machines belonging to the same processing stage. The operator remains unavailable until the recovery condition is satisfied, after which the workstation is released and becomes available again. The recovery condition is triggered by a time period of resting finishing. The resulting decision logic can be expressed as follows:

$$fatigue_i(t) \geq F_{\text{th}} \quad \Rightarrow \quad state_i(t) = \text{Rest}, \quad (8)$$

where $fatigue_i(t)$ is the fatigue level of operator i at time t, and F_{th} is the predefined fatigue threshold. The transition back to the working state is expressed as:

$$fatigue_i(t) \leq F_{\text{rec}} \Rightarrow state_i(t) = \text{Work}, \qquad F_{\text{rec}} < F_{\text{th}}, \quad (9)$$

where F_{rec} denotes the recovery threshold. The recovery threshold must be lower than the fatigue threshold to prevent frequent switching between the working and resting. This strategy introduces fatigue as a direct reconfiguration trigger. Its main objective is to prevent operators from continuing to work under excessive fatigue, thereby limiting the associated increase in operation time and error probability. However, removing an operator from production also reduces the available processing capacity. If several operators reach the threshold simultaneously, the remaining operators may receive an additional workload and the system may experience temporary congestion.

Strategy II: Dynamic Routing Based on Operator Fatigue

The second strategy (Fig.4) maintains the predefined work-and-rest schedule and does not introduce additional fatigue-triggered breaks. Instead, it modifies the

routing of parts in real time. At each routing decision, the fatigue levels of the available operators assigned to the considered processing stage are evaluated. The incoming part is then directed toward the machine operated by the least fatigued available operator.

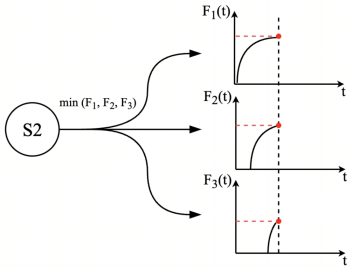

Figure 4: Illustration of Strategy II: Dynamic Routing Based on Operator Fatigue

For a set of available operators $\mathcal{O}_j(t)$ at processing stage j, the selected operator is given by:

$$i^* = \underset{i \in \mathcal{O}_j(t)}{\arg\min} \{fatigue_i(t)\}. \tag{10}$$

The corresponding part is assigned to the machine operated by i^*. If only one machine is available, the part follows the remaining feasible route.
This strategy aims to distribute the workload more evenly among operators and to avoid the excessive solicitation of a single operator. Unlike Strategy I, it preserves the number of active workstations and therefore limits production interruptions. Nevertheless, under sustained production demand, the continuous distribution of work may progressively increase the fatigue of all operators, since no additional recovery period is introduced.

Strategy III: Bottleneck-Based Operator Reallocation

The third strategy (Fig.5) focuses on the regulation of production bottlenecks. The system continuously observes the queues located upstream of the machines. When one workstation is identified as a bottleneck, an operator permutation is performed within the corresponding processing stage in order to improve the flow through the constrained resource.
The bottleneck machine is identified according to the largest observed queue:

$$m^* = \underset{m \in \mathcal{M}_j}{\arg\max} \{Q_m(t)\}, \tag{11}$$

where $\mathcal{M}_j$ is the set of machines belonging to processing stage j, and $Q_m(t)$ is the queue length upstream of machine m at time t.

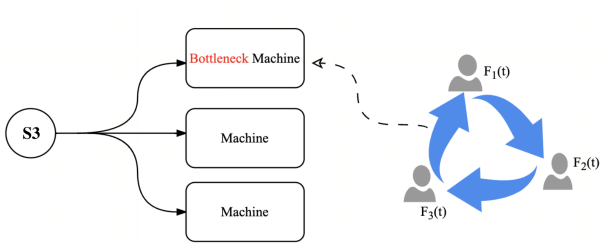

Figure 5: Illustration of Strategy III: Bottleneck-Based Operator Reallocation

Once the bottleneck is detected, the operator assigned to the constrained machine is exchanged with another operator from the same stage. To prevent excessive instability, two successive permutations are separated by a minimum time interval:

$$t - t_{\text{last}} \geq \Delta t_{\text{perm}}, \tag{12}$$

where t_{last} denotes the time of the previous permutation and Δt_{perm} is the minimum permitted interval between two reallocations.
In contrast with Strategies I and II, this strategy is primarily driven by the physical state of the production flow rather than directly by the operator's fatigue level. It seeks to reduce queue accumulation and restore a balanced production flow. However, because fatigue is not used as the principal selection criterion, an already fatigued operator may be reassigned to a critical workstation.

Simulation Results

Each scenario, i.e. the baseline model and the three reconfiguration strategies, is simulated once over a single simulated working day, using generated input parameters. The data were selected solely for model execution and do not refer to any specific industrial case and will be provided at request. Please refer to the Appendix for the detailed daily work-rest schedule used as simulation input (Table 3).

Baseline Model Results
First, the performance of the baseline model without reconfiguration is analyzed. The corresponding results are presented in Table 2. All the other input parameter settings are provided at request.
The baseline system achieves a high production volume of 1455 units, while maintaining an overall error rate of 1.2%. However, the combined analysis of the performance indicators reveals a major workload-balancing problem associated with the presence of a bottleneck. Although the average fatigue level remains at 0.38, the high fatigue dispersion of 0.63 indicates substantial dif-

Table 2: Comparison of the Baseline Model and the Three Reconfiguration Strategies

Indicator	Baseline model	Fatigue threshold	Dynamic routing	Bottleneck management
Average fatigue	0.38	0.326	0.358	0.35
Fatigue dispersion	0.63	0.49	0.665	0.60
Error probability	0.012	0.0094	0.0013	0.008
WIP	21	24.2	19	13
MLT	149 min	163 min	138 min	84 min
Total production	1455	1399	1466	1490

ferences in workload among operators. Some operators are therefore considerably more solicited than others. These disparities negatively affect the production flow, shown by WIP=21 and MLT=149 minutes. The disparities may also contribute to a deterioration in production quality. Therefore, the baseline results have proven the need for applying reconfiguration strategies aiming to improve workload distribution while not decreasing industrial performance too much.

Strategy-Based Model Results

The three implemented reconfiguration strategies are compared with the baseline model in Table 2. These strategies include the fatigue-threshold strategy, the dynamic-routing strategy, and the bottleneck-management strategy.

Strategy I: Compared with the baseline model, the average fatigue decreases from 0.38 to 0.326, corresponding to a reduction of 14.2%. Fatigue dispersion (the standard deviation of the operators' fatigue levels) decreases from 0.63 to 0.49, corresponding to a reduction of 22.2%, while the error probability decreases from 0.012 to 0.0094, corresponding to a reduction of 21.7%. WIP increases from 21 to 24.2, MLT increases from 149 to 163 minutes, and total production decreases from 1455 to 1399 units. The immediate interruption of operators reduces production-flow continuity and increases congestion. This strategy is consequently more suitable for environments in which the operators' fatigue level is strictly prioritized over production rate.

Strategy II: Compared with the baseline model, average fatigue decreases from 0.38 to 0.358, corresponding to a reduction of 5.8%. However, fatigue dispersion increases from 0.63 to 0.665, corresponding to an increase of 5.6%. This indicates that the strategy reduces overall fatigue but may create an imbalance in workload distribution, as some operators are used more intensively than others. The error probability decreases significantly compared with the baseline model, by 89.2%. From an industrial perspective, the strategy improves system fluidity. WIP decreases from 21 to 19, MLT decreases from 149 to 138 minutes, and total production increases from 1455 to 1466 units. These results reflect reduced congestion, better use of resources, and shorter waiting times. The strategy therefore represents a favorable compromise between system performance and implementation, although it requires real-time operator monitoring and a more complex decision-making system.

Strategy III: The bottleneck-management strategy dynamically reassigns operators when a queue becomes full. Compared with the baseline model, average fatigue decreases from 0.38 to 0.35, fatigue dispersion decreases from 0.63 to 0.60, and error probability decreases from 0.012 to 0.008. The industrial indicators also improve substantially. WIP decreases from 21 to 13, MLT decreases from 149 to 84 minutes, and total production increases from 1455 to 1490 units. The strong reduction in error probability of 33.3% is associated with the nonlinear relationship between fatigue and error probability introduced in the model.

Among the three strategies, bottleneck management provides the most balanced results. It simultaneously improves operator-related indicators, reduces congestion and production lead time, and increases throughput. It therefore appears to be the most effective approach for a human-aware RMS.

Furthermore, a practical consideration is that implementing these decision-making strategies requires real-time fatigue monitoring, either through wearable sensors or self-reporting, raising potential privacy concerns, particularly in regions subject to the General Data Protection Regulation (GDPR). This further underscores the significance of the simulation-based approach adopted in this study.

CONCLUSION

This research aims to build a human-aware RMS and make reconfiguration decisions human-centric. The mixed-model simulation is built for modeling the proposed problem: using DES to present all the discrete events during the operation of the RMS and using ABM to present all the agent states and behaviors. The human states are dynamically captured and have a direct impact on the system. Data were generated to run the simulation, and comparisons among the baseline model and strategy-based models were then analyzed. The baseline model shows higher average fatigue, higher fatigue dispersion, and higher error probability compared with the three strategies, whose results reveal clear trade-offs. The fatigue threshold strategy reduces operators' fatigue level but weakens production performance, while the dynamic routing strategy improves flow and minimizes error probability the most, at the cost of a higher fatigue dispersion. The bottleneck management strategy achieves the best balance among fatigue, error probability, and flow and has the highest production in the end. These results have shown that prioritizing human well-being does not necessarily sacrifice production performance.

The contribution of this paper lies in the unique decision-making that prioritizes human factors in an RMS. Unlike the existing human-factor-integrated re-

search where it is merely an index, this research manages to fulfill human-centric RMS.
This research proposed three strategies for human-aware decision-making in RMS, based on the common approaches and difficulty for implementation in the simulation software. But there could be more possibilities for the strategies, and it could get more complicated in a real scenario. This is a limitation of this research and could be further developed to increase the applicability level. It is the same limitation regarding the simulation part, a set of data from real industrial case and a sensitivity analysis would have added value to the interpretations.
Furthermore, human factors are not limited to fatigue level and skill level. Some research divides human factors into categories of cognitive and physical, presenting a more realistic model of the operators.

APPENDIX

Table 3 shows the daily routine of an operator in the mixed-model simulation, as a set of input parameters, and are part of the discrete event triggers. The data are generated by assumptions for simulation, and can be adjusted according to real cases: The model file and source code are available upon reasonable request for research and reproducibility purposes.

Table 3: Daily Work-Rest Schedule

Period	Duration
Work period	4×1 h 52 min 30 s
Short breaks	2×15 min
Lunch break	1 h 30 min
Long rest	14 h 30 min

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AGENT-BASED SIMULATION OF HUMAN-ROBOT INTERACTION IN ASSISTED LIVING ENVIRONMENTS: TRUST, EMOTION AND ADAPTIVE ROBOT BEHAVIOUR

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KEYWORDS
Agent-based simulation, human-robot interaction, older adults, socially assistive robots, trust, emotion, assisted living, reinforcement learning

ABSTRACT
This paper presents a Mesa agent-based model of repeated interaction among an older adult, a socially assistive robot, and a caregiver. Robot actions update trust, emotion, technology acceptance, and engagement under fixed, reactive, and online tabular Q-learning control. A factorial Monte Carlo experiment combines four synthetic user profiles, three reliability levels, two caregiver-availability levels, and 50 replications per cell, yielding 3,600 runs and 6.912 million transitions. In the reference condition, reactive and adaptive control improve trust and acceptance over fixed scheduling. Online Q-learning does not consistently exceed reactive control and uses caregiver escalation more often. Because the state equations and profiles are modelling assumptions rather than empirical estimates, the results demonstrate mechanisms and parameter sensitivity, not clinical effects.

INTRODUCTION

Background and Motivation

The aging population and pressure on care systems motivate socially assistive robots (SARs) for reminders, companionship, and everyday support. Unlike robots designed mainly for physical assistance, SARs act through social interaction (Feil-Seifer and Mataric 2005). Older adults value practical help, controllability, voice interaction, and companionship, but preferences vary widely (Heerink et al. 2010; Wu et al. 2026). Technical performance alone therefore does not ensure acceptance; personalization, communication style, and perceived usefulness also matter (Cho et al. 2024).

Trust, technophobia, perceived intrusiveness, and prior experience shape acceptance (Zafrani et al. 2023). Evidence on trust in SARs still relies heavily on laboratory studies and self-report, with limited standardization and contextual coverage (Gul et al. 2025). Reviews accordingly recommend personalized robot behaviour embedded in the wider care system (Jones et al. 2025; Olatunji et al. 2025).

Research Gap and Contribution

Long-term HRI experiments are costly and cannot readily cover combinations of user traits, robot reliability, and caregiver availability. Agent-based modelling (ABM) represents individual states, local decisions, and stochastic events across counterfactual scenarios; the ODD protocol supports transparent specification (Grimm et al. 2020). ABM has long been used to study emergent human behaviour (Bonabeau 2002), and recent work models trust in human-robot teams (Kalluri 2026). This study contributes an assisted-living triad in which user state changes over 60 simulated days, the robot follows one of three policies, and the caregiver may rescue or assume control.

The study asks: RQ1: How do robot control strategy and reliability affect long-term trust, acceptance, and emotional state? RQ2: Does the relative performance of the strategies depend on the older-adult profile? RQ3: What trade-offs arise among robot adaptability, unresolved needs, and caregiver involvement?

The model is exploratory: its parameters encode plausible HRI mechanisms but are not empirical psychological constants.

RELATED WORK

Socially Assistive Robots and Older Adults

The Almere model links acceptance to usefulness, ease of use, and interaction-related constructs (Heerink et al. 2010). Reviews add privacy, autonomy, communication style, social role, and personal characteristics (Jones et al. 2025; Wu et al. 2026). Technophobia may outweigh the positive effect of trust (Zafrani et al. 2023), while trust research needs objective and subjective measures in realistic settings (Gul et al. 2025).

Benefits are context-dependent. A meta-analysis of 19 studies and 1,083 participants found reduced loneliness under some intervention settings (Mehrabi and Ghezelbash 2025), and home studies demonstrate cognitive and affective support (Cho et al. 2024). These findings motivate models of repeated interaction rather than isolated encounters.

Agent-Based Modelling and Adaptive Decision Making

ABM is suited to heterogeneous actors whose repeated local interactions produce system-level outcomes (Bonabeau 2002). The present model follows ODD and uses Mesa 3 for execution and data collection (Grimm et al. 2020; ter Hoeven et al. 2025). Trust-aware HRI research also embeds estimated trust and human preferences in sequential decision making and reward design (Bhat et al. 2024). Here, tabular Q-learning (Watkins and Dayan 1992) is used only as an interpretable online adaptive baseline; coarse discretization limits generalization compared with function-approximation methods.

AGENT-BASED MODEL

Agent States and Interaction Loop

The model contains an older-adult agent, a robot, and a caregiver. The environment generates daily-activity, reminder, and social-support needs. Every 30 min, the robot observes a noisy user state, selects an action, and receives success, failure, or refusal feedback.

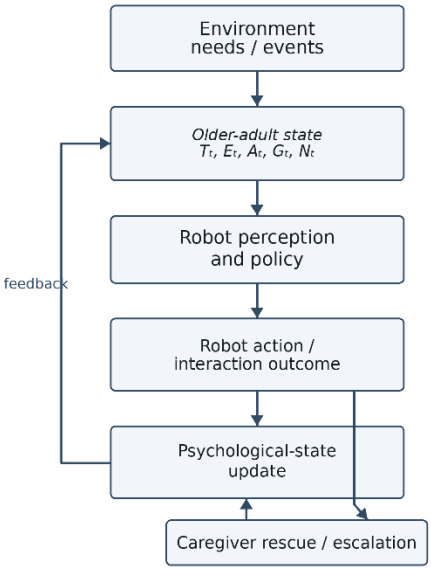

Figure 1: Causal Architecture of the Older-Adult-Robot-Caregiver Simulation Loop

The caregiver may intervene after failure, direct escalation, or excessive waiting. The user's state is then updated before the next period. Equation (1) defines the state vector:

$$\mathbf{x}_t = [T_t, E_t, A_t, G_t, N_t]^{\mathsf{T}} \tag{1}$$

Here, T is trust, E is emotional valence in [-1, 1], A is technology acceptance, G is engagement, and N is current need strength. Table 1 summarizes these variables and the additional bounded parameters. Four synthetic profiles—technology-positive, cautious, technology-anxious, and high-support-need—are experimental parameter sets, not clinical or demographic groups.

Table 1: Main Model Variables

Symbol	Range	Interpretation
T	[0,1]	trust in robot
E	[-1,1]	emotional valence
A	[0,1]	technology acceptance
G	[0,1]	interaction engagement
N	[0,1]	current need strength
Z	[0,1]	technology anxiety
Rel	[0,1]	robot reliability
C	{0,1}	caregiver intervention

Perception and Psychological State Update

The robot observes the noisy state in (2), where ε_t is zero-mean Gaussian noise, σ_p is its standard deviation, and I is the identity matrix:

$$\hat{\mathbf{x}}_t = \mathbf{x}_t + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}\left(\mathbf{0}, \sigma_p^2 \mathbf{I}\right) \tag{2}$$

Perceived values are clipped to their valid intervals; σ_p = 0.08 at baseline. Trust is updated asymmetrically by (3): success uses remaining headroom, whereas failure and refusal reduce current trust.

$$T_{t+1} = \text{clip}_{[0,1]}[T_t + \alpha_s(1 - T_t)S_t - \alpha_f T_t F_t - \alpha_r T_t R_t] \tag{3}$$

S_t, F_t, and R_t indicate success, failure, and refusal. In (3), $\alpha_s = 0.032 N_t$, $\alpha_f = 0.075$, and $\alpha_r = 0.008 + 0.016 I_t$, where I_t is intrusiveness. Acceptance uses analogous gains: 0.025 for success, 0.045 for failure, and $(0.010 + 0.020 I_t)A_t$ for refusal. Equation (4) updates emotion through mean reversion and interaction outcomes:

$$E_{t+1} = \text{clip}_{[-1,1]}[(1 - \lambda)E_t + \beta_s S_t - \beta_f F_t - \beta_i I_t + \beta_c C_t] \tag{4}$$

In (4), λ is the mean-reversion rate; β_s, β_f, β_i, and β_c weight success, failure, intrusiveness, and successful caregiver rescue C_t. Robot proposals are accepted stochastically according to (5):

$$\Pr(Y_t = 1) = \sigma(\theta_0 + \theta_T T_t + \theta_A A_t - \theta_Z Z + \theta_N N_t + \theta_E E_t + \theta_L L - \theta_I I_t) \tag{5}$$

In (5), σ is the logistic function, θ_0 is the intercept, and θ_T, θ_A, θ_Z, θ_N, θ_E, θ_L, and θ_I weight trust, acceptance, anxiety, need, emotion, interaction tolerance L, and intrusiveness I_t. Unresolved needs reduce emotion and engagement; caregiver rescue resolves the need and partly offsets emotional cost without restoring robot trust.

Robot Strategies

Fixed control addresses every pending need and issues a social prompt every four hours with probability 0.55. Reactive control acts when a need threshold determined by anxiety and trust is exceeded; otherwise it escalates. Adaptive control applies ε-greedy Q-learning to discretized trust, emotion, need, and acceptance, choosing engagement, caregiver escalation, or waiting. Equation (6) gives the update:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \eta[r_t + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (6)$$

Here, s_t and a_t are the state and action, r_t is the immediate reward, $\eta = 0.18$ is the learning rate, $\gamma = 0.88$ is the discount factor, and the ε-greedy exploration rate decays from 0.24 to 0.05. The reward is

$$r_t = w_s S_t - w_f F_t - w_r R_t - w_c C_t - w_w W_t \quad (7)$$

where W_t marks waiting with an unresolved need, and w_s, w_f, w_r, w_c, and w_w are fixed positive weights. Success receives the largest reward; failure and refusal are penalized, and caregiver use carries a smaller resource cost.

SIMULATION FRAMEWORK AND EXPERIMENT

Implementation and Reproducibility

The Python 3 implementation uses Mesa 3.3.1, NumPy 2.3.5, pandas 2.2.3, and Matplotlib 3.10.8. Independent NumPy streams govern events, robot outcomes, caregiver availability, exploration, and profile variation. Each 60-day run contains 1,920 half-hour steps ordered as need generation, observation, policy choice, outcome, escalation, state update, and logging. Deterministic seed blocks begin at 20261000. Q-learning and ε-greedy exploration continue during these runs; no separately pretrained test policy is evaluated.

Table 2: Experimental Design and Reproducibility Settings

Factor	Values / assumption
Horizon	60 days; 32 steps/day; 1920 steps/run
Profiles	P1 positive; P2 cautious; P3 anxious; P4 high-need
Strategies	fixed; reactive; adaptive Q-learning
Reliability	0.80; 0.90; 0.98
Caregiver availability	low: 0.35; high: 0.75 per opportunity
Perception noise	σ□ = 0.08 baseline
Monte Carlo	50 replications per factorial cell
Total experiment	72 cells; 3600 runs; 6,912,000 transitions
Software	Mesa 3.3.1; NumPy 2.3.5; pandas 2.2.3; Matplotlib 3.10.8

Synthetic Profiles and Event Process

Initial $T_0/A_0/E_0$ values are 0.65/0.72/0.20 (P1), 0.48/0.50/0.05 (P2), 0.32/0.34/−0.05 (P3), and 0.50/0.56/−0.02 (P4). Anxiety is 0.18, 0.45, 0.75, and 0.40; support need is 0.35, 0.48, 0.50, and 0.82. Runs add small within-profile Gaussian variation. Need occurrence follows a Bernoulli process affected by support need, autonomy, and emotion; beta-distributed intensity affects acceptance, trust change, and waiting cost.

Outcome Metrics and Statistical Analysis

Outcomes are final trust and acceptance, mean emotion, interaction acceptance, unresolved needs, robot resolution, and caregiver interventions per 100 needs. The reference scenario uses reliability 0.90, high caregiver availability, and 200 runs per strategy. We report means, standard deviations, paired t-tests on matched profile-seed combinations, and Cohen's d_z. Sensitivity tests vary perception noise and failure-related trust loss. No human participants or personal data are involved.

RESULTS

Strategy Comparison

Figure 2 shows that fixed control remains near initial mean trust because gains from assistance are offset by scheduled intrusions, refusals, and failures.

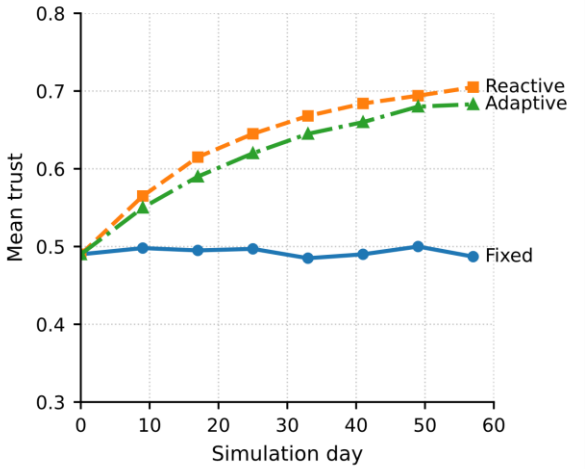

Figure 2: Mean Trust Trajectory in the Reference Condition (*Rel* = 0.90; high caregiver availability; 200 runs/strategy)

Reactive and adaptive control achieve higher trust by acting when assistance is more likely to be useful and accepted.

Table 3: Reference-Condition Results, Mean (SD)

Policy	T final	A final	Mean E	Accept.	Care/100
Fixed	0.487 (.100)	0.419 (.103)	-0.098 (.057)	0.468 (.126)	11.17 (5.63)
Reactive	0.709 (.087)	0.715 (.074)	0.041 (.012)	0.765 (.105)	20.25 (8.54)
Adaptive	0.680 (.128)	0.685 (.122)	0.046 (.011)	0.731 (.158)	31.01 (25.40)

Relative to fixed control, final trust rises by 0.221 under reactive control (paired $t(199) = 26.20$, $p < 0.001$, $d_z = 1.85$) and 0.193 under online Q-learning ($t(199) = 20.77$, $p < 0.001$, $d_z = 1.47$). Acceptance rises by 0.296 and 0.266, respectively. Q-learning produces slightly lower trust than reactive control (difference −0.028, $p = 0.0048$, $d_z = -0.20$), slightly higher mean emotion, and more caregiver interventions (31.0 vs. 20.3 per 100 needs). These results characterize online tabular learning with exploration, not adaptive control in general.

Heterogeneity and Reliability

No policy is optimal for every profile. Q-learning yields slightly higher trust than reactive control for technology-positive (0.736 vs. 0.724) and high-support-need profiles (0.731 vs. 0.721), whereas reactive control leads for anxious (0.685 vs. 0.562) and cautious profiles (0.705 vs. 0.693). Because training and exploration occur during evaluation, the anxious-profile gap may reflect exploration or coarse discretization rather than a general limitation of adaptive policies.

Reliability is the strongest varied driver of trust. For fixed/reactive/adaptive control, mean final trust at $Rel = 0.80$, 0.90, and 0.98 is 0.381/0.559/0.523, 0.492/0.715/0.691, and 0.618/0.873/0.853. Higher caregiver availability nearly eliminates unresolved needs but shifts work to humans, especially under Q-learning.

Sensitivity Analysis

Raising perception noise from 0.08 to 0.12 lowers trust and increases caregiver escalation. Results are highly sensitive to failure-related trust loss: reducing α_f from 0.075 to 0.050 raises final trust to 0.764, while increasing it to 0.100 lowers trust to 0.671.

Table 4: Sensitivity of Reactive Control in the Reference Scenario

Variant	Final T	Final A	Accept.	Care/100
Baseline	0.709	0.715	0.765	20.25
$\sigma_p = 0.04$	0.711	0.718	0.762	18.38
$\sigma_p = 0.12$	0.699	0.707	0.760	22.42
$\alpha_f = 0.050$	0.764	0.720	0.774	19.22
$\alpha_f = 0.100$	0.671	0.717	0.757	20.06

Table 4 therefore demonstrates parameter dependence rather than robustness. Until trust-loss and repair rates are empirically calibrated, numerical outputs should be read as illustrative mechanism tests.

DISCUSSION

Under the stated assumptions, state-aware policies outperform fixed scheduling because they avoid context-insensitive prompts and refusals. Reliability has the largest effect on long-term trust among the varied factors, consistent with HRI evidence (Gul et al. 2025). Online tabular Q-learning does not consistently exceed the transparent reactive rule and may favor caregiver escalation as a low-risk action. Reward design should therefore represent caregiver workload as well as user state. The finding is specific to this discretization, reward structure, and online exploration regime.

The contribution is an integrated, reproducible architecture that links psychological state, robot policy, and caregiver workload. It supports hypothesis generation and identifies parameter regions worth calibrating before field studies. Mesa also permits interactive visualization, although the reported experiments ran in headless mode.

LIMITATIONS AND FUTURE WORK

The psychological coefficients and synthetic profiles are modelling assumptions, not estimates from a clinical sample. Bounded trust, emotion, and acceptance variables cannot represent the full effects of aging, cognition, personality, culture, or relationship development. The environment assumes needs and escalation but omits navigation, collision avoidance, and sensory perception. Adaptive control is limited to tabular Q-learning; reward weights, discretization, and exploration can change caregiver use. Because learning occurs during evaluation and no pretrained test policy is reported, the results cannot establish the performance of a converged policy. Monte Carlo uncertainty does not imply external validity.

Future work should calibrate transitions with longitudinal data and combine objective and subjective trust measures (Gul et al. 2025). The model could include shared caregiver capacity, multiple residents, and navigation. A separate train/test protocol should compare a frozen Q-learning policy with reactive control and function-approximation methods. Contextual bandits or constrained RL could incorporate uncertain emotion recognition, explainability, minimum-trust constraints, and caregiver workload.

CONCLUSION

This reproducible ABM links older-adult state, robot reliability and policy, and caregiver availability. Across 3,600 runs, reactive and online tabular Q-learning policies outperform fixed scheduling in reference conditions, but Q-learning does not consistently exceed reactive control and requires more caregiver escalation. The study therefore supports qualitative claims about mechanism sensitivity, not the general superiority of

adaptive control or real-world effect sizes. Empirical calibration and separate policy training and evaluation are required before validation in HRI settings.

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Manufacturing Optimization Simulation

HOLONIC CONTROL ARCHITECTURE FOR RECONFIGURABLE MANUFACTURING SYSTEMS

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KEYWORDS

Holonic control architecture, ADACOR² model, multi-agent systems, reconfigurable manufacturing systems.

ABSTRACT

In this paper, we aim to model reconfigurable manufacturing systems that incorporate workforce using a holonic control architecture. Our goal is to model and analyze their algorithmic behavior under dynamic and distributed manufacturing conditions, by incorporating coordination mechanisms. Reconfigurable manufacturing systems provide a balance between flexibility and efficiency by enabling real-time physical (hardware) and logical (control, programming) reconfiguration. The selected Holon structure is based on the Adaptive Holonic Control Architecture version 2 (ADACOR²) model implemented through a multi-agent system. ADACOR² addresses limitations of the original model by introducing greater self-organization and dynamic reconfiguration. It is particularly suited to highly dynamic and disruptive production environments, where flexibility and resilience are priorities. Finally, we will present a simulation of the system's deployment using a simple example to validate the proposed approach.

I. INTRODUCTION

This study considers a reconfigurable manufacturing system (RMS) designed for multi-product assembly using a social holonic control architecture. Reconfiguration is generally modeled at three levels: software reconfiguration, involving changes to a station's configuration (e.g., tools or control modes); structural/hardware reconfiguration, involving the addition or removal of stations or modules from a global pool; and operational reallocation, dynamically assigning jobs or operations to available resources (Morgan *et al.*, 2021). System dynamics result from time-varying product demand, in terms of both volume and variant mix, as well as changes in system state caused by reconfiguration delays and task completion. Although extensive research has addressed the characterization, modeling, and optimal control of reconfigurable systems over the past thirty years (Pansare *et al., 2022*), several challenges remain, particularly system complexity, the balance between physical reconfiguration costs and product customization, and the integration of human factors and their interactions with other system components.
These systems typically consist of modular workstations that can be activated, deactivated, configured, or installed according to operational needs, together with cobots and operators. To effectively model and manage these interactions, a Holon-based architecture has been defined. Holons are components of complex systems with self-organized behavior that contribute to the behavior of the whole. In manufacturing systems, a Holon is an autonomous and cooperative component capable of transforming, transporting, storing, and validating information and physical objects (Kruger *et al.*, 2019). Holons may consist of software or combined hardware and software components. The selected holonic architecture is based on the ADACOR², designed to promote modularity, facilitate reconfigurability, and handle disturbances at the architectural level (Leitão *et al.*, 2015). Its implementation relies on the multi-agent systems (MASs) paradigm. Finally, a simulation of a simple reconfigurable manufacturing system is proposed to demonstrate the interest of our approach.
The rest of the paper is organized as follows. The framework description and the interactions and organization of holons are presented in Section 2. Section 3 presents the multi-agent system, and Section 4 presents an application of our holonic control architecture to a reconfigurable manufacturing system. Conclusions drawn from the research findings and the future perspectives are discussed in Section 5.

II. FRAMEWORK DESCRIPTION

In recent years, researchers have highlighted that approaches focused exclusively on technology or, conversely, on humans do not always produce optimal results (Hefied *et al.*, 2023). A component-based development approach, i.e. a more product-centric perspective, enables greater consideration of the reusability of system elements and physical-level reconfigurability, rather than focusing primarily on software reconfiguration, as in much of the existing research. This approach also provides a better representation of the different layers and components of the underlying systems, as well as the interfaces and infrastructure of the manufacturing control architecture. Moreover, it facilitates the design of systems that can evolve and scale more effectively over time.
The physical components of a RMS can be categorized into two general types depending on their characteristics: i) the resources (machines, robots, operators, etc.) which perform the manufacturing operations; ii) the product which accepts the manufacturing treatments. All these physical elements are described by holons (see Figure 1).
The holon structure is then defined on the following basis:

- Order Holons (OHs): they are created per product order and they contain product attributes with a routing control, process control, decision-making and production information.
- Operational Holons (THs): they represent atomic tasks like painting, cutting, etc.

- Resource Holons (RHs): they are mapped to each physical device or operator. RHs can perform their own operations, decision-making and communication ability with the help of its local database.
- Product Holons (PHs): they provide information about the products to be manufactured, including customer requirements, bills of materials, and manufacturing drawings.
- Control Holon (CH): it evaluates global objectives (e.g., throughput, tardiness minimization) and defines plans, which are propagated to product holons.

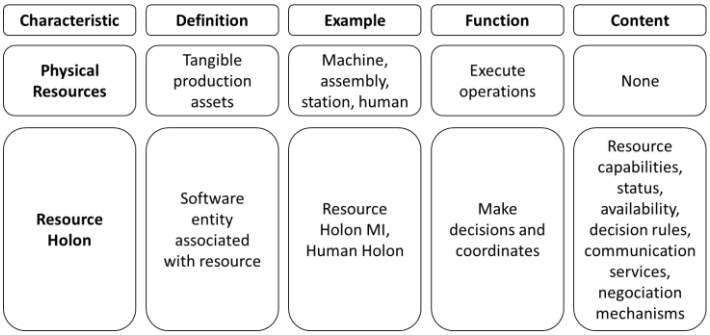

Figure 1- The Holonic Manufacturing System.

Then, two operating modes can be activated:

1. The stationary mode (or normal operation) maintains a decentralized schedule, which consists in the control of the RMS. This mode corresponds to hierarchical control structures → OHs execute steps by requesting service from each resource holon along the assigned path. OHs act as task executors, only initiating requests from a global schedule with minimal autonomy.
2. The disturbance mode (e.g., machine failures, supply disruptions, or resource unavailability) → OHs activate local negotiation (they act as decision-makers to select resource alternatives, adapt sequence logic, and invoke simplified scheduling rules

The advantage of developing this type of mixed architecture, which switches between hierarchical and heterarchical modes, lies in the fact that during normal operation, the best results (values, level, and computation time) are always achieved by a single top-down supervisor. However, in the event of a disruption, a heterarchical mode will provide much better responses (Barbosa *et al.*, 2015). System elements can then be supplied from various sources, enabling simultaneous and distributed design and integration.

Holonic components are organized as follows. The holons corresponding to orders (OHs) request services, negotiate with resources, and monitor their execution. Each resource holon represents a physical resource that manages its local state, negotiates tasks, and makes local decisions. Control holon (the supervisor) coordinates a group of operational holons. Its role includes: overall optimization, conflict resolution, coordination, and supervision. It coordinates resources, assignments, and priorities.

When a disturbance occurs, ADACOR² automatically reduces the influence of the Control holon. The Operational Holons become more autonomous, which corresponds to self-organization. Holons can modify their behavior, alter their connections, and adjust their level of autonomy without manual reconfiguration.

In stable mode, autonomy is limited while coordination remains high. With the disturbed mode, the autonomy is high and the coordination is distributed.

Based on these principles, the resulting multi-agent system is presented below.

III. MULTI-AGENT SYSTEM

A holon becomes an autonomous software agent with communication, decision-making, and negotiation capabilities. The correspondence between the ADACOR² model and its implementation as a multi-agent system is shown in Table 1.

Table 1 – Mapping between ADACOR² and the MAS.

ADACOR²	Multi-agent system
Control holon	Control agent supervises the RMS (specifically in stable mode)
Product holon	Product agent represents product knowledge
Resource holon	Resource agent represents a production resource
Operational holon	Task agent represents the information orders
Order holon	Order agent represents a customer order

For the Order agent, its responsibilities include initiating production, monitoring progress, managing priorities, and negotiating resources. The agent knowledge includes: order ID, due date, priority, customization level, and routing requirements. Its functions: request services, evaluate proposals, award contracts, and monitor implementation. It is a goal-driven agent.

The Resource Agent will execute tasks, monitor the resource's status, report on its availability, and negotiate its schedule. Its capabilities include: capabilities, availability, cycle times, current workload, maintenance status. It is a reactive and deliberative agent. For a "human" Resource agent, the resource agent represents the operator, his knowledge and skills, any certifications he may hold, his availability, his fatigue level, and his experience. His responsibilities include: accepting or rejecting tasks, negotiating workload, and reporting status. For a "cobot", its knowledge includes capabilities, tools, the current task, and safety status. Its responsibilities range from assisting with assembly to collaborating with humans and, of course, executing tasks.

The Control Agent is responsible for coordination, global optimization, conflict resolution, and scheduling support. Their areas of expertise include: system status, resource status, global KPIs, production objectives. The Product Agent is responsible for: process plans, BOMs, configuration, and quality requirements. Their expertise includes: manufacturing knowledge, product variants, and operations sequences.

The negotiation mechanism is as follows:

• Protocol: typically Contract Net Protocol (CNP) or auction-based negotiation (is probably the most natural protocol).

• Negotiation contents: request for proposals (RFPs) sent by OH to eligible Resource Holons. ROs respond with proposals: estimated time to serve, current load, expected delay. OH selects winning offer based on goal function (e.g., minimum delay). This goal function is used to handle task allocation and service binding in absence of CH.

The CNP algorithm is as follows:

a) Primary trading mechanism

Step 1: The Order Agent issues a call.

Step 2: The Resource Agents calculate their bids.

Step 3: The Order Agent evaluates the bids. Possible criteria include: delivery time, cost, energy, workload, etc.
Step 4: Assignment.
Step 5: Execution.

b) Negotiation under disrupted conditions (example involving a perturbation of a resource)

The Resource Agent broadcasts the disruption notification.
The other Resource Agents respond and suggest alternatives.
The Order Agent automatically initiates a new negotiation process.
The reconfigurable manufacturing system reorganizes itself.

c) For human-machine coordination (example)

The Order Agent requests an assembly operation.
The assembly machine (Resource Agent) breaks this down into manual assembly and cobot-assisted assembly.
The Human Resource Agent responds with its availability and skill level.
The Cobot Resource Agent responds with its availability and configured tools.
The assembly machine (Resource Agent) creates a collaborative task plan.

In what follows, we will consider a simple scheduling problem based on a mixed-integer linear problem (MILP). MILP has been a workhorse for formulating RMS scheduling and planning problems (Imsetif *et al.*,2025).

IV. IMPLEMENTATION CASE STUDY

The considered reconfigurable manufacturing system is composed of the following resources:

- M1 and M2 which are CNC cutting machines.
- P1 which is a robotic painting booth,
- A1 is the assembly station composed by manual activities made by an operator and a cobot for the final assembly,
- S is the inspection station which corresponds to an automated quality check.

In fact, the system could be composed of more modular workstations, each of which can be activated, deactivated, configured, or installed based on evolving operational demands.

The reconfiguration is modeled on three levels:

- Software reconfiguration: changing a station's configuration (e.g., switching between tools or control modes).
- Structural/hardware reconfiguration: adding or removing entire stations or modules from a global pool.
- Operational reallocation: dynamically assigning jobs or operations to available resources.

The final products are customizable bicycles based on attributes (frame size, frame type, color, and accessories like the lights, GPS, rack). Each product order is unique, but varies in due date, sequence flexibility, and customization level. For example, we consider 3 types of input orders corresponding to 3 Bikes (B11, B12, B13): B11: red, small frame, GPS, due in 3 hours, B12: blue, medium frame, no accessories, due in 12 hours, and B13: black, large frame, full accessories, due in 6 hours.

So the different holons for our RMS are: i) for Resource holons, we have RH_M1, RH_M2, RH_P1, RH_A1 (which will be described by three types of Resource Agent: a machine, i.e. a classical Resource agent, a human Resource agent and a cobot Resource agent), RH_S, ii) for Order holons for a given bicycle B_i, we define the n possible customizations by $OH_B_i^1$, $OH_B_i^2$ … $OH_B_i^n$, and iii) the Product holon i corresponding to the finished bicycle B_i is PH_B_i.

System dynamics arise from: i) time-varying product demand (in both volume and variant mix), ii) evolving system state due to reconfiguration delays and task completions.

This RMS mathematical formulation will permit to test the ADECOR[2] integrated human factors on the base of the scheduling. The objective function is defined on the basis of a cost function. We have the following assumptions.

- Product recipes are known in advance.
- Reconfiguration actions (e.g., station installs or config changes) incur finite known costs and delays.
- Processing times are deterministic and operation-specific.
- Demand forecasts are available over a finite planning horizon.

Note: While customer demand variation motivates the need for system flexibility, this model does not explicitly penalize unmet demand or forecast satisfaction. Instead, it focuses on minimizing internal system inefficiencies such as operation delays, reconfiguration penalties, resource utilization costs.

Let ($P = \{1, 2, \ldots, P\}$): Set of product types.
Let (I_p): Set of product instances of type (p).
Each product (p) has a sequence of operations (recipe): ($O_p = \{1, 2, \ldots, np\}$).
Let $J \subseteq J*$: Set of currently installed stations; $J*$is the superset of all available installable station types.
Let (K): Set of tools/configurations.
Let ($T = \{0, 1, \ldots, H\}$): discrete time horizon
The decision variables are given by:

1. Job scheduling variables
   - ($x_{i,j,o,t} \in \{0,1\}$): Equals 1 if operation (o) of product instance (i) is assigned to station (j) at time (t)
     - Controls routing and execution time of each task.
     - Captures sequencing logic across products.
2. Station activation variables
   - ($a_{j,t} \in \{0,1\}$): Equals 1 if station (j) is active at time (t)
     - Enables hardware reconfiguration (activating/adding machines).
3. Configuration assignment variables
   - ($y_{j,k,t} \in \{0,1\}$): Equals 1 if station (j) uses configuration (k) at time (t).
     - Controls tooling/software setup for process variation.
4. Reconfiguration transition variables
   - ($z_{j,k,k',t} \in \{0,1\}$): Equals 1 if station (j) switches from config (k) to (k') at time (t).
     - Captures timing and cost of configuration changes.
5. Station installation variables
   - $w_j \in \{0,1\}$: Equals 1 if station (j) is physically present in the system (i.e., installed from the global pool $J*$).

The system aims to minimize a predictive global cost function that integrates multiple operational and reconfiguration factors over a rolling horizon. The objective captures: task lateness penalties for missing production deadlines, energy and operational cost of running active stations, cost of

reconfigurations, such as switching tools or modes, and installation costs for physically adding new stations. These components are grouped in Table 2.

Table 2 – Components of the RMS.

Cost Type	Description	Symbol(s) Used
Task Lateness	Delay of product instance completion past due date	L_i
Station Activation	Cost of keeping station j active	$a_{j,t} f_j$
Processing Energy	Energy cost of executing operations	$x_{i,j,o,t}\ e_{j,o}$
Configuration Switching	Cost incurred when switching a station from configuration k to configuration k'	$z_{j,k,k',t}\ r_{j,k,k'}$
Station Installation	One-time install cost from hardware pool	$w_j\ c_j^{install}$

The objective function is defined over a prediction horizon:

$$\min_{\{x,y,a\}} Z = \sum_{t=0}^{H} \left(\alpha_1 \sum_i L_i + \alpha_2 \sum_j a_{j,t} f_j + \alpha_3 \sum_{j,k,k'} z_{j,k,k',t}\, r_{j,k,k'} + \alpha_4 \sum_{i,j,o,k} x_{i,j,o,t}\, e_{j,o} \right) + \alpha_5 \sum_j w_j\, c_j^{install} \quad (1)$$

Where:
(L_i = max (0, $C_i - d_i$) Lateness cost if job (i) finishes late
(fj): Cost of activating station (j)
($r_{j,k,k'}$): Cost of switching configs
($e_{j,o}$): Energy consumed for operation (o) at station (j)
($c_j^{install}$): One-time cost for installing station (j)
α_i (i from 1 to 5) represent parameters, which allows to normalize variables these parameters are defined on empirical experience).

This objective function (eq.1) is subject to the following constraints:

Precedence within Product Recipes

$$\sum_{j,t} t \times x_{i,j,o,t} \geq \sum_{j,t} \left(t + TP_{j,o-1}\right) \times x_{i,j,o-1,t} + t_{j,o-1} \quad \forall i, o > 1 \quad (2)$$

Where $TP_{j,o}$ refers to the processing time of operation (o) at station (j)

Operation Assignment (Each operation executed once)

$$\sum_{j,t'} x_{i,j,o,t} = 1 \quad \forall i, o \quad (3)$$

Station Capacity (One job per station per time unit)

$$\sum_{i,o} x_{i,j,o,t} \leq 1 \quad \forall j, t \quad (4)$$

Activation Precondition for Job Execution

$$x_{i,j,o,t} \leq a_{j,t} \quad \forall i, j, o, t \quad (5)$$

Tool Compatibility (Station must be configured to required tool)

$$x_{i,j,o,t} \leq \sum_{k \in K_0} y_{j,k,t} \quad \forall i, j, o, t \quad (6)$$

Reconfiguration Delay Enforcement

$$z_{j,k,k',t} \leq y_{j,k,t-1} \Rightarrow z_{j,k,k',t} \leq y_{j,k',t} \text{ and } z_{j,k,k',t} \geq y_{j,k,t-1} + y_{j,k',t} - 1, \sum_i \sum_o x_{i,j,o,\tau} \leq 1 - z_{j,k,k',t} \quad (7)$$

For $\tau = t, \ldots, t + \Delta_{j,k,k'} - 1$

Exclusive Configuration Selection (one config at a time)

$$\sum_k y_{j,k,t} \leq a_{j,t} \quad \forall j, t \quad (8)$$

Station Usage Requires Installation

$$a_{j,t} \leq w_j, y_{j,k,t} \leq w_j \quad \forall j, k, t \quad (9)$$

Maximum Installable Stations Constraint (N_{max})

$$\sum_j w_j \leq N_{max} \quad (10)$$

Completion Time Expression

$$C_i \geq t + PT_{j,o} - M(1 - x_{i,j,0,t}) \quad (11)$$

The presented MILP model is implemented in the GAMS software and solved with Cplex solver. We first conduct a sensitivity analysis based on the α_i parameters. We obtain the following Table 3.

Table 3. Sensitivity Analysis

Scenarios	α_1	α_2	α_3	α_4	α_5	Z*
S1 Balanced	0.2	0.2	0.2	0.2	0.2	99.8
S2 Investment-oriented	0.1	0.1	0.1	0.1	0.6	193.9
S3 Energy-oriented	0.1	0.15	0.15	0.5	0.1	62.3
S4 Reconfiguration-oriented	0.1	0.1	0.6	0.1	0.1	82.9
S5 Activation-oriented	0.1	0.6	0.1	0.1	0.1	95.9
S6 Due-date-oriented	0.6	0.1	0.1	0.1	0.1	49.9
S7 Sustainable production	0.1	0.1	0.15	0.4	0.25	104.6
S8 Industry 4.0 strategy	0.15	0.15	0.35	0.25	0.10	83.8

The results per scenarios are presented in Figure 4. Table 4 presents the results obtained under different weighting configurations of the objective function.

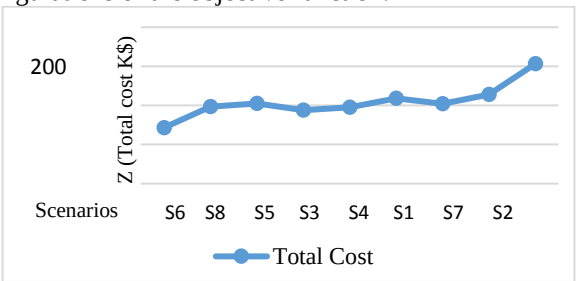

Figure 2- Final Results of the Numerical Application

Table 4. Results

Scenarios	Z*	Makespan	Energy	Reconf.
S1	99.8	4.8	54	6
S2	189.9	5.6	61	8
S3	79.8	5.1	42	7
S4	82.9	5.3	58	3
S5	95.9	5.4	56	5
S7	104.6	4.9	45	4
S8	83.8	4?7	48	2

Scenario S6 becomes practically equivalent to the other scenarios, as it has no influence on the optimal solution (because the lateness term remains equal to zero and therefore has no impact on the optimization outcome).

The findings highlight the trade-offs between operational performance, energy efficiency, investment costs, and system flexibility. The balanced scenario (S1) provides a reasonable compromise among all criteria, resulting in an objective value of 99.8 and a makespan of 4.8 hours. When the optimization emphasizes investment costs (S2), the objective value increases significantly, indicating the strong influence of installation expenses on the overall system cost. Conversely, the energy-oriented scenario (S3) achieves the lowest energy consumption (42 kWh), although this comes at the expense of a slightly longer makespan and a higher number of reconfigurations.

The reconfiguration-oriented strategy (S4) successfully reduces the number of reconfiguration actions to three, improving system stability but leading to a moderate increase in processing time. Similarly, the activation-oriented scenario (S5) minimizes station activation costs while maintaining acceptable operational performance. The sustainable production scenario (S7) balances environmental and economic objectives by reducing energy consumption and reconfiguration frequency, although the corresponding objective value remains relatively high due to installation costs. Finally, the Industry 4.0 strategy (S8) provides one of the best overall performances, combining a low objective value (83.8), the shortest makespan (4.7 hours), and only two reconfigurations. These results demonstrate that the proposed model can effectively support decision-makers in selecting production strategies according to different managerial priorities and operational constraints.

V. CONCLUSION

This paper proposed a holonic control architecture for reconfigurable manufacturing systems integrating both technical resources and workforce considerations. The proposed approach is based on the ADACOR² model and its implementation through a multi-agent system, enabling dynamic adaptation to changing production conditions while preserving overall system coherence. Under normal operating conditions, the Control Holon ensures global coordination and optimization, whereas in disturbed situations, the autonomy of operational entities increases through self-organization and local negotiation mechanisms. This dynamic behavior is particularly suitable for reconfigurable manufacturing systems, where changes in product mix, resource availability, and system configuration frequently occur.

To illustrate the applicability of the proposed approach, a reconfigurable bicycle assembly system is modeled and a scheduling formulation based on MILP is introduced. The obtained results demonstrate the feasibility of combining holonic control principles with optimization-based decision support in a reconfigurable manufacturing environment. The sensitivity analysis further highlights how different managerial priorities, represented through the weighting coefficients, can influence system behavior and lead to distinct operational strategies.

The combination of optimization, holonic control, and agent-based coordination offers a promising framework for managing highly customized production systems such as reconfigurable systems.

Future work will focus on several research directions. First, the proposed MILP model will be extended to explicitly incorporate human factors such as operator skills, fatigue, learning effects, and ergonomic constraints. Second, uncertainty management will be investigated through stochastic or predictive approaches capable of handling demand fluctuations, resource failures, and processing time variability. Third, the interaction between the optimization layer and the multi-agent control layer will be further developed in order to achieve real-time adaptive scheduling. Finally, larger industrial case studies and digital twin implementations will be considered to evaluate the scalability, robustness, and practical applicability of the proposed social holonic control architecture in Industry 5.0 manufacturing environments.

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USING AN OPTIMISATION TOOL FOR THE DESIGN IMPROVEMENT OF PRODUCTION SYSTEMS – AN INTEGRATIVE VIEW OF MATHEMATICAL MODELLING AND DESIGN PROCESS

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Optimisation, Prodcution System, Design, Interactions

ABSTRACT

Mathematical optimisation is widely used to support production system design by evaluating alternative configurations and identifying high-performance solutions. However, numerically optimal solutions are not necessarily feasible or actionable in industrial environments, particularly in brownfield manufacturing contexts. This paper presents an industrial case study illustrating how optimisation results can be integrated into the production system design process to identify critical constraints and explore alternative system architectures. The results show how optimisation can contribute to expanding the feasible design solution space beyond purely numerical performance improvement.

INTRODUCTION

Manufacturing companies increasingly face volatile demand, shorter product life cycles, and growing pressure for operational efficiency and adaptability. In this context, optimisation methods have become important tools for supporting production system design. By systematically evaluating alternative configurations and resource allocations, optimisation enables informed decision-making and rapid exploration of large design spaces.
Reconfigurable Manufacturing Systems (RMS) are particularly concerned by these challenges because their design requires balancing productivity, flexibility, and changeability. Numerous studies have investigated optimisation approaches for RMS configuration and operation, while others have proposed structured methodologies for the design of changeable production systems. However, these research streams remain largely disconnected. Optimisation is often treated as a tool for selecting a configuration, whereas design methodologies provide limited guidance on how optimisation results should influence the evolution of the design itself.
This paper addresses this gap through an industrial case study in the automotive sector. Rather than considering optimisation as an isolated analysis activity, the paper investigates how optimisation results can be translated into design-relevant information, including the identification of critical constraints and the emergence of new architectural alternatives. The objective is to show how optimisation can contribute to the transformation of the feasible design solution space and support production system design decisions beyond purely numerical performance evaluation.

RELATED LITERATURE

RMS have been widely investigated as a means of combining high productivity with adaptability to product and demand changes (Koren and Shpitalni, 2010; Koren et al., 2017). Their design involves complex trade-offs between capacity, functionality, cost, flexibility, and system performance, motivating the extensive use of optimisation approaches for configuration design, resource allocation, and production planning (Dolgui and Proth, 2006; Saxena and Jain, 2012; Moghaddam et al., 2019; Dou et al., 2021). Recent contributions further integrate sustainability, digitalisation, and data-driven decision support into RMS design and operation (Bortolini et al., 2018; Bortolini et al., 2019; Tsutsumi et al., 2018; Yelles-Chaouche et al., 2021).
In parallel, several methodologies have been proposed to support the design of changeable and reconfigurable manufacturing systems. Existing approaches range from systematic engineering design methods (Pahl et al., 2007), axiomatic design principles (Suh et al., 1998), and architecture-oriented frameworks (Deif and ElMaraghy, 2006; Schuh et al., 2009) to more recent lifecycle-oriented development methodologies (Andersen et al., 2018; Andersen et al., 2024). These approaches provide valuable guidance for system development but offer limited support for integrating optimisation results into the evolution of the design process.
Consequently, a gap remains between optimisation models, which identify high-performance solutions, and design methodologies, which guide the transformation of requirements into implementable system architectures. This paper addresses this gap by analysing how optimisation results can be translated into design-relevant information,

enabling the identification of critical constraints and the exploration of new production system configurations.

CONTEXT AND APPLIED METHODOLOGY

This paper proposes a process-oriented integration of optimisation within production system design. Rather than treating optimisation and design as separate activities, optimisation is considered here as an embedded design activity that contributes to the progressive transformation of design information.

The approach is structured around the interaction between the System of Objectives (SoO) and the Object System (OS), where optimisation supports the exploration of feasible configurations, reveals critical constraints applied to elementary functions, and produces results that must be translated into design requirements and implementation constraints. These modifications may affect elementary functions and the sequence of their realisation. The study adapts the SoO-OS framework of Albers and Meboldt (2007), by introducing optimisation as a design-support activity (Figure 1). Optimisation is used to (i) explore alternative configurations, (ii) identify performance-driving constraints, and (iii) provide design-relevant information that can be translated into requirements and architectural decisions.

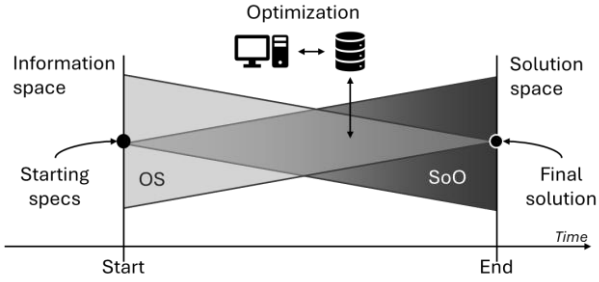

Figure 1: Development Process Overview, adapted from (Albers and Meboldt, 2007)

The results and conclusions presented in the article are based on an industrial research project (project type RDI: Research-Development-Innovation actions) between the LCFC lab at Arts et Metiers and its partner company in the automotive sector, conducted by mixed teams which are composed of academic researchers and industrial engineers. The analysis relies on a previously published MILP model for simultaneous task allocation and sequencing (Stief et al., 2023), solved using CPLEX. It describes in detail the optimisation tool and the mathematical model used during the case study. For reasons of brievety, the mathematical model is not repeated in this paper. The model uses workstation capabilities, task durations, precedence relations and layout constraints as inputs, and generates task-resource allocations, operation sequences and performance indicators as outputs. It allows a parallel task-resource allocation and sequencing, having two objectives: the reduction of the cycle time and the reduction of cost. The objectives can be weighthed depending on the aim of the optimisation. In agreement with the industrial partner, the CPLEX solver has been used.

CASE STUDY, RESULTS AND DESIGN IMPLICATIONS

The aim of the hereafter detailed case study is the analysis of possible gains in cycle time through the elimination of constraints and/or the modification of the system layout. The case study has been realised in strong cooperation with the industrial partner. It concerns mechatronic product assembly on production systems with cycle times ranging between 10s and 60s depending on the products and layouts. Due to confidentiality issues, no absolute values can be indicated and no pictures of the products and production lines can be displayed.

Case Study Description and Optimisation Results

The case study analyses alternative configurations of three workstations (WS) within an existing production line, realising eight different operations, hereafter enumerated Op1 to Op8. The objective is to improve throughput by evaluating different operation placement constraints (free versus constrained) and comparing serial, hybrid, and parallel configurations.

Figure 2 describes the analysed configurations. The current scenario evaluates the performance using the task-resource allocation implemented by the industrial partner. The scenarios ‘A’ and ‘B’ evaluate the performance under the hypothesis that all workstations are universal. ‘Constrained allocation’ supposes that Op1 and Op2 cannot be realised together with Op7 and Op8. ‘Unconstrained allocation’ examines the ideal case that there are no constraints.

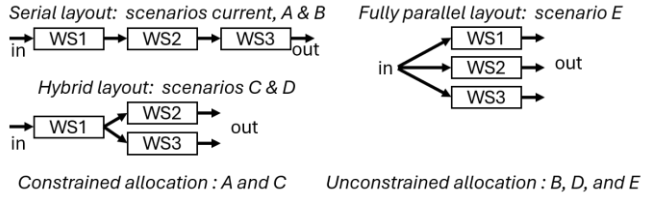

Figure 2: Layout for the Scenarios ‘current’, ‘A’ and ‘B’

In addition, the scenarios C, D and E feature architecture modification. The optimisation results of the different scenarios and configurations are synthesised in Table 1. The cycle time of WS2 is determining the cycle time of the existing serial configuration. To compare the results, all other cycle times are expressed relative to this one.

Table 1: Numerical Results of the Optimisation Tool

Scenario	WS1	WS2	WS3	All	Δ	Unbalancing
Current	74%	100%	60%	100 %	-	40%
A	87%	74%	73%	87 %	13 %	14%
B	73%	87%	74%	87 %	13 %	14%
C	60%	168%	168%	84 %	16 %	24%
D	74%	154%	154%	77 %	23 %	3%
E	222%	222%	222%	74 %	26 %	0%

The columns of WS1 to WS3 detail the individual performance of each workstation. The column “All” indicats the configuration performance. In a serial configuration, the slowest workstation determines the overall cycle time. In a parallel configuration, the cycle time of the parallel workstations is divided by the number of parallel workstations. This is the case for WS2 and WS3 in scenarios ‘C’ and ‘D’ and for all workstations in scenario ‘E’. The delta gives the configuration cycle time improvement compared to the current configuration. The unbalancing value indicates the difference between the slowest and the fastest workstation, highlighting the overall efficiency of the configuration.

It can be seen that scenario ‘E’ obtains the best score having the lowest cycle time and no unbalancing between the workstations. But also a hybrid configuration with one workstation in series and two in parallel improves the cycle time. However, keeping constraints (scenario ‘C’) increases the difficulty to balance the workload of the workstations. The third takeaway from the optimisation results is that even in a serial configuration the deletion of precedence constraints leads to a significant cycle time reduction from more than 10% and an improved workload balancing.

Implications for the Design Process

In the case study, the relaxation of accessibility-related constraints did not only improve performance in the model, but also triggered a redesign of the material flow system, opening a new solution space that would not have been explored otherwise.

The numerical optimisation has been conducted in parallel of the development of a new material flow system. The optimisation highlighted several opportunities to improve the production system compared to its current configuration. Beyond performance evaluation, the optimisation results supported the evolution of the design process by identifying critical constraints and highlighting promising architectural alternatives. Their integration required translating modelling assumptions and numerical outcomes into actionable design requirements.

Not taking into consideration the results during the design process would lead to a suboptimal final solution and generate supplementary development cost and time if adaptation is needed. However, efficient integration to the design process needs a translation of the hypotheses of the mathematical modelling and the numerical outcomes into constraints and/or requirements which are compatible with the information space and the SoO. In the case study example, the impact of the optimisation results has been twice: (i) it has shown the advantage of a parallel system configuration, but (ii) it has also underpinned the necessity to release constraints to achieve the maximum improvements.

In the present case, the constraints corresponded to accessibility constraints for the manufacturing operations. The whole manufacturing process has been conditioned by a sequence of part locatings which have constrained the feasible operations on each workstation. The material flow system had been adapted to this logic as well. In consequence, the simple request of releasing precedence constraints linked to accessibility has oriented the material flow design process to a complete new solution space, as current standard solutions did not satisfy the requirements. Figure 3 illustrates the complexity of the problem and an excerpt from the solution space at an intermediate instance of the development process. The optimisation results did not simply indicate that a parallel configuration would improve throughput. They also revealed that accessibility constraints were the primary factor limiting workload balancing. Consequently, the optimisation process shifted the design effort from workstation configuration towards the redesign of positioning, clamping and material flow solutions. This change expanded the feasible design space by making previously inaccessible system architectures implementable.

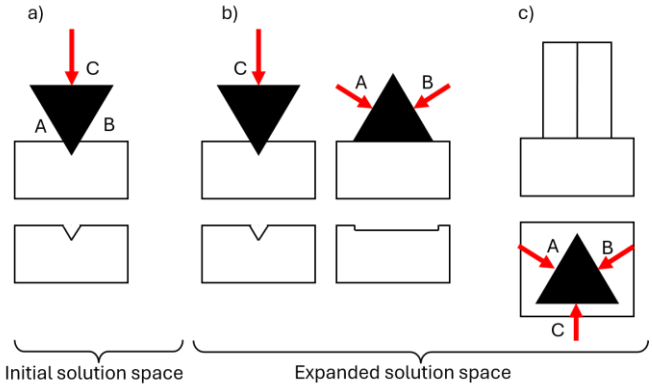

Figure 3: Solution Space at Instance t of the Design Process

Different positioning and clamping solutions allow to satisfy the accessibility constraints (arcs) of each system configuration. The solution a) corresponds to a purely serial layout, b) to a hybrid layout and c) to a complete parallel configuration. The final solution can not be illustrated in this paper due to confidentiality reasons. The implication of optimisation results in the development process has led to a completely new and innovative solution for the transfer and the supply of components to the workstations. This would not have been achieved without.

CONCLUSION AND PERSPECTIVES

This paper proposes a process-oriented perspective on the integration of optimisation in production system design. By considering optimisation as an embedded design activity, the work highlights its role in transforming design information, refining objectives, and guiding the exploration of the solution space. The case study illustrates how optimisation results can reveal critical constraints and trigger the emergence of new design directions, provided that an effective translation into design requirements is ensured. In contrast, considering either activity in isolation leads to closed-loop optimisation with limited impact on real applications or to design solutions driven mainly by empirical knowledge, experience, and subjective preferences of the design team.

Future work will focus on formalising this translation step and integrating data-driven approaches and digital tools to support iterative and adaptive design processes in Industrial contexts.

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TOWARD DEEP ENERGY NEURAL NETWORKS AS DEEP SURROGATES FOR MASS-SPRING MODELS OF TEXTILES

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KEYWORDS
Deep-energy neural networks, surrogate modeling, mass-spring model, textile modelling

ABSTRACT

This paper presents a physics-informed deep learning approach for predicting the static equilibrium of deformable textiles using deep energy-based neural networks as surrogates for mass-spring models. The proposed method leverages the specific structure of the textile deformation problems to define an efficient unsupervised learning methodology.
A reduced task formulation is introduced, which lowers input dimensionality by exploiting physical invariants. This significantly improves the network's generalization capabilities. The loss function also incorporates a term that limits non-physical hyper-elastic deformations. Experimental results demonstrate that the approach yields lower-energy solutions with fewer hyper-elastic violations and provides consistent performance across multiple training runs. The resulting static solutions additionally serve as effective initial configurations that can be refined using standard optimization algorithms.

INTRODUCTION

Numerical models that predict the deformation of a piece of textile under mechanical constraints have found applications in several real-life domains, including computer graphics (Stuyck 2022), textile engineering (Perumalsamy et al. 2014), and robotic textile manipulation (Koessler et al. 2024). These applications exhibit widely varying requirements in terms of computational cost and physical fidelity. For example, computer graphics typically favors fast models capable of producing visually plausible deformations without strict adherence to physical accuracy. Such models often represent textiles as collections of particles whose positions are constrained to preserve structural integrity. Alternatively, textiles have been modeled as meshes of point masses connected by springs and, optionally, dampers (Provot et al. 1995). At the opposite end of the spectrum, finite element models (FEM) are commonly used when high physical fidelity is required and computational constraints are less restrictive.

Recently, data-driven models, particularly deep neural networks, have also gained popularity for applications demanding real-time inference, such as computer graphics or robotic textile manipulation. For instance, the authors of Zhao (2024) proposed a physics-informed neural network (PINN) architecture to model the dynamics of a piece of fabric. Using a different approach, the authors of Li et al. (2025) suggested employing a neural network to predict an oriented bounding box around a moving textile to support dynamic robotic manipulation.
This article focuses on the use of physics-informed deep energy-based neural networks as surrogates for mass–spring models to predict the static equilibrium of a hanging piece of textile. The positions of certain nodes in the folded mesh are constrained, as would occur for a piece of fabric suspended from curtain rods, hooks, or robotic grippers. These models are envisioned as algorithmic building blocks for applications requiring extremely fast inference, where visual realism is not the primary objective. The main motivation for this work is to enable robotic path planning based on random sampling of the robot's joint space, which requires evaluating several thousand potential robot configurations while ensuring collision-free interaction with the textile attached to the end effectors. As demonstrated in this article, the proposed framework can also provide initial deformation estimates that may subsequently be refined to achieve visually realistic graphics.
The main contribution of this work is the introduction of a novel and computationally efficient framework for generating neural-network surrogates of static mass–spring textile models. This framework relies on an energy-based formulation of the problem and leverages physical invariants to simplify the learning task.
The remainder of this article is organized as follows. The next section introduces the governing principles of mass–spring models. The proposed deep energy neural network framework is then presented. A set of numerical experiments follows, illustrating the advantages of the approach. The final section concludes the paper and outlines avenues for future research.

MASS-SPRING NETWORKS

Mass–spring networks are a popular choice for modeling textiles in applications that require computationally ef-

ficient numerical simulations. In these models, a piece of textile is represented as a mesh of point masses connected by massless springs. Although various mesh configurations have been proposed for specific fabrics, this work adopts the approach described in Provot et al. (1995) due to its generality. The mesh incorporates three types of springs that enforce structural cohesion and mimic shear and bending resistance to deformation. The arrangement of these springs within the mesh is illustrated in Figure 1.

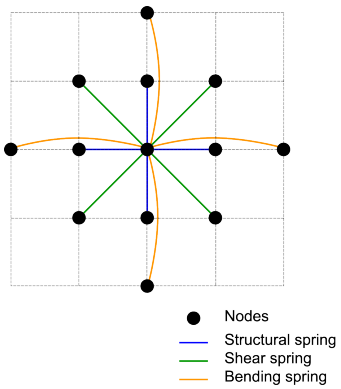

Figure 1: Structure of the mass-spring mesh. The nodes, which correspond to point masses, are organized in a regular grid. They are connected by three types of springs.

Under static assumptions, the deformation of the mesh corresponds to a local minimum of its potential energy, subject to constraints. In general, multiple such local minima may exist, as a piece of textile can settle into different equilibrium configurations. The potential energy of a mesh M is given by:

$$E_p(M) = mg\sum_i z_i + \frac{1}{4}\sum_i \sum_{j\in N(i)} k_{ij}(||p_i - p_j|| - l_{ij})^2, \quad (1)$$

with m the mass of each node, g the gravitational constant, $\mathbf{p}_i = (x_i, y_i, z_i)$ the position of the i^{th} node, $N(i)$ the set of neighboring nodes connected to node i, and k_{ij} and l_{ij} the stiffness and rest length of the spring linking node i to node j.

The foundational work of Provot et al. (1995) highlights an important limitation of this model. By allowing springs to elongate without limit, this model can lead to non-physical textile deformations, referred to as hyper-elastic deformations. To prevent this, the authors propose constraining the maximum extension of each spring such that the ratio of its current length l to its rest length remains below a constant threshold: $\frac{||\mathbf{p}_i - \mathbf{p}_j||}{l_{ij}} < 1 + \tau$ for all springs ij.

The model must also account for interactions between the fabric and its environment. In robotic path-planning scenarios, for example, the positions of certain nodes in the network must be controlled, as they represent the points where the robot grips the fabric.

DEEP ENERGY NEURAL NETWORKS TO ESTIMATE STATIC EQUILIBRIUM

Deep energy neural networks are a type of physic-informed neural networks trained with a loss integrating an energy term which is minimized by the network (Samaniego et al. 2020). This approach has shown promising results in generating solutions for various problems, including material elastic deformation (Roy et al. 2023) and heat transfer (Gao et al. 2022).

The objective of the proposed deep energy neural network is to output the deformation of the mesh given information about the gripped points.

Training Loss

The proposed loss function for training the network comprises two components. The first component is the potential energy of the mesh, defined as $\mathcal{L}_e(M) = E_p(M)$. The second component is a penalty term against hyper-elastic deformations:

$$\mathcal{L}_{he}(M) = \sum_i \sum_{j\in N(i)} (||p_i - p_j|| - (1+\tau)l_{ij})^2, \quad (2)$$

with τ the hyperelastic constant and $(.)_+$ the positive part function. The total training loss is then defined as:

$$\mathcal{L}(M) = \mathcal{L}_e(M) + \alpha\mathcal{L}_{he}(M) \quad (3)$$

Of particular interest to this discussion is the behavior of the loss function when a family of rigid transformations is applied to the mesh, defined as $M = R_z(\theta)M_0 + T$, where T is a translation vector and $R_z(\theta)$ represents a rotation by angle θ about the z-axis.

The hyper-elastic penalty $\mathcal{L}_{he}$ depends only on pairwise distances between points in the mesh, such that $\mathcal{L}_{he}(M) = \mathcal{L}_{he}(M_0)$. Similarly, the only component of the energy loss $\mathcal{L}_e$ affected by this transformation is the gravitational potential energy, resulting in $\mathcal{L}_e(M) = mgNT_z + \mathcal{L}_e(M_0)$, where N is the number of nodes in the mesh and T_z is the z-component of the translation vector.

Therefore, when such a rigid transformation is applied to the mesh, the loss function is translated by a constant that depends solely on the transformation parameters and not on the mesh deformation.

Another advantage of the proposed loss function is that it enables fully unsupervised training of the network. The loss function is based solely on the geometry of the output mesh. There is no need to compute ground-truth data for comparison. Therefore, batches of training examples can be generated on the fly during the training process at negligible computational cost.

Network Architecture

The network architecture is inspired by the FoldingNet decoder introduced by Yang et al. (2018) for shaping 3D point clouds. The underlying principle of this architecture is to fold, a 2D grid into 3D space to create various shapes, such as guitars or chairs. Following this idea, the proposed network applies, independently and in parallel, a sequence of shallow fully connected neural networks (FNNs) to each node of the grid. As shown in Figure 2, each FNN contains a single hidden layer, which folds the initially flat mesh in three dimensions. The mesh undergoes four such folding operations. Between each folding step, the point coordinates are concatenated with additional inputs. These inputs include the relative position of each node within the unfolded grid and information about the positions of the gripped nodes. Figure 2 presents, in addition to a schematic of the network architecture, an example illustrating the effect of each folding operation on the mesh.

Model Inputs and Output

The network outputs a folded configuration of the mesh that minimizes its potential energy while limiting hyper-elastic deformations. The mesh represents a square piece of fabric gripped by two corners on one side. Consequently, the positions of these gripped nodes are assumed to be known and are not predicted by the network.

The model is applied independently to every node in the mesh. As shown in Figure 2, the inputs include the 3D coordinates of each node, along with additional information. These additional inputs consist of the node's position within the flat grid and data describing the positions of the gripped nodes. Two distinct cases are considered for providing this information to the network.

In the first case, the model inputs are augmented with six scalars, representing the 3D positions of the two gripped corners whose positions are controlled. In this scenario, the model directly outputs the position of the node in the world frame.

The second case takes advantage of the structure of the loss function to reduce the number of inputs and simplify the inference task. This is achieved by assuming that the first gripping point is always placed at the origin and that the second gripping point is always aligned along the x-axis. This is akin to applying a rigid transformation to the mesh as defined previously. In this case, only the z-difference between the gripping points and their distance within the xy-plane are used as inputs to the network. The output mesh can then be transformed back to the world frame using simple linear algebra.

NUMERICAL EXPERIMENTS

This section present numerical experiments to evaluate and compare models trained on the tasks.

Mesh definition

The mass-spring network deformed by the proposed neural network model represents a square piece of fabric. The mesh is organized as a 20×20 grid comprising 400 nodes, connected by springs arranged as illustrated in Figure 1.

Table 1: Parameters of the Mass-spring Network

$k_{structure}$	$k_{bending}$	k_{shear}	node mass (kg)
4.258	4.258	16.065	5.0×10^{-4}

The stiffness constants for the springs are adopted from the work of Anatoliotakis et al. (2025). The mass of each node is derived from standard gram-per-square-meter measurements for winter cotton shirts. The values of these parameters are summarized in Table 1. The hyperelastic constant τ is set to 0.1, following Provot et al. (1995).

Results

Figure 3 presents the evolution of both components of the model loss over 200,000 steps of the gradient descent algorithm, for both the reduced and full models. The first component, $\mathcal{L}_e$, corresponds to the potential energy of the folded mesh. The second component, $\mathcal{L}_{he}$, penalizes hyper-elastic behavior of the mesh. For the reduced model, the energy term is corrected to account for the change in gravitational potential energy caused by shifts of the mesh in the z direction.
The training of both models is repeated 30 times to account for stochastic mechanisms in the training process. The solid lines in Figure 3 represent the average validation losses over the 30 training runs. The colored areas around the solid lines indicate ± 1 standard deviation from the mean. Each network is trained over 200 000 steps of the gradient decent algorithm, with a batch size of 32. Generating a supervised training dataset by searching for an explicit static equilibrium every time would have required over 120 000 hours of computation hence the necessity of using an unsupervised method.
As demonstrated in this figure, the reduced model generalizes better than the full model, consistently across multiple training repetitions. Networks trained on the reduced task generate foldings with, on average, lower potential energy. They also exhibit fewer violations of the hyper-elastic constraints enforced by the penalty term. This improved generalization may be due to the fact that the reduced model needs only to learn a simpler version of the target function. In particular, it does

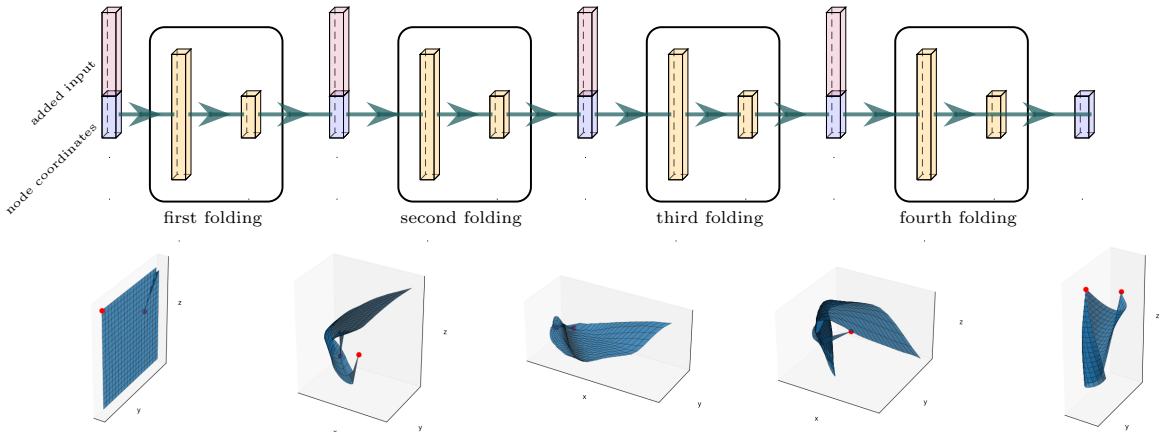

Figure 2: Architecture of the neural network. It consists of four shallow fully connected networks that progressively fold an initially flat mesh into three dimensions. The effects of these folding operations are illustrated below the architecture. Red points represent the gripped nodes.

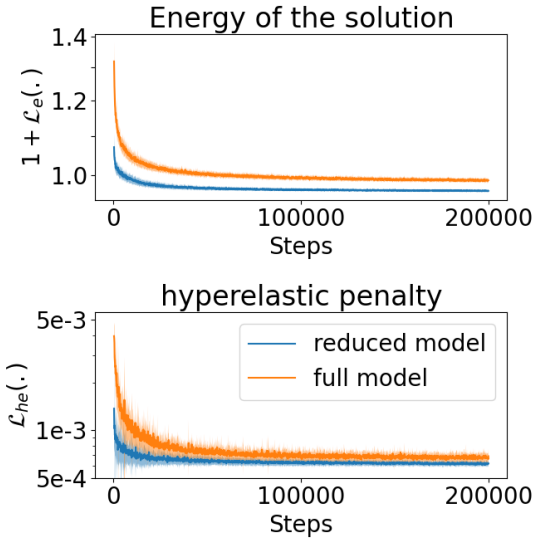

Figure 3: Evolution of $\mathcal{L}_e$ and $\mathcal{L}_{he}$ for the reduced and full models over 200,000 training steps of the gradient descent algorithm. All losses are evaluated on the same validation dataset, independent of the training process.

not need to learn the equivalences that are explicitly accounted for by the frame transformation.

Figure 4 presents examples of the same mesh folded by 4 different networks for the same set of gripped point positions. The first column shows results obtained from the full model, trained independently three times, while the second column presents foldings obtained with the reduced model. To generate the third column, the outputs of the reduced model were refined using a sequential least squares programming optimization algorithm that minimizes $\mathcal{L}$ on a case-by-case basis. In each case, the values of $\mathcal{L}_e$ and $\mathcal{L}_{he}$ associated with the solution are provided.

This figure illustrates the ability of the neural network models to find approximate solutions to the penalized energy minimization problem. The fact that retraining the networks can lead to different equilibrium states is of particular interest for use cases that may require a large, diversified sampling of the solution space, such as robotic path planning.

However, the current architecture is insufficient for computer graphics applications due to the difficulty in predicting small, localized folds of the fabric. These folds are necessary for visually realistic solutions. As shown in Figure 4, the solutions found by the networks can be used as initial positions for standard optimization algorithms. These optimization algorithms can further decrease the loss on a case-by-case basis. Starting from a flat grid as initial solution, the optimization procedure takes, in average, 73 seconds. Starting from the full model solution, this time is reduced to 48 seconds. Starting from the reduced model solution, this time is further reduced to 32 seconds. Inference with the reduced model only takes a few milliseconds. The losses are, in average, reduced by 12.2% and 9.2% respectively by direct optimization.

CONCLUSION AND FUTURE WORKS

This study introduced a novel physics-informed deep learning approach for predicting the static equilibrium of hanging textiles, using energy-based neural networks as surrogates for mass–spring models. The models are trained through an unsupervised methodology that does not rely on experimental data.

By exploiting the structural properties of the problem, the framework simplifies training and reduces both data and computational requirements. By focusing on rigid

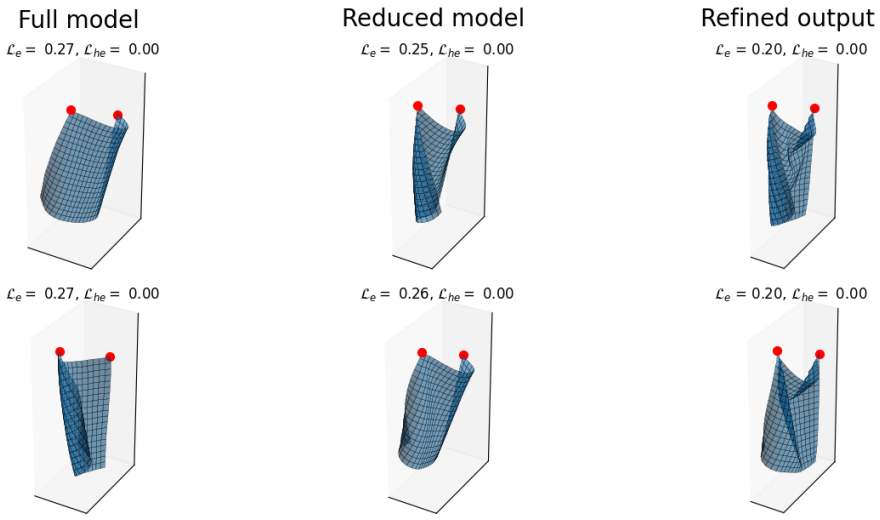

Figure 4: Examples of a mesh folded by networks trained by on the full or reduced task, as well as obtained through direct energy optimization

transformations, the network learns a simplified target function, which improves generalization and consistency across trials. The reduced model produces lower-energy solutions with fewer hyper-elastic violations, and its outputs can serve as effective initialization for higher-precision applications such as computer graphics.

However, limitations remain in capturing fine, localized folds necessary for visually realistic simulations. In addition, this framework does not take collisions into consideration. Future work will address these challenges through architectural improvements or by extending the method to dynamic scenarios.

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Simulation in Smart Manufacturing

TOWARDS SMART INDUSTRY: A MODULAR FRAMEWORK FOR ADAPTIVE MONITORING OF WAAM SYSTEMS

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Smart Manufacturing; WAAM; Reconfigurable Systems; Adaptive Monitoring; Industry 4.0; Axiomatic Design

ABSTRACT

The integration of Internet of Things (IoT) technologies into manufacturing systems is accelerating the transition towards Industry 4.0, where processes become interconnected, adaptive, and data-driven. In this context, Wire Arc Additive Manufacturing (WAAM) faces significant challenges related to process variability, defect control, and real-time monitoring. To address these challenges, this paper proposes an IoT-compatible framework for the design of adaptive monitoring strategies in robotized WAAM systems. Based on axiomatic design principles, the framework structures the relationships between product requirements, process parameters, and monitoring strategies through a modular architecture. The proposed approach supports interoperability, scalability, and hardware/software co-design, providing a foundation for intelligent and reconfigurable monitoring systems aligned with Smart Industry requirements.

INTRODUCTION

The increasing adoption of Industry 4.0 technologies is transforming manufacturing systems into interconnected and data-driven environments. Within this context, Wire Arc Additive Manufacturing (WAAM), and particularly Cold Metal Transfer (CMT)-based WAAM, has emerged as a promising solution for manufacturing and repairing large-scale metallic components. However, process quality remains highly sensitive to variations in thermal, geometric, and deposition conditions, making monitoring and defect detection critical challenges.
Numerous monitoring solutions have been proposed for WAAM. However, most remain application-specific and are typically developed independently from manufacturing system design. In parallel, reconfigurable manufacturing systems (RMS) have demonstrated the benefits of modularity, scalability, and rapid adaptation to changing manufacturing requirements (ElMaraghy, 2006; Bi et al., 2008). Similar principles have been explored for adaptive and multi-sensor monitoring architectures (Mehrabi and Kannatey-Asibu, 2001; Swain et al., 2020). However, these approaches primarily address sensor selection for machining systems. The integration of monitoring strategies within a unified design framework that not only supports sensor selection but also links product requirements, process configuration, sensing technologies, and system evolution remains largely unexplored. Furthermore, Industry 4.0 environments introduce additional challenges related to interoperability, data integration, and dynamic system reconfiguration. To address these limitations, this paper proposes an axiomatic design-inspired framework for the development of adaptive monitoring systems in robotized WAAM environments. The approach structures the relationships between product requirements, process parameters, and monitoring strategies through a modular architecture organized around product, process, and monitoring domains. The objective is to support the design of scalable, interoperable, and reconfigurable monitoring solutions aligned with Smart Industry requirements.

RELATED LITERATURE

Monitoring and control of Wire Arc Additive Manufacturing (WAAM) have been extensively studied to improve process stability, defect mitigation, and product quality (Ding et al., 2015; Xia et al., 2020; Zahidin et al., 2023). Existing approaches focus on monitoring thermal, geometric, and metallurgical variables that directly influence deposition quality and process performance (Mukherjee and DebRoy, 2018; Sreedhar et al., 2012; Franke et al., 2024; Lehmann et al., 2020; Banaee et al., 2023). To support this monitoring, a wide range of sensing technologies has been investigated, including infrared thermography, optical vision systems, laser profilometry, acoustic sensing, ultrasonic inspection, and optical emission spectroscopy.
These technologies are employed to detect common WAAM defects such as porosity, cracks, lack of fusion, residual deformation, and geometric inconsistencies (Albannai, 2022; Wu et al., 2018). Thermal monitoring approaches are widely used to observe melt pool behavior and temperature gradients (Sreedhar et al., 2012), whereas acoustic and ultrasonic techniques support internal defect detection (Chabot et al., 2020; Javadi et al., 2019). Optical emission spectroscopy has also been investigated for detecting gas-related anomalies and porosity formation (Karlina et al., 2024).

Recent studies have explored multi-sensor monitoring strategies combining thermal, optical, acoustic, and ultrasonic measurements to improve process understanding and defect detection accuracy (Gutknecht et al., 2021; Xu et al., 2018). A synthesis of the relationships between defects, monitored parameters, sensing techniques, and sensors is provided in Table 1.

WAAM Defects and Associated Monitoring Devices

The WAAM process is characterized by complex thermophysical phenomena that may lead to various defects affecting the quality and reliability of manufactured components (see Table 1). These defects originate from interactions between thermal, geometric, and material parameters and require appropriate monitoring strategies. Each defect category requires specific sensing technologies, and no single sensor setup is sufficient to fully characterize the process to ensure a zero-defect manufacturing. Consequently, multi-sensor monitoring systems are necessary to enable a comprehensive process monitoring and supervision. In a smart manufacturing context, sensors are interconnected through IoT infrastructures, enabling:

- real-time data acquisition from distributed sources and synchronization of heterogeneous signals
- data fusion for improved defect detection
- continuous feedback for process control
- easy reconfiguration of monitoring configurations

This transition from isolated sensing to connected and intelligent monitoring systems is essential for achieving robust WAAM operation.

Table 1. WAAM Defects and Corresponding Monitoring Strategies

Defect Type	Main Causes	Monitored Parameters	Monitoring Techniques	Typical Sensors
Porosity	Gas entrapment, unstable melt pool, shielding gas issues	Plasma composition, acoustic emissions, thermal stability	Optical emission spectroscopy, acoustic monitoring, ultrasonic testing	Spectrometer, acoustic sensors, ultrasonic probes
Cracks / Lack of fusion	Thermal gradients, residual stress, poor bonding	Temperature distribution, stress evolution	Infrared thermography, acoustic emission, ultrasonic inspection	IR camera, acoustic sensor, ultrasonic sensors
Residual deformation	Heat accumulation, geometric inconsistencies	Displacement, strain fields	Laser interferometry, Digital Image Correlation (DIC)	Interferometer, optical cameras
Distortion / Poor adhesion	Thermal instability, uneven cooling	Thermal gradients, layer geometry	Infrared monitoring, laser profilometry, X-ray tomography	IR cameras, laser scanners, CT systems

CONTEXT AND APPLIED METHODOLOGY

The experimental framework is intended for deployment within the LCFC robotized manufacturing cell, illustrated in Figure 1, which integrates, amongst others, the WAAM process into a robotized and reconfigurable manufacturing environment. The cell is structured around an ABB robot that can be equipped with different manufacturing processes (additive and subtractive), including WAAM through a Fronius CMT welding system, illustrating the high reconfigurability of the manufacturing cell.

It features a closed-loop workflow linking product configurations, process parameters, and monitoring configurations, supported by a set of distributed sensors such as laser tracking, weld station data, robot data, and thermal sensors. These sensing elements are integrated into an IoT-enabled infrastructure connecting heterogeneous sensors to the Promethée server through a Gantner acquisition system, enabling real-time data acquisition, visualization through the Hermes interface, and experiment traceability.

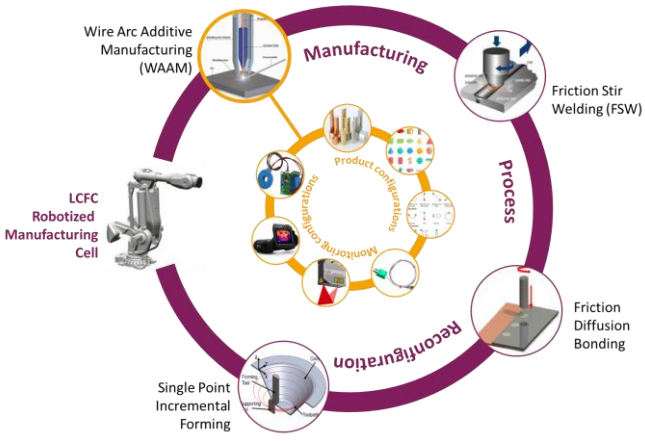

Figure 1. Scope of the Manufacturing Cell

This closed-loop setup ensures continuous interaction between product, process, and monitoring layers while providing a suitable platform for implementing and validating the proposed framework through multi-sensor integration and reconfigurable monitoring configurations.

Axiomatic Design inspired Architecture for Monitoring Development

Axiomatic Design (AD), introduced by Suh (1998; 2001), provides a rigorous framework for the structured development of complex systems by establishing formal relationships between Functional Requirements (FRs), Design Parameters (DPs), and Process Variables (PVs).

The methodology is governed by two fundamental axioms: (i) the independence axiom, which requires maintaining the independence of functional requirements, and (ii) the information axiom, which aims to minimize system complexity by reducing the information content of the design.

In the context of smart manufacturing and IoT-enabled systems, Axiomatic Design is particularly relevant due to its ability to conceive and support modular, reconfigurable, and interoperable architectures. These characteristics are essential for monitoring systems operating in dynamic environments such as reconfigurable manufacturing cells, where process conditions, configurations, and sensing requirements evolve continuously.

In this work, the axiomatic design approach is leveraged to structure a modular and adaptive monitoring framework for WAAM-CMT processes. The proposed model establishes explicit relationships between product requirements, process

parameters, manufacturing objectives, and monitoring strategies. This structured representation enables a comprehensive understanding of how variations in process conditions affect product quality and supports the development of robust and flexible monitoring solutions. The framework adopts a modular scheme, where the whole system is decomposed into the three domains of product, process, and monitoring. These domains are evaluated on two levels: (i) the level of artefacts which represent the hardware and (ii) the level of characteristics which contains data and descriptive information related to the artefacts (Figure 2).

This subdivision enables the rapid integration of new sensors, adaptation to different process configurations, and functional scalability of the monitoring system. Reconfigurability is achieved through the selection and reconfiguration of the monitoring modules according to changing process conditions.

The proposed framework is inspired by classical axiomatic design by introducing a dual mapping structure:

- Horizontal mapping, which represents the functional progression from product requirements to monitoring strategies, ensuring coherence across the design chain
- Vertical mapping, which links each level of the manufacturing system (product, process, monitoring) and which ensures consistency between the artefacts and their characteristics.

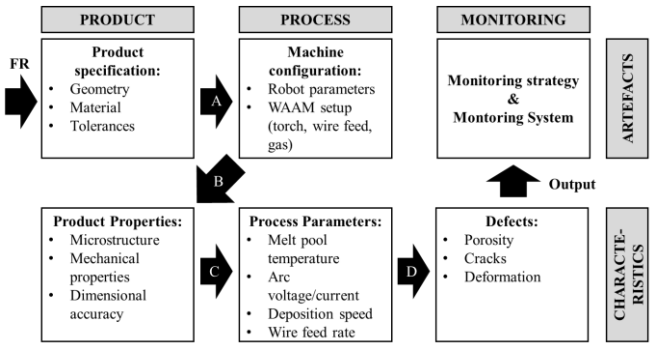

Figure 2. Framework

This cross-layered architecture ensures that monitoring decisions remain consistent, scalable, and adaptable, while explicitly capturing the interdependencies between manufacturing operations and monitoring strategies. As in classical axiomatic design, the relationships between the different domains are established through the so-called zig-zagging process:

- Mapping A – Product to Process Artefacts: This level establishes the relationship between product specifications (e.g. size, shape, material) and machine capabilities. It defines the technical configuration required to meet manufacturing objectives.
- Mapping B – Process Artefacts to Product Properties: This mapping links process variables to resulting product properties (microstructure, mechanical strength, dimensional accuracy), enabling performance evaluation based on process conditions.
- Mapping C – Product Properties to Process Parameters: Product characteristics are translated into physical process variables such as melt pool temperature, arc voltage, wire feed rate, and deposition speed. These parameters directly govern process stability and deposition quality.
- Mapping D – Process Parameters to Defect Detection: This final mapping associates process deviations with potential defects (porosity, cracks, distortions) and identifies the appropriate monitoring methods and sensor technologies required for early detection.

The axiomatic framework will be implemented through a modular architecture, where each mapping step is associated with a dedicated functional module:

- Machine Configuration Module (post-Mapping A): Defines and optimizes machine configurations based on product requirements.
- Product Properties Analysis Module (post-Mapping B): Evaluates the physical and mechanical characteristics of the manufactured part using sensor data and analytical models.
- Process Control Module (post-Mapping C): Integrates IoT-enabled multi-sensor systems for real-time acquisition of process variables.
- Adaptive Monitoring Module (post-Mapping D): Dynamically adjusts monitoring strategies and process parameters according to detected defects and anomalies.

By combining axiomatic design with modularity and IoT concepts, the proposed framework enables the functional independence of monitoring components, the dynamic reconfiguration of monitoring functions and analysis modules, and efficient integration of heterogeneous devices (interoperability).

This approach provides a structured foundation for the development of intelligent, adaptive, and resilient monitoring systems, capable of supporting real-time control and optimisation in smart manufacturing environments.

DISCUSSION, CONCLUSION AND PERSPECTIVES

This paper proposes an axiomatic design-inspired framework for adaptive monitoring in robotized WAAM environments. By structuring the relationships between product requirements, process parameters, monitoring strategies, and defect detection mechanisms, the framework provides a modular and reconfigurable architecture capable of supporting IoT-enabled manufacturing systems. Unlike existing WAAM monitoring approaches, which mainly focus on individual sensing or defect detection solutions, the proposed framework introduces a system-level perspective that explicitly links manufacturing system design and monitoring system configuration. The resulting architecture facilitates interoperability, scalability, and the integration of heterogeneous sensing technologies within a coherent monitoring strategy.

Future work will focus on the implementation and validation of the framework within the LCFC robotized manufacturing cell. Experimental studies will evaluate the integration of multiple sensing technologies, the effectiveness of adaptive monitoring strategies under varying process conditions, and the reconfigurability of the proposed architecture. The generated datasets will also support the development of data-driven approaches, including machine learning and digital twins, for predictive monitoring and autonomous process optimization in smart manufacturing environments.

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SCALABILITY ASSESSMENT OF RECONFIGURABLE MANUFACTURING SYSTEMS UNDER NEW PRODUCT INTEGRATION

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ABSTRACT

Scalability is a key characteristic of Reconfigurable Manufacturing Systems, enabling adjustments in capacity and functionality in response to changing requirements. This paper proposes a framework for assessing RMS scalability under new product integration using a two-stage optimization approach and indicators of residual capacity, reconfiguration effort, and economic feasibility. An initial RMS is optimized for an existing product family, after which the minimum modifications required to integrate a new product are determined. A case study demonstrates the applicability of the proposed framework.

1. INTRODUCTION

Manufacturing companies increasingly face volatile market demands, shorter product life cycles, and growing product variety. It requires production systems that can adapt to change in performance and capacity, without impacting productivity and the high costs related to redesign.

Traditional dedicated manufacturing systems offer high productivity but limited flexibility, while flexible systems provide adaptability at higher investment and operational costs. Reconfigurable Manufacturing Systems (RMS) address this gap through customized flexibility, scalable capacity, and modular architectures (ElMaraghy, 2005; Koren and Shpitalni, 2010).

Core RMS characteristics include modularity, integrability, convertibility, diagnosability, customization, and scalability. Scalability is one of the key characteristics that refers to the ability to adjust production capacity and functionality in response to changing requirements. According to Koren et al. (2016), it enables value creation by changing production capabilities incrementally without replacing the entire system. Consequently, the design of scalable RMS architectures has become a popular topic.

Previous studies have applied optimization to RMS design, configuration, and operational planning to support adaptation to changing production requirements (Deif and ElMaraghy, 2006; Azab and ElMaraghy, 2007; Chaube et al., 2012).

Besides configuration design, researchers have proposed methods for evaluating the reconfigurability and adaptability of manufacturing systems. Youssef and ElMaraghy (2006) introduced the idea of reconfiguration smoothness to assess the effort required to change configurations within the system. Farid (2017) later proposed measures for reconfigurability which are based on the main characteristics of intelligent manufacturing systems. Cerqueus and Delorme (2023) assessed RMS scalability during design by considering future system changes.

Recent studies show growing interest in optimization-based RMS design and changeability evaluation. Research has addressed configuration and resource allocation (Yelles-Chaouche et al., 2021), changeable and reconfigurable system design (Andersen et al., 2024), uncertain future demand (Rezaee and Moghaddam, 2025), and economic and convertibility considerations (Chaikovskaia et al., 2025).

Despite these advances, limited attention has been given to assessing the scalability of an existing RMS when integrating a new product. Existing studies generally assess scalability during the design phase or through capacity expansion analyses, but few approaches explicitly evaluate the ability of an already optimized RMS configuration to absorb a new product while minimizing structural modifications and associated costs.

To address this gap, a progressive optimization approach is proposed to assess RMS scalability under new product integration. The framework first determines an optimized RMS configuration for an existing product family by maximizing throughput while considering routing flexibility. A new product is then introduced, and the minimum-cost reconfiguration required for its integration is determined. Scalability is subsequently assessed using indicators of residual capacity, reconfiguration effort, and economic efficiency.

This paper is organized as follows: Section 2 describes the methodology, Section 3 presents the proposed scalability metrics, Section 4 presents the case study and results, and Section 5 concludes the paper and outlines future research.

2. METHODOLOGY

The proposed methodology assesses RMS scalability by determining the modifications required to integrate a new product. As shown in Figure 1, the framework consists of two optimization stages. First, an optimal RMS configuration is determined for the existing product family and second, the minimum structural changes required for new product integration are identified. Scalability is then assessed using technical and economic indicators.

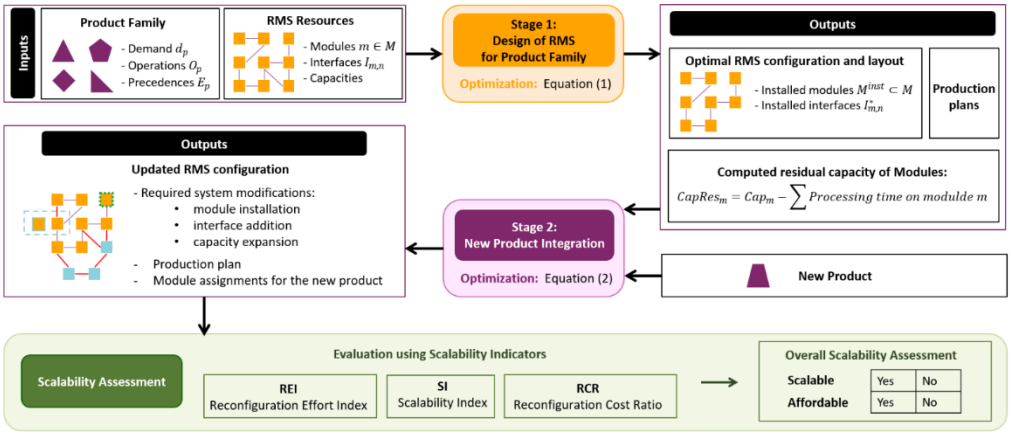

Figure 1: The Proposed Scalability Assessment Framework

This study considers a manufacturing system consisting of a set of modules M interconnected through predefined interfaces. A product family P is characterized by assembly operations O_p and precedence relations $E_p \subseteq O_p \times O_p$, where $(i,j) \in E_p$ indicates that operation i must precede operation j. Operations are assigned to compatible modules while satisfying precedence and capacity constraints to determine feasible production plans.

Stage 1 – Design of RMS for Product Family

The first stage determines the optimal RMS configuration for the existing product family. Routing flexibility allows alternative production plans to improve resource utilization and load balancing. The objective is to maximize total production throughput, as defined in Equation (1):

$$max \sum_{p \in P} \sum_{r \in R} q_{p,r} \qquad (1)$$

where $q_{p,r}$ represents the quantity of product p produced using plan r.

The optimal solution defines the baseline RMS configuration, including the selected modules, interfaces, and production plans, which serves as the reference for new product integration in the second stage. Stage 1 focuses on production performance, while economic considerations are introduced in Stage 2 to evaluate the cost of adapting this baseline configuration.

Stage 2 – New Product Integration

The second stage evaluates the integration of a new product by first utilizing the residual capacity of the baseline RMS. When these capacities are insufficient, the model determines the minimum reconfiguration, including new or replicated modules, additional interfaces, and capacity expansion. The objective minimizes the total reconfiguration cost to integrate the new product, as defined in Equation (2):

$$min \quad c^{mod} \sum_{m \in M} y_m + c^{copy} \sum_{m \in M^{inst}} k_m + c^{I} \sum_{m,n \in M,\, I^*_{m,n}=0} \delta_{m,n} + c^{cap} \sum_{m \in M} e_m \qquad (2)$$

where y_m indicates the installation of a new module, k_m denotes the replication of an existing module, $\delta_{m,n}$ represents the creation of a new interface, e_m denotes additional capacity assigned to module m, $c^{mod}, c^{copy}, c^{I}, c^{cap}$ represent the corresponding modification costs. The optimal solution identifies the minimum cost reconfiguration strategy, providing the basis for the scalability assessment.

3. SCALABILITY ASSESSMENT

In this work, scalability is defined as the ability of an existing RMS to accommodate a new product with limited structural and capacity modifications. The assessment considers residual capacity, structural reconfiguration effort, and economic efficiency.

Capacity Utilization and Residual Capacity

Based on the capacity utilization concept of Kumru (2011), the Residual Capacity Ratio measures the unused capacity of module *m* and is defined as Equation (3):

$$ResidualRatio_m = \frac{CapRes_m}{Cap_m} \qquad (3)$$

where $CapRes_m$ represents the residual capacity of module m after satisfying the production requirements of the existing product family and Cap_m denotes its total capacity. Higher values indicate greater capacity to accommodate additional production demand. At the system level, the ratio is obtained by aggregating the residual and total capacities across all modules and it represents the overall reserve capacity of the RMS before structural modifications are considered.

Reconfiguration Effort and Scalability

Reconfiguration effort has previously been assessed through reconfiguration smoothness (Youssef and ElMaraghy, 2006), considering cost, time, and effort of configuration transitions. This work focuses on structural modifications to the baseline RMS, quantified by the Reconfiguration Effort Index (REI) as defined in Equation (4):

$$REI = w_m \frac{N_{newM}}{N_{newM}^{max}} + w_c \frac{N_{Copy}}{N_{Copy}^{max}} + w_i \frac{N_{newI}}{N_{newI}^{max}} + w_e \frac{\sum_m e_m}{E^{max}} \qquad (4)$$

where N_{newM}, N_{Copy}, N_{newI}, represent the numbers of new modules, module copies, and interfaces, and e_m, denotes the capacity expansion of module m. The corresponding maximum values are used for normalization. The weights w_m,

w_c, w_i and w_e sum to 1 and are set to 0.4, 0.3, 0.2, and 0.1, respectively. The REI is normalized between 0 and 1, with lower values indicating less reconfiguration effort. The Scalability Index (SI) is then defined as Equation (5):

$$SI = 1 - REI \qquad (5)$$

An SI close to 1 indicates minimal modifications, while values near 0 indicate extensive RMS reconfiguration.

Expansion Cost

Economic feasibility is assessed using the Reconfiguration Cost Ratio (RCR), which compares objective of Stage 2 (the minimum reconfiguration cost), with the maximum available investment budget, as defined in Equation (6):

$$RCR = 1 - \frac{Cost_{reconfiguration}}{Budget} \qquad (6)$$

A positive RCR indicates that the required modifications remain within the specified budget, whereas a negative value indicates that the reconfiguration cost exceeds it.

Overall Scalability Evaluation

Overall scalability is evaluated using the SI and RCR, representing structural adaptability and economic feasibility, respectively. An SI threshold of 0.5 is used, with $SI \geq 0.5$ classified as high and $SI < 0.5$ as low. A positive RCR indicates that the reconfiguration remains within the available budget. The resulting classifications are summarized in Table 1.

Table 1: Overall Scalability Evaluation Matrix

SI	RCR	Status	No.
High	Positive	Scalable and affordable	(1)
High	Negative	Scalable but expensive	(2)
Low	Positive	Affordable but structurally less scalable	(3)
Low	Negative	Not scalable and not affordable	(4)

Considering both structural and economic aspects provides a more comprehensive assessment of RMS scalability and supports decisions on new product integration.

4. RESULTS

The proposed framework is evaluated using an RMS dataset with six modules connected through predefined interfaces. The case study includes five products, with P1–P4 forming the initial product family and P5 is introduced in Stage 2. The MILP models are implemented in Python using Pyomo and solved with CBC, in approximately 6s, with most computational effort associated with Stage 1. Larger problem instances are expected to require greater computational effort.

Stage 1 achieves a maximum production of 560 units. Due to routing flexibility, multiple production plans can achieve this throughput with different module utilization levels. The selected optimal solution is used as the baseline for Stage 2.

The M1–M3 interface is removed as it is not required by the selected production plans. For P5 demands of 150 and 210 units, the reconfiguration costs are 160 and 187 cost units, respectively. The corresponding system evolution and reconfiguration actions are shown in Figure 2.

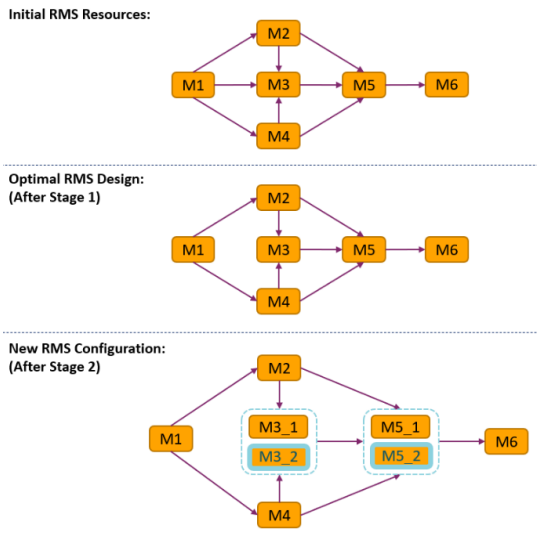

Figure 2: Evolution of the RMS Configuration Throughout the Proposed Framework

Sensitivity to demand, REI weights, and investment budget is reported in Table 2. SI remains above the 0.5 threshold in all investigated cases (0.823–0.9), indicating a stable structural scalability assessment. In contrast, the economic classification changes with the budget. Although these results demonstrate the applicability of the framework, larger RMS instances are required to assess computational scalability.

Table 2: Sensitivity Analysis

D_{P5}	Cost	Mod.	Weights	0.4, 0.3, 0.2, 0.1		0.2, 0.5, 0.1, 0.2	
			Budget	140	250	140	250
150	160	2 Copies (M3, M5)	REI	0.1		0.167	
			SI	0.9		0.833	
			RCR	-0.143	0.36	-0.143	0.36
			Status	(2)	(1)	(2)	(1)
210	187	2 Copies (M3, M5) 27 Extra Cap. (M1)	REI	0.105		0.177	
			SI	0.895		0.823	
			RCR	-0.336	0.252	-0.336	0.252
			Status	(2)	(1)	(2)	(1)

5. CONCLUSION

This paper presents a two-stage MILP framework for assessing RMS scalability under new-product integration. The proposed SI and RCR jointly evaluate structural adaptability and economic feasibility. Results show robust structural scalability across the investigated scenarios, while economic feasibility is more sensitive to the available budget. Future work will evaluate computational scalability on larger industrial scale instances and extend the framework to integrate CSP, transportation costs and module reusability.

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VISION-LANGUAGE MODELS IN SMART INDUSTRY: BENCHMARKING OVOD ARCHITECTURES FOR REAL-WORLD DEPLOYMENT

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KEYWORDS
vision-language models, smart industry, image processing, visual monitoring

ABSTRACT

Modern Smart Industry heavily relies on automated visual monitoring for Occupational Safety and Health (OSH). Traditional closed-set object detectors require costly retraining for new classes. Vision-Language Models (VLMs) and Open-Vocabulary Object Detection (OVOD) offer a scalable, zero-shot alternative, but their practical deployment, constrained by local hardware capabilities and hallucination risks, requires rigorous validation. This paper benchmarks three prominent OVOD architectures (OWL-v2, Grounding DINO, and Florence-2) on a custom dataset comprising specialized industrial tools and protective equipment. Performance is evaluated using Macro-Average Intersection over Union, inference latency, and a custom Over-Prediction Ratio to quantify false-positive hallucinations. The results reveal a critical trade-off: Grounding DINO requires careful threshold calibration to achieve the highest spatial accuracy. In contrast, the generative Florence-2 architecture operates natively without manual thresholding, delivering consistent performance and superior relational reasoning, while its base variants achieve the fastest overall processing speeds at approximately 123 ms per image, effectively outperforming OWL-v2 in both hardware efficiency and baseline accuracy. Ultimately, this study demonstrates that while all evaluated VLMs are highly viable for scalable industrial monitoring, Grounding DINO is preferable for scenarios prioritizing peak localized accuracy, whereas Florence-2 provides a reliable, out-of-the-box alternative that completely eliminates the need for manual threshold tuning.

INTRODUCTION

Modern manufacturing facilities and the broader concept of Smart Industry heavily rely on automation to maximize operational efficiency and maintain rigorous Occupational Safety and Health (OSH) standards (Wang et al. 2018). A critical component of these automated systems is visual monitoring, which includes verifying the compliance of Personal Protective Equipment (PPE), such as safety helmets and goggles, and managing industrial tools like wrenches, screwdrivers, or ladders left in workspaces (Huang et al. 2021).

Historically, automating these visual inspection tasks has relied on traditional, fully supervised object detection architectures, e.g., the YOLO family (Redmon et al. 2016). While highly efficient, these models operate under a closed-set paradigm; they can only detect the specific classes of objects they were explicitly trained on. In a dynamic industrial environment, the continuous introduction of new tools, varying equipment designs, and changing safety regulations makes the repeated collection of annotated data and subsequent model retraining both economically and computationally prohibitive.

Recently, Vision-Language Models (VLMs) and Open-Vocabulary Object Detection (OVOD) frameworks have introduced a paradigm shift in computer vision (Gu et al. 2021). By aligning visual features with natural language embeddings during massive pre-training phases, these models can localize novel, unseen objects based solely on textual prompts (zero-shot detection) (Radford et al. 2021). This capability theoretically eliminates the need for domain-specific retraining, offering a highly scalable solution for industrial monitoring.

Despite their theoretical advantages, the practical deployment of VLMs in real-time industrial scenarios remains a challenge. Whether targeting constrained Edge AI (Chen et al. 2019) or relying on local factory servers, strictly minimizing inference latency remains critical for manufacturing environments. Furthermore, zero-shot models are prone to calibration issues, often generating an excessive number of false-positive bounding boxes when confronted with complex, cluttered backgrounds, leading to severe over-prediction (Li et al. 2023).

This paper investigates whether contemporary VLMs are sufficiently mature for live deployment on the factory floor. The investigation focuses on a comprehensive benchmarking of three OVOD architectures: Google's OWL-v2 (Minderer et al. 2023), IDEA-Research's Grounding DINO (Liu et al. 2024), and Microsoft's generative Florence-2 (Xiao et al. 2024). These models are evaluated on a specialized, custom industrial dataset, deliberately targeting objects not present in the standard COCO dataset (Lin et al. 2014), such as specific PPE and hand tools, as well as testing their capacity for complex spatial reasoning. Finally, a detailed trade-off analysis is presented, examining the relationships between detection accuracy, hardware feasibility, and model calibration, ultimately providing a clear guideline for selecting and deploying VLMs for industrial applications.

MATERIALS AND METHODS

Industrial Dataset Construction

Standard benchmarking collections like COCO (Lin et al. 2014) lack specialized industrial vocabulary. To address this, a custom benchmark was curated from the Open Images Dataset V7 (Kuznetsova et al. 2020), which provides ground-truth bounding box annotations. The compiled corpus consists of 50 selected images for each of the following 10 target classes: Screwdriver, Ladder, Helmet, Glove, Goggles, Wrench, Drill (Tool), Boot, Hammer, and Box. These categories represent key industrial challenges: extreme scale variations (large boxes vs. small hand tools), sparse geometries (ladders), transparency or reflections (goggles), and low-contrast objects blending into cluttered backgrounds (boots, wrenches).

Evaluated Vision-Language Architectures

To assess the readiness of OVOD for industrial applications, three prominent VLM architectures were selected and accessed via the Hugging Face model hub (Wolf et al. 2019):

A. OWL-v2 (Google): A Vision Transformer (ViT) architecture utilizing contrastive image-text pre-training (Radford et al. 2021) to detect objects described by natural language queries (Minderer et al. 2023).
B. Grounding DINO (IDEA-Research): A dual-encoder-decoder transformer model that fuses visual and linguistic features, excelling at fine-grained spatial understanding (Liu et al. 2024).
C. Florence-2 (Microsoft): A generative vision-language model that treats object detection as a sequential text generation problem (Xiao et al. 2024).

Omitting manufacturer prefixes and OWL-v2 patch sizes (16 for base, 14 for large) for brevity, ten variants were evaluated: *owlv2-base*, *owlv2-large*, *owlv2-base-ensemble*, *owlv2-large-ensemble*, *grounding-dino-tiny*, *grounding-dino-base*, *florence-2-base*, *florence-2-large*, *florence-2-base-ft*, and *florence-2-large-ft*. Scale descriptors (*tiny*, *base*, *large*) govern network capacity versus latency trade-offs, *-ft* indicates downstream instruction fine-tuning in generative, threshold-free Florence-2, and *ensemble* indicates multi-prompt embedding aggregation in OWL-v2 to bolster query robustness.

Experimental Setup

The pipeline simulates local industrial REST API deployment, receiving image and text queries to return bounding box JSON payloads. To evaluate true hardware latency without network jitter, benchmarks were executed on an AMD Ryzen 9 3900X CPU, 32GB RAM, and an NVIDIA RTX 3080 GPU. The protocol was split into two phases: a quantitative benchmark measuring baseline detection via single-word prompts (e.g., "helmet", "screwdriver"), and a qualitative evaluation probing relational understanding and attribute binding through descriptive multi-object queries (e.g., "silver wrench next to the red screwdriver").

Evaluation Metrics

Quantitative benchmarking relied on four core metrics:

- Intersection over Union (IoU): Standard spatial overlap ratio between predicted and ground-truth bounding boxes (Everingham et al. 2010).
- Macro-Avg IoU: Unweighted arithmetic mean of IoU scores across all 10 industrial classes, ensuring equal contribution from frequent and sparse categories.
- Avg Latency [ms]: Mean inference time required by the model to process an image and return predictions, critical for assessing real-time application feasibility.
- Over-Prediction Ratio (OPR): Custom ratio of predicted bounding boxes to ground-truth objects, tracking false-positive hallucinations (OPR > 1.0) and missed detections (OPR < 1.0), with 1.0 denoting ideal calibration. Predictions for OWL-v2 and Grounding DINO were evaluated across confidence thresholds from 0.05 to 1.00 (step size 0.05), whereas Florence-2 was evaluated natively without manual thresholding.

For images containing multiple class instances, predictions were paired with ground-truth objects via a greedy highest-IoU assignment, independently matching each ground-truth box to the prediction with maximum spatial overlap. Although this assignment scheme does not inherently penalize duplicate detections of the same instance, the custom OPR metric directly quantifies any resulting over-prediction or hallucination alongside IoU. Complementing these metrics, the qualitative evaluation relied on visual inspection to assess how reliably each architecture resolved complex, relational sentence prompts and spatial constraints.

RESULTS AND DISCUSSION

Quantitative Performance Analysis

This section presents the results of the quantitative performance benchmarking of the selected OVOD architectures. The detection accuracy, measured by the Macro-Avg IoU across all ten industrial classes and a range of confidence thresholds, is illustrated in Figure 1.

The Grounding DINO variants (base and tiny) achieved the highest peak detection accuracy of approximately 0.88–0.90 at very low confidence thresholds. However, their performance degraded rapidly as the threshold increased, indicating a narrow range of optimal calibration. The OWL-v2 variants generally demonstrated lower peak performance than Grounding DINO at low thresholds, but the ensemble variants (e.g., *owlv2-large-ensemble*) significantly outperformed the standard variants across all threshold values, maintaining a balanced IoU over a wider operational range. Conversely, the Microsoft Florence-2 models demonstrated remarkably stable performance. As generative models that natively output detections without requiring an external strict confidence threshold, their Macro-Avg IoU remained constant across the evaluated spectrum at approximately 0.67. While lower than the theoretical peak of Grounding DINO or OWLv2, this stability presents a significant advantage for automated systems where performance predictability is critical.

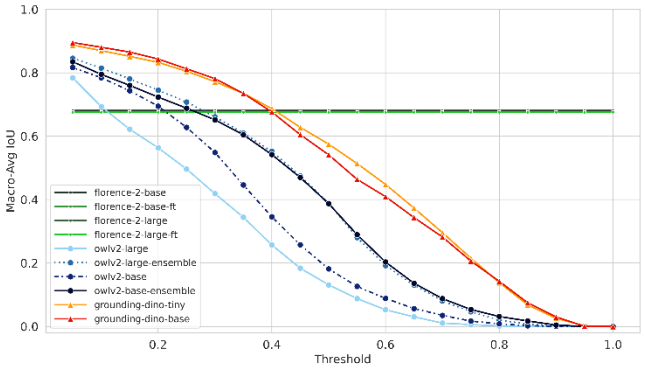

Figure 1: Macro-Avg IoU across confidence thresholds for the evaluated models

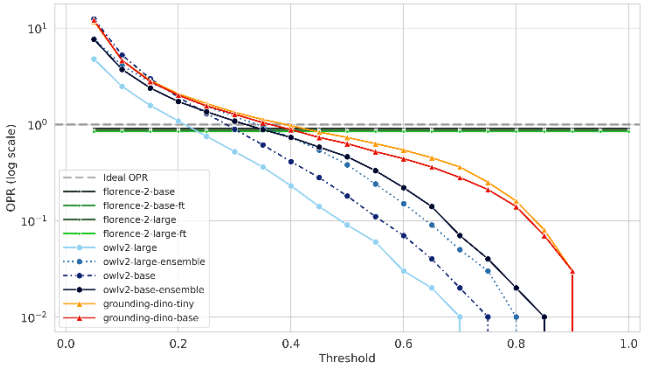

Figure 2: OPR across confidence thresholds for the evaluated models

To fully understand the practical implications of these accuracy scores, it is crucial to examine the corresponding OPR, presented on a logarithmic scale in Figure 2. The data reveals a fundamental trade-off inherent to non-generative open-vocabulary architectures: while the highest Macro-Avg IoU values were achieved at very low confidence thresholds, these settings simultaneously triggered massive over-prediction. For instance, at a threshold of 0.05, models such as *grounding-dino-base* and *owlv2-base-ensemble* generated an excessive number of false-positive detections, reaching OPR values well above 10.0 and 5.0, respectively. In a real-world industrial monitoring system, this behavior would result in a prohibitive number of false alarms (hallucinations). Therefore, the ideal confidence threshold must be carefully isolated within a specific operational window, where the OPR is as close to the 1.0 mark as possible. This ensures a critical balance between correctly detecting actual targets and suppressing spurious bounding boxes.

To provide actionable guidelines for system deployment, Table 1 extracts the performance metrics of the evaluated models at their empirically determined optimal thresholds. For OWL-v2 and Grounding DINO architectures, two specific thresholds were selected per model: one yielding an OPR slightly above 1.0 (favoring sensitivity) and one slightly below 1.0 (favoring precision). This focused analysis allows for a realistic assessment of the models under stable, deployable conditions. It is worth noting that Florence-2 tends to natively slightly under-predict (with OPR ranging from 0.84 to 0.90), occasionally missing some target objects in favor of completely eliminating false positives.

Furthermore, Table 1 details the average inference latency for each architecture. It is important to note that inference latency is fundamentally tied to the model's architectural complexity (parameter count and input resolution) and remains completely independent of the applied confidence threshold. The benchmarking results highlight that the *florence-2-base* is the fastest architecture, requiring approximately 123 ms per image, which makes it highly suitable for high-framerate processing on local factory servers. The *grounding-dino-base* and the *owlv2-base-ensemble* present balanced alternatives, with average latencies of roughly 246 ms and 294 ms, respectively. In contrast, the heavier OWL-v2 large variants demand significantly higher processing times (up to ~778 ms). While robust, this high latency may preclude their use in real-time safety systems operating on constrained factory hardware.

Qualitative Assessment

To evaluate the models' semantic comprehension and capacity for deep spatial reasoning, a qualitative analysis was conducted across multiple challenging industrial scenes. Figure 3 presents a representative example from this testing phase, featuring heavily clustered hand tools.

For this qualitative assessment, the largest available model variant from each architectural family was selected to maximize semantic reasoning and spatial understanding capabilities. The confidence thresholds for non-generative models were set to 0.35, aligning with the empirical data established in Table 1 to maintain an optimal OPR near the 1.0 mark. Consequently, *grounding-dino-base* was utilized as the largest representative of its architecture. For the OWL-v2 family, the *owlv2-large-ensemble* was chosen; the ensemble mechanism internally aggregates multiple prompt embeddings, which significantly enhances robustness and detection stability against complex, multi-word queries. Finally, the *florence-2-large-ft* architecture was evaluated. The *-ft* version was explicitly selected because it underwent extensive instruction tuning, enabling the generative model to natively process and execute full-sentence instructions without the need for manual threshold calibration.

These models were tasked with executing three distinct textual queries of increasing complexity on the representative image:

- P1 (Green): "screwdriver with a red handle" (testing specific attribute filtering).
- P2 (Orange): "pliers with blue handles" (testing complex physical descriptions of degraded tools).
- P3 (Purple): "silver wrench next to the red screwdriver" (testing advanced spatial reasoning and anchor-object relations).

Notably, the target class for prompt P2 ("pliers") was intentionally selected from outside the core 10-class quantitative dataset. This inclusion was designed to explicitly test the architectures' capacity for true zero-shot generalization when confronted with unpredicted, out-of-distribution objects in a real-world industrial setting.

The visual results, presented in Figure 3, illustrate clear differences in the fundamental reasoning paradigms of the evaluated architectures.

Panel A depicts the output of the *owlv2-large-ensemble*. While the model successfully localized the basic attribute prompts (P1 and P2), it failed to resolve the relational query (P3). Operating essentially as a "bag-of-words" detector, it decomposed the complex prompt into independent tags, incorrectly placing purple bounding boxes on a black screwdriver, the red screwdriver itself, and the distant distractor wrench on the far right. It demonstrated an inability to comprehend the spatial constraint "next to".

Panel B shows the performance of the *grounding-dino-base*. This architecture demonstrated significantly superior spatial awareness. It confidently detected the target tools for P1 and P2. Crucially, for the complex relational prompt (P3), it successfully isolated the correct silver wrench immediately adjacent to the red screwdriver, effectively ignoring the visually similar distractor wrench located on the right side of the image.

Panel C illustrates the output of the *florence-2-large-ft*. It is important to note that the confidence scores of 1.00 displayed in the visualization were hardcoded for display purposes, as the specific generative nature of the Florence-2 architecture natively outputs coordinate tokens rather than traditional bounding box probabilities. By processing the text as a cohesive instruction rather than isolated keywords, Florence-2 perfectly understood the anchor object, the target object, and the spatial relationship binding them. Its native capability to execute highly specific visual tasks from complete textual sentences resulted in tight, accurate bounding boxes without any spurious detections or hallucinations. While Figure 3 serves as a representative illustration of these distinct architectural approaches, a comprehensive statistical evaluation of spatial reasoning on dedicated relational datasets remains a subject for future quantitative research.

CONCLUSIONS

This paper investigated the feasibility of deploying OVOD architectures, specifically OWL-v2, Grounding DINO, and Florence-2, for real-time industrial monitoring and OSH compliance. By benchmarking these VLMs on a custom, challenging dataset of industrial tools and protective equipment, the study highlighted the critical trade-offs between detection accuracy, hardware latency, and model calibration.

The quantitative analysis revealed that while traditional contrastive OVOD models, such as Grounding DINO, achieve the highest theoretical peak IoU, they are inherently prone to severe hallucination in cluttered industrial environments. This tendency results in an excessively high OPR, necessitating rigid and often brittle confidence threshold calibration to suppress false positives. In contrast, the generative architecture of Florence-2 demonstrated a distinct operational advantage. By inherently functioning without manual threshold tuning, the fine-tuned Florence-2 models maintained a stable and near-optimal OPR natively. Furthermore, *florence-2-base* exhibited the lowest inference latency (approximately 123 ms), making it highly suitable for real-time monitoring applications on the factory floor.

Table 1: Performance metrics of the evaluated OVOD models at their optimal confidence thresholds (OPR ≈ 1.0). Florence-2 models are evaluated natively without thresholding. Latency values denote means across 10 independent runs.

Model	Th	Avg Latency [ms]	Macro Avg IoU	OPR	IoU									
					Helmet	Screwdriver	Ladder	Glove	Goggles	Wrench	Drill	Boot	Hammer	Box
owlv2-base	.25	297.76	.63	1.29	.55	.72	.69	.48	.56	.73	.70	.45	.82	.59
owlv2-base	.30	297.76	.55	0.89	.46	.64	.64	.40	.48	.65	.59	.35	.78	.49
owlv2-large	.20	778.26	.56	1.08	.22	.74	.68	.28	.34	.81	.71	.51	.87	.49
owlv2-large	.25	778.26	.50	0.75	.16	.69	.61	.22	.27	.75	.62	.44	.84	.38
owlv2-base-ensemble	.30	294.77	.65	1.08	.64	.73	.69	.59	.29	.75	.73	.65	.76	.69
owlv2-base-ensemble	.35	294.77	.60	0.88	.62	.73	.67	.50	.25	.67	.68	.60	.70	.63
owlv2-large-ensemble	.30	779.28	.66	1.20	.58	.73	.74	.44	.32	.81	.75	.71	.88	.65
owlv2-large-ensemble	.35	779.28	.61	0.95	.48	.72	.72	.36	.30	.79	.71	.63	.84	.56
grounding-dino-tiny	.35	186.55	.74	1.12	.72	.59	.74	.75	.75	.68	.70	.82	.89	.71
grounding-dino-tiny	.40	186.55	.69	0.97	.70	.55	.70	.68	.72	.63	.64	.79	.81	.65
grounding-dino-base	.35	246.74	.74	1.04	.71	.64	.76	.73	.52	.73	.80	.83	.89	.74
grounding-dino-base	.40	246.74	.68	0.88	.67	.59	.71	.63	.45	.59	.76	.80	.88	.69
florence-2-base	-	123.42	.68	0.90	.56	.60	.75	.59	.69	.84	.54	.71	.79	.73
florence-2-large	-	263.40	.68	0.85	.52	.65	.74	.58	.67	.82	.63	.67	.76	.72
florence-2-base-ft	-	126.10	.68	0.86	.58	.62	.74	.55	.56	.79	.66	.69	.84	.74
florence-2-large-ft	-	266.86	.68	0.84	.56	.64	.75	.58	.56	.81	.65	.67	.81	.73

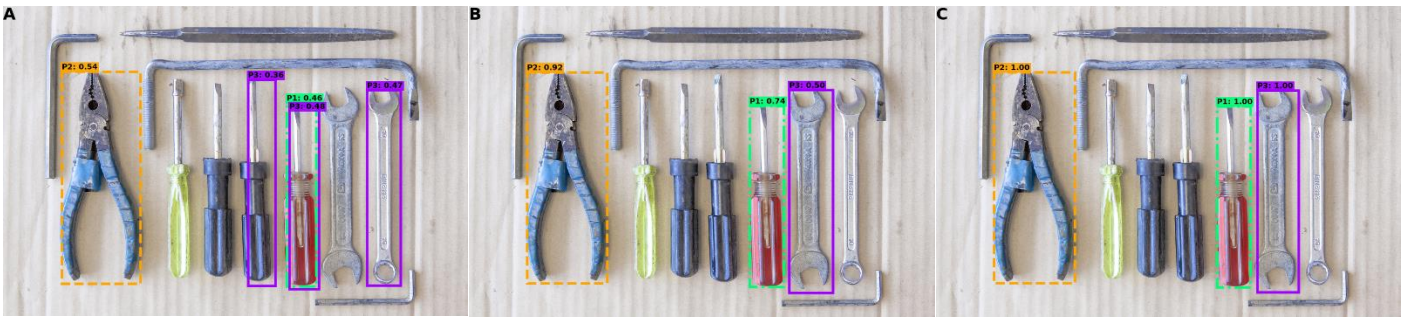

Figure 3: Qualitative comparison of object detection performance. Panels illustrate localization accuracy and confidence scoring for the evaluated models: (A) *owlv2-large-ensemble*, (B) *grounding-dino-base*, and (C) *florence-2-large-ft*

The qualitative assessment further underscored the limitations of simpler contrastive approaches, as OWL-v2 struggled to resolve intricate descriptive prompts involving specific attributes and spatial relationships. Conversely, both Grounding DINO and Florence-2 exhibited advanced scene understanding. They successfully moved beyond simple text-to-image matching, accurately interpreting detailed, full-sentence instructions (e.g., isolating a specific tool based on its proximity to an anchor object).

Ultimately, this study concludes that while all evaluated VLMs offer highly viable, scalable alternatives to closed-set models, their optimal deployment depends on specific operational priorities. Grounding DINO remains the premier choice for scenarios demanding peak localized accuracy, whereas Florence-2 currently provides an out-of-the-box solution for complex Smart Industry applications. While this Open Images-based dataset establishes a robust comparative baseline for OVOD architectures, natural next steps involve in-situ validation on live factory-floor cameras and a direct comparison against a fully supervised closed-set baseline (e.g., YOLO) to precisely quantify the performance trade-offs of zero-shot deployment. Collectively, these models mark a crucial step forward for intelligent, automated inspection and future robotic interaction on the factory floor.

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INDUSTRY 5.0 IN PRODUCTION MANAGEMENT COURSES: INTEGRATION OF OPERATORS' WELL-BEING USING SIMULATION-BASED DECISION-MAKING

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Industry 5.0, Human factors, Operators' well-being, Multi-model simulation, Production management course.

ABSTRACT

Industry 5.0 calls for the development of human-centered approaches and explicit integration of human well-being in all engineering branches. While scientific research on approaches considering human is progressing, the question that arises is how to bring these approaches closer to the education of young engineers. This paper presents how the existing production management simulation course can be enriched with human factors. The integration of learning-forgetting-fatigue-recovery model in discrete event production simulation is presented. The basic course structure based on human factor model is also shown. It is argued that the mentioned organization enables sensibilization of future engineers towards Industry 5.0.

INTRODUCTION

Industry 5.0 shifts the focus from technology-oriented to human-centered, resilient, and sustainable systems. Consequently, the production system needs to be designed and optimized considering human actors as an input.

This human-oriented focus demands paradigm shift in engineering daily work. Hence, all new engineers need to be sensitive to the 5.0 triplet. With this in mind, the goal is not to develop only specialized human-centered education programs, but to bring large population of future engineers towards Industry 5.0 throughout traditional subjects. For the production management courses this means to ask: How to introduce Industry 5.0 and more particularly human factors into production system representations and decision-making processes?

Traditionally, production management proposes performance indicators to evaluate the impact of organizational (takt time, number of operators, task repartitions) and technical (equipment, tools, support) choices. When humans are included, these choices directly impact ergonomics (physical, cognitive, organizational), which further influences operators' behavior, their individual performance, and, consequently, production system performance (Gross et al., 2023). Knowing that these propagation mechanisms can be very complex, the goal is to find a proper model that considers human in production system without demanding too much pieces of information that are not possible to obtain. On the other hand, one should also avoid oversimplification in which human behavior is represented as one parameter.

Finally, while production management can be taught using different means, simulation remains a powerful motor to represent different phenomena observable in real production systems. Simulation-based games are widely used in education (Neriz et al., 2020; Schmuck, 2021). Simulation is also integral part of digital twins and in that sense its importance has only increased with Industry 4.0 and 5.0 (Turner and Garn, 2022). Consequently, this work uses simulation as a base for the human-sensitive production management course.

PROPOSED EDUCATIONAL HUMAN FACTOR SIMULATION MODEL FOR THE SENSIBILIZATION TOWARDS INDUSTRY 5.0

The goal of the presented model is to take into account operators' well-being and skill, but also to be able to estimate the resilience of the production system based on its human resources. Consequently, the proposed educational simulation model uses discrete event simulation of the production system as the base for the integration of operators behavior. This way, the basic production indicators (e.g. manufacturing lead time, work in process) can be enriched with the human-oriented ones. Further, the model should be integrated into discrete event simulation due to its presence in the domain. We discuss the implementation using the AnyLogic PLE, but the same model can be adapted for other software.

Human Factor Models

The human factor models are taken from the literature and, for the pedagogic purpose, they can be divided in 3 categories: well-being model, skill model, and operator-production system model. The first two models are all based on learning-forgetting-fatigue-recovery (LFFR) model (Jaber et al., 2013; Givi, Jaber and Neumann, 2015; Neumann and Medbo, 2017), while the third model extends the previous one towards human error probability (Givi, Jaber and Neumann, 2015).

Well-being model: fatigue-recovery. The fatigue-recovery model assumes that the fatigue is accumulated over time

based on the fatigue rate (λ). Similarly, the operator is recovering from the accumulated fatigue over time and based on recovery rate (μ) that may not be the same as the fatigue rate. This can be represented using the following equations:

$$F(t_i) = 1 - e^{-\lambda t_i} \quad (1)$$

$$R(\tau_i) = F(t)e^{-\mu \tau_i} \quad (2)$$

$$F(t + \Delta t) = F(t) + (1 - F(t))(1 - e^{-\lambda \Delta t}) \quad (3)$$

Fatigue, $F(t_i)$, is calculated using the duration of the production cycle (t_i). Recovery, $R(\tau_i)$, is obtained based on the decrease of the initial fatigue level by the duration of the break (τ_i). The equations (1) and (2) can be adapted to fit the discrete nature of the simulation and consequently to enable fatigue update after every produced part, Δt (3).

Skill model: learning-forgetting. The skill model determines the operation time (T_{ni}) and is based on learning and forgetting. The model assumes that unexperienced worker produces the first unit of the part with the highest time (T_1). This time is then improved using the learning index (b) with every produced part until the minimum possible operation time, M. Similarly, part of this experience is forgotten every time operator breaks the production cycle of the same part (due to the break or because of the production of another part type). The equations (4) to (9) represent the learning and forgetting processes, where u_i is the number of parts remembered at the beginning of the cycle; n_i is the current number of parts in the cycle i; f is the forgetting exponent; s is the number of items that would have been produced if no interruptions occurred; and D is the total forgetting time.

$$T_{ni} = T_1(M + (1 - M)(u_i + n_i)^{-b}) \quad (4)$$

$$T_{ni} = T_1(u_i + n_i)^{-b} \quad (5)$$

$$u_i = (u_{i-1} + n_{i-1})^{1+\frac{f_{i-1}}{b}}(u_{i-1} + n_{i-1} + s_{i-1})^{-\frac{f_{i-1}}{b}} \quad (6)$$

$$f_{i-1} = \frac{b(1-b)log(u_{i-1} + n_{i-1})}{\log(1 + B)} \quad (7)$$

$$B = \frac{D}{t_{ni-1}} = \frac{D}{\int_0^{u_{i-1}+n_{i-1}} T_1 y^{-b} dy} \quad (8)$$

$$s_{i-1} = \left[\frac{1-b}{T_1}\tau_{i-1} + (u_{i-1} + n_{i-1})^{1-b}\right]^{\frac{1}{1-b}} - (u_{i-1} + n_{i-1}). \quad (9)$$

Operator-production system model: human error probability (HEP). In human reliability analysis, numerous factors contribute to human error (Givi, Jaber and Neumann, 2015). With exception to work teams, learning-forgetting-fatigue-recovery model in combination with discrete simulation considers majority of them. Consequently, for the educational purpose, the human error probability model is presented in equation (10).

$$HEP = 10^{\log\left(0.5\left(\frac{T_{ni}}{T_1} + \frac{F(t_i)}{0.99}\right)\right)} \quad (10)$$

The HEP is updated with every change in production time and / or fatigue. This makes it dynamic and easily integrable with discrete event simulation.

Integration of Human Factor Models into a Discrete-Event Production System Simulation

For the integration of human factors, operator's agent should be created. The explicit state chart may or may not be required based on the teacher's choices and software used. The state chart includes 2 states: on break / idle, and working. The operator moves from idle to working state once there are parts to produce or when break ends. Consequently, it goes back to on break / idle state if there are no parts to produce or if there is a scheduled break. The change from one state to another is used to define the working cycles and to enable calculation of fatigue / recovery and production time. In case where students don't have the required knowledge in multi-agent modeling, in software such as AnyLogic, the worker's model can be established using the properties of a Resource pool without explicitly creating a state chart. On seize and On release properties can be used as a triggers for the calculation of fatigue-recovery-learning-forgetting.

HUMAN FACTORS IN THE PRODUCTION MANAGEMENT COURSE

This section presents the example of the production management course including its structure, pedagogic scenarios and evaluation.

Production Modeling and Simulation Course Description

The course is focusing on Production system modeling and simulation. The main learning outcomes include capability to perform physical and information flow management. It is 30-hour long M1 level course. The students have the background in production system design, but no discrete event or any other simulation knowledge. There is no special section dedicated to human factors. Human factors are introduced during the 8-hour long project. The student project represents a group work of 3 to 4 people. It is assumed that the base needed for the modeling of human well-being is acquired during the introduction to production system simulation. The good reflexes regarding the decision-making with or without human factors are acquired during the introduction to the theory of constraints. In the project description, the context regarding the Industry 5.0 and human well-being is provided together with the mathematical model. The students are invited to integrate it in their production system simulation and to provide one or several indicators related to human factors and more particularly well-being. The suggestion of the list of indicators is provided, but the students have the liberty to append and / or modify this list.

The student project is divided into 2 main phases: modeling and decision-making. In the first phase, the students develop human factor production model and propose indicators. In the second phase, based on the proposed indicators, 20 iterations of decision-making should be performed. After each iteration, using the theory of constraints the students try to maximize the performance of the model including human factors.

Project Pedagogic Cases

While the project idea remains the same, the project problem can be tailored to treat different potential solutions related to

human factors. These pedagogic cases can include, but they are not limited to:

Role and trade-off scenario. Inspired by the work of Ivanov (2017), for the decision-making phase every member of the project group can be assigned a role. The role owns and is responsible for one or several performance indicators. With this structure, more dialog and discussion can be imposed to the students. In this case, during every iteration, the point of view of every role should be documented, as well as a justified common decision after the discussion between the roles.

Policy scenario. Introduction of the specific policies, such as margin strategies (Lanfranchi and Duveau, 2008) or any other, might be requested. With this pedagogic case, the students may present their critical opinion on the current practices and promote discussion regarding the efficiency of the proposed solutions.

Reconfigurability scenario. Similarly to the previous scenario, the students may be requested to go beyond the traditional production system organization and to think if / how reconfigurability can be beneficial for the production system performance.

Course Evaluation

The presented course has 4 evaluations: 2 individual knowledge tests, project defense, and project report. The project report has two main goals. Firstly, it should detail modeling choices. Secondly, it should systematically present the decision-making steps. It is recommended to have the structured format for the second part of the report. This can be a table with indicators and the place for the decision and its justification. The project defense includes presentation and Q&A session that promotes discussion.

DISCUSSION

The proposed human factor model has been used in student projects regularly since the establishment of the new course structure in 2023/2024. In majority of cases students were implementing the fatigue-recovery model. Fatigue-recovery-learning-forgetting and HEP models were used in longer student projects related to different pedagogical activities. Certain group of students go beyond the required task and implement (break) policies in order to approach their model to the reality. During the discussions, reconfigurability as a solution for well-being was often mentioned. Students who didn't manage to implement the human factor model tried to estimate well-being using the resource utilization.

While the proposed human factor simulation model suits well the discrete production and short-term estimation of operators' well-being, there are certain limits. The recovery model assumes no accumulation of fatigue after the rest. The fatigue, recovery, and learning exponents are not easy to estimate and there is no detail study on their variation between individuals. There is no effect of fatigue on production time. Finally, it should be stressed, that reflections are needed in order to broaden the proposed pedagogic model. The current focus is only on physical fatigue, while the mental workload is neglected. The choice of the proper model that is not too complex, but at the same time sufficiently precise for the production management domain is still open.

CONCLUSION

The paper presents the model description and course structure for the integration of Industry 5.0 into the production management course. The goal is to show how existing production management simulation and decision-making courses can be enriched to include workers' well-being, skill, and their impacts on the traditional performance indicators. The proposed educational simulation model integrates learning-forgetting-fatigue-recovery and argues that this approach can be used to introduce worker's well-being to an engineering student. Finally, the paper presents the course structure and discusses the main observations and limits.

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Simulation in Logistics

MODELLING OF POTENTIAL SYMBIOSIS NETWORKS BASED ON A FUZZY COGNITIVE MAP: A NIGERIA CASE

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KEYWORDS

Industrial symbiosis, fuzzy cognitive map, quantitative influence analysis, network analysis.

ABSTRACT

Industrial symbiosis promotes resource efficiency through the exchange of materials, energy, and by-products among industrial actors, contributing to the transition toward a circular economy. However, modelling and assessing potential symbiotic networks remains challenging due to the complexity of interactions and the uncertainty affecting decision-making. This study proposes a fuzzy cognitive map to model potential industrial symbiosis networks in a specific case, the Ota Park in Nigeria. Thus, it highlights the importance of advancing circular economy strategies in emerging regions.

I. INTRODUCTION

Industrial symbiosis (IS) has emerged as one of the most promising strategies within industrial ecology to improve resource efficiency and accelerate the transition toward a circular economy (Palagonia et al., 2025; Cisi and Napoli, 2024). Although numerous industrial symbiosis initiatives have been successfully implemented worldwide (Yu et al. 2024; Nouinou et al., 2021; Neves et al. 2020), identifying viable synergies remains a complex decision-making problem involving multiple actors, heterogeneous resources, uncertain information, and dynamic interactions (Hafiz et al., 2021).

In this context, the present study proposes a modelling of potential symbiotic flows based on fuzzy cognitive maps (FCMs). FCMs are particularly valuable for representing complex causal relationships and expert knowledge under conditions of uncertainty and incomplete information (Schuerkamp and Giabbanelli, 2024). They have been widely applied in environmental management and sustainability studies, but their application to industrial symbiosis remains relatively limited (Fons et al., 2004). In our case, the FCM is applied to a specific case study: the Ota Park industrial zone in Nigeria. Indeed, the African continent remains understudied, and its specific characteristics and constraints can complicate the implementation of industrial symbiosis, even though the deployment of industrial symbiosis networks would be highly beneficial for the countries involved (Greenberg & Rogerson, 2014, Negesa et al., 2024).

The proposed methodology first identifies and quantifies the influence of stakeholders, resource flows, and decision variables. It then models potential symbiotic exchanges for establishing sustainable industrial symbiosis networks. The proposed approach integrates expert knowledge to represent causal relationships among key drivers and constraints influencing network development. The results demonstrate that FCMs could provide valuable insights into the dynamics of industrial symbiosis systems.

The rest of this paper is organized as follows. The context is given in Section 2. Section 3 presents the fuzzy cognitive map. Section 4 presents provides a discussion based on these results. Conclusions drawn from the research findings and the future perspectives are discussed in Section 5.

II. CONTEXT

The transition toward more sustainable industrial systems has become a strategic priority for many developing economies, particularly in Africa, where rapid industrialization is accompanied by increasing pressure on natural resources, energy systems, and waste management infrastructures. This context creates a strong need for innovative approaches capable of improving resource efficiency, promoting circular economy principles, and supporting sustainable industrial development. Within this context, evaluating and developing industrial symbioses requires methods capable of addressing numerous heterogeneous actors whose relationships are characterized by uncertainty, incomplete information, and dynamic behaviors.

To address these challenges, fuzzy cognitive maps (FCMs) have attracted increasing attention as an intelligent modeling framework capable of representing complex causal relationships under uncertainty. An FCM models a system as a directed weighted graph in which concepts correspond to relevant variables or stakeholders, while weighted causal links describe the strength and direction of their interactions (Fons et al., 2004). By integrating fuzzy logic with graph-based reasoning, FCMs provide an intuitive and flexible tool for analyzing the propagation of influences throughout complex industrial ecosystems.

The proposed approach is illustrated through a real-world case study conducted in Ota Industrial Park, in Nigeria. Ota is one of Nigeria's largest industrial clusters, hosting numerous manufacturing companies operating in sectors such as food and beverage processing, cement production, chemicals, plastics, packaging, pharmaceuticals, steel manufacturing, and consumer goods. Its high industrial diversity and strategic

location near Lagos create favorable conditions for resource sharing and by-product exchanges.

It is first necessary to identify the relevant stakeholders involved in the industrial ecosystem, as well as the possible resource exchanges and causal relationships that may exist between them (Figure 1).

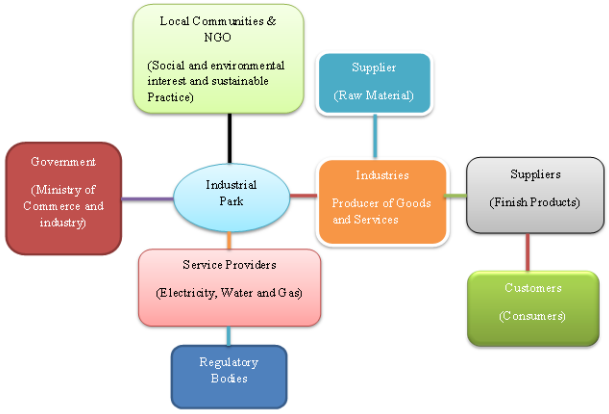

Figure 1- Links between Actors in the Ota Industrial Park.

The following section presents the fuzzy cognitive map and how it could help modeling industrial symbiosis networks.

III. FUZZY COGNITIVE MAP

The first stage consists of developing a cognitive map to qualitatively represent the potential relationships among the industries located within the study area. In this representation, each node corresponds to an industrial facility or sector, whereas the directed links illustrate potential exchanges of waste, by-products, or other resources between generating and receiving industries. The cognitive map is constructed using available information on waste characteristics together with insights derived from documented industrial symbiosis practices and established eco-industrial parks and questionnaires send to stakeholders.

Within the FCM, each causal relationship is assigned a weighted value, typically ranging from −1 to +1, to represent the strength and direction of influence associated with a given resource exchange. These criteria are expressed using qualitative scales (e.g., very high to very low or very easy to very difficult) and subsequently translated into numerical influence values for network analysis. The resulting fuzzy cognitive map is given in Figure 2.

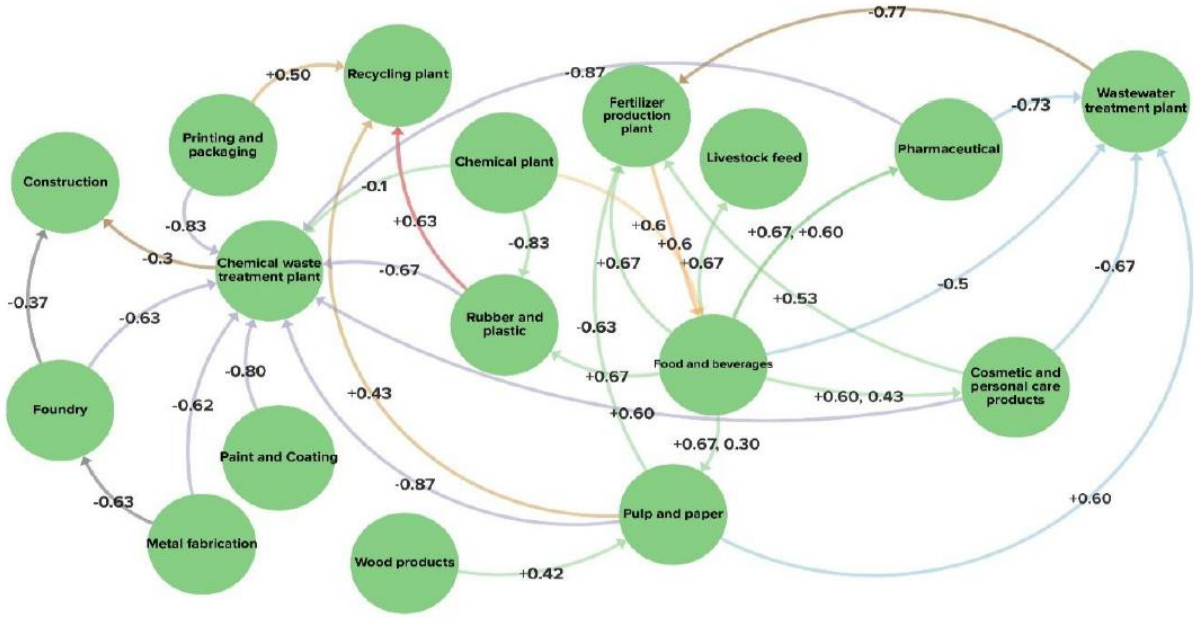

Figure 2- Fuzzy Cognitive Map of the IS Network, with Nodes as Arrows denoting feasible average Influence Weight Score Waste Stream.

The proposed model incorporates the recovery of chemical residues as well as the recycling of plastic and metal waste, thereby reducing the consumption of virgin materials and improving resource efficiency throughout the industrial park. More broadly, the proposed FCM encourages cooperation among companies operating in different sectors, fostering innovation and creating new opportunities for value generation from industrial by-products.

FCMs offer a systematic approach to assess the relative significance and dynamics of concepts within intricate systems. By representing causal relationships and feedback loops, FCM facilitates a detailed understanding of how various components interact and impact one another over time. Based on these results, a discussion within Ota park is presented in the next section.

IV. DISCUSSION

The analysis of our proposed FCM highlights the food and beverage industry as the primary driving force within the proposed industrial symbiosis network. Owing to the diversity and quantity of its by-products, this sector forms the core of several potential symbiotic exchanges identified through the cognitive mapping process. Residual materials such as spent grain, bagasse, fruit processing residues, wastewater, and excess heat can be recovered and reused by neighboring industries for applications including biofertilizer production, chemical processes, and water treatment.

A major symbiotic opportunity identified within the industrial park concerns the recovery of organic residues through a nutrient recycling loop. Organic by-products, including spent

grain and residual yeast, can be valorized as feedstock for the production of organic fertilizers, thereby reducing dependence on conventional synthetic fertilizers and promoting more sustainable agricultural practices. Another promising exchange involves the recovery of carbon dioxide (CO_2) generated during fertilizer production. After capture and purification, this CO_2 can be supplied to beverage manufacturers for carbonation processes, creating an additional circular resource flow between industries. Furthermore, the recovered CO_2 could support greenhouse cultivation in the surrounding agricultural areas of Ota, contributing to enhanced crop production while limiting greenhouse gas emissions.

Another important symbiotic pathway involves the valorization of agro-industrial residues within the manufacturing sector. Fibrous by-products, such as bagasse generated by food processing activities, can be redirected to the pulp and paper industry as an alternative raw material, thereby reducing the demand for virgin wood fibers and contributing to more sustainable resource management. In parallel, surplus thermal energy released during food production may be recovered to support paper manufacturing processes, improving overall energy efficiency. The paper industry, in turn, generates by-products such as white sludge that can be reused by fertilizer manufacturers, creating additional opportunities for material recovery and reinforcing circular exchanges across industrial sectors.

The following section summarizes the main contributions of this work, discusses the insights obtained from the Ota Industrial Park case study, and highlights future research directions for improving the analysis and implementation of industrial symbiosis strategies.

V. CONCLUSION

Industrial symbiosis networks represent complex socio-technical systems in which economic, environmental, technological, and organizational factors interact through numerous direct and indirect relationships. Modeling symbiotic flows with the help of fuzzy coginitive map is very promising for complex socio-technical systems, specifically for emergent countries like Nigeria. The Ota Industrial Park case study demonstrates how coordinated material and energy exchanges can support the transition toward a circular industrial ecosystem while strengthening both environmental and economic sustainability.

Future developments are expected to combine the fuzzy cognitive map with optimization-based models in order to highlight, under different cost and impact scenarios, potential industrial symbiosis implementation pathways and thereby demonstrate its relevance and value to both decision-makers and industrial stakeholders. Artificial intelligence techniques may further enhance influence analysis by automatically generating cognitive maps from operational datasets, while dynamic simulations could improve prediction of long-term network evolution under different policy or market conditions.

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GRADES – A NEW GRAPH BASED SYSTEM DYNAMICS AND DISCRETE EVENT SIMULATION FOR BUSINESS SECTOR SUPPLY CHAINS. PART 1: AGRICULTURE

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KEYWORDS

Event based Simulation, Discrete Event Simulation, Agent based Simulation, Graph Construction, Supply Chain Matching, Synthetic Data Generation

ABSTRACT

Supply chain disruptions driven by geopolitical events, natural hazards, infrastructure, and market volatility have reinforced the interest for realistic business data to support resilience analysis. This paper presents with *GraDES* a simulation framework that combines graph-based network construction, agent-based modeling (ABM), and discrete-event simulation (DES) to generate synthetic invoices and, models realistic supplier–provider ecosystems while avoiding confidentiality constraints associated with real business data utilization. The approach models supply chains as interconnected graphs in which organizations are represented as autonomous agents whose interactions, decisions, and operational dynamics develop through discrete events. The resulting synthetic data are evaluated against publicly available statistical and economic indicators and show close alignment with the aggregate patterns and network characteristics for the selected agricultural business sector.

CREATING SUPPLY CHAIN NETWORKS

A major challenge in supply network research is the scarcity of empirical multi-tier datasets. Because firms often lack visibility beyond their direct suppliers and supply chain relationships are rarely disclosed, comprehensive network data remain limited, constraining the analysis and modeling of real-world supply networks (Brintrup et al. 2018, Ialongo et al. 2022).

GraDES addresses this issue by generating a large-scale synthetic supply chain network from public economic statistics, product classifications, and domain knowledge. Instead of replicating individual firms, it creates a structurally plausible representation of the agricultural sector, capturing heterogeneous firm populations, geographic embeddedness, and multi-tier dependencies.

This graph serves as the foundation for all simulation mechanisms, defining material flows, supplier substitution options, dependency chains, and disruption pathways. Because resilience emerges from the interaction of network topology and actor behavior, a realistic network structure is essential for analyzing cascading failures, bottlenecks, recovery dynamics, and systemic vulnerabilities (Carvalho and Tahbaz-Salehi 2019, Fujiwara and Aoyama 2010). For the agricultural use case, GraDES generates a heterogeneous network of around 750,000 firms and over 1.7 million supplier relationships, providing a scalable environment for evaluating disruptions and adaptation mechanisms through hybrid simulation. These figures describe the scale of the complete generated network. Depending on the research question, selected parts of the network can be used as the simulation environment.

SIMULATION OVER GRAPHS

The generated network is represented as a graph that provides the environment for the simulation. Actors are modeled as nodes, while edges define supplier–customer relationships and store properties relevant to transactions. Such network representations make it possible to seperate disruptions affecting individual firms or relationships from effects emerging at the wider network level (Kim et al. 2015). Hybrid simulation approaches can represent interactions between individual supply-chain entities while also capturing patterns at system level (Brailsford et al. 2019). In GraDES, the graph defines *who can interact* and how dependencies between firms are represented (Kim et al. 2015), ABM represents *how heterogeneous firms behave* (Bonabeau 2002, Clausen et al. 2019), DES governs *when operational processes occur* (Tako and Robinson 2012), and SD represents *how aggregate market conditions feed back into*

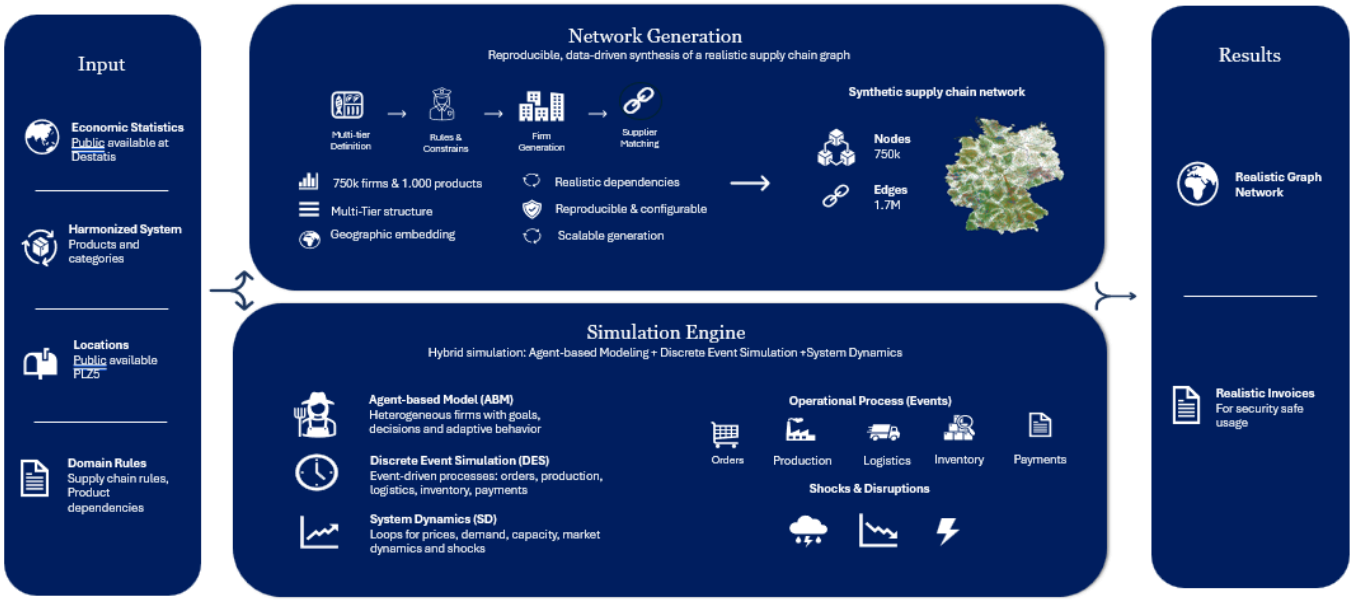

Figure 1: Overview of the Simulation

subsequent interactions (Sterman 2002).

Each simulated transaction is therefore linked to a plausible seller, buyer, product, prices, and tier level in the supply chain. These transactions can subsequently be transformed into invoices, payments, accounting records, and other synthetic business data. The resulting datasets provide a basis for developing and evaluating machine-learning approaches for supply-chain planning, monitoring, and resilience analysis (Kogler and Maxera 2026). Data generation is therefore a consequence of the simulated supply-chain processes rather than an independent process detached from the model. The same environment can also be used to study disruption propagation, dependencies, bottlenecks, recovery dynamics, and market effects.

RESULTS

The end-to-end pipeline produced synthetic transactions whose generated prices followed economically plausible multi-year trajectories. As shown in Figure 2, the simulated price indices reproduce the broad direction compared to DESTATIS (Table 61211) (Statistische Ämter des Bundes und der Länder 2026), timing, and persistence of observed price movements for broiler as example, including responses around major disruption periods. Although differences in magnitude remain, the results demonstrate that realistic aggregate price dynamics can emerge from the interaction of the generated network, firm-level transactions, and market feedback mechanisms.

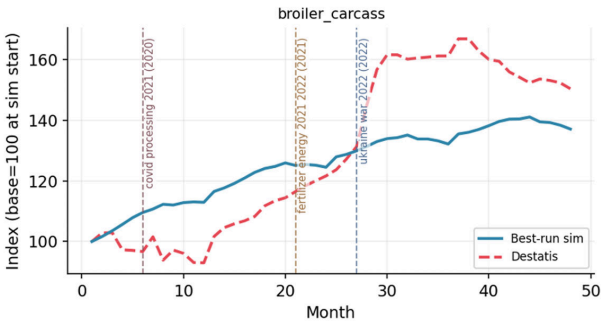

Figure 2: Simulated broiler price index compared with the corresponding DESTATIS producer-price index (Table 61211)

CONCLUSION AND PERSPECTIVES

Providing the GraDES (Graph-based Discrete Event Simulation) framework closes the implementation gap that supply chain research suffers from lack of legitimated and technically available mass invoice data and thus supplier-buyer networks. Consistent synthetic data generation is mandatory to start reliable modeling and forecasting. We first applied this framework to Farmery and Forestry which has some special accounting like credit and natural booking and subsidies which need to be assessed accordingly in later versions for their impact on revenues. The first evaluated results sound promising when compared to DESTATIS and public agricultural data (e.g. BMEL). Identification of regional supplier/ consumer patterns as well as a turnover and quantity comparison are useful to mimic the selected business

sector and others.
In our backlog are the following improvements for better validation results and scalability:

- ML-driven parameter estimation and pricing Layer
- Validation with real-world invoice and accounting data
- Enhanced modeling of regional supplier-consumer networks
- Improved validated accuracy over a longer time range
- Integration of resilience metrics and better shock propagation
- Integration of credit and natural accounting and subsidies
- Improvement of scalability and application beyond agriculture
- Aggregation of invoices to create balance sheets which help to measure a company's financial health.
- Interation of further complex components like diffusion modeling without modifying the base structure of the framework
- Publication as open source software for academic and other non-profit use.

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TACKLING TRANSACTION DATA SCARCITY WITH A MULTI-PARADIGM SIMULATION APPROACH FOR SUPPLY CHAIN NETWORKS

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INTRODUCTION AND RELATED WORK

Machine learning progress on tasks such as demand forecasting, disruption detection, and supplier-risk assessment is limited by the scarcity of high-quality, openly available datasets often due to data privacy and data scarcity Long et al. (2025). Data scarcity is more broadly recognized as "a major challenge when training deep learning models" Alzubaidi et al. (2023), and in supply chains especially the supplier-provider relations matter most. Important artifacts like process lead times, prices, invoices, amounts of goods are often fragmented across firms and tiers, and rarely shared because of competitive and tax sensitivity.

Drechsler notes that these business microdata that are essential for analytics and forecasting purposes are "particularly difficult to protect since a few variables usually suffice to identify larger businesses," (Drechsler 2016). While aggregates e.g. balance sheets and other records are commercially sensitive, this directly applies to invoices, purchase orders, and shipment events, whose disclosure is further restricted by GDPR and national statistics regulations Marwala et al. (2023). The consequences are manyfold: without shared benchmarks, cross-study comparison collapses—an obstacle explicitly cited as motivation for both the SupplyGraph benchmark Wasi et al. (2025) and the SynDelay synthetic dataset Xu et al. (2025)—and reproducibility becomes structurally fragile.

Synthetic data has become the principal answer to this problem. In official business statistics it has been described as "the only viable method" for producing safe public-use establishment-level datasets (Kinney and Reiter 2011), and subsequent work has documented favorable privacy–utility trade-offs for synthetic business microdata (Chien et al. 2021, Marwala et al. 2023, Kiran et al. 2025). In supply-chain settings specifically Long et al. (2025) outline the corresponding research agenda, and Gunn et al. (2025) demonstrate simulation-based synthetic data with two-stage structural and semantic validation, showing how domain constraints can be enforced that purely statistical generators (GANs, VAEs, diffusion models) routinely violate Lee et al. (2013).

A key requirement for synthetic transaction data is that its aggregate patterns remain economically plausible. We therefore validate the generated transactions against public agricultural statistics on regional structure, productivity, revenue, and producer-price dynamics, rather than against confidential firm-level data (Gunn et al. 2025).

We present here for the first time a simulation-based data generation in which network topology, agent decision rules, and event-driven timing are embedded directly into the data-generating process to mimic a special business sector in Germany. Our data validation is performed against publicly available input–output ag-

gregates rather than proprietary holdout data often used for model evaluation.

We model the German agricultural supply chain as a directed, multi-tier graph in which nodes represent companies, products, and locations. This graph provides the structural basis for the simulation by modeling the relationships among actors in the supply-chain.

Building on this graph, we develop a hybrid simulation framework for generating statistically grounded synthetic invoice data for the agricultural sector. The framework combines agent-based modelling (ABM), discrete-event simulation (DES), and system dynamics (SD). The graph captures the structure of the system; ABM represents heterogeneous companies, DES models the temporal order of operational events, and SD provides aggregate market-level price feedback mechanism.

ABM is suited for this use case because the decision-makers are individual firms rather than aggregate sectors and differ in size, location, products, inventory, and behaviour rules. Representing these actors individually allows the model to capture micro-level interactions and heterogeneity, which are central to the analysis of complex supply-chain systems (Clausen et al. 2019). Relatively simple agent-based models can generate emergent behavioural patterns (Bonabeau 2002) and is well suited for studying disruptions because local interactions among firms can produce system-wide effects that cannot be modelled at aggregate level (Borshchev and Filippov 2004, Nabayiga et al. 2025).

DES is incorporated because supply chain operations are event driven. Orders, shipments, deliveries, invoices, payments, and disruptions occur at specific points in time and change the state of the system. This aligns with the standard DES assumption that system evolution is driven by discrete, state-changing events. In logistics and supply chain applications, DES has been widely used to represent operational and tactical processes, while SD is often employed for more aggregate and strategic dynamics (Tako and Robinson 2012, Campuzano and Mula 2011).

The SD component is used to represent market-level feedback in the form of demand- and supply-based price adjustment globally. SD is commonly applied to model system behaviour through stocks, flows, feedback loops, and aggregate dynamic relationships (Campuzano and Mula 2011, Güz and Murakami 2025, Esteso et al. 2023). A similar agricultural supply chain was simulated for resilience studies completely in SD (Esteso et al. 2023), we choose to use a hybrid approach to enable simple extensibility by adding agents or updating policies and keeping the simulation simple to parameterize. More broadly, hybrid simulation approaches allow models to combine features of SD, DES, and ABM to represent different forms of system structure, behavior, and interaction (Borshchev and Filippov 2004).

Network Generation

This section describes the network-generation workflow used to build the heterogeneous supply-chain graph. As shown in Figure 1, a reproducible pipeline transforms product classifications, empirical data and domain knowledge into a fully materialized graph, progressively generating products, dependencies, firms and their relationships.

1. Data Foundation. The generation process combines three complementary input sources. First, products are derived from the Harmonized System (HS) classification maintained by the World Customs Organization (2026) and restricted to the agri-food and forestry domain. Second, a declarative chain specification defines the logical supply chain structure, including admissible upstream relations, position-specific refinements and operating inputs. Third, empirical statistics and geographic data provide calibration information for firm counts, regional distributions and firm-size structures. Official german statistics from DESTATIS (Table 41312 and 41141) (Statistische Ämter des Bundes und der Länder 2026) are transformed from district level to PLZ-2 regions through a dedicated crosswalk, enabling the integration of empirical evidence into the synthetic network without reproducing real firms.

2. Generative Layers. The core of the workflow consists of four consecutive generation layers. The product and dependency layers turn raw HS classifications into typed product nodes and resolve the logical bill-of-materials structure, preserving position-specific subchains and operating-input cycles. The agent layer then generates a synthetic company population, assigning each firm a tier, location, size and operational attributes. Regional calibration artefacts shape the spatial and size distribution of firms, ensuring consistency with observed agricultural and economic patterns. Firms are finally assigned products by category, tier and subchain, from which produced and required product portfolios are derived reproducibly. As indicated on the left of Figure 1, the empirical calibration pipeline supplies the crosswalks, regional clusters, size factors and company-count estimates that drive these steps.

3. Edge Materialisation. Once all nodes have been created, the network is completed through edge generation. Deterministic logical relations are derived directly from node attributes and the dependency specification. These include PRODUCES, REQUIRES, DEPENDS_ON, IN_CHAIN and LOCATED_IN relationships.
In addition, concrete supplier links are created through a spatial matching procedure. Candidate suppliers are identified based on product compatibility, tier constraints and geographic distance. A gravity model (An-

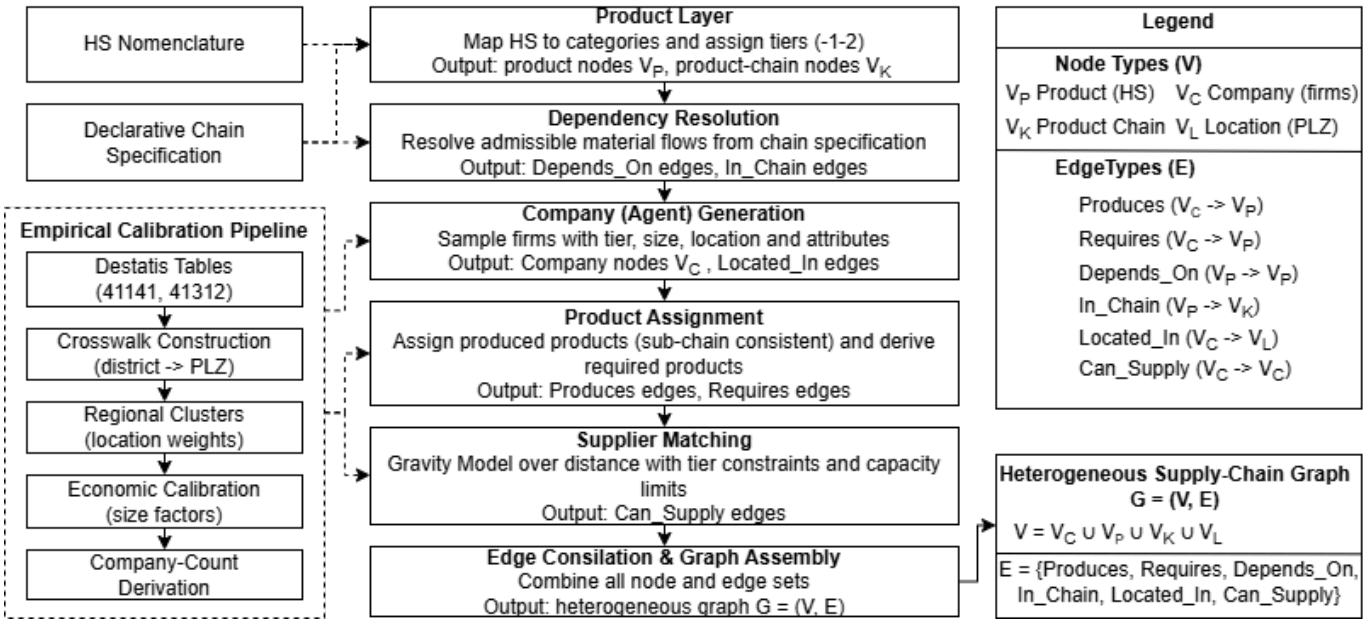

Figure 1: Overview of the Network Generation

derson 2011) that combines firm size and spatial proximity determines the selection probability of each supplier. Weighted sampling without replacement is subsequently performed using the Efraimidis and Spirakis (2006) algorithm, while respecting supplier and customer capacity limits, resulting in the CAN_SUPPLY relationships.

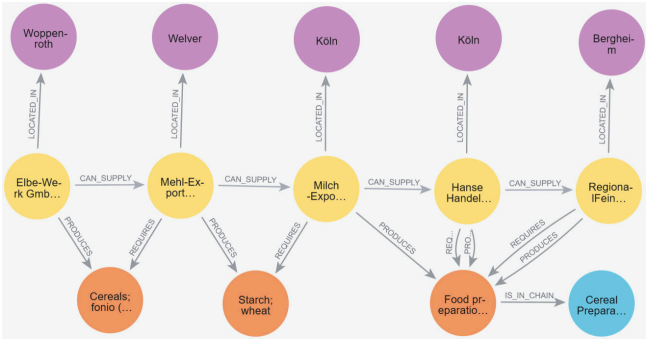

Figure 2: Example of the generated supply chain graph

4. Graph Artefacts. The final output of the workflow is a heterogeneous directed supply chain graph, as shown in figure 2, consisting of company nodes (V_C), product nodes (V_P), product-chain nodes (V_K) and location nodes (V_L). The edge set contains production, consumption, dependency, chain-membership, location and supplier relationships. The resulting graph is stored as a property graph for the simulation and operational analysis.

Simulation Framework

The simulation framework, as shown in Figure 3, comprises three main components: an agent-based layer representing firm behavior, a discrete-event layer governing the temporal and causal ordering of transactions and a system-dynamics layer providing aggregate market-price feedback. The model has monthly time steps and generates synthetic transactions for actors over configurable multi-year horizons.

Data inputs and network representation The simulation layer takes the pre-constructed supply chain graph as input: companies are mapped to internal actor classes and supply relationships define feasible trading partners. Transactions and pricing dynamics emerge from actor policies, shocks, and market feedback.

The model parameters is initialized from public agricultural and economic reference data wherever possible. Producer prices are initialized from BMEL Tabelle 5010600-0000 (Bundesministerium für Ernährung und Landwirtschaft (BMEL) 2026), while input prices such as feed and fertilizer are initialized from BMEL Tabelle 5020200-0000 (Bundesministerium für Ernährung und Landwirtschaft (BMEL) 2026). Prices for processed or downstream products, such as flour, carcass products, and processed dairy, are calculated from the closest primary commodity prices using estimated conversion ratios. Only prices up to simulation start year are used as initial values. The yield coefficients are initialized from DESTATIS Table 41241 (Statistische Ämter des Bundes und der Länder 2026) and are computed as historical moving averages using only time period before

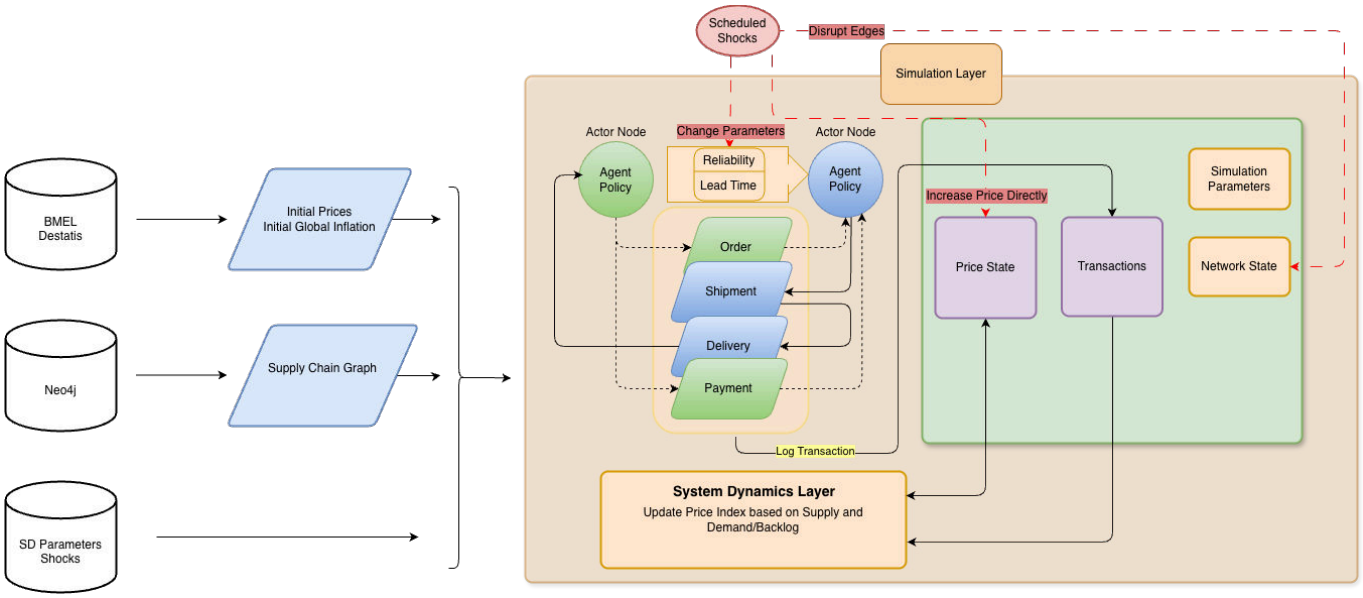

Figure 3: Overview of the Simulation

the simulation start. Livestock feed-conversion ratios and cycle lengths are fixed coefficients and calibrated by sensitivity analysis.

Supply relationships contain product-flow information and operational attributes such as annual volume, reliability, and lead time. Internally, the supply chain is represented as a directed multigraph in which each supply relationship is modeled as an independent edge, so that disruptions can affect individual flows differently. Shocks can reduce capacity, extend lead times, and lower reliability at the edges.

Discrete-event transaction pipeline The discrete-event layer specifies the order and timing in which operational processes are executed. In the proposed framework, transactions follow the sequence *order–shipment–delivery–invoice–payment*. Requested quantities may be modified by demand-side shocks, while delivered quantities are constrained by the runtime state of the supply edge, including its current capacity and reliability multipliers. Unfulfilled orders are recorded as backlog.

Financial transactions are generated only after physical delivery has occurred, so invoices and payments reflect realized delivered quantities rather than planned or requested amounts. This event-first structure enforces causal consistency in the synthetic transaction records.

Agent-based execution and actor archetypes The agent-based layer instantiates companies as rule-based actor nodes in the supply-chain graph. Each actor follows a monthly decision cycle consisting of a planning stage, in which inventory, production capacity, current prices, and available relationships are observed, and an execution stage, in which production, procurement, sales, inventory updates, and financial bookkeeping are carried out. This rule-based implementation is consistent with agricultural ABM models, where farm-level decision-making is often represented through predefined behavioral rules (Ravaioli et al. 2023).

Actors are assigned archetypes that define their state variables and policy. The main farm archetypes are dairy, arable, pig-farming, and broiler production. Non-farm actors, including processors, wholesalers, retailers, and input suppliers, follow generalized forwarding policies constrained by relationships, input availability, capacity, and current disruption states. The archetypes are therefore used as abstractions for individual firm behavior rather than as empirically observed categories.

Dairy farms generate monthly milk output from herd size, annual yield, seasonality, effective capacity, and stochastic yield variation. Herd size responds to milk prices within capacity limits, while feed is replenished through a safety-stock rule. Arable farms produce crops according to cultivation area, yield priors, harvest seasonality, capacity, yield noise, and fertilizer availability; sales respond to inventories and current prices. Pig farms use an expected-margin rule for animal placement, subject to feed and piglet costs and barn capacity, whereas broiler farms follow staggered production cycles that include price-dependent barn filling, mortality, live weight, and feed requirements.

Processing, wholesale, and retail actors follow generalized throughput policies in which procurement and sales are constrained by supply relationships, input availability, capacity, downstream demand, and disruption states. The associated thresholds and archetype parameters are synthetic rather than directly observed firm-level decision rules. They are calibrated iteratively

against aggregate public-data patterns, following the logic of pattern-oriented modeling (Grimm et al. 2005). The resulting parameters therefore represent one plausible configuration of firm behavior rather than a unique optimal solution.

System-dynamics market pricing The system-dynamics layer provides the aggregate endogenous pricing adjustment. The formulation follows the commodity-market system-dynamics logic of Berg (2016), where the relative rate of price change depends on the difference between demand and supply, and is consistent with lagged demand–supply price-adjustment dynamics formalized in delay cobweb models (Guerrini et al. 2024). We adapt this idea to monthly simulation by updating the logarithm of the endogenous market-price index from prior-month supply–demand observations. Input-cost pressure is added as a simplified food-chain price-transmission channel (Meyer and von Cramon-Taubadel 2004, Vavra and Goodwin 2005). The monthly input-price indices for this (fertilizer & compound feed) are currently approximations due to data availability; the relative weighting of these components was calibrated against the co-movement of the official DESTATIS input purchase-price index (Table 61221) (Statistische Ämter des Bundes und der Länder 2026). A mean-reversion term is included as a reduced form of stabilization mechanism, consistent with commodity-price models in which the logarithm of the price follows a mean reversion process (Schwartz 1997). For product p in month t, the transaction price is decomposed as

$$P_{p,t} = B_{p,a(t)} \cdot S_{p,m(t)} \cdot I_t \cdot X_{p,t} \cdot Z_{p,t}, \qquad (1)$$

where $B_{p,a(t)}$ is the initial price in the starting year with $a(t)$ mapping simulation month t to its corresponding calendar year, $S_{p,m(t)}$ is a monthly seasonal multiplier, with $m(t)$ mapping t to its calendar month (1–12), derived by averaging the past known seasonal price variation from reference data, I_t is the external inflation multiplier from starting point, $X_{p,t}$ is the endogenous market-price index, and $Z_{p,t}$ is a temporary exogenous shock multiplier. In synthetic historical rebasing mode, the base price may instead be re-anchored at regular intervals so that generated transactions remain consistent with observed price levels.
The monthly log delta for the index is given by

$$\Delta x_{p,t} = \alpha_p g_{p,t-1} + \eta \pi^{\text{cost}}_{p,t-1} - \lambda x_{p,t-1}, \qquad (2)$$

where $x_{p,t} = \log X_{p,t}$, α_p is the product-specific responsiveness to the lagged demand–supply gap $g_{p,t-1}$, η is the input-cost pass-through gain, $\pi^{\text{cost}}_{p,t-1}$ is the lagged upstream cost-pressure term, and λ is the mean-reversion rate toward the neutral log index. The mean-reversion term prevents temporary price-index deviations from becoming permanent unless supported by continued imbalance or shock cost effects.

Supply shocks enter through the demand–supply gap by reducing effective available supply, while exogenous cost-push shocks affect through the temporary multiplier $Z_{p,t}$, which gradually decays back toward 1 according to the shock recovery rate. Input-cost pressure is computed from weighted upstream product-price indices, allowing increases in inputs such as feed or fertilizer to affect downstream products through a cost-push channel since currently agents do not have dynamic pricing.
All feedback variables entering the market-index update are lagged by one month preventing look-ahead bias between agent decisions and aggregate price adjustment.

RESULTS AND EVALUATION

Evaluation of the simulation focused on verifying if the model can sufficiently model price dynamics particularly with external shocks to later generate invoices. Therefore, we evaluate the model's monthly price dynamics against external benchmarks, reported with explicit error metrics. The section concludes by analyzing the results, assessing risks to their validity, and deriving implications for calibration and future work.

Experimental Setup The results reported in this section are based on a simulated firm population of 10,000 entities evaluated over the 2020–2023 period, covering seven benchmarked products. Since currently not all related industries and products are covered by the simulation, only a subset of these entities are active participants in the simulation. First, the simulation must assign each entity an archetype; this step leaves 3,144 resolved firm archetypes out of the starting population. Second, because relationships whose endpoints failed this mapping are pruned, resolved farms can lose their only trading partners when a downstream processor or wholesaler is itself dropped. Third, of those connected farms, only about 177 (15.7%; 5.6% of the 3,144 resolved farms) are observed generating at least one sales transaction within a given evaluation window. This means the price-dynamics metrics reported below are formed from a substantially sparser set of active market participants than the starting 10,000. Price-dynamics metrics are averaged across three simulation runs with modeled shocks, to account for uncertainty due to random sampling (for instance for shock affectees).

Farm population structure As a structural plausibility check, Figure 5 compares the simulated distribution of utilised agricultural area (UAA) across German counties with the corresponding DESTATIS reference (Table 41141) (Statistische Ämter des Bundes und der Länder 2026). Both distributions place Bavaria and Lower Saxony as the two largest shares of national UAA, with the remaining states following a broadly similar or-

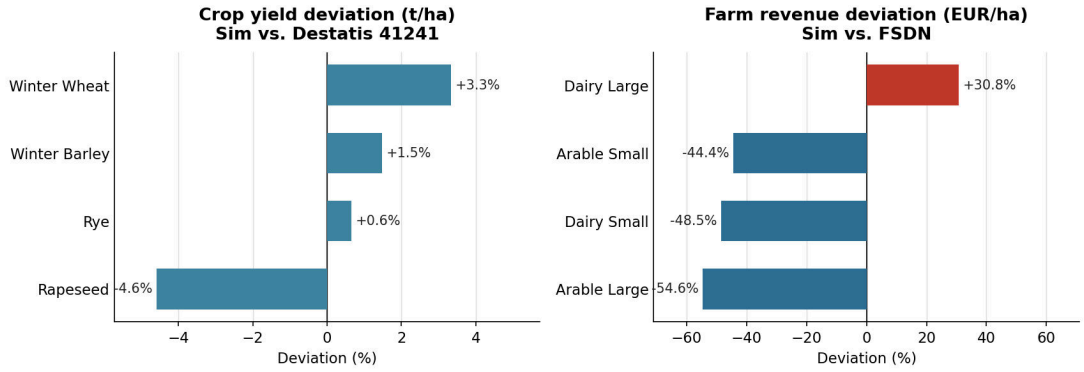

Figure 4: Illustration of the Crop Yield deviation and Farm Revenue Deviation by crop and archetype respectively

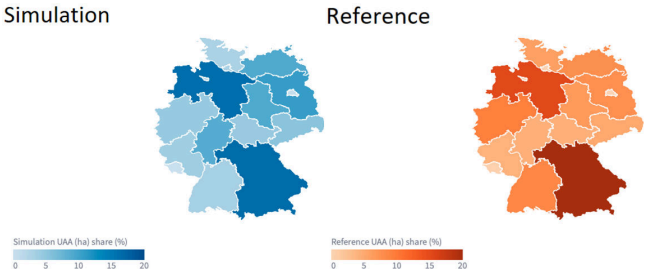

Figure 5: Share of utilised agricultural area (UAA)

dering. This indicates that the filtered simulated population is geographically concentrated in a expectable way.

Productivity and revenue Figure 4 reports crop yield and scaled farm revenue deviations for the four crops and four farm archetypes with sufficient benchmark coverage. Crop yields are closely aligned with the DESTATIS Table 41241 (Statistische Ämter des Bundes und der Länder 2026) reference across all four crops: rapeseed deviates by −4.6%, rye by +0.6%, winter barley by +1.5%, and winter wheat by +3.3%, all within single-digit error. This is expected since the yields are parameterized from past reference values and are not expected to change majorly over the years. Revenue per hectare against FSDN European Commission, Directorate-General for Agriculture and Rural Development (2026) data shows a more mixed pattern: arable large (−54.6%), dairy small (−48.5%), and arable small (−44.4%) are all substantially under-estimated, while dairy large is the only archetype over-estimated, at +30.8%. We believe this is due to the strong filtering of companies at the start which leads to very limited trading partners so even though the crops are being produced they are not sold and hence do not generate revenue. For large dairy farms a plausible explanation is that the archetype's herd-size and per-cow yield parameters scale more steeply with company size than its assigned land area does (i.e. policy shortcoming).

Modelled shock events Three shocks are scheduled during the 2020–2023 run, each targeting specific archetypes. A pandemic shock is applied to dairy-processor, slaughterhouse, and meat-products archetypes (15 targets) starting at absolute month 6 (June 2020), timed according to the documented COVID-19 outbreaks in German meat and poultry processing plants that began in May 2020 (Pokora et al. 2021). A fertilizer-shortage shock is applied to arable, mixed-farm, and pork farm archetypes (20 targets) starting at absolute month 21 (September 2021), anchored to the European natural-gas price surge and resulting ammonia/fertilizer production curtailments that began that month (Stiewe et al. 2022). A war-disruption shock is applied to grain-mill, feed-mill, and dairy-processor archetypes (25 targets) starting at absolute month 27 (March 2022), timed to the first full simulation month after beginning of the conflict (Stiewe et al. 2022).

Price dynamics For price validation, the transaction-derived simulation price index is compared with the corresponding DESTATIS (Table 61211) producer price index (PPI) for agricultural products, which reports monthly product-level price indices (Statistische Ämter des Bundes und der Länder 2026). Let $I^{sim}_{p,t}$ denote the simulated index for product p in month t and $I^{ref}_{p,t}$ the corresponding reference. We report the root mean squared error,

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(I^{sim}_{p,t} - I^{ref}_{p,t}\right)^2}, \tag{3}$$

and the weighted absolute percentage error,

$$\mathrm{WAPE}_p = 100 \cdot \frac{\sum_{t=1}^{T}\left|I^{sim}_{p,t} - I^{ref}_{p,t}\right|}{\sum_{t=1}^{T}\left|I^{ref}_{p,t}\right|}. \tag{4}$$

Because supply-chain resilience is also concerned with directional response, we additionally report a

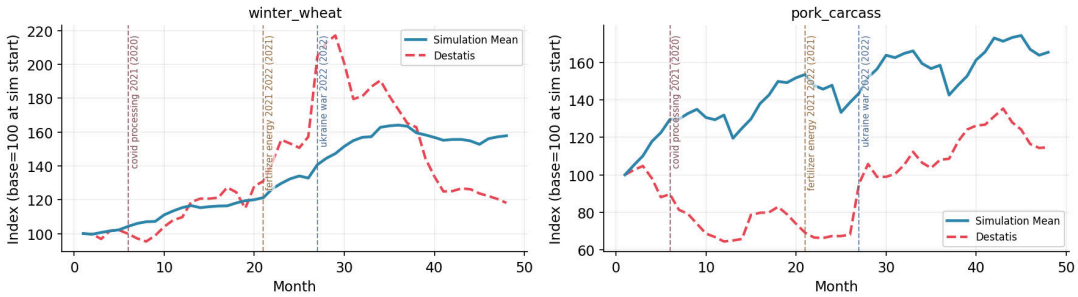

Figure 6: Comparison of Simulated and Reference Price Indices for winter wheat and pork (Jan, 2020 - Dec, 2023)

directional-agreement measure, referred here as turning-point accuracy (TPA). TPA is defined as the share of month-to-month transitions in which the simulated and reference price indices move in the same direction. This follows the general logic of directional predictive-performance evaluation, where performance is assessed from agreement in the signs of two changing series (Pesaran and Timmermann 1992). However, no formal Pesaran–Timmermann test is applied.

Figure 6 illustrates this comparison between products, using winter wheat and pork as representative examples. The figure shows the extent to which the model captures the level and temporal movement of the reference price series particularly in time periods where external shocks were present and are modelled in the simulation.

Table 1: Accuracy per product with modelled shocks

Product	RMSE	WAPE (%)	TPA
Broiler Carcass	17.34	11.6	0.489
Raw Milk	20.88	15.1	0.723
Winter Wheat	25.61	13.6	0.617
Winter Barley	31.26	15.4	0.617
Rapeseed	37.96	22.4	0.545
Pork Carcass	54.62	54.2	0.638
Dairy Processed	327.23	189.7	0.702
Median	31.26	15.4	0.617

Table 1 reports RMSE, WAPE, and TPA per product for the run with modeled shocks, each averaged across three simulation runs. Turning-point accuracy ranges from 0.489 (broiler carcass) to 0.723 (raw milk), indicating that directional alignment varies across products independently of price-level fit. This indicates at times a worse TPA than a completely uninformed baseline. We note this as a potential for improvement in the shock modelling and propagation and the tendancy in the model to only develop positive price indices. This could also be attributed to the small sample of firms taking part in the simulation which limits supply pushing the index to only positive values.

Furthermore, processed dairy shows a significantly larger level error than every other product (RMSE 327.2, WAPE 189.7%), despite a mid-range TPA of 0.702. Because this single product would dominate any unweighted mean across products, we report the median: RMSE 31.26, WAPE 15.4%, and TPA 0.617. Pork carcass also shows a materially larger level error (RMSE 54.6, WAPE 54.2%) than the other five products, despite a TPA (0.638) in the same range as the rest. Part of this level gap is explained by mapping limitations: both products are validated against a DESTATIS proxy series rather than a direct product index (raw milk price for dairy processed, live-animal price for pork carcass), and processor-level prices additionally contain compounded upstream and downstream effects that a farm-level proxy cannot capture. This does not, however, fully account for either case: dairy processed's simulated index reaches roughly seven times the reference level by the final evaluation month, a magnitude consistent with a runaway pricing, and pork carcass's error indicates a calibration regression. We treat proxy mapping as a contributing factor and flag both products for enhanced calibration.

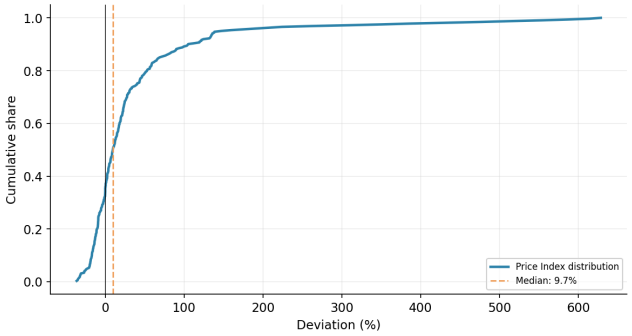

Figure 7: Empirical cumulative distribution of monthly price-index deviation, pooled across all evaluated products and months

Figure 7 shows the combined distribution of monthly price-index deviations across all evaluated products. The median monthly deviation is a modest 9.7%, and the interquartile range spans roughly −8% to +40%, consistent with the reasonable price tracking reported for most of the evaluated products. The distribution's long right tail, however, extends past +600% deviation in a small number of products: the 90th and 95th percentiles (+105% and +147%) already exceed the WAPE

reported for any single better-aligned product, indicating that this tail is driven by a small number of extreme observations — consistent with runaway index for dairy and, to a lesser extent, pork carcass's policy error.

Limitations to validity Three main limitations affecting these results are identified. First, *benchmark coverage* remains incomplete: several processed products are evaluated using primary-product proxies rather than direct DESTATIS series. Second, *resilience evaluation* remains limited: although the framework supports edge disruptions, the current results do not yet include a comprehensive set of evaluated resilience metrics. In addition, the active network remains too sparse to support a detailed analysis of shock and disruption propagation. Therefore, a larger and more densely connected network is required to evaluate cascading effects more systematically. Third, *SD* regressions remain for the price fit of dairy processed products and pork carcass, treated as open calibration issues. Additionally, from an ethical perspective, the framework is intended to support research and scenario analysis rather than operational decision-making. The simulated agent behaviors and pricing rules represent simplified assumptions and may become outdated as market conditions, regulations or business practices evolve. These open issues limit the interpretation of the reported deviations, but do not undermine the evaluation methodology.

Conclusion and Perspectives

The current supply chain modeling and simulation suffers from lack of legitimated and technically available mass invoice data and thus supplier-buyer networks. Consistent synthetic data generation is mandatory to start reliable modelling and forecasting. We first applied this framework to Farmery and Forestry which has some special accounting like credit and natural booking and subsidies which need to be assessed accordingly in later versions for their impact on revenues. The first evaluation results sound promising when compared to DESTATIS and public agricultural data (e.g. BMEL). Identification of regional supplier/ consumer patterns as well as a turnover and quantity comparison are useful to mimic the selected business sector and others. We plan to replace the current extensive parametrization of different impact factors by ML components and the simulation could be improved in terms of validated accuracy over a longer time range. In addition we aim to integrate resilience metrics and better shock propagation to be able to study disruption effects. Once the ground truth invoice data is available we aim for a more detailed evaluation both at invoice and aggregated levels. We developed this extensible modular data generation framework for the agricultural sector but it is suitable for all business sectors and designed for scalability. A rebasing option allows the simulation to utilize all available data to reconcile the price and inflation drifts for generating more aligned historical data. Invoices can be further aggregated to create balance sheets which help to measure a company's financial health. More complex components like diffusion modelling are possible without modifying the base structure of the framework. The achieved regional resolution can simulate the local impact of identified infrastructure breaks, which would be expanded in future work. We named the developed framework *GraDES* and it is planned to publish it as open source for academic and other non-profit use.

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Ontology-based Market Agents: Towards a Dynamic Instantiation of Supply Chain Simulations

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ABSTRACT

This paper describes the PROVIDER Domain Language (PDL), a declarative, YAML-based specification language that serves as the interface between a Knowledge Graph (KG) and an agent-based simulation engine for analyzing supply-chain scenarios. PDL enables the complete description of self-contained supply-chain scenarios including entities, their dependencies, potential shocks, substitution mechanisms, and empirically validated cascade timelines in a format that is simultaneously writable by domain experts, mechanically validatable, and losslessly producible from the KG. Additionally, the paper describes how we build KGs, how we simulate supply chains with agent-based simulation and how the PDL allows for a dynamic simulation with the data stored in the KG.

INTRODUCTION

Modern societies depend on dense, cross-continental supply and service networks whose efficiency leaves little slack. Extreme weather, geopolitical conflict, or market shocks can throw these finely balanced chains out of equilibrium within days. Yet most current tools for supply-chain analysis are reactive: they reconstruct what went wrong after a disruption has already propagated, examine a single, hand-picked value chain, and cannot generalize to the next event. The gap is acute outside the perimeter of formally designated Critical National Infrastructure (CNI): goods chains such as agricultural feed or speciality chemicals carry high societal significance but receive neither the crisis planning nor the continuous monitoring afforded to regulated infrastructure.

A key reason why proactive, anticipatory analysis has remained out of reach is the difficulty of assembling a simulation of sufficient scope. Agent-based supply-chain simulations are typically built and parameterised by hand: domain experts encode each actor, define its initial state, and wire the topology before a single run can start. The modelling effort is therefore proportional to the number of supply chains under study, and the resulting simulation is tailored to one scenario rather than reusable across many. The Proactive Supply-Chain Resilience through Dynamic Simulation with Self-Learning LLM Agents (PROVIDER) project addresses this gap

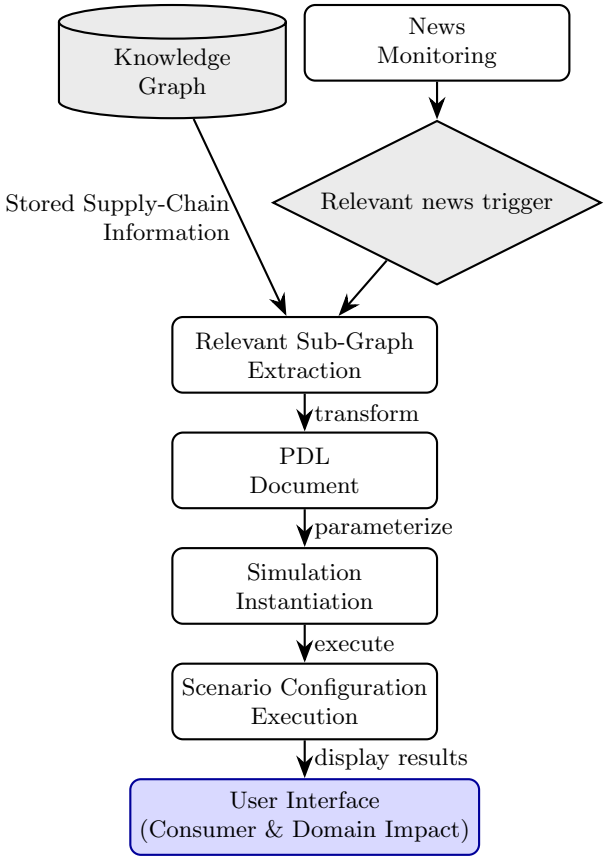

Figure 1: PROVIDER workflow

by building a continuously active, self-learning simulation stack for proactive supply-chain resilience. Its defining architectural idea is that the simulation environment is *generated*, not built: instead of assembling actors and initial conditions by hand, the stack instantiates them dynamically from a Knowledge Graph (KG) that encodes the economic world state in terms of commodities, manufacturers, logistics nodes, trade flows, and disruption events. Large Language Models (LLMs) translate live news into simulation parameters; a Deep Reinforcement Learning (DRL) controller in an adversarial autocurriculum steers the simulation toward worst- and best-case configurations; and the results are returned in natural language to a human operator who retains final decision authority. The designed workflow is displayed in 1.

This paper describes the PROVIDER Domain Language (PDL), the interface between the KG and the simulation engine and how it is used for the simulation. The PDL is a declarative, YAML Ain't Markup Language (YAML)-based specification language that describes a complete, self-contained supply-chain scenario—its entities, dependencies, shocks, substitution mechanisms, and empirically validated cascade timelines—in a form that is simultaneously writable by domain experts, mechanically validatable, and producible from the KG. Its main contribution is to provide the missing contract between knowledge representation and dynamic analysis: the KG materializes a scenario as a PDL document, and the simulation engine consumes that document to instantiate an executable experiment—without hand-coding at any step.

The remainder of this paper is structured as follows. First, we present the relevant related work. Next, we present our methodology by first explaining how we use KGs, secondly what the PDL is and thirdly how we simulate supply chains and how we use the PDL to parameterize the simulation. Lastly, we conclude our methodology and outline how we will expand on it in future work.

RELATED WORK

Supply-chain ontologies. The Supply Chain Operations Reference (SCOR) model, developed by the Supply Chain Council and described by Stewart (1997), is the established cross-industry process reference framework for supply chain management; it defines standard processes, performance metrics, and best practices across the plan, source, make, deliver, and return phases. Several groups have formalized SCOR as Web Ontology Language (OWL). Fayez et al. (2005) showed that a SCOR-grounded ontology can drive the *automatic generation* of supply-chain simulation models at varying levels of detail, reducing definition and execution time significantly.

Sarli et al. (2016) extended this direction by using an ontology network that spans the data-model, federation, supply-chain, and enterprise-modelling domains to semi-automatically generate High Level Architecture (HLA) Federation Object Models for distributed supply-chain simulations. For the commodity and manufacturer layer, Hepp (2008) introduced GoodRelations, a lightweight OWL vocabulary for offers, business entities, prices, and delivery options that is now the official e-commerce extension of Schema.org and informs the PDL entity taxonomy. A systematic review of 22 supply-chain ontologies by Grubic and Fan (2010) confirms that, while the formalisation of SCOR-level processes is mature, no existing ontology couples entity descriptions, conditional disruption events, and resilience mechanisms in a single, reviewable scenario artefact.

Knowledge graphs and simulation. Listl et al. (2024) reviewed 32 papers on knowledge graphs in Digital Twins and found that semantic technologies are used primarily for data unification and semantic modelling of production environments, while automated generation of simulation models from a live, writable knowledge graph remains immature. Listl et al. (2022) proposed an ontological architecture for knowledge-graph-based simulation support in manufacturing, addressing model management and the integration of heterogeneous data sources, but without a full round-trip from world state to executable experiment.

Input–output economics. The PDL parameterizes shocks through a linearised Leontief propagation whose technical coefficients are taken from the Organisation for Economic Co-operation and Development (OECD) Inter-Country Input-Output (ICIO) tables (OECD 2023) and whose bilateral trade flows come from Base pour l'Analyse du Commerce International (BACI) (Gaulier and Zignago 2010). Santos (2006) extended Leontief's framework to the Inoperability Input-Output Model (IIM), which replaces monetary flows with fractional inoperability so that the ripple effects of a disruptive event can be traced across interdependent sectors; the PDL `cascades` block operationalises exactly those validated IIM-style findings as first-class scenario artefacts, preserving their confidence and literature provenance.

Declarative scenario languages. Declarative, human-readable configuration languages have been used to parameterise simulations in several domains without requiring end-users to write code. Hewage et al. (2023) demonstrated YAML-based scenario instantiation for cloud simulations, achieving an 89 % reduction in lines of code compared with conventional CloudSim setups. Jung et al. (2021) developed Configuration and Parametrization Domain-Specific Language (CP-DSL),

a model-agnostic Domain-Specific Language (DSL) for configuring and parameterising ocean models, whose design goals—unified interface for domain experts, grammar-level validation, and separation of scenario concerns from implementation—are directly shared by the PDL. None of these languages, however, covers the combination of probabilistic and conditional event triggers, dependency topology, and substitution dynamics required to instantiate a supply-chain resilience simulation from a knowledge graph.

Research gap. The literature provides the building blocks: supply-chain ontologies, knowledge graphs as simulation substrates, input–output economics, and declarative scenario languages. What is missing is their integration. No prior approach offers a scenario language that is simultaneously losslessly derivable from a live knowledge graph, mechanically validatable, and expressive enough to capture the conditional event logic and substitution dynamics needed to instantiate a supply-chain resilience simulation without hand-coding.

METHODOLOGY

Storing the World State in a Knowledge Graph

The PROVIDER system maintains a single, shared representation of the economic "world state" from which every analysis and simulation is derived. This representation is a KG: an Resource Description Framework (RDF) graph whose nodes are commodities, infrastructure, industries, countries, trade relations and disruptive events, and whose edges encode their dependencies. A central design decision distinguishes our use of the KG from the conventional pattern in which a graph serves only as a static, read-only knowledge base: in PROVIDER the KG is simultaneously the *input*, the *computational substrate* and the *write target* of the analysis pipeline. Events enter the world state by being written into the graph, quantitative effects are computed from the graph, and the results of simulation are written back into it. The project code is available on GitHub.[1] It contains the current state of implementation of the PDL, its schema, the simulation environment as well as data readers for filling the KG. The remainder of this section describes what the graph holds, how the contents are intended to grow over the course of the project, how the data are physically stored, and how they are gathered.

Knowledge Graph Content

The world state is organized into six concentric *shells*, denoted L0–L5, around a set of central reference nodes. The shell model is a deliberate compaction strategy: rather than triplifying the complete bilateral-trade and input–output matrices — which would amount to between 10^5 and 10^9 triples — the graph stores only *precomputed findings* that downstream components actually consume.

- **L0: Reference entities.** This layer contains the canonical entities and concepts reused across datasets and scenarios, such as commodities, product classes, industries, countries, regions, organizations, production sites, ports, logistics infrastructure, and services. For example, natural gas is represented once as a shared commodity node, such as `data:commodity/natural-gas`. These reference nodes provide stable anchors for the graph: events, indicators, and scenario-specific entities can all refer back to them.
- **L1: Bridges.** Concordances between product- and sector-classification systems (HS6 $\leftrightarrow$ ISIC $\leftrightarrow$ ICIO, on the order of several thousand mappings), so that product-level trade data and sector-level input–output data can be joined.
- **L2: Indicators.** Derived, pre-computed economic findings: technical input coefficients a_{ij}, total requirement coefficients $[(I - A)^{-1}]_{ij}$ for the sector pairs referenced by a scenario, supplier-concentration indices (HHI), substitution-coverage ratios and combined risk scores. These are the quantities consumed by the effect computation (Section).
- **L3: Events.** Disruptive events, both historical (from curated event databases) and extracted from news streams, linked to affected reference entities such as commodities, regions, production sites, ports, logistics infrastructure, or services.
- **L4: Scenarios.** Named-graph *overlays* that define a particular analytical situation as a set of local entities linked back to shared L0 nodes; this shell also holds the findings written back by the simulation.
- **L5: Meta.** Catalog, provenance and licensing metadata (e. g., DCAT descriptions of the source datasets), so that every value can be traced to its origin.

The decision to store *derived* indicators in L2, rather than only raw flows, is motivated by the observation that aggregate measures mask product-level vulnerability. For instance, German soybean sourcing appears "competitive" at the sector level (HHI $\approx$ 629), yet only about one fifth of soybean-meal imports can in practice be substituted by alternative suppliers; this fact is not recoverable from raw flows without re-running the upstream analysis, so it is materialized once and stored as a finding.

[1] `https://github.com/PROVIDER-Project`

Knowledge Graph Structure

All data are stored as RDF triples partitioned into *named graphs*, one per shell or per scenario (e.g. `.../trade/baci-indicators/` for indicators, `.../scenario/{id}/` for a scenario overlay, and `.../simulation/findings/` for written-back results). Three representational choices carry most of the method's weight.

First, scenarios are modeled as *overlays rather than copies*. A scenario's local entities are linked to the shared L0 reference nodes by `owl:sameAs`, and events are anchored at the L0 node. As a consequence, a single write at an L0 node is automatically visible in *every* scenario whose overlay references that node, i. e., one event need only be written once to participate in many analyses.

Second, derived economic semantics are expressed through a small RDF vocabulary (the `glm:` "generalized Leontief model" namespace), which reifies the relevant fragments of the input–output structure: a technical input coefficient becomes a `glm:InputLink` carrying a `glm:techCoefficient`; the aggregate response of a commodity to a shock becomes a `glm:ShockPropagation` carrying a `glm:leontiefMultiplier`; and per-importer risk is bundled in a `glm:VulnerabilityAssessment`.

Third, all access is mediated by a single, deliberately endpoint-shaped persistence abstraction that exposes only insert/select operations over named graphs. During the pilot this abstraction wraps an in-memory RDF dataset (configured so that graph queries range over the union of all named graphs); in production it wraps the project's SPARQL endpoint. No other component touches the RDF library directly, which is what makes the storage backend swappable.

Knowledge Graph Data Ingestion

The world state is populated from two complementary streams. The *structural* stream supplies the slowly changing economic backbone: bilateral trade flows from a global trade database, technical coefficients and multipliers from inter-country input–output tables, and historical disruption records from curated event databases. The *event* stream supplies the fast-changing situational layer. News items are reduced by a language-model component into compact fragments of the form *(entity label, event type, impact)*. Each fragment is then *reconciled*—its surface label is mapped to a canonical L0 URI—and written back as an event anchored at that L0 node.

PROVIDER Domain Language (PDL)

The PDL is a declarative, human-readable specification language that sits between the Knowledge Graph and the simulation engine. Its goal is to describe a complete, self-contained supply-chain scenario—its entities, their dependencies, the shocks that may strike it, and the resilience mechanisms available to it—in a form that is at once (i) writable and reviewable by domain experts, (ii) mechanically validatable via its schema, and (iii) producible from, and referring back into, the Knowledge Graph. PDL thus serves as the contract between knowledge representation and dynamic analysis: the KG materializes a scenario *as* PDL, and the simulation consumes that PDL. Because the language is declarative, the same document can be validated, visualized, converted to an executable experiment, and archived as a reproducible artifact.

PDL Structure

A PDL document is a structured (YAML) document with a small number of top-level blocks:

- `scenario` — identifier, name, sector, and a criticality assessment.
- `entities` — the nodes of the supply chain. Each entity has a type (drawn from a fixed enumeration such as *commodity*, *manufacturer*, *infrastructure*, *service*, *region*), a static `vulnerability` in $[0, 1]$, and an open `extra` map carrying KG-enriched attributes (e.g. trade volumes and capacities).
- `supply_chains` — ordered `stages` describing the flow of goods from node to node, together with explicit cross-cutting `dependencies` (e.g. an energy dependency of a processing plant on gas supply) typed and rated by criticality.
- `substitutions` — resilience mechanisms (alternative suppliers, products or routes) characterized by the fraction of a shortfall they can `cover`, a `ramp_up` time, and an `activation` trigger.
- `events` — disruptive shocks. An event has a `type` (from a fixed enumeration such as *natural_disaster*, *market_shock*, *geopolitical*, ...), a `trigger`, and an `impact` expressed as signed percentages with a duration. An event is a *root event* when its trigger carries a probability and no condition; otherwise its trigger carries a boolean `condition` over the state of other events.
- `cascades` — empirically validated event timelines that chain a root event to its downstream consequences, with a confidence and a literature reference.
- `_kg_enrichment` — provenance metadata recording which KG sources populated the entity attributes.

Example: a Soy Feed-Supply Disruption

The following abridged, non-executable excerpt illustrates the structure of a PDL document that specifies a disruption of the Brazil–EU soybean feed chain. A drought reduces Brazilian production; the dependency of EU oil mills on gas supply and the availability of an alternative-protein substitution are made explicit, so that the same document can be both reasoned about and simulated. The example is derived from open data as well as assumptions to create an executable PDL document early in the project.

```
scenario:
  id: soy_feed_disruption
  sector: agriculture
entities:
  - id: brazil_farms
    type: region
    vulnerability: 0.6
    extra: {baci_export_volume_t_year: 1056305.0,
            baci_capacity_t_day: 2893.8}
  - id: eu_oil_mills
    type: manufacturer
    vulnerability: 0.5
supply_chains:
  - id: soy_to_eu_main
    stages:
      - [brazil_farms, santos_port]
      - [santos_port, rotterdam_port]
      - [rotterdam_port, eu_oil_mills]
    dependencies:
      - {from: eu_oil_mills,
         to: gas_supply,
         type: energy,
         criticality: high}
substitutions:
  - id: sub_protein_replacement
    from: eu_oil_mills
    to: alternative_protein_sources
    coverage: 0.2
    ramp_up: 21d
    activation: {trigger: feed_price_spike.active,
                 threshold: {price_increase: 0.25}}
events:
  - id: brazil_drought
    type: natural_disaster
    trigger: {target: brazil_farms,
              probability: 0.15}
    impact:  {supply: "-40%", duration: 90d}
  - id: soy_export_reduction
    type: market_shock
    trigger: {target: santos_port,
              condition: "brazil_drought.active"}
    impact:  {supply: "-30%", duration: 120d}
cascades:
  - id: soy_crisis_cascade
    origin: brazil_drought
    timeline:
      - {at: 0d,  event: brazil_drought}
      - {at: 14d, event: soy_export_reduction}
    validation: {reference: "Soja-Krise 2022",
                 source: "BMEL, FAO GIEWS",
                 confidence: 0.85}
```

Declaratively, the document states the structure of the chain and the conditions under which shocks propagate; read operationally, it provides every parameter the simulation needs: the entities and their vulnerabilities, the topology over which disruptions flow, the probabilistic and conditional triggers, and the substitution dynamics that govern recovery.

Agent-Based Supply Chain Simulation

The goal of the simulation is to anticipate the influence of world events on one or more supply chains. By simulating the behavior of producers, logistics, wholesalers, consumers, and other market actors, supply chains are modeled in an agent-based way. World events can influence individual actors in these chains depending on the geographical location and time. Identifying the influence on later stages of the supply chain and thus highlighting potential shortages is the main goal of the simulation. Events change throughput or price in certain parts of the chain, which influences the entire rest of the chain. By modeling the input and output of each actor, we can simulate the impact of an adverse event anywhere in the chain.

In each simulation step, every agent in the simulation calculates its output and the price of its product or service. An agent is, e.g., a soy farmer or port in Brazil, a wholesaler for soy, or an oil mill in Europe. The behavior of each class of agent or even each individual agent can be set independently to explore different outcomes depending on the modeled market behavior. So far, only basic agent logic is implemented for testing purposes. Part of the projects scope is to develop these agent implementations.

Events represent exogenous shocks that influence different parts of the simulated supply chain. They are modeled by onset and duration, the type of impact (e. g., a capacity constraint or a yield degradation), the scope (influence on a region or industry) and the magnitude. This introduces local (in terms of location or industry) shocks in the simulation of the entire supply chain to determine the impact.

Simulation Parameterization through PDL

Parameterization proceeds in three steps: the KG *materializes* a scenario view, that view is exported as PDL, and the PDL is *converted* into an executable experiment.

Materialization. Given a scenario overlay, a single graph query collects the overlay's entities, follows their `owl:sameAs` links to the shared L0 nodes, gathers the events anchored at those nodes, and joins the L2 indicators. For every affected sector it then computes a quantitative output from the Leontief inverse Miller and Blair (2009). We write A for the matrix of technical coefficients with $A[i,j] = \frac{Z[i,j]}{x[j]}$, so that columns are normalized to unit output and a_{ij} is the direct dependence of sector j on commodity i. The effect of a shock originating at i on sector j is

$$\text{Effect}_j \;=\; \text{shock}_i \,\big|\; \cdot\; [(I-A)^{-1}]_{ij} \qquad (1)$$

where shock_i is the price or supply shock read from the event and $[(I-A)^{-1}]_{ij} =$ `glm:totalRequirement` is the total, direct plus indirect, requirement of sector j for the output of i. Its series expansion $\delta_{ij} + a_{ij} + (A^2)_{ij} + \ldots$ begins with the direct coefficient and adds every further round exactly once. Because events are anchored at shared L0 nodes, the same written event is materialized identically across all scenarios that reference the node: in the pilot, a single "gas price +40 %" event yields the same per-sector effects in both the soy and the fertilizer scenarios without any per-scenario editing.

Export. The materialized view (entities with their vulnerabilities, the events with their (raw-string and decimal) impacts, and the computed effects) is serialized as a PDL document. The materialized PDL is therefore not authored by hand but generated from the world state, while remaining a valid, reviewable PDL artifact.

Conversion to an experiment. A converter turns the PDL document into a complete scenario specification for the agent-based simulation. The supply chain is constructed from the entities and supply chain sections of the PDL by instantiating agents for each entity and linking their inputs and outputs according to the supply chain. Additionally, substitution possibilities are stored in a registry to allow agents to use different suppliers or service providers in scarcity scenarios. The translation from PDL events to simulation parameters proceeds through three key transformations. Impact magnitudes are converted from percentage strings into multiplicative parameters. All temporal intervals are parsed from human-readable duration strings into integer day counts. The PDL loader then constructs a structured event registry containing both an events listing with target entity, impact multipliers, persistence duration, and activation condition for each event, and a timeline indicating at which simulation day each event becomes eligible.

Simulation Execution

At runtime, an EventTracker component manages event lifecycle and condition evaluation. As each simulation timestep progresses, the tracker checks whether any events become newly eligible based on the cascade day schedule. For eligible unconditional events, activation occurs immediately. For eligible conditional events, a simple event calculus evaluates boolean expressions over event states—whether an event is currently active, duration conditions, and standard boolean operators—to determine activation. Once active, events apply their impact multipliers to target entity-parameter pairs for the duration specified in the PDL.

Agents query the current shock environment at each timestep to obtain entity-specific parameter multipliers. Rather than modifying agent parameters directly, the simulation preserves baseline parameters and applies shock multipliers at execution time. This decoupling enables independent scenario specification and permits transparent shock composition.

Simulation outputs are generated as time-resolved records of system state and agent behavior, enabling quantitative analysis of disruption propagation and recovery dynamics. The resulting datasets support direct comparison between baseline and disruption scenarios, estimation of cascade intensity and duration, and reconstruction of causal trajectories from initial shock to downstream market and demand responses.

LIMITATIONS

While this paper introduces the PDL and the core simulation engine, certain limitations currently exist. At present, the PDL and the basic simulation engine and KG are the only implemented artifacts from the PROVIDER project, while all other tools described in this work remain conceptualized and are currently under implementation . Consequently, a full-scale evaluation of the system's performance and effectiveness has not yet been conducted.

CONCLUSION AND OUTLOOK

This paper has outlined how PDL can describe supply-chain scenarios, including entities, dependencies, shocks, and resilience mechanisms, in a form that is readable by domain experts, mechanically validatable, and generated from KG materializations with explicit provenance. The three-step parameterization process—materializing a scenario view from the KG, exporting it as PDL, and converting the PDL into an executable experiment provides a rigorous, automated pipeline from knowledge to simulation. In summary, PDL fulfills its core objective: providing the missing contractual interface between what is known about supply-chain ecosystems and what can be dynamically simulated to understand their

behavior under disruption.
In the future, we will finish the implementation of the described tech-stack and evaluate the system. We will experiment with different agent behavior implementations and methodologies to extract the scenario overlay from the KG. In addition, we will implement a metaheuristic to identify vulnerable scenarios configurations for the simulated supply chain. To achieve this, we will use Adversarial Resilience Learning (ARL) Fischer et al. (2019). ARL uses multiple Reinforcement Learning (RL) agents to test and improve the resilience of Cyber-Physical Systems (CPSs). This approach will be extended to be applicable to this use case by training two agents that test simulation configurations. One agent's goal is to find events that disrupt the supply chain, while the other agent's goal is to find countermeasures to stabilize the system. This will be integrated into the simulation using the established framework *palaestrAI*. Veith et al. (2023)

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SIMULATION AND FACILITY LAYOUT DESIGN FOR INTELLIGENT MANAGEMENT IN FAST-FOOD RESTAURANTS: REDUCING SERVICE TIMES AND MAXIMIZING OPERATIONAL EFFICIENCY THROUGH DISCRETE-EVENT SIMULATION

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KEYWORDS
Simulation, Queueing theory, Bottleneck, Arena Software.

ABSTRACT

This study examines customer flow and service operations in a fast-food restaurant using discrete-event simulation. The objective was to identify bottlenecks and evaluate how different numbers of cashiers and cooks affect service performance. The model, developed in ARENA, represented customer arrival, order placement, food preparation, and order delivery under a First In, First Out discipline. Sixteen staffing scenarios were simulated over a two-hour period. The results showed that four cashiers and four cooks achieved the highest throughput, serving 139 customers. However, the configuration with two cashiers and three cooks served 124 customers with three fewer employees and was therefore selected as a suitable operational alternative among the evaluated scenarios. Based on this configuration, an AutoCAD layout was developed to organize the cashier, kitchen, delivery, and dining areas. The findings show that discrete-event simulation can support staffing and facility-layout decisions, although a detailed cost analysis is still required to determine the economically optimal configuration.

INTRODUCTION

Fast-food restaurants often experience periods of high demand, during which limited service capacity can lead to long queues and delays. These problems affect customer satisfaction and can also reduce the overall efficiency of the restaurant. Previous research has shown that waiting times are commonly related to the availability of cashiers, kitchen staff, and the way customers move through the service process (Afolalu et al., 2020).

Previous studies have also shown that simulation can support capacity planning and the redesign of ordering and production processes (Castillo et al. 2020; Malega and Daneshjo 2024).

Discrete-event simulation provides a practical way to study these situations without disrupting daily operations. It allows different staffing and service alternatives to be tested before changes are introduced in the actual restaurant. This approach has been used to examine queues, processing times, resource utilization, and facility layouts in both service and production environments (Kasemset et al., 2023; Rafly et al., 2024). ARENA is particularly useful because it can represent customer arrivals, waiting lines, service activities, and available resources within the same model (Pappas et al., 2025).

In this study, a discrete-event simulation model was developed to examine sixteen combinations of cashiers and cooks in a fast-food restaurant. The model follows the customer journey from arrival and order placement to food preparation, delivery, and departure. The results were used to identify bottlenecks and compare the performance of the proposed staffing alternatives. Based on the selected configuration, an AutoCAD layout was also developed to improve the organization of the cashier, kitchen, delivery, and dining areas.

MATERIALS AND METHODS

A discrete-event simulation model was developed to examine customer flow and identify bottlenecks in a fast-food restaurant. The model represents the main stages of the service process: customer arrival, menu selection, order placement and payment, food preparation, order delivery, and departure. Customers were served according to a First In, First Out discipline.

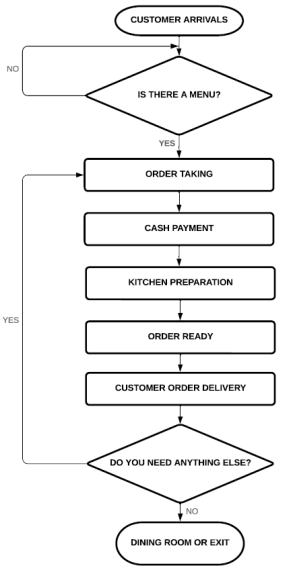

Figure 1: Flow chart of fast-food establishment processes.

The input parameters considered the time between customer arrivals and the service times at the cashier and kitchen areas. Customer interarrival time was represented by an exponential distribution with a mean of 0.9 minutes, cashier service time by a normal distribution with a mean of 1.5 minutes and a standard deviation of 0.2 minutes, and kitchen service time by a uniform distribution ranging from 1.4 to 2.5 minutes.

The model was implemented in ARENA 10.0 and simulated over a two-hour operating period. Sixteen scenarios were evaluated by varying the number of cashiers and cooks from one to four. Each scenario was compared according to the number of customers served and the bottlenecks observed at the cashier and kitchen stages. The resulting staffing configuration was then used as the basis for developing the restaurant layout in AutoCAD.

RESULTS

As shown in Figure 2, block scheduling was carried out in the Arena software, which has a customer entrance, cash register, customer orders, kitchen, order delivery, and finally dining room or exit.

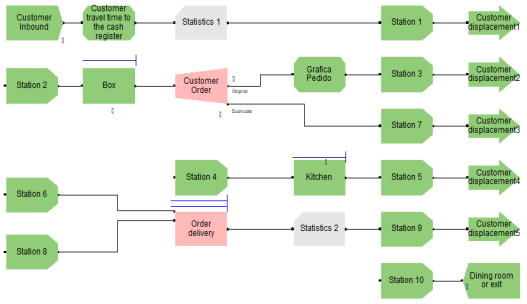

Figure 2: Programming blocks in the Arena software

Therefore, Table 1 shows the data from the tests with cashiers, cooks, and customers served from the 16 tests in the Arena Software simulation.

Table 1: Results of the Sixteen Staffing Scenarios

	Cashier	Cook	Clients served
TEST 1	1	1	60
TEST 2	1	2	76
TEST 3	1	3	77
TEST 4	1	4	77
TEST 5	2	1	62
TEST 6	2	2	104
TEST 7	2	3	124
TEST 8	2	4	105
TEST 9	3	1	58
TEST 10	3	2	104
TEST 11	3	3	121
TEST 12	3	4	115
TEST 13	4	1	ERROR
TEST 14	4	2	102
TEST 15	4	3	126
TEST 16	4	4	139

The simulation results varied according to the number of cashiers and cooks assigned to each scenario. In Test 13, the combination of four cashiers and only one cook caused an excessive accumulation of customers at the kitchen stage. Under these conditions, ARENA returned an execution error and did not provide a valid number of customers served. For this reason, the result was reported as “ERROR” in Table 1 and excluded from the final comparison.

Among the valid scenarios, Test 16 achieved the highest service capacity, with four cashiers and four cooks serving 139 customers. Test 7, however, served 124 customers with only two cashiers and three cooks. This means that it served 15 fewer customers while requiring three fewer employees. Test 15, with four cashiers and three cooks, served 126 customers, only two more than Test 7 despite requiring two additional workers. Test 11 served 121 customers with three cashiers and three cooks. Considering these results, Test 7 was selected as the most suitable staffing alternative among the configurations evaluated. Nevertheless, a detailed economic assessment is still required to determine whether this alternative is also the most cost-effective.

The selected staffing configuration was subsequently used to develop the restaurant layout in AutoCAD. The cashier, kitchen, preparation, delivery, and dining areas were organized according to the sequence of activities represented in the simulation model. The resulting layout aims to maintain a continuous customer flow, reduce unnecessary movement, and improve coordination between the service and food-preparation areas, as shown in Figure 3.

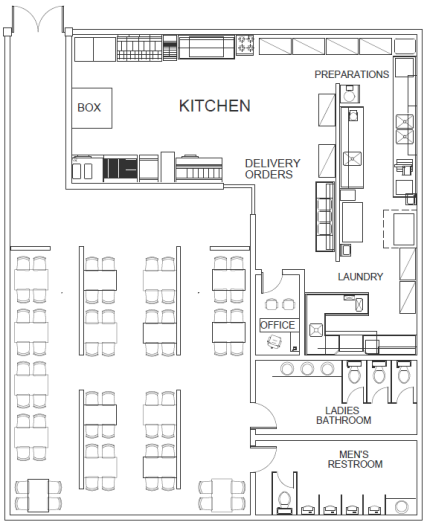

Figure 3: Restaurant Layout Developed in AutoCAD

Overall, the findings suggest that restaurant performance depends not only on the number of employees available, but also on how the main service areas are arranged and connected. The combination of two cashiers and three cooks, supported by the proposed customer and order routes, offered a practical way to organize the restaurant while

maintaining service capacity and avoiding unnecessary movement.

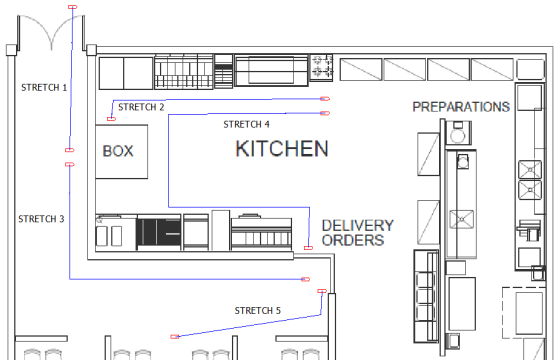

Figure 4: Customer and Order Flow within the Proposed Layout

DISCUSSION

The results follow a similar pattern to previous research on queue management in fast-food restaurants. Rafly et al. (2024) found that customer arrival patterns and the number of available servers directly affect waiting times, particularly during periods of high demand. Afolalu et al. (2020) also showed that reorganizing the queue can improve customer flow and reduce delays. Building on these findings, the present study used ARENA to compare sixteen combinations of cashiers and cooks. The results showed that increasing the number of employees did not always lead to a proportional increase in the number of customers served.

The analysis also showed that staffing decisions are closely related to the physical arrangement of the restaurant. For this reason, the selected configuration was incorporated into an AutoCAD layout connecting the cashier, kitchen, preparation, delivery, and dining areas through defined customer and order routes. However, the model covered only a two-hour operating period and did not include detailed costs or a formal optimization method. Therefore, Test 7 represents the most suitable alternative among the scenarios evaluated, but it should not be considered a definitive optimal solution. Further research should include real waiting-time data, additional simulation runs, and a detailed cost analysis to strengthen the results.

FUTURE RESEARCH

Future work should validate the simulation model using real customer arrival, service, and waiting-time data. Additional simulation runs under different demand conditions would help determine whether the selected configuration remains suitable during peak and low-demand periods. A detailed cost analysis should also be included to evaluate the economic performance of each alternative. In a later stage, point-of-sale data and machine-learning methods could be incorporated to improve demand forecasting and support adaptive staffing decisions.

CONCLUSIONS

The simulation showed that staffing levels have a direct effect on the service capacity of the restaurant. Among the configurations evaluated, Test 7 served 124 customers with two cashiers and three cooks during the two-hour period. Although Test 16 achieved the highest throughput, it required three additional employees to serve 15 more customers. For this reason, Test 7 was selected as a suitable operational alternative rather than a definitive optimal solution. The selected configuration was incorporated into an AutoCAD layout that organized the main service areas and defined the routes followed by customers and orders. Together, the ARENA model and the proposed layout provide a practical basis for supporting staffing and facility-layout decisions in fast-food restaurants.

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Emergency Management Simulation

POTENTIAL OF ARTIFICIAL INTELLIGENCE IN SCENARIO CREATION IN THE CONTEXT OF SIMULATIONS AND TRAINING IN DISASTER MANAGEMENT

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Disaster Management, Artificial Intelligence, Scenario Generation, Simulation, Training, Large Language Models

ABSTRACT

Effective disaster management training depends on scenarios that are realistic, internally consistent and tailored to the hazards, geography and institutional setting of the exercise, yet developing such scenarios traditionally consumes substantial expert time and rarely covers compound or complex events. This paper examines the potential of artificial intelligence (AI), and in particular generative AI and large language models (LLMs), to support scenario creation for simulations and training in disaster management. We review recent prototypes that use AI to generate training scenarios and relate these developments to established simulation approaches.

We discuss how AI can reduce the time and expertise required to build a scenario, increase scenario variability and realism, and enable adaptive exercise design. We further examine current limitations, including the risk of hallucinated or unrealistic content, dependence on data quality and completeness, and the continued need for expert validation. We conclude that AI assisted scenario generation is a promising, but not yet autonomous addition to the disaster management training.

INTRODUCTION

Simulation based exercises are a cornerstone of effective disaster management. They are the primary mechanism through which organisations verify written plans and procedures into tested, rehearsed capability (Perry and Lindell, 2003). Evidence shows that well-designed exercises improve participants' knowledge of emergency procedures, their confidence and judgement under pressure, and their understanding of individual and inter-agency roles, even though evidence on whether these gains persist over time and translate into measurably better real-world response remains more limited (Skryabina et al., 2017).

Every emergency response also leaves a data trail. Crisis-management organisations routinely collect and record large volumes of information about the events they respond to: incident reports, situation logs, damage assessments, statistics and after action reports. In Slovakia, this role is filled in part by the Unified Information System of Economic Mobilization and Development (EPSIS® JISHM), which has already been used to consolidate crisis-relevant information and support preparation for crisis situations at national level (Ristvej et al., 2017). Systems of this kind show that the infrastructure needed to collect and store event-related data typically already exists. However, data in the system are archived rather than actively reused. This growing body of historical event data is, in principle, a valuable resource. It documents how real situations developed, which measures worked, and where the response fell short, and it can in turn be used to inform the design of future exercises rather than being consulted only once, at the time the event occurred.

This potential is greatest for complex, cascading scenarios, in which one triggering event sets off a chain of further, secondary events across different sectors or systems, for example a storm that disables a power substation, which in turn interrupts a hospital operation. Such compound and cascading risks are, by definition, difficult to script from first principles, because their dynamics depend on how multiple systems interact rather than on any single hazard in isolation (Pescaroli and Alexander, 2018). Systematically studying historical events of this kind, including how they unfolded, which dependencies proved critical, and what their downstream consequences were can give exercise designers an empirical basis for constructing scenarios that reflect real interdependencies.

Simulation exercises themselves need to meet certain design conditions. They should be grounded in a plausible, internally consistent situation, they should be matched to the specific hazards, procedures and organisational

structure of the participating audience, they should include a defined set of learning objectives against which performance can be evaluated and their content should be varied enough across repeated exercises to avoid predictability once participants become familiar with recurring scenarios (Dausey, Buehler and Lurie, 2007; Skryabina et al., 2017). Meeting all of these conditions consistently is demanding in practice, particularly for the compound and cascading scenarios discussed above, and it is precisely this combination of requirements that motivates interest in AI assisted approaches to scenario development.

Artificial intelligence, and large language models in particular, can support several distinct stages of this process: analysing complex, multi-source situational data to surface patterns and cascading dependencies, preparing and writing the resulting scenario as a structured narrative with a timeline of injects and generating multiple variants of the same base scenario under different assumptions, to explore how a situation might have evolved had other influencing factors changed. Figure 1 summarises these three areas of application.

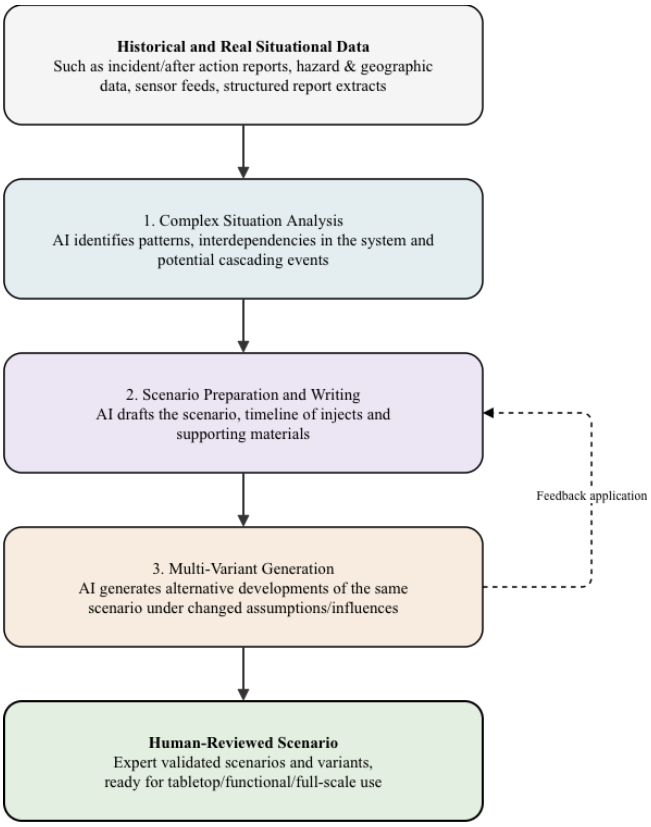

Figure 1 Core Areas of AI Application in Scenario Development.

SCENARIO DEVELOPMENT IN DISASTER MANAGEMENT TRAINING

Exercise programmes typically span a spectrum from discussion-based formats such as seminars, workshops and tabletop exercises to operations-based formats such as drills, functional exercises and full-scale exercises, each demanding a progressively higher level of scenario fidelity and logistical complexity. Across this spectrum, the core planning artefact is the master scenario events list (MSEL): a chronologically sequenced set of injects, together with their intended participant response and mapping to specific exercise objectives, that controllers use to drive play. Although the MSEL is described as the central blueprint of an exercise, Renger et al. (2009) note that neither the exercise-development literature nor established frameworks such as the US Homeland Security Exercise and Evaluation Program provide substantial guidance on how to actually construct an effective one, so in practice its quality depends heavily on the individual experience of the facilitators involved.

Tabletop and functional exercises typically rely on a small team of such facilitators, who must translate hazard data, procedures and lessons learned into an internally consistent MSEL, write the accompanying narrative injects, and prepare supporting artefacts such as maps and schedules. Producing a single, well-tested scenario can take weeks of expert effort, and the resulting content is often reused across multiple training cycles, which reduces its realism and predictive value once participants become familiar with recurring patterns. Exercise designers also frequently lack the time or resources to generate the volume of supporting media, such as simulated social-media posts, press releases, compound events or traffic, that would make an exercise feel realistic under real operational tempo. A recent review of public health emergency-preparedness exercises similarly identifies scenario development, facilitator selection and training, and the alignment of evaluation methods with exercise objectives as recurring weak points across the wider exercise lifecycle, rather than isolated shortcomings of individual programmes (Chambers et al., 2025). These constraints make it particularly difficult to test rare, compound or cascading hazards, which are precisely the scenarios that expose the largest gaps in organisational preparedness.

ARTIFICIAL INTELLIGENCE APPROACHES TO SCENARIO GENERATION

Generative content and text-based scenario writing: large language models can read free-text source material such as threat reports, incident summaries or procedural documents and turn it into the structured blocks an exercise needs: actors and responsibilities, a timeline of events, and specific injects. AiCEF shows this concretely for cybersecurity exercises, it scans open-source cyber-threat reports, automatically identifies key entities such as attacker groups, malware families and targeted systems, and uses them to assemble a structured scenario (Zacharis and Patsakis, 2023). The same approach is directly transferable to disaster management, where an LLM can extract recurring elements from hazard, situation and after-action reports and turn them into draft injects.

Retrieval-grounded scenario generation: to reduce the risk of factually implausible content, several systems combine LLMs with retrieval-augmented generation. Asami, Hiroi and Hatayama (2025) built such a system with the support of Hirakata City and Kumamoto Prefecture in Japan: it retrieves the training scenarios those authorities actually use, together with geographical information on the area, and generates new scenarios through a chatbot interface with the exercise designer kept in the loop. Evaluation by interviews with the same municipal officials confirmed that the retrieved context allowed the system to tailor generated scenarios to specific disaster types and regional characteristics that a general-purpose LLM could not reproduce reliably on its own. A comparable case outside disaster management is AiCEF, whose generated content was reviewed by experts and used in a real awareness tabletop exercise (Zacharis and Patsakis, 2023).

Multi-agent and decision-support frameworks: beyond static scenario text, LLMs are increasingly embedded in multi-agent simulation frameworks that couple scenario generation with adaptive learning and decision-making. Wang et al. (2025) proposed a Scenario–Learning–Decision framework in which LLM-driven agents, combined with prospect-theory-based behavioural modelling, perceive an evolving emergency scenario and support collective human–AI decision making within a simulation platform. Complementary tools such as DisasterResponseGPT (Goecks and Waytowich, 2023) generate rapid, guideline-consistent plans of action from a described disaster situation, and could be paired with scenario-generation systems to produce, in near real time, both an exercise inject and a reference response against which decisions can be benchmarked.

Domain-specific applications: feasibility studies in specific training domains support the general viability of this approach. Navarro et al. (2025) used a general-purpose generative AI platform to draft mass-casualty-incident trauma simulation scenarios and scored them with a validated simulation-quality instrument, finding that AI-generated scenarios were a feasible starting point but still required structured expert review before use. In the military domain, Hans et al. (2025) report that LLM-based tools can produce numerous, varied tactical training vignettes, including adaptive adversary behaviour, at a fraction of the time required for manual scenario design, though the authors caution against overtrusting AI generated content that is not grounded in a credible knowledge base.

Digital twins and adaptive serious games: a complementary line of research links AI-based scenario generation to digital twins and serious games. Fan et al. (2021) outlined a vision for a disaster-city digital twin that fuses real-time and historical data with human and artificial intelligence to support disaster management, including training applications where the twin can generate and adapt scenarios to reflect current infrastructure and population conditions. Loizou and Andreou (2025) proposed a standardised framework for building serious games with real-time self-adaptation capabilities, in which digital twin derived semantic descriptions allow a training scenario to reconfigure itself during play in response to trainee decisions. For Slovak conditions, a maturity model for flood digital twins has recently been proposed, specifying the data layers and sensing requirements a twin must reach before it can support prediction, situational awareness and decision making (Ristvej et al., 2026).

Mining historical incident and after-action reports: a currently under-exploited grounding source is the narrative reporting that authorities routinely produce after every emergency response. In Slovakia, fire and rescue units file a structured intervention report governed by Ministry of Interior decrees (decree MI SR n. 611/2006 and n. 169/2002), and municipalities separately compile an after-action report and continuous report submitted to the district crisis-management authority. Aggregated statistics from these reports are published, but, unlike the United States publicly searchable Lessons Learned Information Sharing repository, individual narrative reports are not compiled into a public, machine-readable archive. NLP information extraction pipelines have already been demonstrated on comparable material: Mirończuk (2020) converted unstructured Polish fire-brigade reports into structured fields using a dedicated extraction system, and Lensing et al. (2025) text-mined a corpus of after-action reports, response plans and impact assessments for the Cascadia earthquake to surface recurring preparedness lessons. Applying an analogous LLM-based extraction pipeline to Slovak intervention and after-action reports, following a schema such as the one summarised in Figure 2, would let scenario generators be grounded in how local incidents have actually unfolded, and would echo the European Union Civil Protection Mechanism's own Lessons Learnt Programme (Decision No. 1313/2013/EU, Art. 13d), which mandates comparable analysis at EU level (Lei et al., 2025).

Figure 2 proposes a concrete extraction schema for this purpose, grouping report content into seven categories: event identification, response timeline, actors and resources, escalation and cascading effects, impact, decisions and actions, and lessons learned. Each schema category maps directly onto a component of the master scenario events list: event identification fixes the initiating hazard, the response timeline supplies inject timing, actors and resources define the participating agencies, and escalation and cascading effects mark the branching points at which a scenario can diverge. Extraction yields exactly the fields an MSEL requires, which is what separates automated scenario drafting from mere summarisation of past events.

Figure 2 Extraction Schema for AI Scenario Generation.

POTENTIAL BENEFITS AND LIMITATIONS

Taken together, there are several concrete benefits of AI assisted scenario creation for disaster management training: a substantial reduction in the time and expert effort needed to prepare an exercise, a larger and more diverse pool of scenarios, including rare or compound events that are difficult to script manually, and closer tailoring to specific hazards.

These benefits are accompanied by well-documented limitations. Generated content remains only as reliable as the data on which it is grounded, and poorly retrieved or ungrounded content can still produce plausible but inaccurate scenario elements (Lewis et al., 2020). Ethical concerns, limited transparency and the difficulty of evaluating the practical realism of AI-generated content remain open problems across application areas (Saengtabtim et al., 2025). Evidence from the mass-casualty and military domains further indicates that human experts must still validate generated scenarios before use, both to correct factual or procedural errors and to guard against over-reliance on automation that could, over time, erode own scenario design and decision making skills (Navarro et al., 2025).

CONCLUSION

The evidence indicates that AI, particularly LLMs combined with retrieval-augmented and multi-agent architectures, has clear and increasingly validated potential to support scenario creation for simulations and training in disaster management. AI can shorten scenario-development time, increase scenario diversity and realism, and, when coupled with digital twins, enable exercises that adapt in time.

Future work should focus on grounding scenario generators in national and regional multi-hazard risk assessments, such as the spatial, multi-level hazard and vulnerable-group mapping already developed for cities and national risk assessments (Fekete et al., 2025) rather than in generic or foreign case material, and on tighter integration between scenario-generation tools and digital twin based simulation platforms that enable adaptive simulations in time.

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A MULTI-SECTOR NETWORK MODEL FOR ASSESSING CRITICAL INFRASTRUCTURE RESILIENCE

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ABSTRACT

This paper presents a multi-sector network model for resilience assessment of interdependent electricity, gas and water infrastructures. Producers, network nodes, consumers and cross-sector links are integrated in one optimization structure. Electricity is represented by DC load flow, while simplified Hazen–Williams and Weymouth relations for water and gas are piecewise-linearized with SOS2 constraints, yielding a MILP formulation. The model includes electricity consumption by pumps, compressors and gas/water producers, gas and water consumption by power plants, and VOLL penalties for unserved demand. A Java/Gurobi prototype solves normal and failure states. A synthetic case study shows how generation failures and gas, water and pipeline restrictions propagate across sectors and create direct and indirect shortages.

INTRODUCTION

Critical infrastructures are not isolated technical objects. They form a system of systems in which the continuity of one service depends on the availability of several others. Electricity, gas and water systems are a particularly important example of this phenomenon. Electricity is needed to operate pumping stations, water intakes, gas extraction facilities and compressor stations. Gas may be used as a fuel for power generation. Water is required by selected power plants for cooling and technological processes. Consumers, including households, industry and critical consumers such as hospitals, depend simultaneously on all of these services. This mutual dependence creates a risk of cascading effects. A failure of a power plant may reduce electricity generation and force the remaining plants to increase production. However, this re-dispatch can also change gas and water consumption. Similarly, a limitation of gas supply may affect electricity generation when gas-fired power plants compete with consumers for the same gas resource. A reduction of water availability may limit the cooling capacity of power plants, which in turn can create electricity shortages even if no electrical line or generator is physically damaged. These examples show that the assessment of critical infrastructure resilience requires more than a separate reliability analysis of each sector.

The aim of this paper is to present a multi-sector network model that can be used to assess the resilience of interdependent critical infrastructures through optimization and simulation. The research direction is also consistent with earlier work on risk and continuity-of-services analysis for critical infrastructure operators (Najgebauer et al. 2022), (Najgebauer and Antkiewicz 2022). The model represents electricity, gas and water sectors as physically interpretable networks. It includes producers, distribution nodes, arcs, consumers and cross-sector links. It also includes functional dependencies such as electricity consumption by pumps and compressors, electricity consumption by gas and water producers, water and gas consumption by power plants, and differentiated consumer demand. The resulting model can be used to solve normal-state and failure-state optimization problems and to compare how alternative incidents influence the service level delivered to consumers.

The approach extends an earlier hybrid simulation-analytical method for assessing the resilience of critical infrastructures (Stąpor and Antkiewicz 2025). The earlier method combined discrete-time simulation, service allocation optimization and resilience indicators. However, it represented service transfer at a relatively aggregate level. The new model introduces explicit network topology and sector-specific physical constraints. This is an important step from a general interdependency model toward an engineering model suitable for scenario analysis, vulnerability detection and decision support. The main contributions of this paper are as follows. First, it proposes an integrated graph representation of electricity, gas, water and consumer layers.

Second, it formulates a *Mixed-Integer Linear Programming* (MILP) optimization model that includes node balances, capacities, directions of flow and physical state variables. Third, it applies piecewise linear approximation with *Special Ordered Sets of type 2* (SOS2) constraints to preserve a tractable MILP formulation while approximating nonlinear gas and water physics. Fourth, it introduces a unified objective function based on production costs, transmission costs, pump and compressor operation costs, and *Value of Lost Load* (VOLL) penalties. Fifth, it demonstrates the use of the model on a four-layer test network and interprets several failure scenarios.

BACKGROUND AND RELATED WORK

Critical-infrastructure resilience concerns the ability to withstand, adapt to and recover from disruptions while maintaining essential services, consistent with Directive (EU) 2022/2557 (European Parliament and Council of the European Union 2022). Interdependencies are commonly classified as physical, cyber, geographic or logical; this work focuses on physical and functional service dependencies that can be expressed by flows and consumption coefficients.

Modeling approaches include empirical, agent-based, system-dynamics, economic input-output and network methods (Ouyang 2014). Input-output models capture sectoral dependence but usually omit network topology, whereas network models represent connectivity, capacities and flow redistribution (Magoua et al. 2022, Wang et al. 2022). The proposed model belongs to the flow-based network group and uses input-output coefficients for cross-sector consumption. It deliberately favors a common MILP representation over detailed sector-specific simulators: DC flow and piecewise-linear gas/water approximations retain the main operational couplings while permitting efficient multi-scenario optimization.

MULTI-SECTOR NETWORK REPRESENTATION

Layers and Elements

The integrated infrastructure is represented as a graph with four conceptual layers: consumers, electricity, gas and water. The consumer layer contains consumer requiring one or more services. The electricity layer contains power plants, electrical buses and substations. The gas layer contains gas producers, gas pipes and gas distribution nodes. The water layer contains water producers, water pipes and water distribution nodes. Arcs represent physical transport possibilities.

Sets

Let $K = \{E, G, W\}$ denote the set of modeled sectors: electricity, gas and water. For each sector $k \in K$, N^k denotes the set of distribution nodes (substations for E) and L^k denotes the set of network arcs between nodes. The set C denotes consumers. The sets P, G and W denote power plants, gas producers and water producers, respectively. Additional sets define links from producers to sectoral networks, from sectoral networks to consumers and from one sector to another.

Table 1: Main Sets Used in the Multi-Sector Network Model

Symbol	Meaning
L_C^k	Links from sectoral nodes to consumers.
L_P^E, L_G^G, L_W^W	Producer-to-network links for power, gas and water.
L_P^G, L_P^W	Gas and water supply links to power plants.
L_G^E, L_W^E	Electricity links to gas and water producers.
L_{NG}^E, L_{NW}^E	Electricity links to gas and water nodes for compressors and pumps.

Example Four-layer Network

The synthetic test network (Figure **??**) contains three power plants, four electrical substations and five lines; three gas producers, three gas nodes and three pipelines; three water producers, three water nodes and three pipelines; and five consumers. C 5 represents a hospital and may receive a higher VOLL.
Each sector is linked to consumers. Cross-sector arcs deliver electricity to gas/water producers and pumps/compressors, and gas or water to power plants. These links create cascade paths: electricity shortages can constrain gas/water transport or production, while gas/water shortages can reduce generation. All case-study data are synthetic.

Parameters

The model uses demand, cost, capacity and physical parameters. Demand D_c^k represents the demand of consumer c for service k. $VOLL_c^k$ is the Value of Lost Load for service k at consumer c and expresses the penalty for one unit of unserved demand. Production costs (c_p^P, c_i^G, c_i^W) are assigned to power plants, gas producers and water producers.
The following By using c_{ij}^E, c_{ic}^E, c_{pi}^E we denote costs of transmitting power through electricity links (substation to substation, substation to costumer, power plant to substation). Transmission of water and gas through pipes requires the usage of pumps and compressors. Theses devices utilize electric energy. The costs of using a unit of energy by one of pumps/compressors are noted

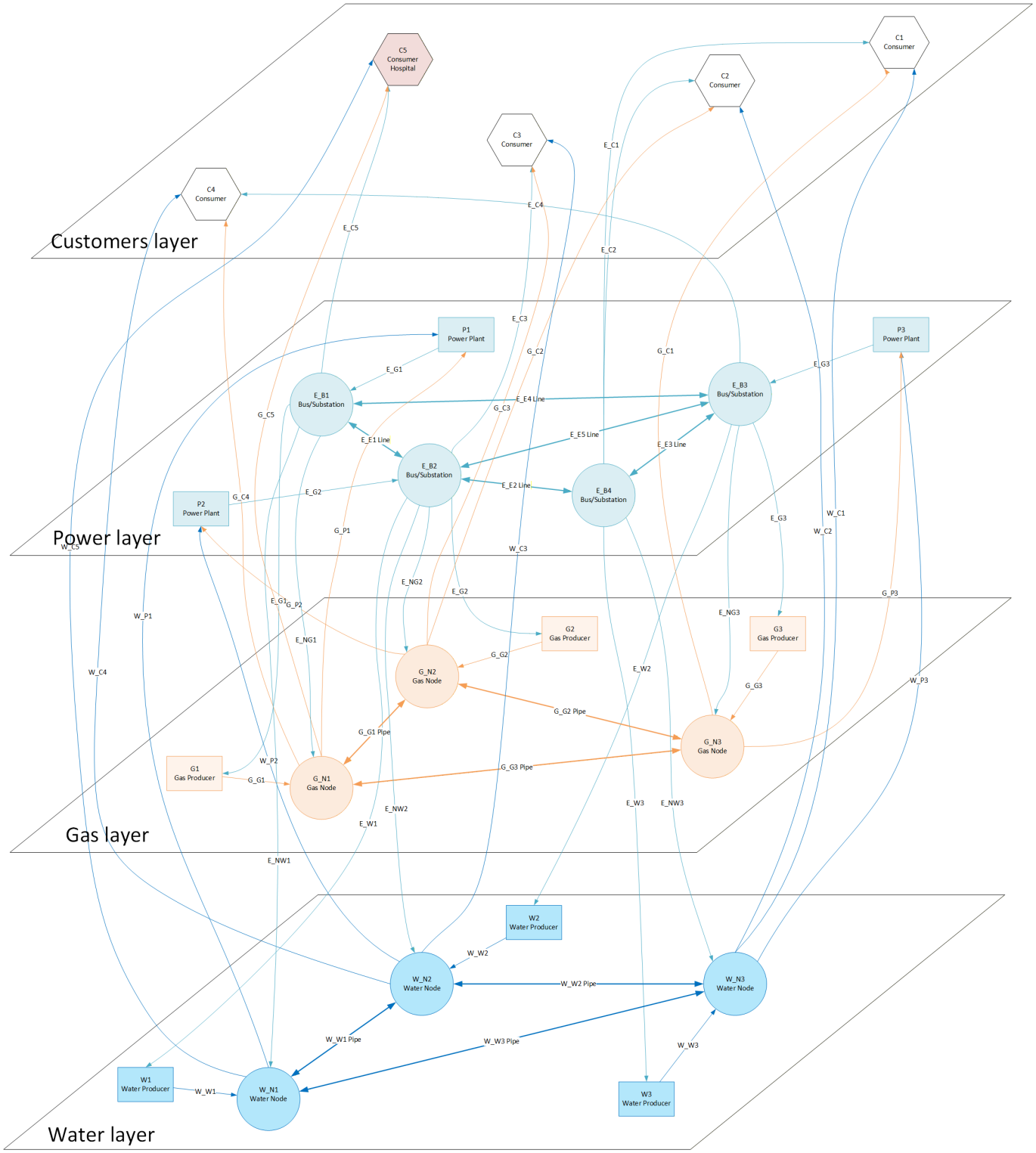

Figure 1: Example Four-layer Electricity-gas-water-consumer Network

Table 2: Flows

Symbol	Meaning
F_{ij}^{k}	$F_{ij}^{k} = F_{ij}^{k,+} - F_{ij}^{k,-}$
$F_{ij}^{k,+}$	Fl. of k from i to j through arc (i,j); $F_{ij}^{k,+} \geq 0$
$F_{ij}^{k,-}$	Fl. of k from j to i through arc (i,j); $F_{ij}^{k,-} \geq 0$
F_{ic}^{k}	Fl. of service k from a node i to consumer c.
F_{pi}^{E}	Fl. of E from power plant p to subst. node i.
F_{qi}^{G}	Fl. of G from gas prod. g to gas node i.
F_{wi}^{W}	Fl. of W from water prod. w to water node i.
F_{ig}^{E}	Fl. of E from subst. i to gas prod. g.
F_{iw}^{E}	Fl. of E from subst. i to water prod. w.
$F_{ij}^{E,pump}$	Fl. of E from substation i to pumps in water node j.
$F_{ij}^{E,comp}$	Fl. of E from substation i to compressors in gas node j.

by respectively c^{pump} and c^{comp}. Please note that electric cost of pumps and compressors are paid by the node from which the flow of water or gas originates. The only exception are transmissions from producers to nodes - in that case the cost is still payed by the node.

Physical parameters of linkages include electrical susceptance (B_{ij}), water-pipe coefficients (K_{ij}^{W}) and gas-pipe coefficients (K_{ij}^{G}). Parameters k_p^G,k_p^W denote how much water or gas is consumed by power plants per unit of electricity generation, while α_g, α_w stand for electricity consumption by gas/water production.

The mathematical model takes into account limits of production ($P_p^{\max}, F_g^{\max}, Q_w^{\max}$) and transportation ($F_{ij}^{k,\max}$) as well as maximum power usage of pumps/compressors in a given node - respectively $E_i^{pump,max}$ and $E_i^{comp,max}$.

Decision and State Variables

The model contains production variables, flow variables, served and unserved demand variables, physical state variables and auxiliary variables. Production variables include P_p for electricity generation, F_g for gas production and Q_w for water production. Flow variables have been described in Table 2. Served demand S_c^k and unserved demand U_c^k are used to measure service delivery. Physical state variables include voltage angles θ_i in the electricity layer, gas pressure potentials π_i and water hydraulic heads h_i. Binary direction variables y_{ij}^k are used when an arc can carry flow in either direction, but not simultaneously in both directions.

MATHEMATICAL FORMULATION

Objective Function

The failure-state optimization task minimizes the total cost of operation and shortage. The objective contains VOLL penalties, production costs, transport costs and auxiliary-device operating costs:

$$
\begin{aligned}
Z = min \sum_{c \in C} \left(\sum_{k \in \{E,G,W\}} VOLL_c^k U_c^k \right) + \sum_{p \in P} c_p^P P_p \\
+ \sum_{g \in G} c_g^G F_g + \sum_{w \in W} c_w^W Q_w + \sum_{i \in N^W} c^{pump} E_i^{pump} \\
+ \sum_{i \in N^G} c^{comp} E_i^{comp} + \sum_{(i,j) \in L^E} c_{ij}^E \left(F_{ij}^{E,+} + F_{ij}^{E,-} \right) \\
+ \sum_{(i,c) \in L_C^E} c_{ic}^E F_{ic}^E + \sum_{(p,i) \in L_p^E} c_{pi}^E F_{pi}^E
\end{aligned}
\tag{1}
$$

The VOLL term is central to the model. It allows the optimization problem to compare shortages of electricity, gas and water in a common monetary unit. Without such a term, it would be difficult to combine megawatts, cubic meters per hour of gas and cubic meters per hour of water in one objective function. A high VOLL value can be assigned to a critical consumer or to a particularly important service, making the optimizer prefer other costs over shortage at that location.
In the normal-state task, shortages are not allowed and all demand must be fully satisfied. The objective then minimizes operating cost subject to full service. In the failure-state task, shortages are allowed but penalized. This distinction reflects operational practice: under normal conditions the system is expected to serve all demand, whereas under emergency conditions the decision maker may need to allocate insufficient resources in the least harmful way.

Consumer Service Balance

The served quantity S_c^k is determined by the sum of all incoming flows of service k to consumer c. The shortage variable U_c^k is the unmet part of demand D_c^k. It is not an independent modelling assumption; it results from the feasibility of production, transmission and cross-sector constraints. $S_c^k + U_c^k = D_c^k,\ S_c^k \geq 0,\ U_c^k \geq 0,\ c \in C,\ k \in K$ Additionally: $S_c^k = \sum_{i:(i,c) \in L_C^k} F_{ic}^k$.

Direction and Capacity of Flows

Some arcs represent physical lines or pipelines that may carry flow in either direction depending on system conditions. However, simultaneous positive flow in both directions on the same physical arc is not possible. The model therefore decomposes flow into positive and negative parts and uses a binary direction variable y_{ij}^k: $F_{ij}^{k,+} \leq F_{ij}^{k,max} y_{ij}^k$ and $F_{ij}^{k,-} \leq F_{ij}^{k,max} \left(1 - y_{ij}^k\right)$. For one-way links, such as producer-to-node, node-to-consumer or electricity-to-(pump/compressor) supply links, only one nonnegative flow variable is used. Failed or disabled arcs are represented by setting their upper capacity bound to zero. Partial degradation is represented by reducing the upper bound.

Electricity Network Equations

The electricity layer uses a DC load-flow approximation in which, active power flow on an electrical line is proportional to the difference of voltage angles at its endpoints: $F_{ij}^{E} = B_{ij}\left(\theta_i - \theta_j\right)$ Reactive power, voltage magnitude changes and losses are not represented. This simplification is common in optimization models when the focus is active-power dispatch and network feasibility rather than detailed AC analysis (Hinneck and Pozo 2023). One electrical reference node is selected and its voltage angle is fixed to zero. This condition prevents an infinite family of equivalent angle solutions shifted by a constant. It does not represent a physical grounding assumption; it is a mathematical normalization.

Water Network Equations

The nonlinear gas and water relations are approximated by *piecewise-linear functions* using convex-combination variables constrained as *Special Ordered Sets of type 2* (SOS2). (Beale and Tomlin 1970, Gurobi Optimization, LLC 2026). The accuracy of the approximation is the subject of further research. Water flow is represented by a relation based on the Hazen-Williams equation. (Williams and Hazen 1905, American Water Works Association 2012). The Hazen–Williams equation can be written as:

$$h_f = 10.67 \cdot \frac{L \cdot Q^{1.852}}{C^{1.852} \cdot D^{4.87}}, \tag{2}$$

where h_f denotes the head loss, L is the pipe length, Q is the volumetric flow rate, C is the Hazen–Williams roughness coefficient, and D is the pipe diameter. Ultimately, it can be simplified to: $h_f = K \cdot Q^{1.852}$, where:

$$K = 10.67 \cdot \frac{L}{C^{1.852} D^{4.87}} \tag{3}$$

and corresponds to K_{ij}^{W} in our model, while Q corresponds to our F_{ij}^{W}.

Gas Network Equations

The physical relation in the gas layer is based on the Weymouth equation. (Weymouth 1912, Menon 2005). In a simplified steady-state form, it can be written as:

$$\left(F_{ij}^{G}\right)^2 = K_{ij}\left(\pi_i^2 - \pi_j^2\right), \tag{4}$$

where F_{ij}^{G} denotes gas flow through pipeline (i,j), π_i, π_j stand for gas pressures at nodes i and j; finally K_{ij} is a pipeline parameter. Our K_{ij}^{G} is different from K_{ij}, but it can be derived from K_{ij}.

A gas reference node is fixed to zero pressure potential. Similarly, one water reference node is fixed for hydraulic head. These reference conditions are numerical anchors analogous to the reference voltage angle in DC load flow.

Nodal Balances

At every distribution node, incoming service flows and local production must balance outgoing flows, consumer delivery and cross-sector consumption. The electricity node balance is the richest one because electricity is used not only by consumers, but also by pumps, compressors, gas producers and water producers. For every $i \in N^{E}$:

$$\begin{aligned} \sum_{j:(j,i)} F_{ji}^{E} - \sum_{j:(i,j)} F_{ij}^{E} + \sum_{p \in P(i)} F_{pi}^{E} = \\ \sum_{c \in C^{E}(i)} F_{ic}^{E} + \sum_{(i,g) \in L_G^{E}} F_{ig}^{E} + \sum_{(i,w) \in L_W^{E}} F_{iw}^{E} \\ + \sum_{j:(i,j) \in L_{NW}^{E}} F_{ij}^{E,pump} + \sum_{j:(i,j) \in L_{NG}^{E}} F_{ij}^{E,comp} \end{aligned} \tag{5}$$

where:

- $P(i)$ stands for a set of power plants connected to the i-th node,
- $C^{E}(i)$ stands for a set of energy consumers connected to the i-th node.

The gas and water balances are analogous. They include producer inflows, network inflows and outflows, deliveries to consumers and deliveries to power plants. A gas node may send gas both to consumers and to power plants. A water node may send water both to consumers and to power plants. This is what creates competition between direct consumer demand and production-support demand during emergencies.

Cross-sector Coupling

Cross-sector dependencies are represented by explicit consumption equations. Power plants consume gas and water according to their generation levels. The coefficients may differ between plants, allowing the representation of different technologies.

$$G_p^{cons} = \sum_{i:(g,p) \in L_P^{G}} F_{gp}^{G} = k_p^{G} P_p, \quad G_p^{cons} \geq 0 \tag{6}$$

$$W_p^{cons} = \sum_{i:(w,p) \in L_P^{W}} F_{wp}^{W} = k_p^{W} P_p, \quad W_p^{cons} \geq 0 \tag{7}$$

Gas and water producers consume electricity according to their production levels:

$$\forall_{g \in G}: \sum_{i:(i,g) \in L_G^{E}} F_{ig}^{E} = \alpha_g F_g \tag{8}$$

$$\forall_{w \in W}: \sum_{i:(i,w) \in L_W^{E}} F_{iw}^{E} = \alpha_w Q_w \tag{9}$$

Power consumed by water pumps in electrical node i (analogous for compressors):

$$\forall_{j \in N^{W}}: \sum_{i:(i,j) \in L_{NW}^{E}} F_{ij}^{E,pump} = E_j^{pump} \tag{10}$$

Pumps and compressors consume electricity according to the amount of transported water (analogous for compressors):

$$E_i^{pump} = \sum_{j:(i,j)\in L^W} k_{ij}^{pump} \left|F_{ij}^W\right| + \sum_{w:(w,i)\in L_W^W} k_{wi}^{pump} F_{wi}^W. \quad (11)$$

The absolute value can be expressed here as: $\left|F_{ij}^W\right| = F_{ij}^{W,+} + F_{ij}^{W,-}$
This approximation deliberately ignores the dependence of pump and compressor energy on pressure or head differences. It is a simplification, but it captures the most important functional relation needed for cascade analysis: moving more water or gas requires more electricity.

Production and Device Limits

Failures modify upper bounds: unavailable plants or arcs receive zero capacity and partial degradation lowers the corresponding limit. Producer outputs equal their outgoing network flows:

$$\sum_{i:(p,i)\in L_P^E} F_{pi}^E = P_p, \quad \sum_{i:(g,i)\in L_G^G} F_{gi}^G = F_g \quad (12)$$

$$\sum_{i:(w,i)\in L_W^W} F_{wi}^W = Q_w \quad (13)$$

OPTIMIZATION AND SIMULATION PROCEDURE

Normal and Failure States

The reference problem assumes full availability, $S_c^k = D_c^k$ and $U_c^k = 0$, with an optional normal-operation utilization factor ρ preserving production reserve (e.g. 80%):

$$P_p \le \rho P_p^{\max}, \quad F_g \le \rho F_g^{\max}, \quad Q_w \le \rho Q_w^{\max}. \quad (14)$$

A failure state changes component capacities; shortages then satisfy $0 \le S_c^k \le D_c^k$ and $U_c^k = D_c^k - S_c^k$ and are penalized by VOLL.

Monte Carlo and Implementation

Deterministic scenarios can be embedded in Monte Carlo experiments by sampling component states, modifying capacities and resolving the MILP. Repetition yields shortage probabilities, expected unserved service, affected sectors and component-impact rankings; for component x and metric y, a useful measure is $\mathrm{AVG}(y|x \text{ failed}) - \mathrm{AVG}(y|x \text{ not failed})$.
The research prototype constructs the model in Java, solves it with Gurobi, reads XLSX input and exports

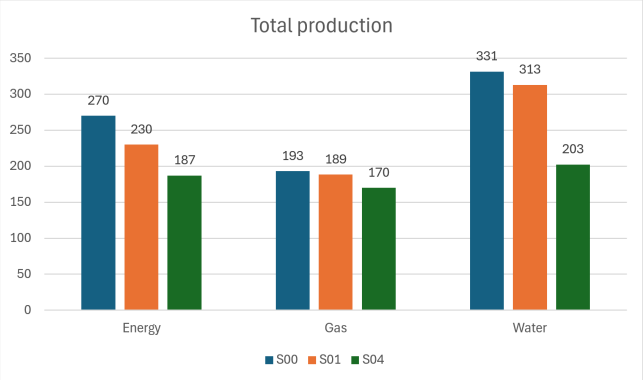

Figure 2: S00, S01 and S04 Total Production Chart

results using Apache POI. The workflow is: load network data, select a normal/failure state, build variables and constraints, solve the MILP, and export flows, production and shortages. These functions are sufficient for the scenarios reported here.

RESILIENCE INDICATORS

Operational outputs can be aggregated as total unserved service $\mathrm{TAUS}^k = \sum_{c\in C} U_c^k$ and VOLL-weighted shortage cost $SC = \sum_{c\in C}\sum_{k\in K} \mathrm{VOLL}_c^k U_c^k$. Repeated simulations additionally provide shortage probabilities, threshold exceedance, expected/maximum shortages, affected consumers/sectors and production changes relative to the reference state.

CASE STUDY AND SCENARIO RESULTS

Reference State S00

In S00 all demand is served. Production is approximately 269 electricity, 193 gas and 331 water units, with objective 19,423; this is the comparison baseline.

Scenario S01: Failure of Power Plant P2

With P2 unavailable, P1 and P3 increase output but cannot fully compensate. Electricity production falls to 230 units and shortage reaches 39 units at C1/C4; gas and water remain fully served because the failed plant no longer consumes them. (Figure 3)

Scenario S02: 50% Gas Availability Limit

A 50% gas-capacity limit reduces gas production to 165 units, causing 15 units of gas shortage (mainly C1/C4) and a 30-unit electricity shortage, while water is fully served. The result demonstrates a gas-to-electricity cascade caused by competition between consumers and gas-fired generation. (Figure 4)

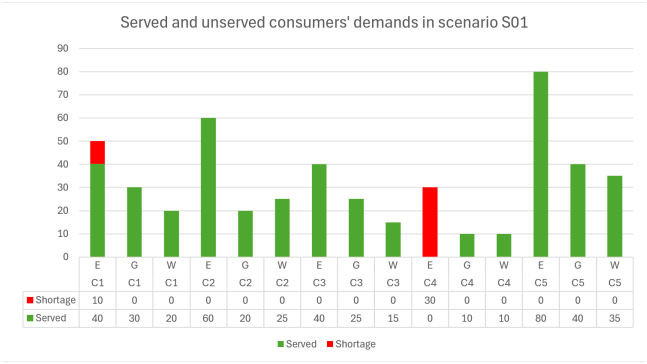

Figure 3: Served and Unserved Consumers' Demands in Scenario S01

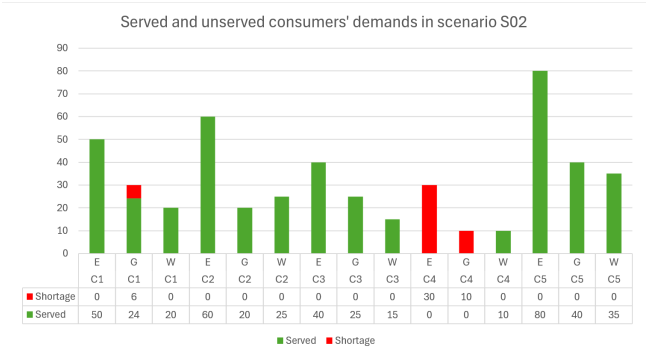

Figure 4: Served and Unserved Consumers' Demands in Scenario S02

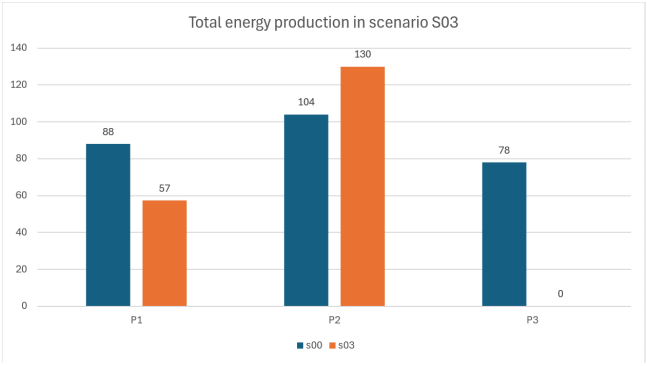

Figure 5: S00 and S03 Energy Production Chart

Scenario S03: 50% Water Availability Limit

A 50% water-capacity limit produces 33 units of water shortage (C1/C2) and 80 units of electricity shortage. P3 stops and P1 is constrained, showing a strong water-to-electricity cascade through power-plant cooling/technology demand. (Figure 5 and Figure 6)

Scenario S04: Failure of Water Pipelines

Failure of water pipelines W_W2 and W_W3 reduces deliverability: water production falls to 203 units, water shortage reaches 55, electricity shortage 80, and electricity production 187. The scenario shows that transport bottlenecks can propagate to power generation even when production resources remain available. (Figure 7)

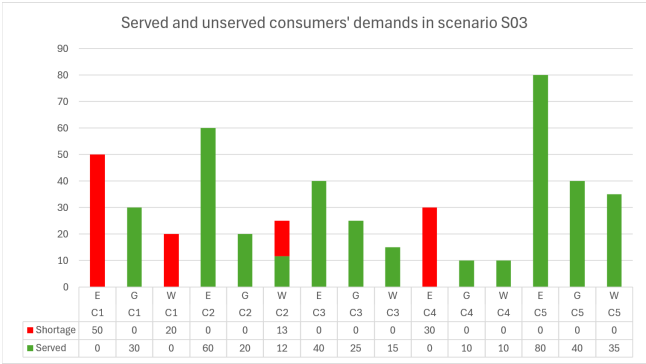

Figure 6: Served and Unserved Consumers' Demands in Scenario S03

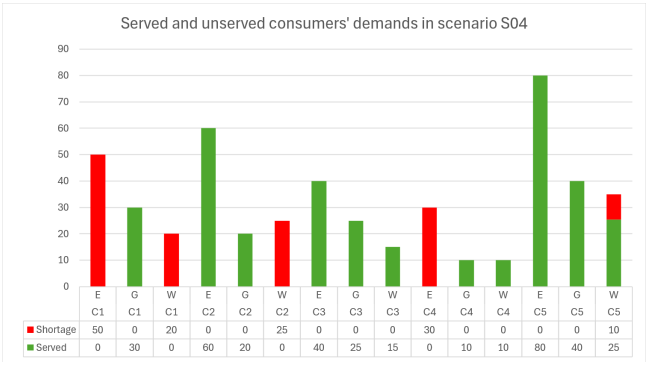

Figure 7: Served and Unserved Consumers' Demands in Scenario S04

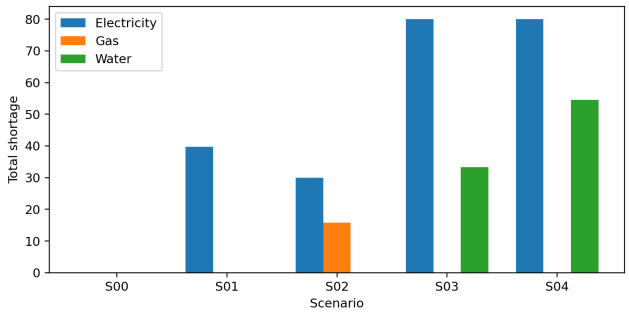

Figure 8: Total Shortages by Sector in the Reference and Failure Scenarios

Summary of Scenario Results

The scenarios produce distinct direct and indirect shortage patterns. Gas/water restrictions and water-pipeline failures propagate to electricity, while the VOLL objective provides a common severity measure, rising from 19,423 in S00 to 354,839 in the most severe case, S04.

Table 3: Scenario Outputs Used for Resilience Interpretation

Scenario	Shortage E	Shortage G	Shortage W	Production E	Production G	Production W	Objective
S00	0.000	0.000	0.000	269.953	193.266	331.353	19423.478
S01	39.769	0.000	0.000	230.000	188.500	313.000	67152.949
S02	30.000	15.736	0.000	238.546	165.000	289.336	125018.607
S03	80.000	0.000	33.356	187.321	167.930	215.000	242746.412
S04	80.000	0.000	54.588	186.985	170.096	202.500	354839.428

DISCUSSION

The model combines five useful properties for multi-sector resilience analysis. *First*, its main variables retain a direct engineering interpretation: production, flow, voltage angle, pressure, hydraulic head, served demand and shortage. *Second*, electricity, gas and water are optimized jointly, so a change in one sector immediately affects feasibility in the others and cascade effects emerge within the same problem. *Third*, DC load flow and SOS2 approximations keep the formulation in the MILP class, enabling repeated scenario and Monte Carlo analyses with mature solvers such as Gurobi. *Fourth*, VOLL converts shortages from physically different sectors into a common economic penalty and allows higher priority to be assigned to critical consumers. Finally, *fifth* failures are represented transparently by changes in component availability or capacity, so their consequences can be traced through flows, production and shortages.

The formulation intentionally simplifies sector physics. DC load flow does not replace detailed AC analysis; water and gas relations are approximated; and pump/compressor electricity demand is modeled linearly with flow rather than explicitly with pressure or head differences. These assumptions are suitable for integrated resilience screening, but not for detailed hydraulic, gas-dispatch or electrical-design studies. Future work should therefore improve physical detail while preserving computational tractability.

For decision makers, the model provides not only total shortages but also diagnostic information: affected consumers, initiating sectors, bottlenecks, alternative paths and producers that mitigate failures. This supports scenario comparison and vulnerability ranking of concrete infrastructure components.

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Optimisation and Stochastic Simulation Methods to Reduce Uncertainty in Assessing the Impact of EDT on Military Operations

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KEYWORDS

Strategic wargaming, stochastic optimisation, approximation algorithms, emerging and disruptive technologies, multidomain operations

ABSTRACT

The article is continuation of researches presented in ESM'24, ESM'25. It presents a progress of the verification of methods to support strategic and operational planning. The tools have an open architecture and were extended to stochastic analysis of impact the new technologies and capabilities in strategic planning process of armed forces development. A new aspect of the discussion is the assumption that the effect of emerging disruptive technologies (EDT) on the battlefield is subject to uncertainty (i.e., it is random). It is hypothesized that the use of stochastic optimization methods will reduce the degree of uncertainty. The verification of the hypothesis concerning reduction of uncertainty in evaluation of impact particular EDTs on the battlefield is the main achievement of the paper. For the purposes of the conference, new operation scenario was introduced and tested in the game.

1. INTRODUCTION

A set of analytical and simulation tools equipped with artificial intelligence algorithms supporting decision-making, as an environment for experimenting with Multi-Domain Operations (MDO), including new possibilities – EDT is being developed by the Military University of Technology (WAT), Faculty of Cybernetics. MDO refer to the synchronized employment of military and non-military capabilities across land, air, maritime, space, cyberspace, and the information environment to outpace and overwhelm adversariesAs part of the ongoing scientific cooperation between NATO Allied Command Transformation (ACT) and WAT, the subsequent experiment was conducted in the form of tabletop exercises. The scenario of the experiment to study impact of EDTs on MDO and to reduce the uncertainty in the assessment is derived from the project designated as ACT INN-01062 - Wargame for Collective Defence Operational emerging disruptivePlanning Process of EDT Supported MDO. The analysis, modelling and experimental results presented here are a continuation of the research demonstrated in the ESM'24 publication entitled Strategic Wargame Toolset for the Investigation of Emerging Disruptive Technologies Impact in Multidomain Operations (Najgebauer A., Antkiewicz R., Maślanik D., 2024) and The Simulation and Analytical Toolset for Multidirectional Allocation Problem of Operational Analysis Informing Strategy Planning Process. (Najgebauer A., Antkiewicz R., Maślanik D., ESM 2025) and Computational Intelligence Used for Strategic Considerations on Disruptive and Emerging Technologies for Multidomain Operations (Najgebauer A., Antkiewicz R., Maślanik D., MilCom 2025). The problem of studying the impact of an EDT in MDOs have been solved with the following key points of the method: (1) A modular approach to forming response forces for potential threats is proposed. (2) The introduction of a division into kinetic and non-kinetic modules, including EDT modules. (3) EDT modules (Command and control (C2) technologies including artificial intelligence (AI), Cyber, Electronic Warfare (EW) and Space technologies, Unmanned Aerial Vehicle (UAV) and Unmanned Aerial Assault Vehicle (UAAV) technologies) with different levels of maturity. (4) Proposals concerning the impact of non-kinetic EDT modules on kinetic operations. (5) Various tasks related to the allocation of forces are being developed and solved in order to respond to potential military threats. (6) The participation of EDT modules in response forces, as well as their technological maturity, forms the basis for evaluating their influence on the outcomes of military operations. The particular hypothesis were verified in the experimental wargame: The impact of EDT on military operations was assessed with the aid of stochastic simulation and optimisation methods, which were implemented and used to reduce uncertainty. Source of uncertenity is described in the next chapter.

2. ASSUMPTIONS TO ALLOCATION OF FORCES

The verification of EDT effectiveness in the realisation of multi-domain operations was carried out for conflict scenario variants defined by:

1. EDT availability;
   a. no EDT
   b. EDT available
2. time to crisis occurrence:
   a. short time to conflict
   b. long time to conflict.

It is assumed that a short time to conflict means financial constraints and limitations in the ability to acquire new reference modules. In the long term, it is possible to consider the issue of no budgetary constraints and focus on dimensioning the armed forces in such a way as to ensure the

fulfilment of the tasks at hand, given the known strength of the enemy.

Let us define the following denotations:

- $W_C = \{1,..,N_C\}$, W_C - the set of number of types of weapon system, N_C - number of weapon system types (for all potential members of analyzed conflict)
- $cpi_w, w = 1,..,W$ - where cpi_w is a combat power index for weapon system w.

In order to formulate some success conditions of military operation, we define category of weapon systems as follows: tanks - W_T, infantry fighting vehicle (BMP) - W_{AV}, long-range anti-armour weapon (LR) - W_{LR}, short-range anti-armour weapon (SR) - W_{SR} short-range artillery - W_{SArt}, multiple launch rocket system - W_{MLRS}, tactical ballistic missiles - W_{TBM}, infantry weapon (mortars, small arms) - W_{inf}, attack helicopters - W_H, air defence weapons - W_{AD}. We assume that:

$W_{WS} = \bigcup_{n \in WC} W_n$

$W_C = \{T, AV, BMP, LR, SR, SArt, MLRS, TBM, inf, H, AD\}$.

$\forall n_1 \neq n_2 : W_{n_1} \cap W_{n_2} = \varnothing$

Now, we can define vector of combat power: $\overline{CP}_{F(E)}(i) = \left(CP_{F(E)}(i), (CP^n_{F(E)}(i))_{n \in W_C}\right)$, where:

- $CP_{F(E)}(i) = \sum_{w=1}^{W} cpi_w \cdot n^w_{F(E)}(i)$ - the general relative combat power of the reference module type i of side F or E is calculated as follows;
- $CP^n_{F(E)}(i) = \sum_{w \in W_n} cpi_w \cdot n^w_{F(E)}(i), n \in W_C$ - the relative combat potential of weapons category $n \in W_C$

where $n^w_{A(B)}(i)$ is a number of weapon system type w in the reference module type i of side F(E).

We assume knowledge of the structure of the opponent's forces in the form of: $\bar{X}_E = \left(x_E(i)\right)_{i=1,..N^E_{RMd}}$ where $x_E(i)$ – number of reference modules of type i and opponent forces: $x_E(i) \in \{0,1,...,I\}, i = 1,..N^E_{RMd}, I$ – max number of modules

Commander decision:

$$\bar{X}_F = \left(x_F(i)\right)_{i=1,..N^F_{RMd}} \quad (1)$$

– the structure of the armed forces (AF) of side F;

where: $x_F(i)$ – number of reference modules of type i for own forces.

, Given the aforementioned assumptions and notation, we can formulate two variants of the problem of allocating our own forces.

Variant I

The problem: Given the enemy's specified forces, determine the necessary own forces to hold the enemy at a depth no greater than g_{def}, while maintaining the ability to continue operations.The problem can be formulated as the following optimisation task:

$$\min_{\bar{X}_F}(F_1(\bar{X}_F), F_2(\bar{X}_F), F_3(\bar{X}_F))$$

where values of $\bar{X}_E, top$ are fixed, and *top* means type of terrain, subject to the constraints:

$S^T_E(\bar{X}_F, \bar{X}_E, T_D) \leq g_{def}$

$g_F(\bar{X}_F, \bar{X}_E) = Fl\left(D; A; X_D; X_A; top\right)[I] \leq d_F$

$g_n(\bar{X}_F, \bar{X}_E) = \frac{CP^n_F(\bar{X}_F)}{CP^n_E(\bar{X}_E)} \geq d_n, n \in W_C$

$g_{aA}(\bar{X}_F, \bar{X}_E) - \frac{CP^T_F(\bar{X}_F) + CP^{LR}_F(\bar{X}_F) + CP^{SR}_F(\bar{X}_F)}{CP^T_E(\bar{X}_E) + CP^{AV}_E(\bar{X}_E) + CP^{BMP}_E(\bar{X}_E)} \geq d_{aA}$

$g_{AD|AIR}(\bar{X}_F, \bar{X}_E) = \frac{CP^{AD}_F(\bar{X}_F)}{CP^H_E(\bar{X}_E)} \geq d_{AD|AIR}, n \in W_C$

$\bar{X}_F, \bar{X}_E \geq 0$ where M is a very large value.

$\bar{X}_F, \bar{X}_E \leq M$

$F_1(\bar{X}_F)$ - maintenance cost

$F_2(\bar{X}_F)$ - set-up cost

$F_3(\bar{X}_F)$ - potential/personnel losses

$S^T_E(\bar{X}_F, \bar{X}_E, T_D)$ - the depth to which the opponent has penetrated our defense

Variant II

The problem: determining own forces with a given ENEMY troop structure, a given desired defence time at a given depth, with constraints on force availability and cost constraints.

$$\min_{\bar{X}_F}\left(\sum_{m \in WC \cup \{F,E,aA,AD|H\}} max\{0, d_m - g_m(\bar{X}_F, \bar{X}_E)\} + max\left\{0, \frac{S^T_E(\bar{X}_F, \bar{X}_E, T_D) - g_{def}}{g_{def}}\right\}\right)$$

$\bar{X}_F, \bar{X}_E \geq 0$

$\bar{X}_F, \bar{X}_E \leq M$

$\bar{F}_1(\bar{X}_F, k) + \bar{F}_2(\bar{X}_F, k) \leq B_F(k)$

$\bar{F}_1(\bar{X}_F, k)$ - the cost of acquiring new units over the *k* next years

$\bar{F}_2(\bar{X}_F, k)$ - the cost of maintaining units owned and acquired in the *k* next years.

$B_F(k)$ - the amount of the total budget for k the coming years.

3. HYPOTHESIS INTERPRETATION AND VERIFICATION METHOD

In this paper, we assume that the effect of EDT on the outcome of a fight is not deterministic. We assume that it is uncertain. It follows from this assumption that, for every feasible solution $\bar{X}_F$, the values of the function:

- $F_3(\bar{X}_F)$
- $g_F(\bar{X}_F, \bar{X}_E)$
- $S^T_E\left(\bar{X}_F, \bar{X}_E, T_D\right)$

are random variables.

Use of optimization and stochastic simulation methods will reduce uncertainty when assessing the impact of EDT on military operations.

Source of Uncertainty

- Lack of reliable data on the parameters of the impact of the latest technologies on the course of warfare.

- Uncertainty regarding the effectiveness of the latest technologies:
    - CYBER – high uncertainty regarding the disruption of information in terms of content and availability, confidentiality, and reliability of data depending on the attack used, as well as its use in the appropriate phase of combat.
    - EW – moderate uncertainty of the results obtained in terms of jamming/interfering with radar, communications, guidance, fire control and aircraft on-board systems.
    - GEOINT – moderate uncertainty resulting from various delays in the delivery of information depending on the type of reconnaissance used.
    - UAV (recon) – moderate uncertainty resulting from the capabilities of the reconnaissance technologies used, technical data of solutions, weather conditions.

Reduction of Uncertainty

Stochastic methods are used to reduce the sensitivity of the obtained solutions to the non-deterministic variability of conditions described in sources of uncertainty, compared to solutions obtained using deterministic methods.

Hypothesis Verification Method

1. The hypothesis was based on the high uncertainty about the effectiveness of EDTs on the outcome of armed (kinetic) combat. The way to account for this uncertainty was to use stochastic models.
2. Two problems can be considered, given the financial resources available to build up the armed forces:
    a. No budget constraints;
    b. Existence of budgetary constraints.
3. In the absence of budget constraints, a force dimensioning problem was formulated in the form of a deterministic and a stochastic optimisation task. The stochastic problem was formulated as follows:

$$\min_{\bar{X}_F}(F_1(\bar{X}_F), F_2(\bar{X}_F), E\{F_3(\bar{X}_F)\})$$
$$P\{g_F(\bar{X}_F, \bar{X}_E) \le d_F\} \ge G_F$$
$$P\{g_n(\bar{X}_F, \bar{X}_E) \ge d_n\} \ge D_n, n \in W_C$$
$$P\{g_{aA}(\bar{X}_F, \bar{X}_E) \ge d_{aA}\} \ge D_{aA}$$
$$P\{g_{AD|AIR}(\bar{X}_F, \bar{X}_E) \ge d_{AD|AIR}\} \ge D_{AD|AIR}, n \in W_C$$

other constraints are the same as in Variant I.

$E\{F_3(\bar{X}_F)\}$ - expected value of friendly force losses.

    a. To compare the solutions to both tasks (deterministic and stochastic version of Variant I), a series (100) of combat simulations and start results were performed, and the expected value of losses was assessed.
    b. The comparison has shown that solving the stochastic problem does not significantly improve the solution obtained from solving the deterministic problem. In our opinion, this is due to the fact that, in the absence of budget constraints, it is possible to design response forces with such combat potential that the introduction of EDT does not cause significant deviations in combat results.
4. In the event of budgetary constraints, a solution to the problem of determining the size of the armed forces has been proposed according to the following procedure:
    a. Using the developed heuristic algorithm, we determine a series of suboptimal solutions to the discrete optimisation task. Considering different solutions results from the properties of heuristic algorithms, which in subsequent attempts may determine different suboptimal (approximate solutions). In this paper, to simplify our analysis, we consider the case of 10 suboptimal solutions.
    b. We simulate the outcome of combat for designated armed forces variants using random (randomly selected) EDT parameters.
    c. We compare variants by determining probability values $P\{F_4(\bar{X}_F^i) < F_4(\bar{X}_F^j)\}, i \ne j, i, j = 1,..., NoTrials$.
    d. We determine this solution $\bar{X}_F^*$, for which $P\{F_4(\bar{X}_F^*) < F_4(\bar{X}_F^j)\} > 0,5,\ j \in \{1,.., NoTrials\} \setminus \{*\}$ where
$$F_4(\bar{X}_F, \bar{X}_E, T_D) = max\left\{0, \frac{S_E^T(\bar{X}_F, \bar{X}_E, T_D) - g_{def}}{g_{def}}\right\}$$
    e. The solutions determined in this way should be the least sensitive to random deviations in EDT performance results.

4. SCENARIO CONDUCTING AND RESULTS

In this point we consider task with budget constrains. Using heuristic algorithms 10 suboptimal solutions were generated, which were then subjected to a random selection procedure for EDT parameters, which was performed 1000 times. Next, the value of the objective function was calculated 1,000 times for each solution based on the independently selected EDT module parameters. By comparing the calculated objective function values for a given pair of solutions, the likelihood of one solution dominating the other was determined. The probability of dominance was calculated as the ratio of the number of dominance instances to the total number of instances (1,000 in this article). Table 1 presents the dominance probabilities for the 10 solutions identified in the tests.It shows, for example, that solution number 2 was not able to dominate solutions 1, 4, 7, 8, and 9 in any of the 1,000 cases of random parameter generation. Solution 2 always dominated solutions 5 and 6, rarely dominated solution 3, and often dominated solution 10.

Table 1. Probabilities of Dominance

p[i>j]	1	2	3	4	5	6	7	8	9	10
1	-	1	1	0	1	1	0	0	0	1
2	0	-	0.01	0	1	1	0	0	0	0.94
3	0	0.98	-	0	1	1	0	0	0	1
4	0	1	1	-	1	1	0	0	0	1
5	0	0	0	0	-	0.99	0	0	0	0.79
6	0	0	0	0	0.01	-	0	0	0	0.06
7	0	1	1	0	1	1	-	0	0	1

8	0	1	1	0	1	1	0	-	0	1
9	0	1	1	0	1	1	0	0	-	1
10	0	0.06	0	0	0.20	0.93	0	0	0	-

On this basis, it was possible to create a table of solution dominance, which is presented below. A value of 1 means that solution i (rows) dominates (it tends to dominate rather than be dominated) solution j (columns). A value of 0 means that solution i dominates solution j, while no value indicates that there is no dominance relationship between the given solutions.

Table 2. Solution Dominance Table

D: i>j (p[i>j] > p[j>i])	1	2	3	4	5	6	7	8	9	10
1		1	1		1	1				1
2	0		0	0	1	1	0	0	0	1
3	0	1		0	1	1	0	0	0	1
4		1	1		1	1				1
5	0	0	0	0		1	0	0	0	1
6	0	0	0	0	0		0	0	0	0
7		1	1		1	1				1
8		1	1		1	1				1
9		1	1		1	1				1
10	0	0	0	0	0	1	0	0	0	

It also made it possible to illustrate the dominance of solutions using the directed graph presented below, where each node represents forces allocation vector and each edge represents relation of dominance.

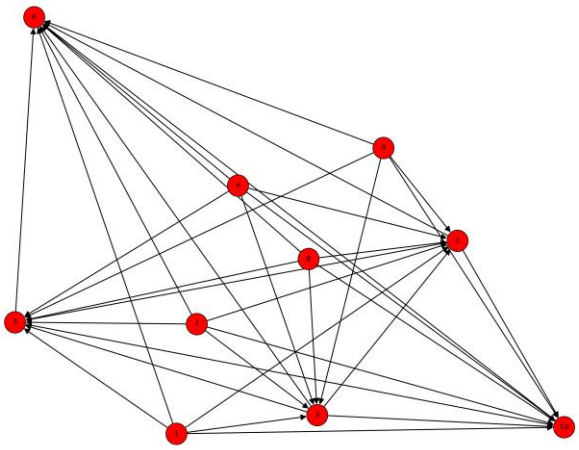

Fig1. Stochastic Dominance Graph

The dominance table and dominance graph, presented above, clearly show the chain of dominance presented below.

$$1 \sim 4 \sim 7 \sim 8 \sim 9 > 3 > 2 > 5 > 10 > 6$$

This means that equivalent solutions 1, 4, 7, 8, 9 dominates solution 3, which dominates solution 2, which dominate solution 5, and so on. Solution 6 proved to be the least stable to changes in the randomly selected EDT parameters. Since the five solutions proved to be equally good, the first one—Solution 1—was used for further analysis.

The pie chart below illustrate the allocated EDT modules of the dominant solution.

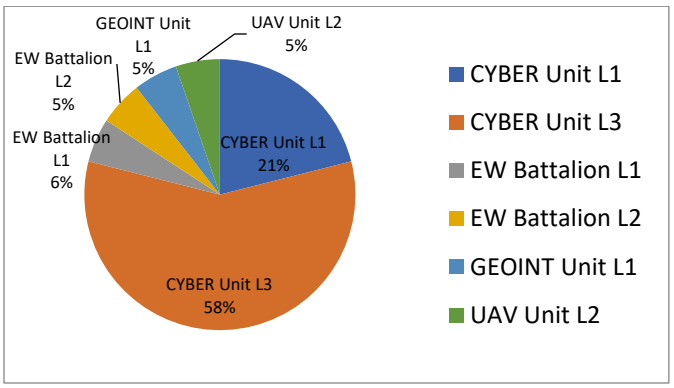

Fig 2. Allocated BLUE EDT Modules.

The symbols L1, L2, and L3 denote the respective maturity levels of the EDT modules. The higher the number, the higher the maturity level.

The table below illustrate BLUE and RED side potential vectors.

Table 3. Combined Potential Vectors

Potential type	BLUE	RED
IFV/APC with AT capability	1 147.43	2 632.85
IFV/APC without AT capability	49.92	114.66
Armored	2 565.10	5 469.48
AT	2 472.74	446.41
SR Artillery	171.29	318.72
MR Artillery	572.58	2 491.96
AA	711.24	1 361.04
Helicopter	0.00	0.00
INT aviation	36.07	163.11
TAC aviation	0.00	0.00
Infantry	17.65	0.00
TBM	26.75	62.65
UCAV	0.00	0.00

Finally, the table below shows the total potential of both sides (affected by terrain and EDTs influence), as well as the initial ratio of RED to BLUE potentials. Potential ratio means quotient of total RED potential divided by total BLUE potential.

Table 4. Total Potentials and initial Force Ratio

Total BLUE potential	9 584.23
Total RED potential	13 635.21
RED to BLUE initial force ratio	1.42

SUMMARY

The particular hypothesis were verified in the experimental wargame - the impact of EDT on military operations was assessed with the aid of stochastic simulation and optimisation methods, which were implemented and used to reduce uncertainty.

This article presents the modified set of software tools based on practical feedback from participants in previous war game held in La Spezia, Italy in 2024 under the auspices of NATO Allied Command Transformation (NATO ACT) and Tuzla , Turkey in 2025. Using a scenario and hypothetical data, the

article allows for the development and examination of simulation based stochastic method. The allocation algorithm was refined and analysed in detail and, finally, the results of this analysis were summarised, confirming its effectiveness and usefulness of the method. Military expert confirmed the possibility of use the method for real strategic planning process.

The results obtained suggest that, for a problem without budget constraints, formulating the optimization problem as a stochastic one does not improve the results significantly. Considering the stochastic version of the optimization problem becomes significant in the case of a problem with budget constraints. The proposed method for selecting dominant solutions allows for the determination of solutions that are least sensitive to random fluctuations in the values of the EDT parameters.

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WHEN WILDFIRE BREAKS THE GRID: A VALIDATED EXTREME-EVENT TESTBED FOR INFRASTRUCTURE RESILIENCE

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ABSTRACT

Wind-driven wildfires are one of California's most serious reliability threats: the January 2025 Eaton and Palisades fires showed how fast a Santa Ana-driven front can overwhelm terrain, damage distribution assets, and cascade into grid failures. We introduce a calibrated fire–grid co-simulation testbed, validated against the official CAL FIRE/NIFC perimeters with Dice coefficients of 0.906 and 0.952 and burned-area errors of -2.2% and $+3.7\%$. The testbed couples a Rothermel/Cell2Fire-style cellular automaton, driven by real terrain, fuel, wind, and slope data, to a pandapower distribution-grid model through a radiant-heat damage mapper, and evaluates four scripted suppression tiers as counterfactuals, producing IEEE 1366 resilience metrics – including SAIDI and fraction of customers connected – across all interventions. Fully open source, built entirely from open geospatial and grid data, and validated by 181 automated tests, the platform gives emergency planners and resilience researchers a practical, evidence-based way to examine rare but high-impact wildfire–grid cascades before the next major wind event arrives.

INTRODUCTION

Wind-driven wildfires are no longer rare anomalies in California, or anywhere else. Climate attribution studies now tie this shift directly to global warming: the record-breaking 2025 European fire season has been formally attributed to climate change (Keeping et al. 2026), and projections for Australia's fire-prone eucalyptus ecosystems show 20- and 50-year return-interval extreme fire-weather events becoming 2–4 times more likely under 3 °C of warming (McGloin et al. 2026). California's grid is squarely within this intensifying, multi-continental trend: wildfire is now one of the most serious and predictable threats to grid reliability. The January 2025 Eaton and Palisades fires made this unmistakably clear when fast-moving flames damaged and de-energized distribution infrastructure even without a preemptive shutoff, becoming the primary driver of grid failure and compounding an already severe disaster. These events exposed a critical gap: utilities and regulators still lack a simulation environment that can show, with evidence, how a real California fire behaves and how those dynamics cascade into electrical grid degradation.

Studying these cascades requires a testbed that is physically grounded in observed fire behavior, coupled end to end to a resolvable grid model, and validated against what actually happened on the ground. Existing fire-grid studies tend to fall short on one side or the other: grid-centric approaches treat fire-caused outages as external, unvalidated scenarios, while fire-centric approaches simulate spread without a standards-based, resolvable grid-impact model; very few combine both.

This paper presents such a testbed, evaluated as a modelling and simulation artifact: a Rothermel/Cell2Fire-style wildfire cellular-automaton model with a wind- and slope-modulated rate of spread over a Moore neighbourhood, calibrated against the official CAL FIRE/NIFC perimeters of the 2025 Eaton and Palisades fires at Dice coefficients of 0.906 and 0.952; a fire-to-grid coupling that maps burned cells to de-energized overhead lines and buses and re-solves the resulting network topology with a power-flow solver at every step, so that fire fidelity and grid-impact fidelity cannot silently diverge; four scripted firefighting-suppression tiers, run as counterfactuals on identical fire scenarios, that quantify how progressively richer intervention bends both the fire-growth and grid-impact trajectories; and an open, fully tested, open-data-only implementation released under an open source license.

BACKGROUND AND RELATED WORK

Semi-empirical fire-spread models remain the workhorse of operational and research fire simulation. Rothermel (1972) expresses the rate of spread of a quasi-steady fire front as a function of fuel-bed properties, fuel moisture, wind speed, and slope, underlying most

subsequent cellular and vector-propagation fire simulators, including the FARSITE family of tools. Pais et al. (2021) reformulate Rothermel-style spread on a raster cellular automaton with Cell2Fire, designed for fast, repeated simulation for strategic landscape and fuel-management planning. The testbed described here follows the same Rothermel/Cell2Fire lineage—a raster cellular automaton with a per-neighbour, wind- and slope-modulated ignition probability over a Moore neighbourhood (cf. Section 4)—but adds an explicit suppression layer and a downstream, timestep-synchronous coupling to an electrical distribution-grid solver, neither native to Cell2Fire.

Prior work shows that California's fire behavior is strongly shaped by terrain, vegetation structure, and post-fire recovery dynamics: LiDAR-derived asset vulnerability and structure proximity (Sultan et al. 2022, Sultan and Pezzillo 2024), post-fire vegetation regrowth (Sultan and Kom 2025), and socio-environmental and climatic impact patterns (Sultan and Kom 2024, Sultan et al. 2026) together indicate that fire spread models must integrate real terrain, fuels, and infrastructure conditions at a spatial resolution fine enough to capture asset-level vulnerability – a requirement directly addressed here.

Utilities in fire-prone regions increasingly rely on Public Safety Power Shutoffs (PSPS)—pre-emptive de-energization of lines forecast to lie in the path of critical fire weather—to reduce ignition risk during high-wind conditions; Huang et al. (2023) survey the policies, decision models, and data behind this practice, in use by California utilities since 2012 and at scale since 2019. CPUC's Wildfire Mitigation Plans further require utilities to demonstrate risk-informed decision making for PSPS, asset hardening, and operational response, but existing tools do not provide a perimeter-validated, grid-coupled fire model capable of supporting these requirements – a gap a calibrated testbed can fill.

This mitigation, however, addresses only the grid-to-fire direction of a two-way hazard chain. Mao et al. (2026) synthesize five domains of evidence for the reciprocal, fire-to-grid direction—physical damage from flame, radiant heat, and smoke; infrastructure-initiated ignition; cascading outages; and multi-scale monitoring—and call for "feedback-coupled fire/grid simulations" as a research priority. Post-event evidence from the January 2025 Eaton and Palisades fires substantiates both directions within a single incident: electrical fault-sensor data placed a transmission-line disturbance seconds before ignition (The New York Times 2025), the same windstorm's fire subsequently damaged Southern California Edison's own distribution facilities independently of any shutoff decision (California Public Utilities Commission, Wildfire Safety Enforcement Branch 2025), and a separate investigation found dozens of distribution-line faults within the burning area on circuits the utility did not de-energize despite a red-flag warning (NPR 2025). Coupled fire/grid simulation studies exist, but are frequently either grid-centric, treating fire-caused outages as an exogenous, unvalidated input, as in the risk-rating framework of Sohrabi et al. (2024), which couples a wildfire-progression simulator to power-flow analysis to rank line/node vulnerability but does not resolve grid topology against a validated, evolving perimeter; or fire-centric, simulating spread without a resolvable, standards-based grid-impact model, as in the adversarial-wildfire formulation of Brun et al. (2026), which builds a Rothermel-based fire-spread *set* calibrated against the Eaton and Palisades perimeters to stress-test contingency screening but treats fire as an optimization input rather than an independently validated simulation state. A closely related, physics-based effort by Nematshahi et al. (2025) – sharing authors with Sohrabi et al. (2024) – couples wildfire simulation to power-system operation via dynamic line rating, extending that group's risk-rating work toward continuous line-capacity adjustment rather than the perimeter-driven topology removal used here; both remain grid-centric in that the fire input is not independently perimeter-validated. The testbed presented here is built end-to-end: the same fire-spread state validated against a real perimeter also drives the damage mapper and grid re-solve, so fire fidelity and grid-impact fidelity cannot silently diverge, reinforcing the need for a coupled model where validated fire behaviour directly drives grid degradation rather than assumed or exogenous outage scenarios.

A separate line of reinforcement-learning work addresses generating rare or adversarial training and evaluation scenarios by construction (Baker et al. 2020, Dennis et al. 2020), motivated by the finding that policies trained only on typical conditions degrade under distribution shift (Kirkpatrick et al. 2017). The present testbed does not implement such a generator – its ignition point, wind vector, fuel-moisture offset, and rate-of-spread multiplier are calibrated, fixed parameters (Section 4.5) – but positions itself as the validated substrate for a future learned-scenario-generation loop.

We quantify grid impact using the standard reliability index SAIDI (IEEE 2012), computed at simulation-timestep resolution together with the instantaneous fraction of customers connected, placing our results in the resilience-trapezoid metric space of Panteli et al. (2017).

CO-SIMULATION ARCHITECTURE

The testbed is organized as a linear co-simulation pipeline executed once per discrete timestep: a wildfire cellular automaton produces an updated burn state; a damage mapper translates newly burned cells into removed grid elements; a power-flow solver re-solves the resulting network topology under the current load profile; and a metrics stage records the standards-based re-

silience indicators for that step. All four stages share a single geospatial reference frame, so that a burned raster cell and the electrical asset it may contain are always co-registered.

The simulation is executed using the palaestrAI software package (Veith et al. 2023), which allows for reproducible experimentation: all experiments are defined as experiment documents, from which palaestrAI sets up environments and agents as independent, re-runnable and comparable modules. All code is available from `https://gitlab.com/arl-experiments/socal-wildfires`.

Each simulation run is driven by a fixed scenario-parameter vector $\Theta = \{\text{ignition point}, \text{wind vector}, \text{fuel-moisture offset}, \kappa\}$, where κ is a multiplicative rate-of-spread scaling factor. Θ is set once at the start of a run from the calibration values described in Section 5.5; in the present testbed it is a calibrated constant rather than a learned or adversarially optimized quantity (see Section 7). Grid-side load is driven by scenario profiles from MIDAS, the OFFIS co-simulation scenario framework, and wind input is sourced from NOAA HRRR reanalysis fields for the simulated period.

All spatial inputs are open data. Terrain is taken from the USGS 3DEP / SRTM GL3 digital elevation model, retrieved via the OpenTopography Global DEM API and cropped to each fire's calibration extent; fuel classification uses coarse classes derived from elevation and, where available, the LANDFIRE 40 Fire Behavior Fuel Model (FBFM40) product, with an optional fuel-reclassification pass (Section 5.5). The electrical network is a 2,294-bus/2,595-line pandapower model spanning the Southern California footprint, rescaled per fire's calibration extent; it is an illustrative distribution topology used to demonstrate the fire-grid coupling rather than a utility-provided feeder model of the exact service areas. No proprietary data source is required to reproduce any result in this paper.

WILDFIRE CELLULAR MODEL AND SUPPRESSION

The wildfire domain is a regular raster grid whose extent and resolution are tied to the underlying DEM tile for the fire being simulated (130×182 cells for Eaton, 159×219 cells for Palisades; Section 5.5). Each cell carries one of three states, $s \in \{\textit{unburned}, \textit{burning}, \textit{burned-out}\}$, together with static per-cell attributes (fuel class, elevation, slope-gradient components) and one time-varying attribute, fuel moisture, initialized from the calibrated fuel-moisture offset (Section 4.6) and held constant thereafter. A cell transitions from unburned to burning when the cumulative fire-spread contribution from burning neighbours exceeds a threshold, and from burning to burned-out once its local fuel is exhausted, subject to the containment margin described below. The automaton updates synchronously: at every sub-step, every burning cell evaluates its outgoing spread probability to each of its eight Moore neighbours *before* any state is written back, keeping the stochastic spread process (Section 4.5) well-defined independent of neighbour iteration order.

The **rate of spread (ROS)** from a burning cell to an unburned Moore neighbour follows the multiplicative structure of the Rothermel surface fire spread model (Rothermel 1972), in which the wind-and-slope-free ROS R^0 is scaled by dimensionless wind and slope coefficients:

$$R = \kappa \cdot R^0 \cdot \varphi_w(u, \theta) \cdot \varphi_s(z, \theta) \ . \tag{1}$$

This is a multiplicative reformulation of Rothermel's original additive form $R = R^0(1 + \phi_w + \phi_s)$, equivalent up to a constant offset and adopted by Cell2Fire (Pais et al. 2021) and most raster fire-spread implementations, since it guarantees $R \geq 0$ and composes naturally with the cell-by-cell κ scaling used during calibration (Section 4.6). R^0 is the no-wind, no-slope base ROS for the cell's fuel class, the outcome of Rothermel's reaction-intensity/heat-sink balance

$$R^0 = \frac{I_R \, \xi}{\rho_b \, \varepsilon \, Q_{ig}} \ , \tag{2}$$

where I_R is the reaction intensity, ξ the propagating flux ratio, ρ_b the fuel bed's oven-dry bulk density, ε the effective heating number, and Q_{ig} the heat of pre-ignition; the testbed looks R^0 up per fuel class from tabulated Rothermel fuel-model parameters rather than re-deriving Equation 2 at runtime. R^0 is further damped by a Rothermel dead-fuel-moisture coefficient $\eta_M(M_f/M_x)$; above a fuel-specific moisture of extinction M_x, $\eta_M = 0$ and $R^0 = 0$, so fire cannot spread through that cell regardless of wind or slope. Fuel classes range from non-burnable (water, urban, barren) through grass and grass-shrub mixes to chaparral shrub—the dominant fuel type in the Eaton and Palisades footprints—and timber, each with its own base ROS and moisture of extinction, derived from LANDFIRE FBFM40 where available or a coarse elevation-based classification otherwise, with an optional reclassification pass for Palisades (Section 5.5). κ is the global rate-of-spread multiplier from Θ (Section 3.1).

The **wind factor** is the single input to which simulated ROS is most sensitive: at the sustained 15–20 m/s gusts characteristic of a Santa Ana event, the wind term dominates R^0 and φ_s combined, and a heading error of even 15° in the wind vector visibly rotates the simulated fire's long axis away from the true one. The wind factor plays two coupled roles—a scalar multiplier on ROS magnitude and the directional driver of an elliptical spread footprint—detailed below together with its calibration against the real fires.

In the standard Rothermel formulation, the wind factor along the downwind bearing is a power-law function of *mid-flame* wind speed U,

$$\phi_{w,R} = C\, U^{B} \left(\frac{\beta}{\beta_{op}} \right)^{-E} , \tag{3}$$

where β is the fuel bed's packing ratio, β_{op} its fuel-specific optimum packing ratio, and C, B, E are fuel-dependent coefficients. Mid-flame wind speed is obtained from the 10 m NOAA HRRR field via a wind adjustment factor (WAF) accounting for logarithmic decay through the fuel canopy, $U_{\text{midflame}} = \text{WAF}(\delta_{\text{fuel}}) \cdot U_{10\text{m}}$, using the unsheltered-fuel WAF appropriate to the open chaparral and grass-shrub classes dominant in both footprints. Since $\phi_{w,R}$ grows without bound as U^B with $B \approx 1$–2, Rothermel's original formulation imposes a wind-speed limit $U_{\text{limit}} \propto I_R^{1/3}$ above which $\phi_{w,R}$ is capped, reflecting the empirical observation (Rothermel 1972) that spread saturates once wind-driven convective preheating no longer scales with wind energy; our implementation applies the same treatment so extreme gusts in the calibration wind schedule do not produce an unbounded ROS.

Eq. 3 gives the wind factor along the downwind bearing only. A wind-driven fire front spreads in every compass direction simultaneously and is well known empirically to trace an elongated ellipse rather than a circle, with the long axis aligned to the wind. To reproduce this on the raster CA, the per-neighbour wind factor $\varphi_w(u, \theta)$ is evaluated separately for each of the eight Moore-neighbour bearings θ_k ($k = 0, \ldots, 7$, in 45° steps):

$$\varphi_w(u, \theta_k) = 1 + \big(\varphi_{w,R}(u) - 1\big) \cdot \max\big(0,\, \cos(\theta_k - \theta_u)\big)^{\gamma} , \tag{4}$$

where θ_u is the wind's bearing and $\gamma > 1$ is an eccentricity exponent controlling how sharply the directional cosine attenuates spread away from the downwind bearing. This is a per-Moore-neighbour discretization of the elliptical propagation intuition behind FARSITE and Cell2Fire (Pais et al. 2021), evaluated directly rather than by fitting and rasterizing an explicit ellipse at every sub-step. Directly downwind, Eq. 4 reduces to the full Rothermel wind factor $\varphi_{w,R}(u)$; at 90° to it, it reduces to 1. This is the mechanism giving the simulated fire its elongated, wind-aligned footprint, and the primary lever by which the eccentricity of the Eaton and Palisades simulated perimeters is brought into agreement with the true, wind-elongated CAL FIRE perimeters (Section 5.6).

A single scalar wind speed and direction suffices for exploratory scenario runs under Θ (Section 4.6). For the calibrated runs in Section 5, however, a static wind vector is not adequate, since both fires changed dominant run direction over the multi-day incident as the driving Santa Ana gradient relaxed and diurnal sea-breeze effects intervened. We therefore derive a time- and space-varying wind schedule directly from the geometry of the real fire perimeter being calibrated against, extracting the medial growth axis of the historical burned-area raster at successive time slices and converting each segment's direction into the wind-vector input for that portion of the run; wind magnitude remains folded into u in Eq. 3 and tuned, together with κ, so that an extreme Santa Ana wind speed (∼20 m/s) reproduces the 60–80 m/min fire-front advance rates reported for the January 2025 Los Angeles fires. We flag this as a scope limitation: perimeter-derived wind direction is calibration information, not an independent forecast input, and deployment to a hypothetical fire would have to supply wind from NOAA HRRR throughout, without the final perimeter's geometry.

The **slope factor** φ_s is computed from the local DEM gradient and aspect, following the same power-law construction as $\varphi_{w,R}$, $\varphi_s = \varphi_{s,0} \cdot \max(0, \tan\alpha)^2$ with α the terrain slope angle along the bearing to the candidate neighbour and $\varphi_{s,0}$ a fuel-dependent coefficient; the $\max(0, \cdot)$ clamp means downslope spread receives no enhancement, while upslope spread is scaled upward, reflecting that a flame front preheats fuel more effectively when it leans into the slope ahead of it. Because φ_s is evaluated cell-by-cell from real terrain and combines with the directional wind factor of Eq. 4 at the same per-neighbour granularity, the resulting perimeter reflects both the topographic irregularity of the Eaton and Palisades canyons and cases where wind and slope reinforce or oppose one another – necessary for the perimeter-level agreement in Section 5.6, since both fires include canyon segments where terrain- and wind-driven preferred directions diverge by 30–60°.

Given R for a candidate burning-to-unburned transition, the **per-sub-step ignition probability** is

$$p = 1 - \exp(-\,R \cdot \Delta t_{\text{CMA}} \,/\, \delta) \ , \tag{5}$$

where δ is the cell size in metres (scaled by $\sqrt{2}$ for diagonal neighbours) and Δt_{CMA} is the cellular-automaton sub-step duration. This is the standard exponential-waiting-time discretization of a rate process: treating R/δ as an instantaneous ignition hazard rate, Eq. 5 is the probability that a Poisson process with that rate produces at least one ignition event within Δt_{CMA}, correctly saturating to $p \to 1$ at high ROS rather than the un-physical $p > 1$ a naive linear discretization would produce. A neighbour ignites when a draw from a seeded random-number generator falls below p; this is the model's only stochastic element and is fully reproducible given the seed (all calibration and validation results in Section 5 use a fixed seed of 47). A separate containment-margin parameter governs how readily a burning cell transitions to burned-out, controlling fire-front residence time and how quickly the fire can be brought under control.

Fire suppression is modelled as four cumulative,

scripted intervention tiers, run as counterfactuals against an identical Θ for each fire: *phase_0_no_ff*, the unconstrained baseline used for calibration and validation (Section 5), with no firefighting response; *phase_1_air*, in which three aerial tankers apply retardant drops that locally reduce spread probability along treated fronts; *phase_2_air_ground*, which adds ground crews that construct firebreaks blocking spread deterministically along their path; and *phase_3_full_triage*, the full intervention set, which further adds triage prioritization that concentrates suppression resources on fronts threatening the highest-value grid assets.

Mechanically, suppression acts on the automaton through three channels. Retardant drops act multiplicatively on R: a treated cell's effective ROS is scaled by a suppression-strength factor $\rho \in [0, 1]$ for the duration the retardant remains effective, propagating through Eq. 5 into a reduced ignition probability along the treated front; retardant does not extinguish an already-burning cell, only slows its outward spread. Ground-crew firebreaks act structurally: a firebreak cell is marked non-burnable for the remainder of the run (equivalent to forcing $R = 0$ for any transition into it), acting as a hard barrier independent of scenario conditions. Both tiers additionally widen the containment margin on treated cells, shortening the residence time during which a treated cell can still ignite untreated neighbours. Because all four tiers are evaluated against the same ignition, wind, and fuel-moisture scenario, differences in the resulting trajectories (Section 6) can be attributed to the intervention itself rather than scenario variance. The complete state of each phase is persisted to a PostGIS/TimescaleDB store, enabling post-hoc replay at any timestep.

GRID COUPLING, DAMAGE MAPPING, AND VALIDATION

The **damage mapper** converts a raster fire state into a graph-topology change, and is the layer that makes the fire–grid coupling more than a nominal juxtaposition of two independently running simulators.

At initialization, every bus and line endpoint of the pandapower network is projected into the same geographic reference system as the wildfire raster and assigned the index of the raster cell it falls into; line geometries spanning multiple cells are rasterized into the ordered sequence of cells they traverse via a standard grid-traversal (Bresenham-style) algorithm, so that a line crossing a canyon is correctly exposed to fire in an interior cell even if both substations lie outside the burned area. This cell-to-asset index is built once and reused at every timestep, letting the per-step damage scan run in time proportional to the number of newly burning cells rather than network size.

At every timestep, the mapper scans only cells that transitioned to *burning* during that step and, for each, looks up grid elements co-registered to it or to any cell within a fixed radiant-heat clearance buffer. This buffer models the standard utility vegetation-clearance distance: radiant heat flux within it is assumed sufficient to either directly damage the conductor or trigger the preemptive de-energization a utility would perform ahead of an advancing front (a Public Safety Power Shutoff). We do not distinguish these two mechanisms; both collapse into a single removal event once a grid element's clearance buffer is engulfed, since disaggregating them would require utility-side switching-policy data the testbed's open-data constraint (Section 3.2) does not provide.

A grid element identified by the damage criterion is removed from the topology for the remainder of the run; there is no repair or re-energization model, consistent with the timescale of a single incident, over which utility repair crews cannot safely re-enter an active fire perimeter.

The resulting, potentially disconnected network is resolved from scratch at every timestep using pandapower's power-flow solver, with nodal loads supplied by MIDAS scenario profiles (driven by NOAA weather data). Re-solving rather than incrementally patching the network is deliberate: as the fire removes assets, islanding and radial re-configuration can change which buses are served at all, and only a full re-solve correctly identifies which customers remain connected to any energized source.

From each timestep's solved network state we compute the standard IEEE 1366 SAIDI (IEEE 2012), accumulated over the run, together with the instantaneous fraction of customers connected, served and consumed power, and intertie flow between the affected feeder and the rest of the network. Reported at simulation-timestep resolution rather than as an annual utility figure, these metrics trace out a resilience trajectory in the sense of Panteli et al. (2017): a pre-event baseline, a disruption phase as the fire advances, and a recovery phase as suppression contains the front and de-energized segments can potentially be restored.

To establish that the wildfire component reproduces real fire behaviour, and that downstream grid-impact numbers are therefore trustworthy rather than an artifact of an uncalibrated model, we calibrate the no-firefighting baseline (*phase_0_no_ff*) against the official CAL FIRE/NIFC final perimeters of the January 2025 Eaton and Palisades fires. Acceptance requires a Dice similarity coefficient of at least 0.8 between simulated and real burned-area rasters, and a burned-area error within $\pm 10\%$ of the reported real acreage.

Table 1 lists the calibrated scenario constants for each fire together with the resulting validation outcome against CAL FIRE/NIFC ground truth. Both runs use a fixed random seed of 47 for reproducibility and a containment margin of 2; Palisades additionally enables the fuel-reclassification pass referenced in Section 4.2. Both runs pass the acceptance criteria, with the Palisades

Table 1: Calibration constants and fire-perimeter validation against CAL FIRE/NIFC ground truth.

Parameter	Eaton	Palisades
Grid size (cells)	130 × 182	159 × 219
Ignition (lon, lat)	−118.0936, 34.1860	−118.5426, 34.0781
Random seed	47	47
Base speed	14	16
Boundary gain	0.3	0.6
Fuel moisture	0.13	0.08
ROS multiplier κ	1.5	1.5
Fuel reclass.	No	Yes
Containment margin	2	2
Real area	∼14,102 ac	∼23,799 ac
Area error	−2.2%	+3.7%
Dice	0.906	0.952

run—the larger and more topographically complex of the two fires—achieving the higher Dice coefficient of the two.

Both results exceed the Dice ≥ 0.8 acceptance threshold by a wide margin and keep burned-area error to within half of the ±10% tolerance, confirmed to hold not only at the final simulated step but at each fire's peak burned extent. Because the fire-spread state driving these results is the same state that feeds the damage mapper (Section 5.1), this level of perimeter agreement is the direct evidential basis for treating the grid-impact trajectories reported in Section 6 as reflective of what the real events plausibly did to the distribution network, rather than as an artifact of an arbitrarily parameterized fire model.

A key design decision underlying this level of agreement is *burnable-footprint containment*: run without constraint, the calibrated rate-of-spread model overshoots the real perimeter substantially (by +145% for Eaton and +129% for Palisades in burned area), because a pure spread model has no representation of the fuel breaks, terrain features, roads, and active suppression that arrested each real fire at its boundary. We address this by dilating the rasterized real perimeter by a small *containment margin* (2 cells, ≈180 m) and marking fuel outside that dilated boundary as non-burnable, so the fire spreads realistically inside the observed footprint—which is what the Dice score measures—without the free-running model's tendency to run past the true boundary. This is a scenario constraint on the calibration run, not a change to the underlying spread physics of Section 4.

RESULTS AND DISCUSSION

Figure 1 summarizes the calibration and validation outcome from Section 5: for both fires, the simulated burned area tracks the CAL FIRE/NIFC ground truth

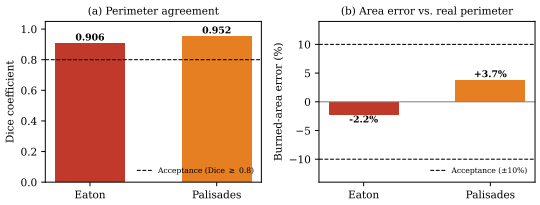

Figure 1: Dice coefficient and burned-area error of the calibrated no-firefighting baseline against the official CAL FIRE/NIFC perimeters.

closely, with Dice coefficients above 0.9 and area errors within a few percent in either direction.

Figure 2 shows the effect of the four suppression tiers on grid resilience and on the fire footprint itself. For Eaton, moving from the no-firefighting baseline to the full-triage tier reduces simulated SAIDI from 1.61 to 0.24 and keeps roughly 5,588 additional acres from burning; for Palisades, the baseline SAIDI is 0.25 and full triage keeps roughly 1,811 additional acres from burning. The magnitude of the SAIDI reduction for Eaton in particular illustrates why a validated, end-to-end coupling matters: because the underlying fire model is perimeter-validated (Section 5.6), the counterfactual comparison across suppression tiers reflects a plausible range of grid outcomes rather than an artifact of an arbitrarily parameterized fire.

These results should be read with two scope caveats in mind. First, the underlying grid model is an illustrative, rescaled 2,294-bus/2,595-line pandapower network (Section 3.3), not a utility-provided feeder model of the exact Eaton or Palisades service areas; absolute SAIDI values are therefore indicative of the qualitative effect of suppression tiering rather than a prediction of the real utilities' historical outage statistics. Second, the four suppression tiers are scripted and hand-designed rather than learned or optimized (Section 4.7); a different allocation of the same aerial and ground assets could plausibly shift the intermediate tiers' outcomes, though the monotonic trend from no suppression to full triage is expected to be robust. Given these caveats, the testbed enables counterfactual, standards-based resilience comparisons that only a perimeter-validated fire model can support with confidence, letting a researcher vary suppression policy, Θ, or grid topology and read off the same SAIDI/customers-connected trajectory used here.

CONCLUSION AND FUTURE WORK

We introduced a calibrated, open-data cellular-automaton co-simulation testbed that captures how a real Southern California wind-driven wildfire behaves and how those dynamics cascade into grid degradation, showing that physically grounded wildfire behavior can drive standards-based IEEE 1366 resilience metrics with

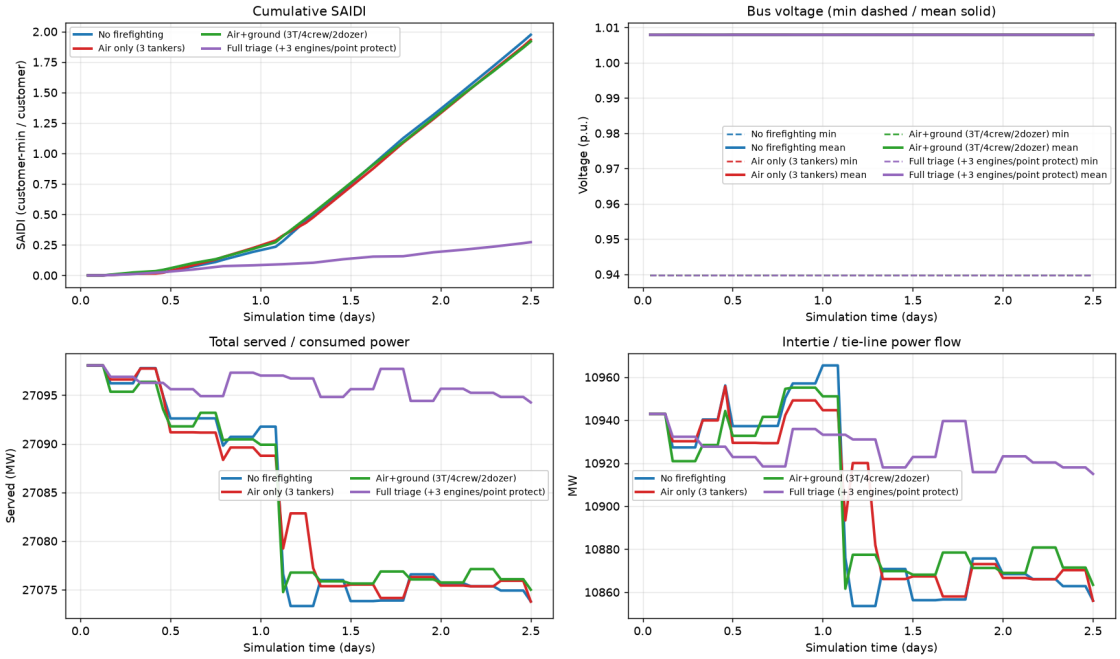

Figure 2: Grid metrics for the four-phases simulation runs.

utility-grade fidelity (Dice coefficients of 0.906 and 0.952 against the January 2025 Eaton and Palisades perimeters). Built entirely from open geospatial and grid data and verified by 181 automated tests, the platform is a reusable, transparent resource for studying rare, high-impact wildfire cascades.

Two extensions follow naturally from the scope boundaries identified in Section 6. First, the global rate-of-spread multiplier Θ is currently a calibrated constant; replacing it with a learned or adversarially optimized scenario generator, drawing on autocurriculum and unsupervised environment-design methods, would let the testbed synthesize physically plausible worst-case scenarios that push restoration policies beyond the historical envelope. Second, extending calibration and validation beyond the January 2025 Southern California fires is essential before any cross-region generalization claim can be supported, enabling comparative studies of terrain, fuel structure, and grid topology as a foundation for future work on resilience planning under extreme conditions.

A further, more fundamental extension addresses the testbed's current scope as a retrospective, rather than predictive, tool: both the burnable-footprint containment used during calibration (Section 5.6) and the perimeter-derived wind schedule (Section 4.2) rely on the final, already-known fire perimeter, which would be unavailable for an in-progress or hypothetical fire. Turning the testbed into a genuinely predictive instrument – an increasingly urgent need given the year-over-year growth in wildfire frequency and severity – requires replacing both calibration aids with forward-only inputs: a forecast-driven wind schedule sourced continuously from NOAA HRRR (or another numerical weather product) in place of the perimeter-derived schedule, and an unconstrained fuel layer in place of the containment margin, with fire arrest emerging endogenously from fuel discontinuities, suppression actions, and terrain rather than a pre-known boundary. We see this as the natural next step, building directly on the validated fire-spread and grid-coupling components already in place.

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MAPPING THE NEXT LINE OF WILDFIRE DEFENSE: GIS-BASED REFCL PRIORITIZATION IN SOUTHERN CALIFORNIA

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KEYWORDS

Wildfire planning, GIS, data-driven prioritization.

ABSTRACT

The Eaton Fire highlighted electrical-infrastructure faults in high-hazard environments as a source of ignition. Rapid earth-fault current limiters (REFCL) can reduce fault energy but are impractical for deployment everywhere. Geographic Information Systems and public data can help identify where to prioritize REFCL. Focused on areas surrounding five recent SoCal wildfire perimeters, our data includes CAL FIRE's Fire Hazard Severity Zones, fire perimeters, California Energy Commission transmission lines, and California substation point data. Results show wildfire exposure was spatially uneven, with clusters of high exposure concentrated around wildland–urban interface areas. With this framework, planners can narrow the search for future REFCL deployment studies and data collection.

INTRODUCTION/LITERATURE REVIEW

Wildfires threaten the public, infrastructure, and the environment: Dry vegetation, steep terrain, electrical systems, transportation, regional economies, and expanding development support large, destructive fires. UCLA's Anderson Forecast estimated billions of dollars of economic losses and long-term infrastructure damage in Los Angeles. Wildfire mitigation is a critical fire-management, electrical-infrastructure, and energy-planning issue.

Electrical failures can become ignition sources during extreme weather. Dense urban areas face lower wildfire exposure than foothill or wildland–urban interfaces (WUI), where dry vegetation and electrical infrastructure overlap. This spatial distribution supports the use of geographic information systems (GIS) because wildfire hazards, transmission infrastructure, and vulnerable landscapes are not evenly distributed.

Rapid earth-fault current limiters (REFCL) can limit current and reduce fault energy during line-to-ground faults on medium- and high-voltage lines, reducing the chance of sparks or arcing from damaged electrical lines and improving supply reliability. Fault energy becomes especially dangerous during dry and windy conditions.

REFCL usefulness depends on conditions, infrastructure, and wildfire exposure. Sathaye et al. (2012) proposed advanced directional protection methods to support resonant grounded networks using REFCLs. The planning question is not whether REFCL should be used everywhere, but where utilities should prioritize deeper feasibility investigation.

Infrastructure, hazards, and land-use patterns connect spatially, so GIS can facilitate utility planning and risk analysis by mapping wildfire risk, transmission exposure, and substation vulnerability to identify overlapping hazards and electrical systems (Alhazmi et al. 2025). Many REFCL studies are highly technical and focus on electrical protection performance instead of geographic planning, creating a research gap between wildfire-hazard mapping and REFCL-deployment planning.

CAL FIRE's Fire Hazard Severity Zones (FHSZ) identify areas with elevated wildfire hazard, and their fire-perimeter datasets show historically burned landscapes. California Energy Commission infrastructure layers include major transmission lines and substation locations revealing infrastructure exposure. However, public datasets lack medium-voltage feeder routes, circuit ownership, fault histories, and detailed protection zones. Thus, GIS analysis cannot fully determine REFCL deployment, but it can support prioritization by identifying areas where wildfire hazard, electrical infrastructure, and WUI-like conditions overlap strongly enough to justify deeper analysis.

Because wildfire risk to electrical infrastructure is spatially uneven and REFCL technology reduces ignition potential from certain electrical faults, GIS can identify where hazards and assets overlap. Limited public research connects these systems with REFCL screening. This project addresses that gap by using publicly available GIS layers to identify where wildfire hazards and electrical infrastructure overlap, warranting more detailed feeder-level analysis.

PROBLEM DEFINITION

This research addresses power-line failures that can ignite wildfires in areas with extreme fire conditions. Wind and dryness can stress power systems, causing them to contact nearby vegetation. The resulting arcing can ignite fires (Mitchell 2013). Though such faults cause a small share of wildfires, they are disproportionately destructive throughout Southern California and conductor lines' exposure to fire is forecast to increase by 40% (Sathaye et al. 2012). California utility companies are already required to assess and mitigate wildfire risk, balancing ignition risk, consequences of power loss, public safety, reliability, and cost (Mitchell 2023). REFCL technology mounted at substations of medium- and high-voltage networks can detect and respond to line-to-ground faults quickly enough to reduce the risk of arcing from a faulted line and has seen limited deployment by Southern California Edison (2022).

From the utility perspective, REFCL deployment must be prioritized. An initial engineering review of a substation and the feeders it supplies assesses existing grounding and protection systems for compatibility. REFCL changes how faulted circuits behave by reducing current during line-to-ground faults, reducing ignition risk for wildfires but requiring upgrades to substation infrastructure. Substations may require compatible automatic reclosers, fault-finding devices, and network upgrades to ensure reliability (Power Systems Consultants Australia 2020). Circuit designs may require other approaches, making a single cost-effective application

incompatible across substations. REFCL technology may require costly four-wire–to–three-wire conversion (Southern California Edison, 2022). Companies also must account for the costs of installation, operational testing, crew training, and ongoing operational oversight. Additionally, substation vary in wildfire exposure and expected risk-reduction value, making deployment at certain sites a higher priority than others.

REFCL must be evaluated alongside covered conductors, undergrounding, vegetation management, sectionalizing, and de-energization strategies. California's Wildfire Mitigation plan requires consideration of strategies to construct, maintain, and operate lines and equipment to minimize wildfire risk (Office of Energy Infrastructure Safety 2026). Utility companies must consider safety, reliability, resiliency, cost, time, and mitigation alternatives (*Cal. Pub. Util. Code § 8386* 2025), highlighting the value of GIS-based screening to narrow the candidate set before expensive engineering and cost–benefit analysis begins.

This study shows how GIS can support electric-distribution planning by organizing data around substations. GIS can support decision making because REFCL value depends on substation location and how their distribution lines intersect with high-hazard landscapes. Here, the primary research question is, where in Southern California do wildfire hazards overlap electrical infrastructure to warrant REFCL review?

METHODOLOGY

Data Selection and Acquisition

Data represents wildfire-hazard zones, recent wildfire perimeters, transmission infrastructure, and substations (Table 1). Transmission lines and substation locations served as a proxy for unavailable public data to develop this screening-level prioritization model. CAL FIRE's FHSZ was the base wildfire-hazard layer.

Table 1. Spatial Datasets used in the analysis

Layer	Source	Year	Shape	Offered	Role	Limitation
Fire Hazard Severity Zones	CAL FIRE LRA/SRA	2026	Poly	Spatial zones	Baseline hazard: wildfire exposure	No ignition probability
California Fire Perimeters	CAL FIRE GIS	2026	Poly	Historical wildfire burn perimeters in California	Five recent fires and surrounding areas	Only final burn extent, no ignition point or conditions
California Transmission Lines	California Energy Commission	2024	Line	High-voltage transmission corridors	Buffer analysis for infrastructure proximity	Less relevant than medium-voltage distribution lines
California Electric Substations	Data Basin	2022	Point	Approximate locations	Proxy nodes during screening	Limited data

Analytical Design and Model Scope

A vector GIS overlay model evaluated overlap between hazard landscapes defined by FHSZ and electrical-infrastructure proxies. Public datasets lacked critical information of feeder routes, circuit topology, protection zones, fault history, ignition records, load data, ownership, and engineering feasibility data. so publicly available transmission-line and substation layers indicated electrical infrastructure exposure.

First, we analyzed corridors using distance-based buffer zones to measure how mapped fire hazards overlapped electrical corridors. Second, we evaluated substation points using fixed 2km buffers to estimate local hazard exposure, proximity to electrical infrastructure, and WUI-like conditions. We translated these spatial measurements into rule-based exposure scores to find where prime REFCL candidates.

Spatial Reference and Data Harmonization

We standardized analyses to a common spatial reference, using area measurements, nearest-feature calculations, and distance buffers. Table 1 shows the datasets used; harmonization ensured all measurements used a consistent coordinate system and unit framework (Esri n.d.). To spatially compare datasets, we standardized all analysis layers to NAD 1983 (2011) StatePlane California V FIPS 0405 (US Feet), WKID 6424 using a Lambert Conformal Conic projection and US Survey Feet as its native linear unit. We projected source layers using Web Mercator into the StatePlane coordinate system using the WGS_1984_(ITRF08)_To_NAD_1983_2011 transformation.

Study-area buffers were 20km; line: 100m, 250m, and 500m; substation: 2km. ArcGIS converted metric to StatePlane units. We stored area outputs in square feet, manually converted to square kilometers using "!SUM_Shape_Area! * 0.000000092903."

We harmonized the FHSZ to classify varying hazard levels. HAZ_CLASS from this layer served as the categorical wildfire hazard variable. We coded nonwildland/nonurban areas, urban unzoned, -999, and unmatched as 0; Moderate as 1; High as 2; and Very High as 3. Recoding provided a consistent numerical representation of mapped hazard categories for further analysis.

Regional Study Area Construction

CAL FIRE wildfire perimeter records provided not a wildfire-risk layer, but a focused analysis envelope (Table 2). Projecting selected records into the StatePlane for final analysis, we dissolved projected perimeters into multipart features, creating a combined regional seed geometry and generating a 20km planar buffer around the fire perimeter, the study-area boundary (Figure 1).

Table 2. Selected Five Fires

Fire name	Year	Alarm date	Containment date	GIS acres
BOBCAT	2020	9/6/2020	10/19/2020	115,998
BRIDGE	2024	9/8/2024	12/31/2024	55,724.199
LINE	2024	9/6/2024	12/23/2024	43,975.602
PALISADES	2025	1/7/2025	1/31/2025	23,448.9
EATON	2025	1/8/2025	1/31/2025	14,056.3

The boundary served as the spatial mask for clipping FHSZ and transmission-line layers before overlay and area calculations. In the substation-exposure model, we applied the boundary after creating the 2km substation buffers to preserve the portion of the substation analysis zones overlapping the study area.

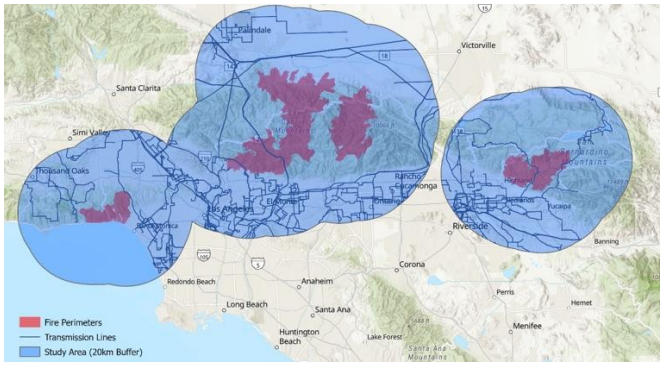

Figure 1. Study Area

Transmission-Corridor Exposure Analysis

Although REFCL commonly applies to medium-voltage networks, it can also apply to high-voltage transmission lines. However, this analysis does not determine whether REFCL is technically feasible on any specific transmission line.

We used distance-based proximity zones around the clipped transmission features as spatial-sensitivity thresholds to examine immediate corridor-adjacent exposure (Figure 2) by calculating the area of each overlay fragment from the projected geometry and summarizing by the established ordinal hazard code for each buffer distance. The main wildfire-hazard exposure chart focused on moderate, high, and very high zones (Esri n.d.).

The buffer distances should not be interpreted as wildfire-spread distances, REFCL protection ranges, or ignition-probability zones. They are a spatial-sensitivity framework for evaluating how much wildfire-prone landscape surrounds transmission corridors.

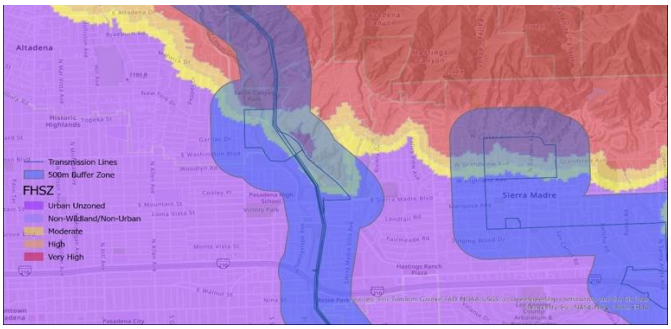

Figure 2. 500m Transmission Line Buffer

Substation Exposure Scoring

We designed the substation-exposure model to highlight substations whose surrounding landscape context suggests wildfire-hazard exposure, calling for a deeper REFCL review. We converted each imported substation point to a local analysis zone so we could measure the surrounding hazard, infrastructure proximity, and WUI conditions consistently. Resulting scores helped prioritize REFCL review.

We calculated wildfire hazard exposure as an area-weighted proportion. We divided each buffer by FHSZ class, and each hazard-class fragment by the total clipped area of that substation: B_i is the clipped 2km analysis zone around substation *i,* and H_j is FHSZ hazard class *j.* The proportion of substation zone *i* located in hazard class *j* is

$$P_{ij} = \frac{A(B_i \cap H_j)}{A(B_i)}$$

$A(B_i \cap H_j)$ is the area of overlap (intersection) between the substation zone and hazard class *j*. $A(B_i)$ is the total clipped area of a substation analysis zone. This process yielded separate proportions in each zone: moderate, high, and very high FHSZ zones. If a zone had no overlap with a measured class of FHSZ, we applied a score of 0 rather than missing data.

We used a WUI-like proxy to represent substation zones containing mapped wildfire hazard and urbanized area, of particular interest to protection initiatives such as REFCL deployment. Wildfires here easily spread to communities. We assigned a proxy score when a buffer contained a combined moderate, high, and very high FHSZ proportion greater than 0 but less than 0.95 to identify zones that could be classified as WUI region proxies. This variable should be interpreted only as a screening proxy.

The final substation exposure score combined three spatial indicators: area-weighted FHSZ exposure, transmission-infrastructure proximity, and WUI-like transition conditions. We assigned heaviest weight to FHSZ exposure with transmission proximity and WUI-like conditions as contextual modifiers. Total score was calculated as *Total Exposure = FHSZ + Transmission Proximity + WUI Proxy*.

Reproducibility and Open-Source Implementation

Rather than depending on ArcGIS functionality, analysis was a sequence of standard vector GIS operations: coordinate transformation, attribute filtering, feature aggregation, distance buffering, spatial clipping, polygon overlay, area calculation, proximity measurement, grouped tabular aggregation, and rule-based classification. We could implement the methods with open-source alternatives: QGIS, GeoPandas, and PostGIS.

We used ArcGIS Pro for analysis, spatial data processing, and map creation: clipping, reprojection, buffering, intersecting, distance-calculation, summary statistics, and map-export. We used Microsoft Excel to create graphs using exported transmission-line exposure data. The Windows desktop was a custom-built workstation, including a Ryzen 7 9800X3D processor, 64GB DDR5 RAM, and an NVIDIA 5070ti GPU to process polygon, line, and point datasets.

RESULTS/DISCUSSION

Transmission Line Buffer Results

The transmission line buffer analysis showed that exposure for each FHSZ increased with buffer distance. This does not mean that risk increases but rather that transmission lines are embedded in broader high-hazard landscapes and ignitions here could rapidly spread. While these results alone tell little about REFCL siting, the same method could be applied to medium-voltage lines. Ultimately, infrastructure exposure should be evaluated spatially, particularly in high-hazard areas (Figure 3).

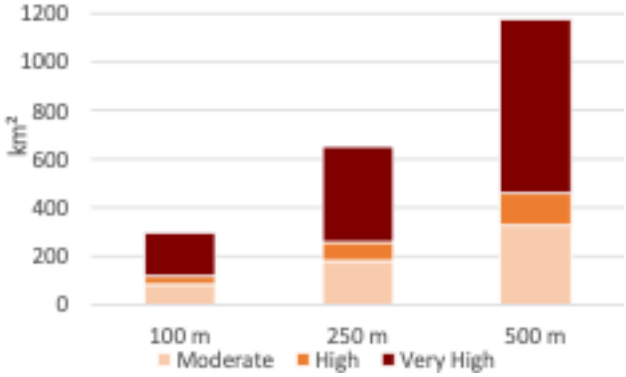

Figure 3. Transmission Line Wildfire Hazard Exposure

Substation Exposure Scoring Results

The substation exposure model produced a spatial gradient, showing lower exposure in urban basin areas with clusters of high exposure near foothill areas with WUI conditions and transmission infrastructure. The inset shows how the Pasadena/ Altadena corridor highlights this trend by providing a representative sample of how substation proximity, WUI conditions, and high fire hazards combine in a higher screening score for nearby substations. The map visualization shows that the scoring framework successfully differentiates

substations surrounded by high fire hazard and substations with limited hazard exposure (with limitations, Figure 4).

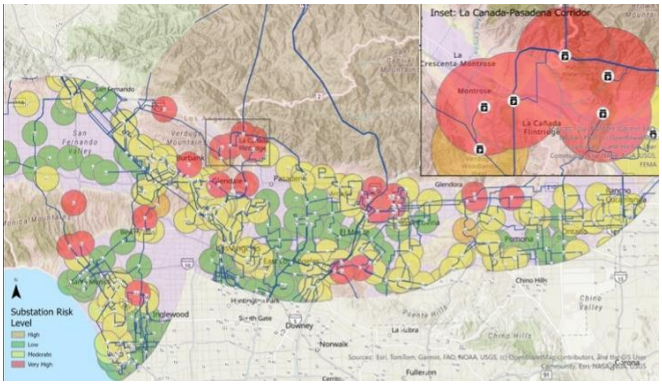

Figure 4. Substation Scoring Exposure Results

Model Behavior and Outliers

Results from the model are useful mainly because they expose limitations. Some areas show high risk, as expected, like substation clusters around foothills, areas already understood to be high-hazard zones. In contrast are a large number of moderate-risk substations in urban areas and even a high-risk example in Inglewood. These substations are not expected to be classified as high risk and likely should not be, but the model recognizes them as high risk likely due to scoring-structure issues, also present when we look at the three substations deep in Angeles Crest, an area known for wildfires, where REFCL would likely be very useful. These issues highlight the problem of attempting to understand substation risk using limited public data without actual distribution lines and other protected-data sources. Reliance on simplified radial buffers misses important context when evaluating REFCL sites. Outliers show this model serves for exploratory screening, not actual guidance on REFCL siting decisions.

Discussion: What Results Mean for REFCL Planning

GIS can identify areas where wildfire hazard, infrastructure proximity, and WUI conditions overlap, creating candidate areas for deeper REFCL feasibility reviews. REFCL prioritization should not be based only on the substation location; many factors contribute to the value calculation at a specific substation. The model attempts to address some factors using proxies but lacks the precision of actual data. Real deployment would require feeder routes, circuit topology, protection zones, fault history, ignition records, load data, ownership, and engineering feasibility data. The model does help narrow the search space of where to look when more sensitive data sets are available.

CONCLUSION

Wildfire risk is a grid-planning problem, highlighted by recent fires. REFCL can reduce fault-related ignition risk and has been deployed in small numbers, but is impractical to deploy everywhere. REFCL value depends on intersections of infrastructure and hazards. Using publicly available data, this project analyzed FHSZ overlap around transmission lines to measure corridor exposure and substation exposure.

These results are not final REFCL siting recommendations and do not consider distribution networks, fault history, and other constraints. Compromises made to develop the model, such as the use of radial buffers, greatly simplify the actual structure of distribution networks.

Future research has great potential to build on the ideas explored here. The integration of utility-provided data, such as distribution networks and precise weather and fuel-stress variables, could help create a much stronger screening tool. With better data and scoring weights, additional refinement could provide a more grounded decision-support model for REFCL planning.

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Simulation in Energy Environments

A CALIBRATED SUBSTATION-BUILDING SIMULATOR FOR DISTRICT-HEATED RESIDENTIAL BUILDINGS AND EVALUATION OF CONTROL STRATEGIES

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Control systems, Energy, Model design, Lumped parameter, Differential equations

ABSTRACT

Evaluating control strategies directly in district-heated buildings is challenging because experimenting with control actions may compromise indoor comfort and operational reliability. This work develops and calibrates a dynamic grey-box simulator of a district-heated residential building using operational data. The simulator couples a thermal resistance–capacitance building model with heat-exchanger and hydronic radiator dynamics. Calibration and validation against historical data demonstrate the model's ability to approximate space-heating demand and provide a computationally efficient environment for comparative controller evaluation. Its use is demonstrated by comparing rule-based and proportional-integral control strategies and by establishing the feasibility of training and evaluating a reinforcement learning agent within the simulated environment, while providing a basis for future extension to building portfolios.

INTRODUCTION

The building sector accounts for approximately 27% of global final energy demand, while space heating (SH) and domestic hot water (DHW) account for nearly 70% of building energy use in advanced economies (International Energy Agency (2025)). With carbon neutrality targeted by 2050 (Rodrigues et al. (2026)), reducing building energy demand and associated emissions is therefore crucial. District heating (DH) has been identified as a promising solution for meeting future heating needs within this transition (Mathiesen et al. (2026)). A DH network distributes centrally produced hot water to connected buildings (see Fig. 1) which are equipped with a DH substation, providing an interface between the network, referred to as the primary side, and the building heating system, referred to as the secondary side. This coupled system exhibits complex transient dynamics driven by weather, occupancy, and heat demand. Consequently, control strategies are required to regulate heat delivery while maintaining indoor thermal comfort and limiting energy use. Traditionally, weather-compensation curves and proportional-integral (PI) controllers have been used to regulate building heating systems through supply-temperature control (Sleptsov et al. (2021)). More recently, supervisory control strategies, including model predictive control (Chaudhry et al. (2021)) and data-driven approaches (Zhang et al. (2023)), have been investigated as alternatives for improving operational performance. However, evaluating such strategies in operational systems requires accurate and computationally tractable models of the coupled substation–building dynamics. To address this, we develop a lightweight grey-box model of a district-heated residential building in Sweden and calibrate it using operational measurements. The model couples substation, hydronic radiator, and building-envelope dynamics while balancing computational efficiency and model fidelity. The resulting simulator is used to compare rule-based and PI control strategies, while reinforcement learning is included as a proof of concept to demonstrate the feasibility of training agents within the environment. Evaluation of the latter serves to verify learning and training practicality rather than to benchmark a fully optimized control strategy. These use cases provide a foundation for future portfolio-level applications and the continued development of reinforcement learning formulations within the simulator.

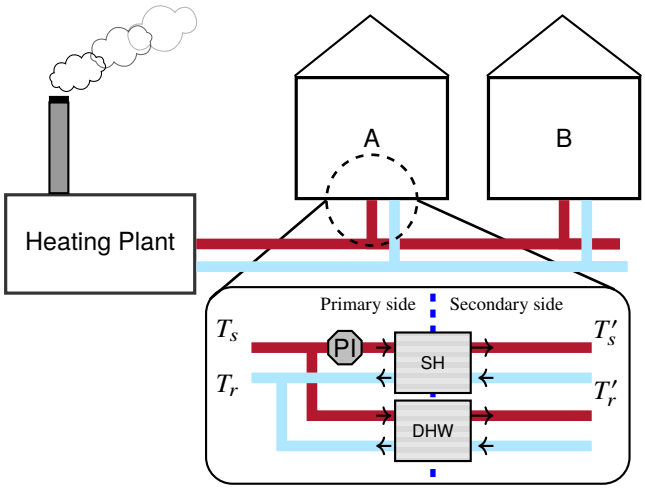

Figure 1: DH Network and a Building Substation

METHODOLOGY

Buildings in a DH network exert a heat load $\dot{Q}_L$, typically being a sum of space-heating, $\dot{Q}_{\text{SH}}$, and DHW $\dot{Q}_{\text{DHW}}$. The former is largely driven by heat losses through the building envelope while the latter is primarily dependent on occupant behavior (Lundström and Akander (2019)). In this work, we focus on modeling the SH component and the substation under consideration records only $\dot{Q}_L$. To formulate and calibrate a physical model for the SH demand, $\dot{Q}_{\text{SH}}$ must be separated from $\dot{Q}_L$ and to this end, the heat demand associated with DHW is modeled as

$$\dot{Q}_{\text{DHW}} = c_p \left(\dot{m}^{\text{dhw}} \Delta T_{\text{circ}} + \alpha \dot{m}^{\text{dcw}} \Delta T_{\text{DHW}} \right), \tag{1}$$

where c_p denotes the specific heat capacity of water, $\dot{m}^{\text{dhw}}$ is the mass flow in the DHW circulation loop, and $\dot{m}^{\text{dcw}}$ is the mass flow of domestic cold water (DCW). The term ΔT_{circ} represents the effective temperature difference associated with circulation losses, while ΔT_{DHW} is the temperature increase from DCW to DHW. The parameter α represents the fraction of DCW used for DHW preparation and is on average in the range 44 – 50% (Lundström and Akander (2019)). A representative DHW temperature difference of $\Delta T_{\text{DHW}} = 45°\text{C}$ is assumed.

Substation and Hydronic Radiator Modeling

The first interaction between the DH substation and the secondary side occurs at a heat exchanger (see Fig. 1). Given a primary-side mass flow $\dot{m}$ and supply temperature T_s, the heat transfer is represented using an effective heat exchanger (Lienhard and Lienhard (2019))

$$\dot{Q}_{\text{SH}} = \epsilon \cdot \min\left(\dot{m}, \dot{m}'\right) c_p \left(T_s - T_r'\right), \tag{2}$$

where $\epsilon \in [0, 1]$ denotes the heat exchanger effectiveness. Simultaneously, invoking energy balance across the exchanger allows us to write

$$\dot{Q}_{\text{SH}} = \dot{m}' c_p \left(T_s' - T_r'\right), \tag{3}$$

where $\dot{m}', T_s', T_r'$ is the mass flow, supply and return temperature on the secondary-side, respectively. Heat is assumed to be delivered to the indoor air through an effective hydronic radiator system. The corresponding heat output is written as (Muniak (2017))

$$\dot{Q} = K \left(\Delta T_{\text{LMTD}}\right)^n = K \left(\frac{T_s' - T_r'}{\ln\left(\frac{T_s' - T_i}{T_r' - T_i}\right)} \right)^n, \tag{4}$$

where K is the heat transfer coefficient of the radiator, n is the radiator exponent, ΔT_{LMTD} is the logarithmic mean temperature difference and T_i denotes the indoor air temperature. When coupling this to a building model driving a SH load, energy balance is enforced such that T_r' dictates that the heat transferred by Eq. (2) equals the heat emitted by Eq. (4). To close the simulation loop, the primary-side mass flow $\dot{m}$ passing through the heat exchanger is modeled by a PI-controlled actuator valve as shown in Fig. 1 (Sleptsov et al. (2021)). The controller tracks the deviation between a setpoint and measurements and adjusts its output by weighting the error by a proportional and integral part denoted K_p and K_i, respectively. A discretization of the PI controller with time step Δt is applied here.

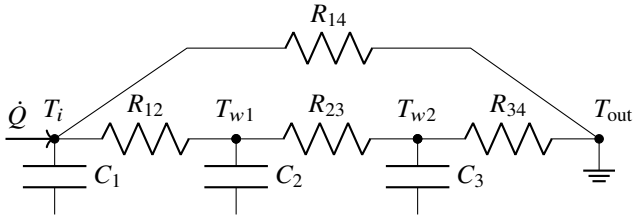

Figure 2: RC Representation of the Building Envelope

Building Grey-Box Modeling

Given substation meter data and limited knowledge of the building properties and occupancy, the grey-box modeling approach known as lumped thermal resistance–capacitance (RC) models (Boodi et al. (2022)) is adopted to represent the building thermal response. Specifically, a 4R3C structure (Wang et al. (2026)) as shown in Fig. 2 is chosen in order to balance model complexity while capturing multiple thermal timescales. To assess the impact of model complexity, a reduced 3R2C formulation is also considered, obtained by removing the state T_{w2} together with its associated capacitance C_3 and resistance R_{34}. Consequently, R_{23} then directly connects the remaining wall node to T_{out}. These models can be written in discrete state-space form

$$s_{t+1} = \mathcal{A} s_t + \mathcal{B} u_t, \tag{5}$$

where $\mathcal{A}, \mathcal{B}$ are the state update matrices, constructed using a zero-order hold assumption on the input vector $u = \left(\dot{Q}, T_{\text{out}}\right)^\top$ wherein T_{out} is the outdoor temperature and $\dot{Q}$ is given by Eq. (4). Here, $s = \left(T_i, T_{w1}, T_{w2}\right)^\top$ is the state vector, representing one indoor state and two hidden wall states. Heat transfer through the walls are governed by R_{12}, R_{23}, R_{34} whereas R_{14} represents ventilation losses.

CALIBRATION & CASE STUDY

Calibration makes use of two datasets, hereafter denoted as D1 and D2, from a nine-story residential building comprising 35 apartments in Sweden. D1 provides high-resolution measurements over a limited time span, whereas D2 contains lower-resolution data spanning a substantially longer period. To ensure sufficient temporal coverage, the RC model is calibrated and validated using D2, while the heat-exchanger, radiator, and operational parameters are identified using D1. D2 contains approximately ten years of hourly measurements of primary-side mass flow, supply and return temperatures,

outdoor temperature and cold-water consumption. Due to the latter, the DHW preparation term in Eq. (1) is estimated and subtracted from the total heat consumption, while circulation losses cannot be explicitly separated. By comparing $\dot{Q}_L$ and $\dot{m}^{\mathrm{dcw}}$ during summer periods, for which $\dot{Q}_L \approx \dot{Q}_{\mathrm{DHW}}$, we estimate $\alpha \approx 43.85\%$. The resulting SH demand is used as input to the RC calibration, spanning the period January 2022 to January 2025, using a chronological 80/20 split for parameter estimation and validation.

D1 contains approximately one year of 10-minute measurements. In addition to most variables available in D2, it includes secondary-side supply and return temperatures, and the mean indoor temperature from 14 distributed sensors. Measurements of the DHW circulation circuit include $\dot{m}^{\mathrm{dhw}}$ and ΔT_{circ} but not $\dot{m}^{\mathrm{dcw}}$, thereby enabling estimation of the first term in Eq. (1) but not the second.

Substation Parameter Identification

Ordinary least squares is used to estimate ϵ in Eq. (2), with the fitted value constrained to the range $[0, 1]$. Data from the heating season is used, since estimating $\dot{m}'$ from Eq. (3) is less sensitive to measurement noise when $T'_s - T'_r$ is not close to zero. The parameters in Eq. (4) are estimated using nonlinear least squares. Since less than half of the apartments have indoor temperature sensors, a single aggregate radiator response is fitted using secondary-side temperatures and the mean indoor temperature $\bar{T}$ computed from the available sensors. Consequently, n is treated as a free effective parameter rather than being constrained to the typical range of 1.3–1.5 (Muniak (2017)). Its fitted value should not be interpreted as the physical exponent of an individual radiator, as the aggregate response also absorbs unmodeled pipe losses, spatial temperature variations, heterogeneous radiator characteristics, and measurement limitations. Finally, the mass flow controller parameters are selected by taking K_p directly from D1. Since the deployed PI controller operates at a higher update frequency than the simulator, K_i is scaled to the simulation time step of $\Delta t = 10$ min so that the integral contribution remains consistent with the original controller.

RC Parameter Optimization

Currently, the building operates under a proprietary control scheme that maintains the indoor temperature close to a prescribed setpoint. Consequently, indoor temperature reveals limited information on the building envelope dynamics and since D2 lacks these measurements, the model is calibrated using inferred space-heating demand rather than indoor temperature trajectories. The RC parameters θ are identified using differential evolution by minimizing the mean absolute error

$$\theta^* = \arg\min_{\theta} \left(N^{-1} \sum_{t=1}^{N} \left| \dot{Q}_{\mathrm{SH},t} - \hat{\dot{Q}}_{\mathrm{SH},t} \right| \right), \tag{6}$$

where N is the number of time steps used during calibration whereas $\dot{Q}_{\mathrm{SH},t}$ and $\hat{\dot{Q}}_{\mathrm{SH},t}$ denote measured and model-predicted demand, respectively. The latter is obtained by first propagating the RC model one time step with $\dot{Q} = 0$, yielding the next indoor state $T_{i,t+1}$, followed by computing the non-negative heating demand required to restore the indoor temperature setpoint $\hat{T}_i = 21°\mathrm{C}$.

CONTROL OBJECTIVE FORMULATION AND EVALUATION

One objective for a building operator is to reduce heat use without substantially compromising occupant comfort. As a demonstration case, we consider night setback, wherein heat supply is reduced during nighttime hours when occupants may be less sensitive to temporary comfort deviations (Dolinay and Vasek (2019)). Since night setback is not currently employed in the studied building, it provides a suitable test case for evaluating control strategies in the proposed simulator. Here, strategy performance is evaluated by recording the total consumption Q_{SH}, pumped water mass m_{tot}, peak mass flow $\dot{m}_{\mathrm{peak}}$, and comfort-violation severity S. Specifically,

$$S = \int_0^T \left(T_{\min}(t) - T_i(t)\right)^+ + \left(T_i(t) - T_{\max}(t)\right)^+ dt, \tag{7}$$

where $(\cdot)^+ = \max(\cdot, 0)$ denotes the positive part of the argument and $[T_{\min}(t), T_{\max}(t)]$ is the comfort temperature band being $[20.5, 21.5]°\mathrm{C}$ during daytime hours (06:00–22:00) and $[20.0, 21.5]°\mathrm{C}$ during nighttime hours. Concretely, all control approaches modify the setpoint value of T'_s:

$$\hat{T}'_s = f\left(T_{\mathrm{out}}\right) + g\left(T_i - \hat{T}_i\right) + \Delta\hat{T}'_s, \tag{8}$$

where $f(\cdot)$ and $g(\cdot)$ are the weather-compensation (WC) and indoor-compensation (IC) curves, respectively, identified from D1. As a refinement to these rule-based functions, a PI controller is used to generate the bounded trim $\Delta\hat{T}'_s \in [-10, 10]°\mathrm{C}$ from the indoor-temperature tracking error. The controller parameters are reported in Table 1, chosen to provide stable responsiveness in simulations.

Table 1: Identified and Calibrated Model Parameters

$R_{14} = 0.5, R_{12} = 0.03, R_{23} = 0.03, R_{34} = 92.94$ [KkW^{-1}]
$C_1 = 80.66, C_2 = 480.72, C_3 = 814.43$ [kWh K^{-1}]
$\epsilon = 1, n = 0.71, K = 7.39$ [kW K^{-n}]
$K_p = 1.67$ [%/K], $K_i = 1.5$ [%/(K · h)] (Mass flow cont.)
$K_p = 7, K_i = 1.25$ [1/h] (Setpoint trim cont.)

Reinforcement Learning

To demonstrate the simulator's applicability to learning-based control, a Soft Actor-Critic (SAC) agent (Haarnoja et al. (2018)) is trained to adjust $\Delta\hat{T}'_s$ via the agent action a_t. The

Table 2: Comparison of Control Approaches for Night Setbacks

Approach	**Setpoint** [°C]	Q_{SH} **[MWh]**	m_{tot} **[tons]**	$\dot{m}_{\mathrm{peak}}$ **[kg/h]**	S [°C-days]
WC	–	67.06	1166.83	1123.93	61.70
WC + IC	21.0	59.02	967.78	1001.47	1.05
WC + IC + PI-Trim	21.0	58.48	995.97	1238.29	2.04
WC + IC	20.8	58.42	951.33	992.68	1.59
WC + IC + PI-Trim	20.8	57.86	973.79	1233.17	2.46
WC + IC	20.6	57.84	935.07	986.98	2.59
WC + IC + PI-Trim	20.6	57.13	948.21	1220.57	3.24
WC + IC + RL-Trim	20.6	58.81	965.47	1326.49	0.30

agent observes current, 2-, 4-, and 6-hour look-ahead values of T_{out} and $\hat{T}_i$, a 24-hour moving average of T_{out}, current values of T_i, $\dot{T}_i$, T_s, and T_r', and two time-of-day features. The reinforcement learning (RL) reward penalizes comfort violations, heat use, infeasible setpoints, and abrupt setpoint changes:

$$r_t = -500 d_t \left(1 + 2.4 d_t^2\right) - 2.4\dot{Q}_t - \left|\hat{T}'_{s,t} - T'_{s,t}\right| - 6\Delta a_t, \quad (9)$$

where d_t denotes the instantaneous comfort-violation severity and $\Delta a_t = |a_t - a_{t-1}|$. We remark that Eq. (9) does not directly penalize e.g. cumulative or peak mass flow, as this experiment is intended to demonstrate that the simulator allows for RL training rather than providing a deployment-ready control policy. Here, both actor and critic networks are multilayer perceptrons trained for 7×10^5 interaction steps. The agent is trained using data from D1 over the period from August 1, 2025 to February 27, 2026. All control strategies are then evaluated on the subsequent unseen period from February 28, 2026 to May 18, 2026, which serves as the validation period for the controller comparison.

RESULTS AND DISCUSSION

After calibration, the identified model parameters are reported in Table 1, and validation results are shown in Fig. 3. The fitted effectiveness reaches the imposed upper bound, $\epsilon = 1$, which is adopted as an effective approximation in the simulations. The identified radiator parameters should be interpreted as effective aggregate quantities. Because, calibration is based on secondary-side temperatures and $\bar{T}$, while the apartment-level number and placement of radiators, their characteristics, and complete DHW separation (c.f. Eq. (1)) are not available. Consequently, the fitted radiator model should be viewed as an aggregate representation of heat transfer to the indoor air mass.

The calibrated RC parameters cannot be directly mapped to physical properties of the building envelope, since multiple parameter combinations may yield similar values of Eq. (6) and several physical effects are represented through lumped states. Qualitatively, the fitted parameters indicate faster heat transfer through ventilation and internal thermal pathways than through the outer wall envelope, while the wall states exhibit substantially higher thermal storage capacity than the indoor air node. Despite its simplicity, the 4R3C model achieves a MAE of 5.8 kW and $R^2 = 0.78$ while remaining discrepancies are partly attributed to incomplete DHW load separation and the use of static values for ΔT_{DHW} and α. In comparison, the 3R2C model achieves an MAE of 5.83 kW and $R^2 = 0.77$, while repeated timing measurements on the same computing platform indicate an approximately 29% reduction in the RC-model runtime per time step. Although this advocates for simpler topologies, no measurable runtime reduction is observed when benchmarking the complete simulation loop, whose cost is not dominated by the RC state update. We therefore retain the 4R3C topology in the present study, while future work should address computational bottlenecks in the overall simulator.

The simulated night-setback control approaches are summarized in Table 2. For $\hat{T}_i = 21°\mathrm{C}$, indoor compensation substantially reduces comfort violations relative to the weather-compensation baseline. Adding a PI trim further reduces heat use at the expense of a modest increase in comfort violations. Lower daytime setpoints (20.8°C and 20.6°C) reduce heat use further, but increase comfort violations, as neither the indoor-compensation curve nor the PI trim explicitly accounts for the time-varying comfort bands. Because PI control is reactive, heat supply tends to increase immediately before and after the morning transition, reducing the benefit of a night setback and this contributes to a larger peak mass flow. When SAC trimming is applied, the comfort severity S is the lowest among all evaluated approaches. However, heat use, pumped mass, and peak mass flow remain comparable to, or higher than, those obtained using PI trimming. We reiterate that this is due to the fact that Eq. (9) penalizes the latter only indirectly such that comfort is prioritized over energy-related objectives. Consequently, the SAC results should be interpreted as a demonstration of RL training and evaluation within the simulator, rather than evidence of a deployment-ready controller.

CONCLUSIONS AND FUTURE WORK

This work presented and calibrated a lightweight grey-box simulator for district-heated residential buildings using op-

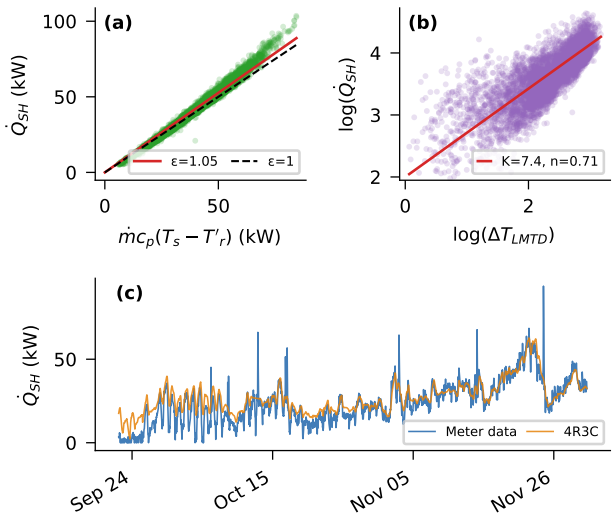

Figure 3: Calibration and Validation Results for the Coupled Substation–Building Model

erational measurements from a real building. The resulting simulator provides a computationally efficient aggregate representation of substation, radiator, and building envelope dynamics and was used to evaluate control strategies for heat savings under comfort constraints. The results indicate that rule-based control can maintain stable indoor temperatures, while PI-based setpoint trimming further reduces heat use with comparable comfort performance. The simulator also provides a platform for training and evaluating reinforcement learning controllers. Future work will focus on improving model fidelity through additional disturbance and occupancy modeling, incorporating an explicit DHW representation, and study alternative reward formulations that explicitly account for peak demand and cumulative mass flow.

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AN ITERATIVE CO-SIMULATION PIPELINE FOR COUPLED MODELS

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ABSTRACT

The study of complex energy systems requires models that capture both system-level dynamics and individual behavior. Energy system optimization models identify least-cost technology pathways but represent human behavior through aggregate projections, abstracting individual decisions. Agent-based behavioral models capture this heterogeneity, but do not see the energy system. Coupling the two model types is non-trivial as they operate at different time resolutions and conceptual paradigms. This paper presents an iterative co-simulation pipeline that couples a national energy system optimization model with a behavior-driven agent-based model of household mode choice and electric vehicle adoption. A shared version-controlled repository serves as the data exchange hub, with automated execution triggered by commits. The proposed pipeline enables reproducible, automated bidirectional coupling while preserving the independence of both models.

INTRODUCTION

Energy systems are complex socio-technical systems where technological infrastructure and human behavior interact continuously. The low-carbon energy transition amplifies this complexity and demands realistic modeling across scales. Large techno-economic energy system models (ESMs) such as STEM (Panos et al. 2023), MARKAL/TIMES (Loulou et al. 2016) and OSeMOSYS (Howells et al. 2011) capture this complexity at the system level. These models usually use a set of policy targets and external constraints to identify the least-cost technology pathway for the energy system. However, because they are computationally intensive, they typically represent behavior at an aggregate level rather than capturing individual heterogeneity. Behavioral aspects are especially important in the mobility domain (Luh et al. 2022), where individual decisions often depend on more than cost alone. The mode choice tends to be influenced by individual habits, while the adoption of an electric vehicle (EV) can be influenced by different aspects such as bounded rationality (Nguyen et al. 2023). Agent-based models (ABMs) can capture this heterogeneity by simulating a population of individual agents. For example, BedDeM simulates Swiss households as a population of individual agents, each making mobility decisions based on their personal attributes, habits, and social influences, grounded in the established psychological theory (Nguyen et al. 2023).

Coupling specialized models, rather than increasing their complexity, can provide a bridge between different modeling approaches and scales, creating more realistic representations (Filatova et al. 2025). The ESM and behavioral ABM coupling should be bidirectional, as fuel prices influence individual choices, while aggregated demand affects the design and cost of the energy system. This coupling can be realized through a co-simulation pipeline, which is the coordinated execution of different models (Gomes et al. 2018). This process is often challenging, as it involves dealing with interdisciplinary design and different conceptual and time scales (Adelt et al. 2023, Filatova et al. 2025). For example, STEM operates over decades with low resolution time steps (Panos et al. 2023), making close synchronization unsuitable and favoring asynchronous approaches such as file exchange (Luh et al. 2022). However, existing model coupling solutions often rely on manual file-based exchange between modeling teams (Luh et al. 2022, Blanco et al. 2019, Torralba-Díaz et al. 2024) demanding effort and domain expertise, which limits automation, reproducibility and traceability of such pipelines (Keppo et al. 2026). Although automated iterative coupling (Torralba-Díaz et al. 2024) and ontology-driven frameworks (Torlini et al. 2026) exist and can provide reproducibility, our approach aims to add lightweight and model-agnostic artifact exchange with minimal manual intervention, providing automated and re-constructable coupling.

This paper presents an automated, iterative co-simulation pipeline for the ESM–ABM coupling, developed within the PROBOUND project (Schumann et al. 2021). The pipeline provides a reproducible Git-based workflow for traceable co-simulation. We demonstrate the approach for a Swiss transport policy scenario, where we show the iterations of both models.

RELATED WORK

Efforts to incorporate consumer behavior into energy system optimization models range from travel time constraints (Daly

et al. 2014), substitution elasticities (Salvucci et al. 2018), and heterogeneous consumer segments (Tattini et al. 2018), to external model coupling (Luh et al. 2022). Coupling approaches include soft-linking transport behavior models with JRC-EU-TIMES (Blanco et al. 2019), linking ABMs with computable general equilibrium models (Niamir et al. 2020), and iterative coupling between electricity market models and ABMs (Torralba-Díaz et al. 2024). To the best of our knowledge, no published work couples an ESM with a psychologically grounded ABM, simulating both mode choice and EV adoption in a single iterative loop. Co-simulation frameworks such as Mosaik (Ofenloch et al. 2022) and HELICS (Hardy et al. 2024), and standards such as FMI (Junghanns et al. 2021), are designed for runtime synchronization between models operating at the same temporal resolution. STEM and BedDeM operate at decadal and annual resolutions, making this solution unfeasible. Our approach instead uses sequential file exchange via a shared Git repository, with a webhook-triggered pipeline automating each iteration.

Another widely recognized challenge in energy system modeling is reproducibility (Pfenninger et al. 2017, Morrison 2018). Workflow tools such as Snakemake (Mölder et al. 2021) address this for single-model pipelines. Our approach applies similar principles to a two-model iterative coupling, providing scenario-level traceability through a shared repository structure and covering the interactions of the models during their joint execution. To the best of our knowledge, reproducible, iterative, and bidirectional coupling has not yet been addressed with a model-agnostic version-controlled architecture.

CO-SIMULATION PIPELINE DESIGN

The pipeline couples two models with different purposes, time resolutions and modeling approaches: the STEM energy system optimization model (Panos et al. 2023) and the agent-based mobility model BedDeM (Nguyen et al. 2023). STEM operates at a decadal resolution to estimate system-level outcomes, while BedDeM simulates annual household decisions influenced by behavioral factors such as costs, travel time, and social interactions. Within BedDeM, mobility behavior is represented through two coupled interacting ABM models: an annual mode choice model and a vehicle purchase model. BedDeM simulates around 3000 agents representing the Swiss household population. Agents' decision-making is based on Triandis' Theory of Interpersonal Behavior. Each agent evaluates available transport options based on cost, travel time, and other attributes, while vehicle replacement and adoption decisions are triggered by vehicle lifetime or changing attractiveness of alternative options (Nguyen et al. 2023). These decisions update the fleet composition and feed back into future mode choices. Aggregated outputs such as trip counts, charging demand, and total mobility demand are produced annually and form the input to STEM for re-optimization of the overall energy system under these constraints.

The interaction between the two models is shown in Figure 1.

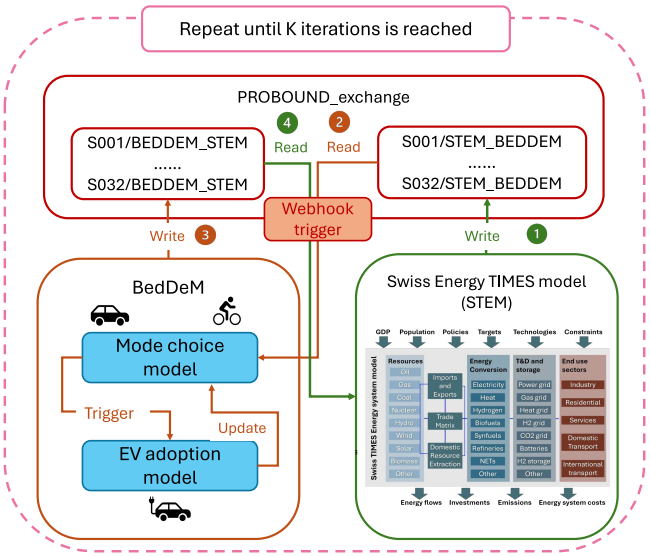

Figure 1: Co-simulation architecture: STEM (Panos et al. 2023) and BedDeM (Nguyen et al. 2023) exchange data via the shared repository. ① STEM commits fleet clusters and fuel prices to `STEM_BEDDEM/`; ② git webhook triggers BedDeM to pull and run; ③ BedDeM commits trip counts and charging demand to `BEDDEM_STEM/`; ④ git webhook triggers STEM to pull and re-run.

STEM provides fuel prices and initial fleet compositions, which are used as input to BedDeM, while BedDeM returns aggregated mobility demand and fleet changes to STEM. Data exchange is implemented via a shared Git repository **PROBOUND_exchange**, which cannot be released publicly because it contains data derived from sources under a non-disclosure agreement. The repository is organized by the scenarios to be investigated by the coupled models (S001–S032). Each scenario contains two interface directories:

- `STEM_BEDDEM/`: decadal STEM outputs (fuel prices and fleet clusters) used as input to BedDeM
- `BEDDEM_STEM/`: annual BedDeM outputs (trip counts, charging demand, and aggregated mobility demand including fleet changes) used as input to STEM

All data artifacts are versioned within the repository, allowing the exchange process to be tracked across scenarios and model runs. Repository events trigger subsequent model executions, enabling automated co-simulation.

Model Input and Output Mapping

Coupling requires alignment of time resolution and model semantics. STEM provides outputs at decadal checkpoint years (2020–2050), while BedDeM requires annual inputs over the simulation horizon. This mismatch is resolved on the BedDeM side by linearly interpolating STEM outputs into annual time series. Fuel prices provided by STEM at decadal

Algorithm 1: STEM–BedDeM Coupling

Input: Scenarios $\mathscr{S}$, shared Git repository $\mathscr{R}$
Output: Fleet composition, fuel prices, mobility demand and CO_2 emissions

```
// Initial setup
```
Run STEM for fleet composition and fuel prices
Map STEM fleet clusters to BedDeM agents
Each side (STEM, BedDeM) registers a webhook on $\mathscr{R}$
```
// On each webhook trigger
```
foreach $s \in \mathscr{S}$ *with new commits from the other model* **do**
  Pull updated results for s from $\mathscr{R}$
  Update local inputs (fuel prices for BedDeM; fleet/demand for STEM)
  Run own model for s
  Commit and push results to $\mathscr{R}$
  if *K iterations reached or other criteria* **then**
    Stop coupling for scenario s

checkpoint years are linearly interpolated to produce annual values written to the BedDeM database according to Equation (1):

$$p(y) = p(y_{\text{start}}) + \frac{p(y_{\text{end}}) - p(y_{\text{start}})}{y_{\text{end}} - y_{\text{start}}} \cdot (y - y_{\text{start}}) \quad (1)$$

where $p(y_{\text{start}})$ and $p(y_{\text{end}})$ are the STEM provided prices at the first and last checkpoint year, respectively, and $y \in \{y_{\text{start}} + 1, \ldots, y_{\text{end}}\}$. Differences in model representation also require harmonization. Vehicle fleets are represented as technology clusters in STEM, but as individual household choices in BedDeM, requiring a translation between aggregate shares and agent-level attributes. Fuel prices differ conceptually, with system-level values in STEM and household cluster level prices in BedDeM that may include taxes and behavioral perceptions. In addition, BedDeM produces agent-level mobility and charging data that are aggregated into sector-level inputs for STEM. These transformation choices may influence the behavior of the iterative loop. All mapping assumptions and file structures are documented and versioned in the shared repository, allowing changes to be tracked across iterations.

Pipeline Automation

On the BedDeM side, the process is automated via a Git webhook that triggers a BedDeM's CI pipeline when new files are pushed to `STEM_BEDDEM/`. The pipeline retrieves the updated data, interpolates the decadal inputs to annual values, executes the BedDeM model, and commits the yearly outputs (Trips, MobilityDemand, E-Charging) to `BEDDEM_STEM/`, which are then pushed back. This incoming update to the STEM model inputs triggers another webhook, which then promptly re-executes the model for the given scenario.

Together, these mechanisms create a fully automated iterative loop, requiring no manual intervention between STEM and BedDeM executions. All exchanged artifacts are versioned under a consistent scenario identifier, allowing previous states to be reconstructed from the repository history.

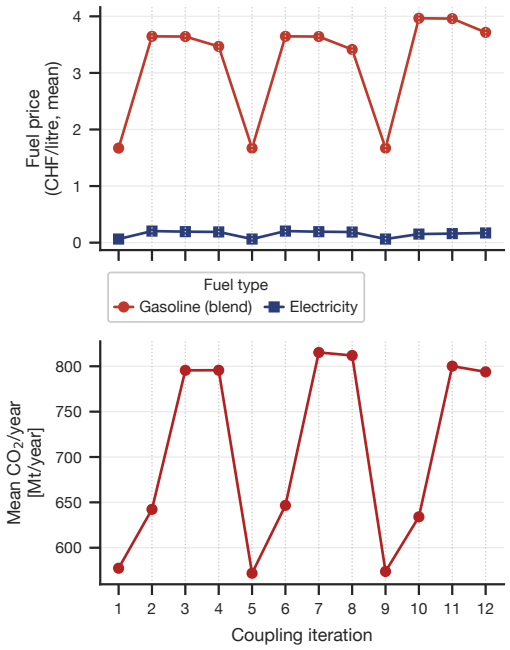

Figure 2: STEM-BedDeM coupling iterations

ITERATION DEMONSTRATION

In the following, we demonstrate the application of our approach in a policy scenario S009, detailed in Schumann et al. (2021). This scenario investigates the effect of increased fuel taxes on household mobility behavior, reflected through changes in the total cost of ownership including purchase costs, fuel costs, and ownership-related expenses as produced by STEM and consumed by BedDeM.

STEM commits updated fuel prices and fleet clusters to the repository. The webhook triggers BedDeM, which returns an updated mobility demand. Figure 2 shows the resulting trajectories across coupling iterations. The top graph gives the mean price of gasoline and electricity per iteration, and the bottom panel the corresponding mean annual CO_2 emissions. Gasoline prices and CO_2 emissions oscillate between iterations, alternating between lower and higher taxes as STEM and BedDeM exchange updated inputs. This comes from the different modeling approaches of both models. STEM reacts to adjusted behavior, and people react to adjusted prices. This pattern illustrates the iterative nature of the coupling. Each model responds to the other's previous output; the loop runs until K iterations or until the exchanged aggregates stabilize. In this case, the K was reached, as the values kept oscillating.

The computational cost of running the optimization model can be high, especially when multiple iterations are required. The automated pipeline executes the full iteration sequence without manual intervention, reducing user interaction in

long-running simulations. In Table 1, we present the total run times of each model and the number of iterations. Out of 313 recorded model run iterations, 64 were removed as statistical outliers, excluding test runs and development commits that do not reflect production run times. Without the automated pipeline, each coupling iteration would have required a manual trigger on both sides, resulting in 498 manual interventions across the 249 clean iterations. Each introduces an additional delay on top of the model runtime, which the git webhook removes entirely.

Table 1: Coupling Iteration Run Times

Model	Model runs	Median	Mean	Total
STEM	249	2h 23m	2h 56m	731h 17m
BedDeM	249	18 min	18 min	75h 06m
Total	498	2h 41m	3h 14m	806h 23m

CONCLUSION

This paper presents a co-simulation pipeline, coupling an energy system optimization model and an agent-based mobility model in an iterative and bidirectional way. Behavioral responses, such as mode choice and vehicle adoption, influence system-level outcomes, which can affect future decisions. This interaction can offer additional insights, although convergence remains dependent on modeling assumptions, aggregation choices, and interpolation over time.

The exchange is implemented through an automated Git workflow in which model outputs are treated as versioned artifacts, reducing the need for manual intervention during co-simulation (e.g., eliminating 498 manual interventions in our demonstration). This supports structured scenario exploration and traceability while introducing challenges, such as large data volumes in file exchange, which can be handled with Git Large File Storage (LFS) or Data Version Control (DVC).

The approach involves simplifying assumptions in time interpolation and semantic mapping between models. Future work may include recalibration of both models to more recent conditions (e.g. a new baseline), integration with extended versions of BedDeM incorporating more detailed representations of perception (Liffey and Schumann 2025) and potential coupling with additional Energy system models, as well as quantifying the coupling effect on the system-level outcomes.

Overall, the pipeline provides a basis for exploring interactions between system-level optimization and individual decision-making, and may offer a transdisciplinary, model-agnostic approach to coupling different model types without having to adjust the models.

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Author Index

Author Index

Abbou R. **211**
Adesanmi A. **249/283**
Alfaro M. **96**
Andrassy V. **313**
Antkiewicz R. **318/326**
Antunes B. **176**
Atanasova T. **91**

Babiarz A. **197**
Balland R. **234**
Becker M. **187**
Bednarska-Czerwinska A. **119**
Beinert D. **286/290**
Bhatia V. **96**
Bodzioch M. **71/77**
Boltuć A. **29/46**
Bühmann L. **300**

Chabanet S. **258**
Charraoui M. **211**
Chávez-Castillo R. **307**
Chevret S. **265**
Chmielewski L.J. **101**
Chovanec D. **313**
Choínski M. **19/61**
Chuquillanqui-Mantari E. **307**
Colloc J. **5**
Cummins E. **96**

Dachkinov P. **111**
Dantan J.-Y. **81/254/269/278**
Dermy O. **258**
Douache M. **234**

Etienne A. **278**

Fagas R. **145**
Faraj A. **249**
Forrestal P. **96**

Gałuszka A. **153/242**
Gałuszka An. **242**
Godot X. **254**
Gomez Echeverri J.-C. . **254**
Grandidier N. **234**
Grzejszczak T. **205**

Haider A.A. **286/290**
Hammer L. **300**
Hansson C. **347**
Hennequin S. **249/283**
Hill D.R.C. **176**
Hinc A. **101**

Jaskot K. **202**

Kalisz S. **145**
Klement N. **269**
Koïta M. **125**
Koller J.A. **286/290**
Kopanov K. **91**

Krajewski J.K. **101**
Kubás J. **313**
Kumar A.-A. **258**
Kumar S. **106**
Kuure P. **161**
Kuźelewski A. **40**

Lanting C.J.M. **54**
Lasak M. **205**
Launay J. **211**
Lee M. **300**
Lienhard J. **187**
Liffey A. **352**
Lokshina I.V. **54**
Luh S. **352**

Maïga O.Y. **125**
Maize M.Y. **125**
Maklavani L.K.D. **269**
Marczyk M. **119/145**
Maślanik D. **326**
Matse D.T. **96**
Mbiydzenyuy G. **347**
Moldovan-Mauer A. **286/290**
Morawiec E. **119**

Najgebauer A. **326**
Nawayseh N. **219**
Nguyen K. **352**

Ochoa A. **340**
Oliver I. **161**
Orłowski A. **101**

Author Index

Petronijevic J. **234/278**
Probierz E. **153/229/242**
Prokop D. **202**
Pudelko A. **119**

Raharijaona T. **258**
Ratby M. **234**
Ristvej J. **313**
Rodriguez C. **340**
Roy D. **249/283**
Rudnicka M. **205**

Sadiki S. **265**
Saggar S. **96**
Sarkar S. **106**
Sava A. **211/265**
Schäfer A. **300**
Schumacker J. **254**
Schumann R. **352**
Siadat A. **254/278**
Skowroński K. **153/242**
Śliwa P. **192**
Stachowiak S. **61**
Stąpor P. **318**
Stief P. **254/265/269**
Struniawski K. **132/137**
Strzoda T. **273**
Sultan V. **332/340**
Swami A. **106**
Szerszeń K. **35**

Tafti M.P.M. **254**
Topçu O. **171**

Veith E.M.S.P. **300/332/340**

Wartel T. **176**
Wellßow A. **300**
Westphal P. **300**
Wojciechowski S. **326**

Yon L. **176**

Zamojski D. **119**
Zhang T. **234/278**
Zieniuk E. **29/35/40/46**